Farthest vertices and a dimension bound for compact hyperbolic Coxeter polytopes

CoxeterPolytopes project

25 October 2026

Abstract

We study a compact hyperbolic Coxeter polytope from one vertex farthest from a fixed interior point. Its incident spherical diagram gives an orthogonal decomposition of a single Euclidean vector of norm less than one. Actual opposite walls, and points obtained from positive combinations of actual walls, impose incompatible lower bounds on that vector. The method has two branches: concave interpolation for A-chains, and prefix interpolation followed by terminal-wall analysis for B- and D-components. We prove that the dimension is at most 17. Common geometric lemmas supply the component estimates; complete actual-rank forest covers then reduce the proof to a small number of whole-component configurations. The last obstruction, a whole D_{18} diagram, is treated by a uniform dotted-head pressure, legally funded reference vectors, the two genuine signed fork directions, and a local calculation at an adjacent A_{18} vertex. Arithmeticity is nowhere assumed.

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1 The theorem and the organization of its proof

Our main result is the following dimension bound.

Theorem 1.1 (Dimension bound)**.** *Every compact real hyperbolic Coxeter polytope has dimension at most* 17*.*

Here compact means that every vertex is finite. The theorem concerns arbitrary Coxeter reflection groups; there is no ground field, lattice, integrality condition, or arithmetic hypothesis. It asserts an upper bound, not sharpness or the existence of an example in dimension 17.

The central object is one vector x, obtained from an original farthest vertex p. Energies of orthogonal components of its incident diagram can be added because they are projections of this same vector. Pressures arising from different auxiliary points cannot in general be optimized separately and then added. Instead, we combine them in one reference vector and retain its complete squared norm.

The case split occurs at the finite irreducible components of the diagram at p. Small components pay uniform energy debits. A large A-component is handled in its zero-sum chain coordinates. A large B- or D-component has a long common prefix; its terminal nodes then require separate analysis. Section 5 records a fresh complete cover at dimension 18, and the separate higher-dimensional cover. A *cover* always means necessary possibilities, not realized examples. Section 6 explains the whole-B_{18} terminal partition. Sections 7–10 give the whole-D_{18} closure.

The final composition has two parts. We exclude every dimension at least 19 by its actual-rank forest cover and local consumers. At dimension 18, the complete cover leaves only a whole D_{18} diagram at the same selected farthest vertex. Neither boundary is obtained by taking a face or deleting a node from a higher-dimensional list. We do not assume that a face is Coxeter, or that the property under investigation is inherited by faces.

2 One original vector and actual geometric points

Use the hyperboloid model with Lorentz signature $(-+\cdots+)$. Every wall has an outward unit spacelike normal n, and the polytope is given by $\langle r, n\rangle \le 0$. Distinct actual wall normals have nonpositive products. Fix a strict interior point z, choose an actual vertex p farthest from z, and write

$$z = E(p - x), \qquad E = -\langle z, p\rangle > 0, \qquad x \in p^{\perp}, \qquad \|x\|^2 = 1 - E^{-2} < 1. \tag{1}$$

The space $p^{\perp}$ is Euclidean. The incident normals at p form a positive definite spherical diagram; compact acute simplicity implies that there are exactly n of them in dimension n. We retain these original normals and this original choice of p, z, x, E throughout an energy argument.

The function $r \mapsto -\langle z, r\rangle$ takes its maximum on a compact hyperbolic polytope at a vertex. Its restriction to a unit speed geodesic has positive second derivative equal to the function itself, so a maximum on a segment occurs at an endpoint; successively passing to proper faces proves the assertion. Consequently

$$-\langle z, r\rangle \le E \quad \text{for every actual point } r \in P. \tag{2}$$

At an adjacent vertex q, write $z = E_q(q - y)$. Equation (2) gives a comparison factor E/E_q, not a claim that q is farthest.

2.1 Positive inverse matrices and feasible projections

Lemma 2.1 (Inverse positivity)**.** *If a positive definite symmetric matrix B has nonpositive off-diagonal entries, then B^{-1} is entrywise nonnegative.*

Proof. Choose s greater than the largest eigenvalue of B and all its diagonal entries. Then $M = sI - B$ is entrywise nonnegative and has spectral radius $s - \lambda_{\min}(B) < s$. Thus $B^{-1} = s^{-1}\sum_{k\ge 0}(M/s)^k$ is entrywise nonnegative. □

Lemma 2.2 (An actual positive projection)**.** *Let n_j, $j \in J$, be actual wall normals with positive definite Gram matrix B_J, and let z be strict interior. Put*

$$q_0 = z - \sum_{i,j\in J} (B_J^{-1})_{ij}\langle z, n_j\rangle n_i.$$

Then q_0 is future timelike, orthogonal to all selected walls, and has negative product with every unselected actual wall. Its normalization is a point of the original polytope.

Proof. The added coefficients are nonnegative by Lemma 2.1. For an unselected wall every added product is nonpositive and the original product is negative; the selected products vanish. Moreover

$$q_0^2 = z^2 - (\langle z, n_i\rangle)_{i\in J}^{T} B_J^{-1} (\langle z, n_i\rangle)_{i\in J} < 0.$$

Since $\langle z, q_0\rangle = q_0^2 < 0$, it has future orientation. All original wall inequalities survive normalization. If fewer than n walls are selected this is a point on a face, not an asserted full-dimensional vertex. □

The same sign check applies to positive combinations of actual normals below: selected products are calculated explicitly; every unknown original wall has nonpositive product with every summand. A nonzero feasible future null vector would give a point of the ideal boundary of P, contradicting compactness. A positive affine Coxeter kernel on actual walls is an obstruction after its future sign and all remaining wall products have been checked.

Lemma 2.3 (Future-point pressure)**.** *Suppose $U \in p^\perp$, $\xi = Ap - U$ is future timelike with $\xi^2 = -k < 0$, and its normalization is feasible against every actual wall. Then the same original vector satisfies*

$$\langle x, U\rangle \geq A - \sqrt{k}.$$

At a different actual vertex q, use the decomposition $\xi = Aq - U$ with $U \in q^\perp$. With all-point comparison factor κ, the corresponding inequality is $\langle y, U\rangle \geq A - \kappa\sqrt{k}$.

Proof. The spatial hypothesis gives $A = -\langle p, \xi\rangle > 0$. Apply (2) to $\xi/\sqrt{k}$: $-\langle z, \xi/\sqrt{k}\rangle = E(A - \langle x, U\rangle)/\sqrt{k} \leq E$. The modified comparison is the identical calculation. □

If $U \neq 0$, its support lies in actual orthogonal components and $\|U\|^2 = A^2 - k$; Cauchy therefore also gives

$$\|x_{\operatorname{supp} U}\|^2 \geq \frac{A - \sqrt{k}}{A + \sqrt{k}}. \tag{3}$$

Normalize a future vector, not a spacelike correction. For a cap $c = Ap - v$, clear a positive definite old block J:

$$w = \sum_{i,j\in J} (B_J^{-1})_{ij}\beta_j n_i, \quad \beta_j = -\langle c, n_j\rangle \geq 0, \quad \eta = c + w, \quad \delta = \eta^2 = 1 - \beta^T B_J^{-1}\beta.$$

If $\delta < 0$, η is the future vector to normalize. If $\delta > 0$, the actual finite projection instead uses

$$p + \frac{A}{\delta}\eta, \qquad \left(p + \frac{A}{\delta}\eta\right)^2 = -1 - \frac{A^2}{\delta}.$$

When its selected products are nonpositive, its positive coefficients check all unselected walls. The pressures are

$$\langle x, v - w\rangle \geq A - \sqrt{-\delta}, \qquad \langle x, v - w\rangle \geq \frac{A}{1 + \sqrt{\delta/(A^2 + \delta)}}. \tag{4}$$

Old normals lie in $p^\perp$, so $v - w$ denotes the spatial part of the corresponding displayed Lorentz vector.

2.2 Opposite walls and common components

Let B be the original incident Gram matrix, $C = B^{-1}$, and

$$\omega_i = \sum_j C_{ij} n_j, \qquad d_i = C_{ii}, \qquad u_i = \omega_i/\sqrt{d_i}.$$

An actual opposite wall has normal

$$c = Ap - v, \quad A > 0, \quad v = \sum_i \beta_i\omega_i, \quad \beta_i = -\langle c, n_i\rangle \geq 0, \quad A^2 = \beta^T C\beta - 1.$$

Strict interiority gives

$$\langle x, v\rangle < A. \tag{5}$$

Along the original edge in direction $-u_i$, of length L_i, the original farthest comparison gives

$$\langle x, u_i \rangle \geq \tanh(L_i/2). \tag{6}$$

Whole-component projection energies sum to $\|x\|^2$.

Proposition 2.4 (Reciprocal shadows; internal geometric interface)**.** *For an actual cap put* $T_i = (C\beta)_i/(A\sqrt{d_i})$*. If* $T_i > 1$*, it is the opposite wall of edge* i*, and* $\tanh L_i = 1/T_i$*. If it is not that opposite wall, then* $\tanh L_i \geq T_i$*. The value* $T_i = 1$ *is incompatible with a finite endpoint. For distinct actual opposite walls,*

$$\tanh L_i \tanh L_j \geq K_{ij}, \qquad K_{ij} = C_{ij}/\sqrt{d_i d_j}. \tag{7}$$

Opposite walls of orthogonal original directions cannot coincide.

Proof. The Schur identity says that $C - C_{\cdot i}C_{i\cdot}/d_i$ vanishes on row and column i; elsewhere it is the inverse of the retained old Gram matrix. It is entrywise nonnegative. Let $d = Dp - \sum_j \gamma_j \omega_j$ be the actual opposite of edge i, distinct from c. Acuteness and the Schur identity give

$$AD \geq \beta^T C\gamma \geq (C\beta)_i (C\gamma)_i/d_i.$$

The actual endpoint equation is $\tanh L_i = D\sqrt{d_i}/(C\gamma)_i$, so $\tanh L_i \geq T_i$. If $T_i > 1$ this alternative is impossible, forcing $c = d$; its endpoint equation gives $\tanh L_i = 1/T_i$. If $T_i = 1$, the distinct alternative is impossible and the same-wall equation would give an ideal endpoint. This is excluded by compactness. For the pair inequality, first use the contacts of the actual opposite of i to get $T_j \geq K_{ij}/\tanh L_i$; the distinct alternative at edge j yields the assertion.

Finally suppose the same wall closes orthogonal directions i, j. Clearing every other old normal adds positive coefficients and gives

$$c_F = Ap - \frac{A}{r_i}u_i - \frac{A}{r_j}u_j, \qquad r_i = \tanh L_i, \quad r_j = \tanh L_j.$$

Its product with the strict interior point remains negative, so $\langle x, u_i\rangle/r_i + \langle x, u_j\rangle/r_j < 1$. But (6) makes this sum at least $1 + (\tau_i^2 + \tau_j^2)/2 > 1$, where $\tau_i = \tanh(L_i/2)$. This contradiction proves the orthogonal assertion. □

The displayed pair inequality concerns full tangents, not half tangents. For an opposite wall, its retained foreign component can have at most one contact: two would make a cycle in its actual finite replacement. A positive foreign cost is $q = \beta_j^2 C_{jj} \geq 1/4$. In a nontrivial component $C_{jj} \geq 4/3$, so the only cost below $1/3$ is a simple 3-contact on an A_1, of cost $1/4$. Moreover

$$q < A^2. \tag{8}$$

Otherwise the actual reciprocal test would close that foreign direction too, violating the orthogonal assertion, with the equality case giving a null endpoint. Set $F(t) = 2t/(1+t^2)$. For unit directions of correlation k in a component of energy at most e, positive heights satisfy

$$\tau_i + \tau_j \leq \sqrt{2e(1+k)}, \qquad F(\tau_i)F(\tau_j) \leq \frac{4s}{(1+s)^2}, \quad s = \frac{e(1+k)}{2}. \tag{9}$$

The second inequality follows from concavity of $\log F$ on $(0,1)$. If its upper bound is smaller than K_{ij}, Proposition 2.4 forces a common actual opposite. Its finite-white extension or positive affine kernel then gives additional actual pressures.

3 Reference vectors, individual prices, and missing floors

The elementary estimate behind every certificate is

$$\|x\|^2 \geq 2\langle x, W\rangle - \|W\|^2. \tag{10}$$

Every term in $\langle x, W\rangle$ must be justified. Write $w = \sum_i b_i P_i$, $b_i \geq 0$, with actual pressures $\langle x, P_i\rangle > a_i$, and demand $A_w = \sum_i b_i a_i$. For a new actual pressure $\langle x, Z\rangle \geq h$, choose

$$Q = Z - \sum_i t_i P_i, \qquad W = w + \mu Q.$$

Before using pressures require $\mu \geq 0$ and check every $b_i - \mu t_i \geq 0$. Then

$$\langle x, W\rangle \geq A_w + \mu(h - \sum_i t_i a_i), \quad \|W\|^2 = \|w\|^2 + 2\mu\langle w, Q\rangle + \mu^2\|Q\|^2. \tag{11}$$

The available price b_i is individual to anchor i. The first inequality is strict if some strictly constrained old anchor retains a positive coefficient, or if $\mu > 0$ and the new pressure is strict. For multiple corrections every new weight must be nonnegative; their combined debit at each anchor and all mutual cross terms must be retained.

Put $P_i = e_1 + \cdots + e_i$, $S_i = \langle x, P_i\rangle$. If selected indices satisfy $S_i > f_i$, concave interpolation through $(0, 0)$ and the selected (i, f_i) gives a reference whose coordinates are successive slopes. Its P_i coefficients are successive slope decreases, hence nonnegative; its demand is its squared norm. Omitting one unknown internal floor replaces two adjacent slopes by their average, losing half their squared difference. For $\sqrt{t}$ at $t \geq 5$, the loss is less than $1/3200$. Nonadjacent omissions have disjoint open intervals, so losses add. A missing floor has coefficient zero. References are Euclidean vectors, not asserted actual points: actual points prove pressures, then positive coefficients combine them in (10).

4 Common component bounds and the precise branch point

4.1 Ordinary components and exceptional actual defects

All bounds below concern the same original projection, with arbitrary finite spectators.

Component	Energy greater than	Component	Energy greater than
ordinary nonshort A_1	$1/10$	ordinary A_2	$319/2000$
ordinary $I_2(m), m \geq 4$	$1/7$	A_3	$1/4$
B_3, H_3	$2/7$	A_4	$2/7$
B_4, D_4, F_4	$1/3$	H_4	$11/36$
A_5	$3/8$	A_6	$3/7$
B_5, B_6	$3/7$	D_5, D_6	$5/14,\ 3/7$
B_7, D_7	$1/2$	B_8, D_8	$4/7$
E_6, E_7, E_8	$9/20,\ 5/9,\ 2/3$		

An actual isolated dotted short A_1 is denoted I. A second short-wall profile J uses two old A_1 factors whose joint energy exceeds .34. The two typed rank-two exceptions, each of energy $> .09$, are

$$N_7 = (\text{old } A_2,\ \text{new isolated native } 7), \qquad O_7 = (\text{old } I_2(7),\ \text{new isolated native } 3).$$

A bare $I_2(7)$ is not automatically O_7. There is at most one short I/J profile or one typed exception; they do not coexist. Acuteness of their actual isolated walls is essential: two on orthogonal supports would have product $-\sqrt{h_1 h_2}$ strictly between $-1/2$ and 0. Appendix A proves the endpoint estimates, the J case and the ordinary rates.

Useful joint bounds, counting I only once, are

$$I + A_1 > 1/5, \quad I + 2A_1 > 3/10, \quad I + T_2 > 21/100, \quad I + T_3 > 2/7. \tag{12}$$

Here T_j is ordinary irreducible of rank j. Both typed exceptions force another T_2 to cost $> 1/5$. Another A_1 costs $> 1/5$ beside N_7 and $> 2/15$ beside O_7. For the latter, isolated height $h = \frac{1}{4}\csc^2(\pi/7) - 1 < 1/3$ gives squared full tangent greater than $3/7$. Squared half tangent exceeds $2/15$, since $4(2/15)/(1+2/15)^2 = 120/289 < 3/7$. This is a real geometric argument. The complete contact and defect proofs, including the actual-wall height constraints behind (12), are given in Appendix A. The uniform actual B/D floor is proved in Appendix B.

4.2 The A-chain branch

For A_r, use the zero-sum subspace of $\mathbb{R}^{r+1}$ with roots $(e_i - e_{i+1})/\sqrt{2}$. The inverse Gram matrix is

$$C_{ij} = \frac{2\min(i,j)(r+1-\max(i,j))}{r+1}.$$

A finite-white classification of actual opposite walls gives strong dual floors except for a short list of literal helper caps. Adjacent weak helpers are excluded using their actual mutual product and a positive two-cap projection. Concave interpolation uses only strong floors; helper pressures are paid at surviving anchors. The small end variance yields stronger component bounds:

r	$r \geq 7$	$r \geq 8$	$r \geq 13$	16	17
$e(A_r) >$	1/2	4/7	5/7	199/250	6/7

$e(A_{18}) > 91/100.$

Later local refinements give $e(A_{12}) > 13/20$ and $e(A_{15}) > 413/500$, uniformly with arbitrary spectators. The shared actual feet, common-pair inventories, uniform $1/2, 4/7, 5/7$ bounds, and the $N = 17, 18, 19$ central-window and paid-reference arguments are proved in Appendix C. The newer $A_{16} > .86$ proof in particular implies the older .796 bound without requiring its separate historical reference. The fresh smaller-rank proofs are in Sections D.5 and D.4. Their full native-cosine inventories and individual actual-helper payments are included. Later exact-17 research remains excluded.

4.3 The common B/D prefix branch

For B_r, use long roots $(e_i - e_{i+1})/\sqrt{2}$, $i < r$, and short root e_r. For D_r, use the long chain and the fork root $(e_{r-1} + e_r)/\sqrt{2}$. On a long initial segment, $\omega_i = \sqrt{2}P_i$, $C_{ij} = 2\min(i,j)$. In type D, $x_1 > \cdots > x_{r-1} > |x_r|$; the last coordinate is signed and the forks are distinct actual duals.

Proposition 4.1 (Generic prefix theorem)**.** *Let $6 \leq M \leq 16$. Suppose an actual component is B_r, $r \geq M + 5$, or D_r, $r \geq M + 6$. Put*

$$f_i = \sqrt{i} - \frac{61}{100\sqrt{2}}, \quad E_M = \frac{f_5^2}{5} + \sum_{i=5}^{M-1} (\sqrt{i+1} - \sqrt{i})^2, \quad H_M = \left\lfloor \frac{M-5}{2} \right\rfloor.$$

With arbitrary finite spectators, $\sum_{j=1}^{M+1} x_j^2 > E_M - H_M/3200$.

Proof. Appendix E proves that a weak site $5 \le i \le M+1$ has a pure cap

$$v = \gamma\omega_i + \omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad 101/125 < \gamma < 6/5,$$

with own label 5, finite label at least 7, or dotted own contact, and no spectator contact. Actual weak sites are nonadjacent. This is proved at these ranks, not transferred from a whole-component exclusion. If 5 is strong, choose a strong $K = M$ or $M+1$. Interpolate through actual strong knots. Each missing internal floor loses $< 1/3200$; at $K = M+1$ the extra increment $(\sqrt{M+1} - \sqrt{M})^2 > 1/(4M+2) \ge 1/66$ pays the possible extra omission.

If 5 is weak, the first-six-root point table in Appendix E gives $x_1 > .5965$, and 6 is strong, with $S_6 > 2.01815$. Interpolate through $(1, .5965), (6, 2.01815)$ and a later strong endpoint. The slopes decrease: $.5965, .28433$, then $1/(\sqrt{K} + \sqrt{6}) < 1/5$. Its energy exceeds

$$L_M = .7600299945 + \frac{(\sqrt{M} - \sqrt{6})^2}{M-6},$$

the last term being zero at $M = 6$. Now $E_6 < .7$. The $k = M - 6$ increments from 6 onward lie in $(1/8, 1/5)$, so their variance is at most $k(1/5 - 1/8)^2/4 \le 9/640$. Thus $L_M - E_M > .76 - .7 - 9/640 > 0$, directly also at $M = 6$. This proves the weak-5 branch without first assuming strong 5. □

Appendix E supplies the complete finite-white inventory, zero-own repair, actual two-cap nonadjacency, all four weak-5 pressures and continuous seals, with every spectator retained. Its original rank inequalities are part of the theorem. A short-tail continuation at an excluded rank still requires its own new local proof.

5 Complete forest covers and their consumers

5.1 A fresh cover in exact dimension 18

Call A_r, $r \ge 7$, or B_r, D_r, $r \ge 9$, dominant. Each costs $> 1/2$, so at most one is possible. With none the ordinary rank-over-14 baseline, even allowing the unique defect, exceeds 1. Exactly one is therefore present. The fresh total-rank-18 necessary cover is:

Dominant	All retained spectator alternatives
A_{12}	$I + (A_5$ or $D_5)$, or $N_7/O_7 + T_4$
A_{15}	ordinary A_3; $T_2 + A_1$; $I + T_2$; or $O_7 + A_1$
A_{16}	ordinary T_2; two ordinary A_1; $I + A_1$; or N_7/O_7
A_{17}	one ordinary nonshort A_1, or I
A_{18}	none
B_{16}	N_7/O_7
B_{17}	I
B_{18}	none
D_{16}	N_7/O_7
D_{17}	one ordinary nonshort A_1, or I
D_{18}	none

Here ordinary $T_2 = I_2(m)$, $m \ge 3$, and $T_4 \in \{A_4, B_4, D_4, F_4, H_4\}$. H_4 and native 5 are not discarded by field arguments.

For completeness, the accounting steps are as follows. A_7 leaves too many ordinary ranks; $A_8, \dots, A_{11}$ are excluded by 4/7 plus the rank-over-14 baseline. Beside A_{12}, ordinary or J rank-six spectators cost $> 3/7$. With I, (12) excludes all rank-five partitions except A_5, D_5. With a typed

exception, partition rank 4: two T_2's cost $> .4$, $T_3 + A_1$ costs $> 1/4 + 2/15$, and the other reducible partitions are stronger. Only irreducible T_4 remains. For A_{13}, A_{14}, the 5/7 bound and (12) exclude all spectators. At A_{15}, exclude B_3, H_3, three ordinary A_1's, $I + 2A_1$, $J + A_1$, and $N_7 + A_1$. At A_{16}, .796 excludes J but retains precisely the listed rank-two alternatives. At A_{17} retain the remaining A_1.

For B_r, D_r, the 9/14 bound forces spectator rank at most 5, hence $r \geq 13$. Proposition 4.1, within its rank domain, gives

r	13	14	15	16
$e(B_r), e(D_r) >$	.7349975	.7683275	.797425	.823735.

At $13, 14, 15$, every ordinary, I/J, and typed remainder exceeds the residual budget; $I + T_2 > .21$ closes rank 15. At 16, two ordinary A_1's and $I + A_1$ are excluded; ordinary T_2 and both typed defects initially remain. The B_3, D_4, D_3 layer checks in Appendix F extend helper coverage to 13, giving energy $> .8689525$; ordinary $T_2 > 1/7$ closes that case. The fresh B_{17} layers reach a strong 14/15 endpoint and energy $> .90715$, excluding a nonshort ordinary A_1. This reuses the interpolation proof after new local coverage, not Proposition 4.1 outside its legal ranks.

The first rows close with small shared consumers:

$$A_{12} + I + (A_5/D_5) > 13/20 + 5/14 = 141/140, \quad A_{12} + N_7/O_7 + T_4 > 13/20 + .09 + 2/7 = 359/350.$$

The improved 413/500 bound closes every A_{15} family: ordinary total rank three costs $> 17/70$, $I + T_2 > .21$, and beside $O_7 + A_1$ the total exceeds 787/750. Thus the older separate ordinary-rank-three consumer is unnecessary. The other internal proof homes are:

Row	Excluding project argument
A_{16}	Proposition C.5, ordinary and $I + A_1$; Appendix J.1, both actual typed defects
A_{17}	Appendix J.2, ordinary and actual I
whole A_{18}	Appendix G, full two-end native-gap argument
B_{16}	Appendix J.3, both actual typed defects
B_{17}	Appendix J.5, actual I; local head and helper kernels
D_{16}	Appendix J.3, both actual typed defects
D_{17}	Appendices J.4.1 and J.6, ordinary and actual I
whole B_{18}	Section 6
whole D_{18}	Sections 7–10

Appendix F proves the local layers used in this cover, including the distinct signed D_4/D_3 forks and the actual A_{17} neighbor for B_{17}. The table identifies the complete proof of every retained row, including both short rank-one cases. The independent higher-dimensional exclusions are proved in Appendices K–N. These arguments use no field, lattice or integral dotted-product hypothesis.

5.2 The separate complete higher-dimensional tail

The exact-18 cover does not exclude higher dimensions. The earlier internal package establishes the separate theorem:

Theorem 5.1 (Internal higher-dimensional exclusion)**.** *There is no compact hyperbolic Coxeter polytope of dimension at least* 19*.*

Appendix K proves the component caps, shared windows and complete actual-rank exclusions from 29 through 20. The published ceiling is used only at dimensions at least 30. At dimension nineteen, Proposition K.35 leaves whole A_{19}, B_{19}, D_{19}. Their proofs occupy Appendices L–N: the A proof splits by the three weak-end patterns; the B proof splits by its two exceptional terminal walls; and the D proof splits by six head types after a common centre analysis. Each case keeps one original vector and every spectator in its cover. Section N.22 joins these exclusions.

The higher ranks have their own complete covers in Appendix K. The short B_3/B_2 and D_4/D_3 continuations are proved before the prefix estimates that consume them. Section N.22 combines these actual-rank arguments with the published compact ceiling. This is a logical conjunction of exclusions, not geometric dimension heredity. No arithmetic theorem or field census enters that conjunction.

6 The whole-B_{18} terminal partition

Classify the actual opposite of old node 17 without assuming a strong head. Section H.5 leaves:

(i) a pure far-three helper with finite own 5, finite own at least 7, or dotted own $1 < g < 4/3$;

(ii) near-three with own 6;

(iii) no left contact, with short white label $3, 4, 5, 6, 7, 8$.

All other isolated, left, bridge and affine/null roles are excluded in that inventory. Sections H.6–H.10 exclude the first branch, and Section H.11 excludes the second. In the last branch, Sections H.13–H.14 exclude distinct terminal caps and then every common-cap role. The last common cap is treated at its actual adjacent whole-A_{18} vertex, with comparison $11/8$; that vertex is never assumed farthest.

Thus an original farthest whole-B_{18} vertex cannot occur. Appendix H gives the complete finite-white inventory, each terminal exclusion, all original-wall checks and the individual coefficient payments. Together with the preceding row consumers, this leaves only whole D_{18} at the selected original vertex.

7 The last case: an actual whole-D_{18} vertex

We treat the final dotted-head case using the complete inventory proved in Appendix I. Keep the original p, z, x and use actual old roots

$$n_i = (e_i - e_{i+1})/\sqrt{2} \quad (1 \le i \le 17), \qquad n_{18} = (e_{17} + e_{18})/\sqrt{2}.$$

Write

$$P_i = e_1 + \cdots + e_i, \quad \Omega_i = \sqrt{2}P_i \ (i \le 16), \quad \Omega_{17} = (P_{17} - e_{18})/\sqrt{2}, \quad \Omega_{18} = (P_{17} + e_{18})/\sqrt{2}.$$

In particular x_{18} is signed and $x_1 > \cdots > x_{17} > |x_{18}|$. In this part only, set

$$S_i = \langle x, P_i \rangle, \qquad f_i = \sqrt{i} - \frac{.58}{\sqrt{2}}. \tag{13}$$

All terminating decimals in this paper denote exact rational numbers.

The head is the actual opposite wall of the long-end edge 1. Section I.1 proves that its normal is

$$H = A_H p - \eta\Omega_1, \qquad A_H^2 = 2\eta^2 - 1 > 0.$$

The same section gives the complete own-contact list. Sections I.8 and I.9 prove the strong terminal floor $S_{16} > f_{16}$, its funded reference append and the complete fork-role list. The finite-head cases close in Section I.22. We treat every dotted head $\eta > 1$ without a parameter cutoff.

Appendix I proves these earlier inputs in geometric dependency order: actual head isolation, the fresh threshold window, strong 16, finite-head closure and the every-head pair exclusions. In particular it proves the changed-threshold strong-site statements afresh rather than inheriting them unconditionally.

7.1 A future point valid for every dotted head

Put $u = \eta^{-2} \in (0, 1)$, and consider

$$\Xi = H + n_1/\eta.$$

This is a positive combination of two actual normals. Its products with H, n_1 are respectively 0 and $-\eta + 1/\eta < 0$, and $\Xi^2 = -(1-u) < 0$. Its time is $A_H > 0$. Every other actual wall has nonpositive product with each summand. Hence its normalization is an actual future point of the original polytope, with no unknown wall omitted. Lemma 2.3 gives

$$\langle x, Z_u\rangle \geq h(u), \qquad Z_u = \frac{(2-u)e_1 + ue_2}{2}, \quad h(u) = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}. \tag{14}$$

Indeed the spatial part of Ξ, divided by $\sqrt{2}\eta$, is Z_u. We will use

$$\sum_j (Z_u)_j = 1, \qquad R(u) = \|Z_u\|^2 = 1 - u + u^2/2.$$

Equation (14) is a point pressure, not an energy lower bound, and is valid on the entire open interval.

7.2 The four original reference cases

Section I.2 gives the following complete initial alternatives: strong 5; a weak pure far-left-three helper at 5; or an isolated dotted weak 5 with a chosen strong $J = 6$ or 7. Thus there are four reference rows. Here is why the initial classification has this precise scope. Left own 4, including own 4/white 3, gives a positive actual affine-B null on the cap and old chain $5, \ldots, 18$; the extra left contact preserves the remaining wall signs. The newly possible near-three/own-five at 5 is excluded by its actual right-H_4 point and independent old strong 11/12, before the new references are used. Other finite-white and isolated finite roles are strong. A historical fifth paid row concerned a native-five head and is outside $\eta > 1$.

We emphasize the *bare* references for the three nonhelper rows. They interpolate through the strong knots, are flat on their first J coordinates, and have endpoint sum f_{16}. Sections I.4 and I.8 give

initial case	B_0	first mean $b <$	unused P_J price $>$	J
strong 5	.9565188	.3652	.151	5
isolated dotted 5, strong 6	.9373243	.3399	.1435	6
isolated dotted 5, strong 7	.9196357	.31938	.1366	7

More explicitly, each reference w has

$$\langle x, w\rangle > A_w, \quad 2A_w - \|w\|^2 > B_0, \quad \operatorname{supp} w \subset \{1, \dots, 16\}, \quad \langle w, P_{16}\rangle = f_{16}, \quad \beta_{16} > .127.$$

No head pressure has been paid. Their raw endpoint-15 lower bounds are $.9403998, .9212053, .9035167$. The legal endpoint-16 append adds at least $.127^2 - .00001 = .016119$; if 15 is weak, its fictitious knot is deleted before pressure. All late repairs preserve the flat first block and its individual prices.

In the isolated-5 rows there is also an unused actual pressure

$$U_5 = P_5 + \frac{v}{2}(e_6 - e_5), \quad .4 < v < 1, \qquad \langle x, U_5\rangle > 1.8185.$$

For

$$Q_H = Z_u - P_J/J, \qquad Q_5 = U_5 - (5/J)P_J$$

the exact identities are

$$\|Q_H\|^2 = R - 1/J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J, \quad \|Q_5\|^2 < \begin{cases} 77/150, & J = 6, \\ 194/175, & J = 7. \end{cases} \tag{15}$$

Both residuals have zero product with the bare flat reference. The positive mutual cross in (15) will not be discarded.

For the fourth row, the exact *unheaded* weak-helper reference interpolates from $(1, a_0)$, $a_0 = .5455$, to $(6, f_6)$, then through the later actual strong knots to 16, replacing each missing knot by a chord. Its first six coordinates are

$$w_1 = a_0, \quad w_2 = \cdots = w_6 = b_0 = (f_6 - a_0)/5. \tag{16}$$

The actual helper proves $x_1 > a_0$ and strong 6. The unheaded reference satisfies

$$\langle x, w\rangle > A_w, \quad L_w := 2A_w - \|w\|^2 > .987759, \quad \langle w, P_{16}\rangle = f_{16}, \quad \beta_{16} > .127.$$

Weak 7, later omissions, and the late 15 repair do not alter (16). Again no head pressure has been paid. Appendices I.4, I.8 and I.20 supply the raw-reference increments, omission tables, isolated-5 pressure and individual coefficient prices used here. These are internal geometric lemmas with written proofs, not script-output assumptions.

8 Uniform head payment and the two true fork heights

8.1 The first three rows

In the strong-5 row choose $\mu = .33$ on Q_H. Its debit $.33/5 = .066 < .151$ fits the individual price. Equations (11)–(15) give

$$L_W > .9565188 + .66(h - .3652) - .1089(R - 1/5) = F(u) := .7372668 + .66h(u) - .1089R(u).$$

For isolated 5 with $J = 6$, use weights $\mu = .33, \nu = .10$ on Q_H, Q_5. The combined debit at J is $.83/6 < .1435$. For $J = 7$ use $\mu = .33, \nu = .11$, with debit $.88/7 < .1366$. In each case all remaining old coefficients are nonnegative *before* the new pressures are invoked, and

$$\begin{aligned} L_W > B_0 &+ .66(h - b) + 2\nu(1.8185 - 5b) - .1089(R - 1/J) \\ &- \nu^2\|Q_5\|^2 - .66\nu(1 - 5/J). \end{aligned} \tag{17}$$

The constant terms before $.66h - .1089R$ in the two rows are $22164209/30000000$ and $51729823/70000000$, both greater than $.7372668$. Thus F is a common lower bound. No optimized helper and head energies have been added.

We prove the bound uniformly on $0 < u < 1$. Put $a = \sqrt{2-u}$, $b = \sqrt{u(1-u)}$. Direct differentiation gives

$$h''(u) = \frac{1/(4b^3) - 1/(4a^3)}{\sqrt{2}} > \frac{7}{4\sqrt{2}}, \tag{18}$$

because $b \le 1/2$, $a > 1$. Hence $F'' > .66 \cdot 7/(4\sqrt{2}) - .1089 > .7$, using $\sqrt{2} < 1.415$. At $u_0 = 2/3$,

$$h(u_0) = \sqrt{2/3} - 1/3, \quad R(u_0) = 5/9, \quad F(u_0) > .9956502,$$
$$F'(u_0) = .2013 - .0825\sqrt{6}, \quad |F'(u_0)| < .0008.$$

The value bound uses $.81649^2 = .6666559201 < 2/3$, and the derivative bound uses $2.4494 < \sqrt{6} < 2.4495$. Taylor's inequality and completion of the square on the segment between u_0 and u give

$$F(u) > .9956502 - \frac{.0008^2}{1.4} > .99564 \qquad (0 < u < 1). \tag{19}$$

No endpoint derivative or sampled minimum is used.

8.2 The weak-helper row: two separate anchor prices

The unheaded reference (16) has two available prices

$$\beta_1 = a_0 - b_0 > .24672, \qquad \beta_6 \ge b_0 - (\sqrt{7} - \sqrt{6}) > .1023. \tag{20}$$

Indeed $2.0392 < f_6 < 2.0394$, so $.29874 < b_0 < .29878$, while $\sqrt{7} - \sqrt{6} < 2.6458 - 2.4494 = .1964$. An adjacent omitted knot decreases the next slope and increases β_6. The bounds follow from $2.4494 < \sqrt{6} < 2.4495$ and $.4101 < .58/\sqrt{2} < .4102$, by positive squaring.

Define

$$Q = Z_u - (1 - 3u/5)e_1 - (u/10)P_6.$$

Its first coordinate is zero, its second is $.4u$, and its third through sixth coordinates are $-.1u$. Therefore

$$\|Q\|^2 = u^2/5, \qquad \langle w, Q\rangle = 0. \tag{21}$$

Choose $\mu = .22$. First pay the *separate* anchor debits

$$\mu(1 - .6u) < .22 < \beta_1, \qquad \mu u/10 < .022 < \beta_6 \quad (0 < u < 1).$$

The remaining original coefficients are nonnegative. Only then add the positive new head pressure μZ_u. The terminal coefficient at 16 remains unused. The full one-vector square gives

$$\begin{aligned} L_{w+\mu Q} &> .987759 + .44[h - a_0 + (3a_0/5 - f_6/10)u] - .00968u^2 \\ &> G(u) := .747739 + .44h(u) + .0542784u - .00968u^2, \end{aligned} \tag{22}$$

since $3a_0/5 - f_6/10 > .12336$. This argument rewrites a positive-coefficient pressure decomposition before using it; it does not subtract old lower floors from an unknown signed pressure.

We give the exact small-margin seal. By (18), $G'' > .44 \cdot 7/(4\sqrt{2}) - .01936 > .5$. At $u_1 = .642$, positive square tests give

$$\sqrt{1.358} > 1.1653325, \qquad \sqrt{.229836} < .4794122,$$

whose comparison squares are 1.35799983555625 and .22983605750884. The difference exceeds .6859203. Since

$$.4850188 \cdot 1.4142136 = .68592018321568 < .6859203, \qquad 1.4142136^2 > 2,$$

we have $h(.642) > .4850188$, and hence

$$G(.642) > .99200425728.$$

For the derivative use

$$.858 < 1/\sqrt{1.358} < .859, \quad .592 < .284/\sqrt{.229836} < .593, \quad .353 < 1/(2\sqrt{2}) < .354.$$

They follow by multiplying the positive radical brackets $1.1653 < \sqrt{1.358} < 1.1654$ and $.4794121 < \sqrt{.229836} < .4794122$. Thus

$$-.094518 < h'(.642) < -.093545, \qquad .00026136 < G'(.642) < .00068948 < .0007,$$

because $G' = .44h' + .0542784 - .01936u$. Strong convexity on the entire interval now gives

$$G(u) > .99200425728 - .0007^2 = .99200376728 > .992. \tag{23}$$

Combining (19) and (23), every one of the four original initial cases supplies a single legally funded reference W with

$$L_W > .992, \quad \operatorname{supp} W \subset \{1, \dots, 16\}, \quad \langle W, P_{16} \rangle = f_{16}, \quad \text{unused } \beta_{16} > .127. \tag{24}$$

All corrections have sum zero and support at most 7; the late omissions and the original endpoint-16 price survive.

8.3 Both signed fork directions

Let $\Delta = S_{16} - f_{16} \geq 0$. The pressure decomposition of W keeps its original terminal coefficient. Therefore $\langle x, W \rangle > A_W + \beta_{16}\Delta$, and the same norm identity gives

$$\|x - W\|^2 < .008 - .254\Delta, \qquad x_{17}^2 + x_{18}^2 < .008 - .254\Delta. \tag{25}$$

The two true unit fork duals are

$$\frac{\Omega_f}{3} = \frac{P_{17} \pm e_{18}}{\sqrt{18}}, \qquad f = 17, 18.$$

For either actual fork height,

$$\begin{aligned} t_f := \langle x, \Omega_f/3 \rangle &< \frac{f_{16} + \Delta + \sqrt{.016 - .508\Delta}}{\sqrt{18}} \\ &< \frac{3.59 + .1265}{\sqrt{18}} < .876. \end{aligned} \tag{26}$$

The numerator decreases with Δ: its derivative is $1 - .254/\sqrt{.016 - .508\Delta} < 0$ on its nonempty domain. The endpoint is handled by continuity. The seals are $f_{16} < 3.59$, $\sqrt{.016} < .1265$, $\sqrt{18} > 4.2426$, and $.876 \cdot 4.2426 = 3.7165176 > 3.7165$. No absolute value has replaced the signed last coordinate.

For an actual fork edge with distance cosine C, (6) gives $t_f \geq \sqrt{(C-1)/(C+1)}$. Hence

$$C < \frac{1 + .876^2}{1 - .876^2} < 7.6, \tag{27}$$

since $.876^2 = .767376$ and $7.6 \cdot .232624 = 1.7679424 > 1.767376$. In Section I.9, every far-next-three role has $g \geq \cos(\pi/5) > .809$ and exact cosine $C = 9g + 1 > 8.281$. Equation (27) removes that entire role, including all its previously classified native/null own alternatives.

9 A local A_{18} lemma at an adjacent actual vertex

Lemma 9.1. *Let q be an actual vertex of a compact Coxeter 18-polytope whose whole incident diagram is A_{18}. Write $z = E_q(q - y)$, $\|y\| < 1$. Suppose*

$$-\langle z, r\rangle \le \kappa E_q \quad \text{for every actual point } r \text{ of the original } P, \qquad \kappa < 11/8.$$

Then $\langle y, u_{18}\rangle > .78$ is impossible for the positive normalized endpoint-dual direction.

The lemma is local at q, which is not assumed farthest. Appendix D supplies its local inventory; no original B_{18} hypothesis is used.

Proof. In the zero-sum subspace of $\mathbb{R}^{19}$, for $i \le j$,

$$D_i = \|\omega_i\|^2 = \frac{2i(19-i)}{19}, \qquad C_{ij} = \frac{2i(19-j)}{19}.$$

Call $5 \le i \le 14$ weak if $\langle y, \omega_i\rangle \le \sqrt{D_i} - .87$. The complete inventory at comparison $11/8$, established before any endpoint pressure, leaves only:

(a) near-four, meaning own 4 and white 3 at the adjacent old node, in the four literal side-order-5/6 orientations;

(b) a pure far-end-three helper

$$d_i = A_i q - \gamma_i \omega_i - \omega_e/2, \quad e = 1 \text{ or } 18, \quad \varphi/2 \le \gamma_i < 17/16,$$

with own 5, finite own at least 7, or dotted own.

Here $\varphi = (1+\sqrt{5})/2$. All isolated caps are strong. The inventory includes own zero, every finite own, parallel and unbounded dotted owns. Every positive point checks every unknown original wall.

Some calculations locate the numerical boundary. Every ordinary edge ratio at least $3/2$ is strong: with $a = 3/2, k = 11/8$, the square comparison for $q_0 = \sqrt{D_i} \in [2.7, 3.1]$ is the positive polynomial

$$-.21q_0^3 + 1.1876q_0^2 - 1.74q_0 + .7569.$$

It is concave with endpoint values $.583074, .519626$. The near-four ratio of side order ℓ is

$$\frac{1/\sqrt{2} + (\ell-1)/(2\ell)}{\sqrt{(\ell+1)/(2\ell)}}.$$

It increases in ℓ and exceeds $3/2$ at $\ell = 7$, since $(67 + 42\sqrt{2})/56 > 9/4$. Only $i = 5, 6$ with white $i - 1$, and $i = 14, 13$ with white $i + 1$, can be weak. For pure helpers the identity

$$\frac{(\gamma + 1/(2\ell))^2}{(\ell+1)/(2\ell)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2\ell))^2}{(\ell+1)/(2\ell)} \ge 0$$

gives the stated cutoff at the weakened comparison.

The local input retains genuine helper nonadjacency at this factor. Its proof uses the following exact actual products. For adjacent helper owns $A, B \in [.808, 1.08]$, the mutual product R satisfies

same end	$R \ge 1 - \frac{18}{19}AB - \frac{\lvert A-B\rvert}{19} > -1/2$
outward	$R \ge \frac{1}{2} - AB + \frac{(A-1)(B-1)}{19} > -.7$
reversed	$R \ge \frac{1}{2} - AB + \frac{(A+1)(B+1)}{19} > -.7.$

The native alphabet leaves $R = 0$ in the first case and $R = 0, -1/2$ in the others. Outward $R = 0$ contradicts

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = \frac{2}{5}C_{i,i+1} - \frac{24}{25} + \frac{8}{95} \geq 884/475 > 0.$$

For reversed $R = 0$, set $l = i, r = 18 - i$. The exact determinant would require

$$4lrA^2B^2 - (l+1)(r+1)(A^2 + B^2) - 2AB - (l+1)A - (r+1)B + 21/4 = 0.$$

Its own derivatives are positive on that rectangle. If one own is not 5, it exceeds .9; evaluating the other at .808, the endpoints $l = 5, 13$ give $.9601824, 1.6961824 > 0$, and concavity in l excludes all. For two native-five owns the norm-two calculation in basis $1, \varphi$ has nonzero rational determinant coefficient 475 outward and -361 reversed. This uses only native cosine algebra. Thus remaining products are 0 for a common helper and $-1/2$ for opposite helpers.

After clearing old roots, the actual two-cap Schur matrices are $I - t\mathbf{1}\mathbf{1}^T$, $t = (\ell - 1)/(2\ell)$, or the matrix with diagonal $\delta_l, \delta_r \in (1/2, 3/5]$, off-diagonal $-1/2$. Both are positive with positive inverse. Their positive projections are feasible against every original wall, including the reverse wall at q. The cap-time and weight estimates in Appendix D give weighted mean loss less than

$$1/10 + (11/8)(7/16) = .7015625 \quad \text{or} \quad 1/10 + (11/8)(3/10) = .5125.$$

Both are less than .87, excluding adjacent weak helpers. None of these steps uses endpoint height $> .8$.

Now assume the new height $\langle y, u_{18} \rangle > .78$. Each weak near-four has an actual mate, formed by releasing its white neighbor and retaining its cap. Positive finite Gram inversion gives an actual adjacent vertex feasible against every wall. Its edge cosine exceeds 11; for example the left mates have

$$C_j = \sqrt{2(i-1)/19}\left((20 - i)/2 + (19 - i)/\sqrt{2}\right) > 11, \qquad i = 5, 6.$$

At comparison 11/8, the normalized mate height exceeds 7/8. A left mate has correlation $< .15$ with u_{18}. The sum of these heights exceeds 1.655, whose square exceeds 2.3, while the squared norm of the sum of the directions is < 2.3. This contradicts $\|y\| < 1$. No left near-four remains.

Some $k = 5, 6$ is now strong by helper nonadjacency, with normalized height $> .677$. For a right mate $j = 14, 15$, the correlation formula

$$\rho_{ab} = \sqrt{\frac{a(19-b)}{b(19-a)}} \quad (a \leq b)$$

gives $\rho_{18,j} < .46, \rho_{18,k} < .161, \rho_{j,k} < .407$. For one vector $V = .5u_{18} + .6u_j + .4u_k$,

$$\langle y, V \rangle > .5(.78) + .6(7/8) + .4(.677) = 1.1858, \quad \|V\|^2 < .77 + .276 + .0644 + .19536 = 1.30576.$$

Equation (10) gives $\|y\|^2 > 1.06584$. All near-four orientations are excluded at .78. Only nonadjacent pure weak helpers remain.

Use rational heights $H_{19-i} = H_i$, linear on 0–5 and 14–19, with

$$H_5 = 1.844, \ H_6 = 1.995, \ H_7 = 2.103, \ H_8 = 2.173, \ H_9 = H_{10} = 2.207.$$

Their positive slopes are $.3688, .151, .108, .070, .034$, then zero. The squares $(H_i + .87)^2$, $5 \leq i \leq 9$, are

$$7.365796, \ 8.208225, \ 8.838729, \ 9.259849, \ 9.467929,$$

strictly below $140/19, 156/19, 168/19, 176/19, 180/19$. Thus every retained positive dual is paid by a strong floor. If 5 is weak use chord 0–6, reflect at 14, and omit each internal weak knot by its neighbor chord. After chords set $w_i = (H_i - H_{i-1})/\sqrt{2}$. The reference is zero-sum with demand $E_0 = \|w\|^2$.

Before internal holes, the exact energies for zero, one and two weak ends are .7205882, .70082285, .6810575. Internal curvatures are .043, .038, .036, .034 and reflections; even summing all eight losses gives

$$2(.043^2 + .038^2 + .036^2 + .034^2)/4 = .0028725 < .002873.$$

Nonadjacency makes the losses legitimate. End chords do not change any eligible internal curvature. Hence

$$E_0 > \begin{cases} .6781845, & 14 \text{ weak: } m = 6,\ K = 13,\ H_K = 1.995, \\ .69794985, & 14 \text{ strong: } m = 5,\ K = 14,\ H_K = 1.844. \end{cases} \tag{28}$$

The surviving individual anchor prices are

$$\beta_{13} \geq (7H_6/6 - H_7)/2 = .11225, \qquad \beta_{14} \geq (6H_5/5 - H_6)/2 = .1089.$$

Adjacent holes increase the anchor price. Omitted knots have price zero; no missing strong 7 or 18 floor is paid.

Define

$$G_0 = -e_{19} + \frac{1}{m} \sum_{\text{last } m} e_i,$$

with 19 included. Then

$$\|G_0\|^2 = (m-1)/m, \quad \langle w, G_0 \rangle = 0, \quad G_0 = \omega_{18}/\sqrt{2} - \omega_K/(m\sqrt{2}).$$

Use the new weight $\mu = .6$. The debit $.6/(m\sqrt{2})$ is $< .071$ for $m = 6$, $< .085$ for $m = 5$, below the separate anchor prices. The positive endpoint coefficient is paid by its actual height. Here

$$-y_{19} > .78\sqrt{18/19} > .7591,$$

since $19(.7591)^2 = 10.94842339 < 18(.78)^2 = 10.9512$. The last-block means are $H_K/(m\sqrt{2}) < .2352$ or $< .261$. The same local-y square, retaining $\langle w, G_0 \rangle = 0$, gives

$$\begin{aligned} m = 6: &\quad \|y\|^2 > .6781845 + 1.2(.7591 - .2352) - .36(5/6) = 1.0068645, \\ m = 5: &\quad \|y\|^2 > .69794985 + 1.2(.7591 - .261) - .36(4/5) = 1.00766985. \end{aligned}$$

Both contradict $\|y\| < 1$, proving the lemma with the local inventory established in Appendix D. □

Appendix D supplies the complete local own/white inventory, isolated-cap point bounds, positive mixed weights and native-five determinant expansion used above. It uses only real geometry and comparison 11/8, with no original-B_{18} premise. The original-x energy is never added to this local-y budget.

10 The near/far pair and the final composition

10.1 Every actual near/far fork edge is short

For a near-three or far-three fork role, the complete finite-white classification in Section I.9 gives an actual other endpoint q with whole diagram A_{18}. The positive reverse direction at q is exactly

its normalized new endpoint dual: p has nonzero product with only that new incident root. Let C be the original edge cosine, $\beta = \sqrt{C^2-1}/C$, and $t = t_f < .876$.

Suppose $C \geq 5.6$. Then

$$E_q/E = C - \sqrt{C^2-1}\, t > 5.6 - .876(5.51) = .77324.$$

The function increases for $C \geq 5.6$. Here $5.51^2 = 30.3601 > 30.36$ and $(11/8)(.77324) > 1$, so $E/E_q < 11/8$. Also $\beta > .9839$, because $31.36(1-.9839^2) = 1.0016631744 > 1$. The positive reverse height satisfies

$$\langle y, u_{18}\rangle = \frac{\beta - t}{1-\beta t} > .78.$$

Its denominator is positive and its limiting gap is

$$.9839 - .876 - .78(1 - .9839 \cdot .876) = .000179192 > 0.$$

The original maximum supplies all-point comparison at q. Lemma 9.1 excludes $C \geq 5.6$. Every actual near/far fork edge therefore has $C < 5.6$.

10.2 The entire near/far family

Suppose the actual fork opposites are near and far:

$$c_N = Np - \gamma\Omega_f - \Omega_o/2, \qquad c_F = Fp - \delta\Omega_o - \Omega_1/2,$$

with f, o the different actual forks. Their times and exact edge cosines are

$$N^2 = 9\gamma^2 + 8\gamma + 5/4, \quad F^2 = 9\delta^2 + \delta - 1/2, \quad C_N = \frac{18\gamma+8}{\sqrt{19}}, \quad C_F = \frac{18\delta+1}{\sqrt{19}}.$$

The short-edge bounds and $\sqrt{19} < 4.359$ give $\gamma < .912, \delta < 1.301$. Put $U = 3\gamma + 4/3, V = 3\delta + 1/6, a^2 = 19/36$. Then $N^2 = U^2 - a^2, F^2 = V^2 - a^2$, and

$$(UV - a^2)^2 - N^2F^2 = a^2(U-V)^2 \geq 0, \qquad UV - a^2 > 0.$$

The actual mutual magnitude consequently satisfies

$$\begin{aligned} \lambda = -\langle c_N, c_F\rangle &= NF - 8\gamma\delta - \gamma/2 - (9/2)\delta - 1/4 \\ &\leq \delta(\gamma - 1/2) - 5/9 \\ &< 1.301(.412) - 5/9 = .536012 - 5/9 < 0. \end{aligned}$$

This contradicts all-wall acuteness. If $\gamma \leq 1/2$ the upper bound is already negative; the accepted own list in fact has $\gamma \geq \cos(\pi/4) > 1/2$. Zero, every finite, parallel and unbounded dotted mutual are excluded at once, for all finite or dotted near/far owns.

10.3 Exhausting the actual head and fork covers

For a dotted head, (27) removes every far-next-three role from Section I.9. What remains is isolated, near-three, or far-three. Their six unordered pairs are exhausted by

Actual pair	Complete excluding argument
isolated/isolated	Sections I.14 and I.15
isolated/near	Section I.16
isolated/far	Sections I.17 and I.18
near/near	Section I.9
near/far	The complete argument above
far/far	Section I.18

Both assignments of the genuine signed forks are included. Every pair is impossible, so no dotted head remains.

The head cover also includes own zero, finite and parallel cases. Section I.1 removes own $0, 3, 4$ by time constraints and own 6 by a positive affine $[6, 3]$ kernel. At own 6 an actual null is $\sqrt{3}H + 2n_1 + n_2$; at parallel own it is $H + n_1$. The remaining finite labels $5, 7, 8, 9$ and every integer label at least 10 are excluded by the complete conjunction in Section I.22. Every dotted own has just been excluded. Thus the whole-D_{18} diagram is impossible.

Theorem 5.1 excludes every dimension at least 19. At exact dimension 18, the fresh complete cover and its row consumers, including Section 6, leave only whole D_{18} at the same original farthest vertex. The head/fork argument excludes it. Hence dimension 18 is impossible, proving Theorem 1.1. Each step uses its own actual-rank cover and the same original vector within a case; no dimension heredity is used.

11 Finite algorithms and conditional verification

11.1 The mathematical finite-cover calculation

The rank covers are finite combinatorial arguments supported by real inequalities, not field enumerations. For fixed total rank n, the forest calculation is:

1. List partitions of n into ranks of spherical irreducibles. Retain all exceptional finite types and classical types at their actual ranks. Keep $I_2(m)$ as a symbolic ordinary rank-two family for unbounded m, distinguishing its exceptional actual typed profiles.

2. Mark at most one allowed $I/J/N_7/O_7$ profile by the actual-wall compatibility lemma. Apply joint bounds only to disjoint sets of components, counting each defect once.

3. Sum proved same-vector energy bounds. Discard a forest only when a strict lower bound is at least 1. If a dominant rank needs a fresh local layer, retain it until that actual-rank lemma is established.

4. Return the retained grouped necessary cover and the inequality discarding every other partition.

Termination follows from finitely many rank partitions and finite type choices at each rank. The unbounded native parameter is represented by inequalities proved over its entire domain. Completeness uses spherical classification, compact acute simplicity, complete actual-defect compatibility, and spectator-uniform component bounds. The output is a necessary cover; no realization follows. A program testing only finitely many native labels would not establish it.

Local own/white calculations have a different domain. After deleting a specified original node, classify every finite extension of the retained spherical forest by its actual opposite wall, including own zero, all finite owns, parallel and dotted owns. Finite white extensions are bounded by spherical

classification. Infinite own families are closed by analytic inequalities over complete intervals. Termination of a finite classification does not justify discarding a continuous branch. Completeness must include spectator attachments, affine kernels and every original wall for every auxiliary point.

11.2 The distinct role of scripts

The final continuous seals are proved by displayed rational inequalities and calculus arguments. Ancillary scripts can check finite arithmetic tables, but their execution records are separate from the geometric arguments that make those tables applicable.

For a reproducibility checker attached to a finite table, its mathematical statement must specify the exact input domain, symbolic own parameters, exhaustive loops or recursion, the terminating measure, exact tests and returned cover. Its correctness proof must map every admissible geometric case to a generated entry and each rejected entry to a valid exclusion. The execution record then states what that particular program returned on those inputs. Algorithm theorem, execution claim and geometric consumer are distinct. Historical receipts identified in the ledger are not a new complete replay of this revision.

11.3 The formal boundary

A suitable Lean kernel may take as explicit parameters:

1. precise finite/affine classification and compact acute geometry statements supplied from the literature;

2. precise internal component and local-cap statements whose full written project proofs have been reviewed;

3. a precisely typed necessary-cover output, if exact enumeration is provided as a conditional premise.

It may verify rank sums, disjoint-cover subtraction, one-vector algebra, rational implications and the final composition giving $n \leq 17$. An internal geometric statement remains an internal proof obligation when used as a parameter; it does not become a literature theorem. Logic maps must list parameters as well as printed axioms.

The supplementary Lean sources express selected algebraic and discrete implications with explicit premises. Their logic maps list the geometric, analytic, classification and finite-data statements left outside each kernel. These conditional deductions are not a formal construction of a hyperbolic polytope or a formalization of every geometric lemma. Separate compilation and actual axiom-audit records document the scope of each checked module.

A Ordinary component proofs and actual defects

Throughout the proof appendices, $F(t) = 2t/(1+t^2)$ denotes the full-/half-tangent conversion. This appendix collects the contact calculations behind the small-component rates. Each is an argument on an actual whole component with arbitrary finite spectators, using Section 2. No field condition is present. We use the spherical/affine classification and the standard Lannér classification: a minimal indefinite Coxeter diagram with all proper subdiagrams positive definite has at most five vertices. A positive affine principal is already impossible by the actual null argument in Section 2.

Proposition A.1 (Ordinary costs and the complete small defects)**.** *All the ordinary component bounds in Section 4 hold with arbitrary finite spectators. For the moment use the weaker bound $B_3, H_3 > 1/4$; the improvement to $2/7$ is proved immediately below. The only exceptions to the rank-over-14 small-component baseline are one possible isolated short I, its separately funded J alternative, or one of the two typed rank-two defects.*

Write $m@j$ for actual contact $\cos(\pi/m)$ at old node j. A missing contact is zero. For a wall put $h = A^2$, own cost q and full foreign cost Q, so $h = q + Q - 1 > 0$. The following calculations prove the proposition; their tables are finite mathematical case analyses rather than empirical output.

Lemma A.2 (Common-wall forcing and isolation)**.** *Suppose the energy of a component is at most e. A pair for which the right side of (9) is below K_{ij} has one actual opposite. If one wall closes every direction of the component, then, with $v_i = (C\beta)_i/\sqrt{d_i}$ and $R_e^2 = 4e/(1+e)^2$,*

$$q - 1 \le h \le R_e^2 \min_i v_i^2. \tag{29}$$

If $h < 1/4$, it has no foreign contact. In a path, a wall closing every direction can have multiple contacts only at the two endpoints; in a D_4 star it has at most one.

Proof. The first assertion is (9). For a closed direction, $\tanh L_i = A/v_i$ and $\tau_i^2 \le e$, which gives (29). Every nonzero foreign cost is at least $1/4$ and is strictly below h by (8); hence the isolation assertion. If the old path between two contacts omits a direction closed by the wall, deleting that direction leaves a cycle in an actual finite replacement. This is impossible. The same argument applies to every two contacts in a D_4 star. Finally, a single-contact own cost $q = 1$ gives an affine principal set, impossible even in the presence of further spectators. □

A.1 The short directions and their exact debits

For a short A_1 opposite put $b = \beta^2$, $h = A^2$, and let Q be its full foreign cost. Then

$$h = b + Q - 1, \qquad h < b/3, \qquad Q < 1 - 2b/3.$$

Equation (8) rules out two foreign components. With one, its cost must be the sole value below $1/3$, namely an A_1 contact 3 of cost $1/4$. Thus the only possibilities are

$$I:\ \text{isolated dotted } A_1,\quad 0 < h < 1/2; \qquad J:\ \text{dotted } A_1 + A_{1,3},\quad 1/3 < h < 3/8.$$

For completeness, with one foreign component, $Q < h = b + Q - 1$ forces $b > 1$, and $Q < 1/3$ then forces $Q = 1/4$. For J, the heavy edge contributes more than $A/2$ to $x \cdot v$. The light shadow contributes at least $A(1 - \sqrt{1 - 1/(4h)})$; this is at least $A/2$ if $h \le 1/3$. Strict (5) therefore gives the asserted lower bound. Native acuteness excludes two short walls. Indeed their height product is below $1/2$, so their product must be zero. Disjoint supports give a negative product, heavy overlaps give a spatial product above $1/2$, and a sole light/light overlap of two J walls gives $1/4$, below their height product. None is zero. A J heavy direction has $\tanh^2 L > 4/13$, hence energy $> 9/100$; its light direction has $\tanh^2 L > 2/3 > 16/25$, hence energy $> 1/4$. Thus J funds both A_1 factors above their combined ordinary baseline $2/10$. Every nonshort direction pays at least $5 - 2\sqrt{6} > 1/10$, since $\sqrt{6} < 49/20$. We may conservatively discard the one short direction, but only the isolated type I can create an unfunded debit.

The rank-two exceptions are

old block / new isolated contact	$h = A^2$	energy
$A_2/7$	$(4/3)\cos^2(\pi/7) - 1 \in (0, 1/12)$	$> 9/100$
$I_2(7)/3$	$(1/4)\csc^2(\pi/7) - 1 \in (0, 1/3)$	$> 9/100.$

(30)

Their spatial supports lie entirely in their own blocks. Two defects in different blocks, or a defect and a short A_1, would have actual wall product $-\sqrt{h_1 h_2} \in (-1/2, 0)$. That interval is forbidden: distinct acute unit walls have product 0 or at most $-1/2$. Therefore

$$\text{at most one short } A_1, \quad \text{or one rank-two defect, but never both.} \tag{31}$$

This argument is independent of every other component type.

A.2 The two rank-two estimates

For A_2, set $e_2 = 319/2000$ and $\sinh^2 L_0 = 31/50$. If both edges have length at least L_0, then their half-tangents exceed $173/500$ and energy is greater than $4 \cdot 173^2/(3 \cdot 500^2) > e_2$. A shorter edge cannot have two positive own contacts: $3/3$ is affine, while $3/4$ or larger gives $\sinh^2 L \geq 4\sqrt{2}/9 > 31/50$, increasing with either contact and with Q. If only the other node is contacted, apply the same test to that shorter contacted direction. For one contact of square b, the shorter-edge condition is

$$h = 4b/3 + Q - 1 < 124b/243, \qquad Q < 1 - 200b/243. \tag{32}$$

Every positive foreign cost must be below h. A label 3 gives $h < 1/4$, hence isolation and negative height. For a label 4, $h < 62/243$ forces each foreign cost to be $1/4$; the only positive possibility is $Q = 1/2$, but then $h = 1/6 < 1/4$. A label 5 has $Q < 1/2$, hence at most one foreign component, whose cost exceeds h because its own cost $\varphi^2/3$ is below one. A label 6 is affine. For a finite label $m \geq 7$, $b > .81$ gives $Q < 1/3$; thus $Q = 0$ or $Q = 1/4$. The latter is possible in (32) only for $b < 729/800$, namely $m = 7, 8, 9, 10$. Indeed $\cos^2(\pi/11) > 45/49 > 729/800$, using $\pi < 22/7$. A dotted contact has $Q < 1/4$ and is isolated.

For an isolated finite contact the same wall closes the mate, since its reciprocal ratio squared is $b/(4b-3) > 1$. For $m \geq 8$ that actual mate has $\tanh^2 L \geq 6\sqrt{2} - 8 > 400/841$, so its energy exceeds $4/25 > e_2$. For an isolated dotted contact, $b < 243/200$ and its distinct mate has shadow $\tanh^2 L \geq b/(4b-3) > 81/124 > 16/25$. For a mixed $m = 7, 8, 9, 10$ contact, the mate's white cost is $b + 1/4 > 1$; the distinct shadow instead gives $\tanh^2 L \geq b/(4b - 9/4) > 11/17 > 16/25$. Both give energy greater than $1/4$. Thus the only exception is isolated new 7 at old A_2. The root $t = \cos(2\pi/7)$ of $8t^3 + 4t^2 - 4t - 1$ lies in $(623/1000, 5/8)$: the polynomial changes sign there and is increasing on $[1/2, 1]$. Hence $b > 1623/2000$ and its actual mate satisfies $\tanh^2 L = 4 - 3/b > 164/541 > 3600/11881$, proving its cost $> 9/100$.

For an old $I_2(m)$, $m \geq 4$, put $k = \cos(\pi/m)$ and $d = (1 - k^2)^{-1}$, so $C = d \left(\begin{smallmatrix} 1 & k \\ k & 1 \end{smallmatrix}\right)$. At energy at most $1/7$, (9) forces one common opposite. Two contacts give an actual $\cosh L \geq \cot(\pi/(2m))/\sqrt{3} > 11/8$, with individual energy $> 3/19 > 1/7$. For one contact β,

$$q(1 - 7k^2/16) \leq 1, \qquad q = d\beta^2. \tag{33}$$

Its minimum left side is $(9d + 7)/64$. Since $m \geq 8$ gives $d \geq 4 + 2\sqrt{2} > 19/3$, only four old labels remain:

m	consequence of (33)
4	$\beta^2 \leq 16/25 < \cos^2(\pi/5)$: $3, 4$; $h \leq 7/32 < 1/4$, $q \leq 1$;
5	$d > 25/9$: only 3; $h < 7/32$, $q < 1$;
6	only 3; $q = 1$;
7	$d > 5$: only 3; $Q < 3/16$.

Isolation or the affine principal excludes the first three rows. The last is isolated new 3 at old $I_2(7)$, with $0 < h < 1/3$. Its actual mate has $\cosh L = \cot(\pi/7)/\sqrt{3} > 109/91$ by the same cubic bracket; the squared comparison has positive numerator 2652. Thus its energy also exceeds $9/100$, completing (30) and (31).

A.3 Rank three and four: one common-wall calculation

For energy $1/4$, adjacent correlation at least $1/\sqrt{3}$ forces a common opposite: the lowest cleared comparison is $74/\sqrt{3} > 39$. At energy $1/3$ the threshold $1/\sqrt{2}$ works, since $51/\sqrt{2} > 34$. For A_4 at energy $2/7$, the adjacent correlations are $\sqrt{3/8}, 2/3, \sqrt{3/8}$; the forcing test is $k^3+16k^2+36k-28 > 0$, whose lowest value is positive because $291\sqrt{6} > 704$. Direct inversion verifies the indicated correlation thresholds for A_3, B_3, H_3 and B_4, D_4, F_4. For H_4 use energy $11/36 < 1/3$. Thus a common wall closes all directions in every case.

Number path bonds as $A_3 : (3,3)$, $B_3 : (4,3)$, $H_3 : (5,3)$, $A_4 : (3,3,3)$, $B_4 : (4,3,3)$, $F_4 : (3,4,3)$, and $H_4 : (5,3,3)$. By Lemma A.2, multiple contacts can only be at the endpoints. Two endpoint 3 contacts in A_3, A_4 are affine; the next minimum is $3/4$. In the other rows below even $3/3$ is excluded. The displayed values are lower bounds for $q - R_e^2 v_j^2$, contradicting (29):

T	j	lower bound
A_3	2	$177/200 + 9\sqrt{2}/100 > 1$
A_4	2	$(230 + 15\sqrt{2})/243 > 1$
B_3	2	$(77 + 18\sqrt{2})/100 > 1$
B_4	3	$15/16 + \sqrt{2}/8 > 1$
F_4	2	$15/16 + \sqrt{2}/4 > 1$
H_3	2	$(72\varphi + 95)/100 > 1.$

D_4 already has at most one contact. In H_4, two endpoint contacts retain, after deleting old node 4, a path at least $(3,5,3)$ of determinant $(5-4\varphi)/16 < 0$. Hence all contacts are single.

For a single contact at j let $\sigma_j = \min_i(v_i^2/q)$. Formula (29) gives $q \le (1 - R_e^2\sigma_j)^{-1}$. The complete inverse calculation is as follows; entries are in path order, and D_4 means leaf,center.

T	d_j	σ_j	upper q
A_3	$3/2, 2, 3/2$	$1/9, 1/3, 1/9$	$225/209, 75/59, 225/209$
A_4	$8/5, 12/5, 12/5, 8/5$	$1/16, 1/6, 1/6, 1/16$	$162/155, 243/215, 243/215, 162/155$
B_3	$3, 4, 2$	$1/3, 1/2, 1/3$	$75/59, 25/17, 75/59$
D_4	$2, 4$	$1/4, 1/2$	$16/13, 8/5$
B_4	$4, 6, 4, 2$	$1/4, 1/3, 1/2, 1/4$	$16/13, 4/3, 8/5, 16/13$
F_4	$4, 12, 12, 4$	$1/2, 2/3, 2/3, 1/2$	$8/5, 2, 2, 8/5$

Here R_e^2 is respectively $16/25$, $56/81$ for A_4, and $3/4$ for the $1/3$ rows. Dividing the upper q by d_j gives the finite native lists: A_3 endpoints $3/4/5$, middle $3/4$; A_4 endpoints $3/4$, inner 3; B_3 nodes $3; 3; 3/4$; D_4 leaf $3/4$, center 3; B_4 nodes $3; \varnothing; 3; 3/4$; F_4 endpoints 3, inner $\varnothing$. Each has $q \le 1$. Equality is affine; in every other listed case (29) gives $h < 1/4$ (the largest needed bounds are $16/75, 7/155, 28/405, 3/16$). Isolation then gives $h = q - 1 < 0$. This proves the six crystallographic and A rows.

For H_3, the minimum contact $\beta = 1/2$ at node 1 or 2 gives tests at node 3 equal to $(199\varphi + 263)/500$ and $(61\varphi + 77)/125$, both above one. At node 3, $\beta \ge 1/\sqrt{2}$ gives the test at node 1 at least $(27\varphi + 118)/150 > 1$. Only 3@3 remains, with $q = (\varphi+2)/4 < 1$ and $h \le 4\varphi^4/75 < \varphi/4$. No spectator gives negative height; one has cost $Q > h$; two have $Q \ge 1/2$ and hence $h \ge q - 1/2 = \varphi/4$. All are impossible. For H_4, $R_e^2 = 1584/2209$; at $\beta = 1/2$ the four tests (contact node, test direction) are

$$(1,4) : \frac{4938\varphi + 3688}{2209}, \quad (2,4) : \frac{7209\varphi + 5334}{2209}, \quad (3,1) : \frac{2500\varphi + 2667}{2209}, \quad (4,1) : \frac{625\varphi + 1219}{2209}.$$

All exceed one, the last by $\varphi > 198/125$; larger contacts only increase these positive tests. This proves $H_4 > 11/36$.

A.4 The two intermediate A components

Use $C_{ij} = 2\min(i,j)(N - \max(i,j))/N$, with $N = 6, 7$. For A_5, distinct 2/3 and 3/4 pairs each demand a sum above 109/100. The vector $u_2 + 2u_3 + u_4$ has square $7 + 4\sqrt{2} < 443/35$, and

$$(109/50)^2 - (3/8)(443/35) = 209/35000 > 0.$$

For A_6, the three pair demands are

$$2249/2000, \qquad 11547/10000, \qquad 2249/2000;$$

the vector $u_2 + 2u_3 + 2u_4 + u_5$ has square $< 33781/1250$, with demand 34037/10000 and Cauchy gap 2181583/700000000 at energy 3/7. These pair bounds follow by substitution in $F(s/2)^2 < K$. Thus a common 2/3 occurs for A_5, and a common 2/3 or 3/4 for A_6, up to reversal.

Each retained path allows one contact. The full two-contact list is

Component/pair	positions	necessary finite-white labels
$A_5, 2/3$	$1,3$	3/3 or $4@1 + 3@3$
	$1,4; 1,5; 2,4; 2,5$	both 3
	$2,3$	both in $3,4,5$
$A_6, 2/3$	$1,3$	3/3 or $4@1 + 3@3$
	$1,4; 1,5; 1,6; 2,4; 2,5; 2,6$	both 3
	$2,3$	$3,4,5$ at 2; $3,4$ at 3
$A_6, 3/4$	$3,4$	both in $3,4,5$
	other eight left/right pairs	both 3

Indeed deletion of either closed node leaves a connected finite path or D/E_6 tree: a 4 can only be terminal, and D/E bonds are all 3. All-3 doubles are affine cycles. The adjacent non-3/3 doubles force an actual edge of cosh at least $(3 + \sqrt{2})/2 > 11/5$ in A_5; for A_6 the minimum formulas are $(5 + 4\sqrt{2})/\sqrt{7}$, $(4\sqrt{2} + 6)/\sqrt{14}$, and $(12 + 9\sqrt{2})/\sqrt{42}$, all above 5/2. The A_5 terminal-4 row has $Q = 0$ and cosh $(3 + \sqrt{2})/2$. The A_6 version has white cost 9/10, hence $Q = 0$, and is treated below. These cosh thresholds imply individual energies above 3/8 and 3/7.

For single 3 contacts the complete own costs and strict white bounds are

component/pair	j	q	Q upper
$A_5, 2/3$	$1,2,3,4,5$	$5/12, 2/3, 3/4, 2/3, 5/12$	$2/3, 2/3, 5/8, 1/2, 5/8$
$A_6, 2/3$	$1,2,3,4,5,6$	$3/7, 5/7, 6/7, 6/7, 5/7, 3/7$	$2/3, 2/3, 3/5, 2/5, 2/5, 3/5$
$A_6, 3/4$	$1/6; 2/5; 3/4$	$3/7; 5/7; 6/7$	$5/8; 1/2; 5/8.$

End and small-Q rows have $h < 1/4$, or only one foreign component with cost $Q > h$. In A_5 at $j = 2,3$ two foreign components would each exceed $1/3, 1/4$, respectively, forcing $Q \geq 2/3$; the same gap treats $A_6, j = 2$. In the remaining $A_6, j = 3,4$ rows, two foreign bonds must be 3 since $Q < 5/8$; together with the contacted old node and its two neighbors they form affine two-fork D_5. This excludes all single 3 cases. A single internal 4 gives the affine $(3,3,4)$ star; an end 4 has $Q \leq 1/3$ and $h < 1/4$, again impossible.

A single 5 is at a retained path end; its H_3/H_4 white cost exceeds 3/4, so $Q = 0$. All cases are

covered by these actual positive replacements:

component/pair	contact	actual direction	$\cosh L$
$A_5, 2/3$	1 or 5	4 or 2	$\varphi^3/\sqrt{3}$
	2	3	φ^2
	3	2	$\sqrt{3}\varphi^3$
$A_6, 2/3$	1	4	$3\varphi^3/\sqrt{21}$
	2	3	$4\varphi^2/\sqrt{14}$
$A_6, 3/4$	1 or 6	4 or 3	$3\varphi^3/\sqrt{21}$
	3 or 4	4 or 3	$9\varphi^3/\sqrt{21}$.

The first group exceeds $11/5$, the second $5/2$, using $\varphi > 8/5$, $\sqrt{21} < 23/5$, $\sqrt{14} < 15/4$. No new wall is inferred: each replacement contains the already actual c. Finally the A_6 double $4@1 + 3@3$ has the positive actual future point

$$\xi = c + (\sqrt{2} - 1/2)n_1 + (\sqrt{2} - 1)n_2 + (\sqrt{2}/2)n_3, \qquad \xi^2 = 5/4 - \sqrt{2} = -k.$$

Its products with c, n_1, n_3 vanish and its product with n_2 is $(\sqrt{2}-3)/4 < 0$. Here $A^2 = (5+4\sqrt{2})/7$ and $A^2 - 9k = (335 - 236\sqrt{2})/28 > 0$; (3) gives energy $> (A - \sqrt{k})/(A + \sqrt{k}) > 1/2$.

A.5 The four intermediate B components

Use $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$ and, for $i < r$, $\omega_i = \sqrt{2}(e_1 + \cdots + e_i)$, while $\omega_r = e_1 + \cdots + e_r$. At energies $3/7, 1/2, 4/7$ the relevant forcing signs are $127\sqrt{2/3} - 100$, $(69\sqrt{6/7} - 52)/7$, and $(199\sqrt{6/7} - 176)/7$, all positive. Thus B_5, B_6 have a common $2/3$, and B_7, B_8 a common $6/7$.

For B_5, B_6, deletion 2 retains $A_1 + B_3/B_4$. The right contact is $3@3$, with the sole B_5 exception $3@5$ giving F_4; deletion 3 allows one left $3/4/5$. For $3@3$, own 0 is affine, own 3 is a triangle, and own ≥ 4 gives $\cosh L_3 \geq 2 + 3/\sqrt{2} > 3$. With no right contact the complete choices are $3/4/5$ at 1 or 2: $3@2$ and either 4 are affine; $3@1$ has $h = Q - 1/2 < 1/6$ and closes a forbidden foreign direction. The two 5 cases have $Q = 0$: $5@1$ also closes 4 with $\cosh L_4 = \varphi^3 > 3$, and $5@2$ has $\cosh^2 L_3 = 2\varphi^2/(2 - \varphi) > 9$. For the B_5 short-end $3@5$, finite F_4 forbids left 1 and all spectators; own 0 is affine and the other replacement forces own 3, giving $\cosh L_2 = 2 + \sqrt{2}$. Every branch exceeds the individual threshold $11/5$.

For the $6/7$ pair, a wall without a right contact has a single left 3 or endpoint 4. Apart from far $3@1$ these give an affine fork, double-4, or star. A minimal nonpositive principal for the far branch must contain $c, 7$, since deletion 7 is finite; their connecting path $c, 1, \ldots, 7$ has eight vertices. Spectators attach through c and do not shorten it. This violates the Lannér order bound, excluding the branch with all spectators. In B_7, let β be the contact to the retained right A_1:

β/own 6	left contacts	pressure or rejection
$\beta \geq \cos(\pi/5)$, any	none	$\cosh L_6 \geq \sqrt{6}\cot(\pi/5) > 3$
$\beta_4, 0$	any allowed	affine $(4, 4)$
$\beta_4, > 0$	none	$\cosh L_6 \geq \sqrt{24}$
$\beta_3, > 0$	none	$\cosh L_6 \geq 2 + \sqrt{2}$
$\beta_3, 0$	none or 1	affine F_4
$\beta_3, 0$	one $j = 2, \ldots, 5$	$\cosh L_6 \geq 1 + 3/\sqrt{2} > 3$.

Here $\beta_m = \cos(\pi/m)$. The last row retains the full foreign cost: its white residual is $\Delta \leq 3/4 - j(6 - j)/12 \leq 1/3$, and $v_6 \geq \sqrt{12}(j/12 + 1/(2\sqrt{2}))$. Thus no allowed E_7 extension is silently removed.

In B_8 the retained right B_2 has one simple 3 at either end. At long end 7, own $0/3$ is affine and own 4 gives $\cosh L_7 \geq 6 + 7/\sqrt{2} > 7$. At short end 8, own > 0 gives $\cosh L_6 \geq \sqrt{6} + \sqrt{3} > 11/3$. At own 0, the left-white budget permits only 3@1 or 3@5: none or 1 gives affine F_4, but 3@5 is an allowed bridge, with $\Delta \leq 1/12$ and $v_6 = \sqrt{12}(5/12 + 1/(2\sqrt{2}))$. It has $\cosh L_6 \geq 5 + 3\sqrt{2} > 9$; it is estimated, not removed by an incorrect F_4 sealing claim. These complete lists prove the four asserted B costs.

A.6 Extended stars, D_7, and the D_8 closure

Let S be a central D_4 star with an extended arm. If one wall closes its center and three neighbors, every minimal nonpositive principal contains c and all four nodes of S, since deletion of any of them is an actual finite replacement. Compactness and the Lannér order bound make that principal exactly $S + c$. Two contacts give a nonpositive proper triangle or cycle. A sole center 3 is affine, and a larger center contact has a nonpositive proper star. A sole leaf contact is $3/4/5$ from its finite proper path: 3 gives finite D_5, 4 affine, leaving only 5. If that leaf extends, it retains a nonpositive $(5, 3, 3)$ star; if another arm extends, it retains a nonpositive $(5, 3, 3, 3)$ path. Each omits a supposedly closed S direction, a contradiction. Further negative contacts cannot repair the positive nonpositive quadratic witness. Thus the four opposites cannot all coincide.

For D_5, D_6 the minimum central correlations are $\sqrt{3/5}, \sqrt{2/3}$. At energies $5/14, 3/7$ the forcing signs are $544\sqrt{3/5} - 362$ and $127\sqrt{2/3} - 100$, positive. For E_6, E_7, E_8 the arm lengths are $(1, 2, L)$ with $L = 2, 3, 4$. In raw norm-two coordinates the central Schur residual is $\delta = 2 - [1/2 + 2/3 + L/(L + 1)]$; on an arm of length m, squared central correlation is $t/(\delta + t)$, $t = m/(m + 1)$. The minima are $\sqrt{3}/2, \sqrt{6/7}, \sqrt{15}/4$. At energies $9/20, 5/9, 2/3$ even $4e/(1 + e)^2$ lies below these correlations. All four opposites would coincide, proving the five extended-star costs.

For D_7 write $o = r - 2$, $a = r - 3$, with fork leaves b, f. At energy at most $1/2$, correlation $\sqrt{4/5}$ forces a common o, a. Deletion a retains $A_3 = b - o - f$, so its right contact is none, middle 3, or endpoint $3/4/5$; deletion o allows one left contact. The complete surviving profiles are

$$4@a + 3@o, \qquad \text{isolated } 5@b, \qquad 3@a + 4@b, \tag{34}$$

up to exchanging the fork leaves. To check the list: middle 3 forces $4@a$ (own $0/3$ is affine); endpoint 5 seals all white extensions by H_4, and a contact at the removed a leaves a nonpositive $(3, 3, 3, 5)$ path. Endpoint 4 forces $3@a$, otherwise an affine terminal-4 fork remains. Endpoint 3 excludes every left contact by cycles or retained F_4; a foreign 3 is affine E_7, a foreign 4 is an affine terminal-4 fork, and larger foreign bonds are nonfinite. Its isolated height is negative for $r = 7$ and it contains affine E_8 for $r \geq 8$. With no right contact, internal-left 3 and endpoint 4 are affine; far-left 3 needs a Lannér path of at least six vertices. This accounts for all spectators as well as the removed own contact. The first and third roles have $\cosh L_o = a + (r - 2)/\sqrt{2} > 7$. The isolated 5 has $\tanh^2 L_a = 1 - (9 - 4\sqrt{5})/(r - 3)$; at $r = 7$ its half-tangent square is $4/\sqrt{5} - 1 > 3/4$. Each role therefore contradicts energy at most $1/2$, proving the D_7 cost.

For D_8, common $4/5$ has exactly $4@4 + 3@5$. Indeed deletion 4 permits simple 3 at a right D_4 leaf, and deletion 5 allows one left 3 or endpoint 4. Long-leaf 3 forces $4@4$ and forbids foreign contacts by the retained $(3, 4, 3, 3)$ path. A fork 3 with a left 3 gives an affine cycle; without a left contact it is affine E_8; a left 4 is nonpositive. With no right contact only far $3@1$ survives the affine tests, but its path to 5 has six vertices. Common $5/6$ has the preceding center-neighbor list. Both common pairs thus force an actual half-tangent square above $3/4$. At energy at most $4/7$, their opposites are distinct.

Here is the remaining three-direction calculation. Put $q = \sqrt{4/5}$, $k = \sqrt{5/6}$, $\lambda = (1 + k)/2$, $\alpha = k\sqrt{1 - k}/2$, $e = 4/7$, and $P(t) = t/(1 + \sqrt{1 - t})^2$. Write $h_i = x \cdot u_i$ and decompose $x =$

$aw + bv + zt_0 + y$, where w, v are the normalized sum and difference of u_5, u_6, $u_4 = qu_5 + t_0/\sqrt{5}$, and $y \perp w, v, t_0$. The full-tangent laws imply

$$h_4 h_5 \geq P(q), \qquad T := (1+\alpha)b^2 + \lambda z^2 \leq \delta := \lambda e - P(k) < 1/400.$$

For the second inequality take the tangent line to $4p/k - (1+p)^2$ at its first zero $p = P(k)$, with $p = h_5 h_6 = \lambda a^2 - (1-\lambda)b^2$. Keeping the negative quadratic terms in $h_4 h_5$ gives

$$h_4 h_5 \leq q\lambda e + A_0|b| + B_0|z| - qkb^2 - q\lambda z^2, \qquad A_0^2 = 8/105, \quad B_0^2 = 4/35.$$

Weighted Cauchy bounds the linear part by $\sqrt{C_0 T}$, where $C_0 = (8/105)/(1+\alpha) + (4/35)/\lambda < 47/250$. Also $qk/(1+\alpha) > .71$, $q > .71$, and $q\lambda e < .489$. Consequently

$$h_4 h_5 < .489 + \sqrt{(47/250)T} - .71T < 101781/200000 < .509 < P(q),$$

a contradiction. The rational seals are $.8944 < q < .8945$, $.9128 < k < .9129$, $.29 < \sqrt{1-k} < .2953$, $.13 < \alpha < .14$, and $.95 < \lambda < .95645$. The tight δ comparison reduces to $91280000 \cdot 70000 - 167780209 \cdot 38083 = 26300653 > 0$; the last square-root comparison uses $\sqrt{47/100000} < 271/12500$. This proves $D_8 > 4/7$ and completes Proposition A.1.

A.7 The strengthened rank-three rates

We prove the improvement $B_3, H_3 > 2/7$ used in the forest accounting. Suppose the component energy is at most $2/7$. The adjacent correlations are at least $1/\sqrt{2}$, so the forcing argument above gives one actual opposite for the whole component, with

$$h \leq R^2 \min_i v_i^2, \qquad R^2 = 56/81.$$

All foreign costs remain included in $h = q + Q - 1$. Multiple contacts can occur only at the two path endpoints.

For B_3, endpoint contacts u, w give

$$q - R^2 v_2^2 = (131u^2 + 106w^2 + 50\sqrt{2}uw)/81.$$

If either label is at least 4, its minimum is $13/9$ or $443/324$, both greater than 1. Larger contacts increase the expression. At $u = w = 1/2, Q = 0$, the actual half-tangent squares in directions $2, 3$ are $(2\sqrt{2}-1)/7$; increasing Q increases both lengths. Their correlation is $1/\sqrt{2}$, so the positive two-direction cover costs at least $(10\sqrt{2} - 12)/7 > 2/7$. For one contact the diagonals are $3, 4, 2$, with minimum ratios $v_i^2/q = 1/3, 1/2, 1/3$. Thus the maximum own costs are $243/187, 81/53, 243/187$. Nodes $1, 2$ admit only native 3, and node 3 only $3, 4$, since $243/374 < \varphi^2/4$. Every $q = 1$ case is affine; every remaining $q < 1$ case has $h < 1/4$, forcing $Q = 0$ and $h = q - 1 < 0$.

For H_3, the endpoint quadratic has positive coefficients; its minimum at $u = w = 1/2$ is $(200\varphi + 287)/324 > 1$. At node 1 or 2, even contact $1/2$, tested in direction 3, gives respectively

$$(599\varphi + 823)/1620 > 1, \qquad (181\varphi + 237)/405 > 1.$$

At node 3, contact at least $1/\sqrt{2}$, tested in direction 1, gives $(75\varphi + 374)/486 > 1$. Only contact $1/2$ remains, with

$$q = (\varphi + 2)/4 < 1, \qquad h \leq 14\varphi^4/243.$$

No foreign component gives $h < 0$; exactly one is impossible by $Q < h$. Two or more give

$$h \geq q - 1/2 = \varphi/4 > 14\varphi^4/243,$$

the final difference being $(75\varphi - 112)/972 > 0$. This covers all native/dotted contacts and all spectators.

A.8 A rank-free joint payment for an isolated I

Let the actual isolated short wall be

$$c_0 = \sqrt{h}\,p - \sqrt{1+h}\,e, \qquad 0 < h < 1/2,$$

with e the old unit I normal. For an actual opposite outside I, write its time as A and its contact with e as γ. The finite white endpoint retains e, so $\gamma = 0$ or $1/2 \le \gamma < 1$. These opposites are distinct. For the chosen unit dual $u \perp e$, its spatial derivative has square at most $1 + A^2 - \gamma^2$. If $\gamma = 0$, the native gap gives $A^2 \ge 1/(4h)$; if $\gamma > 0$, acuteness gives $A^2 \ge (1+h)\gamma^2/h$. Since $\gamma^2 < 1$, the derivative comparison yields in both cases

$$\tanh^2 L \ge \frac{1}{1+4h}. \tag{35}$$

For the nonzero case, substitute the lower bound on A^2; the difference from $1/(1+4h)$, after positive denominators are cleared, has numerator $h((5+4h)\gamma^2 - 1) > 0$.

Let q be the actual I-edge half-tangent square. Then $h = 4q/(1-q)^2$, $0 < q < 1/5$. If θ is the half-tangent square of the other edge, the increasing function $4s/(1+s)^2$ and (35) show $\theta > 1/5 - q$. The exact cross numerator for the strict comparison is

$$(1-q)^2(6-5q)^2 - 20(1-5q)(1+14q+q^2) = (39q-4)^2 + q^2(40 - 10q + 25q^2) > 0.$$

Thus the same original vector satisfies $e_I + \langle x, u\rangle^2 > 1/5$. For another whole A_1 this proves $I + A_1 > 1/5$; adding a second ordinary $A_1 > .1$ proves $I + 2A_1 > .3$.

For the sharper $I + T_2$ bound set $s = \sqrt{q} \in (0, 1/3)$ and $t = \sqrt{1+4h} - 2\sqrt{h}$. Equation (35) gives normalized height at least t in every other actual edge direction, and

$$e_I \ge s^2, \qquad (1-s^2)(1-t^2) = 8st.$$

We need $t + \frac{3}{2}s > .767$. At $t_0 = .767 - 1.5s > 0$, the left-minus-right of that relation is

$$P(s) = .411711 - 3.835s + 9.338289s^2 - 2.301s^3 + 2.25s^4.$$

Putting $v = s - .2 \in (-.2, 2/15)$, its coefficients are $.00343456, -.3038044, 8.497689, -.501, 2.25$. Its cubic term is at least $-.0668v^2$, so

$$P(s) > .0034 - .304|v| + 8.43v^2 > 0,$$

as the last quadratic has negative discriminant. The original relation decreases strictly in positive t, proving the barrier. This is a scalar real argument; no arithmetic chapter of its source is used.

The ordinary T_2 energy is greater than $1/7$. If $s \ge 519/2000$, $s^2 + 1/7 > .21$. Below that split, $s^2 + (.767 - 1.5s)^2$ decreases to $.2100353125 > .21$. Thus $I + T_2 > .21$, counting I once.

For $I + T_3 > 2/7$, B_3, H_3 already have the intrinsic stronger bound. For A_3, if $h > 1/5$, the actual I-edge gives $e_I > 3/70$, and $3/70 + 1/4 > 2/7$. If $h \le 1/5$, all three other half-tangent squares are at least $1/5$. The normalized A_3 dual Gram K has $K^{-1}\mathbf{1} > 0$ and $\mathbf{1}^T K^{-1}\mathbf{1} = 5 - 2\sqrt{3}$; positive Cauchy gives energy at least $(5-2\sqrt{3})/5 > 3/10 > 2/7$. The same positive-cover principle will be used below: if $K^{-1}\mathbf{1} > 0$ and every height is at least $\sqrt{\theta}$, energy is at least $\theta\mathbf{1}^T K^{-1}\mathbf{1}$.

A.9 Propagation from each actual typed rank-two defect

For N_7, put $b = \cos^2(\pi/7) < 5/6$ and $h_N = 4b/3 - 1 < 1/12$. An opposite in another component retains the whole old A_2, so has no contact there or one contact of white cost $q \in [1/3, 1)$. In the contact case its spatial product with the isolated wall has square at least $bq/3$. Acuteness and the bound on the chosen derivative give

$$A^2 \geq \frac{bq}{3h_N}, \qquad \tanh^2 L \geq \frac{b}{b + 6h_N} = \frac{b}{9b - 6} > \frac{5}{9}.$$

The first ratio is minimized at $q = 1/3$; all denominators are positive. Without contact the native gap instead gives $\tanh^2 L \geq 1/(1 + 4h_N) > 5/9$. Every other half-tangent square therefore exceeds $1/5$. In particular another A_1 or irreducible T_2 has energy greater than $1/5$.

For O_7, put $d_7 = \csc^2(\pi/7)$ and $h_O = d_7/4 - 1 \in (0, 1/3)$. No opposite in another component can contact the retained old $I_2(7)$: its minimum white cost $d_7/4 > 1$ violates positive definiteness. Its actual product with the isolated typed wall is thus $-\sqrt{h_O}A < 0$, and the native gap gives

$$A^2 \geq 1/(4h_O) > 3/4 \tag{36}$$

for every opposite in the other component. This immediately gives another A_1 cost $> 2/15$ by the full-/half-tangent comparison in the main text.

To prove another irreducible $T_2 = I_2(m)$ costs $> 1/5$, suppose the contrary. With $k = \cos(\pi/m)$ and $d = (1 - k^2)^{-1}$, distinct opposites are impossible: for $m \geq 4$, their tangent product is at most $F(\sqrt{1/5})^2 = 5/9 < k$; for $m = 3$, use the pair sum bound to get $F(\sqrt{3/20})^2 = 240/529 < 1/2$. There is one common actual opposite, and both actual edge cosines are at most $3/2$. For own contacts $b, g \geq 0$ and entire foreign cost Q, the finite endpoint equations are

$$Q + b^2 < 1, \quad Q + g^2 < 1, \quad h = d(b^2 + 2kbg + g^2) + Q - 1 > 3/4,$$
$$C_b = \frac{\sqrt{d}(b + kg)}{\sqrt{1 - Q - g^2}}, \qquad C_g = \frac{\sqrt{d}(g + kb)}{\sqrt{1 - Q - b^2}}.$$

Every nonzero own is consequently a finite native cosine. There is at least one, and $Q = 0$ or $Q \geq 1/4$.

If only $b = \beta > 0$, the mate has $C_g^2 = 1 + h/(1 - Q - \beta^2)$. The bounds $h > 3/4, C_g \leq 3/2$ force $Q + \beta^2 < 2/5$, hence $Q = 0, \beta = 1/2$. Then $h = d/4 - 1 > 3/4$ implies $d > 7 > d_8$, so $m \geq 9$. But $d > 81/\pi^2 > 81/10 > 31/4$, giving $C_g^2 = (d - 1)/3 > 9/4$, a contradiction.

If both contacts are positive and $Q > 0$, either their maximum is at least $1/\sqrt{2}$, giving $C_b^2 \geq 2 + 4\sqrt{2}/3 > 9/4$, or both are $1/2$. For old $m = 3$, the latter gives $h = Q < 3/4$; for $m \geq 4$, it gives $C_b^2 \geq (1 + k)/(2(1 - k)) > 5/2$. If both contacts are positive and $Q = 0$, a contact of label at least 5 gives $C_b^2 > 3$; two 4's give $C_b^2 = (1 + k)/(1 - k) \geq 3$. For 3/4, old $m \geq 4$ gives $C_b \geq 2$, while old $m = 3$ gives $h = \sqrt{2}/3 < 3/4$. For 3/3, old $m \geq 5$ gives $C_b^2 = (1 + k)/(3(1 - k)) > 3$; old $m = 3, 4$ give $h = 0, 1/\sqrt{2} < 3/4$. This exhausts all finite labels and all foreign costs, contradicting the common-wall low-energy branch. Thus $O_7 + T_2$ has its other component energy $> 1/5$.

B The uniform actual B/D component floor

The next proof supplies the 9/14 input used before the long-prefix argument. Its small boundary cases refer only to the complete ordinary contact calculations in Appendix A. All displacements remain in the indicated component, and all positive actual-root constructions retain every original spectator wall.

Proposition B.1 (Actual B/D floor)**.** *Every whole B_r or D_r component with $r \geq 9$ has actual projection energy greater than $9/14$, with arbitrary finite spectators.*

Proof. Assume energy at most $e = 9/14$. In B it suffices to bound the first seven coordinates; all displacements below lie there. The normalized directions in either type are $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$. We first establish distinctness of $4/5, 6/7$, then use those laws to exclude all common $1/2, 2/3$ profiles. This order is noncircular.

For an internal common $i/(i+1)$ with right retained rank at least four, finite classification permits a right long-leaf 3, or none. The first forces own 4: own $0/3$ is affine fork/triangle, and any foreign contact retains the nonpositive $(3,4,3,3)$ path. Without a right contact, internal-left 3 and endpoint 4 are affine. Only far 3@1 remains, but its path to $i+1$ has at least six vertices, contradicting the Lannér bound. Right D-fork contacts give affine E_8 or a cycle with the left contact. Thus the sole common profile is $4@i + 3@(i+1)$, $Q = 0$, with $\cosh L_{i+1} = i + (i+1)/\sqrt{2} > 7$. The D_8 right-D_4 and center-neighbor boundary calculations are in Section A; the center-neighbor calculation also handles D_9 at $6/7$. In B_9, the extra right short-end 3@9 gives F_4, excluding left and foreign white contacts. The removed own 6 remains to check: own 0 is affine F_4, own 4 the affine $(4,3,3)$ star, and own 3 gives $\cosh L_6 = (1 + 1/\sqrt{2})\sqrt{12} > 11/2$ and $\tau_6^2 > 9/13 > e$. Consequently $4/5$ and $6/7$ are distinct.

Write $h_i = x \cdot u_i$. Distinct $6/7$ gives $h_6 h_7 \geq k/(1+\sqrt{1-k})^2 > 9/16$, where $k = \sqrt{6/7} > 576/625$ with squared margin 21318. Hence $x \cdot w > 3/2$ for $w = u_6 + u_7$. We record four local tests with their exact margins.

Edge 2 *with* $\cosh L_2 \geq 1 + \sqrt{2}$. Then $h_2^2 \geq \sqrt{2} - 1$. Projection of w about u_2 has $\rho = 1/\sqrt{3} + \sqrt{2/7} < 1.114$ and residual square $S = 29/21 + 4\sqrt{2/21} < 2.62$. Its Cauchy envelope $\rho t + \sqrt{S(e - t^2)}$ decreases for $t \geq q = \sqrt{\sqrt{2} - 1}$: its derivative condition follows from $\rho^2 e < 13/16 < 6/5 < \|w\|^2 q^2$. Since $q < .644$, $e - q^2 < .23$, and $\sqrt{2.62 \cdot .23} < .777$, $x \cdot w < 1.114(.644) + .777 = 1.494416 < 3/2$. These follow by squaring, using $1.414 < \sqrt{2} < 1.4143$, $1/\sqrt{3} < .578$, $\sqrt{2/7} < .535$, $\sqrt{2/21} < .309$.

Isolated 5@1. Its height square is $(\varphi - 1)/2$ and its actual direction-4 shadow has $\tau_4^2 = \varphi - 1$. For the same w, projection about u_4 has $\rho = 2(1/\sqrt{6} + 1/\sqrt{7}) < 787/500$ and $S = 16/21 + 4/\sqrt{42} < 69/50$. Use $1/\sqrt{6} < .409$, $1/\sqrt{7} < .378$, $\sqrt{42} > 162/25$; the last residual comparison has cleared margin 23. Since $\sqrt{e} < 401/500$, $e - (\varphi - 1) < 1/40$ and $(93/500)^2 > 69/2000$, Cauchy gives $x \cdot w < 362087/250000 < 3/2$.

Isolated 5@2. Its height square is φ. The direction-1 shadow gives $x_1 > 31/50$, since $F(31/50)^2 < 161/200 < \varphi/2$ with cleared margin 6541881. Its actual edge 3 has $\cosh L_3 = \sqrt{2}\varphi^2 > 37/10$; hence $S_3 = x_1 + x_2 + x_3 > 131/100$, using $3\tau_3^2 > 81/47 > (131/100)^2$, of margin 3433. Distinct $4/5$ gives $x \cdot w_0 > 57/40$, $w_0 = u_4 + u_5$. Put $v = e_1 + e_2 + e_3$, $k_0 = 1/2 + 1/\sqrt{5} < 237/250$, and $x_0 = (31/50, 69/200, 69/200, 0, \ldots)$. Then $x_0 = (11/40)e_1 + (69/200)v$, $x \cdot x_0 > x_0^2 > 31/50$. The residual $w_0 - k_0 v$ has square $13/20 + 1/\sqrt{5} < 11/10$ and excess demand $> 9/50$. All coefficients of $x_0 + (w_0 - k_0 v)/6$ remain positive, and its Cauchy margin is at least

$$(31/50 + (9/50)/6)^2 - e(31/50 + (11/10)/36) = 3/700 > 0.$$

Contacts $4@2 + 3@3$. The white A_2 cost is $2/3$, so $Q = 0$ or $1/4$; the latter has $\cosh L_3 = 4 + 3\sqrt{2} > 7$. At $Q = 0$, put $s = \sqrt{2}$. The actual floors satisfy $\tau_2^2 = (1 + 2s)/7$, $\tau_3^2 = (2s - 1)/3$, $\tau_2 \tau_3 = 1/\sqrt{3}$. Their positive equality vector has $E_0 = (6s - 4)/7 > .6407$. The positive old-A_2 correction of c is an actual future point of square $-s/3$, giving pure displacement

$$U = \frac{2+s}{2} e_1 + \frac{1+s}{3}(e_2 + e_3 + e_4), \qquad x \cdot U \geq A - \sqrt{s/3}, \quad A^2 = (5 + 4s)/2.$$

Its projection coefficients on u_2, u_3 are $\rho_2 = (2s+1)/6$, $\rho_3 = (1+s)/\sqrt{3}$; its residual square is $(7+4s)/12 < 1.055$, with excess demand $\Delta > .06$. Indeed $1.4142 < s < 1.4143$ gives $A > 2.3083$, $\sqrt{s/3} < .6867$, $\tau_2 \in (.7395, .7396)$, $\tau_3 \in (.7806, .7808)$, $\rho_2 < .6381$, $\rho_3 < 1.394$, and excess $> .06122604$. The old positive coefficients exceed $.3, .52$; subtracting $(\rho_2, \rho_3)/16$ removes less than $.04, .088$. The corrected dual therefore has Cauchy margin at least

$$(.6407 + .06/16)^2 - e(.6407 + 1.055/256) = 8825113/11200000000 > 0.$$

This is an actual-point constraint, not an edge-only assertion.

Now exhaust common $1/2$. Deletion 1 retains right rank at least eight, so the right contact is none or long-leaf 3@2. The latter forces own $\gamma \geq 1/\sqrt{2}$ (own $0/3$ is affine), and $\cosh L_2 = (1+\gamma)/\sqrt{1-\gamma^2-Q} \geq 1+\sqrt{2}$, already excluded. Without a right contact, $A^2 = 2\gamma^2 + Q - 1 > 0$ and $\gamma^2 + Q < 1$. Own 0 has zero derivative; own 3 has height below $1/4$ and forces a foreign closure; own $4/6$ is affine. Own 5 with $Q > 0$ has $\cosh L_2 \geq \varphi^2 > 1+\sqrt{2}$; at $Q = 0$ use the isolated-5 test. Finite $m \geq 7$ forces $Q = 0$; $m \geq 8$ uses the edge-2 test, leaving only isolated 7.

For common $2/3$, deletion 2 retains right B/D rank at least seven; deletion 3 allows one left $3/4/5$ at 1 or 2. A right long 3@3 forces own $3/4/5$: own 0 is affine, own 3 a triangle, own 4 is the tested $4+3$ profile, and own 5 has $Q = 0$, $\cosh L_3 = (2+5\varphi)/\sqrt{2} > 7$. Without a right contact the full left list is $3@1, 4@1, 5@1, 3@2, 4@2, 5@2$. The first has $A^2 = Q - 1/2 < 1/6$ and forces foreign closure; $4@1, 3@2, 4@2$ are respectively affine terminal-4, two-fork, and $(4,3,3)$ star. The 5 cases have $Q = 0$ by white cost $\varphi^2/3 > 3/4$ and were tested above. For D_9 one extra right-fork 3 gives retained $D_7 + c = E_8$. It excludes old-1 and foreign contacts, but the removed own 2 must still be checked: own $0/3/4$ gives affine E_8/cycle/star, and own 5 retains nonpositive $(3,5,3)$. At higher rank the fork attachment already contains affine E_8. No finite branch is omitted.

For the final isolated 7@1, put $t = \cos(2\pi/7)$, $g = \cos(\pi/7)$, and

$$\xi = c + g^{-1}n_1 + (1-t)g^{-1}n_2 = \sqrt{t}\,p - \frac{V}{\sqrt{1+t}}, \qquad V = t(e_1+e_2) + (1-t)e_3.$$

It has $\xi \cdot c = \xi \cdot n_1 = 0$, $\xi \cdot n_2 = (1/2 - t)/g < 0$ and $\xi^2 = -(1-t)(2t-1)/(1+t) < 0$. All coefficients and its time coefficient are positive, proving actual feasibility and future orientation. Thus $x \cdot V \geq \sqrt{t(1+t)} - \sqrt{(1-t)(2t-1)}$. The increasing cubic $8t^3 + 4t^2 - 4t - 1$ gives $.6234 < t < .6235$, with endpoint values $-.000926392768, .000105223$. Hence the first radical is > 1.0059, the second $< .305$, and $.95 < \|V\| < .9588 < .959$, so $x \cdot \widehat{V} > .73$. The projection of w onto $\widehat{V}$ is $\rho = (1/\sqrt{6} + 1/\sqrt{7})(1+t)/\|V\| \in (1.3, 1.35)$, and residual square is $< 11/5$. The Cauchy envelope decreases above .73, since $\rho^2 e < 6561/5600 < 3(.73)^2 < \|w\|^2(.73)^2$. It gives $x \cdot w < 1.35(.73) + .492 = 591/400 < 3/2$; the final squared radical margin is $277/1750000$. Thus $1/2, 2/3, 6/7$ all have distinct actual opposites.

Their exact pair sums are

$$x \cdot (u_1 + u_2) > 1.091, \quad x \cdot (u_2 + u_3) > 1.265, \quad x \cdot (u_6 + u_7) > 1.512.$$

Set $W = .1(u_1+u_2) + .055(u_2+u_3) + .305(u_6+u_7)$. Direct positive Gram estimates give

$$x \cdot W > .639835, \qquad W^2 < .636435825,$$
$$.639835^2 - \frac{9}{14}(.636435825) = \frac{70423123}{280000000000} > 0.$$

To verify the pair sums, use $F(a)F(b) \leq F((a+b)/2)^2$. The correlation lower bounds

$$7071/10000, \qquad 81649/100000, \qquad 9258/10000$$

have squared margins $959, 322397, 26052$ below the exact values. At sums $m/1000$, $m = 1091, 1265$, 1512, the cleared differences

$$p(4000000 + m^2)^2 - 16dm^2\,1000000$$

are respectively

$$40828209713231, \qquad 358399613480625, \qquad 52443946917888.$$

The six upper Gram entries of the three pair vectors are $3.415, 3.633, 3.852, 3.102, 1.899, 2.474$, obtained by positive squaring of their prefix correlations. Their expansion is the stated bound on W^2. This contradicts Cauchy and proves the proposition. □

C Coupled actual pressures in an A-chain

The crucial feature of this section is that a foot at one end is tested against *actual* opposites at the other end. A collection of separate edge estimates would lose that coupling.

Theorem C.1 (Uniform actual A pressure)**.** *With arbitrary finite spectators,*

$$\|x_{A_r}\|^2 > \frac{1}{2} \qquad (r \geq 7).$$

We first prove the uniform step for rank at least eight, then give the rank-seven base in Subsection C.3. Put $N = r + 1$, and realize

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad C_{ij} = \frac{2\min(i,j)(N - \max(i,j))}{N}, \quad h_i = x_A \cdot u_i.$$

The root space is the sum-zero subspace of $\mathbb{R}^N$. All its inequalities below use the actual projection x_A.

C.1 A forcing window and the complete common-pair inventory

Suppose first that $N \geq 9$ and $\|x_A\|^2 \leq 1/2$. Consider the four pairs $1/2, 3/4, (r-3)/(r-2), (r-1)/r$. If their respective opposites are distinct, the outer pair sums exceed 1, and the inner sums exceed $19/16$. These follow from (7), concavity of $\log F$, and the minimum correlations $\sqrt{3/7}$, $\sqrt{3/5}$ at $N = 8$. For the inner comparison,

$$F(19/32)^2 = \frac{1478656}{1918225} < \frac{387}{500} < \sqrt{3/5}.$$

For the outer one, $F(1/2)^2 = 16/25 < \sqrt{3/7}$. Set

$$L = 2u_1 + 2u_2 + u_3 + u_4, \qquad R = 2u_r + 2u_{r-1} + u_{r-2} + u_{r-3}.$$

The correlations on the left increase to $\sqrt{i/j}$, whence

$$L^2, R^2 < 12 + 6\sqrt{2} + 4/\sqrt{3} + 4\sqrt{2/3} + \sqrt{3} < 28.$$

The last bound follows, for example, by the positive rational upper bound $2351/84$. For $a = (2, 2, 1, 1)$, direct coordinates give

$$L \cdot R = \left(\sum_{i=1}^{4} a_i \sqrt{\frac{i}{N-i}} \right)^2 .$$

This decreases in N; at $N = 9$ its square root is $\sqrt{2} + 2\sqrt{2/7} + 2/\sqrt{5} < 17/5$. Thus $(L+R)^2 < 80$, whereas distinctness of all four pairs gives $x_A \cdot (L+R) > 51/8$. The Cauchy gap is $41/64 > 0$. After reversing the numbering, a common pair is therefore $1/2$ or $3/4$. This is a relabeling, not a symmetry assumption on P.

Here is the finite inventory used for the target in Theorem C.1. It is useful to state it before making any energy-dependent discard. Write Q for the total spectator cost. Every retained finite diagram is a forest, so there is at most one contact on each retained own path. The foreign costs satisfy (8), are at least $1/4$, and have the gap $(1/4, 1/3)$.

For a common $3/4$ opposite, its own contact positions are in $\{1,2,3\}$ and $\{4,\dots,r\}$. A double profile away from positions $3,4$ survives one deletion as a connected rank-r finite diagram. Its two new bonds must be 3: a weighted bond would be internal in a B path, and D, E_8 are simply laced. The two 3's instead give an affine cycle in the full diagram. This also covers the rank-eight E_8 boundary. At positions $3,4$, the labels are

$$p \in \{3,4,5\}, \qquad q \in \{3,4\}.$$

The pair $3,3$ is affine. The two smallest remaining assignments have actual endpoint cosines

$$\sqrt{\frac{3}{2N}}(N - 3 + \sqrt{2}(N-4)), \qquad (3\sqrt{2}+4)\sqrt{\frac{N-4}{2N}}, \tag{37}$$

both greater than 3 at $N = 9$ and increasing thereafter. Increasing a contact or Q strengthens the appropriate endpoint.

A single 3 at positions $3,\dots,r-2$ contains an affine E_8 tree with arms $1,2,5$, or an affine E_7 tree with arms $1,3,3$, obtained by truncation if needed. The other single-3 possibilities satisfy

position	Q bound	A^2
1	$Q < 5/8$	$Q - (N+1)/(2N) < 1/8$
2	$Q < 1/2$	$Q - 2/N$
$r-1$	$Q < 2/(N-3)$	$Q - 2/N$
r	$Q < 1/2 + 1/(2(N-3))$	$Q - (N+1)/(2N) < 3/(2N(N-3))$.

The end heights are below $1/4$; the middle rows allow at most one foreign component and its cost exceeds A^2. All contradict (8). A single 4 is an affine $3,3,4$ star internally; at either end its white condition gives $Q < 1/4$, so $Q = 0$ and its height is negative. A single 5 can occur only at 1 or 3, has $Q = 0$ from its H_4 cost $3\varphi^2/8 > 3/4$, and at 3 closes direction 4 with

$$\cosh L_4 = 3\varphi^3\sqrt{(N-4)/N}. \tag{38}$$

At 1 it gives foot F below. A zero own contact cannot close an own edge. This completes the common $3/4$ list.

For a common $1/2$ opposite there is possibly a contact at 1 and at most one on $2,\dots,r$. For double positions $1,j$, $j \geq 3$, deletion 2 gives a full connected finite path or branch. A branch has only 3's; a path can have a 4 only at its terminal bond $1-c$, with $j = 3$ or r. All 3's give an affine cycle. The profile $4@1 + 3@r$ contains the affine F_4 path $2,1,c,r,r-1$. The remaining $4@1 + 3@3$ has $Q < 1/(2(r-1))$, hence $Q = 0$, and gives foot H. Adjacent contacts $1,2$ have an arbitrary finite native first label and second label 3 or 4; $3,3$ is affine.

For single 3, the same internal affine exclusions apply. At $1, r, r-1$, respectively, the bounds are

$$Q < 3/4, \quad Q < 1/2 + 1/(2r), \quad Q < 2/r.$$

The first two give height below $1/4$, and the last leaves no foreign cost and negative height. At 2, $Q < 1/2 + 1/(2r)$ allows at most two foreign components. One exceeds $A^2 = Q - 2/N$. With two, both contacts are 3; c, n_2, n_1, n_3 and their two foreign neighbors give the exact affine two-fork diagram. A single 4 is affine internally; at 1, $Q < 1/2$ allows at most one cost greater than $A^2 = Q - 1/N$; at r, $Q < 1/r$ leaves negative height. A single label at least 5 is possible only at 1. Label 6 gives the affine $3, 6$ path. Labels at least 7 have $Q < 1 - \beta_1^2 < 1/4$, hence $Q = 0$. Label 5 with $Q = 0$ gives F; with $Q > 0$ it remains in L. Thus the remaining end profiles are

$$\text{L}: \text{ adjacent, not } (3,3);\ \text{isolated native } \geq 7;\ \text{isolated 5 with } Q > 0, \qquad \text{H}, \text{F}.$$

No foreign branch has been suppressed by assuming a full star finite.

C.2 Three feasible feet prove the half bound

For H and F take the following positive actual combinations:

	ξ	$k = -\xi^2$
H	$c + (\sqrt{2} - \frac{1}{2})n_1 + (\sqrt{2} - 1)n_2 + \frac{\sqrt{2}}{2}n_3$	$\sqrt{2} - \frac{5}{4}$
F	$c + \sum_{j=1}^{4}((j+1)\varphi - 2j)n_j$	$34\varphi - 55.$

For H the products with c, n_1, n_3 vanish and the n_2 product is $(\sqrt{2} - 3)/4 < 0$. For F the products with c, n_1, n_2, n_3 vanish and the n_4 product is $3\varphi - 5 < 0$. Every other actual inequality follows from positivity. Both have $Q = 0$, future coefficient $A > 0$, and pure A-spatial part, with

$$A_{\text{H}}^2 = \frac{3}{2} + \sqrt{2} - \frac{11/2 + 3\sqrt{2}}{N}, \qquad A_{\text{F}}^2 = \frac{\varphi^2(N-1)}{2N} - 1.$$

At $N = 9$,

$$A_{\text{H}}^2 - 9k_{\text{H}} = \frac{437 - 300\sqrt{2}}{36} > 0, \qquad A_{\text{F}}^2 - 9k_{\text{F}} = \frac{50(89 - 55\varphi)}{9} > 0.$$

The time norms increase with N. Equation (3) therefore excludes these profiles under energy at most $1/2$. The central doubles and (38) already have $\cosh L > 3$, so are excluded by (6).

For L, first clear all foreign components positively to obtain $c_F = Ap - v_A$, $c_F^2 = 1 - Q$. Put

$$b = \beta_1,\ d = \beta_2,\ B = b + 2d, \quad h = \frac{4}{3}(b^2 + bd + d^2) + Q - 1, \quad z_0 = h/B^2,$$

$$\lambda_1 = (4b + 2d)/3 - h/B, \qquad \lambda_2 = 2B/3 - 2h/B, \qquad \xi = c_F + \lambda_1 n_1 + \lambda_2 n_2 = Ap - U.$$

The endpoint condition gives

$$B\lambda_1 = 2bd + 1 - Q > 0, \quad B\lambda_2 = 2(1 - b^2 - Q) > 0,$$

and direct multiplication gives

$$\xi \cdot c = \xi \cdot n_1 = \xi \cdot S = 0, \quad \xi \cdot n_2 = -3h/(2B) < 0, \quad k = -\xi^2 = B^2(z_0 - 3z_0^2).$$

Here S denotes all cleared spectator roots. The complete L range is

$$\frac{8}{81} < z_0 < \frac{1}{3}. \tag{39}$$

Indeed adjacent contacts with one label at least 4 give $z_0 > 1/10$; isolated labels at least 7 have $b > 9/10$, so $z_0 = 4/3 - b^{-2} > 8/81$; isolated 5 with $Q \geq 1/4$ gives $z_0 \geq 3\varphi - 14/3 > 2/15$. The upper bound is precisely $1 - b^2 - Q > 0$. Thus ξ is actual, future timelike and has no foreign spatial part.

For $a_N = 2/3 - 2/N$, its full vector data are

$$A^2 = B^2(a_N + z_0), \quad U^2 = B^2(a_N + 3z_0^2), \quad U_{\text{raw}} = \frac{B}{\sqrt{2}}(2/3 + z_0, 2/3 + z_0, 2/3 - 2z_0, 0, \ldots, 0). \tag{40}$$

Subtracting $\sqrt{2}B/N$ from every coordinate centers the raw vector. Its normalized actual demand $g = (A - \sqrt{k})/\sqrt{U^2}$ satisfies $g^2 > 9/20$. At $a_N = 4/9$ the equivalent quadratic is $2523z_0^2 - 720z_0 + 484/9 > 0$, of negative discriminant $-72976/3$; increasing N strengthens it.

Let $W = u_{r-1} + u_r$. Direct coordinates give

$$\rho = \frac{U \cdot W}{\sqrt{U^2}} = \frac{2/\sqrt{N(N-2)} + \sqrt{2/(N(N-1))}}{\sqrt{a_N + 3z_0^2}}. \tag{41}$$

The numerator decreases and denominator increases with N. At $N = 9, z_0 > 8/81$, its squared upper seal is $48114/129500 < 3721/10000$, so $\rho < .61$. If the far opposites are distinct, $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{.61}{\sqrt{2}} + \sqrt{(1/2 - 9/20)4} < .89 < 1.$$

If they are common, their inventory supplies another L foot: H and F have already been excluded. Normal reversal changes the raw sum from $\sqrt{2}B$ to $-\sqrt{2}B'$. The disjoint raw supports therefore give the *positive* centered product

$$U \cdot U' = 2BB'/N, \qquad \operatorname{corr}(U, U') < \frac{2/N}{a_N} = \frac{3}{N-3} \leq \frac{1}{2}.$$

The two actual demands force energy greater than $4(9/20)/3 = 3/5$. This proves the half bound.

C.3 The rank-seven base

Set $N = 8$ and suppose $\|x_A\|^2 \leq 1/2$. The correlations of pairs $1/2, 6/7$ are $\sqrt{3/7}$ and those of $3/4, 4/5$ are $\sqrt{3/5}$. The pair-sum calibrations already proved therefore apply. If all four pairs have distinct opposites, put

$$w = 2u_1 + 2u_2 + u_3 + 2u_4 + u_5 + 2u_6 + 2u_7.$$

Then $x_A \cdot w > 51/8$, whereas direct Gram summation gives

$$\begin{aligned} w^2 = {} & \frac{64}{7} + \frac{32}{3} + \frac{16}{5} + 4 \\ & + \frac{64}{\sqrt{21}} + \frac{64}{\sqrt{105}} + \frac{16}{\sqrt{7}} + \frac{64}{3\sqrt{5}} \\ & + \frac{16}{\sqrt{3}} + 8\sqrt{3/5} < \frac{1208}{15}. \end{aligned}$$

The first four terms sum to less than 28, and the last six are respectively less than 14, 32/5, 32/5, 10, 28/3, 32/5; each comparison follows by squaring positive rational bounds. Cauchy is impossible since

$$(51/8)^2 - (1208/15)/2 = 359/960 > 0.$$

Reversing the numbering if necessary, a common pair is 1/2 or 3/4.

The central pair. For a common $3/4$ opposite there is at most one contact on each of the paths $1,2,3$ and $4,5,6,7$. Native bonds have labels $3,4,5$: a larger label with a retained old bond is not finite. For two contacts outside positions $3,4$, the following finite white tests exhaust the position pairs:

$(1,4),(1,7),(2,4),(2,7),(3,5),(3,7)$	a rank-seven path after deleting 3 or 4
$(1,5),(1,6),(2,5),(2,6)$	D_7 or E_7 after deleting 3
$(3,6)$	D_7 after deleting 4.

Both new bonds must be 3 in these rows; their cycle in the full diagram is affine. At positions $3,4$, the $3,3$ pair is affine; every other assignment gives an endpoint cosine greater than 3, already for the minimal $3,4$ and $4,3$ assignments in (37) at $N=8$. It therefore forces energy greater than $1/2$.

For a single 3-contact, the complete own costs and sufficient white bounds are

j	1	2	3	4	5	6	7
$\beta^t C\beta$	7/16	3/4	15/16	1	15/16	3/4	7/16
$Q <$	5/8	1/2	5/8	3/5	2/5	2/5	3/5.

Here $A^2=\beta^tC\beta+Q-1$. One foreign component is impossible by $q<A^2$ in (8). Every row except $j=3,4$ allows at most one, or has $A^2<1/4$. In rows $3,4$, two foreign components must both start with a 3-contact; the new wall, the contacted old node, its two old neighbors and the two foreign neighbors give the exact affine two-fork D_5 diagram. The case $Q=0$ has nonpositive height.

A single 4 is an affine $3,3,4$ star internally. At an end $Q<1/4$, so $Q=0$ and the height is negative. A single 5 is possible only at positions 1 or 3, and its retained H_4 cost $3\varphi^2/8>3/4$ forces $Q=0$. Position 3 gives direction-four cosine greater than 3. Position 1 gives foot F above. At $N=8$ its time norm satisfies

$$A_F^2=(7\varphi-9)/16, \qquad A_F^2-9k_F=(7911-4889\varphi)/16>0.$$

For example $\varphi<16181/10000$, proved by positive squaring, gives $7911-4889\varphi>1091/10000$. Thus the actual-point inequality (3) excludes this last row. The central common-pair case is complete.

The end pair. For common $1/2$, a double profile has positions $1,j$. For $j=2$ its second label is 3 or 4 and its first label is any finite native label. At $j=3,7$ the labels are 3 at j and 3 or 4 at 1. At $j=4,6$, deletion 2 gives E_7 and both labels are 3. At $j=5$, deletion 2 already contains affine E_6. Two 3-labels always give an affine cycle. The $4@1+3@7$ row contains the affine F_4 path $2,1,c,7,6$; the $4@1+3@3$ row has $Q=0$ and is foot H. The displayed H formula gives $A_H^2-9k_H=(193-134\sqrt{2})/16>0$ at $N=8$.

For a single 3, the own costs are the same seven entries above, while the sufficient Q bounds at positions $1,\dots,7$ are

$$3/4,\quad 4/7,\quad 2/7,\quad 1/7,\quad 1/7,\quad 2/7,\quad 4/7.$$

All except position 2 have $A^2<1/4$. At position 2, one foreign component contradicts $q<A^2$; with two, each exceeds $1/4$, so each is at least $1/3$, contrary to $Q<4/7$. A single 4 is affine internally. At 1, $Q<1/2$ permits at most one foreign component, whose cost exceeds $A^2=Q-1/8$; at 7, $Q<1/7$ leaves none and gives negative height.

A single label at least 5 is possible only at 1. Label 6 contains the affine $3,6$ path. A label at least 7 forces $Q<1-\beta_1^2<1/4$, hence $Q=0$. Label 5 with $Q=0$ was excluded by F. Thus the only remaining profiles are exactly L: adjacent contacts other than $3,3$, isolated label at least 7 with $Q=0$, or isolated 5 with $Q>0$.

The far-end comparison. The positive spectator clearing and actual L-foot formulas (39)–(40) remain valid at $N = 8$, now with $a_N = 5/12$. The lower bound $z_0 > 8/81$ follows as before: adjacent contacts give $z_0 > 1/10$, an isolated label at least 7 has $\beta_1 > 9/10$, and isolated 5 with $Q > 0$ has $z_0 \geq 3\varphi - 14/3 > 2/15$. For the label-seven bound, $8t^3 - 4t^2 - 4t + 1$ is $-1/125$ at $9/10$ and strictly increasing from $\sqrt{3}/2$ to 1.

The actual demand now satisfies $g^2 > 11/25$: equivalently $A^2/k > 324/49$, whose cleared difference is

$$972z_0^2 - 275z_0 + 245/12 > 0.$$

Its discriminant is -3755. With $W = u_6 + u_7$, direct coordinates give

$$0 < \rho^2 = \frac{5/42 + 1/(2\sqrt{21})}{5/12 + 3z_0^2} < \frac{603612}{1170300} < \frac{324}{625}, \qquad W^2 = 2 + 2\sqrt{3/7} < 10/3.$$

Here the numerator is less than $23/100$ and $z_0^2 > 64/6561$ gives the stated rational seal. If the far opposites are distinct, their pair law gives $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{18}{25\sqrt{2}} + \sqrt{(1/2 - 11/25)10/3} = \frac{18}{25\sqrt{2}} + \frac{1}{\sqrt{5}} < \frac{24}{25}.$$

If they are common, the reversed end inventory supplies another L foot. Normal reversal is $e_j \mapsto -e_{9-j}$, so its raw sum is $-\sqrt{2}B'$ on the disjoint final three coordinates. Consequently $U \cdot U' = BB'/4 > 0$, and their normalized correlation is less than $3/5$. The sum of these two normalized directions has square less than $16/5$ and actual scalar product greater than $2\sqrt{11/25}$. Cauchy forces

$$\|x_A\|^2 > \frac{44/25}{16/5} = \frac{11}{20} > \frac{1}{2}.$$

Both far-end alternatives are impossible. This proves the rank-seven base and completes Theorem C.1.

C.4 The paired-end Gram and the uniform $4/7$ bound

Proposition C.2. *Every actual A_r component, $r \geq 8$, has energy greater than $4/7$, with arbitrary finite spectators.*

Proof. Put $N = r + 1 \geq 9$ and suppose energy is at most $4/7$. For distinct outer $1/2$ pairs and inner $3/4$ pairs, the height sums exceed $41/40$ and $1219/1000$, and likewise at the opposite end. At $N = 9$ the correlations are $\sqrt{7}/4, \sqrt{5/8}$; the exact tests are

$$F(41/80)^2 = \frac{43033600}{65302561} < 33/50 < \sqrt{7}/4,$$
$$F(1219/2000)^2 = \frac{23775376000000}{30095768093521} < 79/100 < \sqrt{5/8}.$$

The cross-product margins are $3304513, 28079388159$. The correlations increase with N. For $M_i = u_i + u_{N-i}$, $1 \leq i \leq j \leq 4$,

$$\langle M_i, M_j \rangle = 2N\sqrt{\frac{i}{j(N-i)(N-j)}}.$$

Its logarithmic derivative is $1/N - 1/(2(N-i)) - 1/(2(N-j)) < 0$. Thus every paired-end Gram entry decreases with N. For $W = 2M_1 + 2M_2 + M_3 + M_4$,

$$W_9^2 = \frac{1116}{35} + \frac{60}{\sqrt{7}} + \frac{36}{\sqrt{10}} + \frac{72}{\sqrt{70}} < \frac{747}{10}.$$

The last three terms are $< 227/10, 57/5, 87/10$; these imply the sharper $2614/35 < 747/10$. If all four selected pairs were distinct, then

$$\langle x_A, W\rangle > 3269/500, \qquad (3269/500)^2 - (4/7)(747/10) = 104527/1750000 > 0,$$

contradicting Cauchy. Reversing the numbering, a common $1/2$ or $3/4$ therefore occurs.

Use the complete common-pair inventory above before its half-energy discards. At $4/7$, the individual edge cutoff is $C = 11/3$. The two minimal central-double cosines in (37) exceed $11/3$ already at $N = 9$, and increase thereafter. The single central 5 cosine in (38) also exceeds $11/3$. The remaining left profiles are precisely L, H, F. The same finite-white, affine and full-spectator checks apply; no earlier small-target exclusion of H/F is used.

For $\xi = Ap - U$, write $g^2 = (A - \sqrt{k})/(A + \sqrt{k})$ and $d = U^2/B^2$, where $\sqrt{2}B > 0$ is its raw coordinate sum. At $N = 9$ the exact shared feet satisfy

type	$g^2 >$	$d >$	$\rho = \langle U/\|U\|, u_{r-1} + u_r\rangle <$
L	$9/20$	$4/9$	.61
H	$53/100$	$1/3$	$3/4$
F	$11/20$	$2/9$	$9/10.$

For L, its formulas and negative-discriminant seal above prove the first two entries. The correlation numerator has square

$$(2/\sqrt{63} + 1/6)^2 = 23/252 + 2/(3\sqrt{63}) < 22/125,$$

giving $\rho^2 < 48114/129500 < 3721/10000$. For H, $A^2 = (8 + 6\sqrt{2})/9 > 9/5$, $k = \sqrt{2} - 5/4 < 1/6$, so $A^2/k > 54/5 > 23409/2209$. Also

$$d = \frac{77 - 12\sqrt{2}}{9(11 + 6\sqrt{2})} > 1/3$$

by $22 > 15\sqrt{2}$. For F, $A^2 = (4\varphi - 5)/9 > 184/1125$, $k = 34\varphi - 55 < 167/12500$; their ratio exceeds $961/81$, with integer margin 5752125. Its density $d > 2/9$ is equivalent to $979 > 605\varphi$. These seals follow from $809/500 < \varphi < 40451/25000 < 16181/10000$, proved by positive squaring. The correlation bounds then follow from $(22/125)/(1/3) < 9/16$ and $(22/125)/(2/9) < 81/100$. At fixed native/foreign parameters, A^2, d, g^2 increase with N, while k, B stay fixed.

Let $W_f = u_{r-1} + u_r$. Its square is $< 2 + \sqrt{2} < 7/2$. For every foot,

$$\rho = \frac{2/\sqrt{N(N-2)} + \sqrt{2/(N(N-1))}}{\sqrt{d}}.$$

For $N \geq 10$, L has the improved $\rho < 8/15$: the numerator square is at most $5/36$ at 10, and $d > 7/15 + 64/2187 > 49/100$; thus $\rho^2 < 125/441 < 64/225$. If the far pair is distinct, its demand is $> 41/40$. Perpendicular Cauchy, with $v = \langle x_A, U/\|U\|\rangle$, gives

$$\langle x_A, W_f\rangle \leq \rho v + \sqrt{(4/7 - v^2)(W_f^2 - \rho^2)}.$$

For L this decreases with v above its demand floor and increases with ρ up to its bound; the sign condition is $v^2 W_f^2 > (4/7)\rho^2$. At $N \geq 10$ it is

$$< (8/15)\sqrt{9/20} + \sqrt{17/40} < 9/25 + 33/50 = 51/50 < 41/40.$$

At $N = 9$, use $\rho < .61, W_f^2 < 10/3$ instead: the two terms are $< .41, .6$, the latter square being $1510229/4200000 < 9/25$. For H and F, without derivative estimates, use $v < \sqrt{4/7} < 19/25$ and their pressure floors. For $N \geq 10$ the bounds are $.57 + \sqrt{29/200} < .97$ and $.7 + \sqrt{3/40} < .98$. At $N = 9$ they improve to $.95, .97$. Every far-distinct case is impossible.

If the far pair is common, it supplies a right L/H/F foot. Normal reversal gives negative raw sum $-\sqrt{2}B'$. Except F/F at $N = 9$, their raw supports are disjoint and

$$\langle U, U' \rangle = 2BB'/N, \qquad \operatorname{corr}(U, U') = \frac{2}{N\sqrt{dd'}}.$$

The following complete positive-sum Cauchy seals hold at $N = 9$, and improve thereafter:

pair	$\rho <$	sum demand $>$	energy $>$
L/L	$1/2$	$2\sqrt{9/20}$	$3/5$
L/H	$3/5$	$139/100$	$19321/32000$
L/F	$5/7$	$141/100$	$139167/240000$
H/H	$2/3$	$36/25$	$3888/6250$
H/F	$5/6$	$73/50$	$15987/27500.$

Every last entry exceeds $4/7$. For F/F, at any $N \geq 9$ the two actual isolated end-five walls have exact product

$$1 - \varphi^2/2 + \varphi^2/N \in (-1/2, 0).$$

The lower inequality uses $\varphi^2 < 3$; the upper follows at $N = 9$ from $\varphi > 8/5$ and strengthens with N. This violates the native gap without an unsupported overlap formula. All cases are excluded. □

C.5 A fourth actual foot and the rank-13 threshold

Proposition C.3. *Every actual A_r component, $r \geq 13$, has energy greater than $5/7$, with arbitrary finite spectators.*

Proof. Put $N = r + 1 \geq 14$ and suppose the energy is at most $5/7$. The outer and inner selected correlations increase with N. At $N = 14$ the exact pair calibrations are

$$F(263/500)^2 = (263000/319169)^2 < 6791/10000 < \sqrt{6/13},$$

$$F(641/1000)^2 = (1282000/1410881)^2 < 8257/10000 < \sqrt{15/22}.$$

Thus distinct outer and inner pairs have sum demands $> 263/250$ and $> 641/500$. For $M_i = u_i + u_{N-i}$, the paired Gram identity above decreases entrywise with N. At 14, $W = M_1 + M_2 + 2M_3 + 2M_4$ has diagonal contribution $55489/2145 < 25.869$. Its six off-diagonal contributions, in order $12, 13, 14, 23, 24, 34$, are

$$\frac{28}{\sqrt{78}}, \ \frac{112}{\sqrt{429}}, \ \frac{56}{\sqrt{130}}, \ \frac{112}{\sqrt{198}}, \ \frac{28}{\sqrt{15}}, \ 112\sqrt{3/110},$$

respectively less than $3.171, 5.408, 4.912, 7.960, 7.230, 18.497$. Hence $W^2 < 73.047 < 1461/20$ at every $N \geq 14$. Four distinct pairs would give

$$\langle x_A, W \rangle > 904/125, \qquad (904/125)^2 - (5/7)(1461/20) = 53923/437500 > 0.$$

A common left $1/2$ or $3/4$ follows after relabeling.

Recheck the full common-pair inventory at the new individual cutoff $C = 6$. A central double with second label at least 4 has the first cosine in (37), greater than $.3(11 + 14) > 6$ at $N = 14$. The 5/3 central double and isolated central 5 have cosines

$$(3\varphi + 4)\varphi^2\sqrt{(N-4)/N}, \qquad 3\varphi^3\sqrt{(N-4)/N},$$

both > 6 at 14, using $\sqrt{5/7} > 4/5, \varphi > 8/5, \varphi^2 > 13/5, \varphi^3 > 4$. They increase with N. The central 4/3 case must now be retained. Its foreign cost is zero and its actual positive foot is

$$\mathrm{C}: \quad \xi = c + n_3 + (2-\sqrt{2})n_4, \qquad k = 3\sqrt{2} - 4.$$

Products on c, n_3 vanish; the n_4 product is $1 - \sqrt{2} < 0$, and all unknown wall products follow from positivity. The time is positive, with

$$A_C^2 = 4 + 3\sqrt{2} - \frac{17 + 12\sqrt{2}}{N}, \quad U_C^2 = 8 - \frac{17 + 12\sqrt{2}}{N}, \quad B_C = 3\sqrt{2}/2 + 2.$$

The complete surviving profiles are L,H,F,C. All L spectators are cleared in their entirety.

For L, at $N = 14$, $a_N = 11/21$ and $8/81 < z_0 < 1/3$. Its bound $g_L^2 > 12/25$ is equivalent to

$$4107z_0^2 - 1200z_0 + 1859/21 > 0,$$

whose discriminant is $-99884/7$. Also $d_L > 11/21 + 64/2187 > 11/20$, with last positive comparison $26880 > 24057$. The other exact data at 14, with $s = \sqrt{2}$, are

	A_{14}^2	k	B^2	d_{14}
H	$(31 + 22s)/28$	$s - 5/4$	$(11 + 6s)/4$	$(66 - 6s)/(7(11 + 6s))$
F	$(13\varphi - 15)/28$	$34\varphi - 55$	$(\varphi + 1)/4$	$(1525 - 939\varphi)/(7(\varphi + 1))$
C	$(39 + 30s)/14$	$3s - 4$	$(17 + 12s)/2$	$(95 - 12s)/(7(17 + 12s))$.

For H, $A^2 > 11/5, k < 33/200$, so $A^2/k > 40/3$. For F, $A^2 > 21/100, k < 167/12500$, so $A^2/k > 2625/167$. For C, $A^2 > 5, k < 1/4$. All exceed $961/81$, proving $g^2 > 11/20$. The density bounds $d > .3$ reduce to $143 > 62s$, $15229 > 9411\varphi$, $593 > 372s$. Use $s < 3/2$ and $\varphi < 40451/25000$; the cancellation-sensitive middle comparison has integer margin 40639. All these demands and densities increase with N.

If either far selected pair is common, its reflected inventory supplies a right L/H/F/C foot. The raw supports, each of length at most 5, are disjoint. Normal reversal gives positive correlation $\rho = 2/(N\sqrt{dd'})$. The complete sum-direction tests are

pair type	$\rho <$	energy $>$
L/L	13/50	16/21
L/non-L	9/25	20449/27200
non-L/non-L	10/21	231/310.

For the middle row, $\rho^2 < 200/1617 < (9/25)^2$ and the demand sum is $> \sqrt{12/25} + \sqrt{11/20} > 1.43$. Every energy exceeds 5/7.

Otherwise both far selected pairs are distinct. Their weighted demand is

$$\langle x_A, R\rangle > 452/125, \quad R = u_{N-1} + u_{N-2} + 2u_{N-3} + 2u_{N-4}.$$

Direct within-end Gram bounds give $10 < J = R^2 < 29$. For a left foot,

$$\rho = \langle U/\|U\|, R\rangle = \frac{K_N}{\sqrt{d}},$$

$$K_N = 2\sqrt{\frac{8}{N(N-4)}} + 2\sqrt{\frac{6}{N(N-3)}} + \sqrt{\frac{4}{N(N-2)}} + \sqrt{\frac{2}{N(N-1)}}.$$

Its terms decrease with N; at 14 they are $< .479, .395, .155, .105$, so $K_N < 1.134 < 8/7$. Thus $\rho < 31/20$ for L and $< 21/10$ otherwise. The square seals are $1280/539 < (31/20)^2$, $640/147 < (21/10)^2$. Perpendicular Cauchy gives

$$\langle x_A, R\rangle \le \rho v + \sqrt{(5/7 - v^2)(J - \rho^2)}, \qquad v = \langle x_A, U/\|U\|\rangle.$$

On the admissible rectangle it decreases with v above its pressure floor and increases with ρ below its bound, since $v^2 J > (5/7)\rho^2$. This holds already with $J > 10$. Increasing J to 29 is safe. For L the resulting bound is

$$(31/20)\sqrt{12/25} + \sqrt{436199/70000} < 217/200 + 5/2 = 717/200.$$

For every other foot it is

$$(21/10)\sqrt{11/20} + \sqrt{56557/14000} < 63/40 + 201/100 = 717/200.$$

Both are $< 452/125$, contradicting the far demand. This exhausts all actual profiles and proves the nonarithmetic component theorem. □

C.6 One central-window lemma for $N = 17, 18, 19$

The next lemma separates the geometric inventory from the reference chosen for a particular rank.

Lemma C.4. *Let $N \in \{17, 18, 19\}$, and let an actual A_{N-1} component have the original farthest-vector projection x_A. On $5 \le i \le N-5$, call i weak when $\langle x_A, \omega_i\rangle \le \sqrt{D_i} - .61$. Before any contrary energy bound, every weak opposite is:*

$$\begin{array}{ll} \textit{near-four:} & v = \omega_i/\sqrt{2} + \omega_{i-1}/2,\ i = 5, 6,\ \textit{or a reflected role}; \\ \textit{pure helper:} & v = \gamma\omega_i + \omega_e/2,\ e = 1, N-1, \quad Q = 0, \\ & \gamma \ge \varphi/2, \quad \gamma < 17/16,\ \textit{own } 5,\ \textit{finite } \ge 7,\ \textit{or dotted}. \end{array}$$

Literal weak helpers are nonadjacent, with all spectators retained. If the energy is at most .86, 6/7, or .91, respectively for $N = 17, 18, 19$, no near-four role remains either.

Proof. Both retained old paths have rank at least 4. The complete finite-white list is: none; a simple 3 at an end or an admissible D/E position; an endpoint 4; or two simple end 3's, one on each path. An H/F completion has insufficient rank, and a joined completion has rank $N - 1 > 8$. Finite classification forces the ordinary A-path bridge. An additional branch there cannot be finite. This classification is applied to the actual endpoint, with the entire spectator forest.

For zero own, internal 3/4 and double-white roles give affine stars or cycles. A far 3 has time square $< 1/(2m)$, $m \ge 5$; a next-to-far 3 has $Q < 2/m$ and time square less than its positive foreign cost. Either forces an orthogonal closure. End 4 has $Q < 1/m \le 1/5$, hence no spectator and negative time square. Thus own contact is positive.

For an isolated own, clear all spectators as in the prefix proof, using the sharper A-finite pressure

$$h_i > \sqrt{D_i - 1} - \sqrt{u - 1} \quad (u > 1), \qquad h_i > \sqrt{D_i - u} - \sqrt{u(1-u)} \quad (0 < u < 1).$$

At $u = 1$ there is an actual feasible null. Here $D_i \geq 120/17$, and

$$a_i = \frac{1}{\sqrt{D_i} + \sqrt{D_i - 1}} < .1955.$$

Indeed the denominator exceeds $2.656 + 2.461 = 5.117$ and $5.117 \cdot 391 > 2000$. For $u < 1$ the loss is $< (.1955{+}1.019)/2 = .60725 < .61$. For foreign $u > 1$, $u{-}1 \leq (2{-}\varphi)^2$, so the loss is $< .1955{+}2{-}\varphi < .5775$. For pure finite owns at least 8, singleton loss is $< .1955 + \tan(\pi/8) < .1955 + .4145 = .61$, since $(2829/2000)^2 > 2$. Own 5 has its actual H_4 loss $< 25/76 + 5/24 < .61$; own 7 has its actual A_2-core loss $< 2000/8181 + 9/25 < .61$. The latter uses $\sqrt{D_i} > 53/20$, $\sqrt{D_i - 100/81} > 12/5$. Own $3, 4, 6$ and parallel retain their affine/null principals. Thus every isolated cap is strong.

The strength ratio $T(q)$ increases here, and

$$q \leq \sqrt{19/2} < 25/8, \qquad T(25/8) = 3/2 - 1441/430050 < 3/2.$$

The residual table of the preceding finite-white classification gives ordinary ratios $> 761/500$: near simple-3/own at least 5 gives at least $(101/125{+}2/5)/\sqrt{3/5}$, D-position gives $(101/125{+}1/5)/\sqrt{2/5}$, and E, end-4 and two-white rows are stronger. Smaller owns have the same affine principals. Foreign far helpers have ratio $> 227/(25\sqrt{35}) > 761/500$; foreign near-four has ratio square $> 121/35$. Consequently every weak cap has $Q = 0$. The near-four ratio increases with side order m, and at $m = 7$ has square $(67 + 42\sqrt{2})/56 > 9/4$. Thus only side orders $5, 6$ remain. For a pure helper the residual identity

$$\frac{(\gamma + 1/(2m))^2}{(m+1)/(2m)} - (4\gamma - 2) = \frac{(\gamma - 1 - 1/(2m))^2}{(m+1)/(2m)}$$

and $T < 3/2$ give $\gamma < 17/16$.

For adjacent literal helpers, the general-N AM–GM product bounds in the common-pair calculation leave mutual 0 for the same helper end, and $0, -1/2$ for opposite ends. Outward mutual 0 is excluded by the positive projected determinant

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = \tfrac{2}{5} C_{i,i+1} - 24/25 + 8/(5N) > 0.$$

For reversed mutual 0, put $l = i, s = N - i - 1$. The required determinant is

$$P = 4lsA^2B^2 - (l+1)(s+1)(A^2 + B^2) - 2AB - (l+1)A - (s+1)B + (N+2)/4 = 0.$$

Its own derivatives are positive on $[.8, 1.08]$, and it is concave in l, $5 \leq l \leq N - 6$. At $N = 17$, both owns at least 7 give $P(.9, .9) \geq 14.632 > 0$. One 5 and one own at least 8 give, at $(.809, .923)$ or its reverse, minimum $1.53076164678 > 0$. These endpoint values increase for $N = 18, 19$: at $l = 5$ the coefficient of N is $20A^2B^2 - 6(A^2 + B^2) - B + 1/4 > 0$ at both comparison pairs and their reversals. If owns are $5, 7$, the determinant is a polynomial of degree at most 2 in $\cos(\pi/7)$ over $\mathbb{Q}(\sqrt{5})$, with nonzero linear coefficient. But $2\cos(\pi/7)$ has irreducible polynomial $t^3 - t^2 - 2t + 1$, and degree 3 over $\mathbb{Q}(\sqrt{5})$. Thus it cannot vanish.

For two native-five helpers, use raw norm-two duals $C_{ij} = \min(i, j) - ij/N$ and $v_i = \varphi\omega_i + \omega_e$. Same-end adjacent difference square is $\varphi^2(N{-}1)/N < 4$, forcing positive wall product. For opposite ends, with orientation $\epsilon = \pm 1$,

$$\|v_i - v_{i+1}\|^2 = \varphi^2(N-1)/N + 2(N-2)/N - 4\epsilon\varphi/N < 6.$$

The raw native gap forces mutual 0. For $H_i = v_i^2 - 2$, $S = v_i \cdot v_{i+1}$, the rational coefficient of $N^2(H_iH_{i+1} - S^2)$ in basis $1, \varphi$ is $N(N+6)$ outward and $-N^2$ reversed. It cannot vanish. Hence opposite helpers have mutual exactly $-1/2$ whenever they survive.

The retained two-cap matrices are the common-helper $I - t\mathbf{1}\mathbf{1}^T$, or the opposite-helper matrix with diagonal in $(1/2, 3/5]$, cross $-1/2$. They have positive inverses. Positive actual projection retains the entire old spectator block. Put $\lambda = H^{-1}(A_i, A_j)^T > 0$, $\mathcal{D} = A_i\lambda_i + A_j\lambda_j$ and $Z = W_i\omega_i + W_j\omega_j$. For a common left helper, $W_i = \lambda_i\gamma_i + (\lambda_i + \lambda_j)/(2i)$, $W_j = \lambda_j\gamma_j$; reflect for a common right helper. Outward helpers give $W_i = \lambda_i(\gamma_i + 1/(2i))$, $W_j = \lambda_j(\gamma_j + 1/(2(N-j)))$. Reversed helpers give $W_i = \lambda_i\gamma_i + \lambda_j/(2i)$, $W_j = \lambda_j\gamma_j + \lambda_i/(2(N-j))$. Thus $W = W_i + W_j > .8L$, $L = \lambda_i + \lambda_j$. All cap times exceed 2: the smallest time square is $> (16/25)(120/17) + (4/5)(10/17) + 8/17 - 1 = 379/85 > 4$. For a common helper the same explicit inverse as in the prefix proof gives $\sqrt{1+\mathcal{D}}/W < 7/16$. For opposite helpers, $5/6 < \lambda_i/\lambda_j < 6/5$, and

$$\mathcal{D}/L^2 \le 3/55, \quad A_i + A_j \le L/10, \quad L > 40.$$

The first inequality follows by maximizing $(.6(t^2+1) - t)/(t+1)^2$ on $[5/6, 6/5]$. Consequently $\sqrt{1+\mathcal{D}}/W < \sqrt{15/176 + 1/1024} < 3/10$. The spatial rationalization loss is $< W/10$: use $\sqrt{120/17} > 2.65$, $\sqrt{110/17} > 2.54$. The weighted mean losses are thus $< 43/80$ or $< 2/5$, both $< .61$. Adjacent weak helpers are impossible independently of an energy target.

A near-four has an actual positive finite mate with cosine > 7, hence normalized height $> .86$. At the opposite end there is another mate, or a strong anchor of normalized height $> .77$. The correlations are bounded respectively by $< .5, < .44, < .41$ at $N = 17, 18, 19$. They follow by substituting mate index at most 5 and anchor index at least $N - 6$ in $\sqrt{j(N-k)/(k(N-j))}$. At the three contrary energies, Cauchy would give

$$1.63^2 < .86(3), \quad 1.63^2 < (6/7)(2.88), \quad 1.63^2 < .91(2.82),$$

each false. The reflected roles satisfy the same bounds. This removes all near-fours. □

C.7 Shared actual endpoint cores and three reference tables

For a weak end $i = 5$ with helper $e = 1, N-1$, let j be the old neighbor of the helper end. All three old selected roles are present in each $N = 17, 18, 19$. Set $v = \gamma\omega_5 + \omega_e/2$. The actual time squares are

$$A^2_{\text{far}} = \frac{10(N-5)\gamma^2 + 10\gamma - (N+1)/2}{N}, \qquad A^2_{\text{near}} = A^2_{\text{far}} + \frac{2(N-10)}{N}\gamma. \tag{42}$$

These follow directly from the actual dual Gram. Reflected weak ends have the reflected formula.

There are three positive actual constructions:

own	old correction b_c	residual
5	$\gamma n_5 + (2/3)n_e + (1/3)n_j$	$\delta = 2/3 - \gamma^2 > 0$
finite ≥ 7	$\gamma n_5 + 2(1-\gamma^2)n_e$	$-k,\ k = (1-\gamma^2)(4\gamma^2 - 3) > 0$
dotted	n_5/γ	$-k,\ k = 1 - \gamma^{-2} > 0.$

For own 5, $\eta = c + b_c$ is spacelike and the actual future finite point is $p + (A/\delta)\eta$, of square $-1 - A^2/\delta$. It is orthogonal to all four selected roots, while all other wall products are nonpositive. For finite owns at least 7, put $d = 1 - \gamma^2$. The core $\xi = c + b_c$ has products zero with c, n_5 and $-1/2 + 2d < 0$ with n_e. The dotted core has products zero with c and $-\gamma + \gamma^{-1} < 0$ with n_5. All time coefficients

are $A > 0$. Positivity checks every other actual wall, including the other helper, every spectator and all unknown walls. Thus, for $U = v - b_c$, the actual point pressures give

$$\langle x_A, U\rangle > A - \sqrt{\delta} \quad \text{(own 5)}, \qquad \langle x_A, U\rangle \geq A - \sqrt{k} \quad \text{(other rows)}. \tag{43}$$

Dropping the positive old-normal products yields the same lower bounds for $\langle x_A, v\rangle$.

The rational concave references needed below have linear end segments, reflection $H_{N-i} = H_i$, and the following central heights:

N	H_5	H_6	H_7	H_8	H_9	H_{10}
17	2.046	2.176	2.259	2.300	2.300	
18	2.077	2.218	2.314	2.371	2.389	
19	2.104	2.255	2.363	2.433	2.467	2.467.

Only the independent entries before reflection are used. Positive square tests $(H_i + .61)^2 < D_i$ pay every retained strong floor. If 5 is weak use its chord to 6; reflect at the other end; omit every internal weak knot by its neighbor chord. Nonadjacency proves that all losses add. The complete zero/one/two-weak-end tables are

N	number of weak ends	initial energy	maximum internal loss
17	0	.8626932	$3(.047^2/4) = .00165675$
17	1	.8302129333…	.00165675
17	2	.7977326666…	$2(.042^2/4) = .000882$
18	0	.8954558	.002025
18	1	.8640827333…	.00151875
18	2	$H_6^2/6 + .012789$	$3(.039^2/4)$
19	0	.9258842	$4(.043^2/4) = .001849$
19	1	.8955541833…	.001849
19	2	.8652241666…	$3(.038^2/4) = .001083$.

The repeating decimals abbreviate exact rational sums of squared rational slopes. These tables are complete: after zero/one/two end chords, the eligible internal weak intervals and nonadjacent cardinalities give the indicated maximal counts. Each loss is one quarter of the squared second difference. End chords preserve the neighboring strong anchor; an adjacent omission can only increase its available individual price. Every omitted knot still has price zero.

Proposition C.5 (The $N = 17$ paid reference)**.** *An actual A_{16} component has energy $> 43/50$ with arbitrary finite spectators.*

Proof. Assume contrary $e \leq .86$. The central-window lemma removes near-fours. Use the $N = 17$ table. Zero weak ends already give $e \geq .86103645$. For two weak ends the reference is flat on both six-coordinate end blocks, has energy $E \geq 1195276/1500000$, and separate anchor capacities

$$\beta_6, \beta_{11} \geq (7H_6/6 - H_7)/2 = 839/6000.$$

For one weak end the block lengths are $6, 5$, $E > .8285$, and $\beta_6 \geq 839/6000, \beta_{12} \geq 349/2500$.

For a left weak helper, project its core displacement U onto vectors constant on each end block. In the six/six case,

$$\Pi U = (5\gamma/6)\omega_6 + \omega_{11}/12 \quad \text{(far)}, \qquad \Pi U = (5\gamma/6 + 1/12)\omega_6 \quad \text{(near)}.$$

All old-root corrections have mean zero in their own block. Thus $Y = U - \Pi U$ sums to zero on each block, $\langle w, Y\rangle = 0$, and its square is

own	$\|Y\|^2$ in the far orientation
5	$(2/3)\gamma^2 + 1/12$
finite ≥ 7	$4\gamma^4 - (16/3)\gamma^2 + 29/12$
dotted	$(5/3)\gamma^2 - 19/12 + \gamma^{-2}$.

The near orientation adds $\gamma/3$. These identities follow by expanding the actual $N = 17$ Gram and

$$b_c^2 - 2\langle v, b_c\rangle = k_c^2 + l_c^2 + m_c^2 - l_c m_c - 2\gamma k_c - l_c,$$

where $b_c = k_c n_5 + l_c n_e + m_c n_j$. The finite polynomial increases for $\gamma^2 > .81$ and is $< 13/12$. The dotted one increases for $1 < \gamma^2 < 289/256$ and is $< 229/768 + 256/289 < 148/125$, with cleared margin 271. Consequently every orientation has $\|Y\|^2 < 148/125 + 17/48 < 8/5$.

The projection demand is $H_6(5\gamma/6 + 1/12)$. Let Δ be the actual core demand minus this quantity. For native 5, the fresh time bound $A_{\text{far}} > 2.136$ and radical $< .111$ give

$$\Delta > 2.136 - .111 - H_6(91/120) > .327.$$

For finite owns at least 7, use $A_{\text{far}}(.9) > 2.391, \sqrt{k} \leq 1/4$; for dotted, use $A_{\text{far}}(1) > 2.667$ and radical $< .338$. They give, respectively,

$$\Delta > 2.391 - .25 - 5H_6/6 = 983/3000 > .327, \quad \Delta > 2.667 - .338 - 11H_6/12 > .327.$$

These bounds subtract fixed radical caps from an increasing time-minus-linear envelope; they do not assert that the actual deficit, with its varying radical, increases. Indeed $A_{\text{far}} - H_6(5\gamma/6 + 1/12)$ increases because

$$(A'_{\text{far}})^2 - 120/17 = 1105/(289A_{\text{far}}^2) > 0, \qquad \sqrt{120/17} > 5H_6/6.$$

Subtract the uniform cap $1/4$ from that envelope starting at $\gamma = .9$, or the uniform cap .338 starting at $\gamma = 1$, respectively. The near time is larger with the same cap and reference pairing. This proves the displayed lower bounds on all finite/dotted intervals and both orientations.

For one weak end, the far projection instead uses $(5\gamma/6)\omega_6 + \omega_{12}/10$. Its residual square decreases by $1/60$, and its deficit decreases by exactly $H_5/10 - H_6/12 = 349/15000$. Hence $\|Y\|^2 < 8/5, \Delta > .303$.

Use weight $\mu = .1$, paying each original anchor before adding its actual core pressure. One near foot spends at most $31/320$ at 6; a far foot spends less there and $1/100$ at 12. For two ends the combined debit at each $6/11$ is at most

$$.1(5(17/16)/6 + 1/6) = 101/960 < 839/6000.$$

All surviving original coefficients are nonnegative. With $W = w + \mu \sum Y$, one original-vector square gives

$$e \geq E + 2\mu \sum \Delta - \mu^2 \|\sum Y\|^2.$$

For one weak end the gain exceeds $.2(.303) - .01(8/5) = .0446$, giving $e > .8731$. For two ends retain the full cross by $\|Y_L + Y_R\|^2 \leq 2\|Y_L\|^2 + 2\|Y_R\|^2 < 32/5$. The gain exceeds $.2(2 \cdot .327) - .01(32/5) = 167/2500$, and

$$e > 1195276/1500000 + 167/2500 = 1295476/1500000 > .86.$$

All three cases contradict the contrary bound. □

C.8 Raw helper-end pressures at $N = 18, 19$

For weak 5, put $z = x_1$ for helper 1 and $z = -x_N$ for helper $N-1$. Then $\langle x_A, v\rangle = \gamma h_5 + z/\sqrt{2}$. The shared actual cores give deficits $> .313$ at $N = 18$ and $> .306$ at $N = 19$. Here are the complete seals. The weak floors are $h_5 < 2.078$ and $h_5 < 2.105$. For native 5, radical is $< .111$, own $< .81$, and times are $> 2.155, > 2.17$; the deficits exceed $.36082, .35395$. For finite owns at least 7, the core radical has square $k = (1-\gamma^2)(4\gamma^2 - 3) \le 1/16$:

own	γ bracket	$\sqrt{k} <$	$A_{18,\mathrm{far}} >$	$A_{19,\mathrm{far}} >$	deficit at 18; 19
7	$(.9009, .901)$	.2158	2.4154	2.434	.327322; .321595
8	$(.9238, .9239)$	.2464	2.4795	2.499	.3132358; .3077905
≥ 9	$\gamma > .93$	1/4	2.497	2.517	.31446; .30935.

The native-seven polynomial $8\gamma^3 - 4\gamma^2 - 4\gamma + 1$ increases above $\sqrt{3}/2$; its signs at $.9009, .901$ have numerators $-569738168, 257608$ and denominators $10^{12}, 10^9$. Native eight is $\sqrt{2+\sqrt{2}}/2$. The residual polynomial increases on both brackets; substitution of upper endpoints gives the radical bounds. The tight eight comparison is $14640879 \cdot 41436484 < 607129600000000$. Times come from (42) at the lower endpoints. Every finite label at least 9 has $\gamma > .93$, by monotonicity and $8(.93)^3 - 6(.93) - 1 < 0$. The functions $A_{\mathrm{far}} - 2.078\gamma$ or $A_{\mathrm{far}} - 2.105\gamma$ increase, since $A'_{\mathrm{far}} > \sqrt{D_5}$. Thus the last row covers every finite label.

For dotted $1 < \gamma < 17/16$, retain the exact floor $h_5 \le \sqrt{D_5} - .61$. At 18,

$$A_{\mathrm{far}} - \sqrt{D_5}\gamma = \frac{(5/9)\gamma - 19/36}{A_{\mathrm{far}} + \sqrt{D_5}\gamma} > 1/216.$$

Its numerator is $> 1/36$, denominator < 6. At 19 the same difference is positive. The function $.61\gamma - \sqrt{1-\gamma^{-2}}$ decreases throughout the interval: its radical derivative exceeds 1. Using $\sqrt{33}/17 < .338$, its endpoint lower bound is $> .310125$, and adding $1/216$ at 18 gives $> .313$. Consequently

$$z > .442 \quad (N = 18), \qquad z > .432 \quad (N = 19).$$

A near helper at 19 has the same radical and time larger by

$$A_{\mathrm{near}} - A_{\mathrm{far}} > \frac{(18/19)(.8)}{7} > .108,$$

using $A_{\mathrm{far}} < 3, A_{\mathrm{near}} < 4$. Its deficit is $> .414$, so $z > .58$.

Proposition C.6 (The last two ordinary A-rates)**.** *With arbitrary finite spectators, $e(A_{17}) > 6/7$ and $e(A_{18}) > 91/100$.*

Proof. At $N = 18$, assume $e \le 6/7$. The central-window lemma removes near-fours. The reference table excludes zero and one weak end: the retained energies are $.8934308$ and $> .8625639833\ldots$. With two weak ends, the reference is flat on both six-coordinate end blocks, of signed mean b with $1/4 < b < .262$, and

$$E \ge H_6^2/6 + .012789 - 3(.039^2/4) = 9978827/12000000.$$

One actual weak helper supplies endpoint $z > .442$ at either end. Change that signed endpoint of the reference to .442 and its other five block coordinates to $(6b - .442)/5$. The block sum is preserved; its last coordinate is $> .2116 > .096$, above the next coordinate. All new positive duals

are paid by the old strong floors and this actual endpoint pressure. The single reference's energy gain is

$$(6/5)(.442 - b)^2 > (6/5)(.442 - .262)^2 = .03888.$$

Hence

$$e > 9978827/12000000 + .03888 = 10445387/12000000 > 6/7.$$

Only one helper is used; either orientation works.

At $N = 19$, assume $e \le .91$; no near-four remains. The reference table gives

$$E_0 \ge .9240352, \quad E_1 > .8937, \quad E_2 > .86414.$$

A strong end is a length-five block with $.29 < b_0 < .298$; a weak end is a length-six block with $1/4 < b_1 < .266$. Changing a signed endpoint to $z_0 > b$ raises energy exactly $m(z_0 - b)^2/(m-1)$, preserving the block sum. For $m = 5, z_0 = .432$, the other coordinates exceed .2545, above $.151/\sqrt{2}$; for $m = 6, z_0 \le .58$, they exceed .184, above $.108/\sqrt{2}$. Neighbor omissions only decrease the next coordinate. Opposite end blocks are disjoint.

Zero weak ends already exceed .91. With one weak end and a near helper,

$$e > .8937 + (6/5)(.58 - .266)^2 = 1.0120152.$$

Otherwise its helper is the opposite strong end:

$$e > .8937 + (5/4)(.432 - .298)^2 = .916145.$$

With two weak ends, any near helper gives $.86414 + (6/5)(.58 - .266)^2 = .9824552$. If both helpers are far, they use distinct opposite endpoints. Paying both in the disjoint length-six blocks gives

$$e > .86414 + 2(6/5)(.432 - .266)^2 = .9302744.$$

These exhaust the end patterns and orientations, always on one modified reference for the same x_A. □

D The local A_{18} inventory at comparison $11/8$

This appendix supplies the earlier geometric body used in Lemma 9.1; it does not use that lemma's endpoint pressure or its original B_{18} application. We work at the actual whole-A_{18} vertex q with the same strict interior point and comparison $-\langle z, r\rangle \le \kappa E_q$, $\kappa < 11/8$.

D.1 All zero-own and isolated boundaries

On $5 \le i \le 14$, deletion leaves path orders $\ell = i, r = 19 - i$, both at least 5. The complete one-node finite-white list is none; one simple 3 at an end, a D position or an admissible E position; one terminal 4; or two simple end 3's joining the two paths. Two contacts in one path make a nonfinite cycle. A joined completion has rank 18, hence is classical with only new simple end bonds. There are no foreign components at this vertex.

At zero own, a global endpoint 3, next-to-end 3, or endpoint 4 has time square $9/19 - 1, 17/19 - 1, 18/19 - 1 < 0$, respectively. Other simple-3 placements retain affine E_8 arms $1, 2, 5$ or E_7 arms $1, 3, 3$. Internal 4 gives the 4/3/3 star; two simple white contacts close an all-three cycle. No contact gives zero derivative and time square -1. The affine kernels are positive: for E_7 take center 4 and arms $2; (3, 2, 1); (3, 2, 1)$; for E_8, center 6, arms $3; (4, 2); (5, 4, 3, 2, 1)$. For a 4/3/3 star, take

center $\sqrt{2}$, doubled leaf 1, simple leaves $1/\sqrt{2}$; all-three cycles have coefficients 1. Positive cap time makes every such null nonzero future, and all unknown original wall products are nonpositive. Compactness excludes them.

Isolated own 3 retains the affine E_7 star with the cap as short arm; own 4 has the 4/3/3 star; own 6 the 6/3 chain, with kernel $(\gamma, 1, 1/2)$; parallel own has null $c + n_i$. White 3/own 3 closes an all-three cycle. Far white 4/own 3 retains the E_7 star and far white 4/own 4 the doubled star. For pure helpers own $3, 4, 6$ or parallel retains respectively the path cycle, doubled star, 6/3 chain or $c + n_i$. Additional unselected contacts preserve all wall signs in these actual null arguments.

For isolated positive own put $\tau = \gamma^{-2}$. For $0 < \tau < 1$, the actual positive core $c + \gamma\tau n_i$ gives

$$h_i > \sqrt{D_i - \tau} - \kappa\sqrt{\tau(1-\tau)}.$$

At $\tau = 1$ it is a feasible null. For $\tau > 1$, positive finite projection on c, n_i gives the exact pressure

$$h_i \geq \sqrt{\frac{D_i - 1}{D_i - \tau}}\,(\sqrt{D_i - 1} - \kappa\sqrt{\tau - 1}).$$

The future point, not its spacelike correction, is normalized. Every unknown wall is checked by the positive selected-root coefficients. Here $D_i \geq 140/19$, $a_i = (\sqrt{D_i} + \sqrt{D_i - 1})^{-1} < 5/26 < .193$. For $0 < \tau < 1$, the loss is

$$< a_i\tau + \kappa\sqrt{\tau(1-\tau)} < (.193 + 1.389)/2 = .791 < .87.$$

The radical seal is $.193^2 + (11/8)^2 < 1.389^2$. For finite owns at least 8, the positive-bracket finite formula has loss $< 5/26 + (11/8)(.415) < .764$.

Native 5 uses its positive H_4 projection, old coefficients $\gamma(3/2, 1, 1/2)$, residual $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$, and cap time square > 3.7. The finite-point radical factor $\sqrt{1 + \delta_5/A^2} < 1.01$ gives loss

$$< 25/76 + (11/8)\frac{1.01}{6(.8)} < .619.$$

Native 7 uses the actual negative core $c + (4\gamma/3)n_i + (2\gamma/3)n_{\text{neighbor}}$, whose loss is

$$< 10000/41877 + (11/8)(9/25) < .735.$$

Use $.9 < \cos(\pi/7) < 10/11$, verified by its increasing cubic, and $\sqrt{D_i - 100/81} > 2.47$. Thus every isolated finite or dotted cap is strong at loss .87 and comparison 11/8.

D.2 Every nonisolated ratio and the surviving helpers

The ordinary ratio threshold 3/2 at this comparison is proved in Lemma 9.1 by its explicit concave polynomial. For one white 3 at side position t in a path of order ℓ,

$$R = \frac{\gamma + t/(2\ell)}{\sqrt{1 - t(\ell - t)/(2\ell)}}.$$

Near 3, after the own-3 cycle, has ratio $> (.808 + 2/5)/\sqrt{3/5} > 3/2$, except the own-4 near-four role. At D positions $t = 2, \ell - 2$, after the own-4 star, ratio is

$$\geq \frac{\gamma + 1/\ell}{\sqrt{2/\ell}} > \frac{.808 + 1/5}{\sqrt{2/5}} > 3/2.$$

The lower expression increases for $\ell \geq 5$. At the allowed E positions $\ell = 6, 7, 8$, denominator is at most $1/2$ and own $> .8$. Near endpoint 4 exceeds $3/2$ even at own 3. Far endpoint 4 has own $3, 4$ nulls, and otherwise ratio $\geq \gamma\sqrt{\ell} > 3/2$. Two-white-3 has ratio at least $\gamma\sqrt{D_i} + 1/\sqrt{D_i} > 3/2$ already at $\gamma = 1/2$. All larger finite and dotted owns strengthen these tests. This exhausts the finite-white table.

Only near-four with side order $5, 6$, or pure far helpers, remain weak. The near-four and pure-helper formulas, the order-7 seal, and the entire helper cutoff $\gamma < 17/16$ are derived in the main proof. This completes its own/white inventory without a farthest hypothesis at q.

D.3 Native-five determinant and positive mixed weights

For two adjacent native-five helpers use raw norm-two duals $C_{ij} = \min(i, j) - ij/N$. Put $H_i = v_i^2 - 2, S = v_i \cdot v_j, U = H_i - S, V = H_j - S$ and $P = i(N - i - 1) + 1$, with $j = i + 1$. For outward helpers, the exact expansions are

$$NS = P + (P + N - 2)\varphi, \quad NU = (i - N - 2) + (N - i + 1)\varphi, \quad NV = -i - 3 + (i + 2)\varphi.$$

For reversed helpers they are

$$NS = P + (P + N)\varphi, \quad NU = (i - N - 2) + (3i - N - 1)\varphi, \quad NV = -i - 3 + (2N - 3i - 4)\varphi.$$

Since $H_iH_j - S^2 = S(U + V) + UV$, substituting $\varphi^2 = \varphi + 1$ gives rational coefficients $N(N + 6)$ and $-N^2$, respectively. At $N = 19$ these are $475, -361$, both nonzero. The raw difference-square argument in the central-window lemma forces mutual zero first, so neither determinant can satisfy that zero mutual. No arithmetic criterion is used.

For the surviving mixed pairs, use the explicit positive Schur matrices, coefficients $\lambda = H^{-1}(A_i, A_j)^T$, and all four orientation formulas for W_i, W_j in the central-window lemma. They are purely geometric, so remain valid at comparison κ. Cap times still exceed 2; hence $\sqrt{1 + \mathcal{D}}/W < 7/16$ for a common helper and $< 3/10$ for opposite helpers. With $N = 19$, $\sqrt{D_i} > 2.7$ and $\sqrt{C_{i,i+1}} > 2.6$. Rationalizing $E_c - \|Z\|$, as in the prefix proof, bounds the spatial loss by $W/10$. The actual mixed-point pressure now is

$$\langle y, Z\rangle \geq 1 + \mathcal{D} - \kappa\sqrt{1 + \mathcal{D}}.$$

Thus its weighted mean loss is $< .7015625$ or $< .5125$, respectively. Both are less than .87, proving the required nonadjacency against every original wall. Together with the new endpoint/mate/reference argument already written, this completes all earlier local inputs of Lemma 9.1.

D.4 The fresh $N = 16$ inventory and $A_{15} > .826$

Proposition D.1. *With arbitrary finite spectators, an actual A_{15} component has energy greater than $413/500$.*

Proof. Use the exact interval $5 \leq i \leq 11$, loss .61, and $55/8 \leq D_i \leq 8$. The finite-white list is the two-path list already proved: each retained path has rank at least 4, and a joined completion has rank $15 > 8$. The zero-own affine and foreign-shadow repairs need only side order at least 5, so give precisely the same exhaustive cases at this rank.

For isolated caps, put $a = (\sqrt{D} + \sqrt{D-1})^{-1} < 1/5$. The cleared $0 < u < 1$ loss is $< (.2 + \sqrt{1.04})/2 < .61$. For foreign $u > 1$ it is $< .2 + 2 - \varphi < .582$. For pure finite labels at least 8, use the exact singleton pressure

$$T = r\sqrt{\frac{r-s}{r+s}}, \quad r = \sqrt{q^2-1}, \quad s = \tan(\pi/m) \le \sqrt{2} - 1 < .415.$$

Its loss $q - T$ decreases with q and increases with s. For the first monotonicity, the positive derivative comparison reduces to

$$(1-s^2)r^4 + 2sr^3 + (2s^4 - s^2)r^2 - 2s^3 r + s^4 \ge r^4/4 > 0$$

on $r \ge 1, 0 \le s \le 1/2$. At $q_0 = \sqrt{55/8} < 2.623$, $2.423 < r_0 = \sqrt{47/8} < 2.425$, and

$$T^2 > (47/8)(2.008/2.84) = 11797/2840 > (2.03)^2.$$

The loss is $< .593 < .61$. Own 5 has its positive H_4 loss $< 25/76 + 5/24$; own 7 has $< 10000/40257 + 9/25 < .61$. Own $3, 4, 6$, parallel and the $u = 1$ case have their actual affine/null exclusions. Thus all isolated caps are strong. The ordinary white ratios exceed $3/2 > T(q)$ using only side order at least 5. The complete weak list is near-four at $5, 6$ or its reflection, and pure helpers $\gamma\omega_i + \omega_e/2$, $e = 1, 15,\ Q = 0,\ \varphi/2 \le \gamma < 17/16$. All positive-foreign roles are strong.

For adjacent helpers, the same actual-product bounds leave mutual 0 for a common helper and $0, -1/2$ otherwise. Outward zero contradicts

$$(D_i - 4/5)(D_{i+1} - 4/5) - C_{i,i+1}^2 = (2/5)C_{i,i+1} - 24/25 + 1/10 \ge 41/25 > 0.$$

For reversed zero, the polynomial of the central-window proof has $l + s = 15$ and constant $9/2$. It increases in each own on $[.8, 1.08]$ and is concave in l. At its endpoint $l = 5$ or 10, both owns at least 7 give $P(.9, .9) = 11.88 > 0$. One 5 and an own at least 8 give the exact minimum $P(.809, .923) = .0907947698 > 0$; the reflected endpoint is larger by $5(.923 - .809)$. Two native-five helpers are impossible by the raw determinant proof, valid for $N \ge 15$. Native $5/7$ is impossible by the same nonzero quadratic over $\mathbb{Q}(\sqrt{5})$ versus cubic degree argument. No field restriction is imposed on any other wall. The surviving opposite-helper mutual is therefore $-1/2$. The positive mixed Schur point appends all spectators; its times satisfy $A_i^2 > 699/160 > 4$. The explicit inverse weights give the same mean losses $< 43/80$ or $< 2/5$. The spatial term is $< W/10$, since $\sqrt{55/8} > 2.6$, $\sqrt{25/4} = 2.5$. Helpers are consequently nonadjacent before any energy target is used.

Assume now $e \le 413/500$. A near-four's actual mate has demand $> .86$. At the other end, a mate or a strong $10/11$ has demand $> .76$; their correlation is at most $\sqrt{3/11} < 11/20$. Cauchy would give $1.62^2 < (413/500)(31/10) = 2.5606$, whereas $1.62^2 = 2.6244$. Thus all weak sites are pure helpers.

Use reflected rational heights

$$H_5 = H_{11} = a = 2.012, \quad H_6 = H_{10} = C = 2.128, \quad H_7 = H_9 = 2.196, \quad H_8 = 2.218$$

and linear ends. Flatten an outer weak end to its strong neighbor; omit all internal weak knots by chords. The exact complete bounds are

weak ends	maximum internal loss	$E >$
0	$3(.048^2/4) = .001728$	.8264648
1	$2(.048^2/4) = .001152$	.7928637333...
2	$2(.046^2/4) = .001058$	1138171/1500000.

Every positive dual uses an actual strong floor. For one weak end the block lengths are $6, 5$, with separate prices $\beta_6 \geq 43/300, \beta_{11} \geq 179/1250$. For two weak ends they are $6, 6$, and $\beta_6, \beta_{10} \geq 43/300$. Adjacent omissions increase these capacities.

Use the same three actual endpoint-core constructions of Section C.7, now with the freshly computed times

$$A^2_{\mathrm{far}} = (220\gamma^2 + 20\gamma - 17)/32, \quad A^2_{\mathrm{near}} = A^2_{\mathrm{far}} + 3\gamma/4.$$

In the six/six reference, the core's block projection is $(5\gamma/6)\omega_6 + \omega_{10}/12$ far, and $(5\gamma/6 + 1/12)\omega_6$ near. Fresh substitution of the $N = 16$ dual Gram gives exactly the three residual polynomials in the $N = 17$ payment proof, with the same near addition $\gamma/3$. This is the fixed-block variance identity, not an imported different-rank energy bound. Thus $Y = U - \Pi U$ is orthogonal to the reference and $\|Y\|^2 < 8/5$.

Let Δ be the core demand minus $C(5\gamma/6 + 1/12)$. All orientations and own intervals have $\Delta > 1/3$. Indeed $(A'_{\mathrm{far}})^2 - 55/8 = 15/(4A^2_{\mathrm{far}}) > 0$. At native 5, use time > 2.11, radical $< .111$, projection $< C(91/120)$; the deficit is $> 5779/15000$. For finite owns at least 7, use $A_{\mathrm{far}}(.9) = \sqrt{28/5} > 2.366$ and radical $\leq 1/4$, giving $> 257/750$. For dotted owns, use $A_{\mathrm{far}}(1) = \sqrt{223/32} > 2.639$ and radical $< .338$, giving $> 1051/3000$. Monotonicity of A_{far} minus its linear projection makes these full-interval bounds; near times are larger. In the six/five case, a far projection uses $\omega_{11}/10$ instead. Its variance decreases $1/60$, and its deficit decreases $a/10 - C/12 = 179/7500$. Hence $\Delta > 2321/7500 > .309$.

Pay $\mu = .1$ before using each core pressure. One foot costs at most $31/320 < 43/300$ at the weak-side strong anchor and, if far, $1/100 < 179/1250$ at the opposite anchor. For two feet, the combined cost at each $6/10$ is at most $101/960 < 43/300$. Thus $W = (w - \mu \sum \Pi U) + \mu \sum U$ has a nonnegative old strong-floor part and the actual core pressures, giving

$$e \geq E + 2\mu \sum \Delta - \mu^2 \|\sum Y\|^2.$$

One weak end gains $> .0458$, hence $e > .7928 + .0458 = .8386$. For two ends retain the full cross through $\|Y_L + Y_R\|^2 < 32/5$. The gain exceeds $26/375$, so

$$e > 1138171/1500000 + 26/375 = 1242171/1500000 = .828114 > 413/500.$$

Zero weak ends already exceed the target. This proves the component theorem.

Every ordinary rank-three forest costs $> 17/70$, so this new component rate also excludes its ordinary spectators directly. The other retained spectators $I + T_2$ and $O_7 + A_1$ cost $> .21$ and $> .09 + 2/15$; each sum likewise exceeds 1. A separate historical ordinary-rank-three consumer is therefore unnecessary. □

D.5 The fresh $N = 13$ component and its native-cosine split

Proposition D.2. *With arbitrary finite spectators, an actual A_{12} component has energy greater than $13/20$.*

Proof. Use $N = 13$, the exact window $5 \leq i \leq 8$, loss .65, and $80/13 \leq D_i \leq 84/13$. The same complete two-path finite-white classification applies at these actual ranks: each path has rank at least 4, a joined extension has rank $12 > 8$, and its only connected finite completion is the ordinary path. The zero-own repairs are unchanged: internal 3/4 and two-white roles have affine principals; endpoint 3 or next-to-end 3 has time square below its positive foreign cost, forcing an orthogonal closure; endpoint 4 has $Q < 1/5$, hence negative time square. Every own contact is positive.

The strength ratio now satisfies, for the entire window,

$$T(q) = 20/13 - \frac{q - .65}{q(2q - .65)} < T(3) = 5809/4173 < 7/5.$$

Near simple 3, D/E positions and endpoint 4 have the same larger ordinary residual ratios as before. Even the near-four exception has minimum $(1/\sqrt{2} + 2/5)/\sqrt{3/5} > 7/5$, so it is strong. Far white 4/own 3 retains affine F_4. Two-white-3 has ratio at least $q/2 + 1/q > 7/5$ at positive own; zero own is affine. Positive foreign cost strengthens these tests; the exceptional foreign far/helper ratios remain $> 761/500$ or have square $> 121/35$.

For isolated owns put $a_0 = (\sqrt{D} + \sqrt{D-1})^{-1} < .212$. The actual cleared $0 < u < 1$ pressure has loss $< (.212 + 1.023)/2 = .6175$; foreign $u > 1$ has loss $< .212 + .383 = .595$. Pure finite owns at least 8 have loss $< .212 + \tan(\pi/8) < .627$. The native-five H_4 point has loss $< 125/352 + 5/24 = 595/1056 < .57$. Native seven's A_2 core has loss $< .264 + \sqrt{13/120} < .601$, using $.9 < \gamma < .902$ and $\gamma^{-2} > 49/40$. Own $3, 4, 6$, parallel and $u = 1$ have their actual affine/null exclusions. All spectators are cleared positively and every unknown wall is checked as in the shared proofs. Thus the complete weak list is pure

$$v = \gamma\omega_i + \omega_e/2, \quad e = 1, 12, \quad Q = 0.$$

The identity $\sigma^2/\delta \geq 4\gamma - 2$ and $T < 7/5$ force $\varphi/2 \leq \gamma < .99 < 1$. Only own 5 or finite own at least 7 remains; dotted owns have been excluded by the actual edge inequality.

We prove fresh nonadjacency, without importing a larger-rank determinant. For adjacent owns A, B, the same-helper product satisfies

$$R \geq 1 - AB + (AB - |A - B|)/13 > 1 - .99^2 > 0,$$

contradicting acuteness. Both opposite-helper AM–GM bounds are $> -1/2$, so native gap forces mutual $R = 0$. For outward helpers, the retained parameters obey $\eta < 1/(2\gamma^2) < 25/32 < 4/5$, whereas

$$(D_i - .8)(D_{i+1} - .8) - C_{i,i+1}^2 = (2/5)C_{i,i+1} - 24/25 + 8/65 \geq 428/325 > 0.$$

This makes zero mutual impossible by the actual time/spatial determinant identity.

For reversed helpers put $l = i, s = 12 - i$; $l + s = 12$, $ls = 35$ or 36. The exact time and spatial formulas are

$$A_c^2 = \frac{2l(s+1)A^2 + 2lA - 7}{13}, \quad B_c^2 = \frac{2(l+1)sB^2 + 2sB - 7}{13},$$

$$S = \frac{2lsAB + (s+1)A + (l+1)B + 1/2}{13}, \qquad A_c^2 B_c^2 - S^2 = P/13,$$

$$P = 4lsA^2B^2 - (l+1)(s+1)(A^2 + B^2) - 2AB - (l+1)A - (s+1)B + 15/4.$$

Zero mutual requires $P = 0$. The polynomial increases in either variable on $[.8, 17/16]$: its derivative bracket is at least $1.56ls - 13 \geq 41.6$. If neither own is 5, $A, B > .9$, and the minimum $P(.9, .9)$ at $ls = 35$ is $3.624 > 0$. If both are 5, direct reduction gives $P = ((ls - 56)\varphi - 13)/4 < 0$, also nonzero.

It remains $A = \varphi/2$, $B = \cos(\pi/m)$, $m \geq 7$, including its reversed placement. As a polynomial in B, P has leading coefficient $a = ls\varphi - 13 \neq 0$ and linear coefficient $b = -\varphi - s - 1 \neq 0$. If it vanished, B would have degree at most 2 over $\mathbb{Q}(\varphi)$, hence degree at most 4 over $\mathbb{Q}$. The real cyclotomic degree formula gives $\phi(2m) \leq 8$. Its complete even list with $2m \geq 14$ is $14, 16, 18, 20, 24, 30$. Indeed prime divisors can only be $2, 3, 5, 7$; the totient product bounds leave exactly these even integers.

Thus $m = 7, 8, 9, 10, 12, 15$. For $7, 9$, the native cubic degree remains 3 over the quadratic $\mathbb{Q}(\varphi)$. For 8, the quartic real cyclotomic field has unique quadratic subfield $\mathbb{Q}(\sqrt{2})$: its generator $\cos(\pi/8)$ has a cyclic order-four conjugation, whose square fixes $\sqrt{2}$. For 12, the field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, with quadratic subfields $\sqrt{2}, \sqrt{3}, \sqrt{6}$. Neither contains $\sqrt{5}$; both native degrees remain 4 after adjoining φ. For 10, $B^2 = (\varphi+2)/4$ has nonsquare rational norm $5/16$; its minimal quadratic has zero linear coefficient, unlike P. For 15, the minimal quadratic is

$$4B^2 - 2(\varphi - 1)B - (\varphi + 1),$$

whose discriminant squareclass $3(2 + \varphi)$ has nonsquare rational norm 45. Proportionality to P would require

$$2b + a(\varphi - 1) = ls + 11 - 2s - 15\varphi = 0,$$

impossible. All native cases are excluded. This is algebra of literal finite cosines, not an arithmeticity restriction on the polytope. Weak sites are consequently nonadjacent.

Use heights $H_5 = H_8 = a = 1.830$, $H_6 = H_7 = b = 1.891$, linear ends and reflection. Their strong-floor tests are $2.480^2 < 80/13$, $2.541^2 < 84/13$. Flatten a weak outer end and omit internal weak knots by neighbor chords. The complete reference bounds are

weak outer sites $5, 8$	internal omissions	$E \geq$
0	at most one, loss $.061^2/4$	2690283/4000000
1	at most one, same loss	1901431/3000000
2	$6, 7$ both strong	3575881/6000000.

Zero outer weak sites already gives $e > .65$. For one outer weak site, orient it to 5. The block lengths are $6, 5$, with prices $\beta_6 \geq b/12$, $\beta_8 \geq .1525$. For two outer weak sites they are $6, 6$, with $\beta_6 = \beta_7 = b/12$. Neighbor omissions only increase capacities; no weak floor is charged.

Choose one actual weak outer helper. Use the finite-five or finite-at-least-seven actual core in Section C.7; their selected roots are still present at $N = 13$, with all-wall signs and future orientations unchanged. The freshly computed times are

$$A^2_{\text{far}} = (80\gamma^2 + 10\gamma - 7)/13, \qquad A^2_{\text{near}} = A^2_{\text{far}} + 6\gamma/13.$$

Project its displacement onto the constant end blocks. In the six/six reference,

$$\Pi U = (5\gamma/6)\omega_6 + \omega_7/12 \quad \text{far}, \qquad \Pi U = (5\gamma/6 + 1/12)\omega_6 \quad \text{near}.$$

For six/five, replace the far $\omega_7/12$ by $\omega_8/10$. The residual $Y = U - \Pi U$ has zero mean in each block and $\langle w, Y\rangle = 0$. Fresh block variance gives

$$\|Y\|^2 = (2/3)\gamma^2 + 1/12 \quad (5), \qquad 4\gamma^4 - (16/3)\gamma^2 + 29/12 \quad (\text{finite } \geq 7),$$

with near addition $\gamma/3$. A six/five far residual decreases $1/60$. Thus every case has $\|Y\|^2 < 17/12 < 8/5$. Both reference pairings obey

$$\langle w, \Pi U\rangle \leq b(5\gamma/6) + a/10.$$

For native five, $A_{\text{far}} > 2.027$, radical $< .111$, pairing $< 1.459425 < 1.46$, so its deficit exceeds .456. For finite owns at least seven, $A_{\text{far}}(.9)^2 = 334/65 > 2.266^2$; the pairing envelope at $\gamma = .9$ is 1.60125. The envelope $A_{\text{far}} - (5b/6)\gamma - a/10$ increases because

$$(A'_{\text{far}})^2 - 80/13 = 585/(169A^2_{\text{far}}) > 0, \qquad \sqrt{80/13} > 5b/6.$$

Subtracting the uniform radical cap $1/4$ gives deficit $> 2.266 - .25 - 1.60125 = .41475$ throughout this finite-own interval. Monotonicity of the actual radical-dependent deficit is not required. Near times are larger and pairings smaller. Uniformly $\Delta > .4$. In particular the far-native-five pairing is bounded by 1.46, not the false value 1.438.

Pay $\mu = .1$ for this one helper. A far foot spends $< .0825$ at ω_6 and at most .01 at the opposite strong anchor; a near foot spends $< .090834$ at ω_6. They fit $b/12 > .15758$ and .1525 separately before pressure is used. The single vector $W = w + .1Y$ gives

$$e > E + 2(.1)(.4) - (.1)^2(8/5) = E + .064.$$

Even the worst reference yields

$$e > 3575881/6000000 + 8/125 = 3959881/6000000 = 13/20 + 59881/6000000.$$

All weak sets, orientations and native labels are exhausted, with one original-vector budget. □

E The actual long-prefix inventory and its point pressures

We supply the geometric input of Proposition 4.1. Its exact domain is $6 \le M \le 16$, B_r with $r \ge M+5$, or D_r with $r \ge M+6$. For $5 \le i \le M+1$, deletion of old i leaves a left A_{i-1} of rank at least 4, and a right B_s, $s \ge 4$, or D_s, $s \ge 5$. All roots used at 5 and its neighbors are actual long roots. No short-tail exclusion is imported. Write $D_i = 2i, q_i = \sqrt{2i}$ and $h_i = \langle x, \omega_i \rangle = \sqrt{2}S_i$.

E.1 The strong-ratio test

An edge is strong if

$$\frac{\cosh L_i}{q_i} > T(q_i), \qquad T(q) = \frac{1 + (1 - .61/q)^2}{q(1 - (1 - .61/q)^2)} = \frac{100}{61} - \frac{q - .61}{q(2q - .61)}.$$

Its half-tangent then exceeds $1 - .61/q_i$. In particular $T(q) < 100/61$. For the small left layers, clearing positive denominators gives $T(q) < 38/25$ on $[5/2, 10/3]$, and $T(q) < 8/5$ on $[3, 4]$. The resulting quadratics are concave and positive at both endpoints.

E.2 Clearing all spectators from an isolated own

For an isolated own γ, retain all contacted spectator blocks, with spatial sum v_S and full cost Q. Put

$$c = Ap - \gamma\omega_i - v_S, \quad u = (1 - Q)/\gamma^2 > 0, \quad A^2 = \gamma^2(D_i - u).$$

If $u < 1$, the positive actual-normal combination

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has products zero on c, S, product $\gamma(u - 1) < 0$ on n_i, square $-\gamma^2 u(1 - u)$, and time $A > 0$. Inverse positivity expands v_S in actual spectator normals with nonnegative coefficients. Every unknown wall product is nonpositive. Its actual point gives

$$h_i > \sqrt{D_i - u} - \sqrt{u(1 - u)}.$$

At $u = 1$ it is a nonzero feasible future null. For $u > 1$, the positive finite projection on $S + n_i + c$, with residual $\gamma^2(u-1)$, gives

$$h_i > \sqrt{D_i - u} - \sqrt{u-1}.$$

Own $3, 4, 6$ retain an affine fork, the $4/3/3$ star, or the $6/3$ path; a parallel own has null $c + n_i$. Thus $\gamma \geq \varphi/2$. If $Q > 0$, $u \leq 3/\varphi^2 = 6 - 3\varphi < 7/6$. For $u > 1$ the loss is $< 1/5 + 2/5 < .61$. For $u < 1$ it is at most

$$u/6 + \sqrt{u(1-u)} \leq (1 + \sqrt{37})/12 < 3/5.$$

Every foreign isolated cap is strong.

At $Q = 0$, isolated own 5 has an actual H_4 correction on three consecutive old nodes, with $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$. Concavity bounds its corrected spatial product by γh_i; its loss is $< 25/76 + 5/24 < .61$. Own 7 has the negative A_2 core, of loss $< 400/1701 + 9/25 < .61$, using $.9 < \gamma < 10/11$. Finite own at least 8 has singleton pressure $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$, of loss $< 4/21 + 5/12 < .61$. For dotted own use $c + n_i/\gamma$ and $u = \gamma^{-2}$. Each correction is positive in actual roots. No projected mirror is declared native.

E.3 The complete finite-white lists

A left A_{i-1} permits one contact: endpoint 3, endpoint 4, next-to-end 3, or an admissible E_6, E_7, E_8 position 3. Their Schur residuals are

$$(i+1)/(2i), \quad 1/i, \quad 2/i, \quad 1/4,\ 1/7,\ 1/16.$$

For a simple contact at position j from the far end, $\sigma = \gamma + j/(2i)$; for endpoint 4, $\sigma = \gamma + j/(\sqrt{2}i)$. The actual edge ratio is $\sigma/\sqrt{\delta - Q}$. The right list is

$B_s,\ s \geq 4$	long-leaf 3
$D_s,\ s \geq 5$	long-leaf 3; fork 3 only for $s = 5, 6, 7$.

Two contacts to a retained component create a nonfinite cycle. A white bridge joining left and right has rank $r > 8$; classification forces finite type B or D, with simple endpoint attachments. A further branch at its internal new node is not finite. Exceptional fork pieces are retained with their full allowed foreign costs.

Zero own needs a separate test. Internal simple/doubled left contacts, right long contacts and full bridges retain affine forks, stars or cycles. Right D-fork contact retains affine E_8. For the separate far-3 case,

$$A^2 = Q - \tfrac{1}{2} < 1/(2i) \leq 1/10.$$

Any foreign contact has derivative square at least $1/4 > A^2$, forcing an orthogonal closure; at $Q = 0$ time square is negative. Zero own has therefore not been omitted.

For positive own, the complete direct-strength check uses the displayed residuals. Near simple-3 has $\delta \leq 3/5, \sigma \geq \gamma + 2/5$; next-to-end has $\delta = 2/i, \sigma \geq \gamma + 1/i$; E-rows have $\delta \leq 1/4$. For $\gamma > 101/125$, near simple-3 exceeds $100/61$ from $i = 8$, and next-to-end from $i = 7$. At $i = 5, 6, 7$, the smaller interval tests for T give the same conclusion. Own $3, 4, 6$ are affine except near-4/white-3, whose B cosine $(i-1) + i/\sqrt{2}$ is already strong for $i \geq 5$. In D, own 4 without right contact is affine on the actual right fork. White 4 is stronger, except far white 4/own 3, with affine $3, 4, 3, 3$ path.

Right long 3 has $\delta = 1/2 - Q, \sigma = \gamma + 1/2$. Own $0, 3$ is affine; the next contact has ratio at least $1 + 1/\sqrt{2} > 100/61$. For right fork 3 in D_s, $s = 5, 6, 7$,

$$\delta = 1 - s/8 - Q, \qquad \sigma = \gamma + 1/4.$$

Own $0,3,4,6$ retains affine E_8, a cycle, the $3,3,4,3$ path, or the $6,3$ path. The next own gives ratio $(\cos(\pi/5)+1/4)/\sqrt{3/8} > 1.72 > 100/61$. All finite foreign extensions of the E_6, E_7, E_8 piece remain included through Q. A full bridge has residual at most $1/(2i)$, numerator at least $\gamma + 1/2 + 1/(2i)$; positive own is strong, and zero own gives an affine cycle.

A far simple-3 cap with $Q > 0$ has

$$\cosh L_i = \frac{2i\gamma + 1}{\sqrt{i + 1 - 2iQ}}, \quad \cosh L_i/q_i > R_0(i) := \frac{(809/500)i + 1}{\sqrt{i(i+2)}}.$$

Both $R_0(i)$ and $T(\sqrt{2i})$ increase; the first derivative has sign $(309/500)i - 1 > 0$. The exact intervals cover $5 \le i \le 17$:

i	q_i upper	$R_0(i)$ lower	$T(q_i)$ upper
5–8	4	1.536	1.530
9–12	5	1.563	1.550
13–29	7.62	1.577	1.577.

The squared lower margins are $.052740, .321913, .546001$. The upper comparisons follow by multiplying the numerator/ denominator pairs $(27.4921, 18.0316), (44.2721, 28.6395)$, $(107.2045, 68.003166)$. Every foreign far cap is strong.

Consequently every weak cap has exactly

$$v = \gamma\omega_i + \omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad Q = 0, \quad 101/125 < \gamma < 6/5. \tag{44}$$

Its own is 5, finite at least 7, or dotted. For the upper cutoff, $\sigma^2/\delta \ge 4\gamma - 2$, with difference numerator $[2i(\gamma - 1) - 1]^2$. At $\gamma = 6/5$ this is $14/5 > (100/61)^2$.

E.4 Nonadjacency from one actual mixed point

For adjacent weak owns a, b, direct weighted AM–GM gives

$$\langle c_i, c_{i+1}\rangle \ge \tfrac{1}{2} - ab + \tfrac{1}{4}(a/b + b/a) \ge 1 - ab > -11/25 > -1/2.$$

The actual product must be zero. Clearing all old roots except $i, i+1$ gives the two-cap Schur matrix

$$H = I - t\mathbf{1}\mathbf{1}^T, \qquad t = (i-1)/(2i) > 0.$$

It is positive with positive inverse. Put $\lambda = H^{-1}(A_i, A_{i+1})^T > 0$ and $\mathcal{D} = A_i\lambda_i + A_{i+1}\lambda_{i+1}$. The actual positive projection has displacement $Z = W_i\omega_i + W_{i+1}\omega_{i+1}$, with

$$W_i = \lambda_i a + (\lambda_i + \lambda_{i+1})/(2i), \quad W_{i+1} = \lambda_{i+1} b, \quad Z^2 = \mathcal{D}(1 + \mathcal{D}).$$

Since $Q = 0$, all spectator blocks remain unchanged positive orthogonal blocks; positivity checks every unselected original wall. Write $W = W_i + W_{i+1}$, $A_+ = A_i + A_{i+1} > 4$. Then

$$\lambda_i + \lambda_{i+1} = iA_+, \quad \mathcal{D} = [iA_+^2 + (A_i - A_{i+1})^2]/2, \quad W > (4/5)iA_+.$$

It follows that

$$\frac{1 + \mathcal{D}}{W^2} < \frac{(i+1)/2 + 1/16}{(16/25)i^2} \le 49/256,$$

the last expression decreasing from $i = 5$. For $E_c = W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2}$,

$$E_c - \|Z\| = \frac{2W_iW_{i+1}(\sqrt{2i(2i+2)} - 2i)}{E_c + \|Z\|} < W/(4\sqrt{10}).$$

The actual point gives $\langle x, Z\rangle \ge 1 + \mathcal{D} - \sqrt{1 + \mathcal{D}}$, and $\|Z\| < 1 + \mathcal{D}$. The weighted mean loss is therefore less than $7/16 + 1/(4\sqrt{10}) < .61$, contradicting weakness at both sites. This proves nonadjacency before using any component-energy contradiction.

E.5 The four actual weak-five pressures

At $i = 5$, put $A^2 = 10\gamma^2+2\gamma-1/2$ and $H(A,\delta) = A/(1+\sqrt{\delta/(A^2+\delta)})$. The actual point pressures $\langle x, U\rangle \geq T$ are

own	U	T
5	$\gamma(\omega_4+\omega_6)/2+\omega_3/6$	$H(A, 2/3-\gamma^2)$
$7,8$	$\gamma(\omega_4+\omega_6)/2+\omega_2/4$	$A-\sqrt{\gamma^2-3/4}$
finite ≥ 9	$\gamma(\omega_4+\omega_6)/2+\omega_1/2$	$H(A, 1-\gamma^2)$
dotted	$(\gamma-\gamma^{-1})\omega_5+(\omega_4+\omega_6)/(2\gamma)+\omega_1/2$	$A-\sqrt{1-\gamma^{-2}}$.

The selected old sets are $\{5,1,2\}, \{5,1\}, \{5\}$, and the dotted pair. In the first three cases clear those roots with their positive inverse coefficients. For the dotted pair use $c+n_5/\gamma$. The selected products are zero or nonpositive and time is positive. All unknown actual-wall products are nonpositive by positivity; $Q = 0$ leaves pure component displacement. Finite points are normalized as in (4), not as spacelike corrections.

Concavity gives $\langle x, \omega_4+\omega_6\rangle \leq 2\langle x, \omega_5\rangle$, $\langle x, \omega_2\rangle \leq 2\langle x, \omega_1\rangle$, $\langle x, \omega_3\rangle \leq 3\langle x, \omega_1\rangle$. Weak 5 therefore implies

$$x_1 > \sqrt{2}\{T-\gamma(\sqrt{10}-.61)\} > .5965.$$

For own $5,7,8$, the lower bounds exceed $.8, .7, .65$, using $.809 < \cos(\pi/5) < .81$, $.9 < \cos(\pi/7) < .902$, $.923 < \cos(\pi/8) < .9239$. For finite owns at least 9, the deficit increases from $\gamma = .939$: $A' > \sqrt{10}, H_A > 9/10, H_\delta < 0$. At that endpoint $A > 3.1929, \sqrt{A^2+\delta} > 3.211, \sqrt{\delta} < .344$; thus $H > 2.88$, giving the lower bound $> .65$. For $1 < \gamma < 6/5$, the dotted deficit decreases: $A' < 13/4$, the radical derivative exceeds 1, and $\sqrt{10}-.61 > 5/2$. Its endpoint lower bound is

$$\sqrt{2}\{\sqrt{163/10}-\sqrt{11}/6-(6/5)(\sqrt{10}-.61)\} > .5965.$$

Positive square tests $\sqrt{163/10} > 4.03732, \sqrt{11}/6 < .552771$, $\sqrt{10} < 3.162278, \sqrt{2} > 1.41421$ prove it. Together with nonadjacency this supplies both initial branches of Proposition 4.1 without assuming strong 5.

F Fresh exact-18 classical layers

This appendix supplies the short layers used in the fresh rank-18 cover, at their actual ranks rather than by shifting a larger whole-component theorem. All prices and energies concern the original vector.

F.1 A local reference payment

The long-prefix inventory already reaches 12 in B_{16}, and 11 in D_{16}. Any actual strong endpoint K supplies the interpolation proof, in either initial-5 branch. Its endpoint sum is at most $f_K = \sqrt{K}-.61/\sqrt{2}$. Its individual terminal coefficient is $> 1/7$ for $K = 8, \ldots, 12$, and $> 7/50$ for $K = 13$. The ordinary coefficient is at least $\sqrt{K}-\sqrt{K-1}$; omitting the preceding knot increases it. The weak-5 reference ends with slope $1/(\sqrt{K}+\sqrt{6}) > 1/7$. The exact conservative data are

K	8	9	10	11	12	13
$q_K >$	.7683275	.797425	.823735	.8472225	.8689525	.88864
$f_K <$	2.398	2.570	2.732	2.886	3.034	3.175.

Here q_K is the certified effective reference energy; the entries follow by adding the positive rational increment floors and permitted omission losses in the prefix proof.

For $K = 8, \dots, 12$, if an actual point/core gives $\langle x, V\rangle \geq G$, V equals 1 on the first K coordinates, and $V^2 \leq R$, pay $\mu = 1/7$ at that individual terminal price. The residual $V - P_K$ has zero product with the reference and square at most $R - K$. One square gives

$$e_{\text{classical}} > q_K + 2(G - f_K)/7 - (R - K)/49. \tag{45}$$

The weak-5 interpolation has higher energy and no larger endpoint. No extra overlapping first-coordinate gain is added to it. At $K = 13$ use only a nonnegative weight μ within its actual individual capacity β_K; the corresponding formula is

$$e_{\text{classical}} > q_K + 2\mu(G - f_K) - \mu^2(R - K), \qquad 0 \leq \mu \leq \beta_K.$$

The other strong reference anchors retain positive coefficients, which supplies strictness. The $K = 13$ data above are used in the later B_{17} extension with its separately funded new slope $< .39/3 = .13 < 7/50$, not with weight $1/7$.

F.2 B_{16} with ordinary T_2

At $i = 13$, deletion retains $A_{12} + B_3$. The right finite extensions are long 3 or short 3, the latter giving F_4. Long 3 has $\sigma = \gamma + 1/2, \delta = 1/2 - Q$; own $0, 3$ is affine and larger owns are directly strong. Short 3 has $\sigma = \gamma + 1/(2\sqrt{2}), \delta = 1/4 - Q$. At zero own, the actual path $c, n_{16}, n_{15}, n_{14}, n_{13}$ has affine labels $3, 4, 3, 3$. Positive own has ratio at least $1 + 1/\sqrt{2}$. The F_4 white block has no finite foreign arm. No middle or doubled extra attachment to B_3 is finite. The full B_{16} bridge has internal c and is foreign-sealed; own zero is the cycle, and positive owns have the direct strong ratio. All left and isolated tests retain their already proved actual matrices and full spectator. Thus pure-helper coverage reaches 13. The interpolation proof now gives $e(B_{16}) > .8689525$, using the actual long neighbor 14. Adding ordinary $T_2 > 1/7$ contradicts the norm. This is a fresh local extension, not an out-of-domain application of Proposition 4.1.

F.3 D_{16} with ordinary T_2

At $i = 12$, the right white D_4 has three simple-3 leaf extensions: long leaf 13, and the distinct signed forks $15, 16$. Long 13 has $\sigma = \gamma + 1/2, \delta = 1/2 - Q$; its zero-own two-fork affine tree and own-3 triangle leave only directly strong cases. A fork 3 has $\sigma = \gamma + 1/4, \delta = 1/2 - Q$. Own $0, 3, 4, 6$ gives respectively affine E_8 arms $5, 2, 1$, the five-cycle, the $4/3/3$ star, or the $6/3$ path; parallel gives $c + n_{12}$. All remaining $Q > 0$ cases have ratio > 2.1; pure finite owns at least 7 or dotted have ratio $> 1.15\sqrt{2} > 25/16$. Only $Q = 0$, own 5/fork-3 remains. Full bridges have residual $1/24$; zero own is a cycle and positive owns are strong. Their internal new node cannot take a finite foreign branch.

For own 5, put $\gamma = \varphi/2$ and $\alpha^2 = \varphi + 5/8 > 56/25$. The actual left H_4 correction is $\gamma(3n_{12}/2 + n_{11} + n_{10}/2)$, with residual $\delta_5 = (5 - 3\varphi)/8$. The finite future point is the positive projection, not this spacelike correction. Its pure scaled displacement is flat through 9, with

$$A^2 = 12\alpha^2 - \tfrac{1}{2}, \quad R < 12, \quad G > \sqrt{12 - 25/112} - .091 > 3.34.$$

The radical bound is $\sqrt{\delta_5}/\alpha < .091$. It is feasible against both actual forks, the entire ordinary T_2 and every unknown wall by positive actual-root coefficients. Choose a strong $K = 8$ or 9. Equation (45), plus $1/7$, gives totals greater than

$$.7683275 + 2(3.34 - 2.398)/7 - 4/49 + 1/7 > 1.098,$$

or

$$.797425 + 2(3.34 - 2.570)/7 - 3/49 + 1/7 > 1.098.$$

Thus helper coverage reaches 12 before considering the next layer.

At $i = 13$, the right $D_3 = A_3$ list is middle 3, end 3, end 4, end 5. Middle 3 has the affine four-leaf star at zero own, the triangle at own 3, and direct strength otherwise. End 4 at zero own has the actual null

$$\sqrt{2}c + 2n_f + 2n_{14} + n_{13} + n_h,$$

where h is the other actual fork. Positive owns are directly strong. End 5 is the sealed finite H_4 white block, of ratio > 2.99 even at zero own. The only full bridges are middle-3/full D_{16}, and end-3/full A_{16}, with residuals $1/26$ and $1/8+1/26$. Zero own gives a cycle; positive owns are strong. At rank $16 > 8$, their internal node cannot take another finite branch.

For no-left end-3, $\sigma = \gamma + 1/4, \delta = 5/8 - Q$. The zero-own, own $3, 4, 6$ and parallel principals are the same E_8, four-cycle, doubled star, $6/3$ path and null. Positive foreign cases are strong, and $\gamma \geq 1$ has ratio $\geq \sqrt{5/2} > 25/16$. The actual strength threshold obeys $T(\sqrt{26}) < T(6) < 25/16$. Only $Q = 0$, own 5 or finite own at least 7, remains. Own 5 has its left H_4 point, now flat through 10, with

$$R < 13, \qquad G > \sqrt{13 - 125/448} - .091 > 3.47.$$

For finite $.9 < \gamma < 1$, use both actual negative A_2 cores, corrections $\gamma(4n_{13}/3 + 2n_{\text{side}}/3)$ for sides $12, 14$, and average their spatial pressures. With $\alpha = \sqrt{2}(\gamma + 1/4)$, $d = 4\gamma^2/3 - 1 > 0$, direct products give

$$A^2 = 13\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14\gamma^2/9 < 13\alpha^2, \quad \langle x, Y\rangle \geq A - \sqrt{d}.$$

The average is flat through 11. Both original cores satisfy every wall, including both forks and all spectators. Uniformly $G > \sqrt{13 - 125/529} - 1/3 > 3.23$, $R < 13$. Choose strong $K = 9$ or 10; even this weaker G, with ordinary T_2, gives totals > 1.047 in (45). All extra D_3 roles are excluded. Coverage reaches 13, so interpolation gives $e(D_{16}) > .8689525$, again contradicting its ordinary T_2. No typed defect is charged as ordinary.

F.4 B_{17} with a nonshort A_1

The generic inventory reaches 13. At 14, the preceding complete B_3 layer reaches 14, with the actual old long and short roots retained. Choose strong $K = 12$ or 13. At the right B_2 layer 15, the extra short 3 has its affine F_4 path at zero own; own $4, 6$ gives a doubled star or $6/3$ path; parallel gives a feasible null. Every own $\gamma \geq \cos(\pi/5)$ is strong. Positive foreign cost makes own 3 strong as well. Bridges are internal and foreign-sealed. The sole extra weak case is therefore

$$Q = 0, \qquad c = Ap - (\omega_{15} + \omega_{17})/2.$$

Releasing old 16 and retaining this actual c gives finite A_{17}: old A_{15}, c, and the true short old 17 form its chain. Positive inversion gives its actual future endpoint, feasible against released 16 and every other original wall. Its exact cosine and prefix consequence are

$$C_{16} = (16 + 15\sqrt{2})/\sqrt{18} > 8.7, \quad S_{16}^2 > 16(77/97) > (3.56)^2.$$

Append $S_{16} > 3.56$ to the strong $K = 12/13$ reference, skipping $14, 15$. New slopes $< .53/4, .39/3$ fit the individual capacities $> 1/7, > 7/50$. The weak-five endpoint differs from f_K by less than

.00001, so the same strict junction bounds apply. The two totals, including nonshort $A_1 > .1$, exceed

$$.8689525 + .526^2/4 + .1 > 1.038, \qquad .88864 + .385^2/3 + .1 > 1.038.$$

Thus this case is impossible. Pure-helper coverage reaches 15; interpolation gives $e(B_{17}) > .90715$, whose addition to .1 again contradicts 1. Actual I has not been assigned a nonshort debit, so $B_{17} + I$ remains for its separate consumer.

G The whole-A_{18} endpoint argument

Here the entire diagram at the original farthest vertex is A_{18}, with no spectators. Fix that vertex p and the original strict interior point z, and write

$$p^2 = -1, \qquad z = E(p - x), \qquad E > 0, \quad x \perp p, \quad \|x\|^2 < 1.$$

The spatial space is the zero-sum subspace of $\mathbb{R}^{19}$. Its unit incident normals and duals satisfy

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \qquad \langle \omega_i, \omega_j \rangle = \frac{2\min(i,j)(19 - \max(i,j))}{19}, \qquad D_0 = \omega_1^2 = \omega_{18}^2 = 36/19.$$

Interiority gives $x_1 > \cdots > x_{19}$. Let the actual endpoint opposites be the walls terminating the edges obtained by releasing n_1 and n_{18}. Compactness makes these endpoints finite. For each opposite c, its positive time is $A = -\langle p, c \rangle > 0$. We retain every original wall throughout. Reversing the coordinate formulas gives their right-end versions; no chamber symmetry or arithmeticity is assumed. Distinctness of the two opposites, when their mutual native gap is used, will follow from their actual contacts.

G.1 The reference under the full norm bound

The $N = 19$ central inventory and helper nonadjacency above precede any contrary energy assumption. Its near-four roles can also be excluded under only $\|x\| < 1$, using their sharper actual mates. For a left near-four at $i = 5, 6$, release $j = i - 1$. The positive replacement has diagram $B_{20-i} + A_{i-2}$ and cosine

$$C_j = \sqrt{2(i-1)/19}((20 - i)/2 + (19 - i)/\sqrt{2}) > 11.$$

Its normalized height exceeds $\sqrt{5/6} > .91$. At the other end a mate or a strong 13/14 has demand $> 209/270 > .774$. Their correlation is at most $\sqrt{15/91} < .41$. Cauchy contradicts $1.684^2 = 2.835856 > 2.82$. Thus all central weak sites are nonadjacent pure helpers already under the full norm bound.

Use the same $N = 19$ rational heights and chord omissions. If k is the number of weak end sites $5, 14$, one reference w has energy

$$E_0 \geq .9240352, \qquad E_1 > .8937, \qquad E_2 > .86414.$$

Its left and right block lengths m are 5 or 6, with signed means

$$b_5 = H_5/(5\sqrt{2}) \in (.29, .298), \qquad b_6 = H_6/(6\sqrt{2}) \in (.25, .266).$$

The separate boundary dual prices satisfy

$$a_5 \geq (.4208 - .151)/2 = .1349, \qquad a_6 \geq (451/1200 - .108)/2 > .1339.$$

Every omission is included and can only increase the adjacent available price.

For an actual endpoint pressure $\langle x, V\rangle \geq K$, subtract $B = \sum b_i\omega_i$, with $b_i \geq 0$ supported on the retained boundary duals, so that $Y = V - B$ has zero mean on each end block. Write $w = \sum a_i\omega_i$, with $a_i \geq 0$. Every retained height satisfies $\langle x, \omega_i\rangle \geq H_i = \langle w, \omega_i\rangle$, and $\langle w, Y\rangle = 0$. For $\mu \geq 0$, the individual conditions $\mu b_i \leq a_i$ give

$$\langle x, w + \mu Y\rangle = \sum_i (a_i - \mu b_i)\langle x, \omega_i\rangle + \mu\langle x, V\rangle \geq w^2 + \mu(K - \langle w, B\rangle).$$

Applying $\|x - (w + \mu Y)\|^2 \geq 0$ therefore gives the reference energy plus the gain

$$2\mu(K - \langle w, B\rangle) - \mu^2 Y^2.$$

Opposite-block residuals have zero cross. For a coordinate pressure $x_{\text{end}} \geq t$, the legally optimized reference gain is $m(t - b_m)^2/(m-1)$. For all thresholds used below, the debit is less than .061, below each separate price. If either coordinate were $\geq .603$, the three energy bounds plus this gain would exceed 1; the tight case is

$$.86414 + (6/5)(.603 - .266)^2 = 1.0004228.$$

Therefore $x_1, -x_{19} < .603$. The genuine endpoint-edge inequality and $.603^2 < 90/247 = (18/19)(5/13)$ imply

$$C_L, C_R < 9/4. \tag{46}$$

This bounds actual edge cosines, not cap owns.

G.2 Complete local isolated-endpoint gains

An isolated left endpoint has $c = Ap - g\omega_1,\ A^2 = D_0 g^2 - 1 > 0$. Own zero, $3, 4$ has nonpositive time; own 6 and parallel have the positive actual nulls $\sqrt{3}c + 2n_1 + n_2$ and $c + n_1$. The full remaining list is finite 5, finite 7, finite at least 8, or dotted. For a pressure normalized to raw coordinate sum $\sqrt{2}$, subtract ω_m/m; the residual is zero-sum on the relevant block. Every price below is paid before applying its actual pressure. The resulting complete gain table is

own	μ	gain on $m = 5 >$	gain on $m = 6 >$
5	.6	.02904	.059
7	.4	.05152	.07677
finite ≥ 8	1/4	.0471	157/2400
dotted	1/4	.06	4699/60000.

We prove every row, including continuous dotted owns and the unbounded finite-label tail.

For dotted $g > 1$, put $u = g^{-2} \in (0, 1)$. The actual positive core $c + n_1/g$ gives

$$V = \omega_1 - un_1, \quad K(u) = \sqrt{D_0 - u} - \sqrt{u(1-u)}, \quad Y^2 = 2 - 2/m - 2u + u^2.$$

It has square $u - 1 < 0$, positive time, zero cap product and negative old-1 product; positivity checks every unknown wall. At $\mu = 1/4$ the gain is

$$G_m(u) = \tfrac{1}{2}(K(u) - H_m/m) - (2 - 2/m - 2u + u^2)/16.$$

Its shift is exactly $G_6 - G_5 = 1099/60000$. To prove $G_5 > .06$ uniformly, it suffices that

$$K(u) > .7408 - u/4 + u^2/8.$$

The upper bound $\sqrt{u(1-u)} \le 1/2 - (u-1/2)^2$ follows by positive squaring, with residual $(u-1/2)^4$. Use lower radical values $1.376, 1.1809, 1.0699, .9459$ at $u = 0, 1/2, 3/4, 1$; their squares are $1.893376, 1.39452481, 1.14468601, .89472681$, strictly below $D_0 - u$. Concavity places $\sqrt{D_0 - u}$ above the successive chords. The three resulting strict quadratic lower bounds are

$$\begin{array}{c|l} [0,1/2] & .3852 - 1.1402u + (7/8)u^2 \\ [1/2,3/4] & (7/8)(u - 597/875)^2 + 8357/1750000 \\ [3/4,1] & .4511 - 1.246u + (7/8)u^2. \end{array}$$

The first decreases to $677/20000 > 0$; the last increases from $703/80000 > 0$. This proves the entire dotted row.

For finite $g < 1$, put $u = g^{-2}$, $d = D_0 - 1 = 17/19$. The actual finite projection onto c, n_1 gives

$$V = \omega_1 - n_1, \quad K_f = \frac{d - \sqrt{d(u-1)}}{\sqrt{D_0 - u}}, \quad Y^2 = 1 - 2/m.$$

Writing $s = \sqrt{u-1}$, $K_f^2 = d(\sqrt{d} - s)/(\sqrt{d} + s)$, increasing with g. For every native own at least 8, $s \le \sqrt{2} - 1 < .414214$. Using $.8947 < d < .8948$, $\sqrt{d} > .9459$,

$$(d - .59^2)\sqrt{d} > .51702894, \qquad (d + .59^2)s < .5148265806.$$

Hence $K_f > .59$. Quarter-payment gives $G_{f,5} > .0471$ and $G_{f,6} > .0471 + 1099/60000 = 157/2400$.

For native 5, let $u = 6 - 2\sqrt{5}$ and $b = D_0 - 3/2 = 15/38$. Its actual H_4 correction $\eta = c + g(3n_1/2 + n_2 + n_3/2)$ has square $\delta = 1 - 3g^2/2 > 0$. Use the actual finite point $p + (A/\delta)\eta$, not η. The pressure is

$$V = \omega_4/4, \qquad K_5 = \frac{b - \sqrt{b(u - 3/2)}}{\sqrt{D_0 - u}}, \qquad Y^2 = 1/2 - 2/m.$$

For $s = \sqrt{u - 3/2} < .167$, $.3947 < b < .3948$, $\sqrt{b} > .6282$ give

$$(b - .475^2)\sqrt{b} > .106212915, \qquad (b + .475^2)s < .103610975.$$

Thus $K_5 > .475$. The payment $.6/m \le .12$ fits each boundary coefficient and gives

$$G_{5,5} > 1.2(.475 - .4208) - .36(.1) = .02904, \qquad G_{5,6} > 1.2(.475 - 451/1200) - .36/6 = .059.$$

For native 7, $.900968 < g < .900969$ and $1.231 < u = g^{-2} < 1.232$. The positive actual core $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$, divided by g, gives

$$V = \omega_1 - un_1 - 2(u-1)n_2, \quad K_7 = \sqrt{D_0 - u} - \sqrt{7u - 3u^2 - 4},$$

$$Y^2 = 6 - 8u + 3u^2 - 2/m.$$

Its selected products are zero on c, n_1, negative on n_2, n_3, and nonpositive on every unknown wall; time is positive. On the stated interval, $D_0 - u > .814^2$ and $0 < 7u - 3u^2 - 4 < .070917 < .2664^2$, so $K_7 > .545$. The residual bounds are $Y_5^2 < .299$, $Y_6^2 < .366$. Payment $.4/m \le .08$ gives gains

$$.8(.545 - .4208) - .16(.299) = .05152, \qquad .8(.545 - 451/1200) - .16(.366) > .07677.$$

This proves the table and all its actual-wall and individual-price hypotheses.

G.3 The complete two-finite endpoint exclusion

Suppose both isolated endpoint owns are finite. For $D = 36/19, e = 2/19$, their actual mutual magnitude is

$$M(g,h) = \sqrt{Dg^2-1}\sqrt{Dh^2-1} - egh.$$

It increases strictly in each own for $g, h \geq g_5 = \cos(\pi/5)$. For example the derivative sign follows by squaring positive terms from $g/\sqrt{Dg^2-1} > 1/\sqrt{D}$ and $(D^2-e^2)h^2 > D$, since $D^2-e^2 = 68/19$ and $g_5^2 > 9/17$.

If either own is 5, monotonicity gives

$$0 < \frac{17\sqrt{5}-25}{76} = M(g_5,g_5) \leq M(g_5,h) < M(g_5,1) < \sqrt{(9\sqrt{5}-11)/38} < 1/2.$$

That is forbidden for distinct actual walls. Thus both own labels m, n are at least 7. Relabel so $m \leq n$. Now $M > 0$ already at $m = n = 7$, while $M < M(1,1) = 15/19 < \cos(\pi/5)$. The full mutual alphabet therefore leaves exactly native 3 or native 4. Put

$$H(g,h) = 272g^2h^2 - 144(g^2+h^2) - 8gh + 57,$$

$$F_S(g,h) = 136g^2h^2 - 72(g^2+h^2) + 19 - 4Sgh.$$

They have the signs of $M - 1/2$ and $M - 1/\sqrt{2}$, respectively, with $S = \sqrt{2}$. This follows by the difference of squares against egh plus the target; the other factor is positive. Both increase in each argument on $[.9,1]^2$, for $1.4 \leq S \leq 1.5$, and F_S decreases with S.

For mutual 3, $m \geq 8$ gives

$$M(g_8,g_8) = (17\sqrt{2}-4)/38 > 1/2.$$

At $m = 7$ the exact rational native enclosures below give

$$M(g_7,g_8) < 1/2 < M(g_7,g_9).$$

At the upper 7/8 pair the bounds $u < .8117452, v < .8535543$, $u+v > 1.6652993$, $uv < .692869, gh > .832387$ yield $H < -10018272/10^7 < 0$. At the lower 7/9 pair, $u > .8117433, v > .8830210$, $u+v < 1.6947645$, $uv > .716786, gh < .846633$ yield $H > 1146640/10^6 > 0$. Thus no integer $n \geq m$ gives mutual 3.

For mutual 4, if $m \leq 10$, increase g to the upper $g_{10}^+ = .951057$ and h to 1. Since $(g_{10}^+)^2 < .9045095$ and $4(1.414213)g_{10}^+ > 5.37997$,

$$F_{\sqrt{2}}(g,h) \leq F_{1.414213}(g_{10}^+,1) < -.491362 < 0.$$

If $m \geq 15$, monotonicity gives

$$M \geq (34/19)\cos^2(\pi/15) - 1 > 1/\sqrt{2}.$$

Use $\cos(\pi/15) > 10783/11025$, from $\cos x > 1 - x^2/2, \pi < 22/7$, and $\sqrt{2} < 99/70$. The rational remaining margin is

$$(34/19)(10783/11025)^2 - 1 - 99/140 = 1498515515/323324662500 > 0.$$

Only $m = 11, 12, 13, 14$ remain. Their complete consecutive-integer gaps are

m	last n below target	first n above target
11	31	32
12	20	21
13	16	17
14	15	16.

Here is full rational arithmetic for these gaps. For $g = g_m, S = \sqrt{2}$, set

$$A = 136g^2 - 72, \quad B = 4Sg, \quad C = 64g^2 - 53 - B, \quad T = 2A - B.$$

At $h = 1 - \delta$, $F_S(g, h) = C - T\delta + A\delta^2$. The native enclosures give

m	g^2 lower	g^2 upper	B lower	B upper
11	.9206248	.9206269	5.42770	5.42773
12	.9330110	.9330131	5.46409	5.46412
13	.9427264	.9427284	5.49246	5.49249
14	.9504826	.9504847	5.51501	5.51504.

Thus the following looser bounds hold:

m	C_-	C_+	T_-	T_+	A_+
11	.49225	.49243	100.981	100.984	53.2053
12	1.24858	1.24875	104.314	104.317	54.8898
13	1.84199	1.84216	106.928	106.931	56.2111
14	2.31584	2.31602	109.016	109.018	57.2660.

For last-below n, use its upper cosine and $S = 1.414213$. The δ numerators over 10^6 are 5130, $12311, 19214, 21852$; upper numerators of $A\delta^2$ are $1401, 8400, 20800, 27374$, also over 10^6. The exact resulting upper numerators over 10^9 are

$$-24201530, \ -27059654, \ -191554592, \ -38823632.$$

For first-above n, use its lower cosine and $S = 1.414214$. The δ numerators are $4816, 11170, 17027$, 19215; dropping the positive quadratic term gives lower numerators over 10^9

$$5911056, \ 83359110, \ 21275863, \ 221059130.$$

All signs are strict. Monotonicity in g, h proves every integer $n \geq m$ lies on one side of the stated gap, so mutual 4 is impossible as well. This proves the two-finite exclusion.

G.4 The finite rational angle certificate

For completeness, the only native enclosures used in the preceding finite calculation are:

j	$10^6 g_j^-$	$10^6 g_j^+$	j	$10^6 g_j^-$	$10^6 g_j^+$
7	900968	900969	8	923879	923880
9	939692	939693	10	951056	951057
11	959492	959493	12	965925	965926
13	970941	970942	14	974927	974928
15	978147	978148	16	980785	980786
17	982973	982974	20	987688	987689
21	988830	988831	31	994869	994870
32	995184	995185			

These are exact rational inequalities, not floating-point evaluations. Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$ follows by the tangent addition formula and its first-quadrant sign. For

$$A_s = \sum_{r=0}^{s} \frac{(-1)^r}{(2r+1)5^{2r+1}}, \qquad B_s = \sum_{r=0}^{s} \frac{(-1)^r}{(2r+1)239^{2r+1}},$$

alternating remainders and rational cross multiplication give

$$31415926/10^7 < 16A_5 - 4B_2 < \pi < 16A_6 - 4B_1 < 31415927/10^7.$$

For $0 < x < .45$, the alternating cosine Taylor sums give $C_{10}(x) < \cos x < C_8(x)$. Put $a_j = 10^7 j, p_- = 31415926, p_+ = 31415927$, and

$$Q_{10}(a,p) = 3628800a^{10} - 1814400p^2a^8 + 151200p^4a^6 - 5040p^6a^4 + 90p^8a^2 - p^{10}, \qquad Q_8 = Q_{10} + p^{10}.$$

For every displayed row with integers L_j, U_j, the finite exact certificate is

$$10^6 Q_{10}(a_j, p_+) - 3628800L_j a_j^{10} > 0,$$

$$3628800U_j a_j^{10} - 10^6 Q_8(a_j, p_-) > 0.$$

There are exactly thirty integer inequalities; their positivity is obtained by direct integer multiplication of the specified finite expressions. The alternating remainder signs and cosine monotonicity then prove all fifteen rows. This is a terminating exact verification recipe with a complete stated output; no execution of a checker is claimed by this manuscript transcription. The historical unused cyclotomic fallback is not an input. Finally $1.414213 < \sqrt{2} < 1.414214$ follows by squaring those two rationals.

G.5 The complete both-isolated endpoint conclusion

The local isolated gains, with their separate prices and disjoint residual supports, exclude two dotted endpoints: the three reference-plus-gain lower bounds are

$$.9240352 + 2(.06) > 1, \quad .8937 + .06 + 4699/60000 > 1, \quad .86414 + 2(4699/60000) > 1.$$

For one finite and one dotted endpoint, the full four-pattern totals are

finite own	5/5	6/5	5/6	6/6
≥ 8	1.0311352	> 1.01911666	> 1.01911666	> 1.00787333
5	1.0130752	1.0127	> 1.00105666	> 1.00145666
7	1.0355552	> 1.03047	> 1.02353666	$> 1.01922666.$

In the middle two columns the block order is finite/dotted. The gains add within one square because their residuals occupy opposite zero-sum blocks. Together with the two-finite theorem, these exhaust every isolated endpoint own. Thus both actual endpoint opposites cannot be isolated.

G.6 The complete actual endpoint inventory

We now classify an endpoint opposite before using its own contact. Release n_1. The other endpoint is finite, so the actual Gram on $c, n_2, \dots, n_{18}$ is positive definite. The retained diagram is the connected A_{17} path. A one-node finite enlargement is either disconnected or of type A, B, D: exceptional spherical types have rank less than eighteen. Thus the new node is isolated, or has one simple bond to an end or a next-to-end node, or one bond of label four to an end. Two white contacts would create a cycle and cannot occur in a finite diagram. This gives the six nonisolated placements below, without imposing a condition on an arbitrary face.

Write a nonisolated opposite as

$$c = Ap - g\omega_1 - \eta\omega_j, \qquad g = -\langle c, n_1\rangle \geq 0,$$

where $\eta = 1/2$ for a simple white bond and $\eta = 1/\sqrt{2}$ for a white bond of label four. Put $C_{1j} = \langle \omega_1, \omega_j\rangle$ and $D_j = \omega_j^2$. The actual norm and retained projection give

$$A^2 = D_0 g^2 + 2g\eta C_{1j} + \eta^2 D_j - 1, \qquad \delta_{\text{white}} = 1 - \frac{2\eta^2(j-1)(19-j)}{18} > 0.$$

The edge cosine is

$$C = \frac{D_0 g + \eta C_{1j}}{\sqrt{D_0 \delta_{\text{white}}}}.$$

For a simple end bond the retained projection cost is $17/36$ and the residual is $19/36$; they must not be interchanged. The next-to-end and label-four residuals are $1/9$ and $1/18$. Substituting in the displayed formula and using (46) yields

white contact	C	consequence
$j = 2,\ 3$	$(36g + 17)/19$	$g < 103/144$: own $0, 3, 4$
$j = 3,\ 3$	$(36g + 16)/(2\sqrt{19})$	$g < .101$: own 0
$j = 17,\ 3$	$(36g + 2)/(2\sqrt{19})$	$g < .49$: own 0
$j = 18,\ 3$	$(36g + 1)/19$	$g < 167/144 < 7/6$
$j = 2,\ 4$	$(36g + 17\sqrt{2})/\sqrt{38}$	$C \geq 17/\sqrt{19} > 9/4$
$j = 18,\ 4$	$(36g + \sqrt{2})/\sqrt{38}$	$g < .35$: own 0.

The numerical cuts follow by positive squaring from $\sqrt{19} < 4.36, \sqrt{38} < 6.165, \sqrt{2} > 1.414$. A positive own is at least $1/2$; finite owns are $\cos(\pi/m)$, while parallel and dotted owns are respectively one and greater than one.

For near simple-three with own zero, $A^2 = -2/19$; own three has the actual positive null sum $c + n_1 + n_2$. For the next-to-near simple-three with own zero, the actual affine E_8 principal has center n_3, arms

$$(c), \qquad (n_2, n_1), \qquad (n_4, n_5, n_6, n_7, n_8),$$

and positive kernel coefficients

$$6; \qquad 3; \qquad 4, 2; \qquad 5, 4, 3, 2, 1.$$

Each row of its unit Gram annihilates this vector. For the next-to-far simple-three and far label-four with own zero, $A^2 = -2/19$ and $-1/19$, respectively.

For far simple-three, own zero and own three have $A^2 = -10/19$ and zero. Own four has the actual affine path

$$(n_{17}, n_{18}, c, n_1, n_2), \qquad \text{bonds } (3, 3, 4, 3),$$

with positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. Own six has the null sum $\sqrt{3}c+2n_1+n_2$; parallel has $c+n_1$. The same nulls exclude isolated own six and parallel, while isolated owns zero, three and four have negative time square.

Every positive null combination just used contains its actual opposite with a positive coefficient. Its time is consequently positive, so it is a nonzero future null vector. Against any unselected original wall its product is nonpositive by acuteness; against its selected walls the kernel equations give zero. It therefore determines an ideal point of the whole chamber, contradicting compactness. This checks all original walls, including those absent from the selected diagram.

The complete surviving list at either end is therefore:

near:	simple-three at the neighboring node, own four;
far:	simple-three at the opposite end, own five, finite own at least seven, or dotted;
isolated:	own five, own seven, finite own at least eight, or dotted.

In every far case $g < 167/144 < 7/6$. The two copies of this classification concern the actual endpoint walls of the original vertex.

G.7 Removing the near endpoint and the two-far possibility

The remaining near role has

$$C = (18\sqrt{2} + 17)/19, \qquad x_{\text{end}}^2 \geq \frac{18}{19}\frac{18\sqrt{2} - 2}{18\sqrt{2} + 36} > .601^2.$$

After clearing positive denominators, the last inequality is $200.469258\sqrt{2} - 283.061484 > 0$; its lower bound at $\sqrt{2} > 1.4142$ is .4421406636. Thus $x_{\text{end}} > .601$. A strong end gives coordinate gain $(5/4)(.601 - .298)^2 = .11476125$; a weak end gives gain greater than $(6/5)(.601 - .266)^2 = .13467$. Either removes the zero-weak or one-weak reference pattern: even the smaller one-weak total is $.8937 + .11476125 > 1$.

Only two weak ends could remain. A far opposite at the other end has $g \geq 1/\sqrt{2}$, a weaker bound than its actual surviving own list. Its edge inequality gives $x_{\text{other}} > .394$. The cleared comparison is

$$270.909288\sqrt{2} - 382.98968 > 0;$$

at 1.4142 the margin is .1302350896. Its weak-block gain is therefore greater than $(6/5)(.394 - .266)^2 = .0196608$. An isolated opposite gives weak gain greater than .059 by the complete local table; another near opposite gives the larger near gain. These opposite residuals have disjoint support, zero cross, and separately paid nonnegative coefficients. The weakest combined total is

$$.86414 + .13467 + .0196608 = 1.0184708 > 1.$$

Thus no near role survives, on either orientation.

Suppose next that both endpoints are far, and write

$$c = Ap - g\omega_1 - \omega_{18}/2, \qquad d = Bp - h\omega_{18} - \omega_1/2.$$

Their time squares are

$$A^2 = (36g^2 + 2g - 10)/19, \qquad B^2 = (36h^2 + 2h - 10)/19.$$

For $0 < g < 7/6$,

$$\frac{8}{5}g^2 - A^2 = \frac{50 - 10g - 28g^2}{95} > \frac{2}{855} > 0,$$

because the numerator decreases up to $7/6$. The same holds for h. Their spatial product satisfies

$$\frac{2gh+18g+18h+1/2}{19} > \frac{18(g+h)}{19} > \frac{216gh}{133} > \frac{8}{5}gh > AB.$$

Here $1/g+1/h > 12/7$ and $1080 > 1064$. The two caps are distinct: coincidence of their old contact rows would force $g = h = 1/2$, for which both time squares vanish. Their positive mutual product contradicts actual acuteness.

The preceding both-isolated theorem and the endpoint inventory now leave exactly one far endpoint and one isolated endpoint. Reflect the coordinate formulas if necessary so the far cap is on the left. An isolated own five is impossible for every far own. Indeed, with $u_5 = 6 - 2\sqrt{5}$,

$$D_0 - u_5 = 2\sqrt{5} - 78/19 < 178/475 < 3/8, \qquad A^2 < \frac{8}{5}(7/6)^2 = 98/45.$$

Thus $A^2(D_0 - u_5) < 49/60 < (18/19)^2$. For an isolated right cap of own σ, its time is $\sigma\sqrt{D_0 - u_5}$, whereas the spatial product divided by σ is $(18 + 2g)/19 > 18/19$. The actual cap product would be positive. The supports show these caps are distinct, so this too contradicts acuteness.

G.8 Uniform removal of the far finite and dotted tail

A far own at least seven or a dotted own has $g > .9$. Its actual length consequently gives

$$C > (167/95), \qquad x_{\mathrm{far}}^2 > 648/2489 > .510^2.$$

The last positive rational margin is $648 - 2489(.2601) = .6111$. The legal coordinate gains on the strong and weak blocks are

$$G_{\mathrm{far},5} > .05618, \qquad G_{\mathrm{far},6} > .0714432.$$

The other endpoint is isolated with own seven, finite own at least eight, or dotted. The minimum over the entire local table is

$$G_{\mathrm{iso},5} > .0471, \qquad G_{\mathrm{iso},6} > 157/2400.$$

In particular the weak minimum must not be replaced by the larger own-seven estimate. The two individually funded residuals belong to opposite blocks, so their gains add in one square. All four possibilities give

$(m_{\mathrm{far}}, m_{\mathrm{iso}})$	strict lower bound for $\|x\|^2$
$(5,5)$	$.9240352 + .05618 + .0471 = 1.0273152$
$(6,5)$	$.8937 + .0714432 + .0471 = 1.0122432$
$(5,6)$	$.8937 + .05618 + 157/2400 > 1.01529666$
$(6,6)$	$.86414 + .0714432 + 157/2400 > 1.00099986.$

This removes all these far owns, including arbitrary finite labels and the continuous dotted interval.

G.9 The last mutual gap and an actual orthogonal projection

Only far own five remains. Set

$$\phi = (1+\sqrt{5})/2, \quad g = \phi/2,$$
$$A^2 = (10\phi - 1)/19, \quad L = (18+\phi)/19,$$
$$\delta = 1 - 3g^2/2 = (5 - 3\phi)/8 > 0.$$

Substitution in $t^2 - t - 1$ gives $1.618 < \phi < 1.619$. Write the isolated right cap as $d = A_I p - \sigma\omega_{18}$, with $u = \sigma^{-2}$. If it were finite, $u > 1$ and

$$A^2(D_0 - u) < .8(.9) = .72 < 1 < L^2.$$

Its mutual product with the far cap would be positive. Thus $\sigma > 1$ is dotted and $0 < u < 1$. The actual isolated edge has $C_{\text{iso}} = \sqrt{D_0}\sigma < 9/4$, so $u > 64/171$.

The two caps are distinct because the far cap has a positive old contact at n_1 and the isolated right cap has none. Their actual mutual product is

$$\langle c, d\rangle = \frac{L - A\sqrt{D_0 - u}}{\sqrt{u}} \le 0.$$

Since $A < .9, L > 1, D_0 - u < 260/171 < 25/16$, and $\sqrt{u} > 3/5$, this product is strictly greater than $-5/24$. The native alphabet for distinct actual walls has no value in $(-1/2, 0)$. Therefore

$$\langle c, d\rangle = 0, \qquad AA_I = \sigma L, \qquad u = D_0 - \frac{L^2}{A^2} = \frac{323\phi - 361}{19(10\phi - 1)}. \tag{47}$$

Orthogonality is proved here before it is used. The rational function has derivative with positive numerator 62453. At $1.618, 1.619$ its values are $80807/144210$ and $161937/288610$. The positive comparisons

$$25(80807) - 14(144210) = 1235, \qquad 281(288610) - 500(161937) = 130910$$

give

$$.560 < u < .562. \tag{48}$$

These are geometric identities; no integrality or field restriction is being inferred from them.

Consider the positive actual correction

$$\eta = c + g(3n_1/2 + n_2 + n_3/2) = Ap - g\omega_4/4 - \omega_{18}/2.$$

Direct products give

$$\langle \eta, n_1\rangle = \langle \eta, n_2\rangle = \langle \eta, n_3\rangle = 0, \qquad \eta^2 = \langle \eta, c\rangle = \delta > 0.$$

Thus η is spacelike. The actual H_4 block c, n_1, n_2, n_3 is positive definite, its Schur complement being δ. Equation (47) and the right cap's isolation make d orthogonal to this block.

The required chamber point is the projection

$$q_0 = p + (A/\delta)\eta + A_I d, \qquad H = A^2/\delta + A_I^2, \qquad q_0^2 = \langle p, q_0\rangle = -1 - H.$$

It has zero product with every selected wall. Every correction coefficient on an actual normal is positive; against every unselected original wall, acuteness and $\langle p, a\rangle \le 0$ therefore give $\langle q_0, a\rangle \le 0$. This includes n_{18} and all unknown facets. Also $\langle p, q_0\rangle < 0$, so q_0 is future timelike. Its normalization is a point of the whole chamber. The spacelike vector η itself was never used as a chamber point.

Put

$$a = \phi^{-1} + \frac{\delta L}{A^2 g u} > 0, \qquad V = \omega_4/4 + a\omega_{18}, \qquad U = A^2 + \delta D_0/u.$$

Using $AA_I = \sigma L$, the point has the full decomposition

$$q_0 = (1 + H)p - (Ag/\delta)V, \qquad \delta(1 + H) = U.$$

Compare its normalization with the same original farthest vertex:

$$-\langle z, q_0/\sqrt{1+H}\rangle \le -\langle z, p\rangle = E.$$

Substitution of $z = E(p-x)$ gives the actual pressure

$$\langle x, V\rangle \ge K, \qquad K = \frac{\delta(1+H-\sqrt{1+H})}{Ag} = \frac{U-\sqrt{\delta U}}{Ag}. \tag{49}$$

No auxiliary point has been declared farthest.

All subsequent estimates have positive denominators. The enclosure of ϕ and (48) give

$$.7989 < A^2 < .7995, \qquad .0178 < \delta < .0183,$$
$$.809 < g < .8095, \quad A < .895, \quad L < 1.033.$$

For the lower estimate retain $D_0 = 36/19$ exactly:

$$\frac{\delta D_0}{u} > \frac{.0178(36/19)}{.562} = \frac{3204}{53390} > .06,$$

where $50(3204) - 3(53390) = 30 > 0$. The upper bound is $\delta D_0/u < .0625$, by positive cross multiplication. Consequently

$$.8589 < U < .862, \quad \sqrt{\delta U} < .126, \quad Ag < .7245025, \quad K > \frac{.7329}{.7245025} > 1.01.$$

For the radical comparison, $.0183(.862) = .0157746 < .126^2$, and $1.01(.7245025) = .731747525 < .7329$. Finally,

$$\delta L < .0189039 < .06(.7989)(.809)(.560) = .02171601936.$$

Since $\phi^{-1} < .619$, this proves $0 < a < .679 < .68$.

G.10 One final funded vector for every end pattern

For the pressure (49), subtract

$$B = \omega_{m_L}/m_L + a\omega_{19-m_R}/m_R, \qquad Y = V - B.$$

The left residual is $\omega_4/4 - \omega_{m_L}/m_L$; the right is a times the reflected $\omega_1 - \omega_{m_R}/m_R$. They are supported on the first m_L and last m_R coordinates, have zero sum on each block, and have zero cross. Their full squared norm is exactly

$$Y^2 = \frac{1}{2} - \frac{2}{m_L} + a^2\left(2 - \frac{2}{m_R}\right).$$

In particular the nonzero cross between the original endpoint duals has disappeared only after the proper block subtractions.

Use the single nonnegative weight $\mu = .4$. It costs $.4/m_L \le .08$ at the left anchor and $.4a/m_R < .0544$ at the right, each below its own available coefficient .1349 or $> .1339$. The all-hole reference and the individual-price hypotheses of the completed-square rule are therefore satisfied simultaneously. Write $\ell_5 = H_5/5 = .4208$ and $\ell_6 = H_6/6 = 451/1200$. The gain is strictly greater than

$$.8(1.01 - \ell_{m_L} - .68\ell_{m_R}) - .16\left[\frac{1}{2} - \frac{2}{m_L} + .68^2\left(2 - \frac{2}{m_R}\right)\right].$$

Replacing a by its upper bound lowers the linear deficit and increases the norm penalty. The four rational budgets are

(m_L, m_R)	gain lower bound	strict lower bound for $\|x\|^2$
$(5,5)$	.1080704	1.0321056
$(6,5)$	$>$ .13337706	$>$ 1.02707706
$(5,6)$	.1276	$>$ 1.0213
$(6,6)$	$>$ .15290666	$>$ 1.01704666.

For example the last bracket equals 703/750 and its linear deficit is .3786; the unrounded gain is $.8(.3786) - .16(703/750)$. Every pattern contradicts $\|x\|^2 < 1$. One original vector, one reference and one set of individually available coefficients were used throughout.

Proposition G.1. *An actual original farthest vertex of a compact hyperbolic Coxeter polytope in dimension eighteen cannot have entire incident diagram A_{18}.*

Proof. The finite-white inventory leaves only near, far or isolated endpoint opposites. The near role is impossible on either side. Both isolated endpoints are excluded by their complete local and finite-pair arguments; both far endpoints violate acuteness. Thus precisely one endpoint is far and the other isolated. The uniform coordinate/local-gain argument removes every far own at least seven and every dotted far own. In the remaining far-own-five case, every finite isolated opposite violates acuteness. The dotted isolated opposite has mutual product zero by the actual native gap, and the actual $H_4 + A_1$ point gives the final simultaneous-payment contradiction above. These alternatives exhaust the actual endpoint walls, without arithmeticity, a finite-label cutoff, arbitrary-face heredity, or a rank decrement. □

H Terminal walls in a whole B_{18} diagram

Fix the original strict interior unit point z and an original farthest vertex p, and assume that the entire incident diagram is B_{18}. Normalize the actual old normals by

$$n_i = (e_i - e_{i+1})/\sqrt{2} \quad (1 \le i \le 17), \qquad n_{18} = e_{18}.$$

Write

$$p^2 = -1, \qquad z = E(p - x), \qquad E > 0, \quad x \perp p, \quad x_1 > \cdots > x_{18} > 0, \quad \|x\|^2 < 1.$$

For $P_i = e_1 + \cdots + e_i$, put

$$S_i = \langle x, P_i \rangle, \qquad f_i = \sqrt{i} - .58/\sqrt{2}, \qquad d_K = \sqrt{K} - \sqrt{K-1}.$$

The actual duals are $\omega_i = \sqrt{2} P_i$ for $i \le 17$ and $\omega_{18} = P_{18}$. In particular $\langle \omega_i, \omega_j \rangle = 2\min(i,j)$ for $i, j \le 17$. An index is strong when $S_i > f_i$; equality belongs to the weak case. There are no old spectator directions. All subsequent references in the original frame concern this one vector x.

We shall prove that these hypotheses are impossible. The proof first constructs unconditional terminal sum bounds, then classifies the actual seventeenth opposite. Its far-helper, near-six and short-contact branches are treated by shared actual-point and neighboring-comparison arguments. No bound on an arithmetic field occurs.

We use the actual positive projection and future-point pressure lemmas already proved in the main text. Their hypotheses will be checked for each point below. A positive correction on actual old normals leaves every unknown wall inequality nonpositive by acuteness. A spacelike corrected normal is used only inside its positive finite projection; it is never normalized as a chamber point.

H.1 The early window and actual head pressures

Deleting i, $5 \le i \le 14$, retains $A_{i-1} + B_{18-i}$, with both ranks at least four. The finite one-node list consists of isolation; one left end bond of label three or four; a left D or permitted E position; one right long-end simple bond; or an end-to-end bridge. Two contacts on one retained component create a cycle. A joined rank-eighteen diagram is classical; exceptional types add no case. The right B_3 at $i = 15$ additionally allows its short-end simple bond, and the right B_2 at $i = 16$ allows either end. These last roles will be checked separately.

For an actual edge with $C = \cosh L = qR$, $q = \sqrt{2i}$, clearing the positive denominators of its directional pressure shows that

$$R > T(q) := \frac{50}{29} - \frac{q - 29/50}{q(2q - 29/50)} \quad \Longrightarrow \quad S_i > f_i.$$

The function increases for $q \ge \sqrt{10}$, and substitution gives

$$T(q) < 83/50 \ (q < 7), \qquad T(q) < 801/500 \ (q < 15/4), \qquad T(q) < 159/100 \ (q < 16/5).$$

The finite-white Schur calculations have the following complete non-helper bounds; $\phi = (1+\sqrt{5})/2$.

role	lower bound for R
left near three, own at least five	$[i(\phi+1) - 1]/\sqrt{2i(i+1)}$
D position	$(i\phi/2 + 1)/\sqrt{2i}$
E position	$2\phi/2 + 3/8$
right long three, positive remaining own	$1 + 1/\sqrt{2}$
right B_3 short three, positive own	$1 + 1/\sqrt{2}$
end-to-end bridge	$(i+1)/\sqrt{2i}$.

The near-three bound increases with i. At $i = 6$ it is $(6\phi+5)/\sqrt{84} > 801/500$, which covers $6, 7$; at $i = 8$ it exceeds $83/50$, covering larger i. At $i = 5$, every own at least seven gives $R > 5/3$. For the D row, $i = 5$ gives $R > 159/100$, and $i \ge 6$ gives $R \ge 2927/1750 > 83/50$. An E position has $i \le 8$ and retained residual at most $1/4$. The bridge residual is $1/(2i)$; its numerator is at least $(i+1)/(2i)$, proving the displayed lower bound, including own zero. Left label-four roles have the affine small-own alternative or the larger direct edge bounds: near own three has $C = (i-1) + i/\sqrt{2}$. The positive-null and impossible-time boundaries in the common prefix inventory apply to the actual retained blocks at these ranks. For the right long-three own-zero boundary one can also see the kernel directly: use coefficient one on the cap and old i, two on old $i+1, \dots, 17$, and $\sqrt{2}$ on old short 18. Every selected product vanishes.

An isolated finite cap has the actual finite/core pressures used in the earlier component lemmas. At own five its loss is less than $25/76 + 5/24 < .58$; own seven has loss $1000/4779 + 9/25 < .57$; the remaining finite interval has its largest loss at $i = 5$, own eight: $\sqrt{10} + \sqrt{2} - 4 < .578$. For an isolated dotted cap put $u = g^{-2}$. Its positive reciprocal core gives

$$\langle x, \omega_i \rangle > \sqrt{2i - u} - \sqrt{u(1-u)}.$$

For $i \ge 6$ the loss is at most

$$a_i u + \sqrt{u(1-u)}, \qquad a_i = \frac{1}{\sqrt{2i} + \sqrt{2i-1}} < 37/250,$$

whose maximum is less than .5795, since $\sqrt{1 + (37/250)^2} < 1011/1000$. Thus only an isolated dotted cap at 5 may remain weak.

There is one early near-three exception, own five at 5. For $g = \phi/2$, the actual cap has displacement $v = g\omega_5 + \omega_4/2$. On the right actual H_4, correct by

$$w = g(3n_5/2 + n_6 + n_7/2), \qquad \delta = (5 - 3\phi)/8 > 0.$$

The corrected normal is spacelike and its finite projection is actual; the extra old 4 product is negative, and all unknown wall signs follow by positivity. Its pure spatial displacement is

$$Y = (1/2 + 3g/4)\omega_4 + (g/4)\omega_8, \qquad A^2 = (7 + 13\phi)/2, \quad Y^2 = (33 + 49\phi)/8.$$

The finite-point pressure gives

$$\langle x, Y\rangle > 18/5, \qquad Y^2 < 281/20, \qquad \sum Y_j < 43/5.$$

Independently, the earlier rank-valid .61 prefix lemma supplies a strong $K = 11$ or 12. Its old floors exceed 72/25 or 303/100. The vector $Y + (3/4)P_K$ therefore has squared demand exceeding squared norm by .0401 or .78625625, respectively, a contradiction. This use of the earlier lemma concerns $B_{18} \geq B_{16}$ and precedes any new .58 or strong-five conclusion.

The two extra B_2 roles at 16 deserve separate treatment. For short-three/own-five put $g = \phi/2$ and $\alpha = \sqrt{2}g + 1/2$. The left actual H_4, on the cap and old $16, 15, 14$, has residual $(5 - 3\phi)/8$, time square $16\alpha^2 - 1/2$, and normalized spatial displacement constant through 13, of norm less than 4. Since $\alpha > 1.643$, its pressure exceeds $3.976 - .083 = 3.893 > 3.89$. The earlier, independently available strong $K = 11, 12$ references have energies $> .8472225, > .8689525$, sums $< 2.886, < 3.034$, and terminal capacities $> 1/7$. Their roots and actual points lie through 14. Paying 1/7 at that terminal anchor gives

$$.8472225 + 2(3.89 - 2.886)/7 - 5/49 > 1.0320, \qquad .8689525 + 2(3.89 - 3.034)/7 - 4/49 > 1.0318.$$

For short-three/own-three the same cap has displacement $v = (\omega_{16} + \omega_{18})/2$. Deleting actual short 18 instead gives a positive full D_{18} projection. Its sum and time square are $9+8\sqrt{2}$ and $23/2+8\sqrt{2}$; the squared sum minus eighteen times the time square is 2. Thus its actual short-edge cosine is $8+9\sqrt{2}/2 > 14$, whence $S_{18} > 3.949$. The unspent early $K = 14, 15$ references, of effective energies $> .9231, > .94006$, have terminal capacities $> 1/9$. Paying 1/9 toward P_{18} gives respectively

$$.9231 + 2(3.949 - 3.332)/9 - 4/81 > 1.0108, \quad .94006 + 2(3.949 - 3.463)/9 - 3/81 > 1.0110.$$

The H_4 construction is a positive finite projection; the second point is the positive projection onto the displayed full finite Gram. All original unselected wall products are therefore nonpositive. Neither new point is asserted farthest. These arguments use the early references before the new B_2 conclusion, so they introduce no circularity.

Consequently every weak ordinary site is a pure far helper

$$c_i = A_i p - g\sqrt{2}P_i - e_1/\sqrt{2}, \qquad A_i^2 = 2ig^2 + 2g - \tfrac{1}{2}, \qquad \phi/2 \leq g < 6/5,$$

apart from the isolated dotted 5. The upper cutoff follows from $R^2 \geq 4g - 2 \geq 14/5 > (83/50)^2$ at $g \geq 6/5$.

For adjacent pure helpers the exact time/dual calculation gives

$$\langle c_i, c_{i+1}\rangle \geq \frac{1}{2} - g_i g_{i+1} + \frac{1}{4}(g_i/g_{i+1} + g_{i+1}/g_i) \geq 1 - g_i g_{i+1} > -11/25.$$

Their actual native gap forces orthogonality. The positive two-cap replacement, with every other old wall retained, has positive inverse Schur matrix; its weighted loss is less than $7/16+1/(4\sqrt{10}) < .58$.

Thus both sites cannot be weak. This uses their actual new parameter range, rather than inclusion in an older weak class. In particular some $K = 14$ or 15 is strong.

The same actual helper feet give $x_1 > .5455$ at a weak helper 5. For an isolated dotted weak 5, its reciprocal core gives

$$U' = P_5 + (u/2)(e_6 - e_5), \qquad \langle x, U' \rangle > T' > 1.8185, \qquad .4 < u < 1.$$

Indeed the loss is bounded by $au + \sqrt{u(1-u)}$, $a = \sqrt{10} - 3 < 13/80$; for $u \le .4$ it is less than $.565 < .58$, and globally it is less than .588. The pressure bound follows from $\sqrt{10} > 3.162, \sqrt{2} < 99/70$. These are actual point pressures, not assigned floors at weak nodes.

H.2 Raw references, head isolation and the uniform paid prefix

A reference w has support through a strong K, sum f_K, actual pressure $\langle x, w \rangle > A_w$, and effective energy $L = 2A_w - w^2$. Point insertion need not leave $L = w^2$. For a new actual pressure $\langle x, V \rangle \ge G$, weight $\mu \ge 0$, and $Y = V - P_K$, the completed-square rule is

$$\|x\|^2 > L + 2\mu(G - f_K - \langle w, Y \rangle) - \mu^2 Y^2, \qquad \mu \le \beta_K.$$

The inequality is strict when the old strong reference retains a positive coefficient. Every use below checks its individual prices.

Let $d_i = \sqrt{i+1} - \sqrt{i}$. The first-five energy exceeds .6668157, and rational lower numerators for $10^7 d_i^2$, $i = 5, \dots, 16$, are

$$455488, 385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248, 172465, 161332, 151549.$$

For a direct verification, put $a = 4i + 2$ and use $d_i^2(a - d_i^2) = 1$, whence $d_i^2 > (a^2 - 1)/(a(a^2 - 2))$. Thus the unomitted energies satisfy

$$E_{14} > .9241299, \quad E_{15} > .9413764, \quad E_{16} > .9575096, \quad E_{17} > .9726645.$$

A missing knot $j \ge 6$ loses less than $1/[32(j-1)^3]$; nonadjacency makes these losses disjoint. Their total is less than $(1/32)\int_4^\infty t^{-3}\,dt = 1/1024$. Each terminal capacity is at least d_K.

Before any head payment, the three initial-five cases give the following full contracts:

case at five	$L_{14} >$	$L_{15} >$	$w_1 <$
strong	.9231	.9403998	.3652
pure helper	.95255	.96895	.5455
isolated dotted weak	.9291	.9455	.377.

In the strong case interpolate through all actual strong floors, skipping every weak knot. The first mean is $f_5/5 < .3652$. For a helper, interpolate through $(1, .5455), (6, f_6), (K, f_K)$. Its first-six energy exceeds $.5455^2 + (f_6 - .5455)^2/5 > .74385$; the tails to 14, 15 exceed .2087, .2251. All slopes are positive decreasing.

For isolated dotted weak 5, choose strong $J = 6$ or 7 and a reference constant first J, then straight to K. Insert $\mu[U' - (5/J)P_J]$, with $\mu = .21, .2$. The residual squares are bounded by

$$5 - u + u^2/2 - 25/J < 77/150,\ 194/175$$

and the deficits exceed .119, .221. Its individual P_J prices are $.175, 1/7$, below the capacities $> .1783, > .1628$ already at $K = 14$. The gains exceed .027342, .044057. The bare first-J energies

exceed $.69312, .7139$; the tails to 14 exceed $.2087, .1715$ and to 15 exceed $.2251, .18825$. The full first coordinates are less than

$$.3399 + .21(1 - 5/6) = .3749, \qquad .31938 + .2(1 - 5/7) < .377.$$

This uses the straight reference for these prices. A curved reference with a smaller P_6 capacity is not substituted.

The actual long-head opposite is isolated. Deleting old 1 retains B_{17}; its finite enlargement is isolation or a simple bond to its long endpoint 2. In the latter case

$$v = g\omega_1 + \omega_2/2, \qquad A^2 = 2g^2 + 2g, \qquad C_1 = 2g + 1.$$

Own zero has zero time; own three is an actual affine triangle. Every other own gives $x_1 > \sqrt{\sqrt{2} - 1} > .643$. If 5 is strong, replace the first-five mean $< .3652$ by this pressure, gaining $(5/4)(.643 - .3652)^2 > .0964$ over $L_{14} > .9231$. If 5 is weak, choose strong $J = 6, 7$ and interpolate through $1, J, K$; the first energies exceed $.8034, .8361$ and the tails exceed $.2087, .1715$. Each case exceeds one, including isolated dotted weak 5. Therefore a head has time $A_H^2 = 2g_H^2 - 1$. Owns zero, three, four have impossible time; six and parallel have the positive actual $6/3$ and parallel nulls. Only own five, finite own at least seven, and dotted own remain.

For later reference, every non-five head has a point pressure $\langle x, Z \rangle \geq h$, with $\sum Z_j = 1$ and the following full local data:

own	Z	h	$R = Z^2$
finite ≥ 7	$(1 - u/2)(e_1 + e_2) + (u - 1)e_3$	$[\sqrt{2 - u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2)e_1 + (u/2)e_2$	$[\sqrt{2 - u} - \sqrt{u(1 - u)}]/\sqrt{2}$	$1 - u + u^2/2.$

Here $u = g_H^{-2}$; the first row has $1 < u \leq \cos(\pi/7)^{-2}$, the second $0 < u < 1$. Their actual positive cores are $c_H + g_H^{-1} n_1 + 2(g_H^{-1} - g_H)n_2$ and $c_H + g_H^{-1} n_1$, respectively. Their cap product is zero, the stated old-root products are nonpositive and all unknown signs follow termwise. They are future, giving these pure spatial pressures. Both obey

$$.8h - .16R > .2893. \tag{50}$$

For native seven, $.9009 < g_H < .901$ gives $h > .4315, R < .349$, and the left side exceeds .28936. For finite labels at least eight, $1 < u < 1.172$. Writing $D = (u - 1)(4 - 3u)$, the derivative is

$$\frac{.8}{\sqrt{2}} \left(-\frac{1}{2\sqrt{2 - u}} - \frac{7 - 6u}{2\sqrt{D}} \right) + .64 - .48u.$$

It is negative on $(1, 7/6]$; on $[7/6, 1.172)$, $\sqrt{D} > .284$, its positive term is $< .032$ and its other terms total less than $-.20$. At own eight $h > .4388, R < .373$, giving $> .29136$. For dotted u, the two chords $\sqrt{2 - u} \geq 1.4142 - .379u$ on $[0, 1/2]$ and $\sqrt{2 - u} \geq 1.4494 - .4494u$ on $[1/2, 1]$ give, respectively,

$$.8h - .16R > .63987152 - .0944u - .5657\sqrt{u(1 - u)} > .30587152,$$

$$.8h - .16R > .69978064 - .2142u - .5657\sqrt{u(1 - u)} > .29023064.$$

Use $bu + c\sqrt{u(1 - u)} \leq (b + \sqrt{b^2 + c^2})/2$, with radical bounds $.5736, .6049$. This proves the entire finite and unbounded dotted head seal.

H.3 One uniform paid prefix, without a terminal premise

If the head has own five, its actual unit normal is

$$c_H = A_H p - (\phi/\sqrt{2})e_1, \qquad A_H^2 = (\phi - 1)/2.$$

A weak pure helper at $i = 5, 6, 7$ would have mutual magnitude

$$M_i(g) = A_H\sqrt{2ig^2 + 2g - \tfrac{1}{2}} - \frac{\phi}{2}(2g + 1).$$

This increases in g, i on the allowed range and $M_7(6/5) < 0$, since the positive squared comparison has difference $(814\phi - 1392)/100 < 0$. An isolated dotted weak 5 has own $h < 2$, by its edge pressure and $f_5^2 < 7/2$. Its increasing mutual magnitude

$$A_H\sqrt{10h^2 - 1} - \phi h$$

lies strictly between zero and $1/4$ for $1 < h < 2$: the endpoint squared comparisons are $(7\phi - 11)/2 > 0$ and $377/16 - (29/2)\phi > 0$. Both cases contradict actual acuteness or the native gap. Thus a native-five head makes $5, 6, 7$ strong independently of any terminal hypothesis.

The head's actual finite H_4 projection gives

$$\langle x, P_4/4\rangle \geq h_0, \qquad h_0^2 = (\phi - 1)/4, \qquad h_0 > .39307.$$

At strong $K = 14, 15$ the curved reference, omitting every nonadjacent internal weak knot, has energies

$$w_{14}^2 > .924016575, \qquad w_{15}^2 > .94125107.$$

Indeed the complete raw energies above lose at most $(.0112^2 + .008^2 + .0061^2)/2 = .000113325$ for $K = 14$, and $< .00012533$ for $K = 15$; these bounds include every permitted hole. The P_5 capacity is $> .151$. Pay weight .56 on $P_4/4 - P_5/5$, costing .112. Its square is $1/20$, it is orthogonal to the raw reference, and the single-square gain exceeds

$$1.12(.39307 - .3652) - .56^2/20 = .0155344.$$

Hence this initial/head case gives

$$L_{14} > .939550975, \qquad L_{15} > .95678547.$$

All other initial/head cases give larger uniform energies. For a non-five head and strong 5, pay $.4(Z - P_5/5)$. Its price is $.08 < .151$, and (50) gives gain $.2893 - .8(.3652) + .032 > .02914$. For a weak helper 5, the raw contracts above already give $L_{14} > .95255, L_{15} > .96895$, with no further head graft.

For isolated dotted weak 5, choose strong $J = 6, 7$ and a curved reference to K, constant first J. Its first mean, individual capacity, and the square of $Y_5 = U' - (5/J)P_J$ obey

J	$m <$	$\beta_J >$	$Y_5^2 <$
6	.3399	.1435	77/150
7	.31938	.1366	194/175.

The raw first-J energies exceed $.69317, .71400$; the complete tails to 14 exceed $.2117654, .1732468$ after omission losses $< .000267, .000181$. Thus the raw energies exceed $.9046684, .8870658$. At 15, the additional increment is $> .0172465$. Let $Y_H = Z - P_J/J$. The full cross is $\langle Y_H, Y_5\rangle = 1 - 5/J$,

while $\langle w, Y_H\rangle = \langle w, Y_5\rangle = 0$. The one vector $w + .4Y_H + .09Y_5$ spends $.85/J$, below each displayed capacity. Its entire gain is

$$(.8h - .16R) - .8m + .16/J + .18(T' - 5m) - .0081Y_5^2 - .072(1 - 5/J).$$

It exceeds $.04930, .06699$, respectively. The resulting $K = 14$ energies exceed $.9539684, .9540558$, and the $K = 15$ energies exceed $.9712149, .9713023$. The earlier $.21/.2$ insertion is not also charged.

In every case, therefore, there is one paid reference V, supported through a strong $K = 14$ or 15, with sum f_K, unspent terminal capacity $\beta_K \geq d_K$, and

$$2A_V - V^2 > L_K, \qquad L_{14} = .939550975, \quad L_{15} = .95678547. \tag{51}$$

All corrections have zero sum and support through seven. Their individual nonnegative coefficients were checked before the terminal pressure is used.

We shall also need the following independent implication: if 17 is strong, the head has own five. For a non-five head and strong 5, the curved reference to 17 has energy $> .97133$, and its head gain $> .02914$ already exceeds one. If 5 is a helper, the full curved reference gains $> .74385 - .7126 = .03125$ in its first six coordinates, yielding $E_{17} + .03125 - 1/1024 > 1.0029$. If 5 is isolated dotted, the analogous curved references to 17 have energies $> .9530, .9354$; the joint head/isolated-five gains just proved exceed $.04930, .06699$. Again the totals exceed one. Thus every non-five head is incompatible with strong 17. This argument precedes any classification of its terminal opposite.

H.4 Unconditional terminal sums and isolated opposites

For $N = 16, 17, 18$, let $Y = P_N - P_K$. It has support after K, square $N - K$, and zero product with the entire corrected reference in (51). Paying the nonnegative weight $d_K \leq \beta_K$ shows that $S_N \geq G$ would imply

$$\|x\|^2 > L_K + 2d_K(G - f_K) - (N - K)d_K^2.$$

The rational root bounds

$$3.60555 < \sqrt{13} < 3.60556, \quad 3.74165 < \sqrt{14} < 3.74166,$$
$$3.87298 < \sqrt{15} < 3.87299$$

give $.1361 < d_{14} < .1362$, $.1313 < d_{15} < .1314$. Every bound follows by positive squares and $d_K = 1/(\sqrt{K} + \sqrt{K-1})$.

For $S_{17} \geq 3.760$, the extra terms are bounded below by

$$2(.1361)(.428) - 3(.1362)^2 = .06085028, \qquad 2(.1313)(.297) - 2(.1314)^2 = .04346028.$$

Their totals exceed $1.000401255, 1.00024575$. For $S_{18} \geq 3.830$, the increments are $.06135384$, $.04457632$, giving totals $> 1.000904815, > 1.00136179$. Thus

$$S_{17} < 3.760, \qquad S_{18} < 3.830, \qquad C_{17} < 10.879, \quad C_{18} < 9.81. \tag{52}$$

The edge cosines use the actual directions $P_N/\sqrt{N}$: $C_N < (N + G_N^2)/(N - G_N^2)$. The two fractions are $31.1376/2.8624 < 10.879$ and $32.6689/3.3311 < 9.81$. These are simultaneous consequences for the same x, not energies added from different references.

An entirely isolated 17-opposite has

$$c = Ap - \sqrt{2}gP_{17}, \qquad A^2 = 34g^2 - 1 > 0.$$

Own zero has negative time. Own three, four, six and parallel have positive actual null combinations, respectively,

$$c + n_{16} + 2n_{17} + \sqrt{2}n_{18}, \qquad c + \sqrt{2}n_{17} + n_{18}, \qquad gc + n_{17} + n_{16}/2, \qquad c + n_{17}.$$

Selected products vanish, cap time is positive, and all unselected wall products are nonpositive. Compactness forbids them. For own five, correct on the actual H_4 by $g(3n_{17}/2 + n_{16} + n_{15}/2)$, with positive residual $\delta = (5-3\phi)/8$. The actual finite projection gives pressure on $V = P_{14} + \frac{3}{4}(e_{15} + \cdots + e_{18})$ strictly greater than $(A - \sqrt{\delta})/(\sqrt{2}g)$. Ordering implies $\langle x, V \rangle \le S_{17}$. Since $A^2 = (17\phi + 15)/2$, the rational bounds $A > 4.61, \sqrt{\delta} < .136, \sqrt{2}g < 1.145$ give $S_{17} > (4.61 - .136)/1.145 > 31/8$. Every finite own at least seven has the positive core $c + g^{-1}n_{17} + 2(1 - g^2)g^{-1}n_{16}$. Its norm is $-(1-g^2)(4g^2-3)/g^2$, selected cap/own products vanish, and the other selected product is negative. All unknown wall signs follow from positive actual coefficients. With $u = g^{-2} \in (1, 4/3)$, divide its displacement by $\sqrt{2}g$. Its coordinates are one through fifteen, $2 - u$ at sixteen, and $u/2$ at seventeen and eighteen. Its product with x is at most S_{17} by ordering. The actual pressure is at least $\sqrt{17 - u/2} - \sqrt{(7u - 4 - 3u^2)/2}$. The first radical exceeds 4.04 and the second is at most $1/\sqrt{24} < .2042$; hence $S_{17} > 34/9$. All these finite cases contradict the same prefix rule: the gains at $G = 34/9$ exceed $.065, .047$.

For every dotted $g > 1$, put $u = g^{-2}$. The actual future reciprocal core $c + n_{17}/g$ has norm $-(1 - u)$. Its displacement is $U = \sqrt{2}gP_{17} - n_{17}/g$. For $r = 17 - K$, choose

$$Y = U - \sqrt{2}gP_K, \qquad \nu = d_K/(\sqrt{2}g), \qquad Y^2 = 2r/u - 2 + u.$$

The residual is supported after K, so is orthogonal to the entire already paid reference. Its price is $\nu\sqrt{2}g = d_K$, allowing equality at the terminal. The gain is exactly

$$G_K(u) = \sqrt{2}d_K\left[\sqrt{34 - u} - \sqrt{2K} + .58 - \sqrt{u(1-u)}\right] - d_K^2(r - u + u^2/2).$$

The chord $\sqrt{34 - u} \ge \sqrt{34} - .0864u$ follows from $\sqrt{34} > 5.8309, \sqrt{33} > 5.7445$. The circle square gives $\sqrt{u(1-u)} \le 1/4 + u - u^2$. Using $\sqrt{28} < 5.2916, \sqrt{30} < 5.4773$, the bracket exceeds $u^2 - 1.0864u + .8693$ or $u^2 - 1.0864u + .6836$, both positive. Also $\sqrt{2}d_{14} > .19247, \sqrt{2}d_{15} > .18568$. Use the upper d_K only in the negative constant and the lower d_K in $d_K^2(u - u^2/2)$. This gives

$$G_{14}(u) > .183208395u^2 - .190576198u + .111662851,$$
$$G_{15}(u) > .177060155u^2 - .184483062u + .092398928.$$

For $F = au^2 - bu + c$, the identity $4a(F - \tau) = (2au - b)^2 + 4a(c - \tau) - b^2$ has positive last constants

$$\frac{808151227201376}{10^{18}} \quad (\tau = .061), \qquad \frac{244086609159516}{10^{18}} \quad (\tau = .044).$$

Thus the full dotted interval gives gains $> .061, > .044$, and totals $> 1.000550975, > 1.00078547$. No isolated 17-opposite survives.

H.5 An exhaustive terminal inventory

Deleting old 17 retains actual A_{16} on $n_1, \ldots, n_{16}$ and independent short A_1 on n_{18}. The actual finite opposite extends this whole white Gram positively. A one-node finite enlargement of A_{16} is

A_{17}, B_{17} or D_{17}; exceptional types have insufficient rank. Thus the complete white list is

left contact	short contact
none	none or one finite label $m \geq 3$
3 at $j = 1, 16$	none, 3, or 4
4 at $j = 1, 16$	none
3 at $j = 2, 15$	none.

Two left contacts create a cycle. Joining the short root to a left-three endpoint gives A_{18} or B_{18}; joining it to a left-four or a next-node branch is not finite. White parallel/dotted contacts are impossible at this finite endpoint. Own contacts still range over zero, every finite label, parallel and dotted.

Write a cap as

$$c = Ap - g\omega_{17} - \eta\omega_j - \delta\omega_{18}, \quad \sigma = g + \eta j/17 + \delta/\sqrt{2}, \quad \Delta = 1 - \frac{2\eta^2 j(17-j)}{17} - \delta^2 > 0.$$

Absent contacts have coefficient zero. Direct retained-block elimination gives

$$A^2 = 34\sigma^2 - \Delta, \qquad C_{17} = \sqrt{34}\,\sigma/\sqrt{\Delta}.$$

There is no residual cross between the actual left and short blocks. Own parallel is excluded everywhere by the actual positive null $c + n_{17}$. Isolation was excluded above.

For left-four alone, the far and near lengths are $17\sqrt{2}g + 1$ and $17\sqrt{2}g + 16$. Far own zero has zero time; every positive far own gives length greater than thirteen, while the near length is already at least sixteen. For left next-three, the lengths are $17g + 1$ at $j = 2$ and $17g + 15/2$ at $j = 15$. Far own zero has zero time; far own three is the actual simple cycle $c, n_2, \ldots, n_{17}$. Every larger positive far own exceeds thirteen. Near positive own gives length at least sixteen; near own zero has the affine positive kernel with coefficients one on c, n_{14}, two on n_{15}, n_{16}, n_{17}, and $\sqrt{2}$ on n_{18}.

Left-three plus short-four has $\Delta = 1/34$ and $C_{17} = 34g + j + 17 \geq 18$. Left-three plus short-three has $\Delta = 19/68$. Own three closes a simple cycle; every larger positive own has $g \geq 1/\sqrt{2}, \sigma > 1$ and length greater than eleven. At own zero, the far bridge contains the actual affine path $(n_{15}, n_{16}, n_{17}, n_{18}, c)$, with positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$. For the near bridge, $v = (\omega_{16} + \omega_{18})/2$. Deleting actual short 18 gives positive finite D_{18}. Positive projection checks the released wall and every unknown wall, so this is the actual short-edge endpoint. Its exact data are

$$A^2 = 23/2 + 8\sqrt{2}, \quad \sum v_i = 9 + 8\sqrt{2}, \quad (\sum v_i)^2 - 18A^2 = 2, \quad C_{18} = 8 + 9\sqrt{2}/2 > 14,$$

contradicting (52).

For pure far-three at $j = 1$, own zero has negative time, own three gives its old simple cycle, own four has null $c + \sqrt{2}n_{17} + n_{18}$, and own six has null $gc + n_{17} + n_{16}/2$. Parallel was already null. The remaining owns are five, finite at least seven, or dotted, with

$$C_{17} = \frac{\sqrt{2}}{3}(17g + 1/2), \qquad g < 4/3.$$

At $g \geq 4/3$ this would be at least $139\sqrt{2}/18 > 10.9$. For pure near-three at $j = 16$, $C_{17} = \sqrt{2}(17g + 8)/3$. Own zero has the positive kernel with coefficients one on c, n_{15}, two on n_{16}, n_{17}, and $\sqrt{2}$ on n_{18}. Own three gives a triangle; four and parallel have the nulls above. Owns at least seven or dotted have $g > .9$ and length > 10.98362. Only owns five and six need remain.

Near own five is impossible independently of any head. For $g = \phi/2$, the positive actual right-B_2 core is

$$\xi = c + 2gn_{17} + \sqrt{2}gn_{18} = Ap - \sqrt{2}(g + 1/2)P_{16}, \quad A^2 = (49\phi + 31)/2, \quad \xi^2 = 1 - 2g^2 < 0.$$

Its cap product is this same negative square and its selected old products vanish. All other original wall products are nonpositive; positive time makes it future. The pressure gives

$$S_{16} > \frac{A - \sqrt{2g^2 - 1}}{\sqrt{2}(g + 1/2)} > 3.7.$$

Indeed $A > 7.42$, $\sqrt{2g^2 - 1} < .556$ and $\sqrt{2}(g + 1/2) < 1.852$, while $7.42 - .556 > 3.7(1.852)$. The uniform prefix rule gives increments .06306872, .04497024, contradicting its norm budget. For near own six, $g = \sqrt{3}/2$ gives $C_{17} > 10.5$, hence $S_{17}^2 > 17(19/23) > f_{17}^2$. Thus this surviving role has strong 17 and the isolated native-five head already proved above.

Finally suppose there is no left contact and $\delta = \cos(\pi/m)$ is the short-white magnitude. The length bound implies

$$\frac{g + \delta/\sqrt{2}}{\sqrt{1 - \delta^2}} < 1.866.$$

For $m \geq 9$, $\pi^2 < 10$ and $\cos t > 1 - t^2/2$ give $\delta > 76/81 > 15/16$; even at $g = 0$ the ratio square exceeds $225/62 > 1.866^2$. The entire conservative own list is

m	own g
3	0, every finite positive own, or $1 < g < 1.264$
4	finite own $3, 4, 5$
5	0 or finite own 3
$6, 7, 8$	0.

For short three, $g < 1.866(.8665) - .3533 < 1.264$. Short four gives $g < .82$; its own zero has positive null $c + n_{17} + \sqrt{2}n_{18}$. Short five gives $g < .53$ from $\delta > .809, \sqrt{1 - \delta^2} < .588, \delta/\sqrt{2} > .571$. For short at least six, $g < .933 - .6 < 1/2$. Every null above is a positive actual-root combination with positive cap time, zero selected products and nonpositive product against every remaining original wall. It is therefore a nonzero feasible future ray, forbidden by compactness.

This proves the exhaustive partition into a far-three helper, near-three/own-six, or the displayed no-left short-contact family.

H.6 Finite far-helper owns from raw, unspent references

The far cap is

$$c = Ap - \sqrt{2}gP_{17} - e_1/\sqrt{2}, \qquad A^2 = 34g^2 + 2g - \tfrac{1}{2}.$$

Use the three raw contracts of Section H.2, before any head graft. Their terminal prices are at least d_K, and their full first coordinates are bounded there. Pay $\mu_{14} = .136, \mu_{15} = .1313$ for a new pressure $\langle x, V\rangle > G$, with $Y = V - P_K$. The full square retains $\langle w, Y\rangle$; no absent P_{14} coefficient is borrowed when the actual strong knot is 15.

For own five, the actual H_4 correction

$$\eta = c + (3g/2)n_{17} + gn_{16} + (g/2)n_{15} = Ap - U$$

has positive square $\delta = (5 - 3\phi)/8$. Use the actual finite point $p + (A/\delta)\eta$. Its selected products vanish, its correction coefficients are positive and every unselected original wall remains nonpositive.

Its pressure exceeds $A-\sqrt{\delta}$. For $b=1/(2g)=1/\phi<.619$, the vector $V=U/(\sqrt{2}g)$ has coordinates $1+b$ at one, one through fourteen, and $3/4$ at fifteen through eighteen. Its pressure exceeds 4.10:

$$A^2=(19\phi+16)/2,\quad A>4.834,\quad \sqrt{\delta}<.136,\quad \sqrt{2}g<1.145.$$

The complete residual data are

K	$Y^2<$	$\langle w,Y\rangle$
14	2.634	bw_1
15	2.134	$bw_1-w_{15}/4$.

Use $w_{15}>.1313$ and the actual raw row's L,w_1. The resulting exact strict norm lower bounds are

initial five	$K=14$	$K=15$
strong	1.0217895424	1.02014330566
helper	1.020882792	1.01938584884
isolated dotted weak	1.0258028	1.02332542274.

For example the first column is obtained from $L+.272(4.10-3.332-.619w_1)-.018496(2.634)$. Every row is the same original square with its entire first-coordinate cross retained.

For every finite own at least seven, put $u=g^{-2}\in(1,4/3)$. The positive actual core

$$\xi=c+g^{-1}n_{17}+2(g^{-1}-g)n_{16}=Ap-U$$

has norm $-(u-1)(4-3u)/u$, zero cap/own products and negative n_{16} product. Its unknown wall signs and future sheet follow from positive actual coefficients and $A>0$. For $b=1/(2g)\in(.5,.556)$, $V=U/(\sqrt{2}g)$ has coordinates

$$1+b,\quad 1\ (2,\dots,15),\quad 2-u,\quad u/2,\ u/2.$$

Its pressure is

$$G=\sqrt{17+2b-b^2}-\sqrt{(7u-4-3u^2)/2}>\sqrt{71/4}-1/\sqrt{24}>4.0088.$$

The residuals satisfy $Y_{15}^2<7/4,\ Y_{14}^2<11/4$; both full crosses are bw_1. Indeed $Y_{15}^2=4-15u/4+3u^2/2$, whose difference from $7/4$ is $(3/2)(u-1)(u-3/2)<0$. The three initial-five rows give

	$K=14$	$K=15$
strong	1.0010956736	1.00023617738
helper	1.003278544	1.0024615677
isolated dotted weak	1.005311136	1.0036135113.

The tight entry uses precisely

$$.9403998+.2626(4.0088-3.463-.556(.3652))-.01723969(7/4)=1.00023617738.$$

The weaker energy .94006 or a terminal weight .13 would not justify it. All finite far-helper owns are excluded without a head or short-opposite assumption.

H.7 A single table for every actual short opposite

Deleting actual 18 retains A_{17}. The finite one-node classification gives precisely isolation, an endpoint bond three or four at $j = 1, 17$, or a next-to-end bond three at $j = 2, 16$. A second contact creates a cycle; the other branching trees and exceptional diagrams have insufficient rank. For the actual short opposite $d = Bp - v$, write $\delta = -\langle d, n_{18}\rangle \geq 0$. At this stage its own contact can be zero, any finite label, parallel, or dotted. The complete data are

white role	v	B^2	C_{18}
isolated	δP_{18}	$18\delta^2 - 1$	$3\sqrt{2}\delta$
end three j	$\delta P_{18} + P_j/\sqrt{2}$	$18\delta^2 + \sqrt{2}j\delta + j/2 - 1$	$(18\sqrt{2}\delta + j)/\sqrt{19}$
end four j	$\delta P_{18} + P_j$	$18\delta^2 + 2j\delta + j - 1$	$18\delta + j$
next three j	$\delta P_{18} + P_j/\sqrt{2}$	$18\delta^2 + \sqrt{2}j\delta + j/2 - 1$	$9\sqrt{2}\delta + j/2.$

For example the old short residual is $1/18$, and clearing a white three or four leaves residual $1 - j(18-j)/36$ or $1 - j(18-j)/18$, respectively. The edge formula gives the last column; thus the table includes the actual short normalization.

The unconditional bound $C_{18} < 9.81$ removes near end-four and every positive-own far end-four. The latter has $B = 0$ at own zero. Near next-three has $C_{18} = 9\sqrt{2}\delta + 8$, so a positive own is already too long; at own zero its null is

$$d + n_{15} + 2n_{16} + 2n_{17} + \sqrt{2}n_{18}.$$

Far next-three has $B = 0$ at own zero and $\delta < .7$ by length, so its only possible positive native own is three. Its null is

$$\sqrt{2}d + 2\sqrt{2}n_{18} + 3n_{17} + 2n_{16} + n_{15}.$$

All coefficients are positive, the cap time is positive, selected products vanish and every other original-wall product is nonpositive termwise. These are forbidden future feasible nulls.

We shall apply the remaining three rows twice. In both applications the selected seventeenth cap clears positively to

$$\bar{c} = Ap - (Q/17)P_{17}, \quad a = A^2, \quad Q^2 = 17a + 9, \quad \bar{c}^2 = r = 9/17, \quad a < 63.$$

For near end-three the actual mutual magnitude is $\lambda = AB - Q(\delta + 1/\sqrt{2}) \geq 0$. The length bound gives $\delta < 1.1$, whereas

$$aB^2 - Q^2(\delta + 1/\sqrt{2})^2 = (a-9)\delta^2 - 9\sqrt{2}\delta - a - \tfrac{9}{2} < .21a - 15.39 < 0.$$

This contradicts acuteness, including own zero and every allowed dotted value. Only far end-three and isolation remain.

H.8 Dotted far helpers: the complete mutual split

Let

$$c = Ap - \sqrt{2}gP_{17} - e_1/\sqrt{2}, \quad 1 < g < 4/3, \quad a = 34g^2 + 2g - \tfrac{1}{2}, \quad Q = (34g+1)/\sqrt{2}.$$

Here $71/2 < a < 1127/18$, and the positive actual correction is $\bar{c} = c + \sum_{i=1}^{16}(17-i)n_i/17$. Put $G_{17} = 3.760,\ G_{18} = 3.830$, and

$$t = Q/A = \sqrt{17 + 9/a}, \quad K = A\sqrt{18} - Q, \quad H = A - G_{17}Q/17, \quad J_0 = (t - G_{18})/K.$$

Then $4.14 < t < 4.18$, $K, H, J_0 > 0$, and

$$4HJ_0 > 1, \qquad H + J_0 y^2 - y > 0 \quad (y \in \mathbb{R}). \tag{53}$$

Indeed the first inequality is equivalent to positivity of $F(t) = 4(t - 3.83)(1 - 94t/425) + t - \sqrt{18}$. Replace $\sqrt{18}$ by its strict upper bound 4.243. The new polynomial has value $3333/2125000 > 0$ at 4.14; its difference from that value is

$$(t - 4.14)\{5 - (376/425)(t + 4.14 - 3.83)\} > 0,$$

since the brace exceeds $10919/10625$. The second inequality is the completed square $(2J_0 y - 1)^2 + 4HJ_0 - 1 > 0$.

We also record a useful full-projection test. Suppose an inverse-positive full actual projection has ray $(1 + T)p - LP_{17} - Me_{18}$, with $T > 200$ and $L > M > 0$. Write $k = M/L$. Its norm and the original farthest comparison give

$$S_{17} + kx_{18} \geq \sqrt{17 + k^2}\frac{\sqrt{T}}{\sqrt{1 + T} + 1} > \sqrt{221/15} > 3.83.$$

For the strict bound use $\sqrt{1 + T} > 14$, so the squared last ratio is $> 13/15$. The left side is $(1 - k)S_{17} + kS_{18} < 3.83$, a contradiction. The existence, future sheet and unknown-wall signs of this point follow from its full positive actual projection.

First suppose that the short opposite is far end-three:

$$d = Bp - \delta P_{18} - e_1/\sqrt{2}, \qquad B^2 = 18\delta^2 + \sqrt{2}\delta - \tfrac{1}{2}.$$

Own zero has negative time. Own three has the affine F_4 null above, and owns four, six and parallel have respectively

$$d + \sqrt{2}n_{18} + n_{17}, \qquad d + (\sqrt{3}/2)n_{18} + n_1/2, \qquad d + n_{18}.$$

These positive actual nulls remain feasible against every original wall. For every remaining own put $\lambda = -\langle c, d\rangle \geq 0$ and $v_0 = 1/(17\sqrt{2})$. Positive clearing gives

$$\bar{d} = Bp - (\delta + v_0)P_{17} - \delta e_{18}, \quad \bar{d}^2 = r, \quad \rho = -\langle \bar{c}, \bar{d}\rangle = \lambda + 8/17 = AB - Q(\delta + v_0).$$

The offset here is $8/17$. If $\lambda > 0$, the actual native gap gives $\lambda \geq 1/2$, hence $\rho > r$. The positive actual core $\Xi = \bar{c} + (\rho/r)\bar{d}$ has square $r - \rho^2/r < 0$, positive time, zero d product, negative c product, and zero old A_{16} products. All other products are nonpositive by acuteness. Its pressure and the two original sum bounds would require

$$H + (\rho/r)J - \sqrt{\rho^2/r - r} \leq 0, \qquad J = B - G_{17}v_0 - G_{18}\delta.$$

But $J = \rho/A + (t - G_{18})\delta + (t - G_{17})v_0 > (t - G_{18})\delta$. Positive squaring gives $B < \sqrt{18}\delta + 1/6$, and

$$36(g + 1/34)^2 - a = 2g^2 + 2g/17 + \tfrac{1}{2} + 9/289 > 0.$$

As $Qv_0 = g + 1/34$, this implies $\rho < K\delta$ and $J > J_0\rho$. With $y = \sqrt{\rho^2/r - r}$, the required gap exceeds $H + J_0(y^2 + r) - y > 0$ by (53).

If $\lambda = 0$, the cleared two-cap Gram has diagonal $9/17$, cross $-8/17$, and inverse $\left(\begin{smallmatrix} 9 & 8 \\ 8 & 9 \end{smallmatrix}\right)$. Its positive projection has

$$\alpha = 9A + 8B, \quad \chi = 8A + 9B,$$
$$T = 9a + 16AB + 9B^2 > 319.5,$$
$$L = \alpha Q/17 + \chi(\delta + v_0) > M = \chi\delta > 0.$$

The full-projection test excludes it. This handles every far-three short own and every unbounded actual mutual range.

Now let $d = Bp - \delta P_{18}$ be isolated. Acuteness $AB - Q\delta \geq 0$ implies $\delta^2 \geq a/(a-9) > 1$, so its own is derived to be dotted. Put $\lambda = AB - Q\delta$. The cleared Gram is $\left(\begin{smallmatrix} r & -\lambda \\ -\lambda & 1 \end{smallmatrix}\right)$. At $\lambda^2 = r$, $\bar{c} + \lambda d$ is a forbidden positive future feasible null. For $\lambda^2 > r$ the same core is negative. Writing $J = B - G_{18}\delta$, acuteness and $B < \sqrt{18}\delta$ give $J > (t - G_{18})\delta$, $\lambda < K\delta$. Its required pressure gap is $H + \lambda J - \sqrt{\lambda^2 - r}$, contradicted by (53). If $0 < \lambda^2 < r$, native separation gives $\lambda \geq 1/2$. The positive full projection has

$$T = \frac{a + 2\lambda AB + rB^2}{r - \lambda^2} > \frac{68.5}{19/68} = \frac{4658}{19} > 200,$$

using $B^2 > 17$, $AB > 24$. Its coefficients $\alpha = (A + \lambda B)/(r - \lambda^2)$, $\chi = (\lambda A + rB)/(r - \lambda^2)$ are positive, with $L = \alpha Q/17 + \chi\delta > M = \chi\delta > 0$. The full-projection test excludes this entire interval too. Only $\lambda = 0$ remains.

H.9 The orthogonal neighbor and its quantitative comparison

For that last mutual value,

$$\delta^2 = \frac{a}{a-9}, \quad B = \frac{Q}{\sqrt{a-9}}, \quad w = 9/a, \quad q = \frac{p - (A/Q)(P_{17} + we_{18})}{\sqrt{D}}, \qquad D = \frac{9(a-9)}{aQ^2}.$$

The positive actual projection has full forest $A_{17}(c, n_1, \ldots, n_{16}) + I(d)$. The other endpoint retaining its whole A_{17} is the positive finite projection on $A_{16} + c + n_{18}$. These endpoints are distinct because $\langle d, n_{18} \rangle = -\delta < -1$. Consequently n_{18} is the actual opposite of d, orthogonal to the entire new A_{17}, with

$$h_I = \delta^2 - 1 = \frac{9}{a-9} < 18/53 < 1/2.$$

All these are full-dimensional points satisfying every original wall; no face heredity is used.

Write the same original $z = E_q(q - y - \eta d)$, where y is in the A_{17} space. We prove the following three bounds, separately for old 17 strong and weak:

$$E/E_q < 1.061, \qquad \eta^2 > .1164, \qquad \text{a signed } A_{17} \text{ endpoint coordinate of } y \text{ exceeds } .51. \tag{54}$$

Every actual chamber point, including each auxiliary point used below, is then compared at q with factor < 1.061, because its height relative to the original z is at most E. The point q is never made farthest.

Suppose first that $S_{17} > f_{17} = s$, so the earlier head argument gives the actual isolated native-five head and strong $5, 6, 7$. The original reference through strong 17, with all internal weak knots omitted, has energy $> .97253117$, first-five mean $b < .3652$, fifth capacity $> .151$, and terminal capacity $b_0 = \sqrt{17} - 4 \in (.1231, .1232)$. The actual head foot is $Z_0 = P_4/4$, with pressure

$h_0 > .39307$. Put $Y_0 = Z_0 - P_5/5$, so $Y_0^2 = 1/20$. Paying .56 costs $.112 < .151$ and gains $> .0155344$. The paid effective energy is $L > .988$. Retaining the terminal surplus in this same square gives

$$2b_0(S_{17} - s) + x_{18}^2 < .012, \qquad S_{18} < s + .012/(2b_0) + b_0/2 < 3.82335.$$

We next show $g > 1.12$. For $1 < g \le 1.12$ the reciprocal core $c + n_{17}/g$ has pressure $A - \sqrt{1 - g^{-2}}$. Its residual after payment from P_{17} and P_5 has square $.4 + g^{-2}$, and cross $1/(20\sqrt{2})$ with Y_0. The pressure-minus-reference deficit is

$$D_H = A - \sqrt{1 - g^{-2}} - g(\sqrt{34} - .58) - (\sqrt{10}/10 - .058) + b_0/(g\sqrt{2}).$$

It decreases on this interval: $A'^2 = 34 + 18/a < 5.875^2$; the radical derivative exceeds 1.5, and $\sqrt{34} - .58 > 5.25$, so $D_H' < -.875$. At $g = 1.1$ use $A > 6.5452$, $\sqrt{1 - g^{-2}} < .4166$, the third term < 5.7761, and the last term $> .079$; at 1.12 use $6.66255, .45035, 5.88112, .07767$. Also $\sqrt{10}/10 - .058 < .258228$. Thus $D_H > .173$ on $(1, 1.1]$, and $> .150$ on $[1.1, 1.12]$. With residual coefficient $\nu = .075, .077$ respectively, the full gain over L is

$$2\nu D_H - \nu^2(.4 + g^{-2}) - .56\nu/(10\sqrt{2}) > .015075, \quad > .0127780815.$$

Both contradict $L > .988$. The individual terminal price $\nu\sqrt{2}g < .1231$ fits b_0, while $.112 + \nu/(5\sqrt{2}) < .123 < .151$ fits the fifth anchor. Hence $a > 44.3896$, and $k = 9/a < .20276$.

Suppose $E_q/E \le 20/21$. The explicit q formula gives

$$S_{17} + kx_{18} \ge R(k) := \sqrt{17 + k} - (20/21)\sqrt{k(1 - k)} > 3.764.$$

Indeed R decreases on $0 < k < .20276$: its positive derivative term is $< .122$ and the magnitude of the negative term is $> (20/21).594$. At the upper endpoint use $\sqrt{17.20276} > 4.1476$, $\sqrt{.20276(1 - .20276)} < .4021$. The independent strong $K = 11, 12$ references through strong 5, 6, 7, after every permitted hole, have raw energies $> .86375298, > .88550238$, terminal capacities $> .1543, > .1474$, and fifth capacity $> .151$. After the same .56 head payment, pay .14 from that terminal toward $P_{17} + ke_{18}$. Its residual is orthogonal to the head residual and to the reference, of square $17 + k^2 - K$, with $k^2 < .0412$. Since $f_{11} < 2.907$, $f_{12} < 3.055$, the full original squares give

$$.86375298 + .0155344 + .28(3.764 - 2.907) - .0196(6.0412) = 1.00083986,$$
$$.88550238 + .0155344 + .28(3.764 - 3.055) - .0196(5.0412) = 1.00074926.$$

Thus $E_q/E > 20/21$, so $E/E_q < 1.05$. Moreover $a < 64$, $\delta > 8/\sqrt{55} > 1.078$, $Q/A > 4.14$ imply

$$\eta = (E/E_q)\delta(Q/A - S_{18}) > 1.078(4.14 - 3.82335) > .3413, \qquad \eta^2 > .1164.$$

The known actual q-edge releasing c retains $A_{16} + d$; its other endpoint is $A_{16} + n_{17} + d$. Clearing that block leaves residuals $9/17$ and cross $-g - 1/34$, so its cosine is $C' = (34g + 1)/18 > 977/450 > 2.17$. The signed endpoint coordinate is therefore greater than

$$\sqrt{17/18}\frac{2.17 - 1.05}{\sqrt{2.17^2 - 1}} > .9718(1.12)/1.926 > .56.$$

This proves (54) in the strong case.

If old 17 is weak, $S_{17} \le f_{17} < 3.713$, and

$$E_q/E = \frac{\sqrt{17 + w} - S_{17} - wx_{18}}{\sqrt{w(1 - w)}}.$$

The two original bounds give $S_{17} + wx_{18} < 3.713 + .117w$. If $g \geq 1.1$, then $w \leq 25/119 < .211$. On this range $\sqrt{17+w} > 4.123 + .12w$, since $\sqrt{17+w} + \sqrt{17} < 25/3$, while $\sqrt{w(1-w)} < .408$. The displayed ratio exceeds $(.410 + .003w)/.408 > 1$, contrary to $E_q \leq E$. Thus $1 < g < 1.1$ and $25/119 < w < 18/71$. The same chord holds, $w > .21$, and the denominator is $< \sqrt{954}/71 < .4351$. Consequently

$$E_q/E > .41063/.4351 > .943, \qquad E/E_q < 1000/943 < 1.061.$$

Here $\delta^2 > 119/94 > 81/64$ and $\sqrt{17+w} > 4.1485$, so $\eta > (9/8)(4.1485 - 3.830) = .3583125$, hence $\eta^2 > .128$. Finally $C' > 35/18 > 1.94$, and its endpoint coordinate exceeds

$$.9718(1.94 - 1.061)/1.663 > .51.$$

All three conclusions of (54) follow.

H.10 One local $A_{17} + I$ contradiction for both height branches

We prove the common consumer of (54). The hypotheses are an actual whole $A_{17} + I$ vertex q, comparison factor $\kappa < 1.061$, a known isolated actual I opposite with parameter $h_I < 1/2$, and

$$\|y\|^2 + \eta^2 < 1, \qquad \eta^2 > .1164, \qquad \text{one signed endpoint coordinate of } y \text{ is } > .51.$$

We use the finite-white geometry proved in Appendices C.6 and C.7, not their farthest-vertex energy conclusions. Here are the geometry and the changed numerical comparisons explicitly.

Realize A_{17} in the zero-sum subspace of $\mathbb{R}^{18}$. For its unit roots m_i and duals Ω_i,

$$D_i = i(18-i)/9, \quad C_{ij} = i(18-j)/9 \ (i \leq j), \quad h_i = \langle y, \Omega_i \rangle, \quad e_A = \tfrac{1}{2}\sum_{i=0}^{17}(h_{i+1} - h_i)^2.$$

The sequence h is strictly concave. A site $5 \leq i \leq 13$ is weak if $h_i \leq \sqrt{D_i} - .65$. Deleting such i retains two paths of ranks at least four and the entire actual I. The finite one-node list is isolation; one simple bond at an end, D or permitted E position; an end-four; or two end-threes, one on each path. A joined component has rank seventeen and is classical. The retained foreign I cost Q_I is less than one and, if nonzero, at least $1/4$.

Own zero is removed geometrically before any comparison: internal-three roles have affine arms $1, 2, 5$ or $1, 3, 3$; internal-four gives the $4, 3, 3$ star; two whites give an affine cycle. A cap without an A contact cannot close the edge. End-four requires $Q_I < 1/m \leq 1/5$, hence $Q_I = 0$ and negative time square. In the remaining end-three/node-two cases the full old-component norms give $A_0^2 < Q_I$. The known actual I opposite is $\sqrt{h_I}q - \sqrt{1+h_I}d$; acuteness would require $A_0\sqrt{h_I} \geq \sqrt{Q_I}\sqrt{1+h_I}$, impossible. These are the zero-own calculations of Appendix C.6, with this actual opposite replacing its farthest-only orthogonal-opposite step.

For a white coefficient vector b, all remaining length data are computed by

$$\Delta = 1 - b^{\mathsf{T}} G_{\text{retained}}^{-1} b - Q_I, \qquad \sigma = \gamma + \langle \Omega_i, \sum_{j \neq i} b_j \Omega_j \rangle / D_i, \qquad C = \sqrt{D_i}\,\sigma/\sqrt{\Delta}.$$

The finite-tree calculation in that appendix gives ratio at least $3/2$ for near-three/own at least five, D/E, end-four and two-white rows, after their actual affine small-own boundaries. Positive foreign cost increases this ratio. Foreign far-three has ratio $> 227/(25\sqrt{35}) > 3/2$; foreign near-four has squared ratio $> 121/35$. A near-four with side order at least eight has ratio $> 761/500$, and side order seven has ratio $> 3/2$. For a pure helper with side order m,

$$\sigma = \gamma + 1/(2m), \quad \Delta = (m+1)/(2m), \quad \sigma^2/\Delta - (4\gamma - 2) = (\gamma - 1 - 1/(2m))^2/\Delta \geq 0.$$

In particular $\gamma \geq 17/16$ also gives ratio at least $3/2$.

These geometric bounds are sufficient under the new comparison. For $u = \sqrt{D_i} \in (\sqrt{65}/3, 3]$, the positive-square test at $C = 3u/2$ is

$$(3u/2 - 1.061)^2 - (1 - .65/u)^2(9u^2/4 - 1) = F(u) = 1.175096 - .258u - 1.3/u + .4225/u^2 > 0.$$

Indeed $F(3) > .0147$, and

$$F(u) - F(3) = (3 - u)\{.258 - 1.3/(3u) + .4225(3 + u)/(9u^2)\} \geq 0,$$

as $u > 8/3$. The actual directional floor $(C - \kappa)/\sqrt{C^2 - 1}$ increases here because $\kappa C > 1$. Every listed ordinary role is therefore strong.

For isolated A contacts, after positive clearing of the whole I, set $U = (1 - Q_I)/\gamma^2$; $A_0^2 = \gamma^2(D_i - U)$. For $0 < U < 1$ the positive reciprocal core gives

$$h_i > \sqrt{D_i - U} - 1.061\sqrt{U(1 - U)}.$$

Its loss is less than $.194U + 1.061\sqrt{U(1 - U)} < (.194 + 1.079)/2 = .6365 < .65$. For $U > 1$ the positive finite projection gives loss $< .194 + 1.061\sqrt{U - 1}$. For foreign $Q_I > 0$, $U - 1 \leq 5 - 3\phi < (5/12)^2$; for pure finite own at least eight, the same bound follows from $\tan(\pi/8) < 5/12$. The loss is $< .637$. The $U = 1$ core is a positive future feasible null. Pure own-five uses the actual H_4 finite point: its old loss is $< 25/76 + 5/24$, and replacing comparison one by 1.061 adds

$$.061\sqrt{\delta_5}\sqrt{1 + \delta_5/A_0^2}/\gamma < .016, \quad \delta_5 = (7 - 3\sqrt{5})/16 < 1/36, \quad \gamma > .8, \quad A_0^2 > 3.6.$$

Its total loss is $< .554$. Pure seven has old loss $< 2500/10368 + .36$ and added loss $< .061(.36) = .02196$, hence total $< .624$. Own three, four, six and parallel have the actual nulls proved in the same appendix. Thus no isolated or foreign role remains weak.

The provisional weak list is exactly pure near-four at $5, 6$ or reflection, or a pure far helper with

$$v = \gamma\Omega_i + \Omega_e/2, \quad e = 1 \text{ or } 17, \quad \phi/2 \leq \gamma < 17/16, \quad Q_I = 0.$$

This cutoff concerns only weak helpers. Their nonadjacency is the original-vector calculation of Appendix C.6. To specify its complete domain, for adjacent sites and owns $\alpha, \beta \in (.809, 1.08)$ the common/outward/reversed products have lower bounds

$$1 - 1.08^2, \qquad \tfrac{1}{2} - \alpha\beta + (\alpha - 1)(\beta - 1)/18, \qquad \tfrac{1}{2} - \alpha\beta + (\alpha + 1)(\beta + 1)/18.$$

Native gaps leave common mutual zero or opposite mutual zero/simple. The reversed zero case has numerator

$$18F = 4\ell s\alpha^2\beta^2 - (\ell + 1)(s + 1)(\alpha^2 + \beta^2) - 2\alpha\beta - (\ell + 1)\alpha - (s + 1)\beta + 5, \quad \ell + s = 17, \quad \ell, s \geq 5.$$

It increases in both owns on the permitted range. Unless both are five, one is at least seven and exceeds .9005; concavity in ℓ reduces to endpoints, whose smaller value is $.05526706486 > 0$ at $(.809, .9005)$. The both-five case is the actual distance-one determinant obstruction proved there, on its declared $N \geq 15$ domain. The outward zero case has determinant at least $404/225 > 0$. The resulting mixed projection is positive with positive inverse; adding the entire old I block leaves it positive. Its actual future finite point is feasible against every original wall. The complete inverse calculation bounds the weighted loss by $\kappa(7/16) + 1/10 < .5641875$ for a common helper and

$\kappa(3/10) + 1/10 < .4183$ for opposite helpers. Both are less than .65, proving nonadjacency before near-four has been removed.

A near-four has an actual mate at $j = 4, 5$, obtained by the full positive $A_{j-1} + B_{18-j} + I$ projection. Its geometric cosine exceeds ten, so its new demand exceeds $(10 - 1.061)/\sqrt{99}$, giving $h_4 > 2.24$ or $h_5 > 2.41$. At the other end there is a mate or a strong 12 or 13 by nonadjacency, with demand > 2.178 or > 2.037. Disjoint end-segment Cauchy has worst case

$$e_A > \tfrac{1}{2}(2.41^2/5 + 2.178^2/6) = .976117 > .8836,$$

contrary to $\eta^2 > .1164$. The other placements are stronger. Every remaining weak site is therefore a nonadjacent pure helper.

Here is the entire reference, including every allowed internal hole. Take symmetric heights

$$H_5 = 2.077,\ H_6 = 2.218,\ H_7 = 2.314,\ H_8 = 2.371,\ H_9 = 2.389, \qquad H_{18-i} = H_i,$$

linear from zero to 5, and scale them by $\rho_0 = .98$. Positive squares $(\rho_0 H_i + .65)^2 < D_i$ put them below the strong floors. Their positive successive slopes before scaling are $.4154, .141, .096, .057, .018$. Flatten the first block through 6 if 5 is weak and reflect if 13 is weak; replace each remaining weak knot by its neighbor chord. Deleting j costs $(H_{j-1} - 2H_j + H_{j+1})^2/4$ in reference energy. Nonadjacency makes these edge pairs disjoint. The following bounds cover all hole sets without enumerating subsets:

weak ends	unscaled initial energy	holes	loss per hole	scaled energy
0	.8954558	≤ 4	$\le .00050625$	$> .858$
1	$.8640827333\ldots$	≤ 3	$\le .00050625$	$> .8284$
2	$.8327096666\ldots$	≤ 3	$\le .00038025$	$> .7986.$

The initial values are exactly, with $a = 2.077,\ b = 2.218,\ T = .096^2 + .057^2 + .018^2$,

$$a^2/5 + (b-a)^2 + T, \quad \tfrac{1}{2}\{b^2/6 + a^2/5 + (b-a)^2 + 2T\}, \quad b^2/6 + T.$$

The available individual end-anchor capacities are at least $\rho_0(821/6000)$ on a six-block or $\rho_0(343/2500)$ on a five-block, both $> .134$. The signed block means satisfy $b_5 < .28812$, $b_6 < .98(.2614)$. Every remaining positive coefficient is on an actual strong site; no weak floor or I energy is in these references.

For completeness we now give the feet, deficits and all mutual residual terms needed to use these capacities. At a weak end let m_i, m_e, m_j be its actual own, far-helper and helper-neighbor roots. Write $U = v - k_0 m_i - \ell m_e - m m_j$. The complete own dispatch is

own	(k_0, ℓ, m)	corrected square	pressure with comparison one
5	$(\gamma, 2/3, 1/3)$	$\delta = 2/3 - \gamma^2 > 0$	$> A - \sqrt{\delta}$
finite ≥ 7	$(\gamma, 2(1-\gamma^2), 0)$	$-(1-\gamma^2)(4\gamma^2 - 3)$	$A - \sqrt{(1-\gamma^2)(4\gamma^2-3)}$
dotted	$(1/\gamma, 0, 0)$	$-(1-\gamma^{-2})$	$A - \sqrt{1-\gamma^{-2}}.$

In the first row the point is $q + (A/\delta)(b + k_0 m_i + \ell m_e + m m_j)$; the spacelike corrected normal itself is not a point. The other rows are future negative cores with zero cap product, nonpositive selected old products and all unknown products nonpositive termwise. Time coefficients are positive. All displacements are pure A_{17}.

For a far orientation the time is $A_0(\gamma)^2 = (65/9)\gamma^2 + (5/9)\gamma - 19/36$; near adds $8\gamma/9$. On two six-blocks the projections are

$$\Pi U = (5\gamma/6)\Omega_6 + \Omega_{12}/12 \quad \text{(far)}, \qquad \Pi U = (5\gamma/6 + 1/12)\Omega_6 \quad \text{(near)}.$$

Let $Y = U - \Pi U$. Its sum is zero separately on each end block, so $\langle w, Y\rangle = 0$ after every internal hole. The far residual squares are exactly

$$Q_5 = \tfrac{2}{3}\gamma^2 + \tfrac{1}{12}, \quad Q_f = 4\gamma^4 - \tfrac{16}{3}\gamma^2 + \tfrac{29}{12}, \quad Q_d = \tfrac{5}{3}\gamma^2 - \tfrac{19}{12} + \gamma^{-2};$$

near adds $\gamma/3$. Every finite far square is $< 13/12$, and every dotted far square is $< 148/125$.

We give rational seals for the pressure deficit against the unscaled pairing $2.218(5\gamma/6 + 1/12)$. For owns five, seven, eight and at least nine the respective lower bounds are

own	own bracket	time/radical bounds	deficit lower bound
5	$\gamma < .81$	$A_0 > 2.155,\ \sqrt{\delta} < .111$	$2.155 - .111 - 2.218(91/120) > .362$
7	$.9009 < \gamma < .901$	$A_0 > 2.4154,\ R < .2158$	$2.4154 - .2158 - 2.218(1001/1200) > .3494$
8	$.9238 < \gamma < .9239$	$A_0 > 2.4795,\ R < .2464$	$2.4795 - .2464 - 2.218(.85325) > .34059$
≥ 9	$\gamma > .93$	$A_0(.93) > 2.497,\ R \leq 1/4$	$2.497 - .25 - 2.218(103/120) > .3432.$

All root bounds are positive-square comparisons, using ϕ, the native-seven cubic $8u^3 - 4u^2 - 4u + 1$, and $\gamma_8^2 = (2 + \sqrt{2})/4$. For the last row the derivative applies to the lower envelope $A_0 - \frac{1}{4} - 2.218(5\gamma/6 + 1/12)$: $A_0' > \sqrt{65}/3 > 5(2.218)/6$. It is not a monotonicity claim for the actual varying-radical deficit. Thus the finite deficit is $> .3405$. For dotted owns split the whole interval $(1, 17/16)$ at $1.03, 837/800, 1687/1600$. The increasing function $A_0 - 2.218(5\gamma/6 + 1/12)$ has derivative $> .839$. Time lower bounds at the first three left endpoints are $2.692, 2.775, 2.82$; radical upper bounds for the four intervals are $.24, .295, .317, .338$. The fourth lower function value is the third plus $.839(13/1600)$. The resulting four deficits exceed $.4188, .3913, .3843, .37016$. This covers the entire dotted range, with deficit $> .37$; near times are larger.

Changing comparison from one to 1.061 costs $< .0071$ for five, at most .01525 for finite at least seven, and $< .0207$ for dotted. Indeed the relevant extra terms are $.061\sqrt{\delta}\sqrt{1 + \delta/A^2}$, $.061/4$, and $.061(.338)$; for five $A > 2, \sqrt{\delta} < .115, \delta < .013$. Scaling the reference down gains at least $.02(2.218)(.7575) = .0336027$. Thus every six/six deficit is, before rounding, strictly greater than

$$.3405 - .01525 + .0336027 = .3588527 > .358.$$

For one weak end, on six/five blocks the exact far projection is

$$\Pi U = (5\gamma/6)\Omega_6 + \Omega_{13}/10.$$

Relative to six/six, the pairing increases by $\rho_0(343/15000) = .022409333\ldots$ and the residual square decreases by $1/60$. The new far deficit is therefore strictly greater than

$$.3588527 - .98(343/15000) = .336443366\ldots > .336.$$

Near is unchanged. In particular the opposite price here is $\mu/10$, not $\mu/12$.

For two feet we retain their entire cross. Within each length-six block decompose the own and helper residuals as

$$a(\gamma, k_0) = \frac{\gamma\sqrt{2}}{6}(P_5 - 5e_6) - k_0 m_5, \qquad b(\ell, m) = \frac{e_1 - P_6/6}{\sqrt{2}} - \ell m_1 - m m_2.$$

Direct products give

$$a^2 = 5\gamma^2/3 + k_0^2 - 2\gamma k_0, \quad b^2 = 5/12 + \ell^2 + m^2 - \ell m - \ell \leq 5/12, \quad \langle a_i, b_j\rangle = \gamma_i/6.$$

For equal orientations, $\|Y_L + Y_R\|^2 = Q_{f,L} + Q_{f,R} + (\gamma_L + \gamma_R)/3$. For mixed orientation it is $Q_{f,L} + Q_{f,R} + 2\langle b_L, b_R\rangle + 2\gamma_{\text{near}}/3$, where $\langle b_L, b_R\rangle \le 5/12$. Thus the equal/mixed bounds are $17/6, 11/3$ for two finite owns, and $3.077, 3.91$ when a dotted own is present. These bounds include unequal parameters and both mixed orientations.

If both ends are weak, pay $\mu = .1$ on each foot. The total individual anchor prices are bounded by $31/320, 101/960, 17/192$, all $< .106 < .134$. For one weak end, pay $\mu = .13$. The near price is at most $.13(31/32) < .126$, with residual square < 1.539. For far, the own-anchor price is at most $.13(85/96) < .116$, the separate opposite price is $.13/10 = .013$, and its residual square is $< 1751/1500 < 1.168$. Each fits its own available anchor; a bound on total price alone would not suffice.

In all cases the payment is one completed square. If the reference energy is E_0, the residuals are Y_j, and each projection has been paid from nonnegative individual strong-site coefficients, then

$$\langle y, w + \mu \sum_j Y_j\rangle \ge E_0 + \mu \sum_j \Delta_j, \qquad \|w + \mu \sum_j Y_j\|^2 = E_0 + \mu^2 \|\sum_j Y_j\|^2.$$

For two weak ends, adding the one I coordinate gives

$$e_A + \eta^2 > .7986 + .2(.358 + .358) - .01(3.91) + .1164 = 1.0191.$$

For one weak end the gain is $> .067$: $.26(.358) - .0169(1.539) = .0670709$ near, and $.26(.336) - .0169(1.168) = .0676208$ far. Thus $e_A + \eta^2 > .8284 + .067 + .1164 = 1.0118$.

If neither end is weak, use the known actual signed endpoint coordinate $> .51$ from (54). On its five-block the mean b_5 satisfies $.28 < b_5 < .28812$. Adding $\lambda(e_{\text{end}} - P_5/5)$, with $\lambda = 5(.51 - b_5)/4$, spends only $(.51 - b_5)/(4\sqrt{2}) < .042 < .134$ at that anchor. It gains $(5/4)(.51 - b_5)^2 > .061538418$. Consequently $e_A + \eta^2 > .858 + .061538418 + .1164 > 1$. All weak-end counts are exhausted. The I energy was counted once and every comparison used the same original farthest vertex. This excludes the final orthogonal branch and hence every dotted far-helper seventeenth cap.

H.11 Near-three/own-six: no short opposite survives

The other left-contact terminal role has

$$g = \sqrt{3}/2, \quad c = Ap - \sqrt{2}gP_{17} - P_{16}/\sqrt{2}, \quad a = 65/2 + 16\sqrt{3}, \quad Q = \sqrt{2}(17g + 8), \quad Q^2 = 17a + 9.$$

The inventory made old 17 strong, so there is the actual isolated native-five head $c_H = A_H p - (\phi/\sqrt{2})e_1$, with $A_H^2 = (\phi - 1)/2$. Use the common short table in Appendix H.7. It already excludes end-four, next-three and near end-three. Far end-three has the same owns zero, three, four, six and parallel nulls as before.

For any remaining finite far-three short own, $\phi/2 \le \delta < 1$, its actual mutual magnitude with the head is

$$M_H(\delta) = A_H\sqrt{18\delta^2 + \sqrt{2}\delta - \tfrac{1}{2}} - \phi\delta/\sqrt{2} - \phi/2.$$

It increases strictly: the derivative of its first radical exceeds $\sqrt{18}$, and $\sqrt{18}A_H > \phi/\sqrt{2}$. At $\delta = .8$, positive squares give time product > 1.937 and spatial product < 1.725, so $M_H > .212$. At 1 the corresponding bounds are < 2.419 and > 1.9530878, hence $M_H < .4659122 < 1/2$. Every finite remaining own has $\delta > .8$, so $0 < M_H < 1/2$ contradicts the actual native gap. The caps are distinct by their original contact sets. Thus a far-three short opposite must have dotted own $\delta > 1$.

The following constants are deliberately coarse:

$$60.212 < a < 60.2136, \quad 7.759 < A < 7.76,$$
$$32.13 < Q < 32.14, \quad .1494 < w := 9/a < .1495,$$
$$Q/A > 4.141, \quad H := A - G_{17}Q/17 > .65,$$
$$0 < K := A\sqrt{18} - Q < .8.$$

They follow by positive squares from $1.732 < \sqrt{3} < 1.7321$; for the last bound use $(Q+.8)^2 - 18a = 1.6Q + 9.64 - a > 0$. Positive actual clearing now uses the opposite inverse column:

$$\bar{c} = c + \sum_{i=1}^{16}(i/17)n_i = Ap - (Q/17)P_{17}, \qquad \bar{c}^2 = r = 9/17.$$

This leaves the true short normal untouched.

We shall repeatedly use one quadratic seal. For $s > 0, \rho > 0, \rho^2 > rs$, let $y_0 = \sqrt{\rho^2/s - r}$. If $\rho < .8\delta$ and $J > .31\delta$, then

$$H + (\rho/s)J - y_0 > \frac{13}{20} + \frac{31}{80s}\rho^2 - y_0 = \frac{31}{80}(y_0 - 40/31)^2 + \frac{8853}{42160} > 0. \tag{55}$$

For an isolated short opposite $d = Bp - \delta P_{18}$, put $\lambda = AB - Q\delta \geq 0$. Acuteness gives $\delta^2 \geq a/(a-9) > 1$, so own is dotted. The cleared Gram is $\left(\begin{smallmatrix} r & -\lambda \\ -\lambda & 1 \end{smallmatrix}\right)$. At $\lambda^2 = r$, $\bar{c} + \lambda d$ is a positive future all-wall feasible null. For $\lambda^2 < r$, including $\lambda = 0$, the full positive projection gives an actual finite point whose equations imply

$$k = w + \frac{Q\lambda}{a\delta} \geq w, \qquad k^2 < w, \qquad S_{17} + kx_{18} \geq Q/A - \sqrt{w - k^2}.$$

The two original sum bounds bound the left side by $3.760 + .07k$. For $k \geq w$, rationalization gives

$$\sqrt{w - w^2} - \sqrt{w - k^2} \geq \sqrt{w}\,(k - w) > .07(k - w) \quad (k > w).$$

Hence the gap is minimized at $k = w$. There it exceeds $4.141 - .357 - 3.760 - .07(.15) = .0135$, since $w - w^2 < .1495 - .1495^2 < .357^2$. Every subcritical mutual is impossible. For $\lambda^2 > r$ the positive actual negative core is $\bar{c} + \lambda d$; its square is $r - \lambda^2$, selected products are $r - \lambda^2, 0$, its time is positive, and all unknown signs hold termwise. Its required pressure gap is $H + \lambda(B - G_{18}\delta) - \sqrt{\lambda^2 - r}$. Acuteness gives $B/\delta \geq Q/A > 4.141$, hence $J = B - G_{18}\delta > .31\delta$; also $\lambda < K\delta < .8\delta$. Equation (55) with $s = 1, \rho = \lambda$ contradicts it on the entire supercritical range.

Finally consider dotted far-three short: $d = Bp - \delta P_{18} - e_1/\sqrt{2}$, $\delta > 1$. Its mutual magnitude is $\lambda = AB - Q\delta - (g + 1/2)$. It is strictly positive on this whole range. Its derivative is positive by the square comparison

$$18(a-9)\delta^2 + \sqrt{2}(a-9)\delta + 9a + \tfrac{9}{2} > 0,$$

and at $\delta = 1$ the time/spatial square gap is $25/4 + 15g - \sqrt{2}(1+g) > 16.50395$. Clear d by the far inverse column to obtain

$$\bar{d} = Bp - (\delta + v_0)P_{17} - \delta e_{18}, \quad \bar{d}^2 = r, \quad \rho = -\langle \bar{c}, \bar{d} \rangle = \lambda + 1/34, \quad v_0 = 1/(17\sqrt{2}).$$

The offset is now $1/34$, not $8/17$. If $\lambda = 1/2$, then $\rho = r$ and $\bar{c} + \bar{d}$ is a forbidden positive future feasible null. The actual native gap then forces $\lambda \geq 1/\sqrt{2}$ and $\rho > r$. Use $\Xi = \bar{c} + (\rho/r)\bar{d}$. It has

square $r - \rho^2/r < 0$, positive time, selected products that same negative square and zero, and zero old A_{16} products. Every unselected original sign holds termwise. The necessary same-p gap is

$$H + (\rho/r)J - \sqrt{\rho^2/r - r}, \qquad J = B - G_{17}v_0 - G_{18}\delta.$$

The actual identity $\rho = AB - Q(\delta + v_0)$ implies $J > .31\delta$. Moreover $B < \sqrt{18}\delta + 1/6$ and $Qv_0 = g + 8/17 > A/6$, so $\rho < K\delta < .8\delta$. Equation (55) with $s = r$ excludes the entire remaining range, including arbitrarily large dotted mutuals. Thus no short opposite survives near-three/own-six.

H.12 A central A_{16} lemma with a rank-two spectator

The no-left terminal roles will lead to three types of actual $A_{16} + I_2(m)$ neighbors. We prove one local lemma with the precise three comparison regimes:

spectator	comparison factor κ	conclusion on the same local A_{16} projection
$I_2(3)$	$\kappa < 67/50$	$e_A > 1/2$
$I_2(m),\ m \geq 4$	$\kappa < 3/2$	$e_A > 7/16$
$I_2(5)$	$\kappa < 157/100$	$e_A > 15/32$.

Let the actual vertex be q, write $z = E_q(q - y)$, and suppose every original chamber point has height at most κE_q. Only the actual opposite b in central direction 8 is needed. In zero-sum $\mathbb{R}^{17}$,

$$\Omega_i = \sqrt{2}(P_i - iP_{17}/17), \quad D_i = 2i(17 - i)/17, \quad C_{ij} = 2i(17 - j)/17 \ (i \leq j).$$

Put $D = D_8 = 144/17$, so $2.91 < \sqrt{D} < 3$ and $\sqrt{D - 1} > 2.73$. Write $b = Hq - v_A - v_S$, own $\gamma = -\langle b, m_8\rangle \geq 0$, and whole-spectator cost $Q = \|v_S\|^2 < 1$.

The complete foreign alphabet follows from the retained dihedral inverse. Two contacts make a nonpositive triangle. For one finite contact ϵ the cost is $\epsilon^2/\sin^2(\pi/m)$, and finiteness requires $\cos^2(\pi/m) + \epsilon^2 < 1$. Thus

m	Q
3	$0,\ 1/3,\ 2/3,\ \phi^2/3$
4	$0,\ 1/2$
5	$0,\ (\phi + 2)/5$
≥ 6	0.

In particular the genuine H_3 cost for $m = 5$ is $(\phi + 2)/5 = (5 + \sqrt{5})/10 > 7/10$. The whole block is cleared by its positive inverse on actual roots.

Deleting 8 retains A_7 and A_8. The complete finite-white overlist is one three at $j = 1, 2, 3, 5, 6, 7, 9, 10, 15, 16$; one four at $j = 1, 7, 9, 16$; or two end-threes, one from $1, 7$ and one from $9, 16$; isolation is also allowed. The left $3, 5$ positions are the E_8 additions, $2, 6, 10, 15$ are D additions. Two contacts on one path make a cycle. A joined rank-sixteen tree is classical and has no new internal four. The exceptional finite classification gives no other extension of either retained full path. Every nonempty white cost is at least $7/16$. Thus its retained Schur remainder is at most $9/16$; for $m = 5$ any white contact forces $Q = 0$.

Here is the complete own-zero boundary. For an endpoint-three the time square is $Q - 9/17$, but retention requires $Q < 9/16$. The displayed alphabets allow only $0, 1/3, 1/2$ under that bound, all too small. For next-to-end three the retained remainder is at most $1/4$, hence $Q = 0$ and time square $-2/17$. For endpoint-four it is at most $1/8$, hence $Q = 0$ and time square $-1/17$. Every other internal-three has a positive affine principal: E_8 with arms $1, 2, 5$ at global $3, 14$, or E_7 with

arms $1,3,3$ at $4,\dots,13$. Their kernel coefficients are respectively center 6, arms 3; $4,2$; $5,4,3,2,1$, and center 4, arms 2; $3,2,1$; $3,2,1$. Interior-four gives the $4,3,3$ star with center $\sqrt{2}$, four-leaf 1, simple leaves $1/\sqrt{2}$. Two whites give an all-three cycle with coefficients one. With no white contact the time square is $Q-1<0$. Every null is a positive actual combination, with positive cap time, selected products zero and every other original product nonpositive. Compactness excludes all of them.

For positive own, an own-three together with white-three closes an actual all-three cycle through 8. For a pure far-three helper, own-four gives the $4,3,3$ star at 8, own-six gives the null $\gamma b+m_8+m_7/2$, and parallel gives $b+m_8$. The same own nulls will be used in isolated cases below. The remaining ordinary edge ratios are obtained from the exact formula

$$R=\frac{C_8}{\sqrt{D}}=\frac{\gamma+\beta C_{8j}/D}{\sqrt{1-\beta^2(D_j-C_{8j}^2/D)-Q}}.$$

For two whites add both numerator terms; their retained cross is zero. We list bounds for every ordinary position, leaving only isolation and pure far-three as separate cases:

white position	numerator lower bound	denominator upper bound before foreign cost
near three left/right	$.7+7/16,\ .7+4/9$	$3/4,\ \sqrt{5}/3$
D left/right	$.7+1/8,\ .7+1/9$	$1/2,\ \sqrt{2}/3$
E left	$.7+3/16$	$1/4$
four left/right	$.5+1/(8\sqrt{2}),\ .5+1/(9\sqrt{2})$	$1/\sqrt{8},\ 1/3$
two threes	$.5+17/144$	$\sqrt{17}/12.$

Smaller owns in these rows are the actual cycle/affine boundaries already disposed of. For a far-three with $Q>0$, the $m=3$ alphabet gives $Q\ge 1/3$; the residual is smaller than $.56^2$ and the numerator exceeds .864. For $m\ge 4$, positive $Q\ge 1/2$ makes this bound stronger. Thus in the first two comparison regimes every ordinary row has $R>3/2$, and $C_8>4.36$. The directional floor is then

$$(C_8-\kappa)/\sqrt{C_8^2-1}>\begin{cases}(4.36-1.34)/4.244>.71,\\(4.36-1.5)/4.244>.67.\end{cases}$$

Its square exceeds $1/2$ or $7/16$, respectively.

For the third regime, white contact already forces $Q=0$. One near-three/own-four needs an actual mate, rather than an incorrect null shortcut. If $v_A=\Omega_8/\sqrt{2}+\Omega_7/2$, delete actual 7; the full finite Gram is $A_6+B_{10}+I_2(5)$. For $\Omega_9/2$ instead, delete 9 and obtain $B_9+A_7+I_2(5)$. Their positive actual projections are feasible against every original wall. The exact edge data are

j	σ_j	Δ_j
7	$1/2+9/(10\sqrt{2})$	$1/10$
9	$1/2+8/(9\sqrt{2})$	$1/9.$

As $\sqrt{D_j}>14/5,\ \sigma_j>9/8,\ \sqrt{\Delta_j}\le 1/3$, their cosines exceed $189/20>9$. The new demand is $>(9-1.57)/\sqrt{80}>.82$, more than sufficient. An own-four with no neighboring single-three has the star null; two-white or white-four roles use their ordinary bounds. Every remaining white-three has own at least five, $>.809$. The same ratio formula gives: near-three $>(\phi/2+7/16)/(3/4)>1.662$; $D>(.809+1/9)/(.5)>1.84$; $E>(.809+3/16)/(.25)$; far-four at least $\sqrt{2}+1/4>1.664$ on the

left and $3/2 + 1/(3\sqrt{2}) > 1.7$ on the right; two whites $> (.7 + 17/144)/.344 > 2$. Thus all these rows have $R > 1.66,\ C_8 > 4.83$, and demand

$$(C_8 - 1.57)/\sqrt{C_8^2 - 1} > (4.83 - 1.57)/4.726 > 11/16.$$

Its square is $121/256 > 15/32$.

It remains to cover isolated A contacts, retaining every permitted foreign cost. Put $\tau = (1 - Q)/\gamma^2$; then $H^2 = \gamma^2(D - \tau)$. Own three, four, six and parallel have the positive actual nulls already listed, with own-three using the E_7 star at 8. For $0 < \tau < 1$, positive clearing of the whole spectator gives

$$\Xi = b + w_S + \gamma\tau m_8, \quad U = \gamma(\Omega_8 - \tau m_8), \quad \Xi^2 = -\gamma^2\tau(1 - \tau), \quad U^2 < H^2.$$

It has zero cap/foreign products, negative own product, positive time and every unknown sign termwise. The comparison gives $\langle y_A, U\rangle \geq H - \kappa\gamma\sqrt{\tau(1 - \tau)}$. If $e_A \leq 1/2$, Cauchy instead forces

$$(H - \langle y_A, U\rangle)/\gamma > (1 - 1/\sqrt{2})\sqrt{D - \tau} > .2928(2.73) = .799344 > \kappa/2,$$

a contradiction for every $\kappa \leq 1.57$. Thus this whole continuous interval gives the stronger $e_A > 1/2$ in all regimes. The boundary $\tau = 1$ is an actual null.

For $\tau > 1$ use the actual positive finite point

$$q_0 = q + \frac{H}{\gamma^2(\tau - 1)}(b + w_S + \gamma m_8).$$

Its selected products vanish, its norm and q product are $-1 - H^2/[\gamma^2(\tau - 1)]$, and every unknown wall sign holds by positive actual coefficients. Normalization gives exactly

$$h_8 \geq \sqrt{\frac{D - 1}{D - \tau}}\,\{\sqrt{D - 1} - \kappa\sqrt{\tau - 1}\}.$$

For $Q > 0$, the native gap gives $\tau - 1 \leq 5 - 3\phi < (5/12)^2$. For pure finite own at least eight use the sharper $\sqrt{\tau - 1} \leq \tan(\pi/8) < .415$. In the first regime the absolute loss in dual height satisfies

$$\sqrt{D} - h_8 < .18 + 1.34(.415) = .7361 < .74.$$

Since $\sqrt{D} > 2.91$, normalization gives

$$h_8/\sqrt{D} > 1 - .74/2.91 > .74,$$

and hence energy $> .5476 > 1/2$; the same bound covers the foreign case with its $\sqrt{5 - 3\phi} < .383 < .415$. In the second, $h_8 > 2.73 - 1.5(.415) = 2.1075$, whose normalized value is $> 2/3$, giving energy $> 4/9 > 7/16$. In the third $h_8 > 2.73 - 1.57(.415) = 2.07845 > 2.07$, giving normalized value $> .69$ and energy $> .4761 > 15/32$.

Pure own-five has the actual H_4 correction $\gamma(3m_8/2 + m_7 + m_6/2)$, with $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36,\ H^2 > 4, \gamma > .8$. Its actual finite point yields

$$h_8 > \sqrt{D - \gamma^{-2}} - \kappa\sqrt{\delta_5}\sqrt{1 + \delta_5/H^2}/\gamma.$$

For $\kappa \leq 1.34$ its loss is $< 25/76 + 1.34(1.004)/(6(.8)) < .611 < .74$, so energy $> .5476$. For 1.5, the first radical is > 2.5 and the last square root < 1.01, giving $h_8 > 699/320$, enough for $4/9$. For

1.57, the first radical is > 2.62, the loss $< 15857/48000$, and $h_8 > 2.07$. Pure seven has the actual core $b + (4\gamma/3)m_8 + (2\gamma/3)m_7$, whose norm and cap product are $1 - 4\gamma^2/3 < 0$, with old products zero. The cubic $8u^3 - 4u^2 - 4u + 1$ is increasing on $[.89, 1]$; its signs at $.9, 10/11$ give $.9 < \gamma < 10/11$. Thus $\gamma^{-2} < 100/81$ and $\sqrt{4/3 - \gamma^{-2}} < 9/25$. The pressure is

$$h_8 > \sqrt{D - \gamma^{-2}} - \kappa\sqrt{4/3 - \gamma^{-2}}.$$

For the three regimes use respectively loss $< .25 + 1.34(.36) < .74$, first radical > 2.6, or first radical > 2.68. The latter gives $h_8 > 2.1148 > 2.07$. All required isolated bounds follow.

Finally consider every pure far-end-three helper, including its unbounded dotted range:

$$v_A = \gamma\Omega_8 + \Omega_e/2, \quad e = 1, 16, \qquad H^2 = D\gamma^2 + C_{8e}\gamma - 9/17 > D\gamma^2, \quad C_{8e} \geq 16/17, \quad \gamma > .8.$$

Own three, four, six and parallel have the actual cycle/star/chain nulls above. For five, the actual correction $\gamma m_8 + (2/3)m_e + (1/3)m_j$, where $j = 2, 15$, has residual $\delta = 2/3 - \gamma^2 \in (0, 1/36)$. The positive finite point gives $\langle y_A, U\rangle > H - \kappa\sqrt{\delta}\sqrt{1 + \delta/H^2}$, with $U^2 = H^2 + \delta$. Even at $\kappa = 1.57$ the loss is $< 15857/60000 < 2/7$; as $H > 2$, $U^2 < (10/9)H^2$, this gives $e_A > 162/245 > 1/2$. For finite owns at least seven let $d_0 = 1 - \gamma^2 \in (0, .19)$. The core $b + \gamma m_8 + 2d_0 m_e$ has square $-d_0(1 - 4d_0)$, zero cap/own products, and helper product $-1/2 + 2d_0 < 0$. Its radical is at most $1/4$, while $H > 2.5$; the demand is $> H - 157/400 > .84H$, giving $e_A > 441/625 > 1/2$. For dotted $\gamma > 1$, the reciprocal core $b + m_8/\gamma$ has square $-(1 - \gamma^{-2})$. Its radical divided by γ is at most $1/2$. As $H > \gamma\sqrt{D}$, the demand/norm ratio is $> 1 - 1.57/(2\sqrt{D}) > .73$, giving $e_A > .5329 > 1/2$. All cores have positive actual coefficients and time; all selected products were checked, and all other original-wall products are nonpositive termwise. This finishes the three-regime local lemma without a new farthest choice, a helper cutoff on arbitrary caps, or an omitted foreign cost.

H.13 No-left caps: actual-neighbor recognition and all common cases

The remaining seventeenth opposite has the form

$$c = Ap - \sqrt{2}gP_{17} - \delta P_{18}, \qquad \delta = \cos(\pi/m), \quad 3 \leq m \leq 8,$$

with the own table in Appendix H.5, and

$$A^2 = 34g^2 + 34\sqrt{2}g\delta + 18\delta^2 - 1.$$

First distinguish actual common and distinct caps. The full Gram on old A_{17} and c has Schur residual $1 - 17g^2/9$. If $g = 0, 1/2, 1/\sqrt{2}$, this is positive; its positive full projection is the actual old short-edge endpoint. The actual seventeenth opposite is therefore also the actual short opposite, by uniqueness of the old edge endpoint. For $g = 1/\sqrt{2}$ the reciprocal short cosine is $18\delta + 17 > 9.81$, excluding these common-own-four roles. For $g \geq \phi/2$ the residual is negative, so the cap cannot also close the short edge. This is the complete distinction for the own values left by the table; no native product is applied to a wall paired with itself.

At $g = 0, m = 3$, the actual affine F_4 path $n_{15}, n_{16}, n_{17}, n_{18}, c$ has positive kernel $(1, 2, 3, 2\sqrt{2}, \sqrt{2})$, already excluded. For $g = \phi/2, m = 4$, positive clearing of the whole old A_{17} gives the actual negative core

$$\Xi = Ap - sP_{18}, \quad A^2 = (51\phi + 33)/2, \quad s = \sqrt{2}(18 + 17\phi)/36, \quad \Xi^2 = -(17\phi - 19)/36.$$

Its selected old products vanish, its cap product is its negative square, its time is positive, and every other original sign holds termwise. Its pressure implies

$$S_{18} > \{A - \sqrt{(17\phi - 19)/36}\}/s > 3.97,$$

using $A > 7.599$, radical $< .487$, $s < 1.79$. This contradicts the unconditional sum bound. Thus the only distinct-cap family has $m = 3$ and $\phi/2 \le g < 1.264$.

For each of these remaining families let q be the actual seventeenth endpoint. Its complete actual forest is $A_{16}(n_1, \dots, n_{16}) + I_2(m)(c, n_{18})$. The cap and short root have their original finite mutual $-\cos(\pi/m)$; the full Gram is positive. Its full positive actual projection checks every unknown wall. Put

$$C = \cosh d(p, q), \quad s = \sqrt{C^2 - 1}, \quad t = S_{17}/\sqrt{17}, \quad q = Cp - sP_{17}/\sqrt{17}, \quad E_q/E = C - st.$$

The reverse unit tangent belongs wholly to the dihedral component. Its pressure is

$$\frac{\beta - t}{1 - \beta t}, \qquad \beta = s/C. \tag{56}$$

It increases in β and decreases in t. Thus the reverse-direction energy and the A_{16} energy from Appendix H.12 are orthogonal projections of the same local vector.

For the distinct family $m = 3, g \ge \phi/2$,

$$C = \sqrt{136/3}\{g + 1/(2\sqrt{2})\} > 7.82, \qquad t < .912.$$

The function $C - .912\sqrt{C^2 - 1}$ increases here, so

$$E_q/E > 7.82 - .912(7.756) = .746528 > 50/67.$$

Also $\beta > .9917$, and (56) is $> .83$. The local $A_{16} + I_2(3)$ lemma gives $e_A > 1/2$, while the actual dihedral projection has energy $> .6889$. Their sum exceeds one, excluding every distinct cap.

Of the seven common cases left by the own table, five are $g = 0, m = 6, 7, 8$ or $g = 1/2, m = 4, 5$. Their exact forward cosine is

$$C = \sqrt{34}\,\frac{g + \delta/\sqrt{2}}{\sqrt{1 - \delta^2}}.$$

For $g = 0, m \ge 6$, $C \ge \sqrt{51} > 7.1$; for $g = 1/2, m \ge 4$, $C \ge \sqrt{68}$. Again $t < .912$, hence

$$E_q/E > 7.1 - .912(7.03) = .68864 > 2/3.$$

Now $\beta > .99$, so the reverse pressure is $> (.99 - .912)/(1 - .99(.912)) > .8$. The second local regime gives $e_A > 7/16$; the dihedral projection costs $> .64$. Their sum is $> 431/400 > 1$. This excludes all five common cases.

For the sixth case, $g = 0, m = 5$, we first improve the original comparison without adding a spent head energy. The actual cap is $c = Ap - (\phi/2)P_{18}$, with $A^2 = (9\phi + 7)/2$. If old 17 were strong the actual head would be native five. The distinct head and cap would then have mutual magnitude

$$M = \sqrt{7\phi + 2}/2 - (\phi + 1)/(2\sqrt{2}).$$

Rational squares from $1.618 < \phi < 1.6181$ and $1.4142 < \sqrt{2} < 1.4143$ give $.899 < M < .900$. But this lies strictly between the native-six and native-seven values, a forbidden actual mutual. Therefore $S_{17} \le f_{17}$, and $t < .901$. For its actual $A_{16} + I_2(5)$ neighbor,

$$C^2 = \frac{17(\phi + 1)}{3 - \phi} > 32.2 > 5.67^2, \quad \beta > .984,$$
$$E_q/E > 5.67 - .901(5.582) = .640618 > 100/157.$$

The reverse pressure is $> (.984 - .901)/(1 - .984(.901)) > .73$. The third local regime gives $e_A > 15/32$, while the orthogonal dihedral energy is $> .5329$. The sum exceeds $20033/20000 > 1$. The head gap was used solely to establish old weakness; no head payment was added to this neighboring energy.

The final common case is $g = 1/2, m = 3$. Here $c = Ap - P_{17}/\sqrt{2} - P_{18}/2$, with $A^2 = 12 + (17/2)\sqrt{2}$. The full positive old-$A_{17}$-plus-$c$ projection is the actual short endpoint q, and its whole incident diagram is A_{18}, not the $A_{16} + A_2$ diagram of the other endpoint. Its Schur residual is $19/36$, and

$$C = \frac{17 + 9\sqrt{2}}{\sqrt{19}}, \quad C^2 = (451 + 306\sqrt{2})/19 > 6.82^2, \quad t = S_{18}/\sqrt{18} < .903.$$

Consequently

$$E_q/E > 6.82 - .903(6.747) = .727459 > 8/11.$$

The reverse unit tangent is the positive normalized dual for the new A_{18} endpoint c. With $\beta > .989$, its pressure is

$$\langle y, u_{18} \rangle > \frac{.989 - .903}{1 - .989(.903)} > .8 > .78.$$

Every original chamber point has height at most $E < (11/8)E_q$. The exact hypotheses of Lemma 9.1 therefore hold at this actual whole-A_{18} vertex. That lemma is local and does not require q farthest. It excludes this last common case. Its endpoint threshold concerns the normalized dual direction; it is not a threshold on a raw signed coordinate. All original wall signs here come from the full positive finite projection.

H.14 The terminal composition

The exhaustive seventeenth inventory has three disjoint branches. A far-left-three helper has finite own five or at least seven, or dotted own $1 < g < 4/3$. Appendix H.6 excludes every finite own, and Appendices H.8–H.10 exclude every dotted own, every actual short-white profile and every zero/native/parallel/unbounded dotted mutual. The near-left-three branch has exactly own six and is excluded by Appendix H.11. The no-left/short-contact branch is exhausted by the distinct/common recognition and the seven common cases above. Every other white role and every isolated seventeenth opposite was removed before this split. It follows that an original farthest whole-B_{18} vertex cannot occur.

All arguments are geometric. Literal finite/affine classification, compact acute simplicity, inverse positivity and the actual unit-wall angle alphabet are the external mathematical inputs. The finite tables here are explicit rational/algebraic inequalities, not new enumeration outputs or runtime receipts. The one original farthest vector is retained throughout each original reference argument. A neighboring vector is introduced only with its derived comparison factor, and all its orthogonal energies are taken at that same actual neighbor. The whole-D_{18} case and the independent higher-dimensional exclusions are proved in the remaining parts of the article.

I The earlier whole-D_{18} geometry

This appendix supplies the head, prefix and fork inputs used before the final dotted-head argument in the main text. We keep its geometric steps in dependency order. In particular an old loss .61 is used only to establish head isolation; the later loss .58 is proved afresh before it is used.

Fix the original interior point and farthest vertex

$$z = E(p - x), \quad p^2 = -1, \quad x \perp p, \quad \|x\| < 1.$$

The entire incident diagram is D_{18}, with actual unit normals

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (1 \le i \le 17), \qquad n_{18} = (e_{17} + e_{18})/\sqrt{2}.$$

For $P_i = \sum_{j \le i} e_j$, the duals are

$$\Omega_i = \sqrt{2} P_i\ (i \le 16), \quad \Omega_{17} = (P_{17} - e_{18})/\sqrt{2}, \quad \Omega_{18} = (P_{17} + e_{18})/\sqrt{2}.$$

Thus $x_1 > \cdots > x_{17} > |x_{18}|$; the last coordinate is signed. The fork duals have squares nine and mutual product eight, and $\langle \Omega_i, \Omega_f \rangle = i$ for $i \le 16$, $f = 17, 18$. All original reference arguments concern this one x.

We use the main actual-point and payment lemmas. A positive correction on actual normals retains every unknown original wall sign by acuteness. Every core below has positive time; its negative or null norm selects the future sheet. A spacelike corrected root is used only through its positive finite projected point. Every coefficient payment is made at its individual reference anchor before its pressure is used. A zero-sum residual need not be orthogonal to the reference; the general completed square retains that full cross.

I.1 An initial window and actual head isolation

Put $f_i^o = \sqrt{i} - .61/\sqrt{2}$. The generic prefix proof in Appendix E applies through 13, with its actual M_{12} rank condition. It gives a complete own/white list there and nonadjacent weak sites; a weak cap is

$$c = Ap - \gamma \Omega_i - \Omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad \phi/2 \le \gamma < 6/5,$$

with own five, finite at least seven or dotted. In particular a strong $K = 10, 11$ exists before either new tail. There are no spectators.

Deleting old 1 retains the full D_{17}. Finite classification leaves an isolated opposite $H = A_H p - \eta \Omega_1$, $A_H^2 = 2\eta^2 - 1$, or one additional simple bond at old 2, with $A_H^2 = 2\eta^2 + 2\eta$. A second contact makes a cycle; any other branch or four gives a nonfinite principal at this rank. In the isolated row owns zero, three and four have nonpositive time; six has its actual $6, 3$ affine chain, and parallel has $H + n_1$. In the nonisolated row own zero has zero time and own three an all-three triangle. Every remaining own is at least $1/\sqrt{2}$, and its actual edge has cosine $2\eta + 1$, so $x_1 > .643$.

We first extend the old-loss window through $14, 15$. Deleting these retains $A_{13}+D_4$ and $A_{14}+D_3$. There is at most one contact on each connected retained piece. A connected full completion has rank eighteen, so is classical; an exceptional finite diagram adds no bridge. The target ratio is

$$T_o(q) = 100/61 - (q - .61)/[q(2q - .61)] < 1.56 \qquad (q = \sqrt{28}, \sqrt{30}).$$

Isolation and left-only contacts use the same Schur formulas as the generic prefix. At isolated own five the left H_4 finite point has loss $< .147 + .174$; every finite own at least seven has loss

$$\frac{1.232}{\sqrt{28} + \sqrt{28 - 1.232}} + 1/\sqrt{12} < .408;$$

dotted loss is $< .096 + .5$. The actual zero/small-own nulls are the double-fork affine D chain, the four/simple star, the $6, 3$ chain and the parallel pair. For a left-only own zero, far-three has

negative time, next-far-three and far-four have zero time; near-three/next-near-three give the affine two-fork D principal and near-four gives affine B. Positive own-three with a left-three closes the intervening cycle. All surviving left-only ratios other than the far helper exceed 1.59, 2, or $\sqrt{14}/2$. These leave exactly the displayed pure helper.

The complete new right-hand data are

i, right contact	σ	Δ
14, long leaf three	$\gamma + 1/2$	$1/2$
14, either fork three	$\gamma + 1/4$	$1/2$
15, middle three	$\gamma + 1/2$	$1/2$
15, end three	$\gamma + 1/4$	$5/8$
15, end four	$\gamma + 1/(2\sqrt{2})$	$1/4$.

At 15 an end-five H_4 is also allowed, with ratio > 2.99 already at own zero. The long/middle-three roles have own-zero double-fork affine principals and own-three triangles; all remaining owns give ratio at least $1 + 1/\sqrt{2}$. The fork/end-three roles have own-zero affine E_8, own-three cycles, own-four stars, own-six chains and the parallel null. Their only extra old-loss cases are own five at either index and finite own at least seven at 15. The end-four own-zero kernel has coefficients $(\sqrt{2}, 2, 2, 1, 1)$ along its actual cap, contacted fork, center, other fork and old 15. Bridges are only left-end-three joined to a D_4 leaf, or to the middle/end of D_3, with residuals $1/28, 1/30, 19/120$. Own zero closes an actual cycle; every positive own gives ratio > 1.96. This exhausts the finite list, including both forks.

We exclude the extra right cases without assuming head isolation. If old 5 is strong, a strong $K = 10, 11$ reference has energies $> .823735, > .8472225$, sums $< 2.732, < 2.886$, terminal capacity $> 1/7$, and fifth capacity $> .146$. Every possible actual head supplies a separately funded pure gain $> .0135$: isolated five by its H_4 point, finite at least seven by $H + n_1/\eta + 2(\eta^{-1} - \eta)n_2$, dotted by $H + n_1/\eta$, and nonisolated by $x_1 > .643$. The exact scalar seals for these rank-free points are proved in Appendix C.7 and the head calculation below; their fifth prices are at most .1, with zero sum on the first five. If old 5 is weak, the actual weak-five point and strong 6 give instead bare first-six energies $> .8870, > .9104$, with no additional head gain. For a pure tail vector V, equal to one through K, with $\langle x, V\rangle > G$, $V^2 < R$, a legal $1/7$ payment gives

$$\|x\|^2 > L_K + 2(G - f_K^o)/7 - (R - K)/49.$$

The extra head gain is included only in the already-strong-five row.

At own five set $\alpha = \sqrt{2}(\phi/2 + 1/4)$; $\alpha^2 = \phi + 5/8 > 56/25$. The positive actual left-H_4 correction is $(\phi/2)(3n_i/2 + n_{i-1} + n_{i-2}/2)$; its residual is $(5 - 3\phi)/8$. The projected point has normalized displacement flat through $i - 3$. It gives $G > 3.62, R = 14$ at 14, and $G > 3.74, R = 15$ at 15. The least strong-five totals exceed 1.009, and the bare weak-five totals are larger. For every finite own at least seven at 15, put $u = \gamma^{-2}$. Use both actual positive cores

$$c + \gamma^{-1}n_{15} + 2(\gamma^{-1} - \gamma)n_{14}, \qquad c + \gamma^{-1}n_{15} + 2(\gamma^{-1} - \gamma)n_{16}.$$

Their square is $-d$, $d = 7 - 3u - 4/u$, and their correction cross is $2 - u$. Averaging their spatial displacements gives $Y^2 = A^2 - 3 + u + 2/u < A^2$. Here $.9 < \gamma < 1$, $d \le 7 - 4\sqrt{3} < .0718$, $\alpha^2 > 2.645$, and $A^2 = 15\alpha^2 - 5/8$. Thus Y/α is flat through 13, with $G > 3.677, R = 15$. The two strong-five totals exceed $1.00519, 1.00508$; the bare weak-five totals exceed 1.054. All coefficients are positive and selected products nonpositive, so these are actual all-wall points or cores.

For adjacent hypothetical weak helpers with owns $a, b < 6/5$, the actual product is at least $\frac{1}{2} - ab + (a/b + b/a)/4 \ge 1 - ab > -11/25$. The native gap forces orthogonality. Clearing the

retained left path gives $I - t\mathbf{1}\mathbf{1}^{\mathsf{T}}$, $t = (i-1)/(2i)$, positive with positive inverse. The two-helper mixed-point calculation of Appendix E uses only this matrix, times > 2 and owns $> 4/5$; its weighted loss is $< 7/16 + 1/(4\sqrt{10}) < .61$. The retained right block at $14, 15$ is actual D_3. Hence helpers are nonadjacent through 15.

We next prove old 5 strong; no head isolation is yet used. If it is weak, 6 is strong and its actual point gives $x_1 > .5965$. A finite weak-five own gives the stronger $x_1 > .65$; its reference through $1, 6$ and a strong $K = 14, 15$ has energy > 1.00845, so the own must be dotted, $1 < a < 6/5$. With $a_0 = .5965$, the reference through $1, 6$, then every actual strong knot, has initial energy $> .7600299945$, capacities $P_1 > .312168$, $P_6 > .08806$, and terminal $P_{14} > .1361$. All omitted weak knots are nonadjacent. Their complete worst debit is

$$[(.013586)^2 + (.009296)^2 + (.006871)^2 + (.005344)^2]/2 < .000174.$$

The underlying squared increments satisfy $\sum_{j=6}^{13}(\sqrt{j+1} - \sqrt{j})^2 > .2117655$ and the next one $> .0172465$, giving raw energies

$$q_{14} > .97162, \qquad q_{15} > .988867.$$

These bounds follow directly by rational upper square bounds for $\sqrt{42}, \dots, \sqrt{210}$, or the increment identity $d_j^2 = 2j + 1 - 2\sqrt{j(j+1)}$. At $a_0 = .65$ the initial energy is $> .7968668845$, which proves the finite-own exclusion above.

For the actual full dotted weak-five foot put

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \qquad T_5 = \sqrt{10a^2 + 2a - \tfrac{1}{2}} - \sqrt{1 - a^{-2}}.$$

Its pressure-minus-THIS-reference deficit is $> .1203$. To check the whole interval, the time derivative is < 3.245, the coefficient of a in $\langle w, U_5\rangle$ is > 2.4519, and the radical derivative is > 1, so the deficit decreases. At $a = 6/5$ use time > 4.03732, radical $< .552771$ and $\langle w, P_5\rangle < 1.733828$, giving $> .120355$. Pay $\mu = .07$, $r = .088$, and $q = \mu(\sqrt{2}a + 1/\sqrt{2}) - r \in (0, .081)$ at the separate P_1, P_6 anchors. For $Y_5 = \mu U_5 - qP_1 - rP_6$,

$$Y_5^2 = .0392a^2 + .0049a^{-2} - .04928\sqrt{2}a + .02892 < .005141.$$

Convexity reduces this bound to the two endpoints. The general completed square, retaining its full reference cross, gains $> .011701$. If 15 is strong this already gives $\|x\|^2 > 1.000568$.

If 15 is weak, 14 is strong and its pure helper has time $A_{15}^2 = 30g^2 + 2g - 1/2$. For finite own at least nine, the actual positive finite pair point has spatial foot $g(\Omega_{14} + \Omega_{16})/2 + \Omega_1/2$. For all $g > \cos(\pi/9) > .939$ its deficit exceeds

$$4.9529 - .939(4.68151) - .42179 = .13517211.$$

Indeed at .939, the exact time and pair normalization give $H > 4.9529$; the deficit increases since $H' > \sqrt{30}/1.066 > 5.13$. Its tail residual has square < 1. Paying .0962 costs $< .136049$ at P_{14} and $< .069$ at P_1, no P_6; its tail is orthogonal to Y_5. Its gain $> .016752673964$, together with the previous one, gives $\|x\|^2 > 1.00007367$. For dotted 15, weakness forces $1 < g < 10/9$; the positive reciprocal core has tail square < 1.28 and deficit $> .16613$. Its .086 payment fits the same two anchors, has no P_6 price and gains $> .0191$. The entire finite and dotted tails are covered by these derivative and endpoint bounds, not by a finite sample.

For owns five, seven and eight do NOT insert the weak-five foot. Their own tail points suffice on raw q_{14}:

own	actual correction	deficit lower bound
5	$gn_{15} + (2/3)n_1 + (1/3)n_2$	.3776769
7	$gn_{15} + n_1/2$	.28605949
8	$gn_{15} + n_1/2$	.232552911.

The first row is the actual H_4 finite point; the other two are negative cores. Their spatial feet are $g(\Omega_{14} + \Omega_{16})/2 + \Omega_3/6$ or $g(\Omega_{14} + \Omega_{16})/2 + \Omega_2/4$. The native brackets $.809 < g_5 < .81$, $.9009 < g_7 < .901$, $.9238 < g_8 < .9239$, time/radical pairs 4.5554/.111, 5.0646/.249, 5.1913/.322 and $\langle w, \Omega_3/6\rangle < .2747, \langle w, \Omega_2/4\rangle < .3115$ give the table by direct subtraction. The residual squares are < 1, including the initial variances $1/15, 1/10$. Payment .09 costs $< .118$ at P_{14} and $< .009$ at P_6, and gains $> .0337$. No weak-five foot is present, so there is no duplicate initial debit. Thus old 5 is strong in every case.

A strong $K = 14, 15$ now has the ordinary old-loss reference energy $> .90715$, first-five mean $< .361$ and fifth capacity $> .146$. A nonisolated head would give $x_1 > .643$, and its legal five-block coordinate graft gains $> (5/4)(.643 - .361)^2 > .0994$, contradicting the norm budget. The head is therefore ACTUALLY isolated, with own five, finite at least seven, or dotted. This conclusion did not use any later .58 statement or center/fork hypothesis.

I.2 The fresh .58 window, with the two changed tail rows

From now on put $f_i = \sqrt{i} - .58/\sqrt{2}$. The head isolation just proved remains valid; old .61-strength is not silently called new strength. At $5 \le i \le 13$, deleting i retains $A_{i-1} + D_{18-i}$. The complete white list is isolation; a left endpoint/D/E three or end-four; right long-three or permitted fork-three; or the full endpoint D_{18} bridge. Every left-only own-four has an ACTUAL affine B principal: coefficients $\sqrt{2}$ on its cap, 2 on the old chain $i, \dots, 16$, and 1 on both forks. Any extra left contact preserves all outside signs. It is a forbidden positive future null, not a role declared strong by an inadequate edge ratio.

The new ratio threshold is

$$T(q) = 50/29 - (q - .58)/[q(2q - .58)].$$

It is increasing, with $T < 83/50$ for $q < 7$, $T < 801/500$ for $q < 15/4$, and $T < 159/100$ for $q < 16/5$. For near-three/own-five the ratio $[i(\phi + 1) - 1]/\sqrt{2i(i + 1)}$ increases with i, exceeds 801/500 at six and 83/50 at eight. At five every own at least seven is stronger. D-position ratio is at least $(i\phi/2 + 1)/\sqrt{2i}$, already $> 159/100$ at five and $> 2927/1750$ from six. E, end-four, long-three and bridge rows are larger. Right-fork-three at retained D_5, D_6, D_7 has residual at most 3/8 and numerator $> \gamma + 1/4$; surviving $\gamma \ge \phi/2$ gives ratio $> 83/50$. The actual affine/time small-own boundaries are unchanged.

The new near-three/own-five at old five is absolutely impossible: its actual right-H_4 point has displacement Y, with $\langle x, Y\rangle > 18/5$, $Y^2 < 281/20$, $\sum Y_j < 43/5$. The independently rank-valid OLD strong $11, 12$ references precede this new window. Cauchy for $Y + (3/4)P_K$ gives strict gaps $.0401, .78625625$. Isolated five/seven/eight-or-larger have losses $< .58$, including the exact singleton maximum $\sqrt{10} + \sqrt{2} - 4 < .578$. For isolated dotted $i \ge 6$, reciprocal-core loss is $< a_i u + \sqrt{u(1 - u)}$, with $a_i < 37/250$; its circle bound is $< .5795$. Only isolated dotted five remains in addition to the pure helper, whose weak own range is again $\phi/2 \le \gamma < 6/5$. The cutoff follows from $R^2 \ge 4\gamma - 2$; it is not a cutoff on arbitrary fork opposites.

At fourteen/fifteen retain the COMPLETE white table above. Now $T(\sqrt{30}) < 1.64$. Isolation, ordinary left-only rows, bridges, long/middle-three, end-four and end-five are still strong by their

displayed exact ratios. Own-four is removed by the actual affine principal, not that comparison. The previously excluded own-five and finite-end-three cases remain absolute contradictions, independent of the numerical threshold. The two genuinely new weak candidates are:

$$\begin{array}{l|l} 14, \text{fork-three} & \gamma = \cos(\pi/7) \\ 15, \text{end-three} & 1 < \gamma < 1.05. \end{array}$$

The second cutoff is $1.64\sqrt{5/8} - 1/4 < 1.05$. No formerly strong dotted interval has been discarded without rechecking.

Bulk nonadjacency supplies a strong $K = 11, 12$ independently of these two candidates. In the strong-five branch the raw energies are $> .8626, > .8843$, fifth capacity $> .151$, and terminal capacity $> 1/7$. Every isolated head gives a pure separate gain $> .0155$: five has $P_4/4$ pressure $> .39307$ and .56 payment, with gain $> .0155344$; other heads have actual sum-one foot Z, with $.8h - .16Z^2 > .2893$, and .4 payment at fifth price .08, gaining $> .02914$. In the weak-helper-five branch $x_1 > .5455$ and strong six give initial energy $> .74385$, stronger than the following bare weak-five bounds. For isolated dotted weak five put

$$U' = P_5 + (u/2)(e_6 - e_5), \quad .4 < u < 1, \quad \langle x, U' \rangle > 1.8185.$$

Choose strong $J = 6, 7$. Its zero-sum residual has norm at most $77/150, 194/175$ and deficits $> .119, > .221$. For these EARLY references pay .19 at $J = 6$ and .2 at $J = 7$. The first debit $19/120$ fits $> .15834$; the old .21 debit would not fit at $K = 11$. The two actual gains exceed $.02668, .04405$. First-J energies $> .69312, > .7139$ and exact straight tails $(\sqrt{K} - \sqrt{J})^2/(K - J)$ give, in every weak-five branch,

$$q_{11} > .8697, \qquad q_{12} > .8912.$$

No additional head gain is charged in those rows.

For the native-seven fourteen candidate use both actual side cores, on side 13 or 15, as above. Let $u = \gamma^{-2}$, $\alpha = \sqrt{2}(\gamma + 1/4)$. The complete rational seal is

$$1.2318 < u < 1.2322, \quad 2.64914162 < \alpha^2 < 2.649602, \quad d = 7 - 3u - 4/u < .05736 < .2395^2.$$

Averaging the displacements with the FULL correction cross gives $Y^2 = A^2 - 3 + u + 2/u$. As $A/\alpha > 3.7163$, $\sqrt{d}/\alpha < .1472$, and $3.5 - u - 2/u > .6438 > .24\alpha^2$, the normalized foot is flat through twelve, with $G > 3.569$, $R < 13.76$. The $1/7$ tail payment gives bare weak-five totals $> 1.0025, > 1.0021$; strong-five plus its separate head gain gives $> 1.0109, > 1.0107$.

For the dotted fifteen candidate use the positive actual core $c + \gamma n_{15}$, with norm and cap product $1 - \gamma^2 < 0$, own product zero, positive time and all other original signs termwise. Here $\alpha = \sqrt{2}(\gamma + 1/4)$, $A^2 = 15\alpha^2 - 5/8$. Its normalized displacement is flat through fourteen and satisfies

$$A/\alpha > \sqrt{14.8} > 3.847, \quad \sqrt{\gamma^2 - 1}/\alpha < .182, \quad R < 15.$$

Thus $G > 3.665$, throughout $1 < \gamma < 1.05$. The same legal payments give bare totals $> 1.0046, > 1.0042$ or strong-five-plus-head totals $> 1.0130, > 1.0128$. Both candidates are impossible. The fresh window through fifteen and helper nonadjacency are proved.

Consequently some $K = 14, 15$ is strong. The all-hole references now give effective energies $> .9231, > .94006$, sums f_K, and terminal capacity $> 1/8$ in all initial cases. The weak-five isolated point may now use the later legal .21, .2 payments; they are not added to the earlier versions in the

same norm budget. For either genuine fork put $V = P_{17} \pm e_{18}$, $V^2 = 18$. If its actual cosine were at least twelve, its pressure would exceed 3.902; a 1/8 terminal payment gives

$$.9231 + (3.902 - 3.332)/4 - 4/64 = 1.0031, \qquad .94006 + (3.902 - 3.463)/4 - 3/64 = 1.002935.$$

Thus BOTH actual fork cosines are less than twelve, without assuming strong five, strong center sixteen or a sign for x_{18}.

I.3 The exact center-sixteen inventory

Deleting actual 16 retains A_{15} and BOTH orthogonal fork normals. Write an actual center opposite as

$$c = Ap - g\Omega_{16} - \eta\Omega_j - \delta\Omega_{17} - \epsilon\Omega_{18},$$

omitting absent terms. Finite classification at the actual endpoint gives exactly the following white possibilities:

left contact	fork contacts
none	none; one arbitrary finite label; or $(3,3), (3,4), (3,5)$
endpoint three at $1, 15$	none; one three/four; or both threes
next-three at $2, 14$	none
endpoint four at $1, 15$	none.

Two left contacts make a cycle. A fork added to a next-three position has excessive arms $1, 2, 13$; an internal four is not a long finite classical tree. The full two-path join has rank eighteen, beyond every exceptional type. The three-node fork list explicitly retains H_3, with labels $3, 5$. Thus no field restriction has silently removed a white possibility.

The exact retained Gram gives

$$q_L = 2\eta^2 j(16 - j)/16, \quad \Delta = 1 - q_L - \delta^2 - \epsilon^2 > 0, \quad \sigma = g + \eta j/16 + (\delta + \epsilon)/2,$$

and $A^2 = 32\sigma^2 - \Delta$, $C_{16} = \sqrt{32}\sigma/\sqrt{\Delta}$. The fresh target satisfies $T(\sqrt{32}) < 1.641$, by positive multiplication using $5.656 < \sqrt{32} < 5.657$.

An isolated center has owns zero/three/four/six/parallel excluded by time, the four-leaf-three star, the four/simple star, the $6, 3$ chain and $c + n_{16}$. The left-H_4 point at five gives $\langle x, \Omega_{16}\rangle > \sqrt{32 - g^{-2}} - .2 > 5.27$. Finite owns at least seven have loss $< .12 + 1/\sqrt{12} < .409$; dotted owns have loss $< .09u + \sqrt{u(1-u)} < .5475$. Thus every isolated center is strong. For left-only own zero, the far-three time is negative, far-next-three and far-four have zero time; the near-three/next-three affine D and near-four affine B kernels use both actual forks. Positive own three closes an all-three cycle, four gives the four/simple star at 16, six the $6, 3$ chain and parallel a null. All surviving endpoint-four, next-three and near-three ratios exceed 1.641. The only no-fork weak role is

$$c = Ap - \sqrt{2}gP_{16} - e_1/\sqrt{2}, \quad a = A^2 = 32g^2 + 2g - \tfrac{1}{2}, \quad \text{own five, finite } \geq 7 \text{ or dotted}, \quad g < 6/5.$$

Every left/fork bridge is strong or affine. Endpoint-three plus one fork-three has residual 9/32; own zero closes the long cycle, own three a triangle, and remaining $g \geq 1/\sqrt{2}$ gives ratio > 1.8. Endpoint-three plus one fork-four has residual 1/32 and even its smallest own-zero ratio exceeds two. Both fork-threes with the left contact have the same residual and smallest numerator 17/32. With no left contact, the pair $3, 3$ has own-zero four-cycle, own-three triangle and all remaining ratios > 1.7. The pair $3, 5$ already has own-zero ratio $\phi^3/2 = \phi + 1/2 > 2.1$. The pair $3, 4$ has

positive-own ratio > 2.2; at own zero its actual reciprocal fork endpoint is the full B_{18} positive projection, with cosine $8 + 9\sqrt{2}/2 > 12$, impossible.

For exactly one fork with finite contact b, own-three with fork-three is affine. With fork-four/five its actual reciprocal fork endpoint is D_{18}, of cosine $8 + 9b > 12$. Higher white labels give a direct ratio > 1.8. Own four, six and parallel have the same actual star/chain/null, using the UNCONTACTED fork in the star. For $g \geq \phi/2$, every fork label at least four has ratio at least $\phi/\sqrt{2} + 1/2 > 23/14 > 1.641$. For fork-three/own-five the actual left-H_4 finite point gives a normalized foot flat through thirteen, with $G > 3.85, R < 16$. The independent strong $K = 11, 12$ references have energies $> .8626, > .8843$ and prices $> 1/7$, so the two legal norm bounds exceed $1.0299, 1.0298$. Thus the only positive-own nonpure weak role is single-fork-three, with finite own at least seven or dotted and $g < 47/40$. The cutoff follows from $1.641\sqrt{3}/2 - 1/4 < 47/40$.

At own zero, single-fork-three/four/six have the actual E_8, four/simple and $6, 3$ nulls. Single-fork-five uses the actual H_4 on $c, f, 16, o$, with positive correction $b(3n_f/2 + n_{16} + n_o/2)$. Its spatial displacement is exactly $(b/\sqrt{2})P_{15}$. The finite point gives

$$S_{15} > \sqrt{2 + 8\phi} - (\sqrt{5} - 2) > 3.628.$$

This makes fifteen strong; upgrading its actual floor in the reference gives

$$\|x\|^2 > .95678 + 2(.1313)(.165) + .165^2/15 > 1.$$

For every remaining single-fork own-zero cap, the fork length below twelve leaves only finite white labels seven through ten. Consequently the COMPLETE weak-center list is:

pure far-three	finite five/at least seven, or dotted
one fork, own zero	white label $7, 8, 9, 10$
one fork-three, positive own	finite ≥ 7 or dotted $1 < g < 47/40$.

Every boundary was dispatched before this necessary list was formed.

I.4 Raw references, legal omissions and the center ellipse

We give the reference construction independently of any selected weak-center hypothesis. Whenever fifteen is strong, interpolate through actual strong knots, omitting every weak knot by its neighbor chord. Use the exact increment lower numerators, at denominator 10^7, for $d_i^2 = (\sqrt{i+1} - \sqrt{i})^2$, $i = 5, \ldots, 16$:

$$455488, 385186, 333704, 294372, 263340, 238230, 217494, 200080, 185248, 172465, 161332, 151549.$$

They follow from $d_i^2(a - d_i^2) = 1$, $a = 4i + 2$, and $d_i^2 > (a^2 - 1)/[a(a^2 - 2)]$. Together with the initial energy $> .6668157$, they give $E_{14} > .9241299$, $E_{15} > .9413764$. The sum of all possible nonadjacent omission costs is $< 1/1024$. With a native-five head, its actual first-seven relations force five, six and seven strong; the sharper complete debit is

$$[(.0112)^2 + (.008)^2 + (.0061)^2 + (.0049)^2]/2,$$

giving $> .94125107$. These are squared-slope formulas for every allowed hole set, not a numerical sample of selected weak sets.

The actual native-five head H_4 foot is $P_4/4$, with pressure $> .39307$, square $1/4$. Its .56 zero-sum correction has fifth debit $.112 < .151$ and gain $> .0155344$. For any other isolated head, its rank-free actual sum-one foot has $.8h - .16R > .2893$, and .4 correction gives gain $> .02914$, at

fifth debit $.08 < .151$. Their complete scalar proof is the head-foot proof in Appendix H.2; it concerns the identical actual first three/four long roots and is independent of the terminal type. When integrated, that shared proof belongs immediately after the generic prefix's four weak-five pressures, rather than being repeated in both classical terminal appendices.

The full endpoint-fifteen contracts are

initial/head case	L_{15} lower	initial price
strong five, native-five head	.95678547	.112 at P_5
strong five, other head	.96953	.08 at P_5
weak helper five	.97164	actual $(1,6)$ reference
isolated dotted five, strong six	.96525	.4/6 at P_6
isolated dotted five, strong seven	.96016	.4/7 at P_7.

The weak-helper row uses $x_1 > .5455$, initial energy $> .74385$, and strong six; its improvement over $E_6 < .7126$ is $> .03125$. For the isolated dotted rows, the BARE curved energies are $> .9212053, > .9035167$, first means $< .3399, < .31938$, and capacities $> .1435, > .1366$. They use only the head foot, gaining $> .0440466, > .0566531$. The isolated-five point U' is not also paid in these rows. The native-five head is incompatible with weak isolated dotted five by its actual local mutual in $(0, 1/4)$.

Under the additional assumption that sixteen is weak and fifteen strong, put

$$d = 4 - \sqrt{15} \in (.1270, .1271), \quad \Delta = S_{15} - f_{15} \geq 0, \quad U = f_{16} - S_{16} \geq 0.$$

The common reference has effective energy $> .95678$ and terminal capacity $\beta \geq \sqrt{15} - \sqrt{14} > d$. Retaining that unspent surplus in ONE square gives

$$2\beta\Delta + x_{16}^2 + x_{17}^2 + x_{18}^2 < .04322, \quad x_{16} = d - U - \Delta.$$

Hence

$$x_{17}^2 + x_{18}^2 < .04322 - d^2 + 2dU - U^2.$$

For either signed $t = x_{17} \pm x_{18}$, ordering gives $0 < t < 2d < .2541$, while a direct quadratic comparison gives $t < .233 + 1.1U$: the difference of squares has constant $> .000107$, linear coefficient $> .0042$, quadratic coefficient 3.21. Thus, for $P > 0, z_0 \geq 0, z_0 \leq 10P$,

$$PS_{16} + z_0(S_{16} + t) \leq f_{16}P + (f_{16} + .233)z_0. \tag{57}$$

This is a consequence of one funded original reference, not a new independently optimized terminal energy.

I.5 Pure weak centers: one cleared Gram and its full scalar dispatch

Suppose the center is weak and its cap is the pure far-three helper. Own three/four/six/parallel were excluded in the inventory. If fifteen were also weak, its actual cap is another pure helper. The actual product is $> 1 - gh > -11/25$, hence zero by the native gap. Clearing actual A_{14} gives the positive two-cap Gram $\left(\begin{smallmatrix} 8/15 & -7/15 \\ -7/15 & 8/15 \end{smallmatrix}\right)$; both original forks remain orthogonal positive blocks. The actual mixed-point loss is $< 7/16 + 1/(4\sqrt{10}) < .58$, contradicting both weak floors. Thus fifteen is strong and (57) is available.

Put

$$a = 32g^2 + 2g - \tfrac{1}{2}, \quad k = 16g + \tfrac{1}{2},$$
$$P = k/(8\sqrt{2}), \quad r = 17/32,$$
$$\bar{c} = c + \sum_{i=1}^{15}(16-i)n_i/16 = Ap - PP_{16}.$$

Then $\bar{c}^2 = r$ and $k^2 = 8a + 17/4$. A finite own has $.8 < g < 1$. A dotted weak own has $1 < g < 7/6$: the exact center cosine is $(32g+1)/\sqrt{17}$, and at $7/6$ its ratio already exceeds $821/500 > T(\sqrt{32})$. Set $H = A - f_{16}P,\ K = 3A - k$. All intervals below have $H, K > 0$.

The seven actual fork white profiles and their lengths can be derived before center strength is known. For either assignment $\{f, o\} = \{17, 18\}$, $d = Bp - v$, own $\delta = -\langle d, n_f\rangle \geq 0$, they are

	v	B^2	C_f
I	$\delta\Omega_f$	$9\delta^2 - 1$	3δ
N	$\delta\Omega_f + \Omega_o/2$	$9\delta^2 + 8\delta + 5/4$	$(18\delta + 8)/\sqrt{19}$
N_4	$\delta\Omega_f + \Omega_o/\sqrt{2}$	$9\delta^2 + 8\sqrt{2}\delta + 7/2$	$9\sqrt{2}\delta + 8$
N_2	$\delta\Omega_f + \Omega_{16}/2$	$9\delta^2 + 16\delta + 7$	$9\delta + 8$
F	$\delta\Omega_f + \Omega_1/2$	$9\delta^2 + \delta - 1/2$	$(18\delta + 1)/\sqrt{19}$
F_4	$\delta\Omega_f + \Omega_1/\sqrt{2}$	$9\delta^2 + \sqrt{2}\delta$	$9\sqrt{2}\delta + 1$
F_2	$\delta\Omega_f + \Omega_2/2$	$9\delta^2 + 2\delta$	$9\delta + 1.$

Indeed deleting f leaves the full actual A_{17} path; its one-node finite list is isolation, either-end three/four, or either next-to-end three. For a white $\beta\Omega_j$, set $L = \langle \Omega_f, \Omega_j\rangle, J = \Omega_j^2$. The edge formula is

$$C_f = (9\delta + \beta L)/(3\sqrt{\Delta_f}), \quad \Delta_f = 1 - \beta^2(J - L^2/9),$$

and $(L, J) = (8, 9), (16, 32), (1, 2), (2, 4)$ gives the table. Both signs are actual, with $C_f < 12$. All own labels are initially retained.

Near-four forces $\delta < 1/2$, hence $\delta = 0$; its acuteness square is $-a/2 - 17/8 < 0$. Near-next-three has $\delta < 4/9$ and square difference $(\delta + 1)[a(\delta - 1) - (17/4)(\delta + 1)] < 0$. Both are impossible. For the other profiles clear only positive actual retained roots:

	v_0	$s = \bar{d}^2$	$\rho = -\langle \bar{c}, \bar{d}\rangle$
I	0	1	λ
N	$1/2$	$3/4$	λ
F	$1/16$	$17/32$	$\lambda + 15/32$
F_4	$\sqrt{2}/16$	$1/16$	$\lambda + 15/(16\sqrt{2})$
F_2	$1/8$	$1/8$	$\lambda + 7/16.$

Here $\lambda = -\langle c, d\rangle \geq 0$ is the ACTUAL mutual magnitude, and $\bar{d} = Bp - [(\delta + v_0)P_{16} + \delta(e_{17} \pm e_{18})]/\sqrt{2}$. For N, add $n_o/2$. For the far rows add the positive A_{15} inverse column. The other fork is retained when cleared. No projected normal is called an actual facet.

If $\rho^2 = rs$, $\bar{c} + (\rho/s)\bar{d}$ is a forbidden positive future feasible null. If $\rho^2 > rs$, it is a negative core with square $r - \rho^2/s$, time $A + \rho B/s > 0$, selected products that negative square and zero, and zero retained old products. Every unknown wall is nonpositive termwise. With $t = x_{17} \pm x_{18}$, its pressure is contradicted when

$$D = H + \frac{\rho}{s}\{B - v_0 f_{16}/\sqrt{2} - (f_{16} + .233)\delta/\sqrt{2}\} - \sqrt{\rho^2/s - r} > 0, \tag{58}$$

provided $z_0 \le 10P_0$, where $P_0 = P + \rho v_0/(s\sqrt{2})$, $z_0 = \rho\delta/(s\sqrt{2})$. This funding condition precedes every use of the ellipse.

If $\rho^2 < rs$, use the positive actual finite projection

$$q_0 = p+\alpha\bar{c}+\chi\bar{d} = (1+T)p-V, \quad \alpha = (sA+\rho B)/(rs-\rho^2),\ \chi = (\rho A+rB)/(rs-\rho^2), \quad T = A\alpha+B\chi.$$

All coefficients are positive, its norm is $-(1+T)$, and its normalization satisfies every original wall. Its exact spatial direction and the SAME original maximum give the subcritical estimates below. Both zero mutual and singular boundaries are kept.

Here are the complete finite-own scalar constants. They follow by positive squares, $A'^2 - 32 = 17/a$, and $H' > .58$.

g interval	H lower	K upper
$(.8, .85]$	.4252	.695
$[.85, .9]$	.4542	.752
$[.9, .95]$	.4832	.809
$[.95, 1)$	.5122	.864.

Also $A/K > 20/3$ throughout. For dotted owns the corresponding complete table is

g interval	H lower	K upper	A/K lower
$(1, 1.05]$	.55235	.920	6.5
$[1.05, 1.1]$	.58135	.974	6.5
$[1.1, 9/8]$	.61035	1	6.5
$[9/8, 1.15]$	.62485	1.028	6.46
$[1.15, 7/6)$	.63935	1.05	6.4.

At a right endpoint the asserted K upper is proved by $(k+K_{\text{upper}})^2 - 9a \ge 0$. The five dotted gaps are $.0484, .037476, 0, .045184, 769/3600$; the four finite gaps are $.012025, .005104, .027081, .008496$. K increases, so these endpoint checks cover the intervals. For A/K, square k/A, using $k^2/a = 8 + 17/(4a)$. These are finite rational certificates for continuous ranges, not sampled bounds.

For isolation, acuteness forces $\delta^2 \ge a/(a - 17/4) > 1$. Positive subcritical mutuals are exactly $1/2, 1/\sqrt{2}$. Put $\zeta = 17/(4a)$ and $j = \zeta + k\lambda/(a\delta)$. The finite projection gives

$$S_{16} + jt \ge \sqrt{16 + 2\zeta} - \sqrt{2(\zeta - j^2)}, \quad j^2 < \zeta.$$

For finite g, put $F(\delta, g) = A\sqrt{9\delta^2 - 1} - k\delta$. It increases in both variables for $\delta > 1$. At $g = .8, \delta = 8/5$, its positive-square comparison with $\lambda = 1/2$ has gap $3137/2500 > 0$. Thus the solution δ_0 for $\lambda = 1/2$ satisfies $\delta_0 < 8/5$. For fixed g, the ratio $\lambda/\delta = A\sqrt{9 - \delta^{-2}} - k$ increases with δ, and the solution δ increases with λ. Consequently BOTH subcritical mutuals satisfy $\lambda/\delta > 5/16$, and hence $j - \zeta > (5/16)(k/a)$. The four finite-g strips therefore have

	ζ upper	$j - \zeta$ lower	S_{16} lower
1	.197	.180	3.611265
2	.175	.170	3.612025
3	.157	.161	3.61531
4	.141	.153	3.62223.

To reproduce each certificate, use $t < .255$, $\sqrt{16 + 2\zeta} > 4 + \zeta/5$; the resulting expression increases with j and decreases with ζ on that strip. The four upper radicals are $.332, .335, .335, .331$, whose squared radicands are $.109742, .11195, .111752, .109128$. For dotted g, the same monotonicity gives

$\delta < 8/5,\ k/a > .41,\ \zeta < .127,\ j > \zeta + .128$; the same rationalized test gives $S_{16} > 3.608275$. All contradict weak $S_{16} < 3.59$.

For isolated supercritical mutuals use

$$3\delta - .172/\delta < B < 3\delta - 1/(6\delta), \quad \lambda < K\delta - A/(6\delta), \quad (f_{16} + .233)/\sqrt{2} < 2.7033.$$

Then (58) exceeds

$$H + (.2967/K)\lambda^2 + \{.2967A/(6K) - .172\}\lambda/\delta - \sqrt{\lambda^2 - r}.$$

For finite g, the actual $C_f < 12$ gives $\delta < 4$ and ensures the ellipse price; $\lambda/\delta > .4$ by the increasing positive-square test at $(g, \delta) = (.8, 2)$. Completing one square gives lower gaps $> .1292, .0931, .0593, .0295$ on the four strips. For dotted g, first take $\lambda \le 4$. Monotonicity gives $\delta < 5$, so $z_0 < 20/\sqrt{2} < 10P$. The ratio lower bounds $.45, .55, .6, .7$, respectively on

$$(\lambda, g) > (\sqrt{r}, 1), (1, 1), (1.2, 1), (1.4, 1.15),$$

follow by contradiction from

$$\lambda^2 \le \frac{t^2 a}{9a - (k+t)^2} < \Lambda^2.$$

The last comparison reduces to the increasing quadratic $9a - (k+t)^2 - (t^2/\Lambda^2)a > 0$. Its lower-bound signs are $> .7586/r, > .6637, .715, 1.89$. With these ratios and the dotted strips, one square yields positive lower gaps $> .0156, .0046, .0075, .0004, .0058$. The omitted small-$\lambda$ portions are decreasing up to $1, 1.2, 1.4$; at those endpoints their gaps exceed $.26, .28, .153, .059$. For $\lambda \ge 4$ no refined ellipse is needed: $t < 2d < .2541$, $K < 1.081$, and the coarse core gap exceeds

$$.55235 - .172(1.081) + .2599\lambda^2 - \lambda > .52.$$

Thus every positive isolated mutual is excluded, including the entire unbounded dotted alphabet. Only $\lambda = 0$ remains.

For near-three, acuteness is

$$Q(\delta, a) = (a - 17/4)\delta^2 - (17/4)\delta - 3a/4 - 17/16 \ge 0.$$

Finite g gives $\delta > 1$, dotted g gives $\delta > 39/40$; the latter follows from $Q(39/40, 817/18) = -2693/19200 < 0$. The actual edge gives $\delta < 23/9 < 3$. The only positive subcritical mutual is $1/2$. Its full actual finite projection has $T > 300, j < 1/2$ in $V = K_0(P_{16} + j(e_{17} \pm e_{18}))$; hence the original maximum gives $S_{16} + jt > 3.76$. For finite g, use $\delta < 2,\ B/A < 2,\ j < 160/333$, and T's positive numerator $> 300(19/128)$. For dotted g, the sharper $13/10 < \delta < 3/2,\ B/A < 1,\ j < 1/3,\ T > 8664/19$ follows by the exact tests $AB - k(\delta + 1/2) = 1/2$ at $(g, \delta) = (1, 3/2)$ and below $1/2$ at $(7/6, 13/10)$. Both contradict $S_{16} + jt < 3.59 + .255/2 = 3.7175$.

Every supercritical near-three core is funded since $z_0 - 10P_0 = (4\lambda/(3\sqrt{2}))(\delta - 5) - 10P < 0$. With $J = k/2 - 4A/3$,

$$B > 3\delta + 4/3 - 19/(216\delta), \quad \lambda < K\delta - J, \quad J/K = A/(6K) - 1/2.$$

In the finite strips the gap exceeds $.3238 + (989/2160)\lambda^2 + .3271\lambda - \sqrt{4\lambda^2/3 - r}$; in the dotted strips it exceeds $.4292 + (989/2625)\lambda^2 + .3095\lambda - \sqrt{4\lambda^2/3 - r}$. For $\lambda \ge 1$, completion of one square gives respectively positive gaps $> .1052, > .00406$. Below one the displayed lower functions decrease and at one are $> .2127, > .2194$. Thus near-three also forces mutual zero. Its unique positive Q-root

lies in $(1, 3/2)$ for finite g, or $(39/40, 33/32)$ for dotted g. Set $\zeta = 17/(4a) < .2$ or $< .13$. An isolated/near pair then has positive actual product $\delta_I[1/2-\zeta(\delta+1/2)]$; two near caps have product $\delta - \zeta(\delta + 1/2)^2 > 0$. Two isolated caps have actual mutual $17/(4a - 17) \in (0, 1/4)$. All caps are distinct by their original fork contacts. Hence at least one fork must have a far profile.

For completeness here is the entire far-profile scalar certificate. The exact positive-root sandwiches are

	M	$M^2 - B^2$
F_2	$3\delta + 1/3$	$1/9$
F_4	$3\delta + \sqrt{2}/6$	$1/18$
F	$3\delta + 1/6$	$19/36$.

Always $B < M$ and $B > M - E_1/\delta$, with $E_1 = 1/54, 1/108, 19/216$, respectively. Both F_2, F_4 are automatically supercritical, even at mutual zero. The actual length gives $\delta < 11/9 < 10(1/8)$ for F_2, and $\delta < 11/(9\sqrt{2}) < 10(\sqrt{2}/16)$ for F_4; thus every such ellipse price fits. Substitution in (58) gives

	$g < 1$	$g > 1$
F_2	$.2972 + (989/360)\rho^2 + .4906\rho - \sqrt{8\rho^2 - r}$	$.3967 + (1978/875)\rho^2 + .4642\rho - \sqrt{8\rho^2 - r}$
F_4	$.2972 + (989/180)\rho^2 + .6941\rho - \sqrt{16\rho^2 - r}$	$.3967 + (3956/875)\rho^2 + .6568\rho - \sqrt{16\rho^2 - r}$.

For positive mutual use $\rho \geq 15/16$ in F_2; for zero use exactly $\rho = 7/16$, whose radical is one. For F_4, $\rho \geq 15/(16\sqrt{2}) > 5/8$. The identity $c\rho^2 - \sqrt{m\rho^2 - r} \geq cr/m - m/(4c)$ gives gaps $> .0376, > .1852$ in the finite case and $> .032475, > .07256$ in the dotted case. It retains every finite, parallel and dotted mutual.

For far-three with positive mutual $\rho \geq 31/32$. The upper root gives $\rho < K\delta$, and the actual $\delta < 3$ funds $z_0 \leq 10P_0$. Positive-square monotonicity forces $\delta > 1$; then the improved upper $B < 3\delta + 1/6 - 1/(12\delta)$ gives

$$D > H + c\rho^2 + L\rho + C_1\rho/\delta - \sqrt{\rho^2/r - r}, \qquad c = .2967/(rK).$$

In finite strips, $L > .0577, C_1 > .1446, \rho/\delta > .48$; in dotted strips $L > .0546, C_1 > .1322, \rho/\delta > .6$. For the last ratio the exact monotone test at $(g, \delta) = (1, 155/96)$ has square gap $37199/12288$. For the finite ratio the corresponding test at $(.8, 2)$ has gap $23399/102400$. One square leaves respectively

$$H + .157621875/K - K/1.1868 + .1252 > .0047$$

and, on the five dotted strips, gaps $> .0805, .0545, .0574, .0440, .0368$ with $.1321$ in place of $.1252$. Thus every positive mutual fails.

At zero far-three mutual the Schur determinant is $1/16$. The full actual $A_{15} + n_o + c + d$ positive projection has

$$\alpha = (17A + 15B)/2, \quad \chi = (15A + 17B)/2, \quad T = [17(a + B^2) + 30AB]/2.$$

Monotonicity of $AB - k\delta - g - 1/2$ gives $\delta < 3/2,\ B/A < 1$. The spatial tail ratio is

$$j = \frac{(\chi/\alpha)\delta}{k/8 + (\chi/\alpha)(\delta + 1/16)} < 1/2.$$

Its $T > 300$ for finite g, $T > 1733/4$ for dotted g. The actual point therefore demands $S_{16} + jt > 3.76$ against the same upper 3.7175. This removes zero mutual too.

Every actual fork profile has now been covered under a weak PURE center: near-four/next-three are impossible; isolated/near profiles must be orthogonal and cannot form a pair; some far profile is necessary, but every far profile fails. All scalar interval checks above use explicitly displayed root sandwiches, monotonicities and completed squares on complete ranges. They do not posit a maximum native label or assume arithmeticity. The pure weak-center branch is excluded.

I.6 Nonpure centers and their actual neighbors

Only the two nonpure weak roles from the complete inventory remain. The own-zero/single-fork labels seven through ten will be excluded directly at their actual neighbors; their older head-specific refinements and separate strong-fifteen bootstrap are unnecessary.

First take single-fork-three with finite own $.9 < g < 1$. We do need fifteen strong for this branch. If fifteen were weak, its actual pure helper has parameter $\phi/2 \le h < 6/5$. Put $y = h + 1/30,\ t = g + 1/4$; the actual times satisfy

$$A_h^2 = 30y^2 - 8/15, \qquad A_c^2 = 32t^2 - 3/4, \qquad M = -\langle c, d\rangle = A_h A_c - 30yt.$$

This retains the full fork contact. It increases in both y, t: positive derivative squaring gives

$$y^2 A_c^2 - t^2 A_h^2 = y^2(2t^2 - 3/4) + (8/15)t^2 > 0,$$

and $1024t^2 A_h^2 - 900y^2 A_c^2 = t^2(1920y^2 - 8192/15) + 675y^2 > 0$. At the lower corner $h = .8, g = .9$ the time/spatial square gap is $17.3085 > 0$; at $h = 6/5, g = 47/40$ the comparison with $M = 11/10$ has gap $2861/8000 > 0$. Thus the entire actual range has $0 < M < 11/10$. Native values leave $M = 1/2$ or $M \ge 1/\sqrt{2}$.

Positive clearing of A_{14} and the contacted actual fork gives

$$\bar d = A_h p - \sqrt{2} y P_{15}, \quad \bar d^2 = 8/15, \qquad \bar c = A_c p - \sqrt{2} t P_{16}, \quad \bar c^2 = 3/4, \quad \langle \bar c, \bar d\rangle = -M.$$

The other fork remains a positive retained unit block. Both weak floors give deficits $D_h = A_h - \sqrt{2} y f_{15} > .43$, $D_c = A_c - \sqrt{2} t f_{16} > .60$. At $M = 1/2$, the full positive actual projection has coefficients $43 < \alpha < 65,\ 37 < \chi < 56$, parameter $T < 959$; its pressure would require $\alpha D_h + \chi D_c \le \sqrt{1+T} - 1 < 31$, whereas the left side exceeds 40.69. For $M \ge 1/\sqrt{2}$, the positive core $\bar d + (4M/3)\bar c$ has negative square $8/15 - 4M^2/3$. Its pressure gap exceeds $.43 + .8M - \sqrt{4M^2/3 - 8/15}$. This decreases on the range and at $11/10$ is $> .27$. All original wall signs hold by actual positive coefficients. Thus fifteen is strong.

Use the one ending-fifteen reference of Appendix I.4, with $L > .95678$ and unpaid terminal coefficient $> .1313$. For finite g the actual positive core is $c + n_{16}/g + n_{15}/(2g)$, of norm $-[1 - 3/(4g^2)]$. Its cap/old-fifteen products vanish, its old-sixteen product is negative, both fork products are negative, and time is positive. With $\alpha = \sqrt{2}(g + 1/4)$, its normalized displacement V has

$$V_j = 1 \ (j \le 14), \quad V_{15} = V_{16} = 1 - a,$$
$$V_{17} = 2a + b, \quad V_{18} = \pm b,$$
$$a = \frac{1}{4g(g + 1/4)}, \quad b = \frac{1}{4(g + 1/4)}.$$

The full residual square is $1 - 2a + 6a^2 + 4ab + 2b^2 < 1.174$. Its reference cross is $-a w_{15}$, and its pressure-minus-reference deficit is $> .24441$: use $A_c/\alpha > 3.964$, $\sqrt{1 - 3/(4g^2)}/\alpha < .28285$, $a > .2,\ w_{15} > .1313,\ f_{15} < 3.463$. A legal .13 terminal payment therefore gives

$$\|x\|^2 > .95678 + .26(.24441) - .0169(1.174) > 1.000486.$$

No absent P_{14} capacity or omitted negative cross is used. This removes every finite own in the nonpure positive-own branch.

For dotted single-fork-three, $g > 1$, the actual center endpoint has full forest $A_{15} + A_2 + A_1$. For own zero and white $\delta = \cos(\pi/m)$, $7 \le m \le 10$, the actual endpoint has $A_{15} + I_2(m) + A_1$. In both cases the positive original incident dual is $P_{16}/4$:

$$q = Cp - \sqrt{C^2 - 1}P_{16}/4, \quad t = S_{16}/4 < 449/500, \quad E_q/E = C - \sqrt{C^2 - 1}\,t.$$

These are actual full finite projections, so every unknown original wall is satisfied. The reverse positive unit dual $u = (Cq - p)/\sqrt{C^2 - 1}$ lies in the dihedral block and satisfies

$$\langle y, u\rangle = \frac{\beta - t}{1 - \beta t}, \qquad \beta = \sqrt{C^2 - 1}/C, \quad z = E_q(q - y).$$

The point q is never asserted farthest; every original chamber point is compared there with factor E/E_q.

For dotted single-fork-three, $C = \sqrt{32}(g + 1/4)/\sqrt{3/4}$, so $C^2 > 200/3 > (204/25)^2$. The increasing function $C - (449/500)\sqrt{C^2 - 1}$ gives

$$E_q/E > 4431/5000 > 7/8, \quad E/E_q < 8/7, \quad \langle y, u\rangle > 5875/6824 > 43/50.$$

For own-zero/single-fork label seven through ten,

$$C = \frac{\sqrt{8}\,\delta}{\sqrt{1 - \delta^2}}, \quad C^2 > 648/19 > (29/5)^2,$$

as $\delta > .9$. Hence

$$E_q/E > 117/175 > 2/3, \quad E/E_q < 3/2, \quad \langle y, u\rangle > 2900/3849 > 3/4.$$

The radical upper at $29/5$ is $40/7$; the bounds $\beta > 197/200$ or $124/125$ give the stated reverse fractions by positive rational multiplication. These two neighbor cases will use the following local facts.

I.7 The two required central A_{15} facts

Use zero-sum $\mathbb{R}^{16}$, with

$$\omega_i = \sqrt{2}(P_i - iP_{16}/16), \quad D_i = i(16 - i)/8, \quad C_{ij} = i(16 - j)/8 \ (i \le j), \quad D_8 = 8.$$

At any actual $A_{15} + A_2 + A_1$ vertex, the actual central edge has $C_8 > 9/4$, without a farthest assumption. At an actual $A_{15} + I_2(m) + A_1$ vertex, $7 \le m \le 10$, with comparison at most $3/2$, the A_{15} energy exceeds $7/16$. We prove both statements from the actual central opposite.

Write it $d = Hq - v_A - v_S$, own $\gamma \ge 0$, whole foreign cost $Q = \|v_S\|^2 < 1$. Deleting eight leaves two A_7 paths. The finite white list is one three at $1, 2, 3, 5, 6, 7$ or reflection; one four at $1, 7$ or reflection; or one three on each path. The third-from-end three is E_8; the middle fourth position is affine E_7. No white five extends a full A_7, and a rank-fifteen join is classical with both new bonds simple. The exact edge data are

$$\sigma = \gamma + \sum_{j<8} \beta_j j/8 + \sum_{j>8} \beta_j(16 - j)/8, \quad \Delta = 1 - Q - \sum_{j<8} \beta_j^2 j(8 - j)/4 - \sum_{j>8} \beta_j^2(16 - j)(j - 8)/4,$$

with $H^2 = 8\sigma^2 - \Delta,\ C_8 = \sqrt{8}\sigma/\sqrt{\Delta}$.

We first remove own zero directly, without a foreign-shadow rule. Two white paths close an actual affine cycle. Single three at global $3, 13$ gives affine E_8 with arms $1, 2, 5$; at $4, \dots, 12$ use E_7 with arms $1, 3, 3$. White four at $7, 9$ has the four/simple star. At endpoint four $\Delta > 0$ requires $Q < 1/8$, so $Q = 0$ and $H^2 = -1/16$; at next-three $Q < 1/4$ forces $Q = 0, H^2 = -1/8$. At endpoint three the retained bound is $Q < 9/16$, but $H^2 = Q - 17/32$. In $A_2 + A_1$, the possible foreign contributions under that bound are $0, 1/3, 1/4, 1/2$; simultaneous contact costs at least $1/3 + 1/4 = 7/12 > 9/16$. Thus $Q \le 1/2$, impossible. For the high dihedral block, ANY positive contact costs

$$[4(1 - \delta^2)]^{-1} > 25/19 > 1,$$

so there is no such contact at all. The remaining A_1 again gives only $Q = 0, 1/4, 1/2$ under the retained bound. With no A contact the time is negative and its derivative along that edge is zero. Every affine kernel has positive actual coefficients and cap time, selected products zero, all unknown products nonpositive; it is a forbidden future feasible null.

For the first local statement, $\gamma \ge \phi/2$ directly gives $C_8 \ge \sqrt{2}\phi > 9/4$. Own four has a four/simple star at eight unless a white contact hits 7 or 9; then $\sigma \ge 1/\sqrt{2} + 7/16, \Delta \le 9/16$, giving $C_8 > 4$. Own three has the actual E_7 star at eight unless it contacts one of the two three-node arms. The remaining allowed contacts have $(\sigma, \Delta) \ge / \le (13/16, 1/16), (7/8, 1/4), (15/16, 9/16)$; the weakest cosine is $5\sqrt{2}/2 > 9/4$. This proves the full central-edge statement.

For the high-dihedral second statement, all foreign cost is in the remaining actual A_1. Own-three plus any white-three closes a cycle. Pure far-three own-four/six/parallel have the actual star, $\gamma d + m_8 + m_7/2$, and $d + m_8$ nulls. The remaining ordinary ratios satisfy

role	σ lower	Δ upper before Q
near three	$.7 + 7/16$	$9/16$
D three	$.7 + 1/8$	$1/4$
E three	$.7 + 3/16$	$1/16$
white four	$1/2 + 1/(8\sqrt{2})$	$1/8$
two threes	$1/2 + 1/8$	$1/8$
far three, $Q > 0$	$.8 + 1/16$	$5/16$.

All exceed $3/2$. Their actual edge demand is therefore $> (3\sqrt{2} - 3/2)/\sqrt{17} > 53/80$, giving energy $> 2809/6400 > 7/16$. Extra foreign cost only increases this demand.

For isolated A roles, own three/four/six/parallel have the E_7, star, chain and pair nulls. Put $\tau = (1 - Q)/\gamma^2,\ H^2 = \gamma^2(8 - \tau)$. For $0 < \tau < 1$ the positive actual core $d + w_S + \gamma\tau m_8$ has spatial part $U = \gamma(\omega_8 - \tau m_8)$, norm $-\gamma^2\tau(1 - \tau)$, and $U^2 < H^2$. If $e_A \le 7/16$, Cauchy forces pressure loss divided by γ greater than $(1 - \sqrt{7}/4)\sqrt{8 - \tau} > 5/6$, against comparison loss at most $3/4$. At $\tau = 1$ it is a forbidden null. For $\tau > 1$, the actual positive finite point gives

$$h_8 \ge \sqrt{\frac{7}{8 - \tau}}\,\{\sqrt{7} - \kappa\sqrt{\tau - 1}\}.$$

Foreign $\sqrt{\tau - 1} \le 2 - \phi < .382$; pure finite owns at least eight have $\sqrt{\tau - 1} \le \tan(\pi/8) < .415$. The bracket is positive, and $h_8 > 13/5 - (3/2)(.415) = 791/400$, whose normalized value is $> 2/3$, giving energy $> 4/9 > 7/16$. Pure five uses the actual H_4 finite point, with $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36,\ H^2 > 4, \gamma > .8$:

$$h_8 > \sqrt{8 - \gamma^{-2}} - \kappa\sqrt{\delta_5}\sqrt{1 + \delta_5/H^2}/\gamma > 699/320.$$

Pure seven uses $d + (4\gamma/3)m_8 + (2\gamma/3)m_7$; the native cubic brackets $.9 < \gamma < 10/11$ give

$$h_8 > \sqrt{8 - \gamma^{-2}} - \kappa\sqrt{4/3 - \gamma^{-2}} > 103/50.$$

Both again give normalized height $> 2/3$.

For every pure far-three helper, without an upper own cutoff,

$$H^2 = 8\gamma^2 + \gamma - 17/32 > 8\gamma^2, \qquad \gamma > .8.$$

Five uses the actual correction $\gamma m_8 + (2/3)m_e + (1/3)m_j$, residual $\delta = 2/3 - \gamma^2 \in (0, 1/36)$; its finite point has $U^2 = H^2 + \delta < (10/9)H^2$ and pressure $> H - .2525 > (6/7)H$, giving energy $> 162/245$. Finite owns at least seven use $d + \gamma m_8 + 2(1-\gamma^2)m_e$, negative square $-(1-\gamma^2)(4\gamma^2 - 3)$, radical at most $1/4$. Since $H > 2.5$, energy exceeds $289/400$. Dotted owns use $d + m_8/\gamma$; its radical divided by γ is at most $1/2$, so the pressure ratio is $> 1 - 3/(4\sqrt{8}) > 2/3$, giving energy $> 4/9$. All selected products and positive time are as displayed; unknown wall signs hold termwise. This completes both local facts.

Return to the original actual neighbor. In the dotted single-fork-three case, $C_8 > 9/4$ and factor $< 8/7$ give central normalized height $> 31/(7\sqrt{65})$, hence $e_{A_{15}} > 961/3185$. The orthogonal reverse A_2 energy exceeds $1849/2500$, and their sum is $1 + 65813/1592500 > 1$. In the own-zero/high-dihedral case, the second local fact gives $e_{A_{15}} > 7/16$, while the reverse dihedral costs $> 9/16$. The remaining actual A_1 is nonnegative in each SAME neighbor vector and is never charged twice. Both weak-center families are excluded.

I.8 Strong sixteen and the legal terminal append

The complete center inventory, the entire pure-center exclusion, and the finite/dotted single-fork-three and own-zero/high-dihedral arguments cover every possible weak center. Therefore $S_{16} > f_{16}$ unconditionally. No statement about strength of either fork was used.

Reassemble each endpoint-fifteen reference through actual strong sixteen, then OMIT weak fifteen before applying any pressure. Appending contributes $(4 - \sqrt{15})^2$; omitting fifteen costs $(2\sqrt{15} - \sqrt{14} - 4)^2/2 < .00001$. Since $4 - \sqrt{15} > .127$, the net gain is $> .016119$. The initial zero-sum corrections are supported through seven, so their individual prices and full crosses are unchanged. The endpoint-fifteen omission debit may conservatively be retained in addition to this new debit; no fictitious floor is ever used. The worst paid ending-sixteen effective energy is

$$L_{16} > .95678547 + .127^2 - .00001 = .97290447 > .9729.$$

The other four rows exceed $.985649, .987759, .981369, .976279$. The unpaid P_{16} coefficient is its true last slope: d_{15} if fifteen is strong or $(d_{14} + d_{15})/2$ if fifteen is weak. It exceeds $.127$ in every case. All three late slopes are at least d_{15}.

For either actual fork $V = P_{17} \pm e_{18}$, the residual $V - P_{16}$ has square two and zero reference/initial cross. If its actual cosine were at least ten, pressure would exceed 3.837. A legal $.12 < \beta_{16}$ payment gives

$$\|x\|^2 > .9729 + .24(3.837 - 3.59) - .0144(2) = 1.00338 > 1.$$

Thus BOTH genuine fork cosines are less than ten for every head and every original initial-five case. The subsequent main-text uniform dotted-head argument strengthens these data; it is not needed to prove them.

I.9 The genuine fork inventory and the all-head near-pair bound

Return to the seven actual fork profiles in Appendix I.5, now with $C_f < 10$. For isolated or far-helper own three, an actual affine E_8 kernel has coefficients 2 on its cap, 4 on f, 6 on center 16, 3 on o, and $5, 4, 3, 2, 1$ along $15, \dots, 11$. Own four has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on its cap, $f, 16, 15, o$; own six has its $6, 3$ chain. An extra far helper lies outside these principals. All have positive cap time and all unknown wall signs termwise. Parallel gives $d + n_f$. For near-three, own zero is isolated own three at the other fork; own three is the actual four-cycle, own six the same chain. Near-four now forces own zero and is the isolated own-four null at the other fork; near-next-three likewise forces own zero and is the four-leaf-three center star. Far-four has no positive time at zero, and its only possible positive own under length ten is three, already excluded. These dispose of ALL formerly allowed near-four/next-three/far-four values, not merely their finite members.

Isolated own five is exactly the actual H_4 point already computed in Appendix I.3. Its displacement gives $S_{15} > 3.628$, hence fifteen is strong. With terminal capacity $> .1313$, the one-reference gain $2(.1313)(.165) + .165^2/15$ contradicts $L_{15} > .95678$. That point's computation did not use weak center sixteen. The complete remainder is therefore

I	finite own ≥ 7 or dotted, $g < 10/3$
N	own four/five, finite ≥ 7 or dotted, $g < (10\sqrt{19} - 8)/18$
F	own five, finite ≥ 7 or dotted, $g < (10\sqrt{19} - 1)/18$
F_2	own five or finite ≥ 7, $g < 1$.

These are necessary roles, not examples.

The original ending-sixteen reference retains its terminal surplus. For $\Delta = S_{16} - f_{16} > 0$,

$$\|x - w\|^2 < .0271 - .254\Delta.$$

Since $w_{17} = w_{18} = 0$, each genuine $t = x_{17} \pm x_{18}$ has $t^2 < .0542 - .508\Delta$. The square comparison $(.233 - \Delta)^2 - (.0542 - .508\Delta) = .000089 + .042\Delta + \Delta^2 > 0$ gives

$$S_{17} + x_{18} < 3.823, \qquad S_{17} - x_{18} < 3.823.$$

At cosine $241/25 = 9.64$ the true fork demand has square $1944/133 > 3.823^2$; hence both fork cosines are < 9.64.

For a near-three or far-three fork cap, its actual other endpoint is whole A_{18}. Its original normalized dual height is $t < 3.823/\sqrt{18} < .903$. If its cosine $C \geq 6.82$, then

$$E_q/E > 6.82 - .903(6.747) = .727459 > 8/11, \qquad \langle y, u_{\text{reverse}} \rangle > .8.$$

The reverse positive dual is the new A_{18} endpoint dual. Lemma 9.1 applies at that actual q, with factor $< 11/8$ and endpoint height $> .78$, and gives a contradiction. That lemma is independent of the original head and never assumes q farthest. Therefore

$$C_N, C_F < 6.82, \qquad g_N < 1.21, \qquad g_F < 1.6.$$

These bounds follow from actual all-wall finite endpoints and the same original maximum, not a whole-A_{18} farthest theorem.

Two near-three caps are impossible for EVERY head. Write their owns $g, h < 1.21 < 3/2$ and times $A_g^2 = 9g^2 + 8g + 5/4$. Their spatial product is $K = 8gh + (9/2)(g+h) + 2$. For $P = gh, S = g + h$,

$$A_g^2 A_h^2 - K^2 = 17P^2 - 9(g^2 + h^2) - (17/2)P - 8S - 39/16 \leq (47/4)P - 8S - 39/16 < 0,$$

because $P < 9/4$, $g^2 + h^2 \geq 2P$, $P < 3S/4$, $S < 3$. Their actual product is positive, violating acuteness. They are distinct: coincidence would force $g = h = 1/2$, whose actual cycle was already excluded. This proof retains both fork signs and unequal owns.

I.10 Finite heads at least nine, from their exact bare references

For a non-native-five head, the three BARE ending-sixteen references are

	B	b upper	P_J capacity	J
strong five	.9565188	.3652	$> .151$	5
isolated five, strong six	.9373243	.3399	$> .1435$	6
isolated five, strong seven	.9196357	.31938	$> .1366$	7.

They are constructed by the same .016119 append/omission from bare fifteen energies .9403998, .9212053, .9035167. No head or isolated-five pressure has yet been paid in these rows. They remain flat first J, have sum f_{16} and unspent terminal price $> .127$.

For an isolated finite head let $u = \eta^{-2} \in (1, 4/3)$. The positive actual core

$$H + \eta u n_1 + 2\eta(u-1)n_2$$

has square $-\eta^2(7u - 4 - 3u^2)$, zero head/first-root products, negative second-root product, positive time and all unknown signs. Its pressure is $\langle x, Z\rangle \geq h$, with

$$\begin{aligned} Z &= (1 - u/2)(e_1 + e_2) + (u-1)e_3, \\ h &= [\sqrt{2-u} - \sqrt{7u - 4 - 3u^2}]/\sqrt{2}, \\ R &= Z^2 = 3 - 4u + 3u^2/2, \qquad \sum Z_j = 1. \end{aligned}$$

If the native head label is at least ten, then $1 < u \leq 2 - (2/5)\sqrt{5} < 1.11$. On this WHOLE interval $F(u) = .8h - .16R$ decreases: $(7u - 4 - 3u^2)' > 0$ and $F' < -.4/\sqrt{2} + .16 < 0$. At 1.11, positive square bounds give $\sqrt{.89} > .94339, \sqrt{.0737} < .27149$; thus $h > .4748$, $R = .40815$, and $F > .314536$. Pay $.4(Z - P_J/J)$ from the individual P_J anchor. The general original square gives $B + F - .8b + .16/J$, which in the three rows exceeds $1.0108948, 1.00660696, 1.00152484$. No old head lower gain has been subtracted from a paid lower bound.

In the fourth, weak-helper-five case the existing unheaded reference has energy $> .987759$, $w_1 = .5455$, $w_2 = w_3 < .299$. The exact head cross is

$$\begin{aligned} \langle w, Y\rangle < M(u) &= (1.091 - .2465u)/\sqrt{2}, \quad Y = \sqrt{2}Z, \\ Q = Y - (\sqrt{2}/16)P_{16}, &\quad Q^2 = T(u) = 47/8 - 8u + 3u^2. \end{aligned}$$

Pay $\mu\sqrt{2}/16 < .018$ at the genuinely unspent terminal, retaining this full cross. The norm lower bound is $.987759 + 2\mu(f - M) - \mu^2 T$, $f = \sqrt{2-u} - \sqrt{7u - 4 - 3u^2}$. All three functions decrease on the required interval. Bounding f at the RIGHT and M, T at the LEFT endpoint gives

u interval	$f >$	$M <$	$T \leq$	μ	norm lower
$(1, 1.05]$	.7684	.5972	.875	.2	1.021239
$[1.05, 1.08]$	.7125	.5885	.7825	.15	1.00735275
$[1.08, 1.1]$	.6840	.5833	.7342	.14	1.00156468
$[1.1, 1.105]$	.6778	.5798	.705	.14	1.001381
$[1.105, 1.11]$	.6719	.5789	.698075	.13	1.0001415325.

The square brackets for the two radicals at the successive right endpoints are, in order,

$$(.9746, .2062), \quad (.9591, .2466), \quad (.9486, .2646),$$
$$(.9460, .2682), \quad (.94339, .27149).$$

These prove every f entry; M uses $\sqrt{2} > 1.4142$ and T is exact substitution. The table is a complete interval certificate, not a sampled search. All finite labels at least ten are excluded.

For native nine use its actual cubic: $8\eta^3 - 6\eta - 1 = 0$ and $.93969 < \eta < .9397$. Exact products give $1.1324 < u < 1.1325$. Now $\sqrt{2-u} > .93139, \sqrt{7u-4-3u^2} < .28255$, so $h > .4587,\ R < .394$. The strong-five bare row paid by .4 gives $.9565188 + .8(.4587 - .3652) - .16(.394 - .2) = 1.0002788$. For isolated-five rows set

$$Q_H = Z - P_J/J, \quad Q_5 = U' - (5/J)P_J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J.$$

Their squares are $R - 1/J$ and at most $77/150, 194/175$. Pay $\mu = .4, \nu = .09, .1$, respectively. The COMBINED anchor debits are $.85/6 < .1435,\ .9/7 < .1366$. The full one-vector energy bound

$$B + .8(h-b) + 2\nu(1.8185 - 5b) - .16(R - 1/J) - \nu^2 Q_5^2 - .8\nu(1 - 5/J)$$

exceeds $1.00125296, 1.00128598$. The positive mutual cross is explicitly subtracted.

A weak pure helper five cannot coexist with the native-nine head. Its own is $g = \phi/2$ or $.9 < g < 1.2$, and its actual head mutual is $\lambda = A_H\sqrt{10g^2 + 2g - 1/2} - \eta(2g+1)$. On the latter entire interval it increases, since the radical derivative exceeds $\sqrt{10}$ and $A_H\sqrt{10} > 2\eta$. Native-nine brackets give $\lambda(.9) > .049875 > 0,\ \lambda(1.2) < .340629 < 1/2$, a forbidden native gap. At $g = \phi/2$, direct positive squares give $\lambda < .8755(2.769) - .9396(2.618) < -.0356$, contrary to acuteness. All four original initial cases fail, excluding native nine. The remaining finite heads are five, seven and eight.

I.11 The all-head face envelope and an isolated-own cutoff

The paid ending-sixteen reference also controls EVERY asymmetric old A_{16}-face direction. For $|v| \le 1$, pay .127 at its unspent terminal toward $Z = P_{17} + ve_{18}$. The residual has square $1 + v^2$ and zero reference cross. If $\langle x, Z\rangle \ge 3.761 + .0635v^2$, its original square exceeds

$$.9729 + .254(.171) - .016129 = 1.000205$$

because $.254(.0635) = .016129$. Thus

$$\langle x, P_{17} + ve_{18}\rangle < 3.761 + .0635v^2, \qquad \frac{\langle x, P_{17} + ve_{18}\rangle}{\sqrt{17 + v^2}} < .9124. \tag{59}$$

For the second inequality the quotient of the upper bound decreases with v^2; its derivative numerator is $-1.602 + .0635v^2 < 0$, and $3.761^2 < 17(.9124)^2$. The two old fork inequalities force the displacement direction of ANY actual point on old A_{16} to have this form. The original all-point maximum therefore gives cosine < 10.94 for every such point.

Every isolated actual fork own g is less than $15/8$. Indeed its actual original fork endpoint is whole $A_{17} + A_1$, with cosine $C = 3g$, the new A_1 being the cap itself. At $v = \pm 1$, (59) gives original normalized height $< .90145$. If $g \ge 15/8$, the actual endpoint q has

$$E_q/E > 45/8 - .90145(5.5355) = .635023525 > 5/8, \qquad E/E_q < 8/5.$$

The reverse unit is exactly its whole A_1 normal and has height

$$> \frac{.984 - .90145}{1 - .984(.90145)} > .73.$$

We prove the required local contradiction, rather than transporting an original-farthest A_{17} theorem.

At an actual $A_{17}+A_1$ vertex with factor $<8/5$, inspect central node eight in zero-sum $\mathbb{R}^{18}$. Here $D=80/9,\ 2.98<\sqrt{D}<3, \sqrt{D-1}>2.8$. Its retained paths are A_7, A_9, with the finite white list of Appendix C.6; explicitly single-three positions $1,2,3,5,6,7,9,10,16,17$, terminal fours $1,7,9,17$, two endpoint-threes or isolation. The foreign whole A_1 cost is $Q=0$ or native $Q\ge 1/4$, with $Q<1$. At own zero the endpoint-three time is $Q-19/36$ with $Q<9/16$, allowing only $0, 1/4, 1/2$, all impossible. Next-three and endpoint-four require $Q<1/4, Q<1/8$, hence zero and negative time. Other zero-own positions have the actual E_8/E_7, four/simple or cycle nulls. All are positive future actual combinations, feasible against every unselected original wall.

For positive owns, ordinary white rows except native-five single-white-three have actual cosine >5. The near-three residual is at most $9/16$, with numerator $>.9+7/16$; D/E residual at most $1/4$, numerator $>.9+1/20$; foreign far-three residual at most $5/16$ and the same numerator. A near-four with foreign cost satisfies this latter bound. A pure near-four uses its ACTUAL mate at seven or nine, of cosines $(77+70\sqrt{2})/(6\sqrt{7})>10$ or $9/2+4\sqrt{2}>10$. White-four own-three at a far end has the affine F_4 path; near own-three/four has cosine >6 by its ratio. Two whites have residual at most $9/80$, giving cosine >5. Own three/four/six/parallel at the other positions have their actual cycle/star/chain/null boundary, with helper contacts outside the selected principal. Hence every ordinary row has component energy

$$e_A>(5-8/5)^2/24=289/600>12/25.$$

For isolated own γ, clear the whole foreign root and put $\tau=(1-Q)/\gamma^2$. The same actual reciprocal core and positive finite point as in Appendix I.7 give

$$h_8>\sqrt{D-\tau}-\kappa\sqrt{\tau(1-\tau)}\quad(0<\tau<1),$$

or

$$h_8\ge\sqrt{\frac{D-1}{D-\tau}}\{\sqrt{D-1}-\kappa\sqrt{\tau-1}\}\quad(\tau>1).$$

The first case contradicts $e_A\le 12/25$: Cauchy would require loss $>.3(2.8)=.84$, but the core costs at most $\kappa/2<.8$. The null boundary is forbidden. In the finite-point case, foreign radical $<.4$ gives $h_8>2.16$; pure finite own at least eight has radical $<.415$, giving $h_8>2.136$. Both normalized squares exceed $12/25$. Pure five uses the actual H_4 finite projection, $\sqrt{\delta_5}<.1352, \gamma>.8, H/\gamma>2.7$, and gives $h_8>2.4296$. Pure seven uses the actual two-root negative core, native $.9<\gamma<10/11$, and gives $h_8>2.76-(8/5)(.36)=2.184$. Every selected product and future sign is the same direct actual-root calculation already given; no native projection is assumed.

Pure far-three owns at least seven have the UNEQUAL exact times

$$H_1^2=D\gamma^2+(10/9)\gamma-19/36,\quad H_{17}^2=D\gamma^2+(8/9)\gamma-19/36.$$

Both exceed $D\gamma^2$ for $\gamma>.9$. The finite core has radical at most $1/4$ and $H>8/3$, giving normalized pressure $>.85$. The dotted core has radical divided by γ at most $1/2$, giving pressure ratio $>1-\kappa/(2\sqrt{D})>.7$. Again energy exceeds $12/25$.

The remaining native-five single-white-three row is paid jointly with the known actual A_1 reverse pressure. Choose the H_4 arm on the side AWAY from the helper. Its positive finite point has full displacement U, including the foreign component, and residual $\delta_5<.01825$. The smaller cross $C_{8j}\ge 8/9$ gives $H>2.45$. Therefore the normalized point pressure exceeds $1-(8/5)(.1352)/2.45>.9$.

The actual foreign coefficient is $< 3/4$, so its correlation with the unit A_1 is $< 1/3$. For the ONE vector

$$W = (387/800)u + (591/800)U/\|U\|$$

the positive pressures $> .73, > .9$ give $\langle y, W\rangle > 1.0180125$, $W^2 < 1.0180125$; hence $\|y\|^2 > 1$. All other rows give the orthogonal sum $e_A + e_{A_1} > .48 + .73^2 > 1$. This proves the complete local contradiction and the original cutoff $g < 15/8$, including every head and both signed fork choices.

I.12 Two standalone local lemmas for a positive two-dual reverse

The distant fork-pair exits below require a reverse direction which can involve TWO positive duals. It cannot be replaced by a large single endpoint coordinate. We prove the two precise local lemmas here.

First let the whole actual forest be $A_{17} + A_1$, comparison $\kappa < 1.262$, and let

$$u = ae + b\omega_{17}, \quad a, b > 0, \quad a^2 + (17/9)b^2 = 1, \qquad \langle y, u\rangle > .8564,$$

where e is the whole A_1 unit. Call only sites eight and nine weak if $h_i \le \sqrt{D_i} - .8$. If either is strong, its normalized dual has height $> .73$ and correlation with u less than $1/4$ (the exact bounds are $\sqrt{8/(17\cdot 10)}, 1/\sqrt{17}$). The single vector $.75u + .55v_i$ has pressure > 1.039 and square < 1.07125, giving norm > 1.00675. Both sites must be weak.

Their finite-white list is the same full $N = 18$ list, with all four retained side orders at least eight. Thus zero-own endpoint-three has time $Q - 19/36$, with native $Q < 9/16$ allowing only $0, 1/4, 1/2$; next-three has strict $Q < 2/m \le 1/4$; endpoint-four has $Q < 1/8$. They all have negative time, and every other zero-own role is the actual positive affine kernel already displayed. Using site seven instead would lose that strict side-order bound.

For positive ordinary ratios $R \ge 3/2$, the correct comparison at the weaker $\kappa = 1.27$ and loss .8 reduces to the polynomial

$$-.21q^3 + 1.1729q^2 - 1.6q + .64 > 0 \qquad (2.7 \le q \le 3.1).$$

It is concave and its endpoint values are $.737011, .695459$. Near-four side orders at least eight also have ratio $> 3/2$. Foreign exceptional helpers have ratio $> 227/(25\sqrt{35})$, so cannot be weak. Every isolated actual cap is strong by the same positive points; their complete conservative losses at factor 1.27 are

dotted	$< (.193 + 1.285)/2 = .739$
finite ≥ 8	$< 5/26 + 1.27(.415) < .720$
5 finite point	$< 25/76 + 1.27(1.01)/(6(.8)) < .597$
7 negative core	$< 10000/41877 + 1.27(.36) < .697$
$Q > 0, \tau > 1$	$< .193 + 1.27(.4) = .701.$

These follow from the displayed τ formulas, with $.193^2 + 1.27^2 < 1.285^2$; the null boundary is unchanged. Thus both weak sites must be pure far helpers with $\phi/2 \le \gamma < 17/16$. Their complete common/outward/reversed actual-product calculation and positive two-cap Schur matrices were proved in Appendix C.6, before its comparison. In particular the reversed numerator has minimum $2763353243/50000000000 > 0$, unless both owns are five, when its separate geometric distance-one obstruction applies. The actual two-cap point has weighted mean loss

$$< .1 + 1.262(7/16) = .652125 \quad \text{or} \quad < .1 + 1.262(3/10) = .4786,$$

both $< .8$. Its entire actual A_1 block is retained and every original wall satisfied. This contradicts the two weak sites and proves the first standalone two-dual lemma.

For the second, let the whole actual forest be A_{18}, comparison $\kappa < 1.27$, and

$$u = a\Omega_{18} + b\Omega_{17}, \quad a, b > 0, \quad \|u\| = 1, \quad L = a + 2b \le \sqrt{19/17} < 1.058, \quad \langle y, u\rangle > .85.$$

The inequality for L is Cauchy in the genuine $N = 19$ inverse end block. The complete own/white/null geometry in Appendix D is independent of its later single-endpoint consumer. At sites five through fourteen, the same finite ratio polynomial and isolated point losses just displayed are $< .8$. Hence possible weak roles are near-four at five/six or reflection, or pure far helpers with $\phi/2 \le \gamma < 17/16$. For two adjacent PURE far helpers, the actual two-cap comparison gives weighted losses $< .655625, .481$, both below .8. Thus pure helpers are nonadjacent. This statement does not yet include any near-four role.

A near-four actual mate has cosine > 11, hence normalized height $> 7/8$. Its reverse correlation is $L\sqrt{2j/[19(19-j)]}$. Left mates $j = 4, 5$ have correlation $< .206$ and their two-direction sum contradicts the norm budget. Every left near-four is therefore excluded first. For a right mate $j = 14, 15$, if both left sites five and six were weak, both would now be pure helpers, contradicting their nonadjacency. Hence some left $k = 5, 6$ is strong, with height $> .703$. The three full correlations are $< .666, .234, .407$. The ONE vector $.5u + .6u_j + .4u_k$ has pressure > 1.2312, square < 1.45856, and gives norm > 1.00384. Thus every weak site is a nonadjacent pure helper.

Take symmetric rational heights $H_{19-i} = H_i$, linear on both ends,

$$H_5 = 1.914, \ H_6 = 2.065, \ H_7 = 2.173, \ H_8 = 2.243, \ H_9 = H_{10} = 2.277.$$

Their positive square tests $(H_i + .8)^2 < D_i$ are the earlier .87-window tests, with heights shifted by .07. Flatten a weak first/last block through six, and omit every remaining weak knot by its chord. The complete nonadjacent omission debit is $2(.043^2 + .038^2 + .036^2 + .034^2)/4 = .0028725$. If fourteen is weak, the relevant last block has length $m = 6, K = 13, H_K = 2.065$, energy $E_0 > .72555$ and anchor $> .11808$. If fourteen is strong, $m = 5, K = 14, H_K = 1.914$, energy $E_0 > .74793$, anchor .1159. The initial energy formulas are ordinary squared slope sums; the worst two-weak-end value before omission is $2.065^2/6 + .01772 = .7284241666\ldots$; the one-weak-end value is $.7508121833\ldots$. Stronger opposite-end patterns only increase these bounds.

Retain the exact reverse residual

$$G = u - (L/m)\Omega_K, \qquad \langle w, G\rangle = 0, \qquad G^2 = 1 - 2KL^2/(19m).$$

Payment $\mu = 1/2$ spends only $L/(2m) < .08817, .1058$ at the appropriate actual strong anchor, fitting its own capacity. One completed square gives

$$\|y\|^2 > E_0 + .85 - (H_K/m)L - 1/4 + KL^2/(38m).$$

The gain decreases up to $L = 1.058$, where it is $> .299695$ or $> .277477$. Thus the norm exceeds 1.025245 or 1.025407. The negative term in G^2 and every individual coefficient are retained. This proves the second standalone lemma.

I.13 The local comparison with two orthogonal spectators

We shall need a comparison at a weaker factor than the dihedral comparison in Appendix H.12. At an actual $A_{16} + 2A_1$ vertex q, suppose that every chamber point is compared with the original maximum at factor $\kappa \le 33/20$. If u is a unit vector in the whole two-A_1 plane, then

$$\langle y, u\rangle > .738 \tag{D2A1}$$

is impossible. Both spectator coordinates are retained in this statement.

Here are the complete local details. In zero-sum $\mathbb{R}^{17}$ put

$$\omega_i = \sqrt{2}(P_i - iP_{17}/17), \quad D_i = 2i(17-i)/17, \quad C_{ij} = 2i(17-j)/17 \quad (i \le j).$$

Thus $D = D_8 = 144/17$, $2.91 < \sqrt{D} < 3$ and $\sqrt{D-1} > 2.73$. Write the actual opposite to eight as $d = Hq - v_A - v_S$, with own γ, and $v_S = \beta_1 a_1 + \beta_2 a_2$. The a_j are the two actual orthogonal unit roots. Their full cost is $Q = \beta_1^2 + \beta_2^2 < 1$. Each nonzero β_j is a finite native cosine, at least $1/2$. Clearing spectators uses the positive actual sum $w_S = \beta_1 a_1 + \beta_2 a_2$.

Deleting eight leaves $A_7 + A_8$. Its complete one-node finite extensions are one 3 at $1, 2, 3, 5, 6$, $7, 9, 10, 15, 16$, one 4 at $1, 7, 9, 16$, or one endpoint 3 in each path. A second contact within either path makes a cycle; the exceptional extensions supply exactly the displayed E_8 positions. For own zero at a global end 3, $H^2 = Q - 9/17$ whereas positivity requires $Q < 9/16$. The entire native two-spectator alphabet below $9/16$ is $Q = 0, 1/4, 1/2$, so the time is negative. The next-3 and end-4 remainders are $1/4, 1/8$; they force $Q = 0$ and again negative time. Isolation gives $H^2 = Q - 1 < 0$. All other own-zero roles have the actual affine E_8/E_7, 4/3/3 star or all-3 cycle. For positive own, isolated $3, 4, 6$, a single-white own 3, the non-near single-white own 4, and own 6 have the corresponding star, cycle or chain null. The near-own-4 roles at seven and nine are retained. Parallel gives $d + n_8$. Every null is a positive sum of actual normals, has positive time through d, and is nonpositive against all omitted walls, including both spectators.

For a single white 3 in a side of order $l = 8$ or 9, at distance t from its remote end, the exact ratio is

$$\frac{C_8}{\sqrt{D}} = \frac{\gamma + t/(2l)}{\sqrt{1 - t(l-t)/(2l) - Q}}.$$

For white 4 replace $t/(2l)$ by $t/(\sqrt{2}l)$ and $t(l-t)/(2l)$ by $t(l-t)/l$. A two-end join adds the two costs. Reserve native-5 single-white-3 for the paired argument below. Every other surviving single-white own is $> .9$, except the retained near-own-4. The following lower squares all exceed 25:

role	C_8^2 lower bound
near 3	11449/425
other non-far D/E 3	51984/1700
left far 3 with $Q > 0$	853776/34000
right far 3 with $Q > 0$	118336/4675
near-own-4 with $Q > 0$	1192464/34000.

In the two far rows $Q \ge 1/4$, with the unequal residuals $5/16, 11/36$ and numerators $77/80, 43/45$. For near-own-4 at $Q = 0$, the same actual opposite is the seven- or nine-edge mate, with finite forests $A_6 + B_{10} + 2A_1$ or $B_9 + A_7 + 2A_1$ and lengths

$$(70 + 63\sqrt{2})/\sqrt{238} > 9, \qquad (18 + 16\sqrt{2})/\sqrt{17} > 9.$$

These are positive projections, so all omitted wall signs are checked termwise. The mate's own normalized height is $> (9 - 1.65)/\sqrt{80} > .8$. To obtain the central height as well, use concavity of the actual A_{16} heights $h_i = \langle y_A, \omega_i \rangle$, with $h_0 = h_{17} = 0$: $h_8 \ge (9/10)h_7$ or $h_8 \ge (8/9)h_9$, respectively. Since $D_7 = 140/17$ and $D_8 = D_9 = 144/17$, this gives

$$\frac{h_8}{\sqrt{D_8}} > .8 \min\{(9/10)\sqrt{140/144},\, 8/9\} > .68.$$

White 4 forces $Q = 0$; its far-own-3 has the affine F_4 path, while the near-own-3 and every own at least 4 have $C_8 > 5$. The two-end-3 join has residual $17/144 = 1/D$ and $C_8 \geq 89/17 > 5$. Consequently all these ordinary roles have

$$\frac{\langle y_A, \omega_8 \rangle}{\sqrt{D}} > \frac{5 - 1.65}{\sqrt{24}} > .68, \qquad e_A > .4624 > .46.$$

For isolated contact put $\tau = (1 - Q)/\gamma^2$. For $0 < \tau < 1$, the actual core

$$\Xi = d + w_S + \gamma\tau n_8, \qquad U = \gamma(\omega_8 - \tau n_8)$$

has square $-\gamma^2\tau(1 - \tau)$, positive time, zero selected products except a negative n_8 product, and all omitted products nonpositive. If $e_A \leq .46$, its pressure would exceed $(1 - .68)\sqrt{D - \tau} > .8736$, whereas comparison bounds it by $\kappa/2 \leq .825$. At $\tau = 1$ it is a forbidden future null. For $\tau > 1$ the positive finite projection

$$q + \frac{H}{\gamma^2(\tau - 1)}(d + w_S + \gamma n_8)$$

gives

$$\langle y_A, \omega_8 \rangle \geq \sqrt{(D - 1)/(D - \tau)}(\sqrt{D - 1} - \kappa\sqrt{\tau - 1}).$$

For $Q > 0$, the radical is at most $2 - \phi < .4$; for $Q = 0$ and finite own at least 8 it is at most $\sqrt{2} - 1 < .415$. The resulting heights are > 2.07 and > 2.04525, respectively, and their normalized values exceed .68. Pure isolated 5 uses the positive H4 correction $\gamma(3n_8/2 + n_7 + n_6/2)$, residual $\delta_5 = (5 - 3\phi)/8 < .01825$; since $H > 2$, its normalized pressure exceeds $1 - 1.65(.1352)/2 > .88$. Pure isolated 7 uses $d + (4\gamma/3)n_8 + (2\gamma/3)n_7$. The cubic $8a^3 - 4a^2 - 4a + 1$ gives $.9 < a < 10/11$, hence $H/\gamma > 13/5$ and the core radical divided by γ is $< 9/25$. Its pressure exceeds $1003/1300 > .68$.

For a pure far-3 helper at either end the exact unequal times are

$$H^2 = D\gamma^2 + C_{8e}\gamma - 9/17 > D\gamma^2, \qquad C_{8e} \in \{18/17, 16/17\}.$$

For finite own at least 7, with $\delta = 1 - \gamma^2 < .19$, the core $d + \gamma n_8 + 2\delta n_e$ has radical at most $1/4$, $H > 2.6$ and pressure $> .84$. For dotted own use $d + n_8/\gamma$; its radical divided by γ is at most $1/2$, giving pressure $> 1 - 1.65/(2\sqrt{D}) > 83/116$. All these cores are positive actual sums and check every wall.

The only exception to $e_A > .46$ is native-5 single-white-3. Choose the H4 correction on the side opposite its helper, and retain the full $U = v_A + v_S - w_5$, rather than calling it pure A. Its actual projected point gives

$$\langle y, U/\|U\| \rangle > 1 - \frac{1.65(.1352)}{H} > .9, \qquad H^2 \geq (44\phi + 27)/17 > 2.4^2.$$

The full white remainder gives $Q < 9/16$, and hence $|\langle u, U/\|U\| \rangle| < .75/2.4 < 1/3$. One vector

$$W = .49275u + .73575\, U/\|U\|$$

has $\langle y, W \rangle > 1.0258245$ and $W^2 < 1.0258245$. Thus $\|y\|^2 \geq 2\langle y, W \rangle - W^2 > 1$. For every other role, $e_A + e_{2A_1} > .46 + .738^2 = 1.004644$. This proves the local statement without adding overlapping energies.

I.14 Isolated pairs: finite and orthogonal mutual products

Write the two actual isolated caps as

$$c = Ap - g\Omega_f, \quad d = Bp - h\Omega_o, \qquad A^2 = 9g^2 - 1, \quad B^2 = 9h^2 - 1,$$

and $\lambda = -\langle c, d\rangle = AB - 8gh \geq 0$. The identity

$$(9gh - 1)^2 - A^2B^2 = 9(g - h)^2$$

gives $\lambda \leq gh - 1$; all square roots here are positive. If $0 < \lambda < 1$, it is a native finite cosine, at least $1/2$. The positive projection onto c, d gives an actual $A_{16} + I_2(m)$ vertex with

$$C^2 = \frac{17g^2h^2}{1 - \lambda^2} \geq \frac{17(1 + \lambda)}{1 - \lambda} \geq 51.$$

Its displacement is a positive multiple of $P_{17} + ve_{18}$, $|v| < 1$. Every released root, the head and every unknown wall has nonpositive product termwise with the projection. The original envelope therefore gives

$$\kappa < \frac{3}{2}, \qquad \langle y, u\rangle > \frac{.99 - .9124}{1 - .99(.9124)} > .8$$

in the entire new $I_2(m)$ plane; the original maximum is unchanged. The local A_{16} comparison in Appendix H.12 gives $e_A > 7/16$ for $m \geq 4$.

For completeness the missing spectator $m = 3$ introduces no uncontrolled contact. Its full inverse matrix is $\left(\begin{smallmatrix} 4/3 & 2/3 \\ 2/3 & 4/3 \end{smallmatrix}\right)$. Two positive contacts cost at least one, and a single contact gives exactly

$$Q = 0, \quad 1/3, \quad 2/3, \quad \phi^2/3.$$

The A-contact list is exactly that of Appendix I.13. At own-zero end-3 the only $Q < 9/16$ are zero and $1/3$, both below $9/17$; the next-3 and end-4 rows force zero. All other own-zero and small-own nulls are the same actual principals. Every positive foreign cost is at least $1/3 > 1/4$. Consequently every ordinary ratio, isolated-core estimate, pure-helper estimate and paired H4 estimate in that appendix remains valid with the positive clearing supplied by the A_2 inverse, at the smaller factor $3/2$. In particular the ordinary and isolated energies exceed .46; in the native-5 exception the full foreign correlation with the unit reverse direction remains $< \sqrt{Q}/H < 1/3$, and its same single-vector proof applies since the reverse height exceeds $.8 > .738$. Thus either one square already contradicts the norm bound, or $e_A + e_{I_2} > 7/16 + .64 > 1$. This closes all positive finite mutuals. For $\lambda = 1$, $c + d$ is a nonzero feasible future null.

We next exclude $\lambda = 0$, with no symmetry assumption. The exact relation

$$(17a - 9)b = 9a - 1, \qquad a = g^2, \quad b = h^2$$

shows that exactly one own is finite and one dotted: two finite owns give $\lambda < 0$, and two dotted owns give $\lambda > 0$. Name the finite one $g = \cos(\pi/m)$, $m \geq 7$, so $a < 1 < b$. The actual finite set $n_1, \ldots, n_{15}, n_f, c, d$ has forest $A_{15} + I_2(m) + A_1$. Its positive projection is

$$q_0 = p + \frac{A}{1 - a}c + \frac{gA}{1 - a}n_f + Bd, \qquad C^2 = \frac{8ab}{1 - a}.$$

Its released products are strictly negative, all unknown products are nonpositive, and $q_0^2 = \langle p, q_0\rangle = -C^2$. Write

$$q_0 = C^2p - KV, \quad V = P_{16} + \alpha(e_{17} - \sigma e_{18}), \quad \alpha = \frac{8(1 - a)}{9a - 1} \in (0, 1).$$

The full original reference has $\langle w, V\rangle = f_{16} < 3.5899$. Paying its unspent .127 at the residual $V - P_{16}$, whose square is $2\alpha^2$ and cross with w is zero, proves

$$\frac{\langle x, V\rangle}{\sqrt{16 + 2\alpha^2}} < \frac{3.6967 + .127\alpha^2}{\sqrt{16 + 2\alpha^2}} < .9242.$$

Indeed a violation would give $.9729 + .254(3.6967 - 3.5899) > 1$; the α^2 terms cancel exactly. The displayed envelope decreases with α^2.

The native-7 cubic gives $.9009 < g < .901$. The increasing function

$$C^2 = \frac{9a - 1}{1 - a} + 9\frac{9a - 1}{17a - 9}$$

then exceeds $45.2944 > 6.73^2$, and $\alpha > .238$. The envelope improves to .9228. For $m \geq 8$, $a > (.8535)$, the same function gives $C^2 > 56.49 > 7.5^2$. The exact comparisons are respectively

	E_q/E	$\langle y, u\rangle$
$m = 7$	$> .58848916 > 10/17$	$> (.9888 - .9228)/(1 - .9888(.9228)) > .75$
$m \geq 8$	$> .6294972 > 10/17$	$> (.99 - .9242)/(1 - .99(.9242)) > .75.$

Here $u = (Cq - p)/\sqrt{C^2 - 1}$ belongs to the entire $I_2(m) + A_1$ space. Thus $\kappa < 17/10$ and its energy exceeds 9/16.

We give the local A_{15} argument at this exact factor. In its zero-sum $\mathbb{R}^{16}$ frame, $D_8 = 8$. A finite-white contact to $I_2(m \geq 7)$ costs at least $1/(4\sin^2(\pi/m)) > 25/19 > 1$, so none is possible. The full foreign vector lies in the retained A_1, with cost $Q = \beta^2$, each positive cost at least 1/4. Deleting eight leaves $A_7 + A_7$, with single-3 positions $1, 2, 3, 5, 6, 7, 9, 10, 11, 13, 14, 15$, endpoint 4, or one endpoint 3 in each path. For own zero, the end-3 time is $Q - 17/32$ with $Q < 9/16$, so $Q = 0, 1/4, 1/2$ gives negative time. Next-3 and end-4 force zero with times $-1/8, -1/16$. The remaining own-zero and small-own roles have the same actual affine stars, chains and cycles as the central inventories above. Keep near-own-4 and native-5 single-white-3.

For positive single-white-3 other than native 5, the own is $> .9$. The residual/numerator bounds are

near 3	9/16	$.9 + 7/16$
D position	1/4	$.9 + 1/8$
E position	1/16	$.9 + 3/16$
far 3, $Q > 0$	5/16	$.9 + 1/16.$

The first three give $C_8 > 5$, and the last $C_8^2 > 5929/250 > 4.86^2$. At factor 1.7 the latter normalized height exceeds $3.16/4.757 > .663$, whose square exceeds 7/16. Near-own-4 with foreign cost is stronger. Without foreign cost its actual seven/nine mate has length $\sqrt{63/8}(3/2 + 4\sqrt{2}/3) > 9$. White 4 forces $Q = 0$; the far-own-3 affine F_4 null is excluded, while the retained near-own-3 and every own at least 4 have $C_8 > 6$. The two-end-3 join has $C_8 \geq 8\gamma + 1 \geq 5$.

For isolated contact, $\tau = (1 - Q)/\gamma^2$ gives the same actual core $d + w_S + \gamma\tau n_8$. If $0 < \tau < 1$, energy at most 7/16 would force its pressure $> \sqrt{7} - 7/4 > 17/20$, against $\kappa/2 < 17/20$. The boundary is an actual null. For $\tau > 1$, positive projection gives

$$h_8 \geq \sqrt{7/(8 - \tau)}(\sqrt{7} - \kappa\sqrt{\tau - 1}).$$

For $Q > 0$, $\tau < 75/64$; for pure finite own at least 8, $\sqrt{\tau - 1} < .415$. Thus $h_8 > 1.8945 > 17/9$, giving normalized height $> 2/3$. Pure isolated 5 and 7 use the actual H4 point and the $(4\gamma/3, 2\gamma/3)$

core: their heights exceed 2 and $1.988 > 17/9$. Pure far-3 has $H^2 = 8\gamma^2 + \gamma - 17/32 > 8\gamma^2$. For finite own at least 7 the $d + \gamma n_8 + 2(1-\gamma^2)n_e$ core has radical at most $1/4$, $H > 2.5$, and unit pressure $> .83$. For dotted own $d + n_8/\gamma$ gives pressure $> 1 - 1.7/(2\sqrt{8}) > 39/56$. Both ends and every unbounded interval are included.

Finally native-5 single-white-3 uses the opposite-side H4 point, retaining its full foreign component U. Here $H^2 \geq (5/2)\phi + 47/32 > 11/2$, $\delta_5 < .01825$, so its unit pressure is $> 1 - 1.7(.1352)/2.34 > .9$. Its absolute correlation with the reverse u is $< .75/2.34 < 1/3$. One vector $(81/160)u + (117/160)U/\|U\|$ has pressure and square bounded on opposite sides of 1.0378125; hence $\|y\|^2 > 1$. All other roles have $e_A > 7/16$, and the orthogonal reverse-space energy is $> 9/16$. This excludes the orthogonal pair. Every local point above is a positive actual projection or core; the wall checks include the old head and all unknown walls.

I.15 The complete dotted isolated pair

Only $\lambda > 1$ remains for two isolated caps. Put $P = gh$. The inequality $\lambda \leq P - 1$ gives $P > 2$; the individual cutoff $g, h < 15/8$ then gives $g, h > 16/15$, so both owns are dotted. There is no need to use the older auxiliary four-thirds gap. Assume $g \geq h$. The actual feet

$$q_c = \frac{d + \lambda c}{\sqrt{\lambda^2 - 1}}, \qquad q_d = \frac{c + \lambda d}{\sqrt{\lambda^2 - 1}}$$

are future and feasible: the selected product is zero, the other is negative, and every omitted wall is checked termwise. Their squared coshes satisfy

$$C_c^2 = 9g^2 + \frac{17P^2}{\lambda^2 - 1}, \qquad C_d^2 = 9h^2 + \frac{17P^2}{\lambda^2 - 1}.$$

Thus

$$C_c^2 \geq 9P + \frac{17P}{P - 2} > \frac{441225}{6208} > 71.$$

The last expression decreases on $2 < P < 225/64$, so the strict individual cutoffs justify its endpoint evaluation.

Retain the actual edge $A_{16} + c$ obtained by releasing the other fork at the $A_{17} + A_1$ neighbour of c. The perpendicular foot of p is $(p + Ac)/(3g)$, with zero product against that released fork. The chosen q_c has negative product against it. In the retained Lorentz plane this signed product is a positive multiple of $-\sinh t$; hence q_c is on the forward ray. Compactness supplies the next actual endpoint R, at least as far from p. The old A_{16} envelope gives $C_R < 10.94 < 11$. For every such endpoint,

$$\sqrt{71} < C_R < 11, \quad \kappa < 1.262, \quad \langle y, u \rangle > .8564, \tag{DIIdot}$$

where $u = (C_R R - p)/\sqrt{C_R^2 - 1}$. For the two latter bounds use $t < .9124$, the increasing function $C - .9124\sqrt{C^2 - 1}$, and $C_R^2 > 441225/6208$, which gives $\sqrt{1 - C_R^{-2}} > .99294$ since $(1 - .99294^2)(441225/6208) > 1$. For example $(.99294 - .9124) - .8564(1 - .99294(.9124)) > .00000282$. All comparisons retain the same original maximum.

The complete finite one-node extensions of the retained A_{16} are $A_{16} + I_2(m)$, including $m = 2$; $A_{17} + A_1, B_{17} + A_1, D_{17} + A_1$; and whole A_{18}, B_{18}, in their allowed end orientations. This is the complete finite classification, not a list of likely endpoint types. The new wall $e = Tp - v$ is distinct from old walls: the released fork vanished at the starting endpoint only, and the other old fork has a dotted product with c. Thus $T > 0$ and every original contact of e belongs to $0, \cos(\pi/m), 1, (1, \infty)$.

If e is orthogonal to A_{16}, the reverse lies in the whole rank-two plane. For $m \geq 4$ the local A_{16} lemma in Appendix H.12, and for $m = 3$ its extension above, give energy $> 7/16$; the reverse energy $> .8564^2$ gives a contradiction. For $m = 2$, Appendix I.13 applies directly. For whole B_{18}, the inverse diagonal at c is 18, so $C_R^2 \geq 1 + 18A^2 > 121$, impossible.

Here are the remaining B/D germs without a purity assumption. Put $x = -\langle e, n_f\rangle$, $y = -\langle e, n_o\rangle$. For $B_{17} + A_1$, $T^2 < 120/17$. The near white 4 already gives $T^2 \geq 15$. For the far white 4,

$$T^2 = \sqrt{2}(x + y) + 9(x^2 + y^2) + 16xy, \qquad \frac{T^2}{g^2} = -\tfrac{1}{2} + \sqrt{2}y + 17y^2.$$

If both contacts are positive then $T^2 \geq 8.5 + \sqrt{2}$. If $y = 0$ the second expression is negative. If $x = 0$, its denominator exceeds the first expression, forcing $g^2 < 1$, again impossible. For $D_{17} + A_1$, $T^2 < 240/17$. At the near white 3,

$$T^2 = 13/2 + 15(x + y) + 9(x^2 + y^2) + 16xy.$$

One positive contact forces $T^2 \geq 65/4$; if both vanish, the actual affine D_5 vector with coefficients $1, 1, 2, 2, 1, 1$ on $e, n_{14}, n_{15}, n_{16}, n_f, n_o$ is a forbidden future null. At the far white 3,

$$T^2 = 2(x + y) + 9(x^2 + y^2) + 16xy, \qquad \frac{T^2}{g^2} = -1 + 2y + 17y^2.$$

The zero-contact cases fail as above. If a positive contact is at least $1/\sqrt{2}$, then $T^2 \geq 31/4 + 5\sqrt{2} > 240/17$. The only remaining pair is $x = y = 1/2$, whose actual e, n_f, n_{16}, n_o all-3 cycle gives a feasible null.

The final $A_{17} + A_1$ and whole-A_{18} germs have

$$u = \frac{T\Omega'_e + A\Omega'_c}{\sqrt{C_R^2 - 1}},$$

with both coefficients positive. These are exactly the two standalone local statements of Appendix I.12, at the factors and pressure in (DIIdot). Thus all actual endpoints are impossible, completing the isolated/isolated exclusion over every head.

I.16 The complete isolated/near-three pair

Let

$$c = Ap - g\Omega_f, \quad d = Bp - h\Omega_o - \Omega_f/2, \quad A^2 = 9g^2 - 1, \quad B^2 = 9h^2 + 8h + 5/4.$$

They are distinct actual walls by their different fork contacts. Write $\lambda = -\langle c, d\rangle \geq 0$. For a positive finite mutual,

$$17g^2h^2 - 9(g^2 + h^2) - 16\lambda gh - 8h - 9\lambda g - 5/4 - \lambda^2 = 0.$$

Consequently $gh > (18 + 16\lambda)/17 > 3/2$. The positive pair projection onto c, d, retaining old A_{16}, is an actual $A_{16} + I_2(m)$ vertex and has exactly

$$C^2 = \frac{17g^2h^2}{1 - \lambda^2} > 51.$$

Its displacement is a positive fork-dual combination. The same envelope gives $\kappa < 3/2$, reverse pressure $> .8$; the complete dihedral local comparison above excludes it. At $\lambda = 1$, $c + d$ is an actual future null.

Suppose next that $\lambda > 1$. The positive core

$$\Xi = d + c/\lambda, \quad k = \sqrt{1 - \lambda^{-2}}$$

has square $-k^2$, zero d product and negative c product; all other wall products are nonpositive. Its time is $B + A/\lambda > 0$. Put

$$D = g/\lambda + h + 1/2, \quad v = \sigma(g/\lambda + 1/2 - h)/D, \quad w = k/D.$$

Comparison at the genuine point Ξ/k gives

$$\langle x, P_{17} + ve_{18} \rangle \geq \sqrt{17 + v^2 + 2w^2} - \sqrt{2}w. \tag{DINcore}$$

We verify the entire continuous bound $w < 1/4$. Put $z = 1/g^2$, $H = h + 1/2$, and $j = \lambda/g = \sqrt{9 - z}B - 8h - 9/2 > 0$. Then

$$w^2 = \frac{j^2 - z}{(1 + jH)^2}, \quad \partial_j = \frac{2(j + Hz)}{(1 + jH)^3} > 0, \quad \partial_z = -\frac{1}{(1 + jH)^2} < 0.$$

Replace $g < 15/8$ by $15/8$ to increase this value. The original $C_{\text{near}} < 10$ gives $h < 2$. Concavity $B'' = -19/(4B^3) < 0$ and its tangent at 2 give

$$B(h) \leq (44h + 37/2)/\sqrt{213}, \qquad j < .901h - .757 =: J(h),$$

using $\sqrt{1961/47925} < .20229$. Since the endpoint $j > 8/15$, this forces $7/5 < h < 2$. For $P = 1 + J(h)(h + 1/2)$ define

$$K = P^2 - 16J(h)^2 + 1024/225.$$

Now $K' = 2PP' - 32(.901)J$ and $K''' = 12(.901)P' > 0$, so K' is convex. Its values at $7/5, 2$ are negative: use $P < 2, P' < 2.22, J > .5$ at the first end, and $P < 3.62, P' < 3.3, J > 1.04$ at the second. Thus K decreases, and $K(2) = 185569/1440000 > 0$. This proves $w^2 < 1/16$, with no native-label sampling. The right side of (DINcore) decreases with w, and after subtracting $.0635v^2$ it increases with v^2. Since $\sqrt{137/8} > 4.138$, $\sqrt{2}/4 < .354$, it exceeds

$$3.784 + .0635v^2,$$

contradicting the original envelope $3.761 + .0635v^2$.

It remains to treat $\lambda = 0$. The exact equation is

$$g^2 = \frac{9h^2 + 8h + 5/4}{17h^2 - 9}.$$

Near own 4 gives a negative denominator. For own 5 it gives $(25\phi + 14)/(17\phi - 19) > 225/64$, contrary to $g < 15/8$. Thus $h > .9$. Since c, d are orthogonal and orthogonal to old A_{16}, the positive projection

$$q_0 = p + Ac + Bd, \qquad C^2 = 1 + A^2 + B^2 = 17g^2h^2$$

is an actual $A_{16} + 2A_1$ vertex. Every released and unknown wall is nonpositive. For all $h > .9$, $C^2 > 37$: after clearing the positive denominator, the difference is

$$f(h) = 153h^4 + 136h^3 - (2431/4)h^2 + 333.$$

Here $f'' > 1000$, $f(1.1) = 2.6458$, $f'(1.1) = -28.798$, so $f(h) \geq 2.6458 - 28.798(h - 1.1) + 500(h - 1.1)^2 > 2.23108$. The displacement again has direction $P_{17} + ve_{18}$. Using $\sqrt{36/37} > .9863$ and $\sqrt{37} > 6.0827$ gives

$$E_q/E > .6083 > 20/33, \qquad \langle y, u \rangle > \frac{.9863 - .9124}{1 - .9863(.9124)} > .738.$$

The whole two-A_1 direction is $u = (Ac+Bd)/\sqrt{A^2 + B^2}$. Appendix I.13 excludes this exact comparison. Finite, parallel, dotted and orthogonal mutuals have now all been covered for isolated/near-three pairs.

I.17 Isolated/far-three: every nondotted mutual

For an isolated cap and a far-three cap write

$$c = Ap - g\Omega_f, \quad d = Fp - h\Omega_o - \Omega_1/2, \quad A^2 = 9g^2 - 1, \quad F^2 = 9h^2 + h - \tfrac{1}{2}.$$

They are distinct actual walls. Here $g > .9$, and h is native 5, finite at least 7, or dotted. Put $\lambda = -\langle c, d \rangle \geq 0$, $s_0 = \sqrt{19}/6$. The positive square identity, with $X = 3g, Y = 3h + 1/6$, is

$$(XY - s_0)^2 - (X^2 - 1)(Y^2 - s_0^2) = (s_0 X - Y)^2.$$

It gives the useful full-domain inequality

$$gh \geq \lambda + s_0, \qquad .726 < s_0 < .727. \tag{DIFproduct}$$

At $\lambda = 0$, direct squaring gives

$$g^2 = \frac{9h^2 + h - 1/2}{17h^2 + h - 19/4}.$$

This function decreases: its derivative numerator is $-8h^2 - (137/2)h - 17/4$. For $h > .9$ it gives $g^2 < 769/992 < .81$. For the remaining own $h = \phi/2$ it gives $(11\phi + 7)/(19\phi - 2)$, strictly between $.86, .8649$. Thus g lies strictly between native labels eight and nine, and is < 1, so cannot be dotted. This excludes mutual zero.

We retain an additional simultaneous coordinate bound only as a constraint on the same x:

$$S_2 < 1.04. \tag{DIFS2}$$

Indeed $S_{16} > 3.589$, and if $S_2 \geq 1.04$, Cauchy on the first two and next fourteen coordinates gives $\|x\|^2 > 1.04^2/2 + 2.549^2/14 > 1$. The right side increases for $S_2 \geq 1.04$; also $S_2 \leq \sqrt{2} < S_{16}$. No extra component energy has been added to a paid reference.

Clear d from old A_{16} by its positive endpoint dual:

$$\bar{d} = d + \omega_1^{A_{16}}/2 = Fp - (h + 1/34)\Omega_o - \Omega_f/34, \qquad \bar{d}^2 = r = 9/17, \quad \langle c, \bar{d} \rangle = -\lambda.$$

For $\lambda = 1/\sqrt{2}$, the positive pair inverse, of determinant $r - \lambda^2 = 1/34$, gives an actual whole-B_{18} point, with

$$C = g(34h + 1)/\sqrt{2} > 12.$$

Its direction is a positive combination of the fork duals; the original envelope gives $E_q/E > .0876C > 1.0512$, contrary to the original maximum.

For $\lambda = \phi/2$, clear only n_1: $d_1 = d + n_1/2$ has square $3/4$. The positive projection on c, d, n_1 has coefficients

$$a = \frac{(3/4)A + \lambda F}{\Delta}, \quad b = \frac{F + \lambda A}{\Delta}, \quad \Delta = (2 - \phi)/4,$$

and $C^2 = 1 + [(3/4)A^2 + 2\lambda AF + F^2]/\Delta > 441$. For this last bound, $AF > 13.089$, $\sqrt{3/4} > .866$, $\lambda > .809$, $\Delta < .0955$. Its full displacement, including the head coordinate, is

$$\frac{D}{\sqrt{2}}(P_{17} + ve_{18} + \epsilon P_2), \quad D = ag + bh, \quad |v| < 1, \quad \epsilon = b/(2D) < 5/22.$$

The last inequality follows from $a/b > \lambda$ and $h + \lambda g > 2.2$. Using (DIFS2) conservatively as $S_2 < 1.1$, the normalized pressure is at most

$$\frac{3.761 + .0635s + 1.1\epsilon}{\sqrt{17 + s + 4\epsilon + 2\epsilon^2}}, \qquad s = v^2.$$

On $0 \le s \le 1$, $0 \le \epsilon \le 5/22$, it decreases in s and increases in ϵ. For the derivative signs use numerator < 4.08, denominator square < 19.1, and $.127(19.1) - 3.761 < 0$, $18.7 - 4.08(2.46) > 0$. Its maximum is $< 4.011/\sqrt{18} < .95$. Thus $E_q/E > .05C > 1.05$, again impossible. All projected points have positive actual coefficients and check the selected products, positive time and every omitted wall.

For $\lambda = \sqrt{3}/2$, the actual sum $d + n_1/2 + \lambda c$ is a nonzero feasible future null. For $\lambda = 1$, use $c + d$. For every remaining finite mutual $.9 < \lambda < 1$, use

$$\Xi = d + n_1/2 + \lambda c, \quad \Xi^2 = 3/4 - \lambda^2 = -k^2 < 0.$$

Its c, n_1 products are zero and its d product is negative; all other products are nonpositive. With $D = h + \lambda g$, $t = 1/D$, $v = \sigma(\lambda g - h)/D$, comparison gives

$$\langle x, P_{17} + ve_{18} + (t/2)P_2\rangle \ge \sqrt{17 + v^2 + 2t + (2\lambda^2 - 1)t^2} - \sqrt{2\lambda^2 - 3/2}\, t.$$

The upper bound is $3.761 + .0635v^2 + .52t$. By (DIFproduct), $t \le t_0 = [2\sqrt{Q}]^{-1} < 5/12$, where $Q = \lambda(\lambda + s_0)$. After subtracting $.0635v^2 + .52t$, the lower bound increases with v^2 and decreases with t: its first derivative is $> 1/9 - .0635$, and its radical t-derivative is at most $17/48$, whereas the subtracted coefficient exceeds .86. Set $v^2 = 0, t = t_0$. The radical square

$$B_0 = 17 + 1/\sqrt{Q} + (2\lambda^2 - 1)/(4Q)$$

decreases on $[.9, 1]$, while

$$O = \frac{\sqrt{2\lambda^2 - 3/2} + .52}{2\sqrt{Q}}$$

increases. For the first derivative the numerator is $2s_0\lambda^2 + 2\lambda + s_0 - 2(2\lambda + s_0)\sqrt{Q}$; its positive part is at most 4.181, and the subtracted part exceeds 6.0624. For the second derivative sign the numerator is $2s_0\lambda^2 + 3\lambda + 3s_0/2 - .52\sqrt{2\lambda^2 - 3/2}(2\lambda + s_0)$; its positive part is at least 4.96512, and its subtracted part is below 1.004. At $\lambda = 1$, $t_0 < .3807$, and direct positive squares give

$$\sqrt{B_0} - O > 4.2315 - 1.2272(.3807) = 3.76430496 > 3.761.$$

This excludes the entire finite tail, including all labels at least seven, without extending the inequality to dotted mutuals.

Finally suppose $\lambda = 1/2$. Positive projection on old A_{16}, c, d has determinant $19/68$ and gives an actual whole A_{18} vertex with

$$C = \frac{g(34h+1)}{\sqrt{19}} > 19/2.$$

The strict bound follows from $gh \geq 1/2 + \sqrt{19}/6$. The displacement has positive fork-dual coefficients; hence

$$E_q/E > 9.5 - .9124(9.448) = .8796448 > 7/8, \quad \kappa < 8/7, \quad \langle y, u\rangle > \frac{.9944 - .9124}{1 - .9944(.9124)} > .884.$$

In the chain $n_{16}, \ldots, n_1, d, c$, the reverse is exactly

$$u = (A\Omega'_{18} + F\Omega'_{17})/\sqrt{C^2 - 1}.$$

Both coefficients are positive. The standalone local A_{18} statement of Appendix I.12 applies at the weaker requirements $\kappa < 1.27, \langle y, u\rangle > .85$. Thus every nondotted mutual of an isolated/far-three pair is excluded.

I.18 Far-three pairs and the remaining isolated/far-three pairs

This argument is uniform over every head. For far-three/far-three caps put

$$c_f = F(g)p - g\Omega_f - \Omega_1/2, \quad c_o = F(h)p - h\Omega_o - \Omega_1/2, \quad F(t)^2 = 9t^2 + t - \tfrac{1}{2},$$

where each own is native 5, finite at least 7, or dotted, and each is below 1.6. Their mutual magnitude is

$$\lambda = F(g)F(h) - 8gh - (g+h)/2 - 1/2.$$

Clear both off the actual old A_{16}:

$$b_f = F(g)p - (g + 1/34)\Omega_f - \Omega_o/34, \quad b_o = F(h)p - (h + 1/34)\Omega_o - \Omega_f/34.$$

These are positive old-root corrections, with $b_f^2 = b_o^2 = r = 9/17$ and $\langle b_f, b_o\rangle = -(\lambda + 8/17)$. If $\lambda = 0$, their Schur inverse is $\left(\begin{smallmatrix} 9 & 8 \\ 8 & 9 \end{smallmatrix}\right)$. Positive projection gives an actual whole-D_{18} vertex with

$$C^2 = 1 + 9(F(g)^2 + F(h)^2) + 16F(g)F(h) > 205,$$

contrary to the all-point old-A_{16} bound $C < 10.94$. For $\lambda = 1/2$ use the actual future affine-A_2 null $c_f + c_o + n_1$; for $\lambda = 1$, use $c_f + c_o$. Thus any remainder has $\lambda \geq 1/\sqrt{2}$. The elementary positive Lorentz square, applied to $3g + 1/6, 3h + 1/6$ with residual $19/36$, gives $\lambda \leq gh - 1$. If either own were finite, $gh < 1.6$ would give $\lambda < .6$. Both owns are therefore dotted.

Set $\mu = \lambda + 8/17 > r$. The positive actual sums

$$\Xi_f = rb_o + \mu b_f, \qquad \Xi_o = rb_f + \mu b_o$$

have square $-r(\mu^2 - r^2)$, positive time, zero products with old A_{16} and the respective selected cap, negative product with the other cap, and all other products nonpositive termwise. Since $F(t) > 3t$, their normalized feet satisfy

$$\max(C_f^2, C_o^2) > \frac{17(\lambda+1)^2}{\lambda - 1/17} \geq 72. \tag{DFFfoot}$$

For the last inequality, put $a = \lambda - 1/17$ and use $a + 36/17 + (18/17)^2/a \geq 72/17$. The preceding inequality retains unequal owns through $F(g) + F(h) > 3(g+h) \geq 6\sqrt{gh} \geq 6\sqrt{\lambda + 1}$.

For the dotted remainder of an isolated/far-three pair, write $c = Ap - g\Omega_f$, $d = F(h)p - h\Omega_o - \Omega_1/2$. Now $\lambda > 1$, $g < G = 15/8$, and $\lambda \le gh - s_0$, $s_0 = \sqrt{19}/6$. In particular $h > (1+s_0)/G > .92$, including possible finite owns. With

$$b = F(h)p - (h + 1/34)\Omega_o - \Omega_f/34, \quad b^2 = r, \quad \langle c, b\rangle = -\lambda,$$

the positive actual foot from $rc + \lambda b$ has

$$C_Q^2 = 1 + \frac{F(h)^2}{r} + \frac{17g^2(h + 1/34)^2}{\lambda^2 - r}.$$

This is feasible against all walls; its selected b product vanishes and its c product is negative. We prove a lower bound on the whole domain $h > .92$, rather than assuming $h > 1$. Replace λ by $gh - s_0$ to decrease the bound. The ratio $g^2/[(gh - s_0)^2 - r]$ decreases with g, because $s_0^2 - r = -1/612$; replace g by G. Put $c_0 = 2s_0/G$, $b_0 = c_0 + 1/17$. Then $.774 < c_0 < .776 < h$, $b_0 > .832$. Polynomial division and the strict negative residual give

$$\begin{aligned} C_Q^2 &> 17\{h^2 + h/9 + 1 + b_0/(h - c_0)\} + 1/18 \\ &> 17\{(25/9)c_0 - 7/9 + 2\sqrt{25b_0/9}\} \\ &> 17(2.15 - 7/9 + 3.04) = 67507/900 > 75. \end{aligned} \tag{DIFfoot}$$

The middle step uses $h^2 \ge (8/3)h - 16/9$ and AM–GM in $h - c_0 > 0$. The endpoint seals are $25(.774)/9 = 2.15$ and $25(.832)/9 = 104/45 > 1.52^2$.

We now justify passage from either foot to an actual endpoint. Choose the cap d owning the larger foot in (DFFfoot), or the far cap in (DIFfoot). Its positive original-edge neighbour is whole A_{18}. Release the other old fork there, retaining the actual white A_{17} path $d, n_1, \dots, n_{16}$. The perpendicular foot of p on this line is the normalization of $p + (F/r)b$. Its product with the released old fork is negative with magnitude less than $1/(34\sqrt{r})$. The selected far/far foot has magnitude

$$\frac{rk + (r + \mu)/34}{\sqrt{r(\mu^2 - r^2)}} > \frac{1}{34\sqrt{r}},$$

where $k > 0$ is the other cap own. The isolated/far foot has magnitude

$$\frac{rg + \lambda/34}{\sqrt{r(\lambda^2 - r)}} > \frac{1}{34\sqrt{r}}.$$

The released root projects to a spacelike normal in the retained Lorentz plane, and its signed product is proportional to $-\sinh t$ on the forward ray. Thus the selected foot is after the perpendicular foot. The next actual endpoint R is at or beyond it, and distance from p increases there. Consequently

$$\sqrt{72} < C_R < 10.94, \qquad \kappa < 1.262, \qquad \langle y, u\rangle > .8564. \tag{DFexit}$$

The latter comparisons use the same original envelope and maximum. No intrinsic Coxeter structure of an arbitrary face is assumed.

The complete finite extensions of the retained white A_{17} are $A_{17} + A_1$, A_{18}, B_{18} at either endpoint, and D_{18} at either next-to-end node: seven oriented germs. Exceptional finite types have rank at most eight, and two contacts to this path make a cycle. Write the new wall $e = Tp - v$,

with full original fork contacts $a = -\langle e, n_f\rangle$, $b = -\langle e, n_o\rangle$. The complete germs are

germ	v	$\langle e, d\rangle$
$A_{17} + A_1$	$a\Omega_f + b\Omega_o$	0
A_{18} at d	$a\Omega_f + b\Omega_o$	$-1/2$
B_{18} at d	$a\Omega_f + b\Omega_o$	$-1/\sqrt{2}$
A_{18} at n_{16}	$a\Omega_f + b\Omega_o + \Omega_{16}/2$	0
B_{18} at n_{16}	$a\Omega_f + b\Omega_o + \Omega_{16}/\sqrt{2}$	0
D_{18} at n_1	$a\Omega_f + b\Omega_o + \Omega_1/2$	0
D_{18} at n_{15}	$a\Omega_f + b\Omega_o + \Omega_{15}/2$	$0.$

Every a, b belongs to the full actual alphabet $0, \cos(\pi/m), 1, (1,\infty)$. The new e is distinct from every old p-wall. The released old fork has its only zero at the starting endpoint; the other old fork joins both ends of the retained path, making a nonpositive cycle even when the far own is finite. Nor is e the other original cap: its corrected two-cap determinant is negative. Thus $T > 0$, and the actual original opposite at either fork is distinct from e.

Inverse positivity at R gives

$$B_{18} : T^2 < 7, \qquad D_{18} : T^2 < 14.$$

Acuteness against each distinct original fork opposite gives both original shadow inequalities $(\langle v, \Omega_f\rangle)/(3T) < 1$ and its reflection. For $v = a\Omega_f + b\Omega_o + \beta\Omega_j$, their exact form is

$$(\langle v, \Omega_f\rangle)^2 - 9T^2 = 9 - 17b^2 - 2\beta jb + \beta^2 j(j-18) < 0. \tag{DFshadow}$$

No opposite-only finite-white restriction has been imposed on e.

For B_{18} at n_{16}, $T^2 \geq 15 > 7$. For B_{18} at d, both contacts exceed $3/\sqrt{17} > 1/\sqrt{2}$, so the native jump gives $a, b \geq \phi/2 > .8$, hence $T^2 > 20.76$. For D_{18} at n_1, (DFshadow) gives $a, b > 1/2$, therefore both at least $1/\sqrt{2}$, and $T^2 \geq 16.5 + \sqrt{2} > 14$. For D_{18} at n_{15},

$$T^2 = 9(a^2 + b^2) + 16ab + 15(a + b) + 13/2.$$

A positive contact gives $T^2 \geq 16.25$. With both zero the positive actual sum $e + n_{14} + 2n_{15} + 2n_{16} + n_f + n_o$ is a feasible future affine-D_5 null.

For A_{18} at n_{16}, the shadow is $(1 + b)(1 - 17b) < 0$, and its reflection; both contacts are at least $1/2$. The value $1/2$ gives the actual affine-A_2 null. Both values $1/\sqrt{2}$ give the affine 4–4 null $\sqrt{2}e + n_f + n_o$. Thus $a, b \geq 1/\sqrt{2}$, at least one strictly larger, and

$$T^2 = 9(a^2 + b^2) + 16ab + 16(a + b) + 7.$$

The two nonzero original time coordinates at this actual whole A_{18} give the ONE full inverse identity

$$C_R^2 = 1 + \frac{36}{19}(T^2 + F^2) + \frac{4}{19}TF. \tag{DFA18cross}$$

The cross is retained; no separate cap energy is added.

In both families $h > .92$, so $F^2 > 8$. If one contact is at least native 6 and the other native 4, then $T^2 > 53.212437$; if both are at least native 5, then $T^2 > 55.140354$. These use only $\cos(\pi/6) > .866$, $1/\sqrt{2} > .707$. In either case $TF > 20.63$ and (DFA18cross) gives $C_R^2 > 121.3246 > 119.6836$, impossible. The only remaining assignments are $\{a, b\} = \{\phi/2, 1/\sqrt{2}\}$, in both orders. If $h \geq 1$, then $F^2 \geq 9.5$, $T^2 > 50.796378$, $TF > 21.96$, and the same identity gives $C_R^2 > 119.8689 > 119.6836$. This closes every far/far case.

For the remaining finite far own $.92 < h < 1$ in isolated/far, actual incidence $\langle e, d\rangle = 0$ is exactly

$$f(h) = TF(h) - (9a + 8b + 8)h - (a + b + 1)/2 = 0.$$

The shadow gives $3T > 9a + 8b + 8$, and $F' > 3$, so f increases strictly. Put $K = 9a + 8b + 8, L = (a + b + 1)/2$. At $h = 1$,

$$4\{(K + L)^2 - (19/2)T^2\} = 19a^2 - 53b^2 + 38ab + 38a - 30b + 23.$$

For the two remaining assignments, rational enclosures $.809 < \phi/2 < .81$, $.707 < 1/\sqrt{2} < .708$ make this respectively > 40.171733 and > 22.027294. Both squared quantities are positive, so $f(1) < 0$. Hence $f(h) < 0$ throughout $h < 1$, contradicting incidence.

The last germs $A_{17} + A_1$ and A_{18} at d have

$$u = (T\Omega'_e + F\Omega'_d)/\sqrt{C_R^2 - 1},$$

with both coefficients positive and no old-A_{16} component. Appendix I.12 excludes both at the exact factor and height in (DFexit). All seven actual endpoint germs have been covered. Thus far-three/far-three is impossible over every head; together with Appendix I.17, so is the whole isolated/far-three family. Every null and projected point used only positive actual normals and checked all original walls.

I.19 The complete native-five head

Assume the actual isolated head is $H = A_H p - (\phi/2)\Omega_1$, where $A_H^2 = (\phi - 1)/2$. Use $1.618 < \phi < 1.61804$ and $.5558 < A_H < .5559$ when the sharper upper bound is needed. All cap comparisons below are with distinct actual walls. The algebraic use of native cosines is elementary: it imposes no arithmeticity, field of definition or integrality on the polytope.

For an isolated fork,

$$M(g) = A_H\sqrt{9g^2 - 1} - (\phi/2)g$$

increases, because its derivative exceeds $3A_H - \phi/2 > 0$. For every remaining finite own, $.9 < g < 1$, $.66574582 < M(g) < .763924 < \phi/2$; hence the only possible native value is $1/\sqrt{2}$. But $\cos(\pi/9) < .94$, $\cos(\pi/10) > .951$, and

$$M(.94) < .705712 < 1/\sqrt{2}, \qquad M(.951) > .71513525 > 1/\sqrt{2}.$$

No finite own lies between labels nine and ten. The entire finite-own range is excluded. For dotted owns the same function gives $M(1) > .76275$, $M(1.02) < .782772 < \phi/2$, so $g > 1.02$.

For far-three, put $M(g) = A_H\sqrt{9g^2 + g - 1/2} - (\phi/2)(g + 1)$. Its derivative is positive. Four times the difference of its two positive squared terms is

$$(17\phi - 19)g^2 - 4g - 2\phi.$$

At native 5 it equals $-(\phi + 2)/4 < 0$, contradicting acuteness. For finite owns at least seven it is positive already at .9 and increases, whereas $M(g) < M(1) < .096148 < 1/2$. These owns also fail. Dotted far owns satisfy $g > 1.4$, since $M(1.4) < .452536 < 1/2$. For far-next-three the analogous squared difference is

$$(17\phi - 19)g^2 + (2\phi - 6)g - \phi - 1.$$

It is positive at $g = \phi/2$ and increasing thereafter, but $M(g) < M(1) < .226252 < 1/2$. Only finite owns were left in this role, so far-next-three is entirely excluded under this head.

Near-three own 4 and 5 require an exact native-cosine check. Write $M = T - b$, where $T = A_H\sqrt{9g^2+8g+5/4}$, $b = \phi(g+1/2)/2$. For $g = 1/\sqrt{2}$, $.89035 < M < .90928$, so it would be a finite cosine. In the totally real biquadratic field $\mathbb{Q}(\phi, \sqrt{2})$,

$$T^2 = \frac{(\phi-1)(23+16\sqrt{2})}{8}, \qquad b = \frac{\phi(\sqrt{2}+1)}{4}.$$

The embedding $\phi \mapsto 1-\phi$ fixing $\sqrt{2}$ makes T^2 negative. If M were a finite cosine, $T = M + b$ would lie in a compositum of totally real cyclotomic fields, impossible. For $g = \phi/2$,

$$T^2 = (11+14\phi)/8, \quad b = (2\phi+1)/4, \quad T^2-b^2 = (17+20\phi)/16 > 0, \quad (b+1)^2-T^2 = (7-4\phi)/16 > 0.$$

Thus $0 < M < 1$. The norm of T^2 from $\mathbb{Q}(\phi)$ is $79/64$, not a rational square. The extension has the embedding $T \mapsto -T$, which sends M to $-T-b < -1$. Every conjugate of a finite cosine lies in $[-1,1]$, a contradiction. The remaining near owns are therefore $> .9$. Their head products are > 1.0721886 and increase; this dotted mutual range is retained, not misclassified as a finite cosine.

It remains to close near/far pairs for this head. Every far own is between 1.4 and 1.6. Its increasing head product satisfies

$$M(1) > .0948756 > 0, \quad M(1.6) < .6282926 < 1/\sqrt{2},$$

so it must equal $1/2$. Since $M(1.45) < .4961522$ and $M(1.46) > .5041674$,

$$1.45 < h < 1.46.$$

For a near own $g \in (.9, 1.21)$ define

$$N(g) = \sqrt{9g^2+8g+5/4}, \quad F(h) = \sqrt{9h^2+h-1/2}, \quad \lambda = N(g)F(h) - 8gh - g/2 - 9h/2 - 1/4.$$

On the entire rectangle, $N', F' > 3$, $N'' = -19/(4N^3)$, $F'' = -19/(4F^3)$, and $\partial_g\partial_h\lambda = N'F'-8 > 1$. The minimum derivatives are bounded by

$$\partial_g\lambda > 3.032(4.457) - 12.1 = 1.413624, \quad \partial_h\lambda > 3.967(3.038) - 11.7 = .351746.$$

Thus corner bounds apply to every parameter:

$$.015919 = 3.967(4.457) - 17.665 < \lambda < 4.91(4.489) - 21.5578 = .48319.$$

This is a forbidden native interval. The native-five head now has only the three individual fork roles isolated, near-three, far-three. Their six unordered pair types are covered: isolated/isolated, isolated/near and isolated/far by Appendices I.14–I.18; near/near by Appendix I.9; far/far by Appendix I.18; and near/far by the present rectangle. This proves the full native-five-head exclusion, without the unused older native-five isolated/far continuation.

I.20 Finite heads seven and eight: one paid reference

For either head put $\eta = \cos(\pi/h)$, $h = 7, 8$, $u = \eta^{-2} \in (1, 4/3)$. The actual positive core

$$\Xi = H + \eta u n_1 + 2\eta(u-1)n_2, \qquad \Xi^2 = -\eta^2 d(u), \quad d(u) = (u-1)(4-3u) > 0$$

has zero H, n_1 products, negative n_2 product, positive time and all other products nonpositive. Its spatial pressure, after division by $\sqrt{2}\eta$, is

$$\langle x, Z\rangle \geq h_0 = \frac{\sqrt{2-u} - \sqrt{d(u)}}{\sqrt{2}}, \quad \sum Z_i = 1, \quad \operatorname{supp} Z \subset \{1,2,3\}, \quad Z^2 = R = 3 - 4u + \tfrac{3}{2}u^2.$$

For head seven the cubic brackets give $1.2318 < u < 1.2322$. The difference of radicals increases on that interval: $(6u-7)^2(2-u) > .116 > d(u)$. At the left end, $\sqrt{.7682} > .87646$, $\sqrt{.07060628} < .26572$. For head eight use $u = 4 - 2\sqrt{2}$, $d = 34\sqrt{2} - 48$, $R = 23 - 16\sqrt{2}$ and $1.4142135 < \sqrt{2} < 1.4142136$. These give the rigorous full-domain bounds

h	h_0 lower	R upper
7	.4318	.3489
8	.43955	.37259.

Use the bare references before any head or isolated-five payment:

initial case	B_0	b upper	P_J capacity	J
5 strong	.9565188	.3652	$> .151$	5
5 isolated weak, 6 strong	.9373243	.3399	$> .1435$	6
5 isolated weak, 7 strong	.9196357	.31938	$> .1366$	7.

These are the raw references already constructed, including the legal strong-sixteen append and every omission before pressure. Their support is at most sixteen, $\langle w, P_{16}\rangle = f_{16}$, and their unspent terminal coefficient exceeds .127.

In the first row pay .4 at $Z - P_5/5$. The debit .08 fits, and the one-square lower bound is

$$B_0 + .8(h_0 - b) - .16(R - 1/5).$$

For isolated weak five retain its actual previously unused pressure $\langle x, U'\rangle > 1.8185$. Set

$$Q_H = Z - P_J/J, \quad Q_5 = U' - (5/J)P_J, \quad \langle Q_H, Q_5\rangle = 1 - 5/J.$$

Their squares are $R - 1/J$ and at most $77/150$ or $194/175$. Pay $.4Q_H + \nu Q_5$, with $\nu = .09, .1$. The COMBINED anchor costs are $.85/6 < .1435$, $.9/7 < .1366$; both are checked before either pressure is inserted. The full lower bound, including the positive cross, is

$$B_0 + .8(h_0 - b) + 2\nu(1.8185 - 5b) - .16(R - 1/J) - \nu^2 Q_5^2 - .8\nu(1 - 5/J).$$

The resulting exact rational lower bounds are

h	5 strong	$J = 6$	$J = 7$
7	.9859748	.98694896	.98698198
8	.9883844	.98935856	.98939158.

The fourth initial case, weak-helper five, already has $L > .987759$, and receives no extra head payment. Thus every initial case supplies ONE reference W with

$$L_W > .9859, \quad \langle W, P_{16}\rangle = f_{16}, \quad \operatorname{supp} W \subset [1, 16], \quad \beta_{16} > .127.$$

Put $\Delta = S_{16} - f_{16} \geq 0$. All new payments were at indices at most seven; the terminal price remains unspent. Hence

$$\|x - W\|^2 < .0141 - .254\Delta, \qquad \Delta < .0556.$$

The squared projection of x onto the complement of old A_{15} is bounded by

$$\frac{S_{16}^2}{16} + x_{17}^2 + x_{18}^2 < \frac{(3.59 + \Delta)^2}{16} + .0141 - .254\Delta \leq .83062756 < .9114^2.$$

The quadratic increases throughout $0 \leq \Delta \leq .0556$, which justifies its endpoint substitution. For ANY actual point on old A_{15}, original maximality therefore gives

$$C < \frac{1 + .9114^2}{1 - .9114^2} < 10.81. \tag{DFiniteface}$$

This is an all-point distance cap on the original face, with both signed last coordinates retained.

I.21 Finite heads seven and eight: the complete fork dispatch

Write $\eta_h = \cos(\pi/h)$, $A_h = \sqrt{2\eta_h^2 - 1}$. The exact brackets, from the native-seven cubic and the eight half-angle formula, are

h	η_h	A_h
7	$(.9009, .901)$	$(.7894, .7897)$
8	$(.9238, .9239)$	$(.8408, .8410)$.

(DFheadbrackets)

For far-next-three the actual time is $B(g) = \sqrt{9g^2 + 2g}$. The terminal bound gives $9g + 1 < 9.64$, hence $g < .96$. The complete finite list is therefore

$$m = 5, 7, 8, 9, 10, 11.$$

Indeed own zero has zero time; own $3, 4, 6$ have the actual affine principals already displayed, parallel has $c + n_f$, and dotted violates the length bound. Also $\cos(\pi/12) = (\sqrt{6} + \sqrt{2})/4 > .96575$.

The head/far-next mutual is

$$M_h(g) = A_h B(g) - \eta_h(g + 1).$$

Since $B' > 3$ and $3A_h - \eta_h > 0$, it increases. The entire six-label list is covered by the following consecutive native gaps:

h	own labels	M_h lower	M_h upper
7	$5, 7, 8$	.5325965	.70481042
7	$9, 10, 11$	.7260396	.77649556
8	5	.63199495	.6340268
8	7	.78086418	.78208658
8	$8, 9, 10$	.81780615	.8629634
8	11	.87450304	.87783466.

These intervals are respectively inside $(1/2, 1/\sqrt{2})$, $(1/\sqrt{2}, \phi/2)$, $(1/2, 1/\sqrt{2})$, $(1/\sqrt{2}, \phi/2)$, $(\phi/2, \sqrt{3}/2)$, and $(\sqrt{3}/2, \cos(\pi/7))$. Here are explicit inputs for every row, so the table is a finite rational certificate rather than sampled labels:

g	$B(g)$ bound
g_5	$(2.740, 2.741)$
g_7	$(3.0176, 3.0180)$
g_8	$(3.0867, 3.0872)$
g_9	> 3.134
g_{10}	< 3.1692
g_{11}	> 3.1938
$g < .96$	< 3.196.

Use $1.618 < \phi < 1.619$, $.9396 < g_9 < .94$, $.9510 < g_{10} < .9511$, $.9593 < g_{11} < .96$. The ninth-own cubic $8t^3 - 6t - 1$ has signs $-.001406950912, .004672$ at $.9396, .94$. The tenth-own identity is $g_{10}^2 = (5 + \sqrt{5})/8$. For eleven, $\pi/11 < 2/7$ and the decreasing alternating lower polynomial give

$$g_{11} > 1 - 2/49 + 2/7203 - 4/5294205 > .9593.$$

The displayed B bounds follow by substituting the appropriate rational endpoints into $9g^2 + 2g$ and squaring. For example the first head-seven lower bound is $.7894(2.740) - .901(1.8095) = .5325965$;

its first upper bound is $.7897(3.0872) - .9009(1.9238) = .70481042$. For the second block use the lower ninth-own and upper .96 endpoint. The four head-eight rows use respectively the fifth, seventh, eighth/tenth and eleventh endpoints. Thus all actual far-next-three caps are excluded for both heads.

For far-three use $F(g) = \sqrt{9g^2 + g - 1/2}$ and the increasing mutual

$$M_h(g) = A_h F(g) - \eta_h(g+1).$$

Native 5 gives

$$M_7(g_5) \in (.33508862, .33701975), \qquad M_8(g_5) \in (.42162679, .4233563),$$

both forbidden. The bounds $F(g_5) \in (2.4898, 2.4905)$ follow from $F(g_5)^2 = (11\phi + 7)/4$. For head seven,

$$0 < M_7(g_7) < .47984433 < 1/2, \quad M_7(g_8) > .51383002 > 1/2, \quad M_7(1) < .6328451 < 1/\sqrt{2}.$$

Monotonicity rejects label seven and every finite label at least eight. For head eight, $.57755874 < M_8(g) < .745203 < \phi/2$ throughout finite owns at least seven, leaving only $1/\sqrt{2}$. But

$$M_8(g_{14}) < .7027251 < 1/\sqrt{2}, \qquad M_8(g_{15}) > .707213408 > .7072 > 1/\sqrt{2}.$$

The half-angle identity gives $g_{14} = \sqrt{(1+g_7)/2} \in (.9749, .97494)$. The exact formula $g_{15} = (\sqrt{5} - 1 + \sqrt{30 + 6\sqrt{5}})/8$ gives $g_{15} \in (.97814, .97816)$. The time bounds are $F(g_{14}) < 3.00492$, $F(g_{15}) > 3.01479$. Use $2.23606 < \sqrt{5} < 2.23607$ and $6.58910 < \sqrt{30 + 6\sqrt{5}} < 6.58912$; their positive squares give all enclosures. No finite label lies between fourteen and fifteen. Thus every far-three own for either head is dotted,

$$1 < g < 1.6.$$

Only near/far remains: all other unordered pair types are already excluded over every head. Put

$$\lambda = N(g)F(h) - 8gh - g/2 - 9h/2 - 1/4, \quad 1/\sqrt{2} \le g < 1.21, \quad 1 < h < 1.6.$$

It increases in both variables on this WHOLE rectangle. Indeed $N', F' > 3$, both are decreasing, and the mixed derivative is > 1. At the worst extremes,

$$\partial_g \lambda > 3.032(3.082) - 8.5 = .844624, \quad \partial_h \lambda > 3.377(3.032) - 10.1576 = .081464.$$

The top corner gives $\lambda < 4.91(4.914) - 23.543 = .58474 < 1/\sqrt{2}$. Thus the actual mutual is zero or $1/2$.

For $1/2$, $g \le 1$ would imply $\lambda < 4.273(4.914) - 20.75 = .247522$, impossible. Hence $g > 1$, $N > 4.272, F > 3.082$. Positive projection on old A_{16} and the two actual caps has Schur matrix $\left(\begin{smallmatrix} 1 & -1/2 \\ -1/2 & 9/17 \end{smallmatrix}\right)$ of determinant $19/68$. It is an actual whole-A_{18} vertex, with

$$C^2 = 1 + (36N^2 + 68NF + 68F^2)/19 > 10.8^2.$$

The projection's two original fork coefficients are positive. Using $S_{17} \pm x_{18} < 3.823$, its normalized pressure is less than $3.823/\sqrt{17} < .928$. Thus

$$\rho = E_q/E > C - .928\sqrt{C^2 - 1} > .072C + .464/C > .82,$$
$$\kappa < 50/41 < 1.22, \quad \langle y, u \rangle > 1 - 1.22/10.8 > .887.$$

The reverse is the positive combination $(N\Omega'_{18} + F\Omega'_{17})/\sqrt{C^2-1}$. Appendix I.12 excludes it.

Suppose finally that $\lambda = 0$. Near owns 4 and 5 would give $\lambda < 3.693(4.914) - 18.223 = -.075598$, contrary to acuteness. For finite near own $g = \cos(\pi/m)$, $m \geq 7$, retain old A_{15}, n_f, the near cap and the far cap. Clearing their positive old-root coefficients gives diagonal Schur matrix $\operatorname{diag}(1-g^2, 17/32)$. The resulting actual vertex is whole $A_{16} + I_2(m)$, with

$$C^2 = 1 + \frac{N^2}{1-g^2} + \frac{32F^2}{17}.$$

For $m = 7$, monotonicity and $\lambda(.901, 1.35) < 3.9705(4.154) - 16.5063 = -.012843$ force $h > 1.35$. Thus $N^2/(1-g^2) > 83.65$, $32F^2/17 > 32.47$, and $C^2 > 117.12 > 10.82^2$. For every $m \geq 8$, the corresponding bounds are $N^2/(1-g^2) > 111.3$, $32F^2/17 > 17.88$, so $C^2 > 130.18 > 11.4^2$. Both contradict (DFiniteface).

For dotted near own, the actual head/far native gaps force $h > 1.03$:

	$M_h(1)$ lower	$M_h(1.03)$ upper
head 7	.6309308	.67823359
head 8	.7435456	.7946087.

They are forbidden intervals between successive native cosines. At the corner $(1.08, 1.03)$, $\lambda > 4.515(3.1746) - 14.3242 = .009119$; therefore orthogonality forces $1 < g < 1.08$. The positive actual core $c_N + gn_f$, square $1 - g^2 < 0$, is a genuine point on all old A_{15}, with

$$C^2 = \frac{N^2}{g^2-1} > \frac{20.3876}{.1664} > 122.5 > 10.81^2.$$

The ratio decreases for $g > 1$, since its derivative numerator is $-8g^2 - (41/2)g - 8 < 0$. This contradicts (DFiniteface). Every own of near/far is now excluded, and hence both finite heads seven and eight are impossible.

I.22 How these foundations enter the final argument

The head-isolation argument leaves an actual isolated head with finite own 5, finite own at least 7, parallel, or dotted. The parallel case has its actual positive future null. Appendix I.10 excludes finite heads at least nine; Appendix I.19 excludes five; Appendices I.20 and I.21 exclude seven and eight. These proofs cover every initial-five reference case with the individual original prices and every omission retained.

For a dotted head, the present appendix supplies precisely the earlier pair exclusions: isolated/isolated, isolated/near-three, isolated/far-three, far-three/far-three, near-three/near-three. The near/far pair and the remaining far-next-three role in that head regime are handled by the separate final uniform dotted-head argument in the main revision. That argument uses the original actual head point and its four paid references, both individual weak-helper anchor prices, the fresh local whole-A_{18} endpoint lemma and its actual length comparison. It is not reproduced here. In particular no conclusion of that final argument was used in the present all-head pair proofs or in the standalone two-dual lemmas. The selected dependency graph is acyclic.

All component energies concern the single x at the original farthest p, or the single explicitly named y at an actual neighbour with factor derived from that same original maximum. Every projected point, future null and edge foot has its selected products and all omitted original walls checked. Finite inventories used here are the ordinary finite Coxeter classification with their displayed rank and contact restrictions; the rational interval tables are explicit finite inequalities, not execution receipts. No field restriction, arbitrary-face arithmeticity, exact-seventeen frontier census or new dimension research enters this appendix.

J The other exact-dimension-eighteen consumers

The necessary forest cover is proved before any row is excluded. This appendix supplies its component-specific consumers, using the same original farthest vertex throughout. A typed exception includes its actual isolated opposite wall; a bare old seven bond is insufficient.

J.1 A_{16} with either actual typed-seven defect

Suppose the entire incident forest is $A_{16} + Z$, where $Z = N_7$ means old A_2 with its new isolated seven opposite, and $Z = O_7$ means old $I_2(7)$ with its new isolated three opposite. There is no arithmeticity hypothesis.

Use zero-sum $\mathbb{R}^{17}$, with actual unit normals and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{17}/17),$$
$$D_i = 2i(17-i)/17, \quad C_{ij} = 2i(17-j)/17 \quad (i \le j).$$

Write the original interior point as $z = E(p-x-y)$, with x in this A_{16} space and y in the orthogonal rank-two space. The previously proved typed-defect floor gives

$$e_A + e_Z = \|x\|^2 + \|y\|^2 < 1, \qquad e_Z > .09, \qquad e_A < .91. \tag{T1}$$

All auxiliary points below are compared with this original maximum.

Put $\phi = (1+\sqrt{5})/2$, $\theta = 2\cos(2\pi/7)$, $g_7 = \cos(\pi/7)$, and

$$h_N = (\theta - 1)/3, \qquad h_O = (\theta-1)/(2-\theta), \qquad k = 1/h_O = \theta^2 + 2\theta - 1.$$

The actual isolated defect wall has time $\sqrt{h_Z}$ and spatial support only in Z. In the N_7 case it is $c_Z = \sqrt{h_N}p - g_7\omega_a^Z$, for an old A_2 dual. The cubic $f(t) = t^3 + t^2 - 2t - 1$, increasing on $[1.24, 1.25]$, has signs

$$f(1.2469) = -410632291/10^{12} < 0, \qquad f(1.247) = 105223/10^9 > 0.$$

Consequently

$$1.2469 < \theta < 1.247, \quad 0 < h_N < 1/12, \qquad 3.0485 < k < 3.0491 < 61/20. \tag{T2}$$

The identity $(\theta-1)k = 2-\theta$ follows by the same cubic. These are real native-angle facts, not field restrictions on the polytope.

J.1.1 The central window at its actual energy budget

Call $5 \le i \le 12$ weak if $h_i = \langle x, \Omega_i\rangle \le \sqrt{D_i} - .61$. The target-independent inventory and helper nonadjacency in Appendix C.6 leave only pure far helpers or near-four roles at $5, 6$ and their reflections. Its older energy-dependent near-four conclusion is not used at the larger budget .91.

A left near-four has $v = \Omega_i/\sqrt{2} + \Omega_{i-1}/2$, $i = 5, 6$. The same actual wall is the opposite in direction $j = i - 1$. Keeping the other original roots and all of Z, positive finite projection gives its actual endpoint, with

$$\Delta_j = 1/(17-j), \qquad C_j = \sqrt{2j/17}\{(17-j)/2 + (16-j)/\sqrt{2}\}.$$

For $j = 4, 5$, their squares are respectively greater than $177608/1700$ and $187690/1700$, both greater than 100. Thus the normalized mate height exceeds $\sqrt{9/11} > .9$. At the opposite end, either a

second mate has height $> .9$, or nonadjacency supplies a strong $11/12$ direction of height $> .77$. Their correlation is at most $\sqrt{5/22} < 1/2$. Cauchy would give

$$1.67^2 < e_A(2 + 2\rho) < 3(.91) = 2.73,$$

whereas $1.67^2 = 2.7889$. Both orientations are impossible. Every remaining weak site is therefore a nonadjacent pure helper.

Use the shared rational reference of Appendix C.7:

$$H_5 = 2.046,\ H_6 = 2.176,\ H_7 = 2.259,\ H_8 = H_9 = 2.300, \qquad H_{17-i} = H_i.$$

The ends are linear. A weak end is flattened through six; every other weak knot is removed by its neighboring chord before pressure. The resulting positive-dual reference w has all coefficients at actual strong floors and obeys

number of weak ends among $5, 12$	$E = \|w\|^2$ lower
0	.86103645
1	$> .8285$
2	$E_{\min} = 1195276/1500000.$

(T3)

The last rows include every permitted internal omission. A weak end block has length $L = 6$, mean $b = H_6/(6\sqrt{2}) < .257$; a strong end block has $L = 5$, mean $b = H_5/(5\sqrt{2}) < .29$. In both cases $b > .25$, the next coordinate is $< .1$, and the individual strong-anchor capacities are at least $839/6000$ and $349/2500$, respectively.

If an original signed endpoint coordinate were at least .565, replace its block by $.565, r, \dots, r$, keeping the block sum. Here $r = (Lb - .565)/(L - 1)$ is greater than .17125 or .187, so every surviving coefficient stays nonnegative. The exact variance gain is $L(.565 - b)^2/(L - 1)$. For a weak end, its total with e_Z exceeds

$$E_{\min} + .09 + (6/5)(.565 - .257)^2 > 1.00068746.$$

For a strong end it exceeds $.8285 + .09 + (5/4)(.565 - .29)^2 = 1.01303125$. Hence both signed endpoint coordinates are $< .565$. The actual edge condition in either direction is

$$x_{\text{end}} \geq \sqrt{16/17}\sqrt{(C - 1)/(C + 1)}.$$

At $C = 41/20$ its square is $336/1037 > .565^2$, with positive cleared numerator 198547 over 41480000. Therefore BOTH actual endpoint cosines satisfy

$$C < 41/20. \tag{T4}$$

This is one reference contradiction per endpoint, not addition of two independently optimized lower energies.

J.1.2 The complete endpoint and foreign-contact list

Release old 1, retaining the full old A_{15} and Z. Its actual opposite is isolated within A_{16}, or has a single three at $j = 2, 3, 15, 16$, or a single four at $j = 2, 16$. Two contacts make a cycle; exceptional finite types have insufficient rank. An old A_2 admits at most one new contact, with complete costs

$$Q = 1/3, \quad 2/3, \quad \phi^2/3.$$

An old $I_2(7)$ admits none, since even its smallest contact costs $1/[4\sin^2(\pi/7)] > 1$. When there is also an A_{15} contact, only an endpoint-three with old A_2 cost $Q = 1/3$ remains finite: the full chain is A_{18}. A foreign arm at a D position has excessive arms; a four or five becomes an internal multiple bond in the long path.

Write $c = Ap - g\Omega_1 - \beta\Omega_j - v_Z$, where $\beta = 1/2$ or $1/\sqrt{2}$. For $Q = 0$, the exact data are

white three at j	2	3	15	16
C	$(32g+15)/17$	$(16g+7)/\sqrt{17}$	$(16g+1)/\sqrt{17}$	$(32g+1)/17$

and white fours at $2, 16$ give $(16g + 15/\sqrt{2})/\sqrt{17/2}$ and $(16g + 1/\sqrt{2})/\sqrt{17/2}$. The residuals are $17/32, 1/8, 1/16$, for endpoint three, D-position three and endpoint four. The one mixed endpoint-three multiplies its pure cosine by $\sqrt{51/19} > 8/5$.

At $j = 3$, own zero has the actual affine-E_8 null

$$2n_1 + 4n_2 + 6n_3 + 5n_4 + 4n_5 + 3n_6 + 2n_7 + n_8 + 3c;$$

every positive own has $C \geq 15/\sqrt{17} > 41/20$. At $j = 15$, own zero has time square $-2/17$, own three has the null $c + n_1 + \cdots + n_{15}$, and owns at least four violate (T4). White four at 2 violates (T4) even at own zero; at 16, own zero has time square $-1/17$, and every positive own violates (T4).

For pure white three at 2, own zero has time square $-2/17$, own three has the actual triangle null, and every own at least four violates (T4). The mixed version retains that triangle; its own-zero null is the actual affine E_7 combination

$$2n_1 + 4n_2 + 3n_3 + 2n_4 + n_5 + 3c + 2z_1 + z_2, \qquad \langle c, z_1\rangle = \langle z_1, z_2\rangle = -1/2.$$

Mixed white three at 16 has own-zero time square $-10/51$, the own-three all-three cycle, and for own at least four has $C > (8/5)(32(.7) + 1)/17 = 936/425 > 41/20$. A parallel own always has the null $c + n_1$. Every displayed null has positive cap time and nonpositive product against every omitted wall. A positive finite projection likewise checks all unselected walls termwise.

The only remaining nonisolated endpoint is consequently PURE far three:

$$c = Ap - g\Omega_1 - \Omega_{16}/2, \quad A^2 = (32g^2 + 2g - 9)/17, \quad 0 < g < 677/640 < 53/50, \quad A^2 < 7/4. \text{ (T5)}$$

Own zero has negative time, own three has zero time, and reflection gives the other end. Finite and dotted own values remain present.

J.1.3 Using the actual defect wall

For N_7, a pure endpoint wall is distinct from c_Z and has actual product $-A\sqrt{h_N}$ with it. The native gap forces $A^2 h_N \geq 1/4$, so $A^2 > 3$, excluding (T5). An isolated-within-A endpoint instead has

$$A^2 = Dg^2 + Q - 1, \quad C^2 = Dg^2/(1 - Q), \quad D = 32/17, \quad u = (1 - Q)/g^2 = D/C^2.$$

Own zero has negative time, since the full white cost satisfies $Q < 1$. At $Q = 0$, the same gap gives $u < D/4 < 1$. At $Q > 0$, a contact β to the old A_2 gives $Q = 4\beta^2/3$ and actual defect cross $q = 4g_7\beta/3$ or $2g_7\beta/3$. Acuteness, including orthogonality, gives

$$A^2 \geq q^2/h_N \geq Q(2 + \theta)/[4(\theta - 1)] > 13Q/4.$$

Since $Q \geq 1/3$, $C^2 > [1 + (9/4)Q]/(1 - Q) \geq 21/8 > D$, so again $0 < u < 1$. This covers every foreign orientation and finite or dotted own.

For O_7 all endpoints are pure. Set $M = 4A^2/k$. It is four times the square of an ACTUAL negative two-wall product. Its positive values therefore begin $1, 2, \phi^2, 3, 2 + \theta$, followed by larger finite values, parallel 4 or dotted values greater than four. For the far helper, own four gives $M < 1$; finite owns at least five give $1 < M < 2$. In the dotted case, (T5) gives $1 < M < 7/3 < \phi^2$, so the only possibility is $M = 2$. It forces

$$A^2 = k/2, \quad 32g^2 + 2g - 9 = 17k/2, \quad 1.013 < g < 1.014, \quad 1.2346 < A < 1.235. \tag{T6}$$

These brackets follow by positive substitution using (T2).

Two far helpers would have this same g. They are distinct because their own $g > 1$ differs from the other-end helper coefficient $1/2$. Their spatial product $(2g^2 + 32g + 1/2)/17 > 69/34$ exceeds their time product $A^2 = k/2 < 61/40$. This violates acuteness. Thus a far helper would face an isolated endpoint. If that isolated endpoint had finite own $\delta \leq 1$, then $0 < M_\delta = 4(D\delta^2 - 1)/k < 20/17 < 2$, so it must equal one. For labels at most twelve, $4(D\delta^2 - 1) \leq (32\sqrt{3} - 4)/17 < 3.0485 < k$. For labels at least thirteen, $\delta > .97$ gives $4(D\delta^2 - 1) > 3.0844 > k$. No finite label works; own zero has negative time and parallel is the actual null pair. The other endpoint is therefore dotted, with $\delta > 1$ and $\sqrt{D}\delta < 41/20$.

Put $B = (2g + 16)/17 \in (1.06, 1.0605)$. Their actual mutual magnitude

$$m(\delta) = A\sqrt{D\delta^2 - 1} - B\delta$$

increases on $\delta \geq 1$, because its derivative exceeds $1.2346(1.37) - 1.0605 > 0$. It is positive at one, and its endpoint upper is $1.235(1.79) - 1.06(1.49) = .63125 < .7$. The native gap thus forces $m = 1/2$. At 1.35 it is $< .4956$, while at 1.358 it is $> .5006322$; hence $1.35 < \delta < 1.358$. But its own ACTUAL defect-wall product now has

$$3 < M_\delta < (1.313)(2.472) = 3.245736 < 2 + \theta,$$

the forbidden native-six/native-seven gap. This excludes the last far helper. Both O_7 endpoints are isolated and dotted.

J.1.4 One simultaneous endpoint payment

For either isolated N_7 endpoint clear its full foreign vector with its positive old-A_2 inverse. For O_7, put $Q = 0$. Each end has its OWN parameter $u \in (0, 1)$, and the actual core

$$\eta = c + v_Z + gun_1 = g\{\sqrt{D - u}\, p - Z(u)\}, \qquad Z(u) = \Omega_1 - un_1$$

has square $-g^2u(1 - u)$, positive time, zero selected cap and foreign products, and negative n_1 product. All other walls, including the opposite endpoint and the defect wall, are nonpositive termwise. Its actual normalization gives

$$\langle x, Z(u)\rangle \geq h(u) = \sqrt{D - u} - \sqrt{u(1 - u)}.$$

At a length-L block put $Y = Z - \Omega_L/L$, with the reflected signed expression at the right end. Then

$$\langle w, Y\rangle = 0, \quad \langle Y_l, Y_r\rangle = 0, \quad \langle w, Z\rangle = H_L/L \leq .4092, \quad Y^2 = 2 - 2/L - 2u + u^2 \leq 5/3 - 2u + u^2.$$

The residuals have zero sum and disjoint first/last block supports. Pay $\mu = .32$ separately at BOTH surviving strong anchors: $\mu/L \leq .064$ is below both capacities in (T3). All old coefficients remain

nonnegative before either pressure is used. For $W = w + .32(Y_l + Y_r)$, one original-vector square yields

$$e_A > E + G(u_l) + G(u_r), \quad G(u) = .64(h(u) - .4092) - .1024(5/3 - 2u + u^2). \tag{T7}$$

The spectator energy has not been added to either gain.

For completeness the following finite scalar certificate covers every independent $0 \le u \le 1$. Write

$$G_0 = .64(h(u) - .4112) - .1024(8/5 - 2u + u^2), \qquad G = G_0 - 208/37500.$$

Positive square tests for $\sqrt{D - u}$ at $0, 1/2, 3/4, 1$ give lower bounds $1.3719, 1.1757, 1.0641, .9393$. Use the root chords and $u^2 \le (l + r)u - lr$ on each interval:

$[l, r]$	root chord	K	a	$\sqrt{a^2 + .64^2}$ upper
$[0, 1/2]$	$1.3719 - .3924u$	.451008	.097536	.6475
$[1/2, 3/4]$	$1.3989 - .4464u$	.506688	.208896	.6733
$[3/4, 1]$	$1.4385 - .4992u$	.570432	.293888	.7043

Here $G_0 > K - au - .64\sqrt{u(1 - u)}$. The four root squares are

$$1.88210961, \quad 1.38227049,$$
$$1.13230881, \quad .88228449,$$

each below $32/17, 47/34, 77/68, 15/17$, respectively. The three upper-radical square gaps are

$$.000142978704, \quad .000095351184,$$
$$.000068333456,$$

and all are positive. Finally,

$$au + .64\sqrt{u(1 - u)} \le \frac{a + \sqrt{a^2 + .64^2}}{2},$$

so the three lower bounds for G_0 are $.07849, .06559, .071338$. Thus, uniformly,

$$G > .06559 - 208/37500 = 18013/300000 > .06.$$

Combining (T1), (T3) and (T7) now gives

$$e_A + e_Z > E_{\min} + .12 + .09 > 1.00685066 > 1,$$

a contradiction. The two end parameters were independent, all actual walls were retained, and the rank-two energy was counted once. This excludes BOTH actual typed-seven alternatives beside A_{16}.

Together with the already proved uniform $A_{16} > .86$ and the ordinary/joint spectator bounds, this closes its entire exact-eighteen row: ordinary T_2, two ordinary A_1's and $I + A_1$ are excluded by those earlier bounds, and the two typed cases by the argument here. It does not assert an arbitrary-spectator $A_{16} > .91$ theorem.

J.2 The two rank-one spectators beside A_{17}

We now exclude the two genuinely different rank-one alternatives. An ordinary A_1 has energy greater than .1; an actual isolated short I need not. The latter supplies a wall and additional actual edge pressures instead. Throughout this subsection the entire incident forest is $A_{17} + A_1$, at the original farthest vertex p. Write $z = E(p - x - ye)$, where e is its unit rank-one normal, x belongs to zero-sum $\mathbb{R}^{18}$, and $\|x\|^2 + y^2 < 1$. Set

$$\Omega_i = \sqrt{2}(P_i - iP_{18}/18), \quad D_i = i(18-i)/9, \quad C_{ij} = i(18-j)/9 \quad (i \le j).$$

There is no arithmetic hypothesis.

J.2.1 Common geometry and the unscaled references

Call $5 \le i \le 13$ weak when $\langle x, \Omega_i \rangle \le \sqrt{D_i} - .61$. The target-independent inventory of Appendix C.6 leaves near-four roles at $5, 6$ or their reflections, and pure helpers

$$c = Ap - g\Omega_i - \Omega_a/2, \quad a = 1 \text{ or } 17, \quad \phi/2 \le g < 17/16,$$

with own five, finite at least seven, or dotted. All foreign and own-zero roles have their dispositions in that inventory. Pure helpers are nonadjacent before near-four caps are removed. The cutoff $17/16$ concerns these weak helpers only, never arbitrary endpoint caps.

We remove near-four caps without any contrary component-energy target. For a left cap its same-wall mate is the actual edge endpoint at $j = 4$ or 5. Its full positive finite projection retains $A_{j-1} + B_{18-j} + A_1$, with

$$\delta = 1/(18-j), \quad \sigma = 1/2 + (17-j)/((18-j)\sqrt{2}), \quad C_{\text{mate}} = \sigma\sqrt{(18-j)D_j} > 10.$$

Indeed $\sigma > 1.14$ and the last root is at least $28/3$. Farthest comparison gives height square greater than $9D_j/11$, so the heights at four and five exceed 2.25 and 2.43. At the opposite end there is a mate or a strong twelve/thirteen: otherwise two adjacent weak pure helpers would remain. Their heights exceed 2.218 and 2.077. Disjoint end-segment Cauchy has worst case

$$\|x\|^2 > \tfrac{1}{2}(2.43^2/5 + 2.218^2/6) = 1.000450333\ldots > 1.$$

The other placements and two mates give larger bounds. Reflection excludes the right near-four. Thus every remaining weak site is a nonadjacent pure helper, at the correct original comparison factor one.

Use the unscaled $N = 18$ reference of Appendix C.7, with

$$H_5 = H_{13} = 2.077, \ H_6 = H_{12} = C = 2.218, \ H_7 = H_{11} = 2.314, \ H_8 = H_{10} = 2.371, \ H_9 = 2.389.$$

Flatten a weak end through six and delete every other weak knot by its neighboring chord. Its positive strong-site coefficients give $\langle x, w \rangle \ge E = \|w\|^2$. The complete omission table is

weak ends $5, 13$	E lower bound	end blocks
0	$E_0 = .8934308$	5/5
1	$E_1 > .86256$	6/5
2	$E_2 = 9978827/12000000$	6/6.

Each six-block anchor has capacity at least $821/6000$, and each five-block anchor at least $343/2500$, after every internal omission. Their signed coordinate means satisfy $b_6 < .2614$, $b_5 < .294$.

We use only the explicit comparison-one core and block algebra printed in Appendices C.7 and H.10, not the latter subsection's neighbouring-vertex conclusion or its 1.061 comparison hypothesis. To specify this reuse, a weak-end foot has $U = v - kn_i - \ell n_a - mn_j$, with corrections

$$(k, \ell, m) = (g, 2/3, 1/3), \quad (g, 2(1-g^2), 0), \quad (1/g, 0, 0)$$

for own five, finite at least seven and dotted, respectively. The first uses its actual finite H_4 projected point; the other two are future negative cores. Their positive original-root coefficients check every omitted wall, including e and its opposite. Their pressures $\langle x, U\rangle \geq H$ have far time $A_0^2 = (65/9)g^2 + (5/9)g - 19/36$; near adds $8g/9$. On two six-blocks the projections are

$$\Pi U = (5g/6)\Omega_6 + \Omega_{12}/12 \quad \text{(far)}, \qquad \Pi U = (5g/6 + 1/12)\Omega_6 \quad \text{(near)}.$$

The residual $Y = U - \Pi U$ has zero sum separately on both end blocks and $\langle w, Y\rangle = 0$. Its far squares are

$$Q_5 = 2g^2/3 + 1/12, \quad Q_f = 4g^4 - 16g^2/3 + 29/12, \quad Q_d = 5g^2/3 - 19/12 + g^{-2};$$

near adds $g/3$. The full scalar seals proved there give the six/six deficit $\Delta = H - C(5g/6 + 1/12) > .3405$ for finite owns and $> .37$ for every $1 < g < 17/16$. For a far six/five foot its deficit decreases by $343/15000$ and its square by $1/60$; near is unchanged.

For clarity, the finite-at-least-nine derivative in that seal is a derivative of the lower surrogate

$$\overline{\Delta} = A_0 - 1/4 - C(5g/6 + 1/12) - 343/15000,$$

not of the varying-radical deficit. It satisfies $\overline{\Delta}' > .839$, $Q_f' \leq 16/3$. At $g = .93$, $\overline{\Delta} > .32035$ and the adjusted square is .77940804. Thus a far finite foot at weight .145 gains more than .076; five/seven/eight give the respective deficit/square pairs $(.33915, .505), (.32655, .74), (.3177248, .763)$, which give the same strict bound directly. A far dotted foot at weight .13 gains more than

$$.26(5207/15000) - .0169(1751/1500) = 1057901/15000000 > .0705267.$$

The full two-foot crosses are also coordinate identities. In one six-block write

$$a = (g\sqrt{2}/6)(P_5 - 5e_6) - kn_5, \qquad b = (e_1 - P_6/6)/\sqrt{2} - \ell n_1 - mn_2.$$

Then $a^2 = 5g^2/3 + k^2 - 2gk$, $b^2 = 5/12 + \ell^2 + m^2 - \ell m - \ell \leq 5/12$, and $\langle a_i, b_j\rangle = g_i/6$. Equal orientations have total square $Q_{f,L} + Q_{f,R} + (g_L + g_R)/3$; mixed orientations have $Q_{f,L} + Q_{f,R} + 2\langle b_L, b_R\rangle + 2g_{\text{near}}/3$. Hence the equal/mixed bounds are $17/6, 11/3$ for two finite owns and $3.077, 3.91$ if a dotted own occurs. Unequal parameters and both mixed orientations are included.

Every use below pays $\mu\Pi U$ from its individual anchors before using the actual pressure. For $W = w + \mu \sum Y$, one completed square therefore gives

$$\|x\|^2 \geq E + 2\mu \sum \Delta - \mu^2 \|\textstyle\sum Y\|^2.$$

No independently optimized component estimates are added.

J.2.2 An ordinary rank-one spectator

Suppose first that $\|x\|^2 \le .9$. With one weak end, weight .1 fits both anchor capacities and gives gain $> .2(.317) - .01(8/5) = .0474$; hence $\|x\|^2 > .90996$. With two weak ends, both individual prices are at most $101/960 < 821/6000$. The full mixed square is less than $32/5$, giving gain $> .4(.34) - .01(32/5) = .072$, and total $> .90356$. Only zero weak ends remain.

Both five-block means are $b < .294$. If one signed endpoint coordinate were at least .4, replace that block by $.4, r, r, r, r$, with $r = (5b - .4)/4 > .2625$. The next coordinate is less than .1, so all surviving coefficients remain nonnegative. The block-sum-preserving gain is $(5/4)(.4 - b)^2 > .014045$, contradicting $E_0 + .014045 > .9$. Thus both signed endpoint coordinates are less than .4, and both actual endpoint cosines are less than $3/2$: otherwise their squared coordinate demands are at least $17/90 > .16$. This temporary graft gain is not reused.

For any endpoint wall $c = Ap - v_A - \beta e$, the actual reciprocal shadow and distinct orthogonal-opposite lemma give $A > \beta$, including exclusion of the boundary shadow one. Thus $\|v_A\|^2 > 1$. Finite classification after deleting old one leaves isolation, a three at $j = 2, 3, 16, 17$, or a four at $j = 2, 17$. For $v_A = g\Omega_1 + \delta\Omega_j$, their zero-foreign cosines, in that order, are

$$(34g + 16)/18, \quad (34g + 15)/(2\sqrt{18}),$$
$$(34g + 2)/(2\sqrt{18}), \quad (34g + 1)/18,$$
$$(34g + 16\sqrt{2})/6, \quad (34g + \sqrt{2})/6.$$

Foreign cost only increases the actual cosine. Own-zero cases violate $\|v_A\|^2 > 1$ or have the actual affine E_8 null at three-at-three. Own three at two has its triangle null. The remaining native substitutions and $C < 3/2$ leave only pure far-three at seventeen, with own four. Indeed its $1/\sqrt{2} \le g < 13/17$ allows only four; a foreign $\beta^2 \ge 1/4$ multiplies its cosine square by at least $36/19$, giving $> 64/19 > 9/4$. Two such opposite pure caps are distinct and have product $8\sqrt{2}/9 - 1/3 > 0$. At least one endpoint is therefore isolated within A_{17}, with foreign contact still allowed.

Put $D = 17/9$, $Q = \beta^2$, $u = (1 - Q)/g^2$. The inequality $Dg^2 > 1$ removes own zero, three and four; six and parallel have actual feasible nulls. Also $C < 3/2$ gives $u > 68/81 > .83$. The following complete point constructions give a pure A displacement $Z = \zeta - P_{18}/18$, $\sum \zeta = 1$, $\langle x, Z\rangle \ge t_0$, $R = \|\zeta\|^2$.

For pure five use the finite H_4 point from $c + (3g/2)n_1 + gn_2 + (g/2)n_3$, with $\delta = 1 - 3g^2/2 > 0$. Here $\zeta = (1/4, 1/4, 1/4, 1/4)$, $R = 1/4$, and $t_0 = AL/(\sqrt{2}g(L + \sqrt{\delta})) > .33$, where $L^2 = 7g^2/18$, $A > .486, L > .504, \sqrt{\delta} < .136$. For pure finite at least seven use $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$. Its selected products are zero, zero, and $(3 - 4g^2)/(2g) < 0$, and

$$\zeta = (1 - u/2, 1 - u/2, u - 1), \quad R = 3 - 4u + 3u^2/2, \quad t_0 = \frac{\sqrt{D - u} - \sqrt{7u - 4 - 3u^2}}{\sqrt{2}}.$$

Seven has $1.23 < u < 1.233$, $R < .35, t_0 > .38$. Finite at least eight has $1 < u < 1.1716$, $R \le .5, t_0 > .39$; the two radicals are respectively $> .8469$ and $< .28868$.

For any Q with $0 < u < 1$, clear the entire foreign contact in $c + \beta e + ((1 - Q)/g)n_1$. Its square is $-g^2u(1 - u)$, its cap/foreign products are zero and its own product is negative. Its positive time and original coefficients check every other wall. It gives

$$\zeta = (1 - u/2, u/2), \quad R = 1 - u + u^2/2, \quad t_0 = \frac{\sqrt{D - u} - \sqrt{u(1 - u)}}{\sqrt{2}} > .4, \quad R < .515.$$

The last bounds use $u > .83$, radicals $> .9428$, $< .376$. The boundary $u = 1$ is an actual future null. For $Q > 0, u > 1$, the native alphabet forces own five, foreign three and $Q = 1/4$. Its positive finite correction $c + gn_1 + e/2$ has square $\delta = 3/4 - g^2 > 0$. Its actual projected point gives $R = 1/2$ and $t_0 > .42$, using $A > .697, L > .762, \sqrt{\delta} < .31$, $L^2 = 8g^2/9$. These are points, not native projected normals.

At the isolated end project Z onto $\Omega_5/(5\sqrt{2})$. Weight $1/2$ costs less than $.071 < 343/2500$, and the residual has square $R - 1/5$, orthogonal to the reference. Its gain is greater than $t_0 - .294 - (R - 1/5)/4$. For the five rows just listed these strict gains exceed $.0235, .0485, .021, .02725, .051$. Each is greater than $.9 - E_0 = .0065692$. This closes the zero-weak-end branch. We have proved $\|x\|^2 > .9$ for the entire rank-one forest. An ordinary spectator has $y^2 > .1$, giving the contradiction.

J.2.3 The actual isolated short spectator

Now let its actual wall be $c_I = \sqrt{h}\, p - \sqrt{1+h}\, e$, $0 < h < 1/2$. Appendix A.8 proves, on this same original vector,

$$y \geq s, \quad x_1, -x_{18} \geq a_I = \sqrt{17/18}\, t, \quad t + \tfrac{3}{2}s > .767.$$

Put $k = \sqrt{17/18}$, $D_0 = .7453706$. Since $k > .9718$, $a_I + (3k/2)s > D_0$. Weighted Cauchy therefore gives, for $c > 0, B < D_0$,

$$s^2 + c(a_I - B)_+^2 > \frac{(D_0 - B)^2}{17/8 + 1/c}. \tag{60}$$

It counts the spectator energy once. An endpoint graft on a block of length L, from its current coordinate B toward a_I, gains $L(a_I - B)_+^2/(L-1)$ and costs $(a_I - B)_+/((L-1)\sqrt{2})$ at that individual anchor. It may be capped at a smaller coordinate when necessary. Its full cross with previously paid helper residuals is included in B.

For two weak ends pay .1 to each foot. Equal orientations spend at most $31/320$ at either anchor and have both endpoint coordinates at most .3454. Mixed orientations spend at most $101/960$ and $17/192$; the unloaded coordinate is at most .2865. If $a_I \leq .6$, the respective additional graft prices are at most $.2546/(5\sqrt{2})$ and $.3135/(5\sqrt{2})$. These fit their separate remaining capacities $821/6000 - 31/320$ and $821/6000 - 17/192$. Equation (60) gives gains $> .0629$ for $c = 12/5, B = .3454$, and $> .0711$ for $c = 6/5, B = .2865$. If $a_I > .6$, cap those grafts at .6; their gains alone exceed $.155, .117$. For two finite owns the base totals exceed $E_2 + .1362 - .01(17/6) > .9394355$ and $E_2 + .1362 - .01(11/3) > .9311022$. A dotted own raises the deficit sum above .7105 and gives larger totals. The appropriate graft therefore makes each full total greater than one.

For one weak end orient it to five. A near finite helper at weight .145 spends less than $.145(11/12) < 821/6000$, has square $< 17/12$ and gain $> .06895$. A near dotted helper at .14 spends less than $.14(31/32)$, has square < 1.539 and gain $> .0734$. In each case the other five-block is unloaded. Its full graft is legal for $a_I < .972$, since $(.972 - .294)/(4\sqrt{2}) < .12 < 343/2500$. Equation (60), $c = 5/4, B = .294$, gives graft plus spectator $> .0696$. Thus even the smaller total is $> .86256 + .06895 + .0696 > 1.0011$.

A far finite helper at weight .145 has gain $> .076$. Its own price leaves more than .016, and its opposite price leaves at least .1227, and its endpoint coordinates are at most $.2956, .3761$. The weak graft is legal up to increment .113; the other graft is legal throughout $a_I < .972$. If $a_I \leq .4086$, the weak graft and (60) give $> .0683$, with $c = 6/5, B = .2956$. Above that split cap the weak gain at $(6/5)(.113)^2 = .0153228$, and use the strong graft with spectator $> .0466$, $c = 5/4, B = .3761$. Both give an extra $> .0619$ and total > 1.00046.

A far dotted helper at weight .13 has gain $> .0705267$, leaves at least $1043/48000$ and .1242 at the two anchors, and has endpoint coordinates at most $.294, .3676$. The weak graft is legal up to

increment .1535. Below $a_I = .4475$, its graft plus spectator is $> .0688$, with $c = 6/5, B = .294$. Above this split its capped gain is $(6/5)(.1535)^2 = .0282747$; the opposite graft with spectator is $> .0487$, with $c = 5/4, B = .3676$. Both totals exceed one. Reflection covers the other weak end.

Finally suppose neither end is weak. One full five-block graft with the spectator gives $> .0696$. If the other actual endpoint cosine were at least $8/5$, its signed coordinate would exceed $7/15$, since $17/78 > (7/15)^2$. The disjoint grafts give

$$\|x\|^2 + y^2 > E_0 + .0696 + (5/4)(7/15 - .294)^2 > 1.000298.$$

Thus both endpoint cosines are less than $8/5$. The complete foreign-safe list already proved above leaves only isolation or pure far-three with own four/five: now $g < 139/170$, and a foreign contact would still give $C^2 > 64/19 > (8/5)^2$. Two such far caps are distinct; their spatial product exceeds $17(g_L + g_R)/18 > 1.32$, while each time square $(68g^2 + 4g - 19)/36 < .81$. Their actual product is positive. At least one endpoint is isolated within A_{17}.

For that isolated endpoint pure five is impossible: its time square $(17\phi - 19)/36 < .237$ makes its strictly negative product with c_I smaller in magnitude than $1/2$. The foreign five/three row with $u > 1$ is also impossible, since acuteness would require $h(Dg^2 - 1) \geq 1/4$, whereas $h < 1/2, Dg^2 - 1 < .237$. All other rows use the same actual cores as above, without the ordinary branch's temporary endpoint bound. Pure seven gives gain $> .0485$. For finite at least eight, split $1 < u < 1.1716$ at 1.1: the bounds $(R, t_0) \leq (.5,$ lower $.4407)$ and $(.415,$ lower $.3945)$ give gains $> .0717, .04675$.

For every $0 < u < 1$ the cleared-core gain is $> .06$. Here are complete continuous seals, so no sampled parameter is used. Lower chords for $\sqrt{D - u}$ are $1.3743 - .3916u, 1.4012 - .4454u, 1.4402 - .4974u$ on $[0, 1/2], [1/2, 3/4], [3/4, 1]$; positive endpoint squares prove them. Upper chords for R are $1 - .75u, .8125 - .375u, .625 - .125u$. With $.7071 < 1/\sqrt{2} < .7072$, the gain is bounded below by

$$\begin{array}{c|l} [0, 1/2] & .47776753 - .0895u - .7072\sqrt{u(1-u)} \\ [1/2, 3/4] & .54366352 - .2212u - .7072\sqrt{u(1-u)} \\ [3/4, 1] & .61811542 - .3205u - .7072\sqrt{u(1-u)}. \end{array}$$

The inequality $au + b\sqrt{u(1-u)} \leq (a + \sqrt{a^2 + b^2})/2$, with radical upper bounds $.713, .7411, .7765$, gives respectively $> .0765, .0625, .0696$. Thus every surviving isolated endpoint supplies gain $> .0467$, paid at its own anchor by weight $1/2$. At the opposite end the independent full graft plus the one spectator energy gives $> .0696$. Their residuals have disjoint support. One square yields

$$\|x\|^2 + y^2 > E_0 + .0467 + .0696 = 1.0097308 > 1.$$

This excludes the actual isolated short spectator as well. All original foreign contacts, own parameters and wall signs have been retained; the same original farthest vector and separate anchor prices are used throughout. Both rank-one alternatives beside A_{17} are now closed.

J.3 Typed defects beside B_{16} and D_{16}

These cases share the long-prefix and head geometry. They separate at the final short layer: B_{16} has a genuine short normal and a reciprocal endpoint, while D_{16} has two signed forks and a centre calculation. We prove their common input once.

Use the actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$ along the long chain. For B_{16}, $n_{16} = e_{16}$, $\Omega_i = \sqrt{2}P_i$ for $i \leq 15$, and $\Omega_{16} = P_{16}$. For D_{16},

$$n_{16} = (e_{15} + e_{16})/\sqrt{2}, \quad \Omega_i = \sqrt{2}P_i \ (i \leq 14), \quad \Omega_{15,16} = (P_{15} \mp e_{16})/\sqrt{2}.$$

The last D coordinate is signed. Write the one original vector as $z = E(p-x-y)$, with the classical and rank-two projections x, y. The actual typed defect gives $\|y\|^2 > .09$ and $\|x\|^2 + \|y\|^2 < 1$.

Put $c_7 = \cos(\pi/7)$, $s_7 = c_7^2$. The actual isolated defect wall $d = A_T p - v_T$ has

	v_T	$t_T = A_T^2$
N_7	$c_7 \omega_j^{A_2}$	$4s_7/3 - 1$
O_7	$\omega_j^{I_2(7)}/2$	$1/[4(1-s_7)] - 1.$

The native-seven cubic gives $.9009 < c_7 < .901$, hence $0 < t_N < 1/12$, $.327 < t_O < .342$. Every classical opposite retains the whole old rank-two component. An old $I_2(7)$ accepts no contact: already a single three has determinant $1 - s_7 - 1/4 < 0$. An old A_2 accepts at most one contact, with label three, four or five and whole cost $Q = 1/3, 2/3, \phi^2/3$. All these cases remain; no field restriction removes the last one.

J.3.1 Fresh tails, then strong five

Set $f_i = \sqrt{i} - .61/\sqrt{2}$. Appendix E reaches twelve in B_{16} and eleven in D_{16}, leaving only nonadjacent pure far-three helpers. We reuse the actual local matrices and points in Appendix F, before its ordinary-spectator consumer. Their proofs retain all spectators. The positive foreign costs here are at least $1/3$, so none weakens a test that required only $Q \geq 1/4$.

In B, the fresh $A_{12} + B_3$ layer at thirteen has exactly the long-three and short-three finite extensions printed there. The latter is the sealed F_4 white block; own zero has its actual affine $3, 4, 3, 3$ path. The full long bridge has residual $1/26$; its minimum numerator, even at own zero, is $1/26 + 1/2$. No short-end bridge gives a finite long path. Thus strong-or-pure-helper coverage reaches thirteen.

In D, the fresh D_4 layer at twelve retains its long leaf, both distinct fork leaves, bridges and all foreign costs. The only extra own-five/fork-three row has its actual finite H_4 point, flat through nine, with $G > 3.34, R < 12$. The local payment formula with $K = 8, 9$ and the typed cost gives

$$.7683275 + .09 + 2(3.34 - 2.398)/7 - 4/49 > 1.0458,$$
$$.797425 + .09 + 2(3.34 - 2.570)/7 - 3/49 > 1.0457.$$

The D_3 layer at thirteen keeps middle-three and end-three, four and five, including the middle/full-D_{16} bridge. The own-five point has $G > 3.47, R < 13$. For every remaining finite own $.9 < g < 1$, average the two actual A_2 cores whose corrections are $g(4n_{13}/3 + 2n_{12}/3)$ and $g(4n_{13}/3 + 2n_{14}/3)$. Both satisfy every original wall. With $\alpha = \sqrt{2}(g + 1/4)$, $a = 4g^2/3 - 1$, they give

$$A^2 = 13\alpha^2 - 5/8, \quad R < 13, \quad G = (A - \sqrt{a})/\alpha > 13/4.$$

Indeed its derivative has numerator $(5/8)/A - (g/3 + 1)/\sqrt{a} < 0$, and at one its value is $8\sqrt{5}/5 - 2\sqrt{6}/15 > 13/4$, using $\sqrt{5} > 559/250, \sqrt{6} < 49/20$. Choosing actual strong $K = 9$ or 10 gives

$$.797425 + .09 + 2(3.25 - 2.570)/7 - 4/49 > 1.000078,$$
$$.823735 + .09 + 2(3.25 - 2.732)/7 - 3/49 > 1.000510.$$

This sharper comparison is needed for the typed spectator; the weaker ordinary-case 3.23 estimate is not substituted. Coverage reaches thirteen in both types. Helper packing extends because its actual mixed-point proof uses the same inverse $2\min(i, j)$ and positive retained right blocks. It does not require those right blocks to have the generic bulk ranks.

Some twelve/thirteen is strong. If five were weak, the actual weak-five point gives $x_1 > .5965, S_6 > 2.01815$. The legal interpolation through one, six and that strong end has energy at least

$$.7600299945 + (\sqrt{12} - \sqrt{6})^2/6 = 3.7600299945 - 2\sqrt{2} > .93.$$

Adding only $\|y\|^2 > .09$ is impossible. Thus five is strong, before choosing a head or final centre. The ordinary references of Appendix F now apply with every internal helper omitted: $q_{12} > .8689525, q_{13} > .88864$. Their first five coordinates are $b = f_5/5 < .361$, with individual P_5 coefficient $> .146$.

J.3.2 A common head argument without arithmeticity

Deleting long node one retains B_{15} or D_{15}. Finite classification allows only isolation or a simple three at its long endpoint two. The latter has

$$c = Ap - g\Omega_1 - \Omega_2/2 - v_Z, \quad A^2 = 2g^2 + 2g + Q, \quad C_1 = (2g+1)/\sqrt{1-2Q}.$$

At own zero, zero foreign cost has zero time; positive foreign cost has future null $c + \Omega_2/2 = \sqrt{Q}p - v_Z$. The actual positive expansions of $\Omega_2/2$ are $n_1 + 2(n_2 + \cdots + n_{15}) + \sqrt{2}n_{16}$ in B, and $n_1 + 2(n_2 + \cdots + n_{14}) + n_{15} + n_{16}$ in D. Own three has its triangle. All other positive owns force $C_1 \geq 1 + \sqrt{2}$ and $x_1 > .64$. Graft the first five-block to $.64, (5b - .64)/4, \ldots$. Its other coordinates exceed .29, above the next slope; its P_5 debit is less than $.07 < .146$. Its gain $> (5/4)(.279)^2 > .0973$ makes $q_{12} + .0973 + .09 > 1$. The head must be isolated within the classical component, with foreign contacts still present.

Write $c_H = Ap - g\Omega_1 - v_Z$, $A^2 = 2g^2 + Q - 1$. For a finite pure N_7 head its actual product with d lies strictly between $-1/2$ and zero. For a finite pure O_7 head its magnitude is less than $\sqrt{.342} < 1/\sqrt{2}$, so it must equal $1/2$. This forces

$$g^2 = (1 + 1/(4t_O))/2, \quad 148/171 < g^2 < 577/654, \quad .93 < g < .9394.$$

But $\cos(\pi/8) < .925$ and $\cos(\pi/9) > .9395$, the latter by $8s^3 - 6s - 1$. No finite native own lies there.

For a finite mixed N_7 head the product is $q - A\sqrt{t_N}$, with $q = 4\beta c_7/3$ or $2\beta c_7/3$. At $\beta \geq 1/\sqrt{2}$, $q > .42$ but $A\sqrt{t_N} < \sqrt{2/12} < .409$. At $\beta = 1/2$ the same-root value $q > .6$ is also impossible. The other-root product exceeds $-1/2$ and must vanish. It gives $g^2 = (3s_7 - 2)/(8s_7 - 6) = (1 + 1/(4t_O))/2$, the same gap. Parallel has its actual future null. Every finite head is excluded.

For dotted mixed heads, $Q \geq 2/3$ forces $C_1^2 > 6$ and hence $x_1 > .64$. At $Q = 1/3$, same-root acuteness gives $A^2 > 108/25$ and $C_1^2 > 187/25$. In the other orientation

$$A^2 t_N - q^2 > (4/3)t_N - s_7/9 = (5s_7 - 4)/3 > 0.$$

The actual negative product therefore has magnitude at least $1/2$, so $A\sqrt{t_N} > .8$, $A^2 > 192/25$, $C_1^2 > 313/25$. Again $x_1 > .64$. Only a pure dotted head remains.

For it put $u = g^{-2} \in (0, 1)$. The actual positive core $c_H + g^{-1}n_1$ has square $-(1 - u)$, zero cap product, negative own product, positive time and every omitted wall nonpositive. It gives

$$\langle x, Z \rangle \geq \frac{\sqrt{2 - u} - \sqrt{u(1-u)}}{\sqrt{2}}, \quad Z = (1 - u/2)e_1 + (u/2)e_2.$$

Pay weight $\sqrt{2}/5$ on $Z - P_5/5$, costing $\sqrt{2}/25 < .057$ at the individual anchor. The residual is zero-sum on the first five coordinates and orthogonal to the reference. Its completed-square gain is

$$\eta = .4(\sqrt{2 - u} - \sqrt{u(1-u)} - \sqrt{2}b) - .04(1.6 - 2u + u^2) > .03624.$$

For the full interval, $\sqrt{2}b < .5106$. On the two halves use lower root chords $1.4142 - .379u, 1.4494 - .4494u$ and upper chords $u^2 \le u/2, 1.5u - .5$. The lower gains become $.29744 - .0916u - .4\sqrt{u(1-u)}$ and $.33152 - .15976u - .4\sqrt{u(1-u)}$. The circle inequality with radical upper bounds $.4104, .4308$ gives $> .04644, > .03624$. This one paid head can be combined with any later residual vanishing on the first five coordinates.

J.3.3 The B_{16} last-long layer

Delete fourteen, retaining the actual B_2 on long fifteen and short sixteen. A finite contact to it is a three at either end. For long fifteen the Schur data are $\sigma = g + 1/2$, $\Delta = 1/2 - Q$. Own zero has null $c + n_{14} + 2n_{15} + \sqrt{2}n_{16}$; own three has its triangle; every other positive own is strong. For short sixteen the data are $\sigma = g + 1/(2\sqrt{2})$, $\Delta = 1/2 - Q$. Own zero has the actual affine F_4 kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$ on $c, n_{16}, n_{15}, n_{14}, n_{13}$. Own four has null $\sqrt{2}c + n_{13} + 2n_{14} + n_{15}$, six has $\sqrt{3}c + 2n_{14} + n_{13}$, and parallel has $c + n_{14}$. Every remaining own at least five has ratio $\phi/\sqrt{2} + 1/2 > 47/30 > 25/16$. Positive foreign $Q \ge 1/3$ makes own three strong too.

All four left/right endpoint-three bridges are finite B_{16}, with residual $1/28$ and $Q = 0$. Their minimum numerator, including own zero and the short-right bridge, gives ratio

$$\sqrt{28}(1/28 + 1/(2\sqrt{2})) > 27/14 > 25/16.$$

Thus these own-zero bridges are directly estimated, not dismissed by an invalid all-three cycle. Left-only and isolated cases use their actual long-neighbour points through fifteen. The sole extra weak nonhelper is consequently $c = Ap - (\Omega_{14} + \Omega_{16})/2$, with no foreign part.

For this wall put $a = (1 + \sqrt{2})/2$. Its spatial part is $aP_{14} + (e_{15} + e_{16})/2$, and $A^2 = 10 + 7\sqrt{2}$. Release old fifteen instead. The retained chain old one through fourteen, then c, then the actual short sixteen is finite A_{16}, all bonds three. Positive finite projection gives an actual future endpoint with the omitted old fifteen strict: its products with the positive old-fourteen and short-sixteen terms are negative. Every other wall is nonpositive termwise, and the full rank-two block is retained. In the old edge direction $P_{15}/\sqrt{15}$ its data are

$$b_{15} = \frac{15 + 14\sqrt{2}}{2\sqrt{15}}, \quad b_{15}^2 - A^2 = 17/60, \quad C_{15} = \frac{15 + 14\sqrt{2}}{\sqrt{17}} > 42/5.$$

Therefore $S_{15} > \sqrt{555/47} > 3.43$ on the original vector. Append this floor to an actual strong twelve/thirteen reference, skipping intermediate unavailable floors. The slopes are $a_K = (3.43 - f_K)/(15 - K)$. For legality use $f_{12} > 3.03, f_{13} > 3.17$: then $a_{12} < .4/3 < 1/7$, $a_{13} < .26/2 < 7/50$. For energy use the opposite bounds $f_{12} < 3.034, f_{13} < 3.175$, gaining $> .396^2/3$ or $> .255^2/2$. The first five block and its head price are unchanged. One square with the head and the one spectator gives totals > 1.0474645 and > 1.0473925, respectively. This disposes of the extra cap and extends helper coverage through fourteen. Its actual mixed-point packing makes thirteen or fourteen strong. Their energies exceed $.88864$ and $.9084 - 4/3200 = .90715$. The same head gain and spectator then give at least $.88864 + .03624 + .09 > 1$. Thus the entire typed B_{16} row is impossible.

J.3.4 The D_{16} centre and the point where cases split

Delete centre fourteen, retaining A_{13}, both actual fork A_1's and the full rank-two component. For own g, left contact β at j, and fork contacts δ, ϵ, the exact data are

$$\Delta = 1 - Q - q_L - \delta^2 - \epsilon^2 > 0, \quad \sigma = g + \beta j/14 + (\delta + \epsilon)/2, \quad q_L = \beta^2 j(14 - j)/7, \quad C_{14} = \sqrt{28}\sigma/\sqrt{\Delta}.$$

The sufficient strong ratio is less than 25/16. The complete white list is: no left contact; left three at $1, 2, 12, 13$ or left four at $1, 13$; left endpoint-three plus one fork three/four or both forks three. Without a left contact the two-fork pairs are $(3,3), (3,4), (3,5)$, up to exchange; a single fork has any finite label. Full long bridges are foreign-sealed. A permitted extra old-A_2 arm is retained when its full white Gram is positive.

The left-only and isolated roles have their fresh $D_i = 28$ clearings. The isolated $0 < u < 1$ loss is $< (.096 + 1.005)/2 = .5505$. For $Q > 0, u > 1$, $u \le 8/(3\phi^2) < 1.019$. Hence $\sqrt{28} - \sqrt{28 - u} < .1$ and $\sqrt{u - 1} < .138$, so the full loss is $< .238$. These follow from $\phi > 1.618$, $\sqrt{28} > 5.29$, $\sqrt{28 - 1.019} > 5.19$ and $.138^2 > .019$. At $u = 1$ is the actual null. Pure five/seven/at-least-eight have respective loss bounds $25/160 + 5/24$, $10/81 + \sqrt{13/120}$, and .511. The left four, near-three and D-position tests are stronger than 25/16. A positive foreign cost makes far-left three strong by its actual full-foreign clearing. A pure far-left helper at fourteen would make thirteen strong and give $q_{13} + .03624 + .09 > 1$. No field argument enters these tests.

Left-three/fork-three bridges have residual 2/7, numerator $\ge g + 2/7$; own zero/three have their cycles and owns at least four are strong. Left-three/fork-four or two-fork bridges have residual at most 1/28 and are directly strong including own zero. Actual simple zero-own cycles, where present, are retained. No finite foreign arm fits. For one fork-three, positive owns three/four/six/parallel have the actual cycle, weighted-star, $6, 3$, or pair-null tests. With $Q \ge 1/3$, all remaining owns are strong because

$$\Delta \le 5/12, \quad \sigma \ge \phi/2 + 1/4, \quad \sigma^2/\Delta \ge 3(8\phi + 5)/20 > 625/256.$$

Single fork at least four with own at least five is stronger. Own-three/fork-four or five is retained below; foreign cost makes it strong, and larger forks are directly strong. Own-zero/fork-three has the affine E_8 arms $5, 2, 1$, fork-four has null $\sqrt{2}c + 2n_f + 2n_{14} + n_{13} + n_h$, and fork-six its $6, 3$ null. Fork-five permits $Q = 0, 1/3$; fork at least seven forces $Q = 0$. For two forks own three has its triangle, owns at least four are strong, and own-zero/$(3,3)$ is the affine square. This leaves exactly the rows below.

Every remaining row gives a pure D pressure $\langle x, V \rangle \ge G$, flat one on the first eleven coordinates. Choose actual strong $K = 10$, or eleven if ten is weak. The terminal price 1/7 and head price lie at different anchors; their residuals have zero full cross because the centre residual vanishes on the first K coordinates. The one-vector total is

$$\|x\|^2 + \|y\|^2 > q_K + .03624 + .09 + 2(G - f_K)/7 - (R - K)/49. \tag{61}$$

For positive own and single fork-three, $Q = 0$, put $\alpha = \sqrt{2}(g + 1/4)$, $A^2 = 14\alpha^2 - 3/4$. Own five has its positive finite H_4 point from $g(3n_{14}/2 + n_{13} + n_{12}/2)$, giving $G > 3.60, R < 14$. For finite at least seven or dotted, weakness forces $g < 9/8$. Use the actual core $c + g(4n_{14}/3 + 2n_{13}/3)$, of square $-a$, $a = 4g^2/3 - 1$. Its old selected products vanish, its cap product is $-a$, both fork products are negative, and every other wall is nonpositive termwise. Its displacement has

$$R = 14 + (1/4 - 4g^2/3)/\alpha^2 < 14, \quad G = (A - \sqrt{a})/\alpha > \frac{2(\sqrt{835} - \sqrt{11})}{11\sqrt{2}} > 3.28.$$

The same derivative as above, now with 3/4, is negative, so the endpoint 9/8 covers the entire weak interval.

For own zero and fork-five, $Q = 0$, project the actual H_4 on c, n_f, n_{14}, n_{13}. Its correction is $\delta(3n_f/2 + n_{14} + n_{13}/2)$, $\delta = \phi/2$. Dividing its displacement by $\delta/\sqrt{2}$ gives

$$V = P_{12} + (e_{13} + e_{14} + e_{15} \pm e_{16})/2, \quad R = 13, \quad G > \sqrt{8\phi} - \sqrt{13 - 8\phi} > 3.36.$$

For the genuine N_7 foreign-three row $Q = 1/3$, clear the entire old A_2 by its positive inverse combination v_Z. The finite point from $c + v_Z + \delta n_f$, with $\Delta = 2/3 - \delta^2 > 0$, has displacement $\delta P_{14}/\sqrt{2}$. Since $A > 2.137, \sqrt{\Delta} < .111, \delta < .81$, it gives $\langle x, P_{14}\rangle > 3.5$, so $G > 3.5, R = 14$. No foreign coordinate or second spectator energy is used.

For own zero and single fork at least seven, $Q = 0$, use $c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{14}$. Its cap/fork products vanish, its centre and other-fork products are negative and its time is positive. With $z_0 = \delta^{-2} \in (1, 100/81)$ its data are

$$V = (1^{13}, 3 - 2z_0, z_0 - 1, \ \pm(1 - z_0)), \quad G = \sqrt{16 - 2z_0} - \sqrt{-8 + 14z_0 - 6z_0^2}, \quad R = 24 - 16z_0 + 6z_0^2.$$

The full interval has $14G - R > 32.51$. Indeed, for $z = z_0 - 7/6$, use $\sqrt{16 - 2z_0} > 4.011 - .27z_0$ and $\sqrt{1/6 - 6z^2} \le 1/\sqrt{6} - 3\sqrt{6}z^2$. Endpoint squares and $1.78^2 < .036(96.8748)$ give $14G - R > 32.528 - 1.78z + 96.8748z^2 > 32.51$.

Own-three/fork-four or five has, after releasing that fork, the actual positive finite D_{16} replacement and fork cosine $7 + 8\delta > 12$. Own-zero/$(3, 4)$ has the finite B_{16} replacement, releasing the three-contacted fork, with cosine $7 + 4\sqrt{2} > 12$. Positive inversion checks the omitted fork as well as every other wall, so these are actual endpoints. Both give $\langle x, P_{15} \pm e_{16}\rangle > \sqrt{176/13} > 3.67$, $R = 16$. Finally own-zero/$(3, 5)$ uses the actual finite H_3 projection on the cap and both forks. Its displacement is αP_{14}, $\alpha = (1 + \phi)/(2\sqrt{2})$, and

$$A^2 = 3 + 11\phi/2, \quad \Delta = (2 - \phi)/4, \quad G = \frac{14 + 21\phi - \phi\sqrt{7}}{4A\alpha} > 3.4, \quad R = 14.$$

The last strict comparison follows from $1.618 < \phi < 1.619, \sqrt{7} < 2.646, A < 3.451, \alpha < .927$: its numerator exceeds 43.696772, whereas $3.4(4 \cdot 3.451 \cdot .927) = 43.5074472$. Every finite projection uses its positive actual-root inverse, has future time and retains all omitted original wall inequalities.

The smallest constant pressure, $G > 3.28, R < 14$, makes (61) greater than 1.0249 at ten and 1.0248 at eleven. The rows $(3.36, 13), (3.5, 14), (3.67, 16), (3.4, 14)$ give larger margins. The continuous row instead uses $(14G - R - 14f_K + K)/49$, giving $> .949975 + (32.51 - 38.248 + 10)/49 > 1.0369$, or $> .9734625 + (32.51 - 40.404 + 11)/49 > 1.0368$. Thus centre fourteen is strong. The ending-fourteen reference has energy $> .9084 - 4/3200 = .90715$, retaining every permitted internal hole. The same paid head and one typed spectator give $.90715 + .03624 + .09 = 1.03339 > 1$. This excludes the typed D_{16} row as well. The entire argument uses one original farthest vector, true short/fork normals, individual prices and no arithmeticity or face inheritance.

J.4 Rank-one spectators beside the rank-seventeen classical factors

The ordinary rank-one case and the actual isolated-short case use different information. We first treat the ordinary D_{17} alternative without imposing a shape on its rank-one opposite.

J.4.1 D_{17} with an ordinary nonshort A_1

Use the actual unit D_{17} roots, centre fifteen and forks sixteen/seventeen. The long duals are $\Omega_i = \sqrt{2}P_i$, $i \le 15$; the fork duals are $(P_{16} \mp e_{17})/\sqrt{2}$. The last coordinate is signed. For the original decomposition $z = E(p - x - ye)$, ordinary rank one gives $y^2 > .1$, and hence $\|x\|^2 < .9$. Every classical cap retains this old e, with full foreign cost $Q = \beta^2 < 1$, either zero or at least $1/4$.

The valid bulk parameter is $M = 11$, so helper coverage first reaches twelve, with an actual strong eleven/twelve. If five were weak, its actual point supplies $x_1 > .5965, S_6 > 2.01815$. The positive reference through one, six and that strong endpoint has energy

$$> .7600299945 + (\sqrt{11} - \sqrt{6})^2/5 > .9104024765.$$

Its slopes decrease; the endpoint is below its actual strong floor. This contradicts $\|x\|^2 < .9$. Five is strong before any new tail or head argument.

Here are the fresh tail matrices, to specify precisely which local arguments of Appendix F are reused. At thirteen/fourteen deletion retains $A_{12} + D_4$ and $A_{13} + D_3$. In the actual D_4, long-fourteen contact three has $(\Delta, \sigma) = (1/2 - Q, g + 1/2)$; either fork-three has $(1/2 - Q, g + 1/4)$. In the actual D_3, middle-three, end-three, end-four and end-five have respectively

$$(1/2 - Q, g + 1/2), \quad (5/8 - Q, g + 1/4), \quad (1/4 - Q, g + 1/(2\sqrt{2})), \quad (1 - 3\phi^2/8, g + \phi/4).$$

The sufficient strong ratio is $< T(6) < 25/16$. These are the same finite local matrices, not a whole-component theorem with its rank changed. Left-only and isolated caps use their long-root clearings at $D_i = 2i$: every isolated foreign case is strong, the pure five/seven/at-least-eight/dotted points have their .61 loss bounds, and a positive foreign far-three is strong. Own-zero and owns three/four/six/parallel have the actual affine/null or reciprocal-shadow dispositions of that inventory, retaining every foreign cost. Both fork choices remain.

For completeness the right lists leave only these nondirect rows. At thirteen, a fork-three with $Q = 0$ and own five survives; all positive foreign cases have ratio > 2.1, and finite at least seven or dotted has ratio $> 1.15\sqrt{2}$. At fourteen, middle-three is direct. End-four at own zero has kernel $(\sqrt{2}, 2, 2, 1, 1)$ on the cap, contacted fork, centre, other fork and old fourteen; all positive owns are direct. The sealed end-five H_4 white block has ratio > 2.99 even at own zero. End-three has its own-zero E_8, own-three cycle, own-four star, own-six path and parallel null. Positive foreign cost is direct; $Q = 0, g \geq 1$ has ratio at least $\sqrt{5/2}$. Thus only pure own five and finite at least seven remain there.

Full bridges use simple-three left endpoints only. The D_4/middle-D_3 bridge has residual $1/(2i)$ and numerator at least $g + 1/4 + 1/(2i)$. The end-three bridge at fourteen has residual $1/8 + 1/28 = 9/56$ and numerator at least $g + 1/4 + 1/28$. Their own-zero principals are actual all-three cycles, and every positive own is directly strong. A foreign arm at the internal bridge cannot be finite. No exceptional finite type has the required rank. The surviving weak rows are therefore pure far-left helpers and the explicit right exceptions just listed.

The right own-five exception at $i = 13, 14$ uses its actual left H_4 projected point with correction $g(3n_i/2 + n_{i-1} + n_{i-2}/2)$, $g = \phi/2$. With $\alpha = \sqrt{2}(g + 1/4)$,

$$A^2 = i\alpha^2 - \kappa, \quad \kappa = 1/2 \text{ (13)}, \ 5/8 \text{ (14)}, \quad G > \sqrt{i - \kappa/\alpha^2} - .091, \quad R < i.$$

The normalized displacement is flat through $i - 3$. The finite point has positive time and all original wall signs, including the two forks and old rank-one component. Positive squares give $G > 3.48$ and $G > 3.61$. For thirteen choose actual strong nine/ten; the local one-square formula and the one ordinary charge give totals $> 1.0757, > 1.0762$. Coverage thus reaches thirteen.

For every remaining finite $.9 < g < 1$ at fourteen, average the two actual A_2 cores with corrections $g(4n_{14}/3 + 2n_{13}/3)$ and $g(4n_{14}/3 + 2n_{15}/3)$. Both satisfy every wall inequality. Their correction cross $8g^2/9$ is retained in the average. With $a = 4g^2/3 - 1$, its data are

$$A^2 = 14\alpha^2 - 5/8, \quad R < 14, \quad G = (A - \sqrt{a})/\alpha > \sqrt{69/5} - 2\sqrt{6}/15 > 3.388.$$

The derivative numerator $(5/8)/A - (g/3 + 1)/\sqrt{a}$ is negative on the full interval, proving the endpoint bound. At actual strong ten/eleven the local payment at its individual $1/7$ coefficient, plus $y^2 > .1$, gives totals $> 1.0295, > 1.0294$. No dotted parameter is missing: it was directly strong. Coverage reaches fourteen with only pure helpers.

Their nonadjacency extends by the actual two-cap calculation: adjacent helper products exceed $1-ab > -11/25 > -1/2$, so the native gap forces orthogonality. The retained cap Schur matrix is $I - t\mathbf{1}\mathbf{1}^T$, $t = (i-1)/(2i)$, with positive inverse. The unchanged prefix replacement gives weighted loss $< 7/16 + 1/(4\sqrt{10}) < .61$, contradicting two weak floors. Retained right blocks remain positive; no larger-rank prefix hypothesis is invoked. There is therefore a strong thirteen/fourteen reference with energy $q > .88864$, first-five mean $b < .361$ and individual P_5 coefficient $> .146$. This gives only the safe q_{13} floor; it does not assert strong fourteen or a q_{14} floor.

Delete long one. The full retained D_{16} permits only isolation or a three at its long endpoint two. The latter has $A^2 = 2g^2 + 2g + Q$, $C_1 = (2g+1)/\sqrt{1-2Q}$. Own zero has zero time or its positive full-D null; own three has its triangle. Every other own gives $x_1 > .643$. Its legal first-five graft has gain $(5/4)(.643 - b)^2 > .0994$, already contradictory.

For an isolated-within-D head, $c = Ap - g\Omega_1 - \beta e$, $A^2 = 2g^2 + Q - 1$. The actual foreign shadow and original-farthest orthogonal-opposite obstruction give $A > \beta$, so $2g^2 > 1$. Owns zero/three/four are impossible; six and parallel have actual future nulls. All remaining own and foreign cases have one of the following actual pure-D pressures. A normalized displacement Z, supported in the first five coordinates with sum one, has $\langle x, Z\rangle \geq H$, $R = \|Z\|^2$; weight μ costs $\mu/5$ and gains $2\mu(H-b) - \mu^2(R - 1/5)$ in this one reference.

Pure five uses the finite H_4 correction $g(3n_1/2 + n_2 + n_3/2)$. It gives $Z = P_4/4$, $R = 1/4$, $H^2 = 1/(4\phi) > 2/13 > .392^2$. Weight .4 gives gain $> .0168$. Pure finite at least seven uses $c + g^{-1}n_1 + 2(g^{-1} - g)n_2$, and, for $u = g^{-2} \in (1, 1.232)$,

$$Z = (1 - u/2)(e_1 + e_2) + (u-1)e_3, \quad H = \frac{\sqrt{2-u} - \sqrt{7u - 3u^2 - 4}}{\sqrt{2}}, \quad R = 3 - 4u + 3u^2/2.$$

The complete interval seals for weight $1/2$ are

u	H lower	R upper	gain lower
$[1, 1.1]$	$.6708 - .1871$	.5	.0477
$[1.1, 1.2]$	$.63245 - .204125$	.415	.013575
$[1.2, 1.232]$	$.6196 - .2$	.36	.0186.

They follow by positive squaring and the quadratic radical maximum $1/24$. The native-seven cubic gives the upper 1.232 uniformly for all finite labels at least seven.

For dotted or foreign-cleared $0 < u = (1-Q)/g^2 < 1$, the positive core $c + \beta e + gun_1$ gives exactly the Z, H pressure in the common dotted-head calculation of Appendix J.3. Its scalar proof uses only this pressure, $b < .361$ and the individual first-five capacity. It therefore gives the same legal gain $> .03624$ here at weight $\sqrt{2}/5$; no typed-defect hypothesis or conclusion is used. All original walls are retained by the positive core. At $u = 1$ is a nonzero feasible future null. For $Q > 0, u > 1$, the native alphabet forces own five and foreign three only. Its actual finite H_3 projection on c, n_1, e gives $Z = P_2/2$, $R = 1/2$, $H^2 = 1/(2\sqrt{5}) > 2/9 > .47^2$. Weight $1/2$ gives gain $> .034$. Every foreign finite label and the full dotted own interval is thus included.

Each price is at most $.1 < .146$. Every selected finite point is an actual positive projection; every negative core has future time and nonpositive product with all original walls. All head cases give a legal gain greater than .0135. Hence

$$\|x\|^2 + y^2 > .88864 + .0135 + .1 = 1.00214 > 1.$$

This closes precisely the ordinary D_{17} rank-one case. Tail elimination contradictions have not been added to the final reference, and no short-I energy or arithmeticity is assumed.

J.5 B_{17} with an actual isolated short spectator

We separate the local payment lemmas from the short-tail classification. This makes the order of the argument explicit: the local lemmas first dispose of the exceptional B_2 cap, and only afterwards do we prove the unconditional strong-five bootstrap. None of the local lemmas assumes that bootstrap or an already strong terminal wall.

Throughout, $P_i = e_1 + \cdots + e_i$, $f_i = \sqrt{i} - .61/\sqrt{2}$, and $\phi = (1+\sqrt{5})/2$. The original vector is $z = E(p - x - ye)$, with $\|x\|^2 + y^2 < 1$. The actual isolated I-wall is

$$c_I = \sqrt{h}\, p - \sqrt{1+h}\, e, \qquad 0 < h < 1/2.$$

Put

$$s^2 = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}, \qquad t = \sqrt{1+4h} - 2\sqrt{h}, \qquad h(r) = \frac{4r}{(1-r)^2}.$$

Appendix A.8 proves $y \geq s$, and height at least t in every other actual unit dual direction. At a classical long end with $\Omega_1 = \sqrt{2}e_1$, this says $x_1 \geq t$. In particular

$$0 < s < 1/3, \qquad (1-s^2)(1-t^2) = 8st, \qquad h = h(s^2).$$

These facts concern actual walls and the same original vector. The local lemmas below require only their stated long-root, point and reference hypotheses; a B short root is not among those hypotheses. Where the initial roots are used, they are the actual long roots $n_j = (e_j - e_{j+1})/\sqrt{2}$, $j = 1, 2, 3$, with actual long duals $\Omega_1 = \sqrt{2}P_1, \Omega_2 = \sqrt{2}P_2$.

Lemma J.1 (The additional real I-barrier)**.** *The actual parameters satisfy $t + \frac{5}{2}s > .923$.*

Proof. The test value $t_0 = .923 - 2.5s$ is positive. Substitution in the left-minus-right of the relation gives

$$P(s) = .148071 - 2.769s + 13.601929s^2 - 4.615s^3 + 6.25s^4.$$

For $v = s - .1 \in (-.1, 7/30)$, its coefficients are $.00320029, -.1620642, 12.592429, -2.115, 6.25$. Since $-2.115v^3 \geq -.4935v^2$,

$$P(s) > .0032 - .163|v| + 12v^2 > 0.$$

The last discriminant is $.026569 - .1536 < 0$. The original expression decreases strictly with positive t, proving the claim. This is separate from the earlier .767-barrier. □

J.5.1 Local paid squares, with every initial cross retained

Call W_0 a paid classical vector if its demand D_0 has already been obtained from nonnegative actual-pressure coefficients. Its effective lower energy is $\mathcal{E}_0 = 2D_0 - \|W_0\|^2$. At a retained anchor P_m, let the actual demand be a_m; any debit below is made from its remaining individual coefficient. No I-energy is included in $\mathcal{E}_0$.

Lemma J.2 (A paid head square at an arbitrary initial anchor)**.** *Suppose an actual pure classical pressure is $\langle x, Z\rangle \geq H$, with Z supported in the first m coordinates and $\langle Z, P_m\rangle = 1$. Write $R = Z^2$ and*

$$B_Z = \langle W_0, Z\rangle + \frac{a_m - \langle W_0, P_m\rangle}{m}.$$

For $\nu \geq 0$, with debit ν/m available at P_m, the same square improves $\mathcal{E}_0$ by

$$2\nu(H - B_Z) - \nu^2(R - 1/m).$$

Proof. Use $W_0+\nu(Z-P_m/m)$, after removing ν/m from the anchor coefficient. Its residual square is $R-1/m$; expanding its full cross with W_0 gives the formula. In particular this formula does not assume a weak-five P_1/P_6 reference is flat. □

Lemma J.3 (A joint point and the I-height on a paid reference)**.** *Suppose the actual pressures are $ax_1+vy\ge F$, $y\ge s_0$, with $a>0, v\ge 0$. Put*

$$B_0=(W_0)_1+\frac{a_m-\langle W_0,P_m\rangle}{m}.$$

For $\eta,\tau\ge 0$, write $k=\eta a$, $\ell=\eta v+\tau$, and require the individual anchor debit k/m to be available. The vector $W_0+k(e_1-P_m/m)+\ell e$ improves $\mathcal{E}_0$ by

$$2\eta(F-aB_0)+2\tau s_0-(1-1/m)k^2-\ell^2.$$

If $\ell=s_0$, put $X=(F-vs_0)/a$. The improvement becomes

$$s_0^2+2k(X-B_0)-(1-1/m)k^2,\qquad \tau=s_0-kv/a\ge 0.$$

Proof. Remove k/m at the original anchor before using the two new pressures. Expand the square, retaining $\langle W_0,e_1-P_m/m\rangle$. The old vector has no e-component. This proves both identities and keeps the nonnegative multiplier and individual-price conditions visible, also when k is capped. □

Lemma J.4 (A capped initial graft at a weak-five reference)**.** *Let $a_0=.5965$. Suppose W_0 is already paid, its first coordinate is a_0, and $\langle W_0,P_6\rangle$ equals its retained P_6 demand. If the remaining P_6 price is at least $1/60$, then*

$$\lambda=\min\{(6/5)(t-a_0)_+,.1\}$$

gives, using $e_1-P_6/6$ and the single unused I-height, an improvement

$$s^2+2\lambda(t-a_0)-(5/6)\lambda^2>.015.$$

Proof. The full old-reference cross changes the baseline to a_0; it is not zero in general. In the uncapped branch, weighted Cauchy and Lemma J.1 give

$$s^2+(6/5)(t-a_0)_+^2>\frac{(.923-.5965)^2}{25/4+5/6}>.015.$$

In the capped branch $t>a_0+1/12$. The relation makes t decrease with s, and

$$t(.1)=\frac{\sqrt{11401}-40}{99}<67/99<a_0+1/12.$$

Thus $s<.1$. The .923-barrier then gives

$$s^2+.2(t-a_0)-1/120>(s-.25)^2-83/15000>.0169666.$$

The debit is at most $1/60$. These are one paid square and one I-height, not two independent initial-coordinate gains. □

A flat-five reference below means a nonnegative actual-anchor reference w, with demand $\|w\|^2=q$, first five entries $b=f_5/5<.361$, and remaining P_5 coefficient $>.146$. The anchor demand there is $f_5=5b$. Extra terminal anchors are allowed. Every local gain below preserves their nonnegative coefficients. A head means the actual opposite of the old long node one.

Lemma J.5 (The foreign-cleared head scalar)**.** *On a flat-five reference, suppose an actual pure pressure has*

$$Z = (1 - u/2)e_1 + (u/2)e_2, \quad H = \frac{\sqrt{2-u} - \sqrt{u(1-u)}}{\sqrt{2}}, \quad R = 1 - u + u^2/2, \qquad 0 < u < 1.$$

Weight $\nu = .36$, costing .072 at P_5, gives gain $> .04$.

Proof. Use Lemma J.2. Put $.5091 < .36\sqrt{2} < .5092$. Lower chords for $\sqrt{2-u}$ and upper chords for R, on the three successive intervals, are

$$\begin{array}{c|c|c} [0,.5] & 1.4142 - .379u & 1 - .75u \\ [.5,.75] & 1.4381 - .4268u & .8125 - .375u \\ [.75,1] & 1.4721 - .4721u & .625 - .125u. \end{array}$$

Their endpoint squares and concavity/convexity prove the bounds. After $b < .361$, the gains exceed

$$\begin{array}{c|c} .35636922 - .0958u - .5092\sqrt{u(1-u)} & .04936922 \\ .39283671 - .16873u - .5092\sqrt{u(1-u)} & .04022171 \\ .43444611 - .2242u - .5092\sqrt{u(1-u)} & .04414611. \end{array}$$

The right column follows from $pu + c\sqrt{u(1-u)} \le (p + \sqrt{p^2 + c^2})/2$, using upper radical seals $.5182, .5365, .5564$. □

Lemma J.6 (Finite heads on a flat-five reference)**.** *Suppose the retained classical long-end diagram has only an isolated head or a simple-three extension at old node two. On any flat-five reference, a nonisolated head gives gain $> .0994$. For a pure finite isolated head the head gain alone is $> .0186$, and its gain including the single I-height is $> .0933875$. For a positive foreign finite head the combined gain is $> .0957$. Consequently $q \ge .90715$ excludes every nonisolated or finite isolated head beside actual I, without a terminal-strength hypothesis.*

Proof. For the nonisolated head,

$$A^2 = 2g^2 + 2g + Q, \qquad C_1 = \frac{2g+1}{\sqrt{1-2Q}}.$$

Own zero has zero time or the positive null $c + \Omega_2/2 = \sqrt{Q}p - \beta e$; the dual has its actual positive old-root expansion. Own three has the triangle and parallel the pair null. Every other own gives $x_1 > .643$. Replacing the first five entries by $.643, (5b - .643)/4, \ldots$ is legal: its debit $(.643 - b)/4$ fits the individual P_5 coefficient, and its gain exceeds $(5/4)(.282)^2 > .0994$.

For an isolated finite head $c = Ap - g\sqrt{2}e_1 - \beta e$, $g < 1$, put $Q = \beta^2$. The distinct actual I-wall gives $\lambda = \sqrt{h}A - \sqrt{1+h}\beta \ge 0$. At $\beta = 0$, $0 < \lambda < 1/\sqrt{2}$, so $\lambda = 1/2$. At $\beta > 0$,

$$\lambda < \sqrt{(1+\beta^2)/2} - \beta\sqrt{3/2} \le (\sqrt{10} - \sqrt{6})/4 < 1/2.$$

The middle upper bound decreases with $\beta \ge 1/2$, and its bound before substitution increases with $h < 1/2$; hence $\lambda = 0$. Squaring gives $\beta^2 = h(2g^2 - 1) < 1/2$, so $\beta = 1/2$. Both cases consequently have

$$h(2g^2 - 1) = 1/4, \qquad h > 1/4.$$

Own zero/three/four is impossible, five gives $h = \phi/2 > 1/2$, six gives $h = 1/2$, and parallel has $c + n_1$. Thus a finite isolated own has label at least seven.

If $\beta > 0$, $u = (1-Q)/g^2 < 1$. The actual positive core

$$c + \beta e + gun_1$$

has products zero on c, e, product $g(u-1) < 0$ on n_1, square $-g^2u(1-u)$, and positive time. It yields exactly the pure pressure of Lemma J.5. The I-height costs $> .0557$, by $h > 1/4$, so the gain including I is $> .0957$.

If $\beta = 0$, put $u = g^{-2} \in (1, 1.232)$. The actual core

$$c + g^{-1}n_1 + 2(g^{-1} - g)n_2$$

has zero cap/first-root products, negative second/third-root products and positive time. Its pure data are

$$Z = (1-u/2)(e_1+e_2) + (u-1)e_3, \quad H = \sqrt{1-u/2} - \sqrt{(7u-3u^2-4)/2}, \quad R = 3 - 4u + 3u^2/2.$$

The native-seven cubic gives $g > .90096$, and $1.232(.90096)^2 > 1$. Weight $1/2$ gives

u	head gain lower	h lower	I energy lower
$[1, 1.1]$	.0477	$1/4$	.0557
$[1.1, 1.15]$	.033025	$11/36$	.066
$[1.15, 1.2]$	.0213875	$23/68$	.072
$[1.2, 1.232]$	.0186	$3/8$	.079.

The first radicals are bounded below by $.6708, .6519, .63245, .6196$, the second above by $.1871$, $.204125, .204125, .2$, and R above by $.5, .415, .38375, .36$. Positive squaring proves these; the second radicand has maximum $1/24$ and R decreases. Use $h = u/[4(2-u)]$ and $h(r)$ for the I-column. Every combined gain is $> .0933875 > 1 - .90715$. Every head pressure has positive actual-root coefficients, and all unselected original walls retain nonpositive products. □

Lemma J.7 (An actual joint point for every pure dotted mutual)**.** *For an actual pure dotted head* $c = Ap - g\sqrt{2}e_1$*, put*

$$\begin{aligned} u &= g^{-2} \in (0,1), & h_0 &= \frac{u}{4(2-u)}, \\ a &= \sqrt{1-u/2}, & v_0 &= \sqrt{u/(8-3u)}, \\ d &= \sqrt{1-a^2-v_0^2}, \quad F = 1-d, & s_0 &= s(h_0). \end{aligned}$$

For $u \geq 1/5$*, every actual* $h \geq h_0$ *gives a feasible future point* $(p - ae_1 - v_0 e)/d$*. Consequently*

$$ax_1 + v_0 y \geq F, \quad y \geq s_0, \qquad X = \frac{F - v_0 s_0}{a} = \sqrt{\frac{1-\sqrt{3u/8}}{1+\sqrt{3u/8}}}.$$

Proof. The actual mutual product is $-A\sqrt{h} < 0$, hence $h \geq 1/(4A^2) = h_0$. Put $v = \sqrt{h/(1+h)}$; then $v \geq v_0$, $v < 1/\sqrt{3}$. The denominator $D = u/2 - v_0 v$ is positive because $3u(8-3u) \geq 111/25 > 4$ on this interval. For

$$T = 1/D, \quad \rho = \frac{Ta}{\sqrt{2}g} > 0, \quad \sigma = \frac{Tv_0}{\sqrt{1+h}} > 0$$

direct substitution gives

$$p + \rho c + \sigma c_I = T(p - ae_1 - v_0 e).$$

Its c-product is zero and its c_I-product is $T\sqrt{1+h}(v_0-v)\le 0$. Old wall products are nonpositive by duality; all other actual walls are nonpositive termwise. Its time is positive and $d^2>0$. Thus the normalized point is actual even if the actual c, c_I Gram is not positive. The identities $v_0 s_0 = 1-\sqrt{1-v_0^2}$ and $d/\sqrt{1-v_0^2} = \sqrt{3u/8}$ give X. It decreases with u, whereas s_0^2 increases. The comparison $X > r$ at $u = U$ is the positive rational test

$$8(1-r^2)^2 - 3U(1+r^2)^2 > 0.$$

□

Lemma J.8 (The foreign dotted-head parabola)**.** *For a dotted head with $Q=\beta^2>0$, set $v=(1-Q)/g^2$. Then $0<v<3/4$, and*

$$h \ge h_{\min}(v) := \frac{v}{6-4v}.$$

The actual foreign-cleared core gives the pure pressure

$$Z = (1-v/2)e_1 + (v/2)e_2, \quad H(v) = \sqrt{1-v/2} - \sqrt{v(1-v)/2}, \quad R(v) = 1 - v + v^2/2.$$

For $F(v) = .8H(v) - .16R(v)$, $\zeta = v - 2/3$,

$$F(v) \ge C_0 + A_1\zeta + .448\zeta^2, \quad C_0 > .2976415, \quad 0 < A_1 < .008385, \quad F(v) > .2976.$$

Also $y^2 > .08625v - .014375$ for $v \le .6$, and $y^2 > .0385$ for $v \ge .6$.

Proof. Acuteness with c_I gives $h(2g^2-1) \ge Q$; hence

$$h \ge \frac{Qv}{2(1-Q)-v} \ge \frac{v}{6-4v}.$$

The second expression increases with $Q \ge 1/4$. The core $c + \beta e + ((1-Q)/g)n_1$ has zero cap/old-I products, negative n_1-product, square $-g^2 v(1-v)$ and positive time. It clears the foreign coordinate before producing the displayed pure pressure; every other wall is nonpositive.

Put $r = \sqrt{6}$ and $\ell = r/3 - r\zeta/8 - .09\zeta^2$. On this interval $\ell \ge 7r/24 - .04 > 0$. The exact identity

$$1 - v/2 - \ell^2 = \zeta^2(3r/50 - 3/32 - (9r/400)\zeta - .0081\zeta^2)$$

has nonnegative right side, since the bracket is at least $.045r - .09735 > 0$. Further, $(\sqrt{v(1-v)/2} - 1/3)^2 \ge 0$ gives

$$H(v) \ge (r-1)/3 + (2-r)\zeta/8 + .66\zeta^2.$$

Thus $C_0 = 4(r-1)/15 - 4/45$ and $A_1 = 19/75 - r/10$. Use $2.449489 < r < 2.449490$ and $.008385^2 < 1.792(.2976415 - .2976)$.

For $v \le .6$, $h_{\min} \le 1/6$, and $.23(2.081)^2 < 1$ gives $s(h_{\min})^2 > .23 h_{\min}$. The exact tangent inequality is

$$\frac{v}{6-4v} - \left(\frac{3v}{8} - \frac{1}{16}\right) = \frac{3(2v-1)^2}{8(6-4v)} \ge 0.$$

For $v \ge .6$, $h_{\min} \ge 1/6$, whose I-energy is $> .0385$, again checked using $h(r)$. □

Lemma J.9 (The head on any sufficiently strong flat-five reference)**.** *Assume the actual long-end whitelist of Lemma J.6. On any flat-five reference every actual head gives a legal gain, including I once, greater than .07665. In particular $q \ge .9240775$ excludes all heads. No actual strong-fifteen condition is required.*

Proof. Finite and nonisolated heads are covered by Lemma J.6. For a pure dotted head with $u \le .2$, its actual core $c + g^{-1}n_1$ gives the pressure Z, H, R of Lemma J.5. Here $H > .6658, R \le 1$. Weight .3, with price .06, gives gain $.6(.6658 - .361) - .09(.8) = .11088$.

For pure $.2 \le u \le .5$, use weight $\eta = .33$ of the joint point alone. In Lemma J.7,

$$F > .5839, \quad a < .949, \quad (4/5)a^2 + v_0^2 < .748.$$

The first bound follows at $u = .5$, since d^2 increases; the last follows at $u = .2$, since that expression decreases. The gain is

$$.66(.5839 - .949 \cdot .361) - .1089(.748) = .07780806.$$

Its individual price is $< .066$; no I-energy is added again.

For $.5 \le u \le .8$, use Lemma J.3 with $m = 5, B = .361$, $k = 5(X - B)/4$, $\ell = s_0$. The unused I multiplier is nonnegative because $X - B < .269 < .4a$ and $s_0/v_0 \ge 1/2$. Its price is $< .06725$. Positive rational tests give

u	s_0^2 lower	X lower	gain lower
$[.5, .6]$	.0200	.597	.08962
$[.6, .7]$	.0254	.567	.078445
$[.7, .75]$	.0315	.553	.07758
$[.75, .775]$	.0349	.547	.078145
$[.775, .8]$	.0366	.540	.07665125.

Use the right endpoints for X, and left endpoints in $h(r)$ for s_0^2; for the last two these have $h_0 = 3/20, 31/196$. For $.8 \le u < 1$, the actual pure core gives $H > .491, R \le .52$. Indeed H increases there and R decreases. A half-payment gains $> .05$, and $h_0 \ge 1/6$ contributes $> .0385$.

For a positive foreign dotted contact use Lemma J.8 at weight .4. The head gain is $> F - .2568$, at price .08. For $v \le .6$, adding the I-height gives

$$> .0839665 + (A_1 + .08625)\zeta + .448\zeta^2 > .0789665,$$

using $(.09464)^2 < 1.792(.005)$. For $v \ge .6$ the gain is $> .2976 - .2568 + .0385 = .0793$. All prices fit the single available P_5 coefficient, and every parameter branch gives more than the stated .07665. □

J.5.2 A conditional actual-helper packet at long index fifteen

Lemma J.10 (Actual helper pressures and the retained fifteen packet)**.** *Suppose an actual pure far-three opposite at a long index $i \ge 5$ has*

$$c_i = A_i p - \gamma\sqrt{2}P_i - \Omega_1/2, \qquad A_i^2 = 2i\gamma^2 + 2\gamma - 1/2,$$

where $\sqrt{2}P_i$ is the actual old long dual, $n_i = (e_i - e_{i+1})/\sqrt{2}$, and the old initial roots n_1, n_2 are actual. Its own is assumed to be five, finite at least seven, or dotted; the invoking local classification must establish that list. Write $H_k = P_k/(k\sqrt{2})$. The following are actual pure pressures $\langle x, U\rangle \ge T$:

own	U	T
5	$\gamma(\sqrt{2}P_i - n_i) + H_3$	$H(A_i, 2/3 - \gamma^2)$
7, 8	$\gamma(\sqrt{2}P_i - n_i) + H_2$	$A_i - \sqrt{\gamma^2 - 3/4}$
finite ≥ 9	$\gamma(\sqrt{2}P_i - n_i) + H_1$	$H(A_i, 1 - \gamma^2)$
dotted	$\gamma\sqrt{2}P_i - \gamma^{-1}n_i + H_1$	$A_i - \sqrt{1 - \gamma^{-2}}$,

where $H(A,d) = A/(1+\sqrt{d/(A^2+d)})$.

In particular suppose $i = 15$ *is weak, and the legal reference* w *ends at strong fourteen, is flat-five with* $q > .90715$*, and has individual* P_{14} *coefficient* $> .1361$*. Put* $\alpha = \sqrt{2}\gamma$ *and*

$$\Pi U = \alpha P_{14} + P_5/(5\sqrt{2}), \quad Y = U - \Pi U, \quad \Delta = T - \alpha f_{14} - b/\sqrt{2}.$$

Then $w.Y = 0$*, and the following complete table is valid:*

own	Δ *lower*	Y^2 *upper*	μ	$2\mu\Delta - \mu^2 Y^2$ *lower*
5	.39558	.724	.09	.06534
7	.340518	.963	.09	.05349294
8	.2873002	1.005	.09	.043573536
finite ≥ 9	.299602	1.4	.09	.04258836
dotted	.3307	1.68	.08	.04216.

Thus $W_0 = w + \mu Y$ *has effective energy* $> q + .0421$*. Its debits are* $< .1273$ *at* P_{14} *and* $< .01273$ *at* P_5*. Its initial coordinate is*

$$(W_0)_1 = b + \frac{\mu(1/k - 1/5)}{\sqrt{2}} < .412.$$

This lemma does not assume a B *short normal or that* P_{16} *is an actual long dual.*

Proof. For own five clear old n_i, n_1, n_2, with positive correction $\gamma n_i + 2n_1/3 + n_2/3$; the actual finite block is H_4. Its residual is $2/3 - \gamma^2 > 0$, so use the finite projected point, not the corrected spacelike root. For owns seven/eight clear $\gamma n_i + n_1/2$; its residual is $3/4 - \gamma^2 < 0$. For finite owns at least nine clear γn_i, and for dotted owns use $c_i + n_i/\gamma$. Selected products are zero or negative, time is positive, and all other actual walls are nonpositive termwise. These include the old I, its opposite, any long head and either classical tail. The displayed U's follow by subtraction in the original Euclidean coordinates.

At fifteen the initial residual is $H_k - P_5/(5\sqrt{2})$; the remaining residual has support only in fifteen/sixteen. It is orthogonal to every retained reference anchor. For finite owns,

$$Y^2 = \gamma^2 + 1/(2k) - 1/10;$$

for dotted owns it is $2\gamma^2 - 1.6 + \gamma^{-2}$. Use $\sqrt{28} - .61 < 4.682$, $b/\sqrt{2} < .256$. For finite five/seven/eight, the respective (A_i lower, radical upper, γ upper) are

$$(4.555, .111, .81), \quad (5.064, .249, .901), \quad (5.191, .322, .9239).$$

Their positive squares give the first three deficits. For finite $\gamma \geq .939$, the deficit increases: $H_A > .99$, $A_i' > \sqrt{30}$, $H_d < 0$, and $d = 1 - \gamma^2$ decreases. At .939,

$$A_i > 5.275, \quad \sqrt{A_i^2 + d} > 5.286, \quad \sqrt{d} < .344, \quad H > 4.952,$$

giving $\Delta > .299602$.

For dotted fifteen, weakness implies $\gamma < 10/9$: at $10/9$ its squared edge ratio is $10609/4320 > (39/25)^2$, whereas $T(\sqrt{30}) < 39/25$. The deficit decreases on this interval, since $A_i' < 5.6$, the radical derivative is > 1, and $\sqrt{28} - .61 > 4.681$. At $10/9$, $A_i^2 = 2093/54 > 6.225^2$ and the radical is $< .436$; thus $\Delta > .3307777$. The norm is < 1.68. All entries and prices now follow by rational multiplication. The largest initial coordinate occurs for $k = 1, \mu = .09$, and is below $.361 + .09(.8)/\sqrt{2} < .412$. □

Lemma J.11 (A retained helper, a dotted head and actual I)**.** *Assume the actual helper and reference hypotheses of Lemma J.10, with $f_{14} < 3.31034$, and an actual isolated-within-classical dotted head. For a pure head with $u \geq .2$, a single paid square gives helper gain $> .0421$ and joint head/I gain $> .0515$. For a positive foreign head the corresponding gains are $> .0421$ and $> .0521$. For a pure head with $u \leq .2$, the standalone gain $> .11088$ on w suffices without paying the helper. In particular each case contradicts the original norm budget.*

Proof. For a pure head with $u \geq .2$, take $B = .412$ above the actual initial coordinate B_0. In Lemma J.3, use $m = 5$, $k = 5(X - B)/4$, $\ell = s_0$. Since the helper residual has zero sum on the first five entries, its full cross changes the baseline to precisely B_0; it is not orthogonal to a head. The multipliers are nonnegative: on $[.2, .4]$, $X < .756, a > \sqrt{.8}$, and on $[.4, 1]$, $X < .665, a > 1/\sqrt{2}$. Thus $X - B < .4a$, while $s_0/v_0 \geq 1/2$. The total P_5 price is $< .086 + .01273 < .146$, and the separate P_{14} price fits. The complete continuous table is

u	s_0^2 lower	X lower	$s_0^2 + (5/4)(X - .412)^2$ lower
$[.2, .5]$	0	.629	.05886125
$[.5, .6]$	.0200	.597	.06278125
$[.6, .7]$	.0254	.567	.05543125
$[.7, .8]$	.0315	.540	.05198
$[.8, .9]$	.0385	.514	.051505
$[.9, 1]$	.0464	.490	.054005.

The X-tests are those of Lemma J.7; the I-tests use the left endpoints $h_0 = 1/12, 3/28, 7/52, 1/6, 9/44$. For example the first X-test has positive numerator .0002777057265. This proves gain $> .0515$ on all the displayed intervals. Small u uses the standalone actual core already proved in Lemma J.9.

For a foreign head put $v = (1 - Q)/g^2$, and use its cleared pressure at weight $\nu = .4$. The initial helper products with $Z - P_5/5$, for $k = 3, 2, 1$, are respectively

$$\frac{2/15}{\sqrt{2}}, \quad \frac{3/10}{\sqrt{2}}, \quad \frac{.8 - v/2}{\sqrt{2}}.$$

The last is largest on $0 < v < .75$. Hence the complete adverse cross is at most $(.072/\sqrt{2})(.8 - v/2)$. Using $1/\sqrt{2} < .7072$, the head gain is

$$> F(v) - .29753472 + .0254592v.$$

For $v \leq .6$, adding the single I-height and the parabola gives

$$> .06020458 + (A_1 + .1117092)\zeta + .448\zeta^2 > .05215458,$$

since $.1201^2 < 1.792(.00805)$. For $v \geq .6$ the gain is $> .0538408$. The combined P_5 price is $< .09273$; the P_{14} price is unchanged. No I-energy was included in the pure cleared pressure or the helper gain. The full totals exceed

$$.90715 + .0421 + .0515 = 1.00075, \qquad .90715 + .0421 + .0521 = 1.00135.$$

□

J.5.3 The fresh B_{17} tail and its exceptional actual mate

We now impose the entire $B_{17} + I$ forest, with actual unit roots $n_i = (e_i - e_{i+1})/\sqrt{2}$, $i \le 16$, and genuine short root $n_{17} = e_{17}$. Thus $\Omega_i = \sqrt{2}P_i$ through sixteen and $\Omega_{17} = P_{17}$. All classical opposites retain the old I. The valid bulk parameter is $M = 12$, giving pure-helper coverage through thirteen and the actual mixed-point nonadjacency.

At fourteen deletion retains $A_{13} + B_3$. Its only right finite extensions are long-three at fifteen and short-three at seventeen, the latter the sealed F_4. Their (Δ, σ) are $(1/2 - Q, \gamma + 1/2)$ and $(1/4 - Q, \gamma + 1/(2\sqrt{2}))$. Own zero has the actual affine star or $3, 4, 3, 3$ path; own three on the long side is a triangle, and all other positive cases are directly strong. Full bridges attach by three at left endpoints and the long right endpoint, are finite B_{17} and foreign-sealed. A short-end full bridge would put the old four internally in a long path. Left/isolated/foreign cases retain the complete long-root inventory of Appendix E. Consequently coverage and nonadjacency reach fourteen, before assuming that five is strong.

At fifteen the retained right block is the actual B_2 on long sixteen and short seventeen. A finite new contact is three at either end only. The data are

$$(\Delta, \sigma) = (1/2 - Q, \gamma + 1/2), \quad (1/2 - Q, \gamma + 1/(2\sqrt{2})).$$

The long case has its own-zero star and own-three triangle. In the short case, own zero has the affine F_4 kernel $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$ on $c, n_{17}, n_{16}, n_{15}, n_{14}$. Own four has $\sqrt{2}c + n_{14} + 2n_{15} + n_{16}$, own six has $\sqrt{3}c + 2n_{15} + n_{14}$, and parallel has $c + n_{15}$. For long own zero use $c + n_{15} + 2n_{16} + \sqrt{2}n_{17}$. Direct products give zero selected products, positive cap time and nonpositive products with every remaining original wall. Every other own at least five is strong. Positive foreign $Q \ge 1/4$ makes the short own-three case strong too. All four left/right endpoint bridges are retained, with residual $1/30$; their minimum ratio, including own zero, is

$$\sqrt{30}(1/30 + 1/(2\sqrt{2})) > 100/61.$$

The full bridges are foreign-sealed. The sole additional weak cap is

$$c = Ap - (\Omega_{15} + \Omega_{17})/2, \qquad Q = 0.$$

Release old sixteen and keep this same actual cap. The retained path old one through fifteen, then c, then the actual short seventeen is finite A_{17}, all bonds three. Positive finite projection is future and satisfies every original wall. Omitted sixteen is strict, because it has negative product with the positive old-fifteen and short-seventeen terms. This is the actual other endpoint of old edge sixteen. Its exact data give

$$C_{16} = \frac{16 + 15\sqrt{2}}{\sqrt{18}}, \qquad S_{16} \ge 4\sqrt{\frac{C_{16} - 1}{C_{16} + 1}} = G := \frac{8 + 12\sqrt{2}}{7}, \quad 3.56722 < G < 3.5673 < f_{16}.$$

In particular this floor is not called strong sixteen.

If five is weak, the actual first-six pressure gives $x_1 > a_0 = .5965, S_6 > 2.01815$. The reference through one, six and the actual floor G at sixteen has decreasing slopes and energy

$$> .7600299945 + (3.5672 - 2.01815)^2/10 = .99998558475.$$

Its P_6 coefficient exceeds $.28433 - .154915 > 1/60$. Lemma J.4 therefore gives a contradiction. This uses no head reduction or later strong-five conclusion.

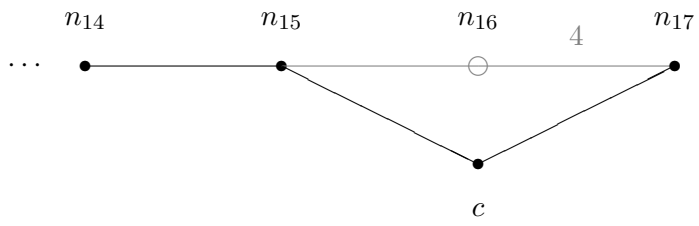


Figure 1: The exceptional cap replaces old sixteen. The solid path through n_{15}, c, n_{17} has only simple bonds. The grey old edges are released; the old four bond is not changed into an invented root.

If five is strong, take the already available strong thirteen/fourteen reference and append the straight chord to G. Its slopes $< .132, < .129$ fit the individual terminal coefficients. The new energy is respectively

$$> .88864 + .39299^2/3 > .9401, \qquad > .90715 + .25688^2/2 > .9401.$$

For the energy bounds use $f_{13} < 3.17423, f_{14} < 3.31034$. For the slope test instead use $f_{13} > 3.1741, f_{14} > 3.3102$ and $G < 3.5673$, all given by positive squaring. The first five block and its full P_5 coefficient are unchanged. Deleting the long head leaves actual B_{16}, whose finite one-node extension is only isolation or simple-three at long two. Thus Lemma J.9 applies to this stronger reference, independently of any strong-fifteen assertion. This contradicts the original norm and excludes the exceptional cap in both initial branches.

J.5.4 Unconditional strong five from two retained actual helpers

Coverage through fifteen now consists only of nonadjacent pure helpers. If five is weak with finite own, the actual first-six table gives $x_1 > .65$. With strong six and an actual strong fourteen/fifteen, positive interpolation gives

$$.65^2 + (2.01815 - .65)^2/5 + 1.2921^2/8 = 1.00555718575 > 1.$$

Thus a weak-five own would be dotted, $1 < a < 6/5$. If fifteen is strong, the first-six/strong-fifteen interpolation has energy $> .985$; its P_6 capacity exceeds $1/60$. The capped initial graft adds $> .015$, again impossible. It remains that both five and fifteen are weak, so fourteen is strong.

Let w interpolate through $(1, a_0), (6, f_6)$ and every actual strong knot through fourteen. Put $b_1 = (f_6 - a_0)/5$; then $.28433 < b_1 < .284332$. The rational increment floors give

$$\|w\|^2 > .9703299945, \quad b_{P_1} > .312168, \quad b_{P_6} > .08806, \quad b_{P_{14}} > .1361.$$

For example $b_{P_6} > b_1 - (\sqrt{7} - \sqrt{6})$, and there are at most four nonadjacent omissions among seven through thirteen.

Retain the complete actual dotted-five foot

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \quad T_5 = \sqrt{10a^2 + 2a - 1/2} - \sqrt{1 - a^{-2}}.$$

Its projection onto the initial reference span is $mP_6 + lP_1$, where

$$m = 4\sqrt{2}a/5 < 1.35766, \qquad l = \sqrt{2}a/5 + 1/\sqrt{2} < 1.04652.$$

The residual Y_5 is orthogonal to w, P_1, P_6, and

$$Y_5^2 = 8a^2/5 - 2 + a^{-2} < 1, \qquad T_5 - \langle w, U_5\rangle > .1203.$$

Indeed $\langle w, P_5\rangle = a_0 + 4b_1 < 1.733828$. The deficit decreases with a: the time derivative is $< 13/4$, the radical derivative is > 1, and $\sqrt{2}\langle w, P_5\rangle > 2.4519$. At $a = 6/5$, use $A > 4.03732, \sqrt{11}/6 < .552771, \sqrt{2} < 1.414214$. Weight $\mu_5 = .05$ consequently gives gain $> .00953$, with debits $< .067883$ at P_6 and $< .052326$ at P_1.

For the complete weak-fifteen packet, project its initial H_k onto P_1, P_6. The projection of U_{15} is

$$\alpha P_{14} + l_k P_1 + m_k P_6,$$

k	l_k	m_k
3	$\sqrt{2}/10$	$\sqrt{2}/15$
2	$\sqrt{2}/5$	$\sqrt{2}/20$
1	$1/\sqrt{2}$	0.

The residual Y_{15} is orthogonal to all original reference anchors. Here $\langle w, H_3\rangle < .275, \langle w, H_2\rangle < .312, \langle w, H_1\rangle < .422$. The proof of Lemma J.10, with these initial demands in place of $b/\sqrt{2}$, gives

own fifteen	$T_{15} - \langle w, U_{15}\rangle$ lower	Y_{15}^2 upper	μ_{15}	gain lower
5	.37658	.824	.09	.06111
7, 8	.2313	1.105	.09	.0326835
finite ≥ 9	.1336	1	.09	.015948
dotted	.1647	1.28	.08	.01816.

All points remain actual; the table uses the pure-helper fifteen classification established before this bootstrap.

The full mixed crosses are

$$\langle Y_5, Y_{15}\rangle = \begin{cases} 2a/15 & k = 3, \\ a/10 & k = 2, \\ 0 & k = 1. \end{cases}$$

Their adverse square terms are at most $.00144, .00108, 0$. The first two rows already exceed $1.0395, 1.0114$, without any I-energy. Their combined P_6 debit is $< .067883 + .008486 < .08806$.

For the last two rows, both residuals are orthogonal to $e_1 - P_6/6$. The already paid vector still has first coordinate a_0 and P_6 demand f_6, so apply Lemma J.4. The total P_6 debit is $< .067883 + 1/60 < .08455 < .08806$; the P_{14} debit is $< .1273$, and P_1 debits are $< .117 < .312168$. The full original-reference/graft cross is retained in that lemma. One square, with I once, therefore exceeds

$$.9703299945 + .00953 + .015948 + .015 = 1.0108079945,$$

or

$$.9703299945 + .00953 + .01816 + .015 = 1.0130199945.$$

Every weak-five case is impossible. Five is now unconditionally strong.

J.5.5 The final composition

Take a strong fourteen/fifteen reference. It has $q > .90715$, first-five mean b, and individual P_5 coefficient $> .146$. The actual B_{16} long-head whitelist, and Lemma J.6, exclude all finite or nonisolated heads. The remaining actual head is isolated-within-B and dotted, retaining every finite contact with old I.

If fifteen is strong, its reference has $q > .9240775$, so Lemma J.9 excludes every such head. If fifteen is weak, fourteen is strong and its actual cap is a pure helper, so Lemma J.11 applies. Its small-parameter branch uses only the standalone actual head point; its other branches retain the full initial helper/head cross and the single I-height. They give the stated strict norm contradictions. This excludes the entire B_{17}+actual-I row.

All local points were positive combinations or positive finite projections of original walls, with every selected sign and future orientation checked before use. The original farthest vector is never replaced. No arbitrary face is declared Coxeter, no disconnected root scale or arithmeticity is imposed, no actual mate floor is promoted to strong sixteen, and no earlier elimination energy is added again.

J.6 D_{17} with an actual short rank-one factor

Keep the actual roots and signed fork coordinates of Appendix J.4.1. The original vector is $z = E(p - x - ye)$, with $\|x\|^2 + y^2 < 1$. Its actual isolated rank-one opposite is

$$c_I = \sqrt{h}\, p - \sqrt{1+h}\, e, \quad 0 < h < 1/2, \qquad s^2 = \frac{\sqrt{1+h} - 1}{\sqrt{1+h} + 1}.$$

We have $y \geq s$, $x_1 \geq t$, where

$$t = \frac{\sqrt{1 + 14s^2 + s^4} - 4s}{1 - s^2}, \qquad t + 2.5s > .923$$

by Lemma J.1. The same y coordinate is counted once; the ordinary debit $y^2 > .1$ is unavailable.

Deleting the head retains actual D_{16}. Its complete extension list is isolation or simple three at long leaf two. The local nulls, points and native-gap arguments from the preceding subsection retain old I and both forks. Thus Lemmas J.6 and J.9 apply on any legally funded reference constant first five, with mean $< .361$ and individual P_5 coefficient $> .146$. Their head-plus-I gain is $> .07665$, supported in first five plus I, without assuming strong five or any B-terminal conclusion. Finite isolated heads have $h > 1/4$, $y^2 > .0557$; pure owns five/six are impossible.

Fresh tails before strong five. The rank-valid $M = 11$ inventory covers five through twelve, with only nonadjacent pure far helpers weak. There is a strong index in each of $9/10, 10/11, 11/12$. The geometric D_4/D_3 lists from Appendix J.4.1 precede its ordinary energy debit. Their only extra weak rows are pure own-five/fork-three at thirteen and pure own-five or finite-own-at-least-seven/end-three at fourteen. Both forks and every foreign contact were retained; dotted end-three owns were directly strong.

The actual left-H_4 finite points are flat through $i-3$, have norm $< i$ and demands $> 3.48, 3.61$ at $i = 13, 14$. If five is strong, their preceding one-square totals before an ordinary debit are $> .9757$; adding $> .07665$ excludes them. If five is weak, the actual pressures $x_1 > .5965, S_6 > 2.01815$ give the stronger bare totals $> 1.039, 1.056$ on the positive P_1, P_6, P_K references. No head gain is needed in that branch.

For every remaining $.9 < g < 1$ end-three own, average the two actual A_2 cores, retaining their full cross $8g^2/9$. The demand is > 3.388, norm < 14, flat through twelve. In the strong-five branch pay $.16/.15$ at actual strong ten/eleven, below the individual terminal prices. With $q_{10} > .823735, q_{11} > .8472225$, the bare totals are

$$.823735 + .32(3.388 - 2.732) - .0256(4) = .931255,$$
$$.8472225 + .30(3.388 - 2.886) - .0225(3) = .9303225.$$

Adding the local head gain closes both. In the weak-five branch the reference norms are $> .8870, .9104$; the capped initial-I graft, Lemma J.4, adds $> .015$. Its support first six plus I is disjoint from the tail residual. Its P_6 price is available, and the totals exceed $1.00952, 1.0085$. This removes the tails before any later floor. The pure-helper two-cap calculation extends nonadjacency through fourteen: the actual inverse is $2\min(i,j)$ and both retained fork blocks remain positive. Its unchanged loss is $< 7/16 + 1/(4\sqrt{10}) < .61$. Thus thirteen or fourteen is strong.

Retained weak-five and weak-fourteen feet. Suppose five weak. For a justified $x_1 > a_0$, interpolate through one, six and actual strong knots up to $K = 13, 14$, omitting weak ones. Put $b_1 = (f_6 - a_0)/5$, $C(a_0) = a_0^2 + 5b_1^2$. The increment sums $> .19304, .21155$, minus at most three/four losses each $< 1/3200$, give $q_{13} > C + .1921025, q_{14} > C + .2103$. The needed capacities are

a_0	$C >$	$q_{13} >$	P_1 coefficient $>$	P_6 coefficient $>$
.5965	.7600299945	.9521324945	.312168	.08806
.643	.7916565045	.9837590045	.367968	.07876
.65	.7968668845	.9889693845	.376368	.07736

Terminal thirteen/fourteen coefficients are $> .14144, .1361$. We never assign a terminal coefficient to an internal knot.

If fourteen is weak, its pure far-helper time is $A_{14}^2 = 28g^2 + 2g - 1/2$. The finite/dotted feet use actual long roots thirteen/fourteen/fifteen and the initial local roots. After projection on P_1, P_6, P_{13}, their initial pieces H_3, H_2, H_1 have respective P_1, P_6 coefficients

$$(\sqrt{2}/10, \sqrt{2}/15), \quad (\sqrt{2}/5, \sqrt{2}/20), \quad (1/\sqrt{2}, 0).$$

The residual norm bounds for own five, seven/eight, finite at least nine, and dotted are $.824, 1.105$, $1, 1.28$. Only the last two residuals are confined to fourteen/fifteen; initial crosses in the other rows must be retained. The actual time/radical bounds are

own	$A_{14} >$	radical $<$
5	4.4095	.111
7	4.9017	.249
8	5.0242	.322

For finite own at least nine the deficit increases with $g \geq .939$. At that endpoint $A > 5.1055$, the other time radical > 5.116 and subtracted radical $< .344$ give demand > 4.7838. Dotted weakness implies $g < 10/9$: the squared ratio there is $83521/34020 > (39/25)^2$, above the exact target, which is $< 39/25$. The deficit decreases on this whole interval, since $A' < 5.4$ and the radical derivative exceeds one. At $10/9$, $A^2 = 5879/162 > 6.023^2$, radical $< .436$, so demand > 5.587. We use $\sqrt{26} - .61 < 4.4891$ throughout.

Every finite-own weak-five foot gives $x_1 > .65$. Fourteen strong then gives $q_{14}(.65) > 1$. Otherwise the weak-fourteen gains are $> .06160, .04163, .03203$ for owns five/seven/eight. For finite at least nine the deficit is $> 4.7838 - .939(4.4891) - .460 = .1085351$, and weight .09 gives gain $> .011436318$, total > 1.0004057. The dotted deficit $> .1391$ gives $> .01467$. Their prices fit the table. Hence weak five is dotted, $1 < a < 6/5$. Retain its full point

$$U_5 = \sqrt{2}aP_5 - a^{-1}n_5 + \Omega_1/2, \quad T_5 = \sqrt{10a^2 + 2a - 1/2} - \sqrt{1 - a^{-2}}.$$

Its projection is $mP_6 + lP_1$, with $m = 4\sqrt{2}a/5 < 1.35766, l = \sqrt{2}a/5 + 1/\sqrt{2} < 1.04652$. The residual has

$$Y_5^2 = 8a^2/5 - 2 + a^{-2} < 1, \quad Y_5 \cdot P_1 = Y_5 \cdot P_6 = 0, \quad (Y_5)_2 < .339412.$$

On the .5965 reference, its deficit $> .1203$, at weight .05, gives gain $> .00953$ and P_1/P_6 debits $< .052326/.067883$.

A nonisolated head forces $x_1 > .643$. If fourteen strong, $q_{14}(.643) > 1$. If weak, on $q_{13}(.643)$ the retained five foot gives $> .00466$, while the fourteen finite-at-least-nine and dotted gains are $> .0126789498, .01557$, with zero cross. For owns five/seven/eight the full cross losses $.00144, .00108$ leave gains $> .06034, .04091, .03131$. Combined P_6 price is $< .076369 < .07876$, so all contradict one. A finite isolated head instead has $y^2 > .0557$, already too much beside $q_{13}(.5965)$. Only isolated dotted heads remain under weak five.

On the .5965 reference the full weak-fourteen accounting is

own	deficit lower	weight	net gain $>$
5	.387329	.09	.06160482
7	.2960209	.09	.043253262
8	.24272051	.09	.0336591918
≥ 9 finite	.1465351	.099	.0192
dotted	.1771	.09	.02151

Here $Y_5 \cdot Y_{14} = 2a/15, a/10, 0$ for five, seven/eight, and the last two rows. The first two totals already exceed one. For own eight add an initial-I graft capped at .08. Its unclipped gain is $> .015$ by weighted Cauchy. If clipped, $t > .5965 + 1/15$; since $t(.11) < .65$, we have $s < .11$ and

$$s^2 + .16(t - .5965) - (5/6)(.08)^2 > (s - .2)^2 + .00690666 > .015.$$

Its residual is orthogonal to both feet because P_1, P_6 are retained. The full P_6 price $< .087582 < .08806$ includes the extra .013334. Thus weak fourteen leaves only the last two rows, each with gain $> .0192$, P_1 price $< .070004$, P_{13} price $< .141422$, no P_6 price and a residual supported only on fourteen/fifteen.

Dotted heads on the retained weak-five reference. For a pure dotted head put $u = g^{-2}$, $h_0 = u/[4(2-u)]$, $s_0 = s(h_0)$. The actual native gap gives $h \geq h_0$. The joint point of Lemma J.7 supplies

$$ax_1 + v_0 y \geq F = 1 - d, \quad a^2 = 1 - u/2, \quad v_0^2 = \frac{u}{8 - 3u}, \quad d^2 = u/2 - v_0^2.$$

Besides its stated all-h domain $u \geq .2$, it is valid for $.1 \leq u \leq .2, h \leq .2$. Indeed $u(8-3u) \geq .77 > 2/3$ and $h/(1+h) \leq 1/6$, so $D_0 = u/2 - v_0\sqrt{h/(1+h)} > 0$. The actual construction is

$$p + \rho c + \sigma c_I = D_0^{-1}(p - ae_1 - v_0 e), \quad \rho = \frac{a}{D_0\sqrt{2}g} > 0, \quad \sigma = \frac{v_0}{D_0\sqrt{1+h}} > 0.$$

The cap product is zero, the c_I product has sign $v_0 - \sqrt{h/(1+h)} \leq 0$, old n_1, e products are negative and every unknown-wall product is negative. Both actual D forks therefore remain satisfied, for every mutual head/I label.

Put $X = (F - v_0 s_0)/a$. Simplification gives

$$X = \sqrt{\frac{1 - \sqrt{3u/8}}{1 + \sqrt{3u/8}}}, \quad (1 - X^2)(1 + s_0^2) = 4\sqrt{3} s_0 X, \qquad X + 2s_0 > .91.$$

For completeness, $0 < s_0 < \sqrt{5} - 2 < .237$. At $X_0 = .91 - 2s_0 > 0$, the left-minus-right expression is greater than $.1719 - 2.66467 s_0 + 10.0283 s_0^2 + 3.64 s_0^3 - 4 s_0^4$. Writing $w = s_0 - .13$, this is exactly

$.00182581 + .092084w + 11.0423w^2 + 1.56w^3 - 4w^4$, bounded below by $.00182581 - .092084|w| + 10.7719w^2 > 0$. The discriminant is negative. The original expression decreases strictly with positive $X < 1$, proving the full barrier without sampling.

Let W_0 be the reference plus $.05Y_5$ and any remaining fourteen foot. Its first coordinate is still $a_0 = .5965$, and the feet are orthogonal to P_1, P_6. Set $\delta = \min((X - a_0)_+, .1)$, $\kappa = 6\delta/5$. The joint paid-reference lemma uses nonnegative multipliers $\eta = \kappa/a$, $\tau = s_0 - \eta v_0$ and gives gain

$$s_0^2 + 2\kappa(X - a_0) - (5/6)\kappa^2.$$

Here $s_0/v_0 > 1/2$, $a \geq 1/\sqrt{2}$, $\kappa \leq .12$ ensure $\tau \geq 0$. The P_6 price $\leq .02$ makes the full cost $< .087883 < .08806$; the other prices still fit. If unclipped, the barrier and weighted Cauchy give gain $> (.91 - .5965)^2/(4 + 5/6) > .02033425$. If clipped, $X > .6965 > X(1/3)$ gives $s_0 < .111$; the gain exceeds $(s_0 - .24)^2 + .00564 > .022281$. For $u \leq .1$, the actual head edge gives $x_1 > .7745$, and an initial graft without an I charge gains $> .03072$. For $.1 \leq u \leq .2, h > .2$, I alone costs $> .0455$. Thus all pure dotted parameters are covered. The least totals are

$$.9703299945 + .00953 + .02033425 > 1.0001942,$$
$$.9521324945 + .00953 + .0192 + .02033425 > 1.0011967.$$

For a positive foreign contact put $Q = \beta^2 \geq 1/4$, $v = (1 - Q)/g^2 \in (0, .75)$. Actual acuteness gives $h \geq Qv/[2(1-Q) - v] \geq v/(6 - 4v)$. The positive core $c + \beta e + ((1-Q)/g)n_1$ clears the whole old I, with

$$x \cdot Z \geq H(v) = \sqrt{1 - v/2} - \sqrt{v(1-v)/2}, \quad Z = (1 - v/2)e_1 + (v/2)e_2.$$

Its projection is $(1 - 3v/5)P_1 + (v/10)P_6$, residual norm $v^2/5$. The full retained initial cross is $W_0 \cdot Z \leq .5965 - .14759v$: use $b_1 < .284332, .05(Y_5)_2 < .016971$. The remaining fourteen foot has zero cross. Pay $\nu = .18$ for $v \leq .2$, $.21$ for $.2 \leq v \leq .4$, and nothing thereafter. The gain is $2\nu[H - .5965 + .14759v] - \nu^2 v^2/5$. Adding I once gives

v interval	$H >$	I energy $>$	total gain $>$
$(0, .2]$	.6658	0	.0246888
$[.2, .25]$	.6292	0	.02557275
$[.25, .3]$	.5979	.0121	.02739115
$[.3, .35]$	.5709	.01515	.02191389
$[.35, .375]$	.5590	.0182	.0229054175
$[.375, .4]$	.5480	.0200	.021464225
$[.4, .75)$	unused	.0217	.0217

Positive squares at right endpoints and monotonicity on $(0, .5)$ prove the H bounds. The left endpoints in the lower bound for h give the I bounds. The largest full P_1 price is $< .052326 + .070004 + .1848 = .30713 < .312168$; the P_6 price is $< .067883 + .0084 < .08806$. The terminal price is unchanged. The uniform gain $> .0214$ contradicts one in both fourteen branches. Every weak-five head is excluded. Five is therefore strong.

The pure-helper coverage through fourteen is now available. If centre fifteen were strong, its reference would have $q_{15} > .9240775$, first-five mean $< .361$ and P_5 coefficient $> .146$. The local head-plus-I gain would give > 1.0007275. Thus centre fifteen is weak.

The complete centre-fifteen list. Deleting fifteen retains actual $A_{14}+2A_1+I$. Its cap has form

$$c = Ap - g\Omega_{15} - \eta\Omega_j - \delta\Omega_{16} - \epsilon\Omega_{17} - \beta e.$$

There is at most one left contact. Put $Q=\beta^2$; the exact data are

$$q_L = 2\eta^2 j(15-j)/15, \quad \Delta = 1-q_L-\delta^2-\epsilon^2-Q>0, \quad \sigma = g+\eta j/15+(\delta+\epsilon)/2, \quad C_{15}=\sqrt{30}\sigma/\sqrt{\Delta}.$$

The sufficient strong ratio is $< T(11/2) < 1.554$. With no left contact, finite classification permits isolation, one arbitrary finite fork label, or pairs $(3,3),(3,4),(3,5)$. Without forks the left possibilities are endpoint three/four at one/fourteen or next-to-end three at two/thirteen. With forks only left endpoint-three with one fork-three/four or both forks-three is finite. Two left contacts make a cycle; other branches or internal multiple bonds are absent from the full finite classification at this rank. Positive I contacts in a single-fork row are precisely $(m,I)=(3,3),(3,4),(3,5),(4,3),(5,3)$; with two forks only $(3,3,I3)$ is possible. Left endpoint-three may carry $I3/I4$, and endpoint-three/fork-three may carry $I3$; other left bridges are sealed. We retain the whole foreign alphabet.

For isolation, own zero has negative time; three/four/six/parallel have the actual star/path/null principals. On $0<u<1$, the cleared core has pressure $\sqrt{30-u}-\sqrt{u(1-u)}$, loss $<.6$. At $u=1$ it is a forbidden feasible future null. On $u>1$, the actual finite projection gives $\sqrt{29}-\sqrt{u-1}$. Foreign cost forces $u<75/64$; pure native at least seven has $u<100/81$, with losses $<.515,.6$. Pure own five uses the actual left-H_4 projection, whose demand is $>5.3/1.04>\sqrt{30}-.61$. Every isolated centre is strong.

For left-only own zero, far-three needs $1/2<Q<8/15$, containing no native foreign cost. Far-next-three and far-four have zero time at $Q=0$; their admissible nonzero costs are too large. The near three/next-three/four rows have actual affine D_5, D_6, B_4 principals. With a simple-three left contact, own three has its actual all-three cycle. At the far endpoint-three, own four/six and parallel have their actual star/path/null dispositions. Every other positive near/next/end-four row is directly strong. In particular, an end-four contact at either end has ratio at least $\sqrt{15}/2+1/\sqrt{30}>2.11$, even at own three. A positive-foreign far-three has ratio $>(.809+1/30)/\sqrt{17/60}>1.57$. For left plus fork-three, $\Delta\le 17/60$: own zero/three has a cycle/triangle; larger owns are strong. With fork-four or both forks-three, $\Delta\le 1/30$ and strength holds even at own zero. Only the pure far helper can remain among left rows.

For own zero with one fork, fork labels three/four/six have the actual affine E_8, B_4 and $6,3$ principals. For finite fork cosine $b\ge\cos(\pi/7)>.9$, whiteness forces $Q=0$. Set $u=b^{-2}\in(1,100/81)$. The actual positive future core $c+b^{-1}n_f+2(b^{-1}-b)n_{15}$, divided by $b/\sqrt{2}$, has signed displacement $(1^{14},3-2u,u-1,\pm(1-u))$, with

$$G=\sqrt{17-2u}-\sqrt{-8+14u-6u^2}, \quad R=25-16u+6u^2.$$

Both fork signs are retained. Complete interval bounds are

u	$G>$	$R\le$
$[1,1.1]$	3.472	15
$[1.1,1.15]$	3.4277	14.66
$[1.15,1.2]$	3.41265	14.535
$[1.2,100/81]$	3.4119	14.44

Use the maximum second radical $1/\sqrt{6}$, positive squares and decreasing R. Pay .14/.13 at actual terminal thirteen/ fourteen, with residual norm $\le R-K$. Adding $>.07665$ makes all eight totals exceed 1.00135.

For fork five put $b = \phi/2$, $\alpha = b/\sqrt{2}$, $\delta_5 = 1 - 3b^2/2$. The actual finite H_4 consists of cap, contacted fork, centre and other fork. Its positive correction is $b(3n_f/2 + n_{15} + n_{\text{other}}/2)$. The corrected vector is spacelike; the point is the normalized positive finite projection of p. Its pure displacement is αP_{14}. For $Q = 0$, demand > 3.49, norm14, and the local head gain close both references. The genuinely permitted $Q = 1/4$ retains I in this same point:

$$x \cdot P_{14} + dy \geq G > 3.61, \qquad d = 1/(2\alpha) < .875, \quad d^2 < .765, \quad G = \frac{\sqrt{5 + 6\phi}}{1 + \sqrt{(13 - 8\phi)/(18 - 2\phi)}}.$$

At $K = 13, 14$, weights $.14, .13$ give bare mixed gains $> .0874216, .0649831$. Every dotted/foreign head has pure gain $> .04$, and a nonisolated head has $> .0994$, already enough. A pure finite head has pure gain $> .0186$ and $s > .236$. Append only the shifted I gain $(s - \mu d)^2$, exceeding $.1135^2, .12225^2$. This replaces the existing coefficient μd by s, using a nonnegative additional multiplier; it does not add a second y^2. The least totals exceed $1.00754385, 1.0056781625$. Head and pure tail residuals are orthogonal, and the displayed mixed norm includes the I coordinate.

For positive own with one fork, own-three/fork-three is an affine triangle. Own-three/fork-at-least-six, or fork four/five with foreign contact, is directly strong. With fork four/five and $Q = 0$, the same cap gives the actual opposite of the contacted fork. The positive finite D_{17} replacement is future feasible against every original wall; the omitted fork is strict. Its actual cosh is $15/2 + (17/2)b > 13$. Thus the signed direction $P_{16} \pm e_{17}$, of norm17, has demand $> \sqrt{102/7} > 3.817$. Either funded reference and the local head gain close it. Own four/six/parallel have the actual star/path/ null principals. For larger owns, every fork label at least four is directly strong.

Only fork-three remains. Own five with $Q \geq 1/2$ is strong. At $Q = 0, 1/4$, use the actual left-H_4 point, or respectively the cleared future core

$$c + g(3n_{15}/2 + n_{14} + n_{13}/2) + e/2, \qquad g = \phi/2.$$

It gives a pure displacement flat through twelve, norm < 15, demand > 3.52. Weights $.15/.14$ at actual strong eleven/ twelve, plus $> .07665$, give totals $> 1.0240725, 1.0229116$. For every larger finite/dotted own, foreign contact is directly strong, as is $g \geq 6/5$. On the full remaining $Q = 0, .9 < g < 6/5$ interval use the actual core

$$\xi = c + g^{-1}n_{15} + (2g)^{-1}n_{14}, \quad k = 1 - 3/(4g^2) > 0, \quad \alpha = \sqrt{2}(g + 1/4).$$

Its cap and fourteen products vanish; its fifteen and both fork products are negative. Its square is $-k$, time positive, and every unknown-wall product is nonpositive by positive actual-wall coefficients. With $A^2 = 15\alpha^2 - 3/4$, $A/\alpha > 3.836$, $k/\alpha^2 \leq 575/5046 < .34^2$, the demand is $G > 3.496 > 3.49$, full norm < 15, flat through thirteen. The quotient derivative reduces to $2g^3 - 3g - 3/8 < 0$ throughout. At thirteen pay .14, residual norm < 2. At fourteen the fourteenth coordinate is $1 - a$, $a = [4g(g + 1/4)]^{-1} < 1/4$. Its projection is $(1 - a/14)P_{14}$, reference pairing $\leq f_{14}$, residual norm $< 1 + 2a < 1.5$. Weight .13 is affordable at that actual terminal; the least total is > 1.0051616. No internal terminal coefficient is assumed.

For two forks the finite pairs are $(3, 3), (3, 4), (3, 5)$. Own three has a triangle; owns at least four are directly strong. Own-zero pair-three-three has the actual affine square, including its foreign case. The other pairs are foreign-sealed. Pair-three-four gives an actual reciprocal finite B_{17} endpoint, with cosh $15/2 + (17/4)\sqrt{2} > 13$, hence the same signed demand above. Pair-three-five uses the actual finite H_3 correction $n_{f_1}/2 + (\phi/2)n_{f_2}$. Its pure displacement is αP_{15}, $\alpha = (1 + \phi)/(2\sqrt{2})$, and the positive finite residual satisfies $q = \delta_H/\alpha^2 = 2\phi^{-6} < 1/8$. Its pressure $\sqrt{15 - q}/[1 + \sqrt{q/15}] > 3.53$, norm15, closes either funded reference with the local head gain. This exhausts the list.

The final pure helper and the actual-wall interface. The weak centre must therefore be the pure far-three helper at fifteen, $A^2 = 30g^2 + 2g - 1/2$. Nonadjacency with fourteen follows from the same actual prefix Schur calculation, retaining both positive fork blocks. Hence fourteen is strong, $q_{14} > .90715$. Nonisolated and finite isolated heads already contradict this reference by their local gains. For dotted heads use Lemma J.11, checking its local geometric interface as follows.

Its finite/dotted helper vectors are $g\sqrt{2}P_{15} - gn_{15} + H_k$ and $g\sqrt{2}P_{15} - g^{-1}n_{15} + H_1$. They use only actual long n_{15}, n_1, n_2; P_{16} in their coordinates is a spatial sum, not a nonexistent long D dual. Both actual D forks have nonpositive product with every positive correction. Their Gram data, fresh times, initial crosses and individual P_5/P_{14} prices are literally those of the local kernel, without requiring a B short normal or a sign for x_{17}.

For pure dotted head $u \le .2$, the kernel bypasses the tail payment and gives gain $> .11088$. For $.2 \le u < 1$, its net tail gain is $> .0421$, including the whole initial residual, and its head-plus-I gain is $> .0515$. Every positive foreign head contact instead gives $> .0521$, with the adverse initial cross and the one I cost included. The least original total is

$$.90715 + .0421 + .0515 = 1.00075 > 1.$$

The P_5 debits $< .09873, .09273$ fit $> .146$, and P_{14} debit $< .1273$ fits $> .1361$. Every actual centre, head, finite foreign label and own range is covered on the same original vector. This excludes $D_{17} + I$ in dimension eighteen, without arithmeticity or an ordinary rank-one debit.

K The independent higher-dimensional tail

The exact-eighteen argument is separate from the exclusion of higher ranks. We start this second argument at the published ceiling twenty-nine. Every remaining dimension is handled by its actual finite forest; no Coxeter structure is inherited by an arbitrary face. The common geometry, ordinary and typed costs, actual A-chain pressures and generic prefix are the lemmas already proved above. Here we supply the additional large-component caps and then join the finite forest cases.

The argument separates at a useful point. An A cap is proved by a central curve and a plateau transfer. A B/D cap is proved by a long prefix and the genuinely different short tails. These two geometric arguments apply to whole components with arbitrary finite spectators. They reunite only in the finite rank and energy accounting. The statements below do not replace an actual component by a convenient abstract completion: every added cap is already an actual wall.

K.1 Common large-component caps

Proposition K.1 (Large-component caps)**.** *At a farthest vertex, with arbitrary finite spectators, the following components are impossible:*

$$A_r \ (21 \le r \le 29), \qquad B_r \ (23 \le r \le 29), \qquad D_r \ (24 \le r \le 29).$$

We give their finite continuation and energy proofs below. All references to old directions concern the chosen farthest vertex, and all added walls are already actual.

K.2 Clearing spectators and the finite continuation list

Put $c_0 = 61/100$. For a long index, write $D_i = \omega_i^2$, $q_i = \sqrt{D_i}$ and $h_i = x \cdot \omega_i$, and call it weak if $h_i \le q_i - c_0$. An actual edge with

$$\frac{\cosh L_i}{q_i} > T(q_i), \qquad T(q) = \frac{1 + (1 - c_0/q)^2}{q[1 - (1 - c_0/q)^2]} = \frac{1}{c_0} - \frac{q - c_0}{q(2q - c_0)},$$

is strong. We use $T(q) < 100/61$ and, on $5/2 \le q \le 31/8$, $T(q) < 761/500$. For the latter, clearing denominators gives the positive concave quadratic $-715800q^2 + 3268319q - 1860500$; check its two endpoints.

First consider an isolated own contact γ with full foreign cost Q. Write $v = \gamma\omega_i + v_S$, $u = (1-Q)/\gamma^2 > 0$, so $A^2 = \gamma^2(D_i - u)$. If $u < 1$, the positive actual combination

$$\xi = c + v_S + \frac{1-Q}{\gamma} n_i$$

has products zero with c, S, product $\gamma(u-1) < 0$ with n_i, and square $-\gamma^2 u(1-u)$. It is future and feasible, giving the pure component pressure

$$h_i > \sqrt{D_i - u} - \sqrt{u(1-u)}. \tag{62}$$

At $u = 1$ the same nonzero feasible vector is null, impossible. For $u > 1$, the actual positive set $S + n_i + c$ has residual $\gamma^2(u-1)$; the finite-projection formula in (4) and $x \cdot n_i > 0$ give

$$h_i > \sqrt{D_i - u} - \sqrt{u-1}. \tag{63}$$

In A the sharper form is $h_i > \sqrt{D_i - 1} - \sqrt{u-1}$: with $B = \sqrt{D_i - 1}$, $t = \sqrt{u-1}$ the exact finite pressure is $\sqrt{B^2 - t^2}/(1 + t/B) > B - t$. No normalized projected mirror is asserted native.

The own labels $3, 4, 6$ and parallel contact are excluded by the local affine fork/star, $(6,3)$ path, and null sum. Thus $\gamma \ge \varphi/2$. If $Q > 0$, $u \le 3/\varphi^2 = 6 - 3\varphi < 7/6$. For A, $D_i \ge 85/11$, $q_i > 11/4$, and $\sqrt{D_i - 1} > 5/2$. Formula (63) has loss $< 4/21 + 5/12 = 17/28 < c_0$, since $u - 1 \le 5 - 3\varphi < 73/500 < (5/12)^2$. Formula (62) has loss at most $2/21 + \sqrt{457}/42 < 127/210 < c_0$. For B/D, $D_i \ge 10$; the respective losses are $< 1/5 + 2/5$ and $u/6 + \sqrt{u(1-u)} \le (1 + \sqrt{37})/12 < 3/5$. Thus every foreign isolated cap is strong.

At $Q = 0$ the isolated estimates are also strong, uniformly for $D_i \ge 85/11$. For own 5, the actual old H_4 segment has $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$ and, by concavity of the old heights, pressure at most γh_i after its correction. Its loss is $< 25/76 + 5/24 < c_0$; here $q_i + \sqrt{D_i - 25/16} > 19/4$ and $\gamma > 4/5$. For own 7, the adjacent negative A_2 core gives loss $< 400/1701 + 9/25 < c_0$, using $.9 < \gamma < 10/11$. For finite own $m \ge 8$, the old singleton projection gives $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$ and loss $< 4/21 + 5/12 < c_0$. For dotted own contact, $c + n_i/\gamma$ gives (62) with $u = \gamma^{-2}$, and the same $127/210$ bound. One left H_4 segment suffices: its displacement is $\gamma(3\omega_{i+1}/4 + \omega_{i-3}/4)$ and concavity bounds its scalar product by γh_i.

Here is the complete finite continuation table at a central release. A retained left path A_{i-1} admits endpoint 3, endpoint 4, next-to-end 3, or an admissible E_6, E_7, E_8 position 3. Their Schur residuals are

$$(i+1)/(2i), \quad 1/i, \quad 2/i, \quad 1/4, 1/7, 1/16. \tag{64}$$

For a left 3 at position j from the far end, $\sigma = \gamma + j/(2i)$; for endpoint 4, $\sigma = \gamma + j/(\sqrt{2}i)$. In each case $\cosh L_i = q_i\sigma/\sqrt{\delta - Q}$. The complete right table is

retained right part	finite attachment
$B_s,\ s \ge 4$	long-leaf 3
B_3	long-leaf 3 or short-leaf 3 (F_4)
B_2	3 at either end
$D_s,\ s \ge 5$	long-leaf 3; fork 3 only for $s = 5, 6, 7$
D_4	3 at any leaf
$D_3 = A_3$	middle 3, end $3, 4, 5$.

Two contacts to a retained component create a cycle. A completion joining both long retained pieces has connected rank at least 21, so finite classification forces the corresponding simple endpoint $A/B/D$ bridge. An additional foreign branch at its internal new node is not finite. The exceptional fork extensions in the table are retained and estimated; an E_7 piece is not declared sealed merely because it is exceptional.

Zero own contact needs a separate check. Internal 3, internal 4, right long 3, and two-piece bridges give the usual affine fork, star or cycle; a right D-fork contact gives affine E_8. A zero-own far 3@1 in B/D satisfies $A^2 = Q - 1/2 < 1/(2i) \le 1/10$; any foreign component then has derivative square at least $1/4 > A^2$, contradicting Proposition 2.4. With $Q = 0$ its height is negative. For A, a zero-own far 3 has height $< 1/(2m)$, a next-to-end 3 has $Q < 2/m \le 2/5$ and height $Q - 2/N < Q$, and an endpoint 4 has $Q < 1/m \le 1/5$ and negative height. A positive Q in the first two cases again forces an orthogonal closure. Interior A contacts retain affine E_8 arms $(1, 2, 5)$ or E_7 arms $(1, 3, 3)$. This proves positivity of the own contact without assuming the whole star has no spectators.

For completeness, all other near-position estimates follow directly from (64). Own 3/4/6 is affine except the near 4/simple-3 role. For own $\gamma > 101/125$, near simple 3 has $\delta \le 3/5$, $\sigma \ge \gamma + 2/5$; the next-to-end row has $\delta = 2/i$, $\sigma \ge \gamma + 1/i$; the E rows have $\delta \le 1/4$. For B/D, the first row exceeds 100/61 from $i = 8$ onward, and the next-to-end row from $i = 7$ onward. At $i = 5$ use $T(q) < 38/25$ on $[5/2, 10/3]$, and at $i = 6, 7$ use $T(q) < 8/5$ on $[3, 4]$; these are positive concave quadratics after clearing denominators. White 4 is stronger, except far 4/own 3, which retains affine $(3, 4, 3, 3)$. The allowed B near 4/own 3 has $\cosh L_i = (i - 1) + i/\sqrt{2}$ and is strong for $i \ge 5$; in D an own 4 with no right contact is already affine on the entire retained right fork. The A near-4 role is kept below.

If a nonisolated far 3 packet has $Q > 0$, the actual data in B/D are

$$\cosh L_i = \frac{2i\gamma + 1}{\sqrt{i + 1 - 2iQ}}, \quad \frac{\cosh L_i}{q_i} > R_0(i) := \frac{(809/500)i + 1}{\sqrt{i(i+2)}}.$$

R_0 increases for $i \ge 5$, with derivative sign $(309/500)i - 1$; T also increases for $q \ge \sqrt{10}$. The three exact comparison rows are

i	q upper	$R_0(i)$ lower	$T(q)$ upper
5–8	4	1.536	1.530
9–12	5	1.563	1.550
13–29	7.62	1.577	1.577.

The lower squared margins are $.052740, .321913, .546001$; the ($T$ numerator, T denominator) pairs are $(27.4921, 18.0316)$, $(44.2721, 28.6395)$, $(107.2045, 68.003166)$, proving each upper comparison by multiplication. In A, a helper-side parameter $m \ge 5$ instead gives ratio $> (2fm+1)/\sqrt{m(m+2)} \ge 227/(25\sqrt{35}) > 761/500$, $f = 101/125$; the derivative sign is $(2f - 1)m - 1 > 0$, with final squared margin 342365. The foreign near-4 role has ratio squared $> 121/35$. Thus every remaining weak packet has $Q = 0$.

K.3 The A_{21} certificate and its plateau transfer

Put $N = r + 1 \in [22, 30]$. Then $D_i = 2i(N - i)/N$, $C_{ij} = 2i(N - j)/N$ for $i < j$, and $\|x_A\|^2 = \frac{1}{2}\sum_{i=0}^{N-1}(h_{i+1} - h_i)^2$, with $h_0 = h_N = 0$ strictly concave. For $5 \le i \le N - 5$, $85/11 \le D_i \le 15$. The preceding inventory leaves a near-4/simple-3 packet or

$$v_i = \gamma_i\omega_i + \omega_{e_i}/2, \quad e_i \in \{1, N - 1\}, \quad Q = 0, \quad f := 101/125 < \gamma_i < g := 27/25, \quad A_i > 2. \tag{65}$$

Allowed own contacts are 5, finite $m \ge 7$, or dotted. The dotted upper bound follows because at $\gamma = g$, $\sigma^2/\delta \ge 4g - 2 = 58/25 > (761/500)^2$; the difference numerator is $[2m(g - 1) - 1]^2$. The

near-4 edge ratio increases with m and at $m = 8$ is $2\sqrt{2}/3+7/12 > 761/500$, so $m \ge 8$ is strong. At $m = 7$, $N = 22, 23$ use $D_7 \le 224/23 < (25/8)^2$ and $T(q) < 3/2$ on $[3, 25/8]$, whose cleared quadratic is $-3400q^2 + 13237q - 7442 > 0$. Every weak near-4 therefore has a mate at index at most 5 for $N = 22, 23$, at most 6 otherwise, or the reversal. That mate is actual by its positive B replacement. Its data are $\delta = 1/j$ and $\cosh L_j = (1/2 + (j-1)/(j\sqrt{2}))\sqrt{jD_j}$. By reversal take $6 \le j \le N-4$; then $jD_j \ge 72(N-6)/N \ge 216/5$ and the coefficient is $> 13/12$. Thus $\cosh^2 L_j > 507/10 > 49$, giving normalized height $> \sqrt{3}/2 > 43/50$.

We next exclude adjacent weak literal packets. Write $j = i+1$, $A = \gamma_i$, $B = \gamma_j$, and $R = c_i \cdot c_j$. Weighted AM–GM of their time coefficients gives

$$\begin{aligned}
\text{same helper}: \quad & R \ge -AB(1-1/N) \ \pm (A-B)/N \\
& \qquad + (N-1)/(2N) + (N+1)(A/B + B/A)/(4N), \\
\text{outward helpers}: \quad & R \ge 1/2 - AB + (A-1)(B-1)/N, \\
\text{reversed helpers}: \quad & R \ge 1/2 - AB + (A+1)(B+1)/N.
\end{aligned}$$

The first is $> 1 - g^2 > -1/2$; the others are $> 1/2 - g^2 - (1-f)(g-1)/22 > -7/10 > -1/\sqrt{2}$. Hence the native gap gives $R = 0$ for one helper and $R \in \{0, -1/2\}$ for opposite helpers. For outward helpers, $R = 0$ is impossible because $\eta_\ell = \delta_\ell/\sigma_\ell^2 < 4/5$ and

$$(D_i - 4/5)(D_j - 4/5) - C_{ij}^2 = (2/5)C_{ij} - 24/25 + 8/(5N) > 0.$$

For reversed helpers, put $l = i$, $s = N - j$, $l + s = N - 1$. The same determinant, multiplied by N, is

$$4lsA^2B^2 - (l+1)(s+1)(A^2+B^2) - 2AB - (l+1)A - (s+1)B + (N+2)/4.$$

It increases in $A, B \ge 4/5$: the derivative uses $4lsB^2 - (l+1)(s+1) \ge (39/25)ls - N > 0$. If either contact is at least .9, concavity in l reduces the minimum to $l = 5$ or $s = 5$; at $(4/5, 9/10)$ it is at least $(509N - 9974)/500 \ge 306/125 > 0$. The last possibility has both contacts $\varphi/2$. In raw norm-two coordinates their difference squared is $\varphi^2(N-1)/N < 4$ for the same helper, so their wall product is positive. For opposite helpers, $R = 0$ would have rational coefficient $N(N+6)$ or $-N^2$ in the expansion of $N^2[(v_i \cdot v_j)^2 - A_i^2 A_j^2]$, according to orientation. Neither vanishes. This proves $R = -1/2$ for opposite helpers.

The retained two-cap Schur matrix is $H = I - t\mathbf{1}\mathbf{1}^T$, $t = (l-1)/(2l)$, for one helper; for opposite helpers it has diagonal residuals $\delta_l, \delta_s \in (1/2, 3/5]$ and off-diagonal $-1/2$. It is positive with positive inverse. Put $\lambda = H^{-1}(A_i, A_j)^T > 0$ and $\mathcal{D} = A_i\lambda_i + A_j\lambda_j$. Its actual mixed point has positive displacement $Z = W_i\omega_i + W_j\omega_j$, and

$$Z^2 = \mathcal{D}(1+\mathcal{D}), \qquad x \cdot Z \ge 1 + \mathcal{D} - \sqrt{1+\mathcal{D}}.$$

For one helper, $W > (4/5)l(A_i + A_j)$ and $\mathcal{D} = [l(A_i+A_j)^2 + (A_i - A_j)^2]/2$; thus $\sqrt{1+\mathcal{D}}/W < 7/16$. For opposite helpers, $5/6 < \lambda_i/\lambda_j < 6/5$, $\mathcal{D}/(\lambda_i + \lambda_j)^2 \le 3/55$, $\lambda_i + \lambda_j > 40$, and $W > 32$; the corresponding bound is $< \sqrt{15/176 + 1/1024} < 3/10$. In either case

$$W_i q_i + W_j q_j - \|Z\| < \frac{W}{2(\sqrt{85/11} + \sqrt{80/11})} < 3W/32.$$

The weighted mean loss of two weak indices is therefore below $17/32$ or $63/160$, both below .61. Adjacency is impossible. Every weak near-4 can now be removed: its mate has normalized height $> 43/50$; an opposite mate or a strong anchor among $N-6, N-5$ has height $> 39/50$ and correlation at most $1/3$. For $N \ge 24$ the latter follows from $6/(N-6) \le 1/3$; for $N = 22, 23$ the mate index is

at most five and the opposite anchor at least 16, giving correlation at most $30/\sqrt{8160} < 1/3$. Their sum contradicts $(41/25)^2 > 8/3$. Thus all weak central indices have (65) and are nonadjacent.

It remains to exclude weak endpoint 5. We give the complete fixed $N = 22$ certificate before its exact lift. Nonadjacency and concavity give strong 6, a middle floor $h_{11} \ge b$, and a strong 16 or 17, where

$$a = \sqrt{96/11} - .61, \quad b = \sqrt{120/11} - .61, \quad d = \sqrt{85/11} - .61,$$
$$2.344 < a < 2.3443, \qquad 2.6928 < b < 2.693, \qquad 2.169 < d < 2.17.$$

If 16 is strong, the minimum reference x_0 through heights a, b, a at $6, 11, 16$ has $E = x_0^2 = a^2/6 + (b-a)^2/5 > 47/50$ and positive coefficients $c_6 = c_{16} = (11a - 6b)/60 > 4/25$, $c_{11} = (b-a)/5 > 0$. Put $K = 5a/6$, $L = a/12$. For a far helper at 21, each pressure vector in the table has projection $(5\gamma/6)\omega_6 + \omega_{16}/12$, scalar product $\gamma K + L$ with x_0, and height $A^2 = (85/11)\gamma^2 + (5/11)\gamma - 23/44$.

own	pure pressure vector U	target T
5	$\gamma(\omega_4 + \omega_6)/2 + \omega_{19}/6$	$H(A, 2/3 - \gamma^2)$
7, 8	$\gamma(\omega_4 + \omega_6)/2 + \omega_{20}/4$	$A - \sqrt{\gamma^2 - 3/4}$
≥ 9	$\gamma(\omega_4 + \omega_6)/2 + \omega_{21}/2$	$H(A, 1 - \gamma^2)$
dotted	$(\gamma - \gamma^{-1})\omega_5 + (\omega_4 + \omega_6)/(2\gamma) + \omega_{21}/2$	$A - \sqrt{1 - \gamma^{-2}}$

Here $x \cdot U \ge T$, $S = \|U - \operatorname{proj} U\|^2$, and $\Delta = T - \gamma K - L$. The residual variances in row order are $2\gamma^2/3 + 1/12$, $2\gamma^2/3 + 1/6$, $2\gamma^2/3 + 5/12$, and $5\gamma^2/3 + \gamma^{-2} - 19/12$. The exact bounds and chosen positive multipliers are

own	5	7	8	≥ 9	dotted
$\Delta >$	3/10	1/4	9/40	13/50	3/10
$S <$	3/4	3/4	3/4	13/12	5/4
μ	1/5	1/5	1/5	24/125	1/6.

The actual sets are respectively $J = \{5, 21, 20\}$, $\{5, 21\}$, $\{5\}$, and the dotted pair; they use the finite-projection formula in (4) or the future-core formula in (4). The following seals cover the full native ranges, not samples. For 5, $\gamma \in (.809, .81)$ gives $A > 2.21$, $\sqrt{2/3 - \gamma^2} < 1/9$, $\gamma K + L < 1.78$. For 7, $\gamma \in (.9, .902)$ gives $A > 2.47$, radical $< .253$, and projection < 1.96. For 8, $\gamma \in (.923, .9239)$ gives $A > 2.5475$, radical $< .322$, projection < 2.0003, so excess $> .2252$. For finite $m \ge 9$, $\gamma > .939$ by the cubic $8\gamma^3 - 6\gamma - 1$; its deficit increases, since $A' > 11/4$, $H_A > 20/23$, $H_\delta < 0$, and $K < 2$. At .939, $A > 2.5917$, $\sqrt{A^2 + \delta} > 2.6144$, $\sqrt{\delta} < .344$, giving $H > 2.29$ and projection < 2.03. The dotted deficit decreases on $(1, 27/25)$ because $A' < 3$, $K > 1.9$, and the radical derivative exceeds $15/8$. At $27/25$, $A > 2.99$, radical $< .38$, projection < 2.306. Its variance increases to a value below $5/4$.

The vector $x_0 + \mu(U - \operatorname{proj} U)$ has all coefficients nonnegative: the largest old-6 removals are $.154, 4/25, 3/20$, and the old-16 removal is at most $1/60$. Its Cauchy margin is $(E + \mu\Delta)^2 - E - \mu^2 S > 0$, increasing in E. The two tight rows have exact lower margins

$$(47/50 + 9/200)^2 - 47/50 - 3/100 = 9/40000,$$
$$(47/50 + (24/125)(13/50))^2 - 47/50 - (24/125)^2(13/12) = 219/39062500.$$

If 17 is strong, the reference through a, b, d has energy $a^2/12 + (b-a)^2/10 + (b-d)^2/12 + d^2/10 > 24/25$. Its old coefficients exceed $4/25$; replace the helper projection by $\omega_{17}/10$. The deficit decreases by less than $1/40$ and variance decreases by $1/60$. The same multipliers still give positive margins (the tight finite-8 margin is already .01). For a helper at 1, reverse the helper term in every vector. The projection joins the ω_6 coefficient; its value on x_0 is unchanged, while A^2 gains $12\gamma/11$. Since

the two time coefficients sum to less than $25/4$, $A_{\text{left}} - A_{\text{right}} > 192/1375$. Targets increase by more than .1, using $H_A > 4/5$. The variance gains $\gamma/3$. All finite rows then have $\Delta > 13/40, S < 3/2$ and work with $\mu = 1/6$; dotted rows have $\Delta > 2/5, S < 5/3$ and work with $\mu = 4/25$. Their old-6 removals are at most $11/72, 59/375 < 4/25$. This treats both helpers and both strong anchors and excludes weak 5.

For $N > 22$, express each reference or dual vector by its potential $t_i = W \cdot \omega_i$ and insert $N - 22$ copies of t_{11} between its two old halves. Its norm $\frac{1}{2}\sum(t_{i+1} - t_i)^2$ is unchanged. The only split coefficient is the positive peak at 11, replaced by nonnegative drops at $11, N - 11$. Residual potentials are supported in the unchanged first and last six indices, so all projection coefficients and variances above are exactly preserved. Concavity and nonadjacency give the required middle heights at $11, N - 11$; the old $6, 16/17$ floors can only increase. For the two helpers the time coefficients are

$$A_N^2 = 10\gamma^2 - 1/2 - (10\gamma - 1)^2/(2N), \quad A_N^2 = 10\gamma^2 + 2\gamma - 1/2 - (10\gamma + 1)^2/(2N).$$

They increase with N, as do $H(A, \delta)$ and $A - \sqrt{-\delta}$. Every lifted positive certificate therefore keeps its strict margin. Reversal also excludes weak $N - 5$.

Finally the $N = 22$ full central reference has

$$E_{\text{full}} = (\sqrt{85/11} - .61)^2/5 + \sum_{i=5}^{10}(\sqrt{D_{i+1}} - \sqrt{D_i})^2 > 1007/1000.$$

Lower/upper pairs for $\sqrt{D_5}, \ldots, \sqrt{D_{11}}$ are

$$\begin{aligned} &(2.7797, 2.7799), \quad (2.9541, 2.9543), \quad (3.0895, 3.0897), \\ &(3.1908, 3.1910), \quad (3.2612, 3.2614), \quad (3.3028, 3.3030), \\ &\qquad\qquad (3.3166, 3.3167); \end{aligned}$$

squaring them gives lower energy 1.007192468. For $N \geq 23$ lift this same curve with a constant plateau between $11, N-11$; these two actual heights are at least $\sqrt{11} - .61$ even if weak, since their strong neighbors have dual square at least $D_{10}(23) = 260/23 > 11$. For $N \geq 23$, only $6, \ldots, 10$ and their reversals may now be omitted, with at most three weak indices per flank. For $N = 22$, the weak indices may instead be any nonadjacent subset of $6, \ldots, 16$, including the middle index 11; this subset has size at most six. Removing one isolated floor, including that middle floor, loses less than $1/1600$, since $q''(t) = -1/q(t)^3$ and $q > 11/4$ on the original central interval. Segment interiors are disjoint; the surviving potential stays concave with nonnegative duals. Consequently $\|x_A\|^2 > E_{\text{full}} - 6/1600 > 4013/4000 > 1$. This proves the A cap, with all spectator pressure already cleared.

K.4 Classical packets and the two short tails

For B/D put $D_i = 2i$ and $S_i = h_i/\sqrt{2}$. The continuation table in Section K.2 makes every bulk direction strong except

$$v_i = \gamma_i\omega_i + \omega_1/2, \qquad A_i^2 = 2i\gamma_i^2 + 2\gamma_i - 1/2, \qquad Q = 0, \quad 101/125 < \gamma_i < 6/5. \tag{66}$$

The upper bound follows from $\sigma^2/\delta \geq 4(6/5) - 2 = 14/5 > (100/61)^2$ at $\gamma = 6/5$; the difference numerator is $[2i(\gamma - 1) - 1]^2$. For a right long 3 the data are $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$; own $0/3$ is affine and the next contact gives $\sigma/\sqrt{\delta} \geq 1 + 1/\sqrt{2} > 100/61$. For a right fork 3 in D_s, $s = 5, 6, 7$,

$\delta = 1 - s/8 - Q$, $\sigma = \gamma + 1/4$. Own 0/3/4/6 retains affine E_8/cycle/$(3,3,4,3)$ path/$(6,3)$; the next contact gives ratio at least $(\cos(\pi/5) + 1/4)/\sqrt{3/8} > 1.72 > 100/61$. This includes every possible foreign extension of the finite E_6, E_7, E_8 piece. A two-piece white bridge has residual $1/(2i)$ and numerator at least $\gamma + 1/2 + 1/(2i)$, hence is strong; zero own instead gives an affine cycle.

Adjacent packets (66) are impossible. Weighted AM–GM gives

$$c_i \cdot c_{i+1} \geq 1/2 - ab + (a/b + b/a)/4 \geq 1 - ab > -11/25 > -1/2,$$

so their actual product is zero. Their retained Schur matrix is the same $I - t\mathbf{1}\mathbf{1}^T$, $t = (i-1)/(2i)$, as in Section K.3. Its positive actual mixed point has $\sqrt{1 + \mathcal{D}}/W < 7/16$ and $W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2} - \|Z\| < W/(4\sqrt{10})$. The derivation uses $A_i > 2$, $\gamma_i > 4/5$ and $i \geq 5$ only: the same explicit inverse and time-coefficient formulas apply. Their sum is below .61, proving nonadjacency.

The index-5 pressure table and its native scalar seals are given in Appendix E. Weak 5 forces $x_1 > .5965$, strong 6, and one of $16, 17$ strong. The positive prefix interpolation through $1, 6, 16$ then has energy

$$.5965^2 + (2.01815 - .5965)^2/5 + 1.55051^2/10 = 1.00043812051 > 1. \tag{67}$$

Endpoint 17 increases the last term. Thus 5 is strong using bulk indices alone, before either short-tail estimate below.

In B, the bulk reaches $r - 4$. At $i = r - 3$, the extra right B_3 short-end 3 gives F_4, forbidding every white left or foreign extension. Own 0 is affine; otherwise $\delta = 1/4$, $\sigma \geq 1/2 + 1/(2\sqrt{2})$, strong. At $i = r - 2$, the extra short-end 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$. Own 0/4/6 is affine, while own ≥ 5 is strong. Own 3 with $Q > 0$ is strong since $\delta \leq 1/4$. The only new weak role is therefore

$$Q = 0, \qquad v = (\omega_{r-2} + \omega_r)/2. \tag{68}$$

If the short-end contact also joins the left path, the full finite white diagram is B_r and has no foreign extension at c. Its residual $1/(2i)$ and numerator $\geq \gamma + 1/(2i) + 1/(2\sqrt{2})$ are strong even at own 0 for $i \geq 21$. This is a retained zero-own branch, not an assumed affine exclusion. For (68), deleting $r - 1$ and adding the already actual c gives positive A_r, so the same wall actually closes $r - 1$. Its exact cosh is

$$C = \frac{(1 + \sqrt{2})i + 1}{\sqrt{i + 3}}, \qquad i = r - 2.$$

For $21 \leq i \leq 23$, $C/\sqrt{2i+2} > 79/50$, $C > 9$, and $\sqrt{2i+2} < 7$. The first comparison reduces to $984i^2 - 19014i - 17498 > 0$, increasing from $i = 21$. Using $\sqrt{C^2 - 1} > C - 1/C$, its normalized loss is $< 1/(79/50 + 4/63) < .61$. Thus this mate is strong.

In D, the bulk reaches $r - 5$. At $i = r - 4$ the extra fork 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/4$. Its own 0/3/4/6 is affine as above; $Q > 0$ is strong. At $Q = 0$, own finite ≥ 7 or dotted has ratio > 1.61, above $T(\sqrt{2i}) < 1.577$ for $19 \leq i \leq 29$. Only own 5 needs the averaged projection below. At $i = r - 3$ the full right $D_3 = A_3$ list is middle 3, end 3/4/5. Middle 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$ and is affine or strong. End 4 has $\delta = 1/4 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$; own 0 is affine, otherwise it is strong. End 5 is finite H_4, forbidding further white contacts, but own 0 is retained: already $\delta = 1 - 3\varphi^2/8$ and $\sigma \geq \varphi/4$ give ratio $> 100/61$. End 3 has $\delta = 5/8 - Q$, $\sigma = \gamma + 1/4$. A left bridge is finite A_r, with no further foreign extension; its residual $1/8 + 1/(2i)$ and numerator $\geq \gamma + 1/4 + 1/(2i)$ are strong for $i \geq 21$ unless own 0, which is an affine cycle. Without that bridge, own 0/3/4/6 is affine; $Q > 0$ and $\gamma \geq 1$ are strong. Precisely own 5 or finite own ≥ 7, $\gamma < 1$, $Q = 0$, remains.

For own 5 in either short D layer, put $\gamma = \varphi/2$ and $v = \gamma\omega_i + \omega_f/2$ at the contacted fork. The heights are $A^2 = i(\varphi + 5/8) - 1/2$ or $i(\varphi + 5/8) - 5/8$. Select old paths $i, i-1, i-2$ and $i, i+1, i+2$; at the D_3 layer the last node is the other, uncontacted fork. Each actual H_4 correction is $w_\pm = \gamma(3n_i/2 + n_{\text{side}} + n_{\text{second}}/2)$, with $\delta_5 = 1 - 3\gamma^2/2$. The average $Y = v - (w_+ + w_-)/2$ satisfies

$$x \cdot Y > A - \sqrt{\delta_5}, \quad Y^2 = A^2 + 1 - 15\gamma^2/8 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

The cross product is $w_+ \cdot w_- = 3\gamma^2/4$. For $i \geq 19$, $A > 6$, $\|Y\| > 6.4$, $\sqrt{\delta_5} < 1/7$ (use $\varphi > 809/500$), and $\widehat{Y} \cdot u_5 < 3.3602/6.4 < .55$. Thus $x \cdot \widehat{Y} > 41/42 > .97$; strong 5 gives $x \cdot u_5 > .8$. Their sum contradicts $1.77^2 > 3.1$.

For the finite-≥ 7 D_3 role, $i \geq 21$ and $.9 < \gamma < 1$. Now $A^2 = i(2\gamma^2 + \gamma + 1/8) - 5/8$ and the two actual negative A_2 corrections are $w_\pm = \gamma(4n_i/3 + 2n_{\text{side}}/3)$, with norm residual $-d$, $d = 4\gamma^2/3 - 1 > 0$. Their cross product is $8\gamma^2/9$; the average has

$$x \cdot Y \geq A - \sqrt{d}, \quad Y^2 = A^2 + 1 - 14\gamma^2/9 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

At $i = 21$, $Y^2 - 40(\gamma + 1/4)^2 = 4\gamma^2/9 + \gamma + 1/2 > 0$; larger i only increases it. Hence correlation is $< 1/2$. The ratio d/A^2 increases in γ and decreases in i; its derivative numerator is $[i(4\gamma^2 + 13\gamma + 3) - 5\gamma]/3 > 0$. Its maximum is $1/195$, giving $x \cdot \widehat{Y} > 1 - 1/\sqrt{195} > .928$. Strong 5 gives $x \cdot u_5 > .807$, so their sum contradicts $1.735^2 = 3.010225 > 3$. Every correction is positive and feasible against all unknown walls. Thus the weak inventory (66) extends through $r - 3$ in D, with the single B terminal role retained.

We finish the prefix closure explicitly. Let $f_j = \sqrt{j} - .61/\sqrt{2}$ and $E_M = f_5^2/5 + \sum_{j=5}^{M-1}(\sqrt{j+1} - \sqrt{j})^2$. The concave reference has nonnegative duals. The exact seals are $E_{21} > 1.009$, $E_{20} > .99752$, $f_5/5 < .361$: $f_5^2/5 > .6514$ and the increment lower bounds for $j = 5, \ldots, 20$, in units 10^{-5}, are

$$4545, 3846, 3333, 2941, 2631, 2380, 2173, 2000, 1851, 1724, 1612, 1515, 1428, 1351, 1282, 1219.$$

They follow from $(\sqrt{j+1} - \sqrt{j})^2 > 1/(4j + 2)$. One isolated missing floor loses $< 1/3200$ by $|(\sqrt{t})''| < 1/40$; two consecutive missing floors lose $< 1/800$, by the variance of their three segment slopes. Disjoint segment interiors make these losses additive.

For a far weak packet at any $5 \leq i \leq 21$, the actual pressure table in Appendix E, with $q = \sqrt{2i}$, gives $x_1 > \sqrt{2}[T - \gamma(q - .61)] > .424$. For finite own $5, 7, 8, \geq 9$, use $A - \gamma q > .079$ and the four bracket lower bounds $.61(.809) + .079 - .111$, $.61(.900) + .079 - .253$, $.61(.923) + .079 - .323$, $.61(.939) + .079 - .344$; each times $7/5$ exceeds $.424$. For dotted $\gamma \in (1, 6/5)$ the bracket decreases in both γ, q: $A' - q < (q^2 + 2)/(4q^3) < .1$, while the radical derivative exceeds 1. Its endpoint bound is

$$\sqrt{2}[\sqrt{3119/50} - (6/5)\sqrt{42} + .732 - \sqrt{11}/6] > .424,$$

sealed by $7.8981 < \sqrt{3119/50}$, $\sqrt{42} < 6.480741$, $\sqrt{11}/6 < .552771$, and $\sqrt{2} > 1.4142$.

In D_r, $r \geq 24$, index 21 is now in range. If strong, $E_{21} - 8/3200 > 1$. Otherwise 20 is strong and $x_1 > .424$; the reference through 20, with this first-coordinate constraint, has energy

$$E_{20} + \frac{5}{4}(.424 - .361)^2 - 7/3200 > .99752 + .00496125 - .0021875 = 1.00029375 > 1. \tag{69}$$

Its slopes still decrease, so all dual coefficients are nonnegative. For B_r, $r \geq 26$, one of $21, 22$ is strong and the same E_{21} argument applies. At $r = 25$ it still applies through $r - 3$. At $r = 24$, choose strong 21 or 22, or the strong mate 23 of (68). At $r = 23$, a terminal role at 21 likewise supplies strong 22; a far weak 21 instead uses (69). In these terminal cases there is at most one two-hole cluster; the excessive loss $9/3200 + 1/800 < .009$ still leaves energy above one. This proves both classical caps and completes Proposition K.1.

K.5 The same central geometry at ranks nineteen and twenty

Lemma K.2 (The actual $N = 20, 21$ central geometry)**.** *Let $N \in \{20, 21\}$, with an actual A_{N-1} component and arbitrary finite spectators. On $5 \le i \le N-5$, define weak by $x_A.\omega_i \le \sqrt{D_i} - .61$. Before any contrary energy target, its complete weak list is a pure near-four cap $v = \omega_i/\sqrt{2}+\omega_{i-1}/2$, $i = 5, 6$ or its reversal, or a pure far-three helper $v = \gamma\omega_i+\omega_e/2$, $e = 1, N-1$, $101/125 < \gamma < 27/25$, with own five, finite at least seven or dotted. Every weak cap has zero spectator cost. Literal helper weak indices are nonadjacent, for both helper assignments. Each near-four has an actual positive finite mate at four/five or its reversal, with normalized height $> 43/50$.*

Proof. The central interval is still $5 \le i \le N-5$, but its smallest dual square is now $15/2$, rather than $85/11$. We check the changes to the inventory before using the preceding two-cap argument. The exact data are

$$D_i \ge 15/2, \quad q_i > 27/10, \quad q_i < 31/8, \quad C_{i,i+1} \ge 7.$$

Both retained paths have rank at least four. Thus finite classification gives precisely the same no-contact, simple-three, endpoint-four and two-end-three possibilities. A joined finite completion has rank at least nineteen, so it is an ordinary A bridge, with no extra foreign branch. The zero-own repair uses only the helper-side length at least five and the actual affine principals. Its far-three and next-to-far rows have time below their positive foreign cost and force an orthogonal closure; the end-four row has no foreign cost and negative time. Every central own contact is therefore positive.

The isolated-own pressures and selected finite or future points in Subsection K.2 are unchanged. Here

$$q_i + \sqrt{D_i - 1} > 26/5.$$

For $u > 1$ with foreign cost, the loss is less than $5/26 + 5/12 = 95/156 < .61$. The same bound holds for pure native labels at least eight, by $\tan(\pi/8) < 5/12$. For $0 < u < 1$, the loss is at most

$$(5/26)u + \sqrt{u(1-u)} \le (5 + \sqrt{701})/52 < 63/104 < .61.$$

The native-seven negative-core loss is less than $20/81 + 9/25 = 1229/2025 < .61$: use $\sqrt{D_i - 100/81} > 5/2$, whose squared margin is $5/324$. The native-five H_4 loss remains less than $25/76 + 5/24 < .61$. The affine owns and the null boundary are unchanged. Thus every isolated cap is strong at these ranks as well.

For the other contact roles the target remains $T(q) < 761/500$ on $[5/2, 31/8]$. All ordinary near-three, next-to-end, exceptional-position and full-bridge estimates above use only helper-side length at least five. Their foreign-cost versions only increase the ratios. The same foreign far-three and near-four comparisons therefore remove every foreign weak role. The complete weak list is a pure near-four/simple-three cap or the literal packet (65), with $101/125 < \gamma < 27/25$. The dotted upper bound uses the same identity with residual square $[2m(\gamma-1)-1]^2$ and only $m \ge 5$. Near-four is strong for helper-side $m \ge 8$. At $m = 7$ the two actual values of q_7 lie in $(3, 25/8)$, where the sharper positive-quadratic seal is $T(q) < 3/2$. Thus a weak near-four has its actual mate at four or five, or a reversal.

We next justify the two-cap argument at these ranks, before removing near-four. Its mutual-product formulas are identities in N. They still force product zero for a common helper and product zero or $-1/2$ for opposite helpers. In the outward case the zero-product determinant is positive because

$$(2/5)C_{i,i+1} - 24/25 + 8/(5N) > 0.$$

In the reversed case the displayed determinant polynomial in Subsection K.3 has positive derivatives in the two own parameters. If one own is at least .9, its endpoint minimum is

$$(509N - 9974)/500 \geq 206/500 > 0.$$

Indeed $l, s \geq 5$, $l + s = N - 1$ and $ls \geq 70$ make the positive derivative bracket $(39/25)ls - N$ sufficient even after subtracting $2(27/25) + (l + 1)$. If both owns are five, the raw difference and the nonzero rational coefficients $N(N+6), -N^2$ in the preceding calculation apply literally. Hence opposite helpers have product exactly $-1/2$.

The positive two-cap Schur matrices and their actual displacements are the same as above. Their time coefficients exceed two. To check this without separating incompatible minima, reverse so $i \leq N/2$. The smaller, far-helper time is exactly

$$A_N^2 = 2i\gamma^2 - \tfrac{1}{2} - \frac{(2i\gamma - 1)^2}{2N}.$$

It increases in N and in i on this half interval, and in $\gamma \geq .8$. Its minimum at $N = 20, i = 5, \gamma = .8$ is $187/40 > 4$. The near-helper time is larger by $2\gamma(1 - 2i/N) \geq 0$. We may therefore use the already printed inverse calculation with $A_i, A_j > 2$. It gives $\sqrt{1 + \mathcal{D}}/W < 7/16$ for a common helper and $< 3/10$ for opposite helpers. The spatial rationalization now gives

$$W_i q_i + W_j q_j - \|Z\| < \frac{W}{2(\sqrt{15/2} + \sqrt{7})} < W/10.$$

Thus the weighted loss of two adjacent weak packets is less than $43/80$ or $2/5$, both below .61. Adjacent literal packets are impossible, retaining the entire spectator block and every unknown wall.

The near-four mate is an actual positive finite replacement, with the same cosine formula printed in the cap proof. Reversing its numbering if needed gives $6 \leq j \leq N - 4$ and

$$jD_j \geq 72(N - 6)/N > 216/5, \qquad \tfrac{1}{2} + \frac{j - 1}{j\sqrt{2}} > 13/12.$$

Its cosine is therefore greater than seven, and its normalized height exceeds $43/50$. None of these steps used an energy target stronger than the original $e_A < 1$. □

Proposition K.3. *An actual A_{19} or A_{20} component has projection energy greater than $6/7$, with arbitrary finite spectators.*

Proof. Suppose $e_A \leq 6/7$ and apply Lemma K.2. At the other end there is another such mate, or, by nonadjacency, a strong anchor in $\{N - 6, N - 5\}$. That anchor has normalized height $> .77$. The correlation is at most

$$\sqrt{\frac{5 \cdot 6}{15 \cdot 14}} = 1/\sqrt{7} < 1/2.$$

The sum of the two heights is > 1.63, while the sum direction has square less than three. This contradicts the assumed energy, since $1.63^2 - 3(6/7) = 5983/70000 > 0$. Reversal removes the other near-four orientation. Only nonadjacent literal helpers remain.

There is now a strong $i \in \{5, 6\}$ and a strong $j \in \{N - 6, N - 5\}$. Their initial and final coordinate intervals are disjoint. The exact height-energy identity and Cauchy give

$$e_A \geq \frac{h_i^2}{2i} + \frac{h_j^2}{2(N - j)}.$$

At $N = 20$, the endpoint floors are already $q_5 - .61 > 2.1$ and $q_6 - .61 > 2.28$, by the positive squares $15/2 > 2.71^2$ and $42/5 > 2.89^2$. Increasing N improves them. Each term is therefore greater than

$$\min\{2.1^2/10, 2.28^2/12\} = 1083/2500.$$

Consequently $e_A > 1083/1250 > 6/7$, with final difference $81/8750 > 0$. This proves the proposition on the same original projection, with no spectator or face-inheritance restriction. □

K.6 The one short spectator beside D_{23}

Lemma K.4. *The entire forest $D_{23} + I$ cannot occur in dimension twenty-four.*

Proof. The forest is exactly $D_{23} + I$, with no further spectators. Proposition K.6 applies because $23 \geq 21$. Its proof uses the rank-valid prefix and local head coupling, independently of the dimension-twenty-four accounting below. □

K.7 The finite accounting in dimensions twenty-four through twenty-nine

Proposition K.5. *No compact hyperbolic Coxeter polytope has dimension at least twenty-four.*

Proof. The published ceiling twenty-nine leaves only $24 \leq n \leq 29$. Call A_r, $r \geq 7$, or B_r, D_r, $r \geq 9$, dominant. If no component is dominant, each ordinary rank pays its strict rank/14 cost. The common defect theorem leaves at most one discarded short rank, or one typed rank-two defect. The respective total costs exceed $(n-1)/14$ or $.09 + (n-2)/14$, both above one. Thus a dominant block exists. Every dominant block costs more than $1/2$, so there is exactly one. Write its rank as r and its spectator rank as $S = n - r$. Only now may every other block be treated by the small ordinary rates and the single permitted debit. Their complete three conservative costs are

$$S/14, \qquad (S-1)/14, \qquad .09 + (S-2)/14.$$

They represent, respectively, no debit, one short I rank discarded, and one typed rank-two defect. A J profile pays both ranks and does not create a worse case. Empty ordinary sums are zero.

For dominant A, its half-cost first gives $S \leq 7$, hence $r \geq 17$. The shared uniform $A_{\geq 13} > 5/7$ cost then gives $S \leq 4$: at $S = 5$ the weakest typed case already costs $5/7 + .09 + 3/14 = 713/700 > 1$, while the short case exceeds one. Thus $r \geq n - 4$. For $n \geq 25$ this contradicts the cap $r \leq 20$. At $n = 24$ only $r = 20, S = 4$ remains. Proposition K.3 and the three spectator costs close it, the weakest short case already giving $6/7 + 3/14 > 1$.

For dominant B/D, the shared $9/14$ floor first gives $S \leq 5$. For $n \geq 25$ we have $r \geq 20$. If $r = 20, 21$, the rank-valid generic prefix gives energy $> .90$, and $S \geq 4$; even the discarded-short cost $3/14$ is too large. If $r = 22, 23$, its energy is $> .94$, with $S \geq 2$. One remaining ordinary rank costs $> 1/14 > .06$; a sole typed rank-two block costs $> .09$. The caps exclude every larger rank. Thus $25 \leq n \leq 29$ fail.

At $n = 24$, $19 \leq r \leq 23$. The exact generic-prefix rows and their full original rank hypotheses give

r	19	20	21
$e_{B/D} >$	.88864	.90715	.9240775

$e_{B/D,\, r=22} > .9401975.$

Here $M = 13, 14, 15, 16$ respectively, so the D hypothesis $r \geq M + 6$ and the B hypothesis $r \geq M + 5$ both hold. The same increment and omission proofs give these rows; they do not apply a prefix below its permitted rank. Together with the three spectator costs at $S = 5, 4, 3, 2$, they exceed one. For $r = 23$ the B cap already applies. The remaining D_{23} has one spectator rank: an ordinary A_1

costs $> .1$, too much beside $> .94$; a J or typed defect needs two ranks. The sole remaining case is the actual isolated I and is exactly Lemma K.4. This exhausts every finite forest, without asserting that an abstract forest is realized. Dimensions above twenty-nine use only the named published starting bound. □

K.8 One isolated-I argument for all the proper large classical rows

Proposition K.6. *An entire farthest-vertex forest $B_r + I$, $r \geq 20$, or $D_r + I$, $r \geq 21$, is impossible. No further spectators are allowed in this statement.*

Proof. Use $M = 15$ in the actual prefix inventory. Its hypotheses are B_r with $r \geq M + 5 = 20$, or D_r with $r \geq M + 6 = 21$; its geometric window is exactly $5 \leq i \leq M + 1 = 16$. Deleting each of those actual long nodes leaves a left path of rank at least four and a right B of rank at least four, or D of rank at least five. The complete white/foreign list and its actual affine or point exclusions leave only the pure far-three helper weak. The actual two-cap argument excludes adjacent weak helpers. These are precisely the stated hypotheses and domain of Appendix E; no $M = 17$ or unproved short-tail continuation is being invoked.

First suppose five weak. Then six is strong, the actual weak-five point gives $x_1 > a_0 = .5965$, and one of fifteen/sixteen is strong. Put $b_0 = 2.01815 < f_6$ and interpolate through

$$(1, a_0), \quad (6, b_0), \quad (K, b_0 + \sqrt{K} - \sqrt{6}), \qquad K = 15 \text{ or } 16.$$

The actual endpoint floor is larger than the displayed surrogate. The slopes are a_0, .28433 repeated five times, and $c_K = 1/(\sqrt{K} + \sqrt{6})$ repeated $K - 6$ times, then zero. They decrease strictly. Thus this is a paid nonnegative-anchor reference W_0, with no I coordinate or previously credited I energy. Its first coordinate is a_0 and its retained P_6 demand and pairing both equal b_0. Explicitly

K	$\|W_0\|^2 >$	$c_K <$	remaining individual P_6 coefficient $>$
15	.9851486	.159	.12533
16	1.000435	.156	.12833.

Indeed the common first-six energy is .7600299945. The two tail contributions are greater than $1.4234^2/9$ and $1.5505^2/10$, using $\sqrt{15} > 3.8729$ and $\sqrt{6} < 2.4495$. The coefficient is $.28433 - c_K$, greater than $1/60$ in both rows. Lemma J.4 applies to this exact paid reference and adds $> .015$ using the single unused actual I height. Already the weaker fifteen total exceeds 1.0001486; the sixteen row exceeds one even before that graft. Consequently five is strong.

Choose an actual strong $K \in \{15, 16\}$ and now use the concave reference through the actual strong floors, omitting nonadjacent weak knots. The common exact rational increment bounds give

K	E_K lower	energy after all permitted holes $>$
15	.92564	.9240775
16	.94176	.9401975.

At most five isolated internal holes cost $5/3200$. The reference is constant on its first five coordinates, with mean $b = f_5/5 < .361$ and actual P_5 demand f_5. Its individual P_5 coefficient is at least $b - (\sqrt{6} - \sqrt{5}) > .36 - .214 = .146$; omitting six only increases that coefficient. All other anchor coefficients remain nonnegative. No I energy has been used by this reference either.

The head deletion retains B_{r-1} or D_{r-1} of rank at least nineteen or twenty. Finite classification permits exactly isolation or a simple three at the retained long leaf two. A short-end exceptional attachment is too small at these ranks. The entire forest hypothesis leaves just the old I as a

possible foreign factor. In the nonisolated row, own zero has zero time or the positive actual null $c + \Omega_2/2$; own three has its triangle. The other own ranges and every isolated foreign/native boundary are exactly the local head inventory of Lemmas J.6 and J.9.

Their selected long roots n_1, n_2, n_3 and actual old I have the same physical Gram and coordinates here. Their finite projections or future cores have positive actual-wall coefficients, retain all unknown walls, and retain both signed forks in type D. Their joint point covers every actual head/I mutual parameter; it does not presume a finite pair Gram. They do not use a terminal B short normal, a positive final D coordinate or a tail conclusion. The uniform gain is therefore $> .07665$ on this paid flat-five reference, including I exactly once. Its price is at most .1, below the available .146. Hence

$$\|x_{B/D}\|^2 + y^2 > .9240775 + .07665 = 1.0007275 > 1.$$

This proves the proposition with precisely its entire-forest scope. In particular it does not exclude $D_{20} + I$, whose $M = 15$ rank hypothesis fails, or supply a theorem with arbitrary extra spectators. □

K.9 One smaller-loss window for whole B_{22}, D_{22}, D_{23}

Lemma K.7 (A local whole-diagram .58 window)**.** *Let an entire actual farthest forest be B_r or D_r, $r \geq 19$. On $5 \leq i \leq \min\{20, r-4\}$ in B, or $5 \leq i \leq \min\{20, r-5\}$ in D, use the loss .58. The complete weak list is the pure far-three helper, with own five, finite at least seven or dotted and $\phi/2 \leq g < 6/5$, except that isolated dotted own at five is also permitted. Literal helper weak indices are nonadjacent. Their actual first coordinate bounds are $> .5455$ at five, $> .378$ through twenty, and $> .45$ through nine. No short-tail continuation or whole-rank exclusion is part of this local statement.*

Proof. This time the whole-diagram hypothesis is essential. There is no foreign inverse-Gram cost. Set $c_0 = 29/50$, $F_i = \sqrt{i} - c_0/\sqrt{2}$, and call a long index c_0-strong when $S_i > F_i$. The old .61 floors are still denoted by f_i. We use actual unit roots, the genuine short root in B, and both signed forks in D.

The new inventory is checked locally. The sufficient ratio for c_0-strength is

$$T_0(q) = \frac{50}{29} - \frac{q - 29/50}{q(2q - 29/50)}.$$

It increases for $q \geq \sqrt{10}$. Positive rational multiplication at the right endpoints gives

$$T_0(q) < \begin{cases} 83/50 & q < 7, \\ 801/500 & q < 15/4, \\ 159/100 & q < 16/5, \\ 33/20 & q < 32/5. \end{cases}$$

The local bulk conditions are exactly a right B of rank at least four, or right D of rank at least five. At the stated ranks this gives 5..18 in B_{22}, 5..17 in D_{22}, and 5..18 in D_{23}. Each deletion leaves A_{i-1} and a right B_s, $s \geq 4$, or D_s, $s \geq 5$. The complete one-node finite-white list is the one proved above: left endpoints, next-to-end and exceptional positions, right long leaf, the finite D fork extensions, and full simple endpoint bridges. No generic prefix theorem is being invoked with $M > 16$.

For clarity, at the additional D_{23} index eighteen the retained blocks are exactly $A_{17}+D_5$. A right long-leaf three is the ordinary D extension. A right fork three gives finite E_6, with $(\Delta,\sigma)=(3/8,g+1/4)$. Own zero has the actual affine E_8 with arms $1,2,5$; own three has its all-three cycle; own four has the actual $3,3,4,3$ principal on the cap, the contacted fork and the three old nodes ending at eighteen; own six has the $6,3$ path; parallel has the pair null. Every other own has ratio $>(.8+.25)/\sqrt{3/8}>83/50$. A long-left/fork bridge with those E_6 arms cannot be finite at rank twenty-three. The permitted full bridge instead meets the long leaf and is finite D_{23}, with the same endpoint ratios. This supplies the local endpoint argument at eighteen explicitly.

The complete .58 scalar checks are as follows. Near-three has least ratio $[i(\phi+1)-1]/\sqrt{2i(i+1)}$. It increases with i. At $i=6$ it exceeds $801/500$, handling six/seven; at eight it exceeds $83/50$, handling every later index. At five, own at least seven gives ratio $>5/3$. Thus only own-five/near-three at five is new. Next-to-end positions have ratio at least $(i\phi/2+1)/\sqrt{2i}$, above $159/100$ at five and $83/50$ from six onward. Exceptional positions have residual at most $1/4$, $j\ge 3, i\le 8$, hence ratio at least $2g+3/8>83/50$. The far-four/own-three principal is affine. Near-four/own-three has actual cosine $(i-1)+i/\sqrt{2}$; its other owns are stronger. Right long-three and the short-three B_3 exception, when used later as a separate short-layer check, have ratio at least $1+1/\sqrt{2}>83/50$. A D_s fork, $s\ge 5$, has the same affine own list just checked. The only finite fork extensions have $s=5,6,7$, with residual $1-s/8\le 3/8$. Own zero uses actual affine arms $1,2,5$ by extending the retained long arm with at most the old $i,i-1,i-2$; all exist because $i\ge 5$. Own three closes the all-three cycle, own four uses the $3,3,4,3$ path on $i-2,i-1,i,c,f$, and own six/parallel use the local path/null. Every other own has remaining ratio $>(.8+.25)/\sqrt{3/8}>83/50$. Full bridges have smaller residuals and larger numerators, with their own-zero cycles retained.

The isolated own-five H_4 loss is $<25/76+5/24<.58$. For seven, the old negative core gives loss $<1000/4779+9/25<.57$: here $q+\sqrt{D-100/81}>3+2.9$. For finite owns at least eight, the loss is largest at $D=10,m=8$, where it is $\sqrt{10}+\sqrt{2}-4<.578$. The monotonic singleton projection proves the whole remaining finite interval. The old owns three/four/six and parallel retain their actual affine/null principals. For isolated dotted own, $u=g^{-2}\in(0,1)$, the actual core gives loss at most

$$\frac{u}{\sqrt{2i}+\sqrt{2i-1}}+\sqrt{u(1-u)}.$$

At $i\ge 6$ the first coefficient is $<37/250$, so this is $<(37/250+1.011)/2=.5795$. Only isolated dotted five remains. Thus, apart from that row and near-five, every weak bulk cap is the pure helper $v=g\omega_i+\omega_1/2$, with own five, finite at least seven or dotted.

The new near-five role. For $v=(\phi/2)\omega_5+\omega_4/2$, put $g=\phi/2$ and use the actual right H_4 correction

$$w=g(3n_5/2+n_6+n_7/2),\qquad \delta=(5-3\phi)/8,\qquad Y=v-w.$$

Its extra old-four product is negative. Its normalized finite projection retains every original wall. Direct substitution gives

$$A^2=(7+13\phi)/2,\quad Y^2=(33+49\phi)/8,\quad \sum Y_j=\sqrt{2}(2+5\phi/2).$$

From $1.618<\phi<1.62$, $A>3.74$ and $\sqrt{\delta}<.136$, we have $x.Y>3.6, Y^2<14.05, \sum Y_j<8.6$. The independently valid old .61 inventory at $M=11$ supplies a strong $K=11$ or twelve, before any new strong-five conclusion. For $Z=Y+(3/4)P_K$ the two demand/square bounds are

K	$x.Z>$	$Z^2<$	squared-demand excess $>$
11	5.76	33.1375	.0401
12	5.8725	33.7	.78625625.

Here $f_{11} > 2.88, f_{12} > 3.03$. Cauchy excludes both, so the new near-five role is impossible without a .58 strong-five premise.

Helper nonadjacency and three actual consequences. Every weak helper has $\phi/2 \le g < 6/5$. The upper bound is reproved at this threshold: at $g \ge 6/5$ the squared ratio is at least $4g - 2 \ge 14/5 > (83/50)^2$. Adjacent helper caps are distinct actual walls, and their product is at least $1 - ab > -11/25 > -1/2$. It must be zero. Their positive Schur inverse and actual mixed point give loss $< 7/16 + 1/(4\sqrt{10}) < .58$, using only $i \ge 5, g > .8, A > 2$. Thus new weak helpers are nonadjacent; we have not presumed they were already old .61-weak. This last two-cap calculation uses only the two displayed pure-helper profiles, their actual long duals and the retained positive old blocks. A later short-layer classification may supply those same profiles; after checking that physical interface it may reuse this calculation, without extending the present classification domain.

The actual helper point table gives $x_1 > \sqrt{2}[T - g(\sqrt{2i} - .58)]$. Its complete native ranges imply

$i = 5$ weak helper	$x_1 > .5455$
$i \le 20$ weak helper	$x_1 > .378$
$i \le 9$ weak helper	$x_1 > .45$.

For the first subtract at most $.03\sqrt{2}(6/5)$ from the old scalar expression $> .5965$. For the second, the four finite lower brackets are $.58(.809) + .079 - .111$, $.58(.900) + .079 - .253$, $.58(.923) + .079 - .323$ and $.58(.939) + .079 - .344$. Their products with $\sqrt{2}$ exceed .378. The dotted difference decreases in g, q and has enlarged endpoint

$$\sqrt{119/2} - (6/5)\sqrt{40} + .696 - \sqrt{11}/6.$$

The bounds $\sqrt{119/2} > 7.71362, \sqrt{40} < 6.32456, \sqrt{11}/6 < .55278, \sqrt{2} > 1.4142$ give $> .3781118$. For $i \le 9$ the finite $A - gq$ term is $> .15$, increasing in g and decreasing in q; the same brackets then exceed .45. For dotted use $\sqrt{1391/50} > 5.274, \sqrt{18} < 4.243, \sqrt{11}/6 < .553, \sqrt{2} > 1.4$, giving $> .45556$. All these pressures use actual positive finite projections or future cores. At a last D long node, $\omega_{21} + \omega_{22} = \sqrt{2}P_{21}$ in D_{22} is only a spatial identity; no extra normal is introduced. □

Lemma K.8 (A conditional early paid-reference contract)**.** *Keep the whole-diagram and threshold conventions of Lemma K.7. Suppose the actual helper-or-strong window, with only its permitted isolated dotted-five exception, has been completed through sixteen. For either actual strong $K \in \{15, 16\}$ there is a paid vector W supported through K, with $\sum W_j = F_K$, remaining individual endpoint-K coefficient $> 1/9$, and effective energy $L_W > Q_K$, where $Q_{15} = .94006, Q_{16} = .95618$. The three initial-five branches, their later conditional extensions to an actual strong nineteen/twenty, and the complete flat-pressure table below have the stated positive-dual and same-vector scope. This lemma supplies no missing short-layer classification or strong endpoint.*

Proof. Use $F_i = \sqrt{i} - .58/\sqrt{2}$ throughout this proof.

Early references in all three five branches. Put $\mathcal{E}_K = F_5^2/5 + \sum_{i=5}^{K-1} d_i^2$. Changing the initial loss from .61 to .58 adds $.03\sqrt{2/5} - .00357 > .0154$. One omitted internal floor j loses $< 1/[32(j-1)^3]$. All holes $j \ge 6$ therefore cost less than $\frac{1}{32}\int_4^\infty t^{-3}dt = 1/1024$.

For either actual strong $K = 15$ or sixteen we need a paid vector W, supported through K, with $\sum W_j = F_K$, endpoint coefficient $> 1/9$ and effective energy

$$L_W = 2D_W - \|W\|^2 > Q_K, \qquad Q_{15} = .94006, \quad Q_{16} = .95618.$$

Here D_W is its pressure demand. Effective energy need not equal the square of the vector. If five strong, the concave reference and the full hole bound give these values; its endpoint coefficient is at least $d_{K-1} > 1/8$. If five a weak helper, interpolate through $(1, .5455), (6, F_6), (K, F_K)$. Its first-six energy exceeds .74385, and its tail energies at fifteen/sixteen/nineteen exceed $.225, .2404, .2804$. The slopes decrease and the endpoint coefficient $1/(\sqrt{K} + \sqrt{6}) > 1/9$; the first two energies exceed the required Q_K.

If five is isolated dotted, its actual core gives

$$U' = P_5 + (u/2)(e_6 - e_5), \qquad x.U' \geq T' = \frac{\sqrt{10 - u} - \sqrt{u(1-u)}}{\sqrt{2}}.$$

Its loss is $< .588$, since with $a = \sqrt{10} - 3 < 13/80$, $au + \sqrt{u(1-u)} < (a + \sqrt{1 + a^2})/2 < .588$. Weakness forces $u > .4$: at $u \leq .4$ the coarser loss $au + .5 < .565$ is too small. In particular $T' > 1.8185$. Choose actual strong $J = 6$ or seven and $K = 15$ or sixteen. Let W_0 interpolate through $(J, F_J), (K, F_K)$, and put

$$V = U' - (5/J)P_J, \qquad W = W_0 + \mu V, \qquad \mu = .21 \ (J = 6), \quad .2 \ (J = 7).$$

The residual has sum zero, support through J, and $W_0.V = 0$. Its squared norms are bounded by $77/150, 194/175$, since $U'^2 = 5 - u + u^2/2$ decreases for $.4 < u < 1$. The deficits exceed $.119, .221$. The J payments are $.175, 1/7$, below the available $.1808, .165$; those capacities only increase when K increases. The endpoint-K coefficient stays unchanged. The complete energy data are

J	$F_J^2/J + 2\mu\Delta - \mu^2 V^2 >$	$\text{tail}_{15} >$	$\text{tail}_{16} >$	$\text{tail}_{19} >$
6	.720462	.225	.2404	.2804
7	.757957	.188	.2037	.2445.

Thus both early columns supply $L_W > Q_K$. Every source coefficient is nonnegative, and the construction remains legal at a later actual strong $K = 19$ or twenty. No strong-five or terminal premise has been used in constructing the early references.

For an actual normalized pressure $x.Z > G$, $Z_1 = \cdots = Z_{16} = 1$ and $Z^2 \leq R$, pay $1/9$ at the remaining endpoint $K = 15$ or sixteen. The residual $Z - P_K$ has zero product with W because $\sum W_j = F_K$, regardless of whether $L_W = \|W\|^2$. One square gives

$$e > L_W + 2(G - F_K)/9 - (R - K)/81.$$

This uses a positive-dual payment, not the invalid subtraction of two lower pressure bounds. With $F_{15} < 3.463, F_{16} < 3.590$, each row below is sufficient in both early references:

G	R	minimum excess above one $>$
3.956	19	.00023
4	19.7	.0013
4.05	20	.0087
4.1	20	.0198
4.15	21	.0186
4.4	22	.0618.

The remaining task is to verify actual point pressures fitting this table; a diagram count or an abstract Gram completion would not suffice. □

Proposition K.9. *None of the entire finite diagrams B_{22}, D_{22}, D_{23} occurs at an actual farthest vertex.*

Proof. The local windows of Lemma K.7 reach eighteen/seventeen/eighteen respectively, and hence already contain sixteen. Actual helper nonadjacency supplies a strong fifteen/sixteen, so Lemma K.8 applies with all its individual price conditions. We now complete the genuinely different short layers and fund their actual pressures by that table.

The actual B_{22} short tails. At nineteen deletion leaves $A_{18}+B_3$. The new short-three F_4 role has residual $1/4$ and numerator $g+1/(2\sqrt{2})$. Own zero has its actual affine F_4 kernel, and every positive own has ratio at least $1+1/\sqrt{2}>83/50$. The long-three and all full bridges are strong. At twenty the right B_2 allows three at either end. The long end is the same local row. The short end has residual $1/2$ and numerator $g+1/(2\sqrt{2})$; own zero/four/six/parallel have their actual affine/null principals, while owns at least seven are directly strong. In particular the own-zero short kernel on $c, n_{22}, n_{21}, n_{20}, n_{19}$ is $(\sqrt{2}, 2\sqrt{2}, 3, 2, 1)$. Own four/six use respectively $\sqrt{2}c+n_{19}+2n_{20}+n_{21}$ and $\sqrt{3}c+2n_{20}+n_{19}$. All four left/right endpoint bridges have residual $1/40$ and minimum ratio $\sqrt{40}(1/40+1/(2\sqrt{2}))>83/50$, including own zero.

Own five uses the actual left H_4 point, with $\alpha=\sqrt{2}g+1/2$, $A^2=20\alpha^2-1/2$, $R<20$. The positive squares $A/\alpha>4.45$ and $\sqrt{\delta_5}/\alpha<.083$ give $G>4.3$. Own three is the actual cap $v=(\omega_{20}+\omega_{22})/2$. Releasing old twenty-one gives a positive finite A_{22} endpoint, with its omitted-wall product strict and every unknown wall retained. Its cosine is $(21+20\sqrt{2})/\sqrt{23}>245/24$. It gives $x.P_{21}>4.15$, $R=21$: the last comparison is $1511(245/24)>15289$. Both points fit the paid table. Thus through twenty only helpers, strong indices and possibly isolated dotted five remain in B_{22}.

The actual D_4 and D_3 tails. For D_{22} these are eighteen and nineteen; for D_{23} they are nineteen and twenty. The D_4 finite list allows three at any leaf. The fork-three residual is $1/2$, numerator $g+1/4$. Own zero/three/four/six/parallel have their exact principals. Own at least eight is strong because $\sqrt{2}(\cos(\pi/8)+1/4)>33/20>T_0(q)$. Retain own five and the newly possible seven. Own five uses its actual left H_4 point, with $A^2=i\alpha^2-1/2$, $\alpha^2=\phi+5/8$ and residual ratio $<.01$. At $i=18$ its first sixteen coordinates are not all flat. Use instead the independently valid old .61 strong-five reference and an old strong eleven/twelve: the first twelve are flat and the old terminal price exceeds $1/7$. The pressure $G>4.1, R<20$ gives energy >1.0102 in both rows. The old strong-five bootstrap here uses only the complete $M=16$ window through seventeen. Own seven uses

$$\xi=c+g^{-1}n_i+(2g)^{-1}n_{i-1}, \qquad k=1-3/(4g^2)>0.$$

Its cap and $i-1$ products vanish; the i product is negative, its square is $-k$ and its time is positive. For $.9<g<.902$, at $i=18$ we have $A/\alpha>4.22$ and $\sqrt{k}/\alpha<.18$, so $G>4.04, R<18$. Both quantities only strengthen in the D_{23} layer $i=19$; its $R<19$ is still covered by the first paid row. All selected displacements are flat through sixteen in this seven case. Bridges and long leaf have their strong or own-zero cycle dispositions. Both forks remain.

At the right $D_3=A_3$ layer, middle-three is direct or affine. End-four retains its actual affine own-zero kernel and is otherwise strong. End-five is sealed H_4 and directly strong even at own zero. Full left bridges retain their own-zero cycle and direct positive-own bounds. Thus only no-left end-three remains, with residual $5/8$, numerator $g+1/4$. Own zero/three/four/six/parallel have the actual E_8/cycle/star/path/null principals. Own five is the actual left H_4 point, $A^2=i\alpha^2-5/8$, giving $G>4.1, R\le 20$. Every $g\ge 53/50$ is directly strong, since $(131/100)^2(8/5)>(33/20)^2$.

For the whole interval $.9<g<53/50$ use both actual A_2 sides, with corrections $g(4n_i/3+2n_{i-1}/3)$ and $g(4n_i/3+2n_{i+1}/3)$. Their positive future cores have square $-d$, $d=4g^2/3-1>0$, and all selected products are zero or negative, including the additional contacted fork. Average

their pressures retaining the full correction cross $8g^2/9$. The displacement is flat through sixteen, and

$$x.Y/\alpha \geq \frac{A-\sqrt{d}}{\alpha}, \quad A^2 = i\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14g^2/9 < i\alpha^2.$$

At $i = 19$, splitting at $g = 26/25$ gives respectively $(A/\alpha, \sqrt{d}/\alpha) > (4.33, 0)$ with radical $< .365$, and $A/\alpha > 4.337$ with radical $< .381$. Thus $G > 3.956, R < 19$. At the actual D_{23} index twenty the same positive squares give $A/\alpha > 4.44$ in the first interval and > 4.45 in the second; the radical bounds are unchanged. Hence $G > 4.069 > 4.05, R < 20$, a separate sufficient table row. The increasing ratio d/α^2 justifies both full intervals, not only the endpoints. This proves the complete window through twenty for whole D_{23} without any claim at its later centre twenty-one.

The remaining actual D_{22} centre twenty. Here the retained white diagram is $A_{19} + 2A_1$. Its complete finite list is one left endpoint-three/four or next-to-end-three; with one fork only a left endpoint-three and fork-three/four; with both forks the all-three endpoint D_{22} bridge. Their residuals are $1/40$ or $11/40$; the directly strong branches include zero own where appropriate, and the left/fork-three zero/three rows retain the cycle/triangle. Internal branches cannot be finite at this rank. With no fork, the locally rechecked bulk inventory applies. The isolated-five correction uses only actual long roots and the spatial sum $\omega_{21} + \omega_{22}$.

With no left contact, one finite fork label or the pairs $(3,3), (3,4), (3,5)$ exhaust the white possibilities. At own zero, fork three/four/six have their affine principals. Fork five uses the actual left H_4 point: $R = 19$, $A^2/\alpha^2 = 6 + 8\phi$, $\sqrt{\delta_5}/\alpha < 15/56$, giving $G > 4.08$. The last bound uses $\phi > 79/49$, because $5 \cdot 49^2 - 109^2 = 124 > 0$; it is not inferred just from $\phi > 8/5$. For fork at least seven put $b = \delta^2 \in (.81, 1)$. The actual core

$$c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{20}$$

has zero cap/fork products, negative centre product, positive time and negative square. Its normalized data are

$$R(b) = 30 - 16/b + 6/b^2, \quad G(b) = \sqrt{22 - 2/b} - \sqrt{-8 + 14/b - 6/b^2}.$$

For $b \leq .9$, $R < 19.7$ and $G > 4.41 - .41 = 4$. For $b \geq .9$, $R \leq 20$ and $G > 4.44 - .39 = 4.05$. The derivatives have numerators $16b - 12$ and $12 - 14b$; the latter radical square has maximum $1/6$, or $4/27$ on the second interval. These are whole-interval positive-square bounds.

At own three, fork three has its triangle. Fork four/five instead gives the actual finite D_{22} contacted-fork endpoint, with cosine $(20 + 22\delta)/2 > 17$, signed direction $\sqrt{2}\omega_f = P_{21} \pm e_{22}$, $R = 22, G > 4.4$. The released-wall product is strict, so this is an actual endpoint. Fork at least six is direct. Own four has its centre/left/other-fork affine star. For own at least five and fork at least four, direct ratios leave only own-five/fork-four; it has the same left-H_4 data as the B_2 own-five case, $G > 4.3, R < 20$. It must not be discarded by the old .61 ratio. For fork-three/own-five, the left-H_4 point has $A^2 = 20\alpha^2 - 3/4$, $G > 4.1, R < 20$. For other finite or dotted owns, $g \geq 6/5$ is direct. On $.9 < g < 6/5$, the actual core above with n_{20}, n_{19} has $k = 1 - 3/(4g^2)$, $A^2 = 20\alpha^2 - 3/4$ and $R < 20$. The full derivative bound $k/\alpha^2 \leq 575/5046 < .34^2$ and $A/\alpha > 4.44$ give $G > 4.1$.

For two forks, own three has its triangle and owns at least four are direct. At own zero, pair-three-three has its square. Pair-three-four has an actual finite B_{22} contacted-fork endpoint, cosine $(40 + 22\sqrt{2})/4 > 17$, again $G > 4.4, R = 22$. This adjacent vertex is not declared farthest. Pair-three-five uses the actual finite H_3 projection, with displacement αP_{20}, $q = 2\phi^{-6} < 1/8$ and pressure

$$\sqrt{20 - q}/(1 + \sqrt{q/20}) > 4.45/1.08 > 4.1, \quad R = 20.$$

Every listed point fits the paid table. This exhausts the actual centre classification and completes the window through twenty in D_{22} as well. All signs of the final coordinate were retained.

The final reference, without an assumed strong five. Nineteen/twenty are now helper-or-strong, so one is strong. If five a weak helper, its earlier interpolation has energy $> .74385 + .2804 > 1$. If five isolated dotted, its same paid reference has effective energy $> .720462 + .2804 > 1$ or $> .757957 + .2445 > 1$. The endpoint-J payment remains legal because its tail coefficient decreases. Thus five is strong, proved only after the terminal classifications.

If twenty strong, the reference energy exceeds $\mathcal{E}_{20} - 1/1024 > .99752 + .0154 - 1/1024 > 1$. Otherwise twenty weak gives strong nineteen, $\mathcal{E}_{19} > .9847 + .0154 = 1.00010$, and $F_5/5 < .366$. If a weak helper occurs at six through nine, its actual $x_1 > .45$ adds $> (5/4)(.45 - .366)^2 = .00882$, more than the full hole loss. If none occurs there, holes start at ten and cost less than $\frac{1}{32}\int_8^\infty t^{-3}dt = 1/4096$. The weak-twenty point gives $x_1 > .378$ and adds $> (5/4)(.378 - .366)^2 = .00018$. The total is

$$e > 1.00010 + .00018 - 1/4096 > 1.0000358 > 1.$$

Each initial adjustment leaves positive decreasing slopes at both junctions. Effective-reference, tail and initial gains above belong to the stated disjoint branches; none is added twice. All three whole diagrams are therefore excluded on the original vector. □

K.10 Two small remainders and one extra prefix gain

Lemma K.10. *An actual A_{20} component cannot have spectator rank two.*

Proof. The arbitrary-spectator $N = 21$ reference in Lemma K.14 gives $e_A > .92274$. An ordinary irreducible rank-two factor costs $> 1/7$; two ordinary A_1's or $I + A_1$ cost $> 1/5$ jointly; a J pair costs $> .34$; and either actual typed defect costs $> .09$. These exhaust rank two and each exceeds .09. Orthogonality on the same original vector gives total energy $> .92274 + .09 = 1.01274 > 1$, a contradiction. □

Lemma K.11 (The final packet preserves an extra initial gain)**.** *With exactly the hypotheses of Proposition 4.1, let $G_0 = 3969/800000 = .00496125$. Then its same actual prefix energy is greater than $E_M - H_M/3200 + G_0$, for every $6 \le M \le 16$. This is a replacement bound, not an independent gain to add to a different initial-coordinate coupling.*

Proof. The weak-packet points and full scalar seals in the cap proof give $x_1 > .424$ for a weak packet at every $5 \le i \le 21$. Here we use only $i = M + 1 \le 17$. Its selected actual roots are $i, 1, 2$ for native five, $i, 1$ for seven/eight, i for the remaining finite owns, or the actual dotted pair. Its converted coordinates use old long neighbours $i - 1, i + 1$, which exist because $i + 1 \le r - 3$ in B and $\le r - 4$ in D. All foreign spectators have already been cleared and all point coefficients and wall signs are those of that pressure proof.

If five weak, the generic-prefix proof gives $L_M - E_M > .76 - .7 - 9/640 = 147/3200 > G_0$; at $M = 6$ use its separate direct $L_6 > .76, E_6 < .7$ comparison. No strong-five conclusion is needed in that branch. Now suppose five strong and put $K = M + 1$. If K strong, its reference gains an extra increment $d_M^2 > 1/(4M + 2) \ge 1/66$. At most one extra hole costs $1/3200$, and $G_0 + 1/3200 < 3/500 < 1/66$. This also accounts for the possible hole at six when $M = 6$. If K weak, M strong and the actual packet gives $x_1 > .424$. Modify only the initial five reference slopes to

$$.424, \quad (f_5 - .424)/4, \quad \dots, \quad (f_5 - .424)/4.$$

They are positive and decreasing at both junctions: $1.8 < f_5 < 1.805$, and the next slope is $< 1/4$ even after omitting its neighbour. The exact gain is $(5/4)(.424 - f_5/5)^2 > G_0$. The old hole intervals lie after five and have disjoint interiors, so their losses do not overlap this change. These are legal reference duals on one vector, with no tail or total-norm bootstrap. □

K.11 The complete forest accounting in dimensions twenty-three and twenty-two

Proposition K.12. *Dimensions twenty-three and twenty-two are impossible.*

Proof. At either dimension there is exactly one dominant component, by the same ordinary/debit and half-cost argument used above. The three spectator costs are still complete. For $S \geq 2$ the typed cost is larger than the short-rank-discard cost by $13/700$. For dominant A, the half-cost gives $S \leq 7$, so $r \geq 16$ at dimension twenty-three and $r \geq 15$ at twenty-two. The caps give $r \leq 20$. The following weaker consequences of already written component theorems suffice:

r	15	16	17	18	19	20
$e_A >$	4/7	.86	6/7	.91	6/7	6/7.

At dimension twenty-three, the weakest short-discard totals for $r = 16, 17, 18, 19, 20$ are respectively $.86 + 6/14$, $6/7 + 5/14$, $.91 + 4/14$, $6/7 + 3/14$ and $6/7 + 2/14$, all at least one and strictly exceeded. Every other debit case is stronger. At twenty-two, ranks fifteen through nineteen similarly use spectator ranks seven through three: $4/7 + 6/14 = 1$, $.86 + 5/14 > 1$, $6/7 + 4/14 > 1$, $.91 + 3/14 > 1$ and $6/7 + 2/14 = 1$, strictly exceeded. The last A_{20} rank-two remainder is exactly Lemma K.10. Thus no A remains.

For dominant B/D, the $9/14$ floor gives $S \leq 5$. At twenty-three this means $r \geq 18$. The generic $M = r - 6$ floors for ranks eighteen/nineteen/twenty, even for the weaker D requirement, are $> .8689525, .88864, .90715$; their spectator ranks five/four/three give totals above one after either debit. At rank twenty-one the floor $> .9240775$ meets a rank-two remainder. A typed block costs $> .09$. An ordinary irreducible rank-two block costs $> 1/7$. Two A_1's have at least one ordinary cost $> .1$, including the possible short-rank discard; J is stronger. Thus all these totals exceed one. Rank twenty-two with its floor $> .9401975$ permits only isolated I, because an ordinary A_1 costs $> .1$. It is excluded by Proposition K.6. At rank twenty-three, B_{23} has its cap and whole D_{23} is excluded by Proposition K.9.

At dimension twenty-two, $r \geq 17$. Ranks seventeen/eighteen/nineteen have floors $> .8472225$, $.8689525, .88864$ and spectator ranks five/four/three. Their weakest costs $4/14, 3/14, 2/14$ suffice. At rank twenty the improved standalone prefix floor is

$$.90715 + G_0 = .91211125.$$

Every rank-two remainder costs $> .09$: a typed exception does so directly; an ordinary irreducible costs $> 1/7$; two A_1's include one nonshort cost $> .1$, and J is stronger. All totals exceed one. Rank twenty-one has only one spectator. An ordinary A_1 contradicts even the original $> .9240775$ floor, while actual I is excluded by Proposition K.6. At rank twenty-two only whole B_{22}, D_{22} remain, excluded by Proposition K.9. Every own-rank finite forest has now been covered. The argument never descends to an arbitrary face or uses a higher-dimensional cover at a different rank. □

K.12 One rational curve for the two remaining large-A couplings

We use the same central inventory with a different reference, rather than starting a new component classification. At $N = 20$ the target will be $e_A < .91$; at $N = 21$ no target beyond $e_A < 1$

is needed. The complete pure-helper inventory and its nonadjacency were proved before the 6/7 contradiction. Only the removal of near-four must be revisited. At $N = 20$ its mate and opposite anchor have height sum > 1.63, correlation $< .4$ and hence sum-direction square < 2.8. This contradicts $e_A < .91$, with margin .1089. At $N = 21$ the opposite anchor is fifteen or sixteen. At fifteen its height exceeds .79, since $D_{15} = 60/7 > (61/21)^2$, and its correlation with the near mate is $\le 1/\sqrt{8} < .354$. Then $1.65^2 > 2.708$. At sixteen its height exceeds .77 and the correlation is $\le 5/16$, so $1.63^2 > 2.625$. An opposite near-four mate has at least the latter height and smaller correlation. All these comparisons contradict $e_A < 1$. Thus the central weak indices are nonadjacent pure helpers on both stated domains, with arbitrary finite spectators retained.

Use the symmetric rational heights $H_{N-i} = H_i$, linear from zero to five and from $N - 5$ to N:

N	H_5	H_6	H_7	H_8	H_9	H_{10}
20	2.128	2.288	2.406	2.488	2.536	2.552
21	2.150	2.317	2.445	2.537	2.597	2.626.

At $N = 21$, $H_{11} = H_{10}$. Every entry is strictly below $\sqrt{D_i} - .61$, by positive squaring, and successive differences decrease. The tight $N = 21, i = 7$ comparison, for example, has margin $28/3 - 3.055^2 = 925/3000000 > 0$. The coordinate reference is $w_j = (H_j - H_{j-1})/\sqrt{2}$. Summation by parts uses nonnegative actual strong-floor coefficients and gives $x_A.w \ge \|w\|^2$ at its available bends.

If end five is weak, replace that entire end by the straight line zero to six; six is strong. Do the reflected replacement at a weak $N - 5$. Delete each remaining weak j by averaging its two neighbours. The exact energy loss is $(H_{j-1} - 2H_j + H_{j+1})^2/4$. Nonadjacency gives disjoint affected edge pairs, so these losses add for every permitted weak set. The following finite table records the full bound, not selected weak patterns. In the loss column, count is the maximum number of nonadjacent internal holes and the second factor is the maximum quarter-square on the remaining interval:

N	weak ends	initial energy	internal loss $\le$	surviving energy $>$
20	0	.9544848	5(.000441)	.9522798
20	1	.9250917333...	4(.000441)	.9233277
20	2	.8956986666...	4(.000324)	.8944026
21	0	.981678	5(.00038025)	.97977675
21	1	.9528575833...	5(.00038025)	.9509563
21	2	.9240371666...	4(.000324)	.9227411.

For an exact rational formula behind every repeating decimal, put $a = H_5, b = H_6, T = \sum_{j=7}^{\lfloor N/2 \rfloor} (H_j - H_{j-1})^2$. The three initial energies are

$$a^2/5 + (b-a)^2 + T, \quad \tfrac{1}{2}[b^2/6 + a^2/5 + (b-a)^2 + 2T], \quad b^2/6 + T.$$

In the two-weak-end rows, six and $N - 6$ are strong and the holes lie in $7..N - 7$, giving four possible holes in either rank. The zero/one-end interval lengths give the other counts. Direct second differences of the displayed rational heights give the per-hole maxima. This proves completeness of the finite loss table without a combinatorial enumeration or an arithmetic packing premise.

An actual endpoint floor $x_1 \ge z$ improves a constant initial block of length m and coordinate mean b_0 by replacing its entries with $z, (mb_0 - z)/(m-1), \ldots$. If $z > b_0$, the gain is

$$\frac{m}{m-1}(z - b_0)^2.$$

If $z \le b_0$, leave the block unchanged. All reference sums are preserved. For every $z \le 1$ the remaining means exceed the next coordinates: at $N = 20$, the first-five and first-six comparisons are $> .126$ versus $< .114$, and $> .123$ versus $< .084$; at $N = 21$, they are $> .130$ versus $< .119$, and $> .127$ versus $< .091$. Deleting an internal neighbour only decreases that next coordinate. Thus the new duals are nonnegative at both junctions. The first and last blocks have disjoint supports, so their changes add without changing any intervening height.

Lemma K.13. *An actual A_{19} cannot coexist with either typed rank-two exception.*

Proof. The typed energy $> .09$ gives $e_A < .91$, the stated $N = 20$ domain. Zero or one weak ends already contradict that target by the table. With two weak ends its reference energy exceeds .8944026. Both typed exceptions give every other actual normalized direction height square greater than $(11 - 4\sqrt{7})/3 > 11/80$. For N_7 the stronger $> 1/5$ propagation suffices; for O_7 use the actual no-contact/native-height proof in Appendix A.9. The rational seal is $\sqrt{7} < 847/320$, with squared margin $609/102400$. At the two actual A_{19} ends this gives

$$x_1^2, x_{20}^2 > (19/20)(11/80) = 209/1600 > .36^2.$$

The first and last six-block means are $< .27$. Their legally funded disjoint modifications add $> 2(6/5)(.36 - .27)^2 = .01944$. The same-vector energy is therefore $> .8944026 + .01944 + .09 = 1.0038426 > 1$. Both typed roles, rather than a bare dihedral label seven, are covered. □

Lemma K.14. *An actual A_{20} has energy $> .92274$ and cannot coexist with an ordinary nonshort A_1 or an actual isolated I.*

Proof. The $N = 21$ window and table apply under $e_A < 1$ with arbitrary finite spectators. Their least energy is $> .9227411 > .92274$. An ordinary nonshort A_1 costs $> .1$, already contradictory.

For actual I, write its universal parameters as s, t: $e_I \ge s^2$, $t + (3/2)s > .767$, $0 < s < 1/3$ and $t < 1$. Both actual normalized end heights are at least t. Put $a = \sqrt{20/21} > .975$; then $x_1, -x_{21} \ge at$. Every endpoint change has $z = at < 1$, so the junction checks just proved apply. In the zero/one-weak-end rows the two absolute block means are $< .305$; in the two-weak-end row they are $< .274$. Each block length is five or six, so their two changes add at least $(12/5)(at - b)_+^2$, using $b = .305$ or $.274$ respectively. Weighted Cauchy, on these two changes and the *single* I cost, gives

$$s^2 + (12/5)(at - b)_+^2 > \frac{(a \cdot .767 - b)^2}{(9/4)a^2 + 5/12}.$$

Indeed $(at - b)_+ + (3/2)as > a \cdot .767 - b$. The denominator is $215/84 < 2.56$. With $b = .305$ the numerator exceeds $.442825^2$, so the gain is $> .076$. With $b = .274$ it exceeds $.473825^2$, so the gain is $> .0875$. The least final total is $> .9227411 + .0875 > 1.01024$; the zero/one-end totals are larger. No separate I charge was added at the second end, and no other component energy was reused. □

K.13 The proper D_{19}/D_{20} continuation

Lemma K.15. *For an actual D_{19} or D_{20} component, the .61 pure-helper window extends through fifteen or sixteen respectively, with arbitrary finite spectators. This continuation does not assume strong five.*

Proof. The valid $M = 13/14$ bulk reaches fourteen/fifteen. Its complete geometry supplies an actual strong eleven/twelve. In either initial-five branch there is a positive paid reference ending there, with

$$q_{11} > .8472225, \quad q_{12} > .8689525, \quad F_{11} \le f_{11} < 2.886, \quad F_{12} \le f_{12} < 3.034,$$

and remaining individual terminal coefficient $> 1/7$. For weak five use the actual $x_1 > .5965, S_6 > 2.01815$ interpolation; it has larger energy and terminal slope $1/(\sqrt{K} + \sqrt{6}) > 1/7$. No ordinary spectator energy has been used in these references.

At $i = r - 4 = 15, 16$, deletion retains $A_{i-1} + D_4$. Its finite extensions are three at any right leaf, or full endpoint D_r bridges. A foreign branch at the internal new node of such a rank-$r > 8$ bridge is not finite. Own zero keeps its actual cycle; positive owns are directly strong. Long-leaf three has its bulk strong/affine disposition. Fork three has residual $1/2 - Q$ and numerator $g + 1/4$; own zero/three/four/six/parallel have the actual principals, all $Q > 0$ cases and native owns at least seven are directly strong. Both physical forks are retained. The only extra role is pure own-five/fork-three.

Put $g = \phi/2$, $\alpha^2 = \phi + 5/8$ and $\delta_5 = (5 - 3\phi)/8$. The actual left H_4 correction is $g(3n_i/2 + n_{i-1} + n_{i-2}/2)$. Its finite projected point has positive time, all original wall signs, and normalized displacement flat through $i - 3$, with

$$A^2 = i\alpha^2 - 1/2, \quad Y^2 < i\alpha^2, \quad A/\alpha > \sqrt{i - 25/112}, \quad \sqrt{\delta_5}/\alpha < .091.$$

Here $\alpha^2 > 56/25$. At fifteen this gives $G > 3.75, R = 15$; at sixteen it gives $G > 3.88, R = 16$. Both are flat through twelve. Pay $1/7$ at the early reference endpoint before adding the actual point. The two lower bounds $q_K + 2(G - f_K)/7 - (R - K)/49$ exceed 1.0122 or 1.029, respectively, for *both* $K = 11, 12$. Thus the extra role is excluded without a strong-five or later-tail assumption. The actual two-helper Schur calculation then extends nonadjacency to the newly supplied index; all right and spectator blocks stay positive. □

Lemma K.16. *The entire forests $D_{19} + N_7$, $D_{19} + O_7$ and $D_{20} + I$ are impossible in dimension twenty-one.*

Proof. For either typed defect beside D_{19}, weak five has actual prefix energy $L_{13} > .7600299945 + 1.156^2/7 > .95$, using $\sqrt{13} > 3.6055, \sqrt{6} < 2.4495$, already too much beside $> .09$. Hence five strong. Lemma K.15 supplies coverage through fifteen. If fifteen strong, its reference has energy $> .9240775$. If fifteen weak, fourteen strong and the actual pure packet gives $x_1 > .424$, so the legally modified initial reference gives $> .90715 + .00496125 = .91211125$. In both branches the separate typed energy $> .09$ contradicts one. This repeats the proved prefix-gain calculation only *after* supplying its missing D4 geometry; it does not invoke $M = 14$ at D_{19} outside that theorem's rank hypothesis.

For $D_{20}+I$, the same lemma first supplies coverage through sixteen in both initial-five branches. The positive actual long-root helper points, nonadjacency and initial I parameters are therefore exactly the geometric hypotheses used in the proof of Proposition K.6. Its weak-five reference at a strong fifteen/sixteen has energy $> .9851486$ and remaining P_6 price $> .12533$; the capped single-I graft gives $> .015$. Thus five strong. Its strong fifteen/sixteen flat-five reference has $q > .9240775$, actual P_5 demand f_5 and remaining price $> .146$. Deleting long one retains actual D_{19}, with just the same isolated/long-two-three head list and only the old I outside. The selected actual head roots and both genuine forks retain the local point signs, so the q-free head/I gain $> .07665$ closes it. We have replaced the missing $M = 15$ bulk justification by a proved actual D4 continuation, without applying its rank-restricted theorem to D_{20}. No D3 or terminal-seventeen floor is needed. □

K.14 One full weak-nineteen payment

Lemma K.17. *Suppose an actual pure helper at long node nineteen has own five, finite at least seven or dotted, $g < 6/5$, with actual long dual $\sqrt{2}P_{19}$ and actual selected roots n_{19}, n_1, n_2. Let a*

paid classical vector W be supported through eighteen, have sum F and retained actual P_{18} demand F, constant first three entries a, no foreign coordinate, effective energy $L > .985613$, and remaining P_{18} coefficient $> .107$. If $\sqrt{2}F \le 38/7$ and $a/\sqrt{2} < .267$, its full actual foot contradicts the norm budget. This is a conditional local pressure lemma, not a classification of node nineteen.

Proof. The helper time is $A^2 = 38g^2 + 2g - 1/2$. Its actual finite or future cores, using the initial and own long roots, give

own	U	T
5	$g(\sqrt{2}P_{19} - n_{19}) + H_3$	$H(A, 2/3 - g^2)$
$7, 8$	$g(\sqrt{2}P_{19} - n_{19}) + H_2$	$A - \sqrt{g^2 - 3/4}$
≥ 9 finite	$g(\sqrt{2}P_{19} - n_{19}) + H_1$	$H(A, 1 - g^2)$
dotted	$g\sqrt{2}P_{19} - g^{-1}n_{19} + H_1$	$A - \sqrt{1 - g^{-2}}$,

where $H_k = P_k/(k\sqrt{2})$ and $x.U \ge T$. These are the actual correction/finite-projection recipes already proved, now at their physical index nineteen. They require no B short normal. In type D, $\sqrt{2}P_{20}$ occurring in the coordinate sum is the sum of the two actual fork duals, and both fork products remain nonpositive under each correction.

Put $\alpha = \sqrt{2}g$ and $V = U - \alpha P_{18}$. The four variances are $g^2 + 1/6, g^2 + 1/4, g^2 + 1/2$ and $2g^2 - 3/2 + g^{-2}$; all are < 2.1, the last increasing to $1867/900$ at $6/5$. The helper mass in the first three coordinates is exactly $1/\sqrt{2}$, so the *full* cross is

$$W.U = \alpha F + a/\sqrt{2}, \qquad \Delta = T - W.U > T - g(38/7) - .267.$$

The following complete scalar seals prove $\Delta > .19$. For own five use $A > 5.097$, $\sqrt{\delta} < .111, g < .81$; for seven use $A > 5.663$, radical $< .253, g < .902$; for eight use $A > 5.8119$, radical $< .323, g < .9239$. Each substituted difference exceeds .19. For finite $g \ge .939$, $H - g(38/7)$ increases because $A' > \sqrt{38}$, $H_A > 47/50$ and $H_\delta < 0$. At .939, $A > 5.906$, the other time radical > 5.916 and $\sqrt{\delta} < .344$ give $H > 5.58$, again sufficient. For dotted owns the deficit decreases on $(1, 6/5)$: $A' - \sqrt{38} < .1$, $\sqrt{38} - 38/7 < .74$ and the subtracted radical derivative exceeds one. At its enlarged endpoint,

$$\Delta > 7.5246 - .55278 - (6/5)(38/7) - .267 > .1905 > .19.$$

These are positive-square and whole-interval derivative bounds, not sampled native labels or a numerical maximum.

Pay $\mu = 1/16$ at the true endpoint-eighteen coefficient. Its price $\mu\alpha < .107$ is available. Adding the actual U pressure and completing the *same* square gives

$$e > L + 2\mu\Delta - \mu^2 V^2 > L + .015546875 > 1.0011598 > 1.$$

The nonzero $W.V = a/\sqrt{2}$ is already included in Δ; no separate early-block or head energy has been added. □

K.15 The whole B_{21} terminal

Proposition K.18. *The entire farthest forest B_{21} is impossible.*

Proof. Lemma K.7 supplies the complete whole .58 bulk through seventeen, with isolated dotted five retained. There is an actual strong fifteen/sixteen, so the full paid-reference contract of Lemma K.8 applies. At eighteen the right B_3 permits long-three or sealed short-three F_4. Own zero has its

actual affine F_4 principal and every positive own has ratio at least $1+1/\sqrt{2}$. Full endpoint bridges are directly strong, including own zero. Thus only pure helpers can be weak there.

At nineteen the right B_2 permits three at either endpoint. The long end is the same local row. At the short end, own zero/four/six/parallel have the actual F_4/star/path/null principals and owns at least seven are strong. Every full bridge has residual 1/38, with minimum ratio $\sqrt{38}(1/38+1/(2\sqrt{2}))>83/50$, including own zero. Own five has its actual left H_4 foot on nineteen/eighteen/seventeen, $\alpha=\sqrt{2}g+1/2$, $A^2=19\alpha^2-1/2$ and $R<19$. The squares $A/\alpha>4.337$ and $\sqrt{\delta_5}/\alpha<.083$ give $G>4.2$; it is flat through sixteen and is excluded by the early paid table.

Own three instead gives the same actual cap $v=(\omega_{19}+\omega_{21})/2$ at the finite A_{21} replacement of old twenty. Every original wall is retained and the released wall is strict. Its exact cosine $(20+19\sqrt{2})/\sqrt{22}>466/47$ gives $x.P_{20}>4.04, R=20$, since $2299(466/47)>22701$. The early fifteen/sixteen paid formula gives respectively $>1.00655, >1.00679$. This is an actual edge from the original vertex, not a farthest assumption at its mate. The complete helper-or-strong window therefore reaches nineteen.

One of eighteen/nineteen is strong. Weak helper five would give the earlier interpolation energy $>.74385+.2679>1$. For isolated dotted five, the same zero-sum residual reference at a strong $K=18,19$ has these effective lower bounds:

J	initial base plus gain	$\text{tail}_{18}>$	$\text{tail}_{19}>$
6	.720462	.2679	.2804
7	.757957	.2318	.2445.

Its endpoint-J payment was legal at fifteen and becomes easier. At strong eighteen it has $L>.988362$ or $>.989757$; at nineteen it already exceeds one. Its sum is F_K, endpoint coefficient $>.14$, and its first three coordinates are constant $a=F_J/J+\mu_0(1-5/J)<.377$.

If nineteen weak, eighteen strong. For strong five its ordinary concave reference has $L>.97119+.0154-1/1024>.985613$, sum F_{18}, first-five mean $<.366$ and endpoint coefficient $>1/9$. The isolated-dotted reference just described is stronger and also has first-three mean $<.377$. Both satisfy $\sqrt{2}F_{18}=5.42<38/7$ and $a/\sqrt{2}<.267$. Every remaining weak-nineteen cap is an actual pure helper with the required physical roots. Lemma K.17 therefore excludes it, keeping the full early residual cross.

It remains strong nineteen; weak helper five and isolated dotted five were already excluded in that branch. For strong five, $\mathcal{E}_{19}(.58)>1.00010$. If an internal helper occurs at six through nine, its actual $x_1>.45$ gain .00882 pays the full hole loss $<1/1024$. Otherwise every hole is at least ten; a helper's $x_1>.378$ gain $>.00018$ pays $<1/4096$. If there are no holes, no correction is needed. In every branch the legal positive reference has energy greater than one. □

K.16 Whole D_{21}: sum the exceptional feet before bounding their errors

Proposition K.19. *The entire farthest forest D_{21} is impossible.*

Proof. Use actual long roots through twenty and the other actual fork $n_{21}=(e_{20}+e_{21})/\sqrt{2}$. The genuine long duals stop at nineteen; the fork duals are $(P_{20}\mp e_{21})/\sqrt{2}$. The last coordinate is signed. This proof changes the loss to $c_1=4/7$ and writes $G_i=\sqrt{i}-c_1/\sqrt{2}$. We reclassify the weak directions at this threshold; no .58 classification is transported unchanged.

The sufficient ratio is

$$T_1(q)=7/4-\frac{q-4/7}{q(2q-4/7)}.$$

It increases on $q\ge\sqrt{10}$; positive-denominator comparisons give $T_1(q)<1.673$ through $q=\sqrt{38}<37/6$, $T_1(q)<1.63$ through $q=\sqrt{14}<15/4$, $T_1(4)<1.64$ and $T_1(\sqrt{12})<1.62$. The actual bulk

5..16 has retained $A_{i-1}+D_{21-i}$, with right rank at least five. Its complete finite-white list, zero-own principals and full rank-twenty-one bridges are already proved local geometric statements. There is no foreign component.

Near-three has the same increasing least ratio $[i(\phi+1)-1]/\sqrt{2i(i+1)}$. At five/six native owns at least seven are strong; at seven the own-five ratio is > 1.63, at eight it is $> 1.66 > T_1(4)$, and from nine it is > 1.68. Thus only near-three/own-five at five and six need points. Next-to-end white-three has least ratio $(i\phi/2+1)/\sqrt{2i}$, strong from six onward. At five retain its actual floor, rather than call it strong:

$$h_5 \geq \sqrt{10} - c_D, \quad c_D < .57549.$$

Indeed its cosine is at least $1 + 5\phi/2$, giving $h_5^2 \geq 10(5\phi)/(5\phi+4) > 6.6915 > 2.58679^2$; use $\phi > 1.61803$ and $\sqrt{10} < 3.16228$. Exceptional positions have ratio $2g + 3/8 > 1.673$. The far-four/near-four, right long-three, finite D fork and bridge ratios or actual principals are unchanged geometric rows; their lower bounds now exceed the displayed T_1 targets. In particular a fork at retained rank at least five has ratio $> (.8+.25)/\sqrt{3/8} > 1.673$ after its own-zero/three/four/six/null cases. Full endpoint bridges retain their positive-ratio own-zero branch or the actual affine cycle of the other orientation.

Isolated finite five and seven have the preceding losses $< 25/76+5/24$ and $< 1000/4779+9/25$, both below $4/7$. For finite owns at least eight, use the exact singleton projection. With $r = \sqrt{q^2-1}$ and $s = \tan(\pi/m) \leq \sqrt{2}-1$, its positive finite correction gives

$$h_i > r\sqrt{(r-s)/(r+s)}.$$

To derive this, select the actual cap and n_i. Their residual is $1-g^2 > 0$, the displacement after correction is $g(\omega_i - n_i)$, and the finite-point formula divided by g is $\sqrt{r^2-s^2}/(1+s/r)$, the displayed expression. Its loss decreases in q and increases in s. At $q = \sqrt{10}$, $r = 3, s < .415$, it exceeds 2.6 by $9(517/683) > 169/25$, giving loss $< 3.163 - 2.6 = .563 < 4/7$. Owns three/four/six/parallel retain their actual affine/null exclusions.

For isolated dotted own, $u = g^{-2} \in (0,1)$, the positive future core gives the actual pressure

$$x.(\omega_i - un_i) \geq T_i(u) = \sqrt{2i-u} - \sqrt{u(1-u)}.$$

At $i \geq 8$ its loss is at most $a_i u + \sqrt{u(1-u)}$, $a_i = 1/(\sqrt{2i}+\sqrt{2i-1}) < .128$, whose maximum is $< .5681 < 4/7$. Retain isolated dotted five/six/seven. Every other weak bulk row is the pure helper, with $\phi/2 \leq g < 6/5$; the upper cutoff is reproved from $4(6/5) - 2 = 14/5 > (1.673)^2$.

The two near-three points. The own-five near role at five was excluded outright by the actual right-H_4/old-floor argument in Lemma K.7. At six use the actual right correction $g(3n_6/2 + n_7 + n_8/2)$, $g = \phi/2$. Its finite point retains old five/nine and every unknown wall. For its displacement Y,

$$A^2 = 9/2 + 8\phi, \quad Y^2 = (41+61\phi)/8, \quad \sum Y_j = \sqrt{2}(5/2+3\phi),$$

giving $x.Y > 4.04, Y^2 < 17.463, \sum Y_j < 10.401$. Here $A > 4.1766$, $\sqrt{\delta_5} < .136$, $1.618 < \phi < 1.61804$ and $\sqrt{2} < 1.414214$ suffice. The valid old .61 bulk supplies a strong twelve/thirteen, with floors $> 3.03, > 3.17$. For $Y + (3/4)P_K$, the squared demands are $39.84765625, 41.18430625$, and the norm squares are $< 39.8145, < 40.377$. Both contradict Cauchy before a new tail or strong-five conclusion is used.

New weak helpers remain nonadjacent: their actual mutual product is $> 1-ab > -11/25$, hence zero, and the positive mixed-point loss is $< 7/16 + 1/(4\sqrt{10}) < 4/7$. Isolated dotted five/six/seven need not be nonadjacent. We now retain all their feet together.

The summed-foot identity and its complete error bounds. Let w be a concave formal prefix reference, with nonnegative coefficients β_i in the spatial duals ω_i. At each selected isolated dotted index use its actual pressure, put $Z = \sum \beta_i u_i n_i$ over those indices and $W = w - Z$. Since $w.n_i = \beta_i$ and distinct old root products are nonpositive, $Z^2 \le \sum \beta_i^2 u_i^2$. Summing the actual pressures and completing the single square gives effective energy

$$L_W \ge w^2 + \sum_i \{2\beta_i[T_i - (q_i - c_1)] + (2u_i - u_i^2)\beta_i^2\}.$$

An actual D-position-five floor instead costs $2\beta_5(c_D - c_1)$. These are pressure certificates for the same W, not separate scalar lower bounds to add later. All dotted corrections have support through eight and sum zero; they preserve the late endpoint coefficient and vector sum. Their mutual cross terms have a favourable sign, even for adjacent dots.

For the retained G_i-floor reference when five is not a weak helper, the actual slope differences give $\beta_5 > .1, \beta_6 > .012, \beta_7 > .0095$; the latter follow from $(d_5 - d_6)/\sqrt{2}$ and $(d_6 - d_7)/\sqrt{2}$. Omitting a neighbour increases them. The separate weak-helper-five interpolation below charges only its dotted-six correction, whose actual coefficient exceeds .10 and hence .012. It charges no dotted-seven pressure. These bounds concern the two specified references, rather than an arbitrary concave vector. Write $r(u) = u - u^2/2$. Each signed summand is $2\beta[c_1 - \mathrm{loss}(u) + \beta r(u)]$. For $u \le 3/4$, $r(u) \ge (5/8)u$. At five, $a_5 < .163$ and $\beta \ge .1$ bound the effective loss by $.1005u + \sqrt{u(1-u)} < .5528 < c_1$. At seven, $a_7 < .137, \beta \ge .0095$ similarly give $.1310625u + \sqrt{u(1-u)} < .56983125 < c_1$. For $u \ge 3/4$ the same expressions decrease, so both summands are nonnegative throughout their actual intervals.

At six the summand increases with $\beta \ge \beta_0 = .012$: its derivative is positive because $\mathrm{loss}(u) - 2\beta_0 r(u)$ is bounded by $.133u + \sqrt{u(1-u)} < .57095 < c_1$ on $u \le 3/4$, and decreases thereafter. At β_0 the effective loss is $< .1405u + \sqrt{u(1-u)} < .5752$, again with a decreasing tail. Its entire debit is therefore

$$2(.012)(.5752 - 4/7) = 99/1093750 < .000091.$$

All maxima used here are the elementary identity $\max[au + \sqrt{u(1-u)}] = (a + \sqrt{1+a^2})/2$. No individual dotted-six floor was assumed.

The early reference and the actual terminal prices. First suppose five is not a weak helper. Keep formal five, whether ordinary, D-position or isolated dotted, and keep dotted six/seven. Omit only nonadjacent weak-helper floors. Through a strong fifteen/ sixteen their entire loss is bounded by the five decreasing terms $j = 6, 8, 10, 12, 14$, whose sum is $< .000422$. For D-position five, $\beta_5 < .11425$: the first mean is $< .366402$ and the next retained slope, at six or seven, is $> .20484$. Thus its separate debit is $< 2(.11425)(.57549 - 4/7) < .000929$. The common initial-block change from .61 adds

$$(27/700)\sqrt{2/5} - [(.61)^2 - (4/7)^2]/10 > .0198377.$$

The old exact $E_{15} > .92564, E_{16} > .94176$, minus all three conservative errors, give a paid W supported through K, of sum G_K and effective energy

$$L_W > Q_{15} = .944035, \qquad L_W > Q_{16} = .960155.$$

Subtracting both dotted and D-position errors is conservative even when their five cases cannot coexist. The true individual P_{15} coefficient is at least $d_{14} > 9/70$, and P_{16} at least $d_{15} > 1/8$; the early corrections do not change them.

If five a weak helper, its actual scalar point gives $x_1 > .530$: subtract at most $(.58 - 4/7)\sqrt{2}(6/5)$ from .5455. Six is strong or isolated dotted. Interpolate through $(1, .530), (6, G_6), (K, G_K)$ and use

its actual summed dotted-six foot if necessary. The first-six energy is $> .74018$ because $G_6 > 2.0454$; the fifteen/sixteen tails are $> .225, > .2404$. Even after .000091 these exceed both Q_K. The endpoint coefficient $1/(\sqrt{K} + \sqrt{6})$ also exceeds the same required prices. Thus every initial branch has been covered.

For an actual normalized pressure flat through sixteen, with demand G and full norm square $\leq R$, the positive-dual payment gives

$$e > L_W + 2\tau(G - G_K) - \tau^2(R - K).$$

Use $G_{15} < 3.468923, G_{16} < 3.595940$. The legal price is $\tau = 1/8$ at both endpoints, except native-eight at the centre, where $\tau = 9/70$ is used at fifteen. Effective energy is never identified with W^2.

Actual D4 at seventeen and D3 at eighteen. At seventeen, the complete D_4 leaf-three/bridge classification and the old affine principals leave fork-three owns five/seven/eight. Owns at least nine have ratio $\sqrt{2}(\cos(\pi/9) + 1/4) > 1.68 > 1.673$. Own five has its actual left H_4 point, $A^2 = 17\alpha^2 - 1/2$, $G > 4, R < 17$, flat through fourteen. It uses the old .61 early eleven/twelve contract in both initial branches, not the new sixteen-flat formula. Its two totals exceed 1.04. For seven/eight use the actual future core

$$c + g^{-1}n_{17} + (2g)^{-1}n_{16}, \quad k = 1 - 3/(4g^2) > 0.$$

The cap/sixteen products vanish and the seventeen product is negative. For $.9 < g < .924$, $A/\alpha > 4.09$, $\sqrt{k}/\alpha < .21$, so $G > 3.88, R < 17$. It is flat through fifteen, and the old eleven/twelve price $1/7$ gives both totals > 1.0086. No incorrect sixteen-flatness has been used.

At eighteen the right D_3 finite list is middle-three, end-three/ four/five, with every full bridge retained. Middle-three and end-four have their actual affine/direct alternatives; end-five is directly strong even at own zero. A full middle bridge has residual $1/36$ and is directly strong at zero own. An end-three bridge has residual $1/8 + 1/36$, zero-own cycle and positive-own ratio > 1.9. The no-left end-three case has the old own-zero/three/four/six/null principals. Own five has its left H_4 point, $G > 4, R < 18$, flat through fifteen; the old early reference gives both totals > 1.022. Owns $g \geq 43/40$ are directly strong. For $.9 < g < 43/40$ use $c + g^{-1}n_{18} + (2g)^{-1}n_{17}$, with

$$A^2 = 18\alpha^2 - 5/8, \quad A/\alpha > 4.21, \quad k/\alpha^2 < 519200/5193841 < .317^2.$$

This actual core is flat through sixteen, $G > 3.893, R < 18$. The new paid formula exceeds 1.0031 at both early endpoints. Thus eighteen has only helper or strong as well.

The complete actual centre nineteen. Deleting nineteen retains $A_{18} + 2A_1$. With own g, one left contact η at j and fork contacts δ, ϵ, the data are

$$\Delta = 1 - q_L - \delta^2 - \epsilon^2, \quad \sigma = g + \eta j/19 + (\delta + \epsilon)/2, \quad C = \sqrt{38}\sigma/\sqrt{\Delta}.$$

The complete finite-white list is the same actual high-rank tree list already proved: left end-three/four or next-to-end-three; with both forks only the full endpoint D_{21} bridge; with one fork only left endpoint-three and fork-three/four. The full and fork-four bridges have residual $1/38$ and minimum numerators $1/2 + 1/38$, $\sqrt{2}/4 + 1/38$, directly strong. Fork-three bridges have residual $21/76$; own zero/three give the cycle/triangle and all larger owns have ratio > 1.8. With no fork contact, the rechecked left and isolated inventory applies. Its helper feet use $\omega_{20} + \omega_{21} = \sqrt{2}P_{20}$ only as a spatial identity, retaining both actual forks.

With no left contact the white list is one arbitrary finite fork, or pairs $(3,3),(3,4),(3,5)$. At own zero, fork-three/four/six have their actual affine principals. Fork-five uses the actual left H_4 point, $R=18$, $A^2/\alpha^2=5+8\phi$ and $\sqrt{\delta_5}/\alpha<15/56$, giving $G>3.96$. Its corrected rational premise is $\phi>79/49$, as proved above.

For fork at least seven, put $b=\delta^2$. The actual core $c+\delta^{-1}n_f+2(\delta^{-1}-\delta)n_{19}$ has selected products zero/negative, positive time and negative square. Its full signed data are

$$R=29-16/b+6/b^2,\quad G=\sqrt{21-2/b}-\sqrt{-8+14/b-6/b^2}.$$

Monotonicity of R and the quadratic-radical maximum give this complete native split:

fork label	b	$G>$	$R<$ or $\leq$
7	$(.81,.82)$	3.91	18.42
8	$(2+\sqrt{2})/4$	3.91128	18.49034
9	$(.88,.89)$	3.925	18.60
≥ 10	$(.9,.95]$	3.948	18.81
≥ 10	$[.95,1)$	4.04	19.

For seven the two roots are $>4.304,<.388$; for nine, $>4.327,<.402$; for the last two ranges, $>13/3,<.385$ and $>4.346,<.298$. The original native cosine bounds follow from their triple-angle polynomials and positive squares. The critical eight row is not replaced by a coarse continuous bound:

$$R=109-64\sqrt{2},\qquad G=\sqrt{13+4\sqrt{2}}-\sqrt{68\sqrt{2}-96}.$$

The rational enclosure $1.41421356<\sqrt{2}<1.41421357$ gives roots $>4.319358,<.408078$ and the displayed R upper bound. At fifteen the price $9/70$ gives energy >1.00008648; at sixteen the price $1/8$ gives >1.00007843. Every other row gives both totals >1.00085.

At own three, fork-three gives the triangle. Fork-four/five instead gives the actual finite D_{21} contacted-fork endpoint, $C_f=(19+21\delta)/2>16$, signed direction $\sqrt{2}\omega_f$, $R=21, G>\sqrt{315/17}>4.30$. Fork at least six is direct. Own four/six/parallel retain their actual principals. For own at least five and fork at least four, only own-five/ fork-four is not directly strong; its left H_4 point has $\alpha=\sqrt{2}g+1/2$, $A^2=19\alpha^2-1/2$, $G>4.25, R<19$. Own-five/fork-three has $A^2=19\alpha^2-3/4$, $G>4.2, R<19$. Their lowest selected old root is seventeen, so both are genuinely flat through sixteen. For remaining finite/dotted own with fork-three, $g\geq 6/5$ is direct; on $.9<g<6/5$ the actual core $c+g^{-1}n_{19}+(2g)^{-1}n_{18}$ has $R<19$, $A/\alpha>4.326$ and $\sqrt{k}/\alpha<.34$, hence $G>3.98$.

For two forks, own three has its triangle and larger owns are direct. At own zero pair-three-three has its square; pair-three-four has an actual finite B_{21} mate with $C_f=(38+21\sqrt{2})/4>16$, again $G>4.30, R=21$. Pair-three-five has its actual finite H_3 foot with $R=19, q=2\phi^{-6}<1/8$ and $G>\sqrt{19-1/8}/(1+\sqrt{1/152})>4$. Every row is excluded by the stated positive-dual early formula. The actual list is exhausted and nineteen is helper or strong.

The final summed reference. One of eighteen/nineteen is strong. Weak helper five gives effective energy $>.74018+.2679-.000091>1.0079$, so is impossible. Otherwise the same formal reference and summed-dot correction apply through that endpoint. Their finite nonadjacent hole loss is $<.00044$, by appending the sixteen/eighteen terms as needed. The same .000091 and .000929 errors remain valid. Consequently

$$L_{18}>.9895677,\qquad L_{19}>1.0030777.$$

Thus nineteen cannot be strong. In the weak-helper case at nineteen, the paid eighteen reference has sum G_{18}, true retained demand G_{18}, first-three mean $a = G_5/5 < .367$ and unchanged endpoint coefficient $> d_{17} > 1/9$. The summed dots only affect coordinates five through eight. Hence $a/\sqrt{2} < .260 < .267$ and $\sqrt{2}G_{18} = 38/7$. The actual helper uses the physical long n_{19} and both signed fork directions; its P_{20} coordinate sum is not declared a long old dual. All hypotheses of Lemma K.17 now hold, with the stronger $L > .9895677$. Its single paid full-foot square contradicts the norm budget. All summed errors and terminal prices have been charged once, with no arithmetic or unproved dotted floor. □

K.17 The complete exact-twenty-one cover and its consumers

Proposition K.20. *Dimension twenty-one is impossible.*

Proof. The common ordinary/debit costs exclude a forest without a dominant block, and two dominant blocks cost more than one. After establishing uniqueness, write the dominant rank as r and spectator rank as $S = 21 - r$. The complete three spectator budgets remain $S/14, (S-1)/14, .09 + (S-2)/14$, with their actual incompatible defect definitions. A dominant A has $S \le 7$ and hence $14 \le r \le 20$. Modern written component bounds make its lower ranks immediate: use $5/7, .826, .86, 6/7, .91$ at $r = 14, 15, 16, 17, 18$ respectively. With the weakest short-discard spectator costs $6/14, 5/14, 4/14, 3/14, 2/14$, every total is strictly greater than one. Typed and fully ordinary cases are stronger. Thus the older detailed rank-fifteen/sixteen spectator partitions need not be reproduced as separate consumer arguments.

At A_{19}, an ordinary irreducible rank-two remainder costs $> 1/7$ beside $> 6/7$. Two ordinary A_1's cost $> .2$; actual $I + A_1 > .2$ is a single joint bound, and J is stronger. Both remaining typed exceptions are exactly Lemma K.13. At A_{20}, the sole ordinary or actual-I rank-one remainder is excluded by Lemma K.14; J and typed exceptions need two ranks. This exhausts the A side at its own rank.

For B/D, $S \le 5$ and $r \ge 16$. The valid D-admissible $M = r - 6$ floors at sixteen/seventeen/eighteen are $> .823735, .8472225, .8689525$. Their weakest short-discard spectators cost $4/14, 3/14, 2/14$, already too much. At B_{19}, its genuinely valid $M = 14$ improved floor $> .91211125$ closes an ordinary rank-two block, a two-A_1 remainder with its at least one nonshort cost $> .1$, or either typed block $> .09$. At D_{19}, its valid $M = 13$ floor $> .88864$ closes an ordinary irreducible remainder. A two-A_1 remainder costs $> .2$ if ordinary, $> .2$ jointly if it contains actual I, or $> .34$ if it is J. The two typed cases are the separately proved actual D4 continuation in Lemma K.16.

At rank twenty even the weaker common floor $> .90715$ contradicts an ordinary nonshort A_1 cost $> .1$. The remaining entire $B_{20} + I$ is covered by Proposition K.6; $D_{20} + I$ is covered by Lemma K.16, after its own missing D4 layer was supplied. At rank twenty-one there are no spectators. Propositions K.18 and K.19 exclude those whole diagrams. Every finite farthest forest of rank twenty-one is therefore impossible. There is no appeal to face inheritance or to a necessary cover computed at another dimension. □

K.18 The proper classical forests in exact dimension twenty

Lemma K.21. *The entire forests $B_{18} + N_7/O_7$, $D_{18} + N_7/O_7$, $B_{19} + I$ and $D_{19} + I$ are impossible in dimension twenty.*

Proof. We use the old loss .61 and true actual-rank continuations. For B_{18}, the valid bulk reaches fourteen. At fifteen the right B_3 permits long-three or the sealed short-three F_4. The latter has

residual $1/4$, numerator $g + 1/(2\sqrt{2})$: own zero has its actual affine principal and every positive own has ratio at least $1 + 1/\sqrt{2}$. All full endpoint bridges are finite B_{18}, foreign-sealed and directly strong, including their zero-own ratio. Thus coverage/nonadjacency reach fifteen.

For D_{18}, the valid bulk reaches thirteen. At fourteen the complete right D_4 list and its affine/direct branches leave only pure own-five/fork-three. Its actual left H_4 point has $\alpha^2 = \phi + 5/8$, $A^2 = 14\alpha^2 - 1/2$, $R < 14$ and $G > 3.62$, using $\alpha^2 > 56/25$ and residual ratio $< .091^2$. It is flat through *eleven*, so choose an actual strong ten/eleven, not twelve. The early positive-dual formula and the separate typed energy $> .09$ give both totals > 1.085. Both initial-five branches admit those same lower reference bounds. The finite white list retains both forks, every foreign cost and every full bridge; the non-direct role has $Q = 0$ by the strong foreign estimate, not by an assumption about the typed block.

At fifteen the right D_3 permits middle-three or end-three/four/five. Middle and end-four have their actual affine/direct branches; sealed end-five is strong even at own zero. Full bridges retain their zero-own cycles and direct positive-own bounds, with no finite extra branch at the internal cap. The end-three no-left role has residual $5/8 - Q$, numerator $g + 1/4$; positive Q and $g \geq 1$ are strong and own zero/three/four/six/parallel have their actual principals. Own five gives its actual left H_4 pressure $G > 3.74, R < 15$, flat through twelve. For finite $.9 < g < 1$, average both actual A_2 cores on sides fourteen/sixteen, retaining their full cross $8g^2/9$. With $d = 4g^2/3 - 1$ their data are

$$A^2 = 15\alpha^2 - 5/8, \quad Y^2 = A^2 + 1 - 14g^2/9 < 15\alpha^2, \quad x.Y \geq A - \sqrt{d}.$$

The bounds $A/\alpha > \sqrt{15 - 125/529} > 3.842$ and $\sqrt{d}/\alpha < 1/3$ give $G > 3.5, R < 15$. At actual strong eleven/twelve, the pure-point formula plus the orthogonal typed cost gives both totals > 1.099 or > 1.030, respectively. Each core is a positive combination of actual walls, future feasible against the entire typed block and every other wall. Thus through fifteen only pure helpers can be weak.

The prefix-gain *calculation*, now with that actual extended coverage, gives $e_{B/D} > .90715 + .00496125 = .91211125$. If five weak its actual reference is stronger; if fifteen strong the extra increment pays the gain; otherwise the actual weak-fifteen packet supplies the legal $x_1 > .424$ modification at strong fourteen. The physical long roots through sixteen exist in both components. We have not applied the old $M = 14$ theorem below its original ranks. Adding either typed cost $> .09$ excludes all four typed forests.

For entire $B_{19} + I$, the valid bulk reaches fifteen. Its actual B_3 layer at sixteen has the same long-three/sealed-F_4/full-bridge list as just proved, now at its physical rank; all remaining weak caps are pure helpers. Their actual nonadjacency extends through sixteen. The scalar argument in Proposition K.6 therefore applies after this independently supplied geometric window: weak five has the paid $> .9851486$ reference and unused-I gain $> .015$; strong five has the flat fifteen/sixteen reference $q > .9240775$, true P_5 demand f_5 and individual price $> .146$. Deleting the head retains actual B_{18}, with precisely the isolated/long-two-three whitelist and only the old I outside. The q-free local head/I gain $> .07665$ gives the contradiction. This is a proof at supplied actual geometry, not the generic $B_r \geq 20$ theorem asserted at $r = 19$.

Finally consider entire $D_{19} + I$. Lemma K.15 first supplies coverage through fifteen. At sixteen the right D_3 has the same complete finite, foreign, affine and bridge list as above. Only pure own-five or finite $.9 < g < 1$ in the end-three row remain. Own five has its actual left H_4 point, $G > 3.87, R < 16$, flat through thirteen; the bare eleven/twelve formula already exceeds 1.026. For finite owns use both actual A_2 cores on sides fifteen/seventeen, with their full cross, $A^2 = 16\alpha^2 - 5/8$, $R < 16$ and $G > 3.63$, since $A/\alpha > \sqrt{16 - 125/529} > 3.97$ and the radical ratio is $< 1/3$. If five weak, its actual eleven/twelve references have energies $> .9104, > .9314$, so their pure-point totals

exceed 1.020. If five strong, the flat-five reference can first use the single initial-I adjustment: from $t + (3/2)s > 3/4$ its joint gain is

$$s^2 + (5/4)(t - .361)_+^2 > 151321/3050000 > .0496,$$

with a legal clip at .6 when necessary. Its support first five plus I is orthogonal to the actual tail residual. It uses only the available P_5 price, leaving the early terminal price intact. The pure-point formula plus this one joint gain gives both totals > 1.007. Thus coverage/nonadjacency reach sixteen in either branch.

The same independently checked scalar bootstrap and flat-reference argument now prove five strong and give $q > .9240775$. Deleting long one retains actual D_{18}, so the same local head/I kernel, its selected long roots and both signed forks apply and give $q + .07665 > 1$. Earlier tail contradictions were used only to remove their branches; no spent I gain is carried into this new final reference. This whole-forest argument stops before centre seventeen; its older mixed-H_4 catalogue remains preserved but is unnecessary for this composition with the later local head theorem. □

K.19 The two rank-one remainders beside A_{19}

Lemma K.22. *An actual A_{19} has projection energy $> 563/625 = .9008$, with arbitrary finite spectators. In particular it cannot coexist with an ordinary nonshort A_1.*

Proof. Suppose $e_A \le .9008 < .91$. The complete $N = 20$ window and near-four comparison above apply. Zero or one weak end already has reference energy $> .9522798$ or $> .9233277$. Thus both five/fifteen are weak. The paid rational reference w has energy $E > .8944 = 559/625$, first six coordinates $H_6/(6\sqrt{2})$ and last six their negatives, where $H_6 = 2.288$. Its internal weak holes have already been paid. For the actual weak cap at five, either helper assignment gives

$$A^2 = (15/2)\gamma^2 + C\gamma - 21/40, \quad C \in \{1/2, 3/2\}, \qquad K = w.v = \frac{143}{750}(10\gamma + 1).$$

The positive corrections below use only old n_3, n_4, n_5, so have zero product with this flat reference. For $Y = v - u$ we therefore have $w.Y = K$.

For a finite selected correction, $S^2 = Y^2 = A^2 + \delta$, and its actual finite projected point gives the normalized pressure $x_A.(Y/S) \ge g_0$, with $g_0^2 = (S - \sqrt{\delta})/(S + \sqrt{\delta})$. For a future core of square $-d$, $S^2 = A^2 - d$ and $g_0 = (A - \sqrt{d})/S$. We prove the full overlap gap $g_0 - K/S > .08$ on every own range.

Native five uses the actual left H_4 correction $\gamma(3n_5/2 + n_4 + n_3/2)$, with $\delta = (5 - 3\phi)/8$. The smaller correlation $C = 1/2$ gives $S^2 = 79/40 + 7\phi/4 > 2.19^2$, $\sqrt{\delta} < .14$ and $K < 13013/7500$. Hence $K/S < .8$ and $g_0^2 > 93/107 > .93^2$, a gap $> .13$. Native seven uses $\gamma(4n_5/3 + 2n_4/3)$, $d = 4\gamma^2/3 - 1 > 0$. Its selected products are zero/negative and its time is positive. From $.9 < \gamma < .902$ we have $A > 2.449$, $\sqrt{d} < .292$, $K < 1.911$ and $S^2 = (37/6)\gamma^2 + C\gamma + 19/40 < 977/120 < 9$. Thus $(A - \sqrt{d} - K)/S > .246/3 > .08$.

Every finite own at least eight uses the actual finite pair correction $u = \gamma n_5$, $\delta = 1 - \gamma^2$. Here $S^2 = (13/2)\gamma^2 + C\gamma + 19/40 > 2.5^2$, $\sqrt{\delta} < .4$, so $g_0^2 > 21/29 > .85^2$. The exact lower overlap polynomial is

$$.77^2 S^2 - K^2 \ge \frac{1573}{7200}\gamma^2 - \frac{387563}{900000}\gamma + \frac{4414927}{18000000} > .218(\gamma - 1)^2 + .005\gamma + .027 > 0.$$

Thus $K/S < .77$ and the gap is $> .08$ on the whole finite interval.

For dotted own $1 < \gamma < 27/25$, use the actual future core $c + \gamma^{-1} n_5$. It has zero cap product, negative old-five product and positive time. Now $S^2 = (15/2)\gamma^2 + C\gamma - 61/40 + \gamma^{-2} > 7.325 > 2.7^2$ and $\sqrt{d} < .38$. Since $A > S$, $g_0^2 > 58/77 > .86^2$. Multiplying the overlap difference by γ^2 gives the bound

$$\gamma^2(.77^2 S^2 - K^2) > .0319 + .069(\gamma - 1) + 2.632(\gamma - 1)^2 + 2.813(\gamma - 1)^3 + .811(\gamma - 1)^4 > 0.$$

The rounded quartic coefficients come from $146047/180000, -387563/900000, -16929473/18000000$, $.5929$, so every coefficient comparison is rational. This gives a gap $> .09$ for the entire dotted interval.

All selected finite points or future cores are feasible against every actual wall by their positive corrections, and have pure A displacement because $Q = 0$ was proved in the inventory. For their unit $y = Y/S$, pay $\mu = .08$ in the same vector:

$$e_A \geq E + 2\mu(g_0 - w.y) - \mu^2 > 559/625 + 4/625 = 563/625.$$

This contradicts the contrary target. A separate nonshort A_1 costs $> .1$, giving total > 1.0008. □

Lemma K.23. *An actual A_{19} cannot coexist with an isolated I, with arbitrary further finite spectators.*

Proof. Write the actual I parameters as s, t, so $y_I \geq s$, $t + (3/2)s > .767$ and both normalized end heights are at least t. Put $a = \sqrt{19/20} > .974$, giving $x_1, -x_{20} \geq at$. We first remove every near-four without assuming $e_A < .91$. Its actual mate has index $j = 4, 5$ and height $> .86$; at the opposite end there is a mate or a strong $k \geq 14$ of height $> .77$. Except $(j, k) = (5, 14)$, the exact dual correlation is at most $1/3$. The positive vector $(543/800)u_j + (435/800)u_k$ has pressure $> 80193/80000$ and norm square at most that same value. One completed square gives energy greater than one.

At the exceptional pair, the actual heights exceed $2.355, 2.231$; their positive squared margins are $.000975, .002999$. The concave reference through those heights at five/fourteen has

$$E = \tfrac{1}{2}[2.355^2/5 + .124^2/9 + 2.231^2/6] > .97.$$

If $s \geq .18$, its orthogonal I energy already gives > 1.0024. If $s < .18$, the barrier gives $t > .497$ and $-x_{20} > .484$. The last-six mean is $< .264$; its legal endpoint modification adds $> (6/5)(.484 - .264)^2 = .05808$. The remaining mean is $> .217$ and the middle slope $< .01$, so the new reference stays concave. This also exceeds one. Reflection treats every orientation. The complete weak window is therefore pure-helper and nonadjacent.

Use the same full rational curve and omission table at $N = 20$. If zero or one ends are weak, its energy is at least the weaker bound $> .9233277$. The two endpoint block means are $< .302$ and each length is five or six. Put $z = (at - .302)_+$. Their disjoint legal changes add at least $(12/5)z^2$. Weighted Cauchy with the single I height gives

$$s^2 + (12/5)z^2 > \frac{(a \cdot .767 - .302)^2}{(9/4)a^2 + 5/12} > \frac{.445058^2}{511/200} > .0775.$$

The exact denominator is $613/240$; the last cleared margin is $.000064123364$. Thus both end indices must be weak.

At five write the actual cap as $v = \gamma\omega_5 + \omega_e/2$, $e = 1$ or nineteen, with $4/5 < \gamma < 27/25$ and $Q = 0$. Its time is $A^2 = (15/2)\gamma^2 + C\gamma - 21/40$, with $C = 3/2$ for the near helper and $1/2$ for the

far helper. Native five uses its actual H_4 projection with pressure $H(A, 2/3 - \gamma^2) > A - .111$. For finite owns at least seven, the positive actual core

$$c + \gamma n_5 + 2(1 - \gamma^2) n_e$$

has cap/five products zero, helper product $3/2 - 2\gamma^2 < 0$, square $-(1 - \gamma^2)(4\gamma^2 - 3)$ and positive time. Its radical is at most $1/4$. For dotted own use $c + \gamma^{-1} n_5$ and radical $\sqrt{1 - \gamma^{-2}}$. Positive old corrections allow the common left pressure $\gamma h_5 + h_e/2$ to retain each actual demand. Every other wall, including the actual I wall, is still satisfied.

A near helper forces $x_1 > .65$ on all three ranges. For native five use $A > 2.365, \gamma < .81$ and $\sqrt{2}[2.365 - .111 - .81(2.129)] > .65$. For finite owns at least seven, $A - \gamma(q - .61)$ increases from $\gamma = .9$, where $A > 2.6267$; subtracting its radical $\le .25$ still gives $x_1 > .65$. For dotted own the deficit decreases through $1 < \gamma < 1.08$: $A' < 3$, the radical derivative exceeds two, and $q - .61 > 2.12$. At 1.08, $A > 3.1373$, radical $< .37771$, so $\sqrt{2}[3.1373 - .37771 - 1.08(2.129)] > .6508$. The weak end has six-block mean $< .270$; this endpoint gain is $> .17328$, impossible beside $> .8944026$. Neither weak cap can therefore have a near helper.

For either far-helper cap, divide its full actual pressure by $\gamma > .8$. Native five gives $(2.188 - .111)/.81 > 2.39$. Finite owns at least seven have increasing A/γ, initially $\sqrt{200/27} > 68/25$, and radical ratio at most $5/18$. For dotted own, $A/\gamma - \sqrt{\gamma^2 - 1}/\gamma^2$ decreases; its first derivative term is $< .11$ and its subtracted derivative > 1. At 1.08 it exceeds $(2.9602 - .37771)/1.08 > 2.39$. Since $h_e > 0$ and $1/(2\gamma) < 5/8$, both actual feet give

$$h_5 + (5/8) h_{19} > 2.39, \qquad h_{15} + (5/8) h_1 > 2.39.$$

The full finite and continuous dotted alphabets have been covered.

To spend these pressures together, set $A_0 = .556, B_0 = 2.0425$, $C_0 = 2.288, r = 5/8$; then $B_0 + r A_0 = 2.39$. Build a symmetric reference with heights A_0, B_0, C_0 at one/five/six, then the old central heights and reflection. Its initial slopes are $.556, .371625$ repeated four times, $.2455, .118, \ldots$, decreasing. Delete actual internal weak knots seven through thirteen; their entire loss is $\le .001296$. The positive end coefficients in the spatial duals are

$$\begin{aligned} \beta_1 &= \frac{5A_0 - B_0}{8} = 59/640, \\ \beta_5 &= \frac{5B_0 - A_0 - 4C_0}{8} = 1009/16000, \\ \eta &= \beta_1 - r\beta_5 = 1351/25600 > 0. \end{aligned}$$

All remaining coefficients use actual strong floors. Direct rational calculation of that same central segment gives energy $\ge .021912$, so the reference w_A has

$$E_A \ge A_0^2 + (B_0 - A_0)^2/4 + (C_0 - B_0)^2 + .021912 = .9437388125.$$

Let $\lambda = \sqrt{19/10} > 1.3784$. The single actual-$I$ barrier gives $h_1 + h_{19} + 3\lambda s > 2\lambda(.767)$. Use the one full vector $W = w_A + 3\eta\lambda e_I$. Its two foot coefficients are β_5, and the remaining end coefficient is exactly η times that one barrier. Consequently

$$x.W > E_A + 2\eta[\lambda(.767) - A_0], \qquad W^2 = E_A + (171/10)\eta^2.$$

The reference values A_0, B_0 are used through these positive combined pressures. No separate endpoint floor is required there. The rational bounds $4\eta[\lambda(.767) - A_0] > .105807111375$ and $(171/10)\eta^2 < .047625$ give

$$\|x\|^2 \ge 2x.W - W^2 > .9437388125 + .105807111375 - .047625 = 1.001920923875 > 1.$$

This is one spatial certificate and one I coordinate. Every internal hole and every foot overlap has been charged in that vector. □

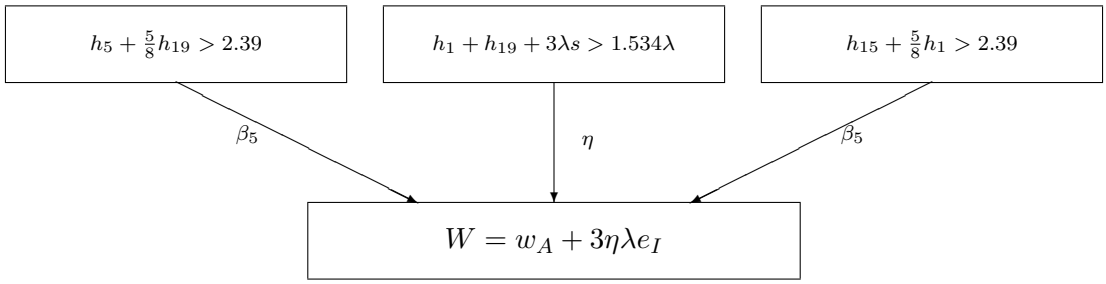


Figure 2: The two far-helper pressures and the single actual-I barrier fund one vector. The arrows record positive pressure coefficients, rather than Coxeter bonds.

K.20 The whole twenty-dimensional A row

Lemma K.24. *The entire vertex forest cannot be A_{20}.*

Proof. Use the $N = 21$ notation of Lemma K.2. That lemma and the rational central curve apply before any contrary energy assumption: the weak internal knots are nonadjacent pure helpers, and the near-four roles have already been excluded by norm. Put $C = 2.317$ and $B = 2.150$. According as the end knots five and sixteen are both weak, just one is weak, or both are strong, the concave reference energies, including every actual internal hole, are

$$E_2 > .92274, \qquad E_1 > .9509563, \qquad E_0 > .97977675.$$

These are the $N = 21$ curve rows above. No nonexistent strong end is inserted into the reference.

An actual weak cap at five has own γ and a simple helper at one or twenty, with

$$A^2 = (160\gamma^2 + c\gamma - 11)/21, \qquad c = 32 \text{ (near)}, \quad c = 10 \text{ (far)}.$$

The native-five correction is $\gamma n_5 + (2/3)n_e + (1/3)n_{e'}$, with finite residual $2/3 - \gamma^2$. For finite owns at least seven use the core $c + \gamma n_5 + 2(1 - \gamma^2)n_e$, of negative square $-(1 - \gamma^2)(4\gamma^2 - 3)$; for dotted own use $c + \gamma^{-1}n_5$. All coefficients are positive. The finite row uses the projected point, the other rows a future negative core. Every original wall inequality therefore holds by acuteness. Their pressures on $U = v - w_{\text{old}}$ exceed, respectively,

$$A - .111, \quad A - \sqrt{(1 - \gamma^2)(4\gamma^2 - 3)}, \quad A - \sqrt{1 - \gamma^{-2}}.$$

The finite and continuous dotted domains are those of the window lemma; the latter has $\gamma < 1.08$.

A near helper forces $x_1 > .64$. For five use $A > 2.365, \gamma < .81$. For finite owns at least seven, $A - 2.151\gamma$ increases from $\gamma = .9$, where $A > 2.649$; its radical is at most .25. For dotted own the deficit decreases through 1.08, where $A > 3.1636$ and the radical is $< .37771$. Substituting these bounds into the actual pressure proves the .64 floor throughout. The weak-end six-block mean is $< .274$. Its endpoint graft gains $> (6/5)(.64 - .274)^2 = .1607472$, already impossible beside E_2. Every weak end thus has a far helper.

Suppose both ends are weak. The reference is flat on its first and last six coordinates, with endpoint coefficients $\geq 1549/12000$. The projection of a far foot is $(5\gamma/6)\omega_6 + \omega_{15}/12$. The residual

Y has zero sum on both blocks and vanishes in the middle, hence $w.Y = 0$. Its variance and projected deficit satisfy

own	Y^2	Δ lower
5	$(2/3)\gamma^2 + 1/12 < 1.1$	.30
finite ≥ 7	$4\gamma^4 - (16/3)\gamma^2 + 29/12 < 1.1$	.30
dotted	$(5/3)\gamma^2 - 19/12 + \gamma^{-2} < 1.22$	.32

For five use $A > 2.2$, radical $< .111$, giving $\Delta > .3319$. For the other finite owns split at $.91, .92$: lower times $2.4649, 2.4937, 2.5225$ and upper radicals $.2318, .2434, .25$ give deficits $.302266, .300158, .30305$. The dotted deficit decreases through 1.08, where $A > 2.9794$ and radical $< .37771$ give $> .3233$. These are complete intervals.

Add .13 times both residuals to one reference. At either anchor the combined cost is $< .13(59/60) = 767/6000 < 1549/12000$. Their cross is retained using $\|Y_L + Y_R\|^2 \leq 2(Y_L^2 + Y_R^2)$. With zero, one or two dotted ends the gains are $> .08164, .082784, .083928$. Each added to E_2 exceeds one.

If just five is weak, the six-block and opposite five-block coefficients are at least $1549/12000$ and .1315. The far-foot projection becomes $(5\gamma/6)\omega_6 + \omega_{16}/10$. Its variance decreases by $1/60$ and its deficit by $B/10 - C/12 = 263/12000$. Thus nondotted owns have $\Delta > .278, Y^2 < 1.1$, and dotted owns have $\Delta > .30, Y^2 < 1.22$. Payment .14 costs $< .126$ and .014 at the two anchors and gains $> .05628$ or .060088. Both contradict E_1.

Both ends are now strong. The reference has first and last five-block value $b < .305$, and both anchor coefficients $> .1315$. Releasing the first root retains A_{19}. Finite classification gives isolated, simple-three at $2, 3, 19, 20$, or four at $2, 20$. Two contacts make a cycle; exceptional finite ranks are too small. The complete nonisolated edge-cosh list is

contact	white residual	$\cosh L_1$
3 at 2	21/40	$(40\gamma + 19)/21$
3 at 3	1/10	$(40\gamma + 18)/(2\sqrt{21})$
3 at 19	1/10	$(40\gamma + 2)/(2\sqrt{21})$
3 at 20	21/40	$(40\gamma + 1)/21$
4 at 2	1/20	$(40\gamma + 19\sqrt{2})/\sqrt{42}$
4 at 20	1/20	$(40\gamma + \sqrt{2})/\sqrt{42}$

The minimum allowed owns are $1/2, 0, 1/2, 1/\sqrt{2}, 0, 1/2$; smaller owns have time-zero or actual affine/null obstructions. Every row forces $x_1 > .396$. All but far-three have cosh > 1.5; that last squared floor is $200(\sqrt{2} - 1)/[21(10\sqrt{2} + 11)] > .396^2$. It reduces to $10441790\sqrt{2} > 14764031$, certified by $\sqrt{2} > 707/500$. If both ends are nonisolated, their disjoint grafts gain $2(5/4)(.396 - .305)^2 = .0207025$, contradicting E_0.

Reflect an isolated end to the left. Its time is $A^2 = (40/21)\gamma^2 - 1$. Own zero, three and four have nonpositive time square; six has the affine six–three path; parallel has the positive feasible null sum. Only five, finite owns at least seven, or dotted remain.

For five take $\gamma(3n_1/2 + n_2 + n_3/2)$. Its finite residual is $1 - 3\gamma^2/2$, and its displacement is $\gamma\omega_4/4$. Its projected-point pressure exceeds .393, since $A > .4966$, the second time root is $> .5146$, and the residual root is $< .1352$. Centering in the first five coordinates leaves deficit $> .0447$ and variance $\gamma^2/10 < .06561$. Payment .6 costs $< .0972$ and gains $> .0300204$.

For finite owns at least seven the actual core is $c + \gamma^{-1}n_1 + 2(\gamma^{-1} - \gamma)n_2$, of square $-[7 - 4\gamma^2 - 3\gamma^{-2}]$. Its centered variance is $(28/5)\gamma^2 - 8 + 3\gamma^{-2} < .6$. For seven, $.9009 < \gamma < .901$ gives $A > .7388$, radical $< .2395$, projected deficit $> .11187$. For larger labels $\gamma > .92$. The radical is

strictly concave, because its squared argument has second derivative $-8-18\gamma^{-4}<0$. Its derivative at .92 is $<.66$, whereas $A'>1.38$; the deficit derivative is therefore $>1.38-.66-.43>0$. Its initial value is $>.1222$. The uniform deficit is $>.111$. Payment .2 costs $<.04$ and gains $>.0204$, contradicting E_0.

For dotted own, $\gamma>1$ is unbounded. The core $c+\gamma^{-1}n_1$ has centered variance $(8/5)\gamma^2-2+\gamma^{-2}$. Put $u=\gamma^{-2}$, $D=40/21$, $q=\sqrt{D}$ and $a=q-\sqrt{D-1}$, so $0<a<.43$. Concavity gives $\sqrt{D-u}\ge q-au$, while

$$au+\sqrt{u(1-u)}\le(a+\sqrt{a^2+1})/2<(.43+1.089)/2=.7595.$$

Since $q>1.38$, the projected deficit divided by γ is $>.1905$, including its .430 block payment. Multiplier $1/(8\gamma)$ costs $1/40$ at the anchor; the variance divided by γ^2 is at most $8/5$. Its gain $>.022625$ contradicts E_0. All corrections have positive coefficients on actual walls; the finite rows use projected points, the negative rows future cores. The full zero-, one- and two-weak-end partition is exhausted. □

K.21 The whole twenty-dimensional B row

Lemma K.25. *The entire vertex forest cannot be B_{20}.*

Proof. Use the genuine unit B_{20} roots, including the short root $n_{20}=e_{20}$. Lemma K.7 covers five through sixteen. At seventeen the retained short piece is B_3: short-three own zero has its affine principal, and positive owns have ratio $\ge 1+1/\sqrt{2}$. At eighteen it is B_2: short-three owns zero, four, six and parallel have their actual weighted-star, affine and null exclusions; finite owns at least seven are directly strong. Endpoint bridges have residual $1/36$; their own-zero cycle and positive-own ratios exclude weakness.

For the remaining own-five/short-three role the left H_4 projection is flat through fifteen, has norm <18 and pressure $G>4.1$. Old strong knots eleven/twelve have endpoint coefficient $>1/7$. Old weak five would give actual $x_1>.5965$, strong six, and a strong endpoint sixteen/seventeen already supplied by the ordinary bulk and B_3 row. Its positive three-knot interpolation then has energy $.7600299945+2.2-(4/5)\sqrt{6}>1.0004379$. Thus old five is strong. The old eleven/twelve references exclude the $G>4.1$ row; even the safe norm bound twenty gives >1.0102.

The exceptional B_2 own-three role has an actual finite D_{20} mate after releasing the *short* root twenty. Positive finite-Gram projection identifies the actual endpoint and gives $\cosh 9+5\sqrt{2}>16$. Its pressure obeys $S_{20}^2>300/17>4.2^2$. Thus $G>4.2, R=20$, flat through both early endpoints, excludes this row by >1.042. The weaker long-edge mate would not suffice. We have complete .58 coverage through eighteen, with only nonadjacent left helpers and the isolated dotted-five exception. The unused last-long-edge catalogue is unnecessary.

Releasing head one retains B_{19}, so its finite-white list is isolated or simple-three at long leaf two. The latter has $A^2=2\gamma^2+2\gamma$, $C_1=2\gamma+1$. Own zero has time zero, three the affine triangle; all remaining owns force $x_1>.643$. The old bulk gives a strong endpoint fourteen/fifteen. If old five is strong, its first-five graft gains $(5/4)(.643-.361)^2=.099405$ beside $q>.90715$. If old five is weak, six is strong and $S_6>2.01815$. Interpolation through $(1,.643),(6,2.01815)$ and that endpoint has decreasing positive slopes, initial energy $\ge .7916565045$, and final chord energy $\ge 2.5-\sqrt{21}/2>.20871$. This too exceeds one. The head is therefore isolated. Time square, affine six–three and positive null sum exclude owns zero, three, four, six and parallel. Only five, finite at least seven and dotted remain.

Their actual head pressures, normalized to sum one, are

own	Z	h with $x.Z \ge h$
5	$P_4/4$	$> .393$
finite ≥ 7	$(1-u/2)(e_1+e_2)+(u-1)e_3$	$[\sqrt{2-u}-\sqrt{(u-1)(4-3u)}]/\sqrt{2}$
dotted	$(1-u/2)e_1+(u/2)e_2$	$[\sqrt{2-u}-\sqrt{u(1-u)}]/\sqrt{2}$

Here $u=\gamma^{-2}$ is respectively greater than one or in $(0,1)$. The first is an actual H_4 projected point; the others are the positive future cores on actual head roots proved above. Write $R=Z^2$ in the last two rows. The uniform scalar

$$.8h-.16R>.2893 \tag{70}$$

holds throughout both domains. For seven, $.9009<\gamma<.901$ gives $h>.4315, R<.349$, hence $>.28936$. For finite labels at least eight, $1<u<4-2\sqrt{2}<1.172$. With $D=(u-1)(4-3u)$ the derivative of the left side is

$$\frac{.8}{\sqrt{2}}\left[-\frac{1}{2\sqrt{2-u}}-\frac{7-6u}{2\sqrt{D}}\right]+.64-.48u.$$

It is negative up to $7/6$; above that point its three contributions are $<-.28,.032,.08$. At the eight-endpoint $\sqrt{2-u}>.910, \sqrt{D}<.289, R<.373$, giving $>.29136$. For dotted owns use the chords $1.4142-.379u$ on $u\le .5$ and $1.4494-.4494u$ on $u\ge .5$. Together with $R\le 1-.75u$ and $R\le .75-.25u$ they give

$$.63987152-.0944u-.5657\sqrt{u(1-u)}>.30587152,$$
$$.69978064-.2142u-.5657\sqrt{u(1-u)}>.29023064.$$

The circle bounds use $\sqrt{.0944^2+.5657^2}<.5736$ and $\sqrt{.2142^2+.5657^2}<.6049$. No sampled minimum is used.

We next prove fresh .58-strong five. An actual weak helper five supplies the curved reference through $(1,.5455)$, six, and all later strong knots, ending at seventeen/eighteen. Its initial energy is $>.74385$, compared with the ordinary .58 six-energy $<.7126$. Since $E_{17}>.97231$ and all holes cost $<1/1024$, its energy is $>.97231+.74385-.7126-1/1024>1.0025$. The only other weak-five role is an isolated dotted cap. A native-five head forbids helper five/six/seven: their actual wall product would be positive, because

$$A_i^2-(\phi^3/2)(2\gamma+1)^2<(28\gamma^2-32\gamma-13)/5<0.$$

Its product with the isolated weak dotted-five cap lies in $(-.3,0)$, also forbidden. Thus only the other heads need be considered with the latter cap.

Choose actual strong $J=6$ or seven and $K=17$ or eighteen. Interpolate their genuine .58 floors and every strong knot between, omitting and charging all weak knots. The resulting data are

J	initial mean m	w^2	available P_J	Q_5
6	$<.3399$	$>.9530$	$>.1435$	$<77/150$
7	$<.31938$	$>.9354$	$>.1366$	$<194/175$

The squared-increment lower numerators from six through sixteen, over 10^6, are

$$38461, 33333, 29411, 26315, 23809, 21739, 20000, 18518, 17241, 16129, 15151.$$

Their sums from six and seven are $.260107, .221646$. Hole costs are $< .000267, .000181$; endpoint eighteen adds $> 1/70$ and pays any seventeenth hole. The terminal slopes give the displayed available coefficients.

The actual dotted-five foot is $Z_5 = P_5 + (v/2)(e_6 - e_5)$ with demand $T_5 > 1.8185, v > .4$; its centered variance is $Q_5 = 5 - 25/J - v + v^2/2$. Set $Y = Z - P_J/J$, $Y_5 = Z_5 - 5P_J/J$. The head support lies in the first three coordinates, so $Z.Z_5 = 1$ and $Y.Y_5 = 1 - 5/J$. The single vector $W = w + .4Y + .09Y_5$ has total anchor cost $.85/J$, below both available coefficients. Its effective energy is

$$w^2 + .2893 - .8m + .16/J + .18(T_5 - 5m) - .0081Q_5 - .072(1 - 5/J).$$

The two gains are $> .04930, .06699$ and both totals exceed one. Fresh strong five has been proved before any assumed strong eighteen floor.

All new references now start at five, with $b < .3652$ and available P_5 coefficient $> .151$. The head gains are $> .015456$ for five (pay .56 on $P_4/4 - P_5/5$), $> .02914$ for the other finite heads (pay .4 and use (70)), and $> .03384$ for dotted heads. For the last, the pure dotted scalar $> .03624$ at $\sqrt{2}b < .5106$ loses $< .0024$ when only its linear block term changes to $\sqrt{2}b < .5166$. No typed theorem is imported. If eighteen is strong, its reference has energy $> .985613$ before the head, and even the least gain contradicts the norm.

Otherwise eighteen is an actual helper and seventeen is strong. The reference has $q > .97133$, first-five mean $b < .3652$, and available P_{17} coefficient $> 1/9$. For that actual helper put $V = U - \sqrt{2}\gamma P_{17}$. Its variance is < 2.1, and its pressure deficit, including the nonzero cross $w.V = b/\sqrt{2}$, is $> .21$. For five, seven, eight and larger finite labels the respective deficit seals are $> .3444, .2853, .2306, .2303$. The last interval starts at $\gamma > .939$; its derivative is positive since $A' > 6$, $H_A > .94$ and its radical decreases. The dotted deficit decreases through $6/5$, where $A > 7.3307$ and radical $< .5528$ give $> .2177$. Thus every continuous parameter is covered.

Payment $\nu = 1/16$ costs $\nu\sqrt{2}\gamma < .107 < 1/9$ and gains $> 2\nu(.21) - \nu^2(2.1) = .018046875$. Combine the head and this foot in one vector. If its initial support has length $s = 3, 2, 1$, the residual product is $(\ell_s - .2)/\sqrt{2}$. For head five $\ell_s = 1/4$, giving total cross cost $< .002476$; for the other finite heads $\ell_s \le 1/2$ gives $< .010608$; for dotted heads $\ell_s \le 1$ gives at most .02 exactly. The final totals are

$$\begin{aligned} .97133 + .015456 + .018046875 - .002476 &= 1.002356875, \\ .97133 + .02914 + .018046875 - .010608 &= 1.007908875, \\ .97133 + .03384 + .018046875 - .02 &= 1.003216875. \end{aligned}$$

Every actual head case is impossible, proving the whole B_{20} row. □

K.22 The whole twenty-dimensional D row

The three possible isolated long heads have different paid references, but share one finite-white inventory at the last centre. We prove that inventory once, retaining the head-dependent cross products in a small table. This also keeps the endpoint choice honest: one of fourteen/fifteen and one of sixteen/seventeen is strong; no proof below assumes all four floors.

Lemma K.26 (The early D_{20} prerequisites)**.** *For the entire forest D_{20} the long-end cap is isolated, of own five, finite at least seven, or dotted. Moreover $S_5 > \sqrt{5} - .61/\sqrt{2}$, without a hypothesis on that cap.*

Proof. Use $n_i = (e_i - e_{i+1})/\sqrt{2}$ for $i < 20$ and $n_{20} = (e_{19} + e_{20})/\sqrt{2}$. Then $\omega_i = \sqrt{2}P_i$ through eighteen and $\omega_{19,20} = (P_{19} \mp e_{20})/\sqrt{2}$. Write $s_i = \sqrt{i} - .61/\sqrt{2}$. The old bulk applies literally

through fifteen: right retained rank is at least five, and all walls are actual. Its only weak rows are nonadjacent pure far helpers. For the head, retained D_{19} admits isolated or simple-three at its long leaf two: other leaves give exceptional finite rank twenty, internal attachments an extra fork, and a four edge is not a large finite D extension. Two contacts give a cycle. The nonisolated row has $A^2 = 2\gamma^2 + 2\gamma$, $C_1 = 2\gamma + 1$, exactly as in the preceding proof. Its $x_1 > .643$ contradicts either the old strong-five graft or the old weak-five three-knot reference there. Those arguments use only the first fifteen long coordinates, so the signed forks do not alter a coefficient. The isolated list follows from $A^2 = 2\gamma^2 - 1$, the actual affine six–three path, and the parallel null sum.

We supply the head-independent strong-five bootstrap before using any head gain. Suppose $S_5 \le s_5$. Its actual far helper gives $x_1 > a = .5965$, $S_6 > B = 2.01815$. Some $K = 11$ or twelve is old-strong. Interpolate through $(1, a), (6, B), (K, B + \sqrt{K} - \sqrt{6})$, with zero slope after K. The slopes $a, .28433, 1/(\sqrt{K} + \sqrt{6})$ strictly decrease. Thus all duals are positive, the endpoint coefficient exceeds $1/7$, and its energy is

$$q_K = .7600299945 + \frac{K - 6}{(\sqrt{K} + \sqrt{6})^2} > .91.$$

The terminal sum g_K is $< s_K < 3.034$. For any actual pressure $x.V \ge G$ with V equal to one on the first K entries and $V^2 \le R$, paying $1/7$ gives

$$\|x\|^2 > q_K + \frac{2(G - g_K)}{7} - \frac{R - K}{49}. \tag{71}$$

This is one square on this reference; no head is paid.

We need a late actual endpoint. Releasing sixteen gives $A_{15} + D_4$; the right long-leaf-three row has numerator $\gamma + 1/2$, residual $1/2$, with own-zero two-fork affine principal, own-three triangle, and all other owns ratio $\ge 1 + 1/\sqrt{2}$. Only the two short leaves use numerator $\gamma + 1/4$. Releasing seventeen gives $A_{16} + A_3$: middle-three has its affine star/triangle, positive ratios $\ge 1 + 1/\sqrt{2}$; end-four the weighted-star/strong rows; end-five is a sealed actual H_4, strong already at own zero. Full bridges have residual $1/32$ at sixteen, and $1/8 + 1/34$ or $1/34$ at seventeen; own zero is the all-three cycle, positive owns are strong. Left/isolated roles have the complete local list of Lemma K.7. Its proof is local at these actual retained pieces: it does not assert a rank condition on a larger-row theorem.

Apart from a pure left far helper, only a right short-leaf/end three remains. Own zero, three, four, six and parallel have respectively the affine E_8, cycle, F_4 path, six–three path and feasible null sum. Own five has the positive left H_4 projection, flat through $i - 3$, $R < i$, with $G > \sqrt{i - 125/448} - .091$. At $i = 16, 17$ this is $> 3.745, 3.9$; inserting the safe $q > .91, g_K < 3.034$ in (71) gives $> 1.0111, 1.0349$.

For finite owns at least seven or dotted, supposed .58 weakness gives $.9 < \gamma < .92$ at sixteen, or $.9 < \gamma < 1.05$ at seventeen. This follows from the exact target $2773138/1687945 < 1.643$ and the residuals $1/2, 5/8$. Use the actual core

$$\xi = c + \gamma^{-1} n_i + (2\gamma)^{-1} n_{i-1}, \quad k = 1 - 3/(4\gamma^2), \quad \alpha = \sqrt{2}(\gamma + 1/4).$$

Its products at c, n_{i-1} vanish, at n_i are negative; its square is $-k < 0$, future time positive. All other wall products are nonpositive term by term. Its displacement is flat through $i - 2$, with

$$G = (A - \sqrt{k})/\alpha, \quad A^2 = i\alpha^2 - (i - 12)/8, \quad R = i - [(i - 12)/8 + k]/\alpha^2.$$

At sixteen, $2.645 < \alpha^2 < 2.738, k < .114$ give $G > 3.768, R < 15.82$. At seventeen, k/α^2 increases through 1.05 to $2350/24843 < .308^2$, while $\sqrt{17 - (5/8)/2.645} > 4.094$. Its derivative sign is

$-4\gamma^4-\gamma^3+6\gamma^2+(9/4)\gamma+3/16>0$ on this interval. Thus $G>3.786, R<16.82$. Formula (71) gives $>1.0213, 1.0060$. Only pure left helpers can therefore be weak at sixteen/seventeen. Their actual mixed replacements forbid adjacency, so at least one of those two is .58-strong, hence old-strong.

Append that actual endpoint J to the same weak-five reference. Its energy is at least

$$.7600299945+\frac{4-\sqrt{6}}{4+\sqrt{6}}=.7600299945+2.2-(4/5)\sqrt{6}>1.0004379945,$$

using $\sqrt{6}<244949/100000$. This contradiction proves old strong five without any head or centre floor. □

Lemma K.27. *The entire vertex forest cannot be D_{20}.*

Proof. Keep the original x and notation of the preceding lemma. There are three head branches, denoted H_5, H_∞, H_f for native five, dotted, and finite label at least seven. We construct their paid references before considering the centre.

For H_5, put $g=\phi/2$, $A_0^2=(\phi-1)/2$. The actual left H_4 projection has displacement $g\omega_4/4$, residual $1-3g^2/2$, and exact pressure gA_0. Hence $S_4\ge 2\sqrt{2}A_0>1.572$. A helper at any of five/six/seven is impossible: its actual head product is positive since $A_i^2-(g^2/A_0^2)(2\gamma+1)^2<(28\gamma^2-32\gamma-13)/5<0$ on $.8<\gamma<1.2$. An isolated weak dotted-five cap has actual product $\phi\gamma-A_0\sqrt{10\gamma^2-1}\in(-.3,0)$, where weakness gives $1<\gamma<2$. Negativity follows from $(7\phi-11)/2>0$; the lower bound from $\phi-\sqrt{5(\phi-1)}>-.15$. Thus fresh .58 five/six/seven are genuinely strong. On any subsequent actual-floor reference pay $.14T$, $T=P_4-(4/5)P_5$. Its anchor cost .112 is below .151, and its effective gain is

$$2(.14)(1.572-.8(1.826))-.8(.14)^2=.015456.$$

For H_∞, $u=g^{-2}\in(0,1)$ and the actual core $c+g^{-1}n_1$ gives $Z=(1-u/2)e_1+(u/2)e_2$ with pressure $h=[\sqrt{2-u}-\sqrt{u(1-u)}]/\sqrt{2}$. On any old reference, flat on the first five with $b<.361$, pay $\lambda(Z-P_5/5)$, $\lambda=\sqrt{2}/5$. Its cost $\lambda/5<.057$ fits the available coefficient $>.146$. The pure scalar proved above gives gain $>.03624$; if $u\le.5$ it gives $>.04644$. The paid coordinates satisfy

$$w.P_3<1.1962,\quad w.P_2<.8918,\quad w_1<.5166\ (u\ge.5),\quad w_1<.5873\ (\text{all } u).$$

These nonflat entries will be used, not replaced by a flat mean.

For H_f, write $g^{-2}=1+v$. The actual three-wall core is

$$c+g^{-1}n_1+2(g^{-1}-g)n_2;$$

its square is $-[7-4g^2-3g^{-2}]<0$ and its displacement is the same $Z=(1-u/2)(e_1+e_2)+(u-1)e_3$ as the finite head in the preceding lemma. This identity uses no short normal. Here $0<v<.234$; the elementary cosine lower bound $\cos(\pi/7)>.9005$ gives $u<617/500$. Pay $.4(Z-P_5/5)$. The q-free scalar (70), at identical actual head hypotheses, gives gain $>.2893-.8(.361)+.032=.0325>.031$. For $v\le.1$, directly

$$\sqrt{1-v}>.9486,\quad \sqrt{v-3v^2}<.2646\quad\Longrightarrow\quad G>.56568(.9486)-.56569(.2646)-.3368>.05.$$

The paid coordinates obey $w.P_3<1.243, w.P_2<.962, w_1<.461$ for $v\ge.1$, and $w_1<.481$ for all v. Only this common local head calculation is reused; the whole B_{20} theorem and its short endpoint are not inputs.

We justify the required late references. The early eleven/twelve old reference, now with old strong five, has $q_{11} > .8472225, q_{12} > .8689525$ and endpoint coefficient $> 1/7$. Pay the appropriate head as just constructed. All own-five right-fork rows at fifteen/sixteen/seventeen are excluded already without that payment: their H_4 pressure is $> \sqrt{i - 125/448} - .091$, flat through $i - 3$, $R < i$, giving $> 1.01101, 1.01087$ at fifteen, and increasing thereafter. The additional finite/dotted rows at sixteen/seventeen have $G > 3.768, R < 15.82$ or $G > 3.786, R < 16.82$ as proved above. Adding even the least head gain .015456 to the old reference excludes them by > 1.01616 and > 1.00089. The remaining weak rows through seventeen are precisely pure left helpers. Their actual nonadjacency gives an actual strong endpoint $K_A \in \{14, 15\}$ and $K_B \in \{16, 17\}$. For H_5 use fresh floors $f_i = \sqrt{i} - .58/\sqrt{2}$; for the other heads use old s_i floors. No individual fifteen/sixteen/seventeen floor has been asserted.

Each missing knot loses $(2\sqrt{j} - \sqrt{j-1} - \sqrt{j+1})^2/2 < 1/[32(j-1)^3]$. In H_5, missing knots start at eight: the total losses at fourteen, fifteen/sixteen and seventeen are $< .000158, .000172, .000182$. In the other heads they start at six; at most $4, 5, 5, 6$ holes cost $1/3200$ each. The old rational increment list gives $E_{14} > .9084, E_{15} > .92564, E_{16} > .94176$, $E_{17} > .94176 + 1/66$. Consequently the same legally paid references have data

	L_{14}	L_{15}	L_{16}	L_{17}	a_{14}	a_{15}	a_{16}	a_{17}
H_5	.939098	.956324	.972444	.9875	3.332	3.463	3.590	3.714
H_∞	.94339	.9603175	.9764375	.991276	3.311	3.442	3.569	3.692
H_f	.93815	.9550775	.9711975	.986036	3.311	3.442	3.569	3.692

Here L_K is a strict lower bound on twice certified pressure minus squared norm, not on that norm itself. The actual coordinate sum is f_K or s_K, strictly below a_K. For small dotted parameter $u \le .5$ add .0102 to its four L entries; for finite $v \le .1$ add .019. Remaining P_{14}, P_{15} coefficients exceed $1/8$; those at sixteen/seventeen exceed .12. The head changes only the first-five coefficients.

If an actual V is flat one through K_A, $x.V \ge G$, $V^2 \le R$, paying $1/8$ gives

$$\|x\|^2 > L_{K_A} + (G - a_{K_A})/4 - (R - K_A)/64. \tag{72}$$

Both actual fork directions are $P_{19} \pm e_{20}$, of square twenty. Cosh $> 41/5$ gives $G > \sqrt{360/23} > 3.955$. Equation (72) exceeds one in every head/endpoint row (the least is 1.001098). Thus both actual fork coshes are at most $41/5$, independently of a centre floor. Reciprocal uses below concern actual endpoints: for a positive finite replacement Gram, its inverse is entrywise nonnegative. Projecting p adds positive multiples of actual normals, preserves every unselected strict inequality and makes the released old one strict. The projected point is therefore the actual neighboring endpoint, with no unseen truncating wall.

The centre inventory. Delete eighteen, retaining A_{17} and two orthogonal forks. If the actual centre wall has own γ, left contact β at j, and fork contacts δ, ϵ, then

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 > 0, \quad \sigma = \gamma + \beta j/18 + (\delta + \epsilon)/2, \quad C_{18} = 6\sigma/\sqrt{\Delta}.$$

For simple three, $q_{\text{left}} = j(18 - j)/36$; endpoint four doubles it. The complete finite-white list is: one left endpoint-three/four at one/seventeen or next-three at two/sixteen; one fork with a left endpoint-three and fork-three/four; both forks with a left endpoint only in the all-three bridge; or, without a left contact, the two-fork pairs $(3, 3), (3, 4), (3, 5)$. More left contacts create a cycle; other branches would require exceptional finite rank twenty. This is classification of the actual white set, not of a face. The sufficient .58 strength target is $< 83/50$.

Mixed left/fork bridges have residual 1/36 (both forks or fork four), with ratios $\geq 19/6$ or $1/6+3/\sqrt{2}$. The one-fork-three bridge has residual 5/18: own zero is the all-three cycle, own three the triangle; other owns give ratio at least $(1/\sqrt{2}+1/36+1/4)/\sqrt{5/18} > 83/50$.

With no fork contact the isolated roles have the established .58 feet and affine/null exclusions. This remains the genuine centre: the identity $\omega_{19}+\omega_{20} = \sqrt{2}P_{19}$ is used only in $n_{18} = \omega_{18} - (\omega_{17}+\omega_{19}+\omega_{20})/2$; no extra root is introduced. For far left-four and next-three, own zero has time zero and positive owns are strong. Near left-four is strong already at own zero. Near next-three own zero has the affine two-fork principal; positive owns have ratio $\geq 5/3$. Near endpoint-three owns zero/three/four/six have respectively the two-fork, triangle, weighted-star and six–three affine principals. All other positive owns have ratio $(36\gamma+17)/(6\sqrt{19}) > 83/50$. Far endpoint-three own zero has $A^2 = -1/2$, three its cycle, four/six the right affine principals, and parallel a null sum. Thus the complete possible weak no-fork list is

$$v = \gamma\omega_{18} + \omega_1/2, \quad A^2 = 36\gamma^2 + 2\gamma - 1/2, \quad \phi/2 \leq \gamma < 6/5,$$

with own five, finite at least seven, or dotted.

For these helpers the positive actual corrections are

own	w_{old}	Y in $V = P_{17} + Y$	G lower, Y^2 upper
5	$\gamma n_{18} + (2/3)n_1 + (1/3)n_2$	$aP_3 + (e_{18}+e_{19})/2$	4.185, .631
7, 8	$\gamma n_{18} + n_1/2$	$aP_2 + (e_{18}+e_{19})/2$	4.08, .655
finite ≥ 9	γn_{18}	$aP_1 + (e_{18}+e_{19})/2$	4.067, .785

Here $V = (v - w_{\text{old}})/(\sqrt{2}\gamma)$ and $a = 1/(6\gamma), 1/(4\gamma), 1/(2\gamma)$ respectively. The finite five and at-least-nine rows use projected points with residuals $2/3-\gamma^2, 1-\gamma^2$; the seven/eight row is a negative core of norm $-(\gamma^2 - 3/4)$. Their common time term is $\sqrt{18 + 1/\gamma - 1/(4\gamma^2)} > 4.33$. The normalized residual bounds are $.145, .25, .263$; $Y^2 = ka^2 + 1/2$ gives the displayed variances. All products against every unselected actual wall are nonpositive by the positive corrections, including the actual head.

For dotted helper put $a = 1/(2\gamma) \in (5/12, 1/2)$ and $t = 4a^2$. The future core $c + \gamma^{-1}n_{18}$ gives

$$Y = aP_1 + (1-t/2)e_{18} + (t/2)e_{19}, \quad Y^2 \leq .75, \quad G = \sqrt{18 + 2a - a^2} - a\sqrt{2 - 8a^2}.$$

For each constant $c \leq .5873$, $(G - ca)' > 7/9 - c > 0$. At $a = 5/12$, positive square bounds $\sqrt{2687} > 51.83$, $\sqrt{22} < 4.691$ give $G - .394a > 3.829$. Hence all head/helper crosses have the bounds

head parameter	$w.Y_5$	$w.Y_{7,8}$	$w.Y_{\geq 9}$	$G - w.Y_{\text{dot}}$
H_5	$< .24625$	$< .219$	$< .211$	> 3.829
$H_\infty, u \geq .5$	$< .2493$	$< .2478$	$< .2754$	> 3.7779
$H_\infty, u \leq .5$	$< .2493$	$< .2478$	$< .3131$	> 3.7484
$H_f, v \geq .1$	$< .259$	$< .2673$	$< .2458$	> 3.8010
$H_f, v \leq .1$	$< .259$	$< .2673$	$< .2564$	> 3.7927

The finite columns follow by multiplying the actual $w.P_k$ bounds above by a; the dotted column uses the increasing $G - ca$ at the common endpoint 5/12. No cross is set to zero.

At $K_B = 17$ pay .12 for Y; at $K_B = 16$ pay for $e_{17} + Y$. The latter has an extra unit square and no extra cross, since the reference stops at sixteen and $e_{17}.Y = 0$. One combined square gives

$$\|x\|^2 > L_{K_B} + .24(G - a_{K_B} - w.Y) - .0144(Y^2 + \varepsilon), \quad \varepsilon = 0 \text{ (17)}, \ 1 \text{ (16)}. \tag{73}$$

The minimum over the four helper rows is > 1.0043 in H_5, > 1.001092 in large-u dotted, and > 1.001396 in large-v finite heads. For the small parameter cases, at $K_B = 17$ the reference alone exceeds one; at sixteen the minimum is > 1.0044935 or > 1.0186855 respectively. For example the tight dotted test is $.991276 + .24(3.7779 - 3.692) - .0144(.75) = 1.001092$. Every no-fork weak centre role is excluded.

The fork rows. For one fork and no left contact put $\alpha = \sqrt{2}(\gamma + \delta/2)$, $A^2 = 18\alpha^2 - (1 - \delta^2)$. At own zero, fork labels three/four/six have their actual affine E_8, weighted-star and six–three principals. Fork five has the positive left H_4 projection with $G = \sqrt{4 + 8\phi} - \phi^{-3} > 3.87$, $R = 17 < 18$. For finite fork labels at least seven use the positive core

$$c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{18}.$$

Its square is $-(1-b)(4b-3)/b < 0$, $b = \delta^2$, and its products at c, n_f vanish and at n_{18} are negative. The displacement is flat through seventeen, with

$$R = 28 - 16/b + 6/b^2, \qquad G = \sqrt{20 - 2/b} - \sqrt{-8 + 14/b - 6/b^2}.$$

Every finite label is in the following table:

fork label/domain	G lower	R upper
5	3.87	18
$7, .81 < b < .82$	3.798	17.412
8	3.79392	17.49034
$\geq 9, .88 < b \leq .9$	3.808	17.63
$\geq 9, .9 \leq b < 1$	3.831	18

For eight the exact correlated values are $R = 108 - 64\sqrt{2}$ and $G = \sqrt{12 + 4\sqrt{2}} - \sqrt{68\sqrt{2} - 96}$. The rational seal $1.414213562 < \sqrt{2} < 1.414213563$ gives the table. In the other rows R increases for $b > .75$, and the second radicand increases up to $6/7$ and decreases thereafter, proving the interval bounds. Labels seven and at least nine satisfy the indicated disjoint native ranges. Equation (72) excludes every row with either actual early endpoint and each head: the least test is the native-five/eight/fourteen value $.939098 + (3.79392 - 3.332)/4 - 3.49034/64 = 1.0000414375$. For finite heads its least test is > 1.0041459; for dotted heads > 1.0093859. No head cross occurs because these displacements are flat throughout the head support.

For positive own three, release the contacted fork. The same actual centre wall completes the retained A_{19} to finite D_{20}, so the reciprocal endpoint has $C_f = 9 + 10\delta \geq 14 > 41/5$, impossible. Own four has the affine star on $c, 18, 17$ and the other fork. Own six/fork three has the six–three principal. Own six/fork at least four and own at least seven/fork at least four are directly strong by ratio $\geq \sqrt{2}\gamma + 1/2 > 83/50$; own five/fork at least five is also directly strong. The two exceptions five/three and five/four have the positive left H_4 correction $\gamma(3n_{18}/2 + n_{17} + n_{16}/2)$, finite residual $(5 - 3\phi)/8$ and $\alpha^2 \geq \phi + 5/8$. Their displacement is flat through fifteen, $R < 18$, and $G > \sqrt{18 - 75/224} - .091 > 4.1$, dominating the common row. For single-fork three with finite at least seven or dotted own, $\gamma \geq 6/5$ is directly strong. Throughout $.9 < \gamma < 6/5$ the actual core $c + \gamma^{-1} n_{18} + (2\gamma)^{-1} n_{17}$ has $k = 1 - 3/(4\gamma^2) > 0$, flat displacement through sixteen, $R < 18$, $A/\alpha > 4.208$ and $\sqrt{k}/\alpha < .338$. Thus $G > 3.87$ and the same common row excludes it. Parallel own is independently null; including it in that continuous core interval would also contradict the reference.

For two forks the complete pairs are three/three, three/four, three/five. Own three has the affine triangle, own at least four ratio $\geq 1 + 1/\sqrt{2}$, and parallel the null sum. At own zero, three/three is the affine square. Three/four has, after releasing the three-contacted fork, the actual finite B_{20} reciprocal endpoint $C_f = 9+5\sqrt{2} > 41/5$, impossible. Three/five has a positive finite H_3 projection, displacement αP_{18}, $\alpha = \phi^2/(2\sqrt{2})$ and $q = 2\phi^{-6} < 1/8$; its pressure is $\sqrt{18-q}/(1+\sqrt{q/18}) > 4.22/(13/12) > 3.87$, $R = 18$. It too is covered by the common row. All original wall inequalities in these finite projections follow from the positive inverse Gram coefficients.

The complete centre domain has been exhausted. Eighteen is therefore genuinely .58-strong in every head branch. Append its fresh floor for H_5, its old floor otherwise, to the reference whose actual endpoint is K_B. At seventeen the added energy is $d_{17}^2 > 1/70$; at sixteen the chord across the possibly weak seventeenth adds $17 - 12\sqrt{2} > .0294$. Both slopes are below the available endpoint coefficient and leave the head payment unchanged. The least resulting totals are the finite-head values $.986036 + 1/70 > 1.0003217$ and $.9711975 + .0294 = 1.0005975$. This contradicts the same original $\|x\| < 1$ in every branch. □

K.23 The complete dimension-twenty conjunction

Proposition K.28. *There is no compact hyperbolic Coxeter polytope of dimension twenty.*

Proof. We exhaust the forest of an actual farthest vertex in this dimension; no descent from a higher-dimensional face is involved. An irreducible A_r with $r \geq 7$, or B_r, D_r with $r \geq 9$, is called dominant. Each costs more than one half, so at most one occurs. If none occurs, the ordinary rank-fourteen rate, even discarding the single possible isolated I rank, gives $19/14 > 1$. A typed rank-two defect instead gives $.09 + 18/14 > 1$; the funded J pair is stronger than its ordinary rank rate. Thus exactly one dominant occurs.

For an A dominant, rank seven is excluded by its half bound and the ordinary spectator rate. Ranks eight through twelve are excluded by $4/7$ and that same rate: even with I there are seven ordinary ranks at twelve, or with a typed defect six ranks beside its .09 cost. Ranks thirteen/fourteen have energy $> 5/7$ and at least five ordinary ranks after discarding I; a typed defect leaves at least four, so both alternatives exceed one. At ranks fifteen, sixteen and seventeen use respectively $> .826, > .86, > 6/7$. The complete minimum spectator comparisons are

r	ordinary or J	I present	typed present
15	$> 5/14$	$> 2/7$	$> .09 + 17/70$
16	$> 2/7$	$> 2/7$	$> .09 + 1/7$
17	$> 17/70$	$> .21$	$> .09 + .1$

Each sum with its actual dominant floor exceeds one. For the middle I entry the ordinary remainder has rank three; the joint I theorem, rather than discarding that rank, supplies $> 2/7$. The rank-two typed remainders are either an ordinary dihedral or two ordinary A_1 factors, both at least the shown charge. There is at most one defect. Rank eighteen uses $> .91$: an ordinary rank-two remainder is $> 1/7$, $I + A_1 > 1/5$, either typed defect $> .09$, and $J > .34$. The typed sum is strictly greater than one as well. At rank nineteen its remaining A_1 is either ordinary or I; Lemmas K.22 and K.23 exclude both. Rank twenty is whole and is excluded by Lemma K.24.

For a B/D dominant its $9/14$ bound forces rank at least fifteen: at rank fourteen even an I leaves five ordinary ranks, while a typed defect leaves four beside its .09. At fifteen and sixteen the valid weaker D prefix floors are $> .797425, > .823735$. The smallest remainders are I+ordinary rank four $> 2/7$, respectively a typed defect plus an ordinary dihedral $> .09 + 1/7$; all other partitions are stronger. Both rows exceed one. At seventeen the weaker floor is $> .8472225$.

Ordinary total rank three costs $> 17/70$, I+dihedral costs $> .21$, I+two ordinary A_1 costs $> .3$, and a typed defect plus ordinary A_1 costs $> .19$. Each closes the row. At eighteen the weaker floor $> .8689525$ excludes ordinary rank two, $I + A_1$, and J. Both typed remainders are excluded by Lemma K.21. At nineteen a nonshort A_1 is excluded by the actual complete coverage through fifteen and its prefix gain $> .91211125$; that same lemma supplies the needed local D_4 continuation before using the prefix proof. Its I alternative is also excluded there. Whole B_{20} and D_{20} are excluded by Lemmas K.25 and K.27. Finite classification and the complete spectator partitions above exhaust the actual rank-twenty forest. Orthogonal component energies use the same original x throughout. □

K.24 The two proper A rows in exact dimension nineteen

Lemma K.29. *An A_{17} component cannot coexist with either actual typed-seven defect as its entire rank-two remainder.*

Proof. The actual defect costs $> .09$, so $e_A < .91$. Use the complete $N = 18$ window of Appendix C.6, before its weaker contrary-energy conclusion. A near-four actual mate has normalized demand $> .86$; the opposite strong anchor or mate has demand $> .77$ and correlation $< .44$. Cauchy would give $1.63^2 < .91(2.88)$, false. All weak knots five through thirteen are thus nonadjacent pure helpers, with $Q = 0$, own five, finite at least seven or dotted, $\phi/2 \le \gamma < 17/16$. The rational reference of Appendix C.7 has end heights $H_5 = 2.077, H_6 = 2.218$ and central heights $2.314, 2.371, 2.389$, reflected. For zero, one or two weak end sites its complete all-hole energies are

$$E_0 \ge .8934308, \quad E_1 > .86256, \quad E_2 \ge 9978827/12000000.$$

Strong ends have five-block mean $.29 < b_0 < .294$; weak ends six-block mean $1/4 < b_1 < .2614$.

Both actual defects force the endpoint floors $x_1 > .392, -x_{18} > .392$. For old A_2/new seven, propagation gives endpoint squared half-tangent $> 1/5$, hence $x_1^2 > 17/90$. For old $I_2(7)$/new three let its isolated opposite have time square $h = (1/4)\csc^2(\pi/7) - 1$. The actual endpoint opposite retains that whole old block. Any finite contact to it would cost at least $1/[4\sin^2(\pi/7)] > 1$, so the endpoint cap has no such contact. Releasing old one leaves A_{16}. Its complete nonisolated contact list is three at $2, 3, 16, 17$, or four at $2, 17$. The corresponding coshes and minimum allowable owns are

$3:2$	$(17\gamma + 8)/9$	$1/2$
$3:3$	$(34\gamma + 15)/(6\sqrt{2})$	0
$3:16$	$(17\gamma + 1)/(3\sqrt{2})$	$1/2$
$3:17$	$(34\gamma + 1)/18$	$1/\sqrt{2}$
$4:2$	$(17\gamma + 8\sqrt{2})/3$	0
$4:17$	$(17\gamma + 1/\sqrt{2})/3$	$1/2$

All but the far-three row have cosh $> 3/2$, giving $x_1^2 > 17/90$. The remaining row gives $289(\sqrt{2} - 1)/[18(17\sqrt{2} + 19)] > (49/125)^2$; clearing denominators leaves $3780919\sqrt{2} > 5336767$, certified by $\sqrt{2} > 707/500$.

For an isolated endpoint $A^2 = (17/9)\gamma^2 - 1$ and its product with the actual defect wall is $-A\sqrt{h}$. For finite own, $hA^2 < 8/27 < 1/2$; the native gap forces $hA^2 = 1/4$. The literal seven algebra gives $7h^3 + 7h^2 - 1 = 0$, $.327 < h < .329$, with degree three over $\mathbb{Q}$. Consequently $.93 < \gamma^2 < .94$. Only label twelve can lie in this interval: for labels at most eleven, $\sin(\pi/11) > 717/2662$ gives squared cosine $< .93$; for labels at least thirteen, $\pi/13 < 22/91$ gives squared cosine $> 7797/8281 > .94$. At twelve equality would imply $h = 9/(17\sqrt{3} - 2)$, impossible for its cubic degree. This is literal-angle algebra, not an arithmetic ground-field restriction. Parallel own is excluded by the same gap and

interval. For dotted own $hA^2 > 1/4$; the next native possibility gives $A^2 \geq 1/(2h) > 3/2$, so its cosh squared exceeds $5/2$ and again supplies the floor. Reflection covers the other endpoint.

Every weak cap gives a stronger floor at its helper end. The complete raw-pressure seals in Appendix C.8 are .36082, .327322, .3132358, .31446 for five, seven, eight and larger finite owns. For dotted own its exact weak floor gives $> .310125 + 1/216 > .3132358$. Thus the helper endpoint exceeds .4429, since $1.4142(.3132358) = .44297806836$. A near helper has larger time square by $8\gamma/9$. As the two times are $< 3, 4$, its time increases by $> 32/315 > .1$, with the same residual. Its endpoint exceeds .58 for every own parameter.

On a constant block of length m and mean b, grafting a signed endpoint value $z > b$ gains $m(z-b)^2/(m-1)$. All the following grafts preserve positive ordered slopes: the other coordinates stay $> .2517$ in a five-block or $> .184$ in a six-block, above the next reference coordinate. Opposite end blocks are disjoint, so they are changed in one reference, including all previous hole charges. With no weak ends use .392 twice: $E_A > .8934308 + 2(5/4)(.392 - .294)^2 = .9174408$. With one weak end a near helper alone gives $> .98$. Otherwise graft .4429 in the opposite strong block and .392 in its weak block, giving $> .86256 + (5/4)(.4429 - .294)^2 + (6/5)(.392 - .2614)^2 = .9107416445$. With two weak ends a near helper alone gives $> .95$; otherwise their far helper endpoints are distinct, giving $> 9978827/12000000 + 2(6/5)(.4429 - .2614)^2 > .91063$. All cases exceed .91, contradicting the orthogonal defect cost. □

Lemma K.30. *An actual A_{18} component cannot coexist with an isolated I; further orthogonal finite spectators are allowed.*

Proof. Use $N = 19$ and the one actual I coordinate $y \geq s$. Its rank-free parameters give both endpoints at least ηt, where $\eta = \sqrt{18/19} > .973$ and $t + (3/2)s > .767$. We first strengthen the near-four removal to the norm-only domain needed here. Its actual finite mate, replacing $j = i - 1$ at $i = 5, 6$, has diagram $B_{20-i} + A_{i-2} + S$ and cosh

$$\sqrt{2(i-1)/19}\,[(20-i)/2 + (19-i)/\sqrt{2}] > 11.$$

Finite positive projection retains every spectator and recognizes this actual endpoint. Its normalized height exceeds .91. The opposite mate or strong knot thirteen/fourteen has height $> .774$ and normalized correlation $< .41$. Thus $1.684^2 = 2.835856 > 2.82$ contradicts $e_A < 1$. The complete shared $N = 19$ window now has only nonadjacent pure helpers, before any other energy assumption.

The shared rational heights are $H_5 = 2.104, H_6 = 2.255, H_7 = 2.363, H_8 = 2.433, H_9 = H_{10} = 2.467$, reflected. With zero, one or two weak end sites the complete energies are $E_0 \geq .9240352, E_1 > .8937, E_2 > .86414$. The strong five-block mean is $b_0 < .298$, weak six-block mean $b_1 < .266$, each $> .29$ or $> 1/4$ respectively. Available dual coefficients are $\beta_6 \geq 1607/12000$ and $\beta_{14} \geq .1349$; deleting the next internal knot only increases them.

A weak-five far helper has $A^2 = (140\gamma^2 + 10\gamma - 10)/19$. Use the full positive actual feet of Appendix C.7, not their raw normal sums. Their corrections are $\gamma n_5 + (2/3)n_e + (1/3)n_{e'}$ for five, $\gamma n_5 + 2(1-\gamma^2)n_e$ for finite owns at least seven, and $\gamma^{-1}n_5$ for dotted own. Their actual pressures are bounded below by A minus the respective finite/core radical; the full I block remains an unselected acute block. When both ends are weak, projecting the spatial foot gives

$$PU = (5\gamma/6)\omega_6 + \omega_{13}/12, \quad Y = U - PU, \quad w.Y = 0.$$

Its variance V_2 is $(2/3)\gamma^2 + 1/12$ for five, $4\gamma^4 - (16/3)\gamma^2 + 29/12$ for other finite owns, and $(5/3)\gamma^2 + \gamma^{-2} - 19/12$ for dotted own. The exact cross between two far feet is $Y_L.Y_R = (\gamma_L + \gamma_R)/6$: each head residual pairs with the opposite helper's mass $1/\sqrt{2}$, and all same-side block sums vanish.

Pay $\mu = .138$ for each far foot. At either anchor the total cost is $< .138(31/32) < 1607/12000$. For $\Delta_2 = H - 2.255(5\gamma/6 + 1/12)$ the gain per foot, including half the exact cross, is $F = 2\mu\Delta_2 - \mu^2(V_2 + \gamma/3)$. The full finite/dotted seals are

own	Δ_2 lower	$V_2 + \gamma/3$ upper	F lower
5	67/192	.791	.081248696
7	.3371541666	1.026	.0735154059
8	.32852125	1.087	.069971037
≥ 9	1591/4800	1.107	.070400792
dotted	.36109375	1.54	.070334115

The first three use actual time/radical bounds $(2.17, .111), (2.434, .2158), (2.499, .2464)$ and upper native cosines $.81, .901, .9239$. For larger finite owns evaluate at .93, where $A > 2.517$, then use $A' > 2.714$ and $V_2' \leq 16/3$ to see that the lower gain increases throughout the remaining finite interval. For dotted owns use $1 < \gamma < 17/16$; its deficit decreases to an endpoint $> .36109375$, while $V_2 < 1.185, \gamma/3 < .355$. This proves every row continuously, not by sampling. Both far weak ends therefore give $E_A > .86414 + 2(.0698) > 1.00374$, without spending I.

For exactly one weak end, a near helper already gives $E_A > 1.0120152$ by the shared raw endpoint floor .58. For a far helper the projection becomes $(5\gamma/6)\omega_6 + \omega_{14}/10$. Its deficit and variance are $\Delta_1 = \Delta_2 - 1349/60000$, $V_1 = V_2 - 1/60$. Payment $\mu = .11$ gives a gain $> .058$ in every row: the conservative five/seven/eight/larger/dotted gains are $.065714, .06064868, .05809604, .0585244, .06034939$. The same finite derivative and dotted endpoint bounds prove the complete intervals. Retain the propagated left endpoint at the same time. Let $R = e_1 - P_6/6$, so $R^2 = 5/6$ and $Y.R = \gamma\sqrt{2}/6$. Put $B = b_1 + \mu\gamma\sqrt{2}/6 < .294$, $z = (\eta t - B)_+$ and $\nu = \min((6/5)z, 1/4)$. The full vector is $w + \mu Y + \nu R$. Its coefficient-six cost is $\mu 5\gamma/6 + \nu/(6\sqrt{2}) < .128 < 1607/12000$; the opposite cost is $.011 < .1349$. Its full mixed square therefore gives

$$E_A + y^2 > E_1 + .058 + s^2 + 2\nu z - (5/6)\nu^2.$$

If not clipped, $z + (3\eta/2)s > .452$ and weighted Cauchy gives $s^2 + (6/5)z^2 > .452^2(57/169) > .068$. If clipped the graft alone is $> 5/96$. Both totals exceed $.8937 + .058 + 5/96 > 1$.

With no weak ends graft both propagated endpoint floors in the two five-blocks. Their gain is $(5/2)z^2$, with $z = (\eta t - b_0)_+$ and $z + (3\eta/2)s > .448$. Its sum with s^2 is $> .448^2(190/481) > .0792$, contradicting E_0. The grafts stay ordered even for $\eta t < 1$, because their other block entries exceed .1125 and the next reference entry is $< .108$. Finally if both ends are weak with a near helper, its .58 graft gives $E_A > .9824552$. If the opposite propagated floor exceeds .58 the other .58 graft already contradicts the norm. Otherwise graft its actual propagated floor there; with $z = (\eta t - b_1)_+$ one has $z + (3\eta/2)s > .480$, so its gain with s^2 is $> .480^2(57/169) > .077$. Both block changes are legal and disjoint. All end patterns, internal weak sets and actual own/foreign possibilities are covered, spending the one I energy only once. □

K.25 The proper classical rows in exact dimension nineteen

Lemma K.31. *The entire forest $B_{18} + I$ is impossible.*

Proof. Use the actual short root $n_{18} = e_{18}$ and long roots before it. The genuine generic bulk covers five through fourteen. At fifteen retained B_3 admits its long-three and short-three F_4 extensions. The latter has residual $1/4$, numerator $\gamma + 1/(2\sqrt{2})$, own-zero affine F_4, and positive-own ratio $\geq 1 + 1/\sqrt{2}$. Its full white F_4 cannot acquire an I arm; full B_{18} bridges have internal new node

and are also foreign-sealed. Left/isolated/foreign estimates use only genuine long roots and retain their full old-I clearing.

At sixteen the right piece is B_2. A second four is not a finite extension, so the only new contacts are a three at either end. Long-three has numerator $\gamma + 1/2$, residual $1/2 - Q$, affine own zero/three and otherwise strong ratio. Short-three has numerator $\gamma + 1/(2\sqrt{2})$, same residual; owns zero/four/six have the affine F_4/star/six–three principals, parallel the actual null sum. Own at least five apart from affine six is strong; at own three a positive $Q \geq 1/4$ is strong. Thus the sole extra row is $Q = 0, v = (\omega_{16} + \omega_{18})/2$. Full endpoint bridges have internal new node, no foreign extension, and direct strong ratio already at own zero.

For the extra row release old seventeen. The same actual wall completes finite A_{18}, with the old I retained. Positive finite projection identifies the actual endpoint and gives $C_{17} = (17 + 16\sqrt{2})/\sqrt{19} > 9.09$, hence $S_{17}^2 > 13753/1009 > 3.6919^2$, stronger than f_{17}. If five is weak the actual interpolation through one, six and seventeen already has energy > 1.0004379945. If five is strong, extend its actual fourteen/fifteen reference by the chord to f_{17}. The energy is $> .9550775$, the first five entries remain $b < .361$ and their coefficient $> .146$. The single actual-I initial graft adds $> .0496$: for a nonclipped five-block, weighted Cauchy with $t + (3/2)s > .767$ gives $s^2 + (5/4)(t - b)_+^2 > (.767 - .361)^2/(61/20) > .0496$; clipping the endpoint at .6 already gives $> .0714$. This rules out the extra row before using sixteen in a bootstrap.

Coverage through sixteen is now precisely nonadjacent pure left helpers. If five is weak, one of fifteen/sixteen is strong. At sixteen its weak-five reference alone exceeds one; at fifteen it has effective energy $> .9851486$, first coordinate .5965 and remaining P_6 coefficient $> 1/60$. Lemma J.4 adds $> .015$ using the unused actual I. Thus five is genuinely strong.

Choose the actual strong endpoint fifteen or sixteen. The ordinary old-floor reference has $q > .9240775$, first five entries $b = f_5/5 < .361$, assigned P_5 pressure f_5 at the old .61 loss, and available P_5 coefficient $> .146$. We now check the local head lemma's actual whitelist, rather than importing a B_{17} row theorem. Deleting one retains $B_{17} + I$. Its only nonisolated finite extension is simple three at old two, with optional I contact only three. The isolated-within-B row may meet I by any finite label. These are exactly the two raw equations

$$A^2 = 2\gamma^2 + Q - 1, \qquad A^2 = 2\gamma^2 + 2\gamma + Q,$$

with Q arbitrary finite in the first row and $Q = 0, 1/4$ in the second. All head points use only genuine long normals one/two/three; old short root eighteen remains an unselected acute wall. The whole head, including positive foreign contact, therefore satisfies Lemma J.9. Its q-free gain $> .07665$, with I once, gives $q + \text{gain} > 1.0007275$. There is no need to prove sixteen individually strong or to repeat a terminal head/I scalar catalogue. □

Lemma K.32. *The entire forest D_{17} with either actual typed-seven defect is impossible.*

Proof. Use the genuine signed forks sixteen/seventeen, centre fifteen, and the one defect projection of energy $e_7 > .09$. For old $I_2(7)$, white positivity prohibits any foreign contact, since its smallest cost is $1/[4\sin^2(\pi/7)] > 1$. For old A_2 both old nodes cannot be contacted (the triangle is affine); a single contact gives precisely $Q \in \{0, 1/3, 2/3, \phi^2/3\}$. In particular every positive foreign cost is at least $1/3$. This is not the I cost $1/4$.

The bulk covers five through twelve, with only nonadjacent pure helpers. Choose actual strong $K = 10$ or eleven, and separately nine or ten. In both initial-five branches a positive reference has endpoint no greater than f_K, energy

$$q_9 > .797425,\ q_{10} > .823735,\ q_{11} > .8472225, \quad f_9 < 2.569, f_{10} < 2.732, f_{11} < 2.886.$$

The endpoint price exceeds .16 at nine/ten or 1/7 at ten/eleven. For every actual pure pressure $x.V > G$ flat one through K, $V^2 \le R$, its one-square payment gives

$$\|x\|^2 + e_7 > q_K + .09 + 2\mu(G - f_K) - \mu^2(R - K). \tag{74}$$

For $K = 10, 11, \mu = 1/7$, it suffices that $14G - R > 32.6$; the two resulting margins above one exceed $25003/9800000, 47961/19600000$.

At thirteen retained D_4 permits three at a leaf. The long leaf fourteen uses numerator $\gamma + 1/2$, not the fork numerator $\gamma + 1/4$. Own zero/three is affine, otherwise the long-leaf row is strong. In the fork row owns zero/three/four/six and parallel have their actual E_8/cycle/weighted-star/six–three/null obstructions; positive Q and pure finite at least seven or dotted are strong. Its only extra is pure own five. The left H_4 correction on thirteen/twelve/eleven gives $\alpha^2 = \phi + 5/8$, $A^2 = 13\alpha^2 - 1/2$, flat displacement through ten, $R < 13$, $G > 3.48$. Pay .16 at actual nine/ten; the totals in (74) exceed $1.076545, 1.076295$. Full bridges are finite D_{17} with internal new node, so cannot acquire a finite foreign arm; residual 1/26 and the own-zero cycle cover them.

At fourteen right A_3 adds middle-three/end-three/four/five. Middle-three/end-four have their affine or direct strong rows, end-five is a sealed actual H_4 strong already at own zero. Full bridges are finite D_{17} or A_{17}, with the new node too far from the ends to add an A_2 arm. The lone end-three has residual $5/8 - Q$, numerator $\gamma + 1/4$. Positive Q is strong; zero/three/four/six and parallel have the actual affine/null obstructions. Pure dotted owns are strong. Own five has the left H_4 foot $G > 3.61, R < 14$, flat through eleven, hence $14G - R > 32.6$. For the remaining $.9 < \gamma < 1$ select both actual A_2 cores with old corrections $\gamma(4n_{14}/3 + 2n_j/3)$, $j = 13, 15$. Their averaged pressure is pure, flat through twelve, with

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad G > \sqrt{14 - 125/529} - .327 > 3.38, \quad R = 14 - \frac{14\gamma^2/9 - 3/8}{2(\gamma + 1/4)^2} < 13.67.$$

The subtracted quotient increases from $177/529 > .33$; the radical quotient increases to $8/75 < .327^2$. Thus $14G - R > 33.65$ on the complete interval. All cores have positive coefficients, negative square, future time and nonpositive products at every original wall, including the two defect walls. Coverage now reaches fourteen.

Delete centre fifteen. Retained whites are A_{14} plus the actual two forks and the defect. With left contact η at j, forks δ, ϵ and total foreign cost Q,

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = \gamma + \eta j/15 + (\delta + \epsilon)/2.$$

The sufficient .61 ratio target is $< 25/16$. The complete old-piece list is left endpoint-three/four, next-three, neither; left plus one fork only endpoint-three with fork-three/four; left plus both forks only the all-three bridge; without left contact, one fork or the pairs $(3,3), (3,4), (3,5)$. Additional cycles, long arms and interior four bonds are not finite. Full bridges are foreign-sealed; the one-fork-three bridge cannot acquire a foreign A_2 arm of lengths fourteen/one/two. Their residuals $1/30, 17/60$ give the affine or direct strong rows. In the no-fork list zero-own far-three would need $1/2 < Q < 8/15$, absent from the exact cost list; far-four/next-three require $Q < 1/15, 2/15$, so have time zero. The other zero-own rows are actual affine principals. The isolated and foreign-cleared feet use genuine old fifteen/fourteen/thirteen, and only the pure left helper remains weak; the sum of the two fork duals is solely a spatial identity.

The complete hard no-left centre rows have the following pure displacements, all flat through

at least eleven:

row	actual construction	G lower	R upper
own 0, fork 5	left H_4	3.49	14
own 0, fork ≥ 7	three-wall core	3.4	15
own 3, fork $4,5$	actual reciprocal D_{17}	3.81	17
own 5, fork 3	left H_4	3.73	15
own ≥ 7/dot, fork 3	special core	3.49	15
own 0, forks $3,4$	actual reciprocal B_{17}	3.81	17
own 0, forks $3,5$	finite H_3	3.53	15

Every row has $Q = 0$: a positive cost makes the five/own-three rows strong and is prohibited by white positivity in the others. For the fork-five row, its normalized time and finite residual are $1 + 8\phi, 13 - 8\phi$, giving $G > 3.49$. For fork at least seven let $z = 1/\delta^2 \in (1, 100/81)$. The positive core is $c + \delta^{-1} n_f + 2(\delta^{-1} - \delta) n_{15}$, with

$$G = \sqrt{17 - 2z} - \sqrt{-8 + 14z - 6z^2} > 3.81 - .41 = 3.4, \quad R = 25 - 16z + 6z^2 < 15.$$

The second radicand is $1/6 - 6(z - 7/6)^2$. For own three and fork four/five the same actual centre wall completes a finite D_{17} endpoint with $\cosh 15/2 + (17/2)\delta > 13$; positive inverse projection retains the defect and recognizes that endpoint. For the three/four pair the finite mate is B_{17} with $\cosh 15/2 + (17\sqrt{2})/4 > 13$. Both give the stated $G > \sqrt{17(6/7)} > 3.81$. Own-five/fork-three has the left H_4 correction and $G > \sqrt{15 - 75/224} - .091 > 3.73$. For the remaining $.9 < \gamma < 6/5$ single-fork-three interval use $c + \gamma^{-1} n_{15} + (2\gamma)^{-1} n_{14}$, of norm $-[1 - 3/(4\gamma^2)]$. Its normalized radical is $< .34$ and time $> \sqrt{15 - 150/529}$, giving $G > 3.49$. Larger owns are directly strong. The three/five pair uses the positive H_3 projection with $q = 2\phi^{-6} < 1/8$ and pressure $\sqrt{15 - q}/(1 + \sqrt{q/15}) > 3.53$. Own zero fork-three/four/six and the two three forks have their affine principals; positive own four/six and parallel their star/path/null exclusions. Remaining positive-own/fork at least four rows have direct ratio $\geq \phi/\sqrt{2} + 1/2$. Thus every own/finite-white/foreign case is either listed, strong, or affine/null impossible. Each listed row satisfies $14G - R > 32.6$, so (74) excludes it.

Only nonadjacent pure helpers remain through fifteen. Weak five would give a strong fourteen/fifteen endpoint and energy $> .968$ from its actual first-six reference; adding the one .09 defect contradicts the norm. Thus five is strong. If fifteen is strong its reference $q > .9240775$ suffices. Otherwise fourteen is strong, $q > .90715$. The actual weak-fifteen feet use only n_{15}, n_1, n_2 and positive corrections, with both fork products separately negative; the spatial sum $\omega_{16} + \omega_{17}$ is not an added wall. More explicitly, put $\Omega_{16} = \omega_{16} + \omega_{17}$ solely as a spatial sum. Their displacements are $\gamma(\omega_{14} + \Omega_{16})/2 + \omega_3/6$ for five, the same with $\omega_2/4$ for seven/eight, or $\omega_1/2$ for finite owns at least nine. For dotted own the displacement is $(\gamma - \gamma^{-1})\omega_{15} + (\omega_{14} + \Omega_{16})/(2\gamma) + \omega_1/2$. Their respective pressures are $H(A, 2/3 - \gamma^2)$, $A - \sqrt{\gamma^2 - 3/4}$, $H(A, 1 - \gamma^2)$, $A - \sqrt{1 - \gamma^{-2}}$, with $A^2 = 30\gamma^2 + 2\gamma - 1/2$. Concavity of the original heights bounds the dual sum by $2h_{15}$ and $h_2 \leq 2h_1, h_3 \leq 3h_1$. The full native/dotted scalar in the proof of Lemma K.11, valid for $\sqrt{10} \leq q \leq \sqrt{42}$, therefore gives $x_1 > \sqrt{2}[T - \gamma(\sqrt{30} - .61)] > .424$. Grafting only the first five slopes gains $(5/4)(.424 - .361)^2 = 3969/800000$. The same reference and one defect therefore give $\|x\|^2 + e_7 > .90715 + 3969/800000 + .09 = 1 + 1689/800000 > 1$. Both typed defects are excluded. □

Lemma K.33. *The entire forest $D_{18} + I$ is impossible.*

Proof. Use the actual centre sixteen and signed forks seventeen/eighteen; $x_1 > \cdots > x_{17} > |x_{18}|$, with no assumed sign of x_{18}. Let s, t be the same rank-free actual-I parameters. We will need the slightly stronger elementary barrier

$$t + (8/5)s > .788. \tag{75}$$

To prove it, substitute $t_0 = .788 - 1.6s > 0$ into $(1-s^2)(1-t^2) - 8st$. At $z = s - 1/5 \in (-1/5, 2/15)$ its value is

$$.00093696 - .0586944z + 8.962384z^2 - .4736z^3 + 2.56z^4 > .0009 - .06|z| + 8z^2 > 0.$$

The cubic is bounded below by $-.064z^2$ and the last quadratic has negative discriminant. Since the original expression decreases with positive t, the barrier follows. On any strong-five reference with $b < .361$, a legal first-five graft and the one I energy consequently add

$$s^2 + (5/4)(t-b)_+^2 > (.788 - .361)^2/[4/5 + (8/5)^2] > .05426.$$

If $t \geq .6$, clip that graft at .6; it alone gains $> .0714$. All other block entries stay above the next slope, so this is an actual positive-dual payment, not an independent norm estimate.

The genuine bulk is complete through thirteen. Choose an actual strong $K = 10$ or eleven before classifying either tail. In the strong-five branch the sharper ordinary references have $q_{10} > .8244, q_{11} > .8482$, endpoints $f_{10} < 2.731, f_{11} < 2.8854$, and available price $> 1/7$. Indeed their unomitted energies exceed $.824619, .848442$; nonadjacent holes at six/eight/ten lose respectively at most half the squares of $.0172, .01111, .008$. The total losses are $< .000209637, .000241637$. In the weak-five branch the actual first/six references give $L_{10} > .8870, L_{11} > .9104$, without any I charge. For every actual pure pressure $x.V \geq G$, flat one through K, $V^2 \leq R$, the two branches therefore give

$$\|x\|^2 + y^2 > \begin{cases} q_K + .05426, & 5 \text{ strong}, \\ L_K, & 5 \text{ weak}, \end{cases} \quad + \frac{2(G - f_K)}{7} - \frac{R - K}{49}. \tag{76}$$

The pure residual vanishes on the initial support, so its cross with the first-five/I adjustment is zero.

At fourteen retained D_4 admits three at a leaf. Distinguish its long-leaf numerator $\gamma + 1/2$ from short-leaf $\gamma + 1/4$. The former is affine or strong; in the latter positive foreign cost, finite at least seven and dotted owns are directly strong. Zero/three/four/six and parallel have their actual affine/null obstructions. The remaining pure five has a positive left H_4 foot, flat through eleven, $G > 3.62, R < 14$, excluded in both branches of (76) by > 1.05. Full bridges have internal new node and are foreign-sealed.

At fifteen the right A_3 admits middle-three, end-three/four/five. Middle-three/end-four are affine or strong; end-five is the actual sealed H_4, strong even at own zero. Their full bridges are finite D_{18} or A_{18}, internal too far from the ends for a foreign I arm. The lone end-three has residual $5/8 - Q$, numerator $\gamma + 1/4$. Positive $Q \geq 1/4$ is strong, as are pure owns $\gamma \geq 1$. Own zero/three/four/six and parallel are affine/null. Pure five has $G > 3.74, R < 15$, flat through twelve, and is excluded by (76). For $.9 < \gamma < 1$ use both actual A_2 cores with corrections $\gamma(4n_{15}/3 + 2n_j/3)$, $j = 14, 16$. Their average has

$$\alpha = \sqrt{2}(\gamma + 1/4), \quad G > \sqrt{15 - 125/529} - .327 > 3.515, \quad R = 15 - \frac{14\gamma^2/9 - 3/8}{2(\gamma + 1/4)^2} < 14.67.$$

The saving increases from $177/529 > .33$ and the radical quotient increases to $8/75$. Inserting the full saving gives > 1.0065 in the strong-five branch and > 1.0152 in the weak branch. All feet are pure D and feasible against the entire actual I block by positive corrections. Thus coverage through fifteen consists only of nonadjacent pure left helpers, before any centre or strong-five floor.

Delete centre sixteen, leaving A_{15}, both forks and I. The exact Schur data are

$$\Delta = 1 - q_{\text{left}} - \delta^2 - \epsilon^2 - Q > 0, \quad \sigma = \gamma + \eta j/16 + (\delta + \epsilon)/2, \quad C_{16} = \sqrt{32}\,\sigma/\sqrt{\Delta},$$

and the sufficient ratio target is < 1.577. Finite classification gives the same complete structural centre list as in Lemma K.32, now with A_{15} and the actual I foreign costs. No-left fork pairs are again $(3,3),(3,4),(3,5)$. A left-plus-both-forks bridge has residual $1/32$ and is foreign-sealed; left-plus-fork-four also has residual $1/32$ and is strong at own zero. Left-plus-fork-three has residual $9/32$ at $Q=0$, and may have the retained I-three extension with $Q=1/4$; its smaller residual $1/32$ only strengthens the affine/direct ratios. It is not falsely declared sealed. With no fork the zero-own far-three would need $1/2 < Q < 17/32$, containing no native I square; far-four/next-three need $Q < 1/16, 1/8$, so have time zero. The other zero-own/isolated/foreign-cleared rows have the actual local affine/foot proofs. Only the pure left helper can remain weak without a fork.

For own zero and one fork, labels three/four/six have the actual affine E_8/weighted-star/six–three principals. Fork five has $Q=0$ or $1/4$. Its positive left H_4 projection, with $\alpha=\phi/(2\sqrt{2})$, has

$$Y_D/\alpha=(1^{14},1/2,1/2,1/2,\pm 1/2),\quad Y_D^2/\alpha^2=15,\quad \delta_5/\alpha^2=13-8\phi.$$

At $Q=0$ its time ratio is $\sqrt{2+8\phi}$, giving $G>3.62, R=15$, enough in (76). At $Q=1/4$ retain the same point's I coordinate rather than clearing and reusing it. Its ambient normalized data are

$$G=\frac{\sqrt{6+6\phi}}{1+\sqrt{(13-8\phi)/(19-2\phi)}}>3.74,\quad R=19-2\phi<15.764,\quad d=\sqrt{2}/\phi<.875,$$

where the pressure is $x.V_D+dy\ge G$, V_D flat through both early endpoints. If five is weak the bare ambient graft already exceeds 1.057. If strong, its bare value exceeds .995 using the sharper q_K, f_K above. For this one mixed row form the ambient vector with classical part $w+(V_D-P_K)/7+\lambda(e_1-P_5/5)$ and I coordinate $d/7+\nu$. Optimizing the unused nonnegative multipliers gives the legal additional gain

$$(5/4)(t-b)_+^2+(s-d/7)_+^2>\frac{[.788-b-(8/5)d/7]^2}{84/25}>\frac{.227^2}{84/25}>.01533.$$

This uses (75) and includes the full shifted I cross. If $t\ge .6$ its clipped initial gain already suffices. The mixed row is therefore excluded.

For own zero and any fork label at least seven, white positivity forces $Q=0$. The positive actual core $c+\delta^{-1}n_f+2(\delta^{-1}-\delta)n_{16}$ gives, with $b=\delta^2\in(.81,1)$,

$$G=\sqrt{18-2/b}-\sqrt{-8+14/b-6/b^2},\quad R=26-16/b+6/b^2.$$

We need the correlated continuous bound $14G-R>34.1904$, not independent coarse bounds for G,R. Set $z=1/b-7/6$. Then the second radicand is $1/6-6z^2$ and $R=15.5-2z+6z^2$. Concavity gives $\sqrt{18-2/b}\ge 4.252-.252/b$; the endpoint at $100/81$ has squared margin $1311/5062500$. The tangent bound on the other root gives

$$14G-R\ge 39.912-14/\sqrt{6}-1.528z+(42\sqrt{6}-6)z^2>34.1965-1.528z+96.8748z^2>34.1904.$$

The last step completes a square, using $1.528^2<.0061(4\cdot 96.8748)$. Equation (76) then exceeds $1.000219, 1.000313$ for strong five, and 1.008 for weak five. Every native label in this continuous interval is covered.

The remaining fork rows have the following actual pure feet:

row	construction	G lower	R upper
own 3, fork $4,5$	reciprocal D_{18}	3.94	18
own 5, fork 3	left H_4 or cleared core	3.64	16
own ≥ 7/dot, fork 3	special core	3.62	16
own 0, forks $3,4$	reciprocal B_{18}	3.94	18
own 0, forks $3,5$	finite H_3	3.66	16

For the first row, positive Q is directly strong; at zero the same cap has actual reciprocal cosh $8+9\delta>14$, giving $G>\sqrt{18(13/15)}>3.94$. For five/three, $Q=0$ or $1/4$ remains, with $A^2=16\alpha^2-3/4+Q$, $\alpha^2=\phi+5/8$. At $Q=1/4$ the actual core is $c+\gamma(3n_{16}/2+n_{15}+n_{14}/2)+e/2$, of square $\delta_5-1/4<0$; it is pure D after clearing. Its time ratio exceeds 3.971 and radical ratio is $<.322$, so $G>3.64$; the zero-cost projected point is stronger. For single-fork-three and $.9<\gamma<6/5$, $Q=0$ remains; the actual special core has norm $-[1-3/(4\gamma^2)]$, time ratio >3.964 and radical ratio $<.34$, hence $G>3.62$. Positive Q and larger owns are strong. The two-fork three/four actual mate has cosh $8+(9/2)\sqrt{2}>14$. The three/five H_3 projection has $q=2\phi^{-6}<1/8$ and pressure $\sqrt{16-q}/(1+\sqrt{q/16})>3.66$. The affine triangles, stars, six–three paths, squares and parallel nulls exclude all remaining small-own rows; positive own with fork at least four has direct ratio $\geq\phi/\sqrt{2}+1/2$. Each displayed pure foot is flat through at least thirteen, has positive actual correction coefficients and is feasible at both actual signed forks and every unknown wall. Its substitution in (76) exceeds one in both initial-five branches. This completes every finite-white/foreign/own case at centre sixteen.

Coverage through sixteen is now exactly the pure left-helper list, with actual nonadjacency. The prefix inverse entries $2\min(i,j)$ hold through the centre; both actual forks are retained positive old blocks in the mixed projections. No spatial dual sum is used as a new root. The same weak-five bootstrap as in Lemma K.31 therefore proves five strong: strong sixteen gives bare energy >1, while strong fifteen gives $>.9851486$ and the capped initial-I gain $>.015$. Choose the actual strong endpoint fifteen or sixteen; it supplies $q>.9240775$, first-five value $b<.361$ and remaining P_5 coefficient $>.146$. Finally delete head one. Retained $D_{17}+I$ has isolated or simple-three at old two, the latter optionally meeting I only by three. Other leaves give exceptional arms too large, and four or a further fork is not finite. Its raw equations are exactly those printed in the preceding lemma, with every pure head point using actual D long roots one/two/three. The actual signed forks remain unselected acute walls. All conditions of the q-free Lemma J.9 hold. Its gain $>.07665$ with the one unused I gives a total >1.0007275. We do not need the older separate proof that sixteen is individually strong. □

Lemma K.34 (The remaining ordinary and B proper rows)**.** *The forests $D_{17}+T_2$, $B_{17}+N_7/O_7$, and B_{18}/D_{18}+ordinary A_1 are impossible. Here T_2 is ordinary; both actual typed defects are distinguished.*

Proof. For $D_{17}+T_2$, the genuine bulk reaches twelve. At thirteen the retained D_4 whitelist is exactly that in Lemma K.32, with a general ordinary dihedral retained. Any positive foreign cost is at least $1/4$ and makes the short-leaf row strong. Pure finite owns at least seven or dotted have ratio $>1.15\sqrt{2}>25/16$, above the old target at $\sqrt{26}$. The sole extra pure five has the actual left H_4 foot $G>3.48, R<13$, flat through ten. Pay $1/7$ at actual nine/ten, in either initial-five branch; the pure D energy exceeds .9757, incompatible with $e_{T_2}>1/7$. Full bridges have internal new node and are foreign-sealed. Hence only nonadjacent pure helpers remain through thirteen. The *proof* of the prefix-gain lemma now applies with $M=12$: its actual weak-thirteen foot uses only old one/two/twelve/thirteen/fourteen, all genuine long roots. It gives $e_D>.8689525+.00496125=.87391375$; the ordinary dihedral again excludes it. No D3 floor is used.

For B_{17} with either typed defect, the bulk reaches thirteen and the right B_3 layer fourteen is completed by its actual long-three or sealed F_4 list. Thus pure-helper coverage reaches fourteen before any bootstrap. Weak five would give an actual strong thirteen/fourteen endpoint and energy $>.7600299945+(19-2\sqrt{78})/7>.95$; the defect's .09 excludes it. Five is strong. At fifteen the new B_2 short-three row is affine at owns zero/four/six, strong for the other owns at least five, and strong at own three with positive foreign cost. Its sole extra pure own-three cap is $v=(\omega_{15}+\omega_{17})/2$.

Release old sixteen: this same cap, with the entire defect retained, completes finite A_{17} and is the actual endpoint by positive projection. Its cosh is $(16+15\sqrt{2})/\sqrt{18} > 8.7$, giving $S_{16} > 3.56$. Choose actual strong $K = 13$ or fourteen; their reference energies are $> .88864, > .90715$, endpoint upper bounds $3.175, 3.311$ and available coefficients $> .14, > .13$. Append the chord to 3.56 at sixteen, skipping fifteen. The slopes $< .13, 1/8$ fit those individual coefficients. The complete totals with the one defect exceed $.88864 + .385^2/3 + .09 > 1.028$ and $.90715 + .249^2/2 + .09 > 1.028$. The extra cap is impossible; coverage now reaches fifteen. The actual prefix-gain proof at $M = 14$ gives $e_B > .91211125$, again incompatible with .09.

For B_{18} with a nonshort A_1, the B_3 layer fifteen is the same sealed affine/strong list as in Lemma K.31; it does not use a short-I parameter. For D_{18} distinguish the D4 long leaf at fourteen, complete its bridges and isolate the pure own-five fork row. Its actual foot has $G > 3.62, R < 14$, flat through eleven. The old ten/eleven references give $e_D > .9957$, so the ordinary A_1 cost $> .1$ excludes this extra. At fifteen the right A_3 middle-three/end-four/five and full bridge lists are those printed above. An ordinary-A_1 contact has cost at least $1/4$, strengthening the direct ratios; full bridges have internal new node with long arms and cannot acquire that foreign arm. The pure end-three own-five has $G > 3.74, R < 15$; other remaining $.9 < \gamma < 1$ owns use the two actual A_2 cores, flat through twelve, $G > 3.5, R < 15$. The old eleven/twelve references give respectively $e_D > 1.0094$ and $e_D > .9408$, so both are excluded by the ordinary .1. Thus both B_{18} and D_{18} have complete pure-helper coverage through fifteen in this spectator setting. The prefix-gain proof at $M = 14$, using actual existing long neighbors rather than a false generic rank invocation, gives $e_{B/D} > .91211125$. Adding the nonshort A_1 is impossible. All pressures in this proof are pure in the classical block; the named orthogonal spectator energy is counted once. □

K.26 The complete dimension-nineteen forest cover

Proposition K.35. *If an actual farthest vertex in compact dimension nineteen exists, its entire forest is one of* A_{19}, B_{19}, D_{19}.

Proof. We supply the own-rank exhaustion, using current component bounds to avoid an unnecessary succession of older weaker frontier tables. As in dimension twenty a dominant component is A_r with $r \geq 7$ or B_r, D_r with $r \geq 9$. Each costs more than one half, so two cannot occur. Without one, discard at most the single I rank: the ordinary cost is $> 18/14$. A typed defect instead gives $.09 + 17/14 > 1$; the funded J pair is stronger than its ordinary rank baseline. Hence exactly one dominant occurs.

An A_7 dominant is excluded by its half bound and remaining ordinary ranks. At eight through twelve use $4/7$: even discarding I leaves six ordinary ranks at twelve; a typed defect leaves five besides its .09. Both costs strictly exceed the complementary budget. At thirteen/fourteen the actual $5/7$ theorem and four ordinary ranks suffice even after discarding I. A typed defect leaves total ordinary rank at least three; its cost $> 17/70$ together with .09 closes this alternative. For rank fifteen use $> .826$. Its ordinary rank-four remainder is $> 2/7$, an I plus ordinary rank-three is $> 2/7$, and a typed defect plus an ordinary dihedral or two ordinary A_1 costs $> .09 + 1/7$. Every sum exceeds one. Rank sixteen has $> .86$; the complete rank-three remainder possibilities are an ordinary irreducible, ordinary dihedral plus A_1, three A_1, I+dihedral, I+two A_1, $J + A_1$, or a typed defect plus ordinary A_1. Their weakest cost is $> .19$, so every sum again exceeds one. This includes the four older residual A16 cases in one line.

At seventeen the ordinary irreducible dihedral costs $> 1/7$, which with $e_{A_{17}} > 6/7$ suffices strictly. Two ordinary A_1, $I + A_1$ and J are stronger. Either actual typed rank-two defect is excluded by Lemma K.29. At eighteen an ordinary A_1 is excluded by $.91 + .1 > 1$; the actual I alternative by Lemma K.30. Rank nineteen is the whole A_{19} row.

For a B/D dominant its 9/14 theorem forces rank at least fourteen: at thirteen the worst I remainder leaves five ordinary ranks, or a typed defect leaves four beside .09. At fourteen/fifteen/sixteen the valid weaker D prefix floors are respectively $> .7683275, > .797425$, $> .823735$. The smallest sufficient spectator charges are respectively $> 2/7$, $> .09 + 1/7$, and $> .19$. These account for all ordinary partitions, I, J, and both distinct typed defects: the rank-five or rank-four I remainders use the proved joint I+rank-three or ordinary-rank rates, not a fictitious ordinary cost for I. Every row exceeds one.

At seventeen a two-A_1 or $I + A_1$ remainder costs $> 1/5$, already sufficient with the weaker prefix $> .8472225$; J is stronger. An ordinary dihedral is excluded for B_{17} by its genuine $M = 12$ floor $> .8689525$, and for D_{17} by Lemma K.34. Both typed defects are excluded for B_{17} by that lemma and for D_{17} by Lemma K.32. At eighteen the remaining rank-one factor is ordinary or I. The ordinary rows are excluded by Lemma K.34; the actual I rows by Lemmas K.31 and K.33. Rank nineteen is whole.

Finite irreducible classification supplies no additional exceptional types beyond the small components already charged in the ordinary rate. Defect incompatibility forbids a second defect; every surviving old rank-two seven without its specified isolated opposite is ordinary. The rank partitions above are therefore exhaustive and necessary, not assertions that the three retained whole forests are realized. □

The final exact-nineteen conjunction uses this complete cover and the three separate whole-case propositions L.1, M.1, and N.3. Each concerns an actual whole forest at this same rank, with its own complete head/end classification. No twenty-dimensional theorem is decremented and no Coxeter structure is inherited by a face.

L The whole A_{19} case

The next three appendices treat the whole classical forests at rank nineteen. In each case every point and pressure belongs to one original farthest vertex p, chosen from one strict interior point. We retain $z = E(p - x)$, $x \in p^{\perp}$ and $\|x\| < 1$. No arithmetic restriction is used.

L.1 A whole A_{19}: the window and the three reference shapes

Use the zero-sum subspace of $\mathbb{R}^{20}$, with actual old roots and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad \Omega_i = \sqrt{2}(P_i - iP_{20}/20), \quad D_i = \Omega_i^2 = i(20 - i)/10.$$

Write $h_i = x \cdot \Omega_i$, $h_0 = h_{20} = 0$. The sequence is strictly concave and

$$\|x\|^2 = \frac{1}{2}\sum_{i=0}^{19}(h_{i+1} - h_i)^2.$$

A central site $5 \le i \le 15$ is weak when $h_i \le \sqrt{D_i} - .61$, equality included.

We use the shared complete .61 window of Lemma K.2 at $N = 20$, before any contrary component-energy bound: every weak cap is pure, with no spectator projection, and is either

$$v_i = g\Omega_i + \Omega_e/2, \quad e = 1 \text{ or } 19, \quad \cos(\pi/5) \le g < 27/25,$$

or a near-four cap at $i = 5, 6$ or a reflected site. The first row has own native five, finite native at least seven, or dotted; the smaller own labels and parallel case have their actual affine or feasible-null obstructions. The literal helpers are nonadjacent. These are the actual finite-white, own-zero,

foreign-clearing and mixed two-cap conclusions of the shared window proof, with retained paths of rank at least four and full connected rank nineteen; they do not follow by applying a higher-rank whole-forest exclusion here. Its full internal proof precedes the higher-tail energy consumers; it is not a published classification or an additional assumed field condition.

There is a useful stronger removal of near four at this rank. For a left near-four cap at $i = 5, 6$, its positive finite replacement at $j = i - 1$ is an actual endpoint. Its white forest is $B_{21-i} + A_{i-2}$; positive inverse projection proves feasibility against every original wall, including the released j. The exact edge cosine is

$$\cosh L_j = \sqrt{2(i-1)/20}\left\{(21-i)/2 + (20-i)/\sqrt{2}\right\} > 11.$$

For $i = 5$, use $.632(8 + 7.5(1.414)) > 11$; for $i = 6$, use $.707(7.5 + 7(1.414)) > 11$. Its normalized dual demand is $> \sqrt{5/6} > .91$. At the opposite end, either a second near-four mate has that same demand, or helper nonadjacency gives a strong $k = 14$ or fifteen, with demand $> 1 - .61/2.7 > .774$. The correct normalized correlation is

$$\rho = \sqrt{\frac{j(20-k)}{(20-j)k}} \le 1/\sqrt{7} < .38.$$

Cauchy would give $(.91 + .774)^2 < 2 + 2(.38) = 2.76$; instead $1.684^2 = 2.835856$. Thus near four is impossible under $\|x\| < 1$, without a stronger component assumption.

All remaining weak sites are therefore nonadjacent pure helpers. Their edge-ratio target is

$$R(q) = 100/61 - \frac{q - .61}{q(2q - .61)}.$$

It increases for $q \ge 5/2$, while $q \le \sqrt{10} < 16/5$ and $R(16/5) = 3/2 - 251/565104 < 3/2$. For side order $m \ge 5$, $\sigma = g + 1/(2m)$, $\delta = (m+1)/(2m)$, and

$$\frac{\sigma^2}{\delta} - (4g - 2) = \frac{(g - 1 - 1/(2m))^2}{\delta} \ge 0.$$

Weakness consequently sharpens the entire helper range to $g < 17/16$. Native five has $.809 < g < .81$; every other finite own has $.9 < g < 1$.

Take rational symmetric heights

$$H_5 = 2.128,\ H_6 = 2.288,\ H_7 = 2.406,\ H_8 = 2.488,\ H_9 = 2.536,\ H_{10} = 2.552, \qquad H_{20-i} = H_i.$$

They lie strictly below the strong floors by positive rational squaring. Linear interpolation on the tails gives nonnegative dual coefficients. Omit every actual internal weak knot by replacing its height by the mean of its neighbours. The holes are nonadjacent, so their open affected edge pairs are disjoint and their exact quarter-square losses add. If an end site five or fifteen is weak, flatten that entire end through six or fourteen instead. The complete reference data are

weak end count	$E = \|w\|^2$ at least	end block lengths	individual capacities
0	$.9544848 - 5(.000441) = .9522798$	$5, 5$	$\beta_5, \beta_{15} \ge 83/625$
1	$3462479/3750000$	$6, 5$	$\beta_6 \ge 79/600,\ \beta_{15} \ge 83/625$
2	$1341604/1500000$	$6, 6$	$\beta_6, \beta_{14} \ge 79/600.$

(A19refs)

For count one orient the weak end to the left by relabelling the old path. This assumes no symmetry of the polytope. There are at most five internal holes in the first row, four in the other rows. The

one-end row loses at most 4(.000441); the two-end row loses at most 4(.000324). Their exact undeleted energies follow from

$$a^2/5+(b-a)^2+T,\quad \tfrac12[b^2/6+a^2/5+(b-a)^2+2T],\quad b^2/6+T,$$

where $a=2.128$, $b=2.288$ and $T=.118^2+.082^2+.048^2+.016^2$. The boundary coefficients are

$$(a/5-(b-a))/2=83/625,\qquad ((7/6)b-H_7)/2=79/600.$$

An adjacent hole increases these capacities. All positive coefficients are paid by actual strong floors, so $x\cdot w\ge E$ in every allowed hole pattern. The constant signed means obey

$$.3<b_5<.302,\qquad 1/4<b_6<.27.$$

L.2 The actual helper feet and their complete continuous bounds

For a weak site five, write $c=Ap-v$, $v=g\Omega_5+\Omega_e/2$. The fresh exact times are

$$A^2_{\text{far}}=(300g^2+20g-21)/40,\qquad A^2_{\text{near}}=(300g^2+60g-21)/40. \tag{A19times}$$

Both formulas concern $N=20$.

The following constructions give actual pure spatial pressures $x\cdot U\ge H$. At native five use the finite H_4 correction

$$k=gn_5+\tfrac23 n_e+\tfrac13 n_{\text{adj}},\qquad \delta=2/3-g^2>0,\quad U=v-k.$$

The normalized finite projected point is obtained from $p+(A/\delta)(c+k)$. Its future time is positive, its square is $-1-A^2/\delta$, and all selected products vanish. Every omitted original-wall product is nonpositive by the positive normal coefficients and acuteness. The pressure is

$$H=A+\delta/A-\sqrt{\delta(A^2+\delta)}/A>A-\sqrt{\delta}>A-.111.$$

The spacelike correction $c+k$ alone is not a hyperbolic point.

For finite own at least seven, put $d=1-g^2$ and use $\xi=c+gn_5+2dn_e$. Its selected products are zero, zero and $-1/2+2d<0$; its square is $-d(1-4d)<0$. It is future and satisfies every other original wall termwise. Thus $U=v-gn_5-2dn_e$, $H=A-\sqrt{d(1-4d)}$, with the last radical at most $1/4$. For dotted own use $\xi=c+g^{-1}n_5$, of square $-(1-g^{-2})$, zero cap product and negative own-root product. It gives $U=v-g^{-1}n_5$, $H=A-\sqrt{1-g^{-2}}$. All vectors U are spatial in the original $p^\perp$.

A near helper forces $x_1>.64$. Indeed weakness gives $h_5<2.129$, and dropping the positive old-root corrections from the pressure yields $x_1\ge\sqrt{2}(H-gh_5)$. At native five, $A>2.365$, giving

$$x_1>1.414(2.365-.111-.81(2.129))>.64.$$

For finite $g\ge.9$, $A'_{\text{near}}(g)>\sqrt{7.5}>2.7$; thus $A_{\text{near}}-2.129g$ increases. At .9, $A>2.626$, giving $1.414(2.626-.25-.9(2.129))=.6502986$. For dotted $1<g<17/16$, $A'_{\text{near}}<3$, whereas

$$\frac{d}{dg}\sqrt{1-g^{-2}}=\frac{1}{g^3\sqrt{1-g^{-2}}}>2.$$

Here $g^3<5/4$, the radical $<2/5$. The difference $A_{\text{near}}-\sqrt{1-g^{-2}}-2.129g$ decreases. At 17/16, its time is >3.0879 and radical $<.338$, which again gives $x_1>.64$. All time/derivative comparisons follow by positive squaring in (A19times).

If site five is weak, its first-six reference block has mean $b_6 < .27$. Grafting .64 into its first coordinate, preserving the sum, raises its energy by $(6/5)(.64 - .27)^2 = .16428$. The other five coordinates remain $> (1.5 - .64)/5 = .172$, above the next old coordinate $< .118$. This is a legal ordered dual reference, and even the weakest row of (A19refs) then exceeds one. Every weak end helper is therefore FAR, in both orientations.

For a far helper on six/six blocks, block averaging gives

$$\Pi U = (5g/6)\Omega_6 + \Omega_{14}/12, \quad Y = U - \Pi U, \quad w \cdot Y = 0, \quad \Delta = H - 2.288(5g/6 + 1/12).$$

The three residual squares are

own	$Q = \|Y\|^2$
5	$(2/3)g^2 + 1/12$
finite ≥ 7	$4g^4 - (16/3)g^2 + 29/12$
dotted	$(5/3)g^2 - 19/12 + g^{-2}$.

(A19Q)

At native five, $A_{\text{far}} > 2.188$, so $\Delta > .34 > .30$, $Q < .521 < 13/12$. For all other finite owns, $A_{\text{far}} - 2.288(5g/6 + 1/12)$ increases, since $A'_{\text{far}} > \sqrt{7.5} > 5(2.288)/6$. The complete radical/time seals are

g interval	A_{far} at left $>$	$\sqrt{(1-g^2)(4g^2-3)} \leq$	$\Delta >$
$(.9, .91]$	2.449	.2318	.3105333...
$[.91, .92]$	2.478	.2434	.3088666...
$[.92, 1)$	2.5065	.25	.3117.

The first two radical bounds use the increasing quartic on those short intervals; its global maximum is $1/16$. The tight middle lower bound is $23165/75000$. The finite residual polynomial increases for $g^2 > .81$ and tends to $13/12$. Thus every finite case has $\Delta > .30$, $Q < 13/12$.

In the dotted interval the same derivative argument makes Δ decrease. At $17/16$, $A_{\text{far}} > 2.9108$, the radical is $< .338$, and $2.288(5g/6+1/12) = 2.2165$. Thus $\Delta > .3563 > .35$. Its Q increases with $g^2 \geq 1$ and at $17/16$ equals $229/768 + 256/289 < 148/125$; the cleared margin is $271 > 0$. These bounds cover the whole continuous dotted interval.

L.3 Two weak ends: the exact nonzero cross and both prices

Suppose both five and fifteen are weak. Use the six/six reference in (A19refs). For a left far-helper foot, its scaled first and last residual blocks are

$$\sqrt{2}Y_{\text{own}} = (g/3, g/3, g/3, g/3, g/3 - k, -5g/3 + k),$$
$$\sqrt{2}Y_{\text{helper}} = (1/6, 1/6, 1/6, 1/6 - m, 1/6 - l + m, -5/6 + l).$$

Here (k, l, m) equals $(g, 2/3, 1/3)$ at native five, $(g, 2(1-g^2), 0)$ for other finite owns, and $(1/g, 0, 0)$ for dotted. The blocks have zero sum and direct expansion gives (A19Q). For the right foot of parameter d, reverse and negate the two blocks. In the product of an own block with a reversed helper block only its final two deviations survive; their scaled product is $-g/3$, independently of k, l, m. Consequently for EVERY type pair,

$$Y_L \cdot Y_R = (g + d)/6, \qquad \|Y_L + Y_R\|^2 = Q_L + Q_R + (g + d)/3.$$

We have not assumed orthogonality of the two actual feet.

Choose $\mu = 13/100$ and form one vector $W = w + \mu(Y_L + Y_R)$. At each of the two individual anchors, the combined debit is

$$\mu(5g/6 + 1/12) < 403/3200 < 79/600.$$

The remaining coefficient is nonnegative before using its original floor. Both actual pressures and one completed square give the following complete three-pair table:

types	$\|Y_L + Y_R\|^2 <$	gain over $E >$
finite/finite	17/6	6487/60000
finite/dotted	17729/6000	.11906
dotted/dotted	9229/3000	.13000.

Native five belongs to the finite column. For example the first gain is $.26(.60) - .0169(17/6)$. All three gains follow from the full cross above and the uniform deficit/norm bounds. Thus

$$\|x\|^2 > \frac{1341604}{1500000} + \frac{6487}{60000} = 1 + \frac{3779}{1500000} > 1.$$

This excludes the two-weak-end case with every internal hole retained.

L.4 Zero or one weak end: the actual endpoints and their feet

For either of the other reference rows, if $x_1 \geq .55$ graft .55 into its constant first block, preserving the block sum. For length five the gain exceeds $(5/4)(.55 - .302)^2 = .07688$; for length six it exceeds $(6/5)(.55 - .27)^2 = .09408$. Both exceed $1 - E$ in their respective rows. The remaining block coordinates stay $> .2375$ or $> .19$, above the next coordinate $< .16$; a neighbouring hole only improves that junction. Reflection gives $x_1, -x_{20} < .55$. The actual endpoint comparison is

$$x_1^2 \geq (19/20)\frac{\cosh L - 1}{\cosh L + 1}.$$

Since $19/60 > .55^2$, both actual endpoint cosines are less than two.

Deleting one retains A_{18}. Finite classification gives exactly isolation; one simple-three at $2, 3, 18, 19$; or one simple-four at two or nineteen. Two contacts make a nonpositive cycle; exceptional finite types cannot contain this rank-eighteen retained path. For $v = g\Omega_1 + \beta\Omega_j$ the complete nonisolated data are

white contact	$\cosh L$	small-own obstruction
3@2	$(38g + 18)/20$	$g = 0:\ A^2 < 0; \quad g = 1/2:$ affine triangle
3@3	$(38g + 17)/(2\sqrt{20})$	$g = 0:$ affine E_8
3@18	$(38g + 2)/(2\sqrt{20})$	$g = 0:\ A^2 < 0$
3@19	$(38g + 1)/20$	$g = 0:\ A^2 < 0; \quad g = 1/2:\ A^2 = 0$
4@2	$(38g + 18\sqrt{2})/\sqrt{40}$	already > 2 at $g = 0$
4@19	$(38g + \sqrt{2})/\sqrt{40}$	$g = 0:\ A^2 < 0.$

For the affine E_8 at old three, the actual arms have lengths two (old two/one), one (the cap), and five (old four through eight). The positive affine combination has future time through the cap and nonpositive omitted-wall products. Every other positive native contact is at least $1/2$; in the first row the next allowed own is at least $1/\sqrt{2}$. Substitution makes every row except 3@19 exceed two.

A remaining nonisolated left endpoint is therefore the far-three cap

$$v_L = g\Omega_1 + \Omega_{19}/2, \quad A_L^2 = (19/10)g^2 + g/10 - 21/40, \quad 1/\sqrt{2} \leq g < 39/38 < 1.1.$$

If both endpoints had this form, their time squares would be $< (8/5)g^2, (8/5)d^2$, since $.3g^2 + .1g - .525 < 0$ for $g < 1.1$. But

$$v_L \cdot v_R = (gd + 1/4)/10 + (19/20)(g + d) > (19/11)gd > (8/5)gd > A_L A_R.$$

The actual walls are distinct: coincidence would force $g = d = 1/2$. Their positive mutual contradicts acuteness. At least one endpoint is isolated.

For an isolated endpoint, $v = \sigma\Omega_1$, $A^2 = (19/10)\sigma^2 - 1 > 0$. Own zero/three/four has nonpositive time, six has its actual affine $6, 3$ principal, and a parallel own has the nonzero feasible future null $c + n_1$. The complete remaining own list is five, finite at least seven, or dotted. For dotted own, the actual edge bound also gives $19/40 < u = \sigma^{-2} < 1$.

Normalize each actual isolated foot by $\sqrt{2}\sigma$. On the left write $Z = \zeta - P_{20}/20$, $\sum \zeta_i = 1$; the right formula is negative reversal. The complete profiles, pressure $x \cdot Z \geq t$ and square $R = \zeta^2$ are

own	ζ	t	R
5	$(1/4, 1/4, 1/4, 1/4)$	$> .34$	$1/4$
finite ≥ 7	$(1 - u/2, 1 - u/2, u - 1)$	$\{\sqrt{1.9 - u} - \sqrt{7u - 4 - 3u^2}\}/\sqrt{2}$	$3 - 4u + 3u^2/2$
dotted	$(1 - u/2, u/2)$	$\{\sqrt{1.9 - u} - \sqrt{u(1 - u)}\}/\sqrt{2}$	$1 - u + u^2/2.$

(A19endfeet)

For native five, use the finite H_4 correction $(3/2)\sigma n_1 + \sigma n_2 + (\sigma/2)n_3$. Its residual $\delta = 1 - (3/2)\sigma^2 > 0$ gives the actual finite projected point $p + (A/\delta)(c + k)$, of square $-1 - A^2/\delta$. It is future and satisfies every original wall by the selected product calculation and positivity. Its raw pressure is $H = AL/(L + \sqrt{\delta})$, $L^2 = A^2 + \delta = .4\sigma^2$. The positive-square bounds $A > .4934, L > .5112, \sqrt{\delta} < .1352$ give $H > .390$ and $t > .34$.

For finite own at least seven, use $c + \sigma^{-1}n_1 + 2(\sigma^{-1} - \sigma)n_2$. Its cap/own products vanish, the next product is $(3 - 4\sigma^2)/(2\sigma) < 0$, and its square is $-(7 - 4\sigma^2 - 3/\sigma^2) < 0$. The dotted core is $c + \sigma^{-1}n_1$, square $-(1 - u)$, zero cap product and negative own product. Both have positive future time, positive coefficients on actual normals, and every omitted-wall product nonpositive. They prove the remaining two profiles, for this same x.

The following scalar seals cover all their parameters. Native seven satisfies $.9009 < \sigma < .901$, hence $1.23 < u < 1.233$,

$$\sqrt{1.9 - u} > .8167, \quad \sqrt{7u - 4 - 3u^2} < .2671, \quad t > .388, \quad R < .34935.$$

The cosine bracket follows by signs of $8\sigma^3 - 4\sigma^2 - 4\sigma + 1$; it asserts no field restriction. For finite own at least eight, $u < 4 - 2\sqrt{2} < 1.1716$, $\sqrt{1.9 - u} > .85346$, and

$$7u - 4 - 3u^2 = 1/12 - 3(u - 7/6)^2 < 1/12.$$

Thus $t > .399$, $R \leq .5$. At $u \geq 1.1$, $R \leq .415$; at $u < 1.1$, the radicals improve to $> .8944$ and $< .265$, giving $t > .4448$.

For dotted own put $q = \sqrt{1.9}$, $a = q - \sqrt{.9} = 1/(q + \sqrt{.9}) < .43$. The lower chord and circle maximum give

$$\sqrt{1.9 - u} - \sqrt{u(1 - u)} \geq q - au - \sqrt{u(1 - u)} > .6185,$$

using $q > 1.378$, $\sqrt{.9} > .948$, $\sqrt{1 + a^2} < 1.089$. Hence $t > .437$. On $u \geq .5$ the sharper chord $\sqrt{1.9 - u} \geq 1.4178 - .4692u$ and $\sqrt{1 + .4692^2} < 1.105$ give $t > .445$. On $u < .6$, use $1.14 - .5 = .64$ to get $t > .4525$. Also $R < .6378125$ throughout, and $R \leq .58$ when $u \geq .6$. All square-root and chord comparisons are strict rational square checks.

On a constant end block of length m, the projection of Z is its boundary dual divided by $m\sqrt{2}$. Thus for $V = Z - \Pi Z$,

$$w \cdot V = 0, \quad \|V\|^2 = R - 1/m, \quad \Delta_h = t - b_m.$$

A paid coefficient $\nu \geq 0$ costs $\nu/(m\sqrt{2})$ at that individual boundary. Combining the actual pressure with the remaining original floors BEFORE its square gives

$$\|x\|^2 \geq E + 2\nu\Delta_h - \nu^2(R - 1/m). \tag{A19endpay}$$

L.5 No weak end: one isolated endpoint already suffices except at five

For zero weak ends, take $\nu = 1/2$ in the isolated finite case. Its price is $< 1/14 < 83/625$. At native seven the gain is $> .086 - .15/4 = .0485$. For all finite labels at least eight, the complete subdivision is

u	$t >$	$R - 1/5 <$	$\Delta_h - (R - 1/5)/4 >$
$(1, 1.1)$	.4448	.3	.0678
$[1.1, 1.14)$	.4122	.215	.05645
$[1.14, 1.1716)$	.399	.1894	.04965.

The middle interval uses radicals $> .8717, < .28868$, with $.4122(1.4143) < .8717 - .28868$. The last uses $> .85346, < .28868$; the polynomial R decreases here, with $R(1.1) = .415, R(1.14) = .3894$. Thus every finite isolated own at least seven gives

$$\|x\|^2 > .9522798 + .0485 > 1.$$

For an isolated dotted own, take $\nu = 1/3$, price $< 1/21 < 83/625$. At $u < .6$ the gain is $> (2/3)(.1505) - .438/9 = 31/600$. At $u \geq .6$ it is $> (2/3)(.143) - .38/9 = 239/4500$. Both gains exceed $1 - .9522798$.

Only isolated native five remains. It cannot coexist with a nonisolated far-three opposite: its time is $< .497$, while the latter has time < 1.257. Their spatial product is $\sigma(d/10 + 19/20) > .95(.809) = .76855$, whereas the time product is $< .497(1.257) = .624729$. Their actual mutual would be positive. Thus the other endpoint is isolated as well, and the preceding cases force it too to have native five.

For each normalized native-five foot, $\Delta_h > .038$, $\|V\|^2 = .05$. Use $\nu = 3/4$ at BOTH ends; each price is $< 3/28 < 83/625$. The two residuals have disjoint first/last five-block support and zero block sum, so their cross is exactly zero. The single vector $w + (3/4)(V_L + V_R)$ gives

$$\|x\|^2 > .9522798 + 2\{(3/2)(.038) - (9/16)(.05)\} = 1.0100298 > 1.$$

This exhausts zero weak ends.

L.6 One weak end: the helper and endpoint must be paid jointly

Orient the weak site to five. It is far by the helper argument. On the six/five reference its block projection and residual are

$$\Pi U = (5g/6)\Omega_6 + \Omega_{15}/10, \quad Y = U - \Pi U, \quad w \cdot Y = 0, \qquad \Delta = H - 2.288(5g/6) - 2.128/10.$$

Relative to six/six, the square decreases by $1/60$ and the deficit decreases by $83/3750$. The exact finite middle-row bound $23165/75000$, and the stronger native-five/dotted bounds, give

$$\Delta > .2867, \qquad Q = \|Y\|^2 < 1.168.$$

The dotted upper square is $148/125 - 1/60 = 1751/1500 < 1.168$.

We must retain the full cross with the isolated endpoint. Write the far-helper correction at old nineteen/eighteen as $(l, m) = (2/3, 1/3)$, $(2(1-g^2), 0)$, or $(0, 0)$. Its scaled last-five residual is

$$\sqrt{2}Y_{\text{helper}} = (1/5, 1/5, 1/5 - m, 1/5 - l + m, -4/5 + l).$$

For an isolated STRONG-side endpoint, its residual there is $1/5$ minus the reversed ζ. Direct multiplication gives

isolated own	$Y \cdot V$
5	$1/(20\sqrt{2})$
finite ≥ 7	$\{.8 - u/2 + m(1.5u - 2)\}/\sqrt{2} \leq (.8 - u/2)/\sqrt{2}$
dotted	$\{.8 - u/2 - l(1-u) - mu/2\}/\sqrt{2} \leq (.8 - u/2)/\sqrt{2}$.

The finite sign uses $u < 1.233 < 4/3$. For an isolated WEAK-side endpoint, its ζ ends before coordinate five; multiplication with the own six-block row printed earlier gives

$$Y \cdot V = g/(3\sqrt{2}) < 85/336.$$

These formulas retain every correction coefficient, including its sign.

For paid nonnegative μ, ν, one vector $W = w + \mu Y + \nu V$ therefore yields

$$\|x\|^2 \geq E + 2\mu\Delta + 2\nu\Delta_h - \mu^2 Q - 2\mu\nu(Y \cdot V) - \nu^2(R - 1/m). \tag{A19joint}$$

Both individual anchor debits must fit before either old floor is used.

If an isolated endpoint is on the strong side, choose $\mu = .148$. Its own-six debit is

$$\mu(5g/6) < 3145/24000 < 79/600.$$

At fifteen, the helper AND endpoint together cost

$$\mu/10 + \nu/(5\sqrt{2}) < 37/2500 + 4/35 < 83/625$$

for all choices $\nu \leq .8$ below. The common helper gain exceeds .059279328. Using the full cross, the complete table is

isolated strong-side own	ν	$\Delta_h >$	$R - 1/5 <$	$Y \cdot V <$	total gain over $E >$
5	.8	.038	.05	1/28	.07962218
7	.3	.086	.15	.133	.085568928
finite ≥ 8, $u \geq 1.1$	.3	.097	.215	.179	.082234128
finite ≥ 8, $u < 1.1$	.3	.1428	.3	.215	.098867328
dotted, $u \geq .6$	.25	.143	.38	.358	.080537328
dotted, $u < .6$	.25	.1505	.438	.402	.077406328.

For instance the last row is exactly $.059279328 + .5(.1505) - .0625(.438) - .074(.402)$. All cross bounds use $\sqrt{2} > 7/5$ and the stated intervals. Every gain is $> .0774$.

If the strong endpoint is not isolated, the weak endpoint is isolated by the actual endpoint classification. It cannot have native five: the same time/spatial comparison with a far-three opposite, just proved, would force that opposite to be isolated. The remaining finite/dotted weak-side cases use $\mu = .1, \nu = .35$. Now BOTH feet spend the SAME six coefficient; the full debit is

$$\mu(5g/6) + \nu/(6\sqrt{2}) < 17/192 + 1/24 = 25/192 < 79/600.$$

At fifteen the cost is only .01. The common helper gain is $> .04566$ and the full cross penalty is $< .07(85/336) = 119/6720$. The complete table is

isolated weak-side own	$\Delta_h >$	$R - 1/6 <$	total gain over $E >$
7	.118	.183	.08813416
finite ≥ 8	.129	1/3	.07741833
dotted	.167	.472	.08703166.

The tight finite row is $.04566 + .7(.129) - .1225/3 - 119/6720 = .077418333\ldots$. All gains again exceed .0774. Thus in every one-weak-end case,

$$\|x\|^2 > \frac{3462479}{3750000} + .0774 = 1 + \frac{2729}{3750000} > 1.$$

Proposition L.1. *A whole A_{19} forest cannot occur at the original farthest vertex of a compact Coxeter nineteen-polytope.*

Proof. The number of weak sites among the two central end sites five/fifteen is exactly zero, one or two. The three preceding arguments exclude these cases, with equality assigned consistently to weak. Their shared local window, endpoint and foot lemmas precede all three consumers and do not use any end-case conclusion. Every remaining internal weak site was retained through its actual omission, and every new pressure was paid inside one original-vector reference. □

M The whole B_{19} case

M.1 The short layers before the terminal analysis

Use actual unit roots

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (i \leq 18), \qquad n_{19} = e_{19}, \quad \omega_i = \sqrt{2}P_i\ (i \leq 18), \quad \omega_{19} = P_{19}.$$

Now $x_1 > \cdots > x_{19} > 0$. Put $S_i = x \cdot P_i$, $f_i = \sqrt{i} - .58/\sqrt{2}$, $d_i = \sqrt{i+1} - \sqrt{i}$, and

$$E_K = f_5^2/5 + \sum_{i=5}^{K-1} d_i^2.$$

Weak includes equality $S_i \leq f_i$. The local whole-diagram lemma K.7 applies literally through fifteen: both retained old components have rank at least four. Its complete weak list is

$$v = g\omega_i + \omega_1/2, \quad A^2 = 2ig^2 + 2g - 1/2, \quad \cos(\pi/5) \leq g < 6/5, \tag{B19packet}$$

with own five, finite at least seven, or dotted, together with isolated dotted five. Pure helpers are nonadjacent. We must extend this at the actual B_3, B_2 boundary.

At sixteen the right B_3 has a long-three extension or the short-three F_4 extension. The latter is foreign-sealed; own zero has the actual affine F_4 principal, whereas every positive own has ratio $\geq 1 + 1/\sqrt{2} > 83/50$. A full bridge is B_{19} and directly strong. Thus the same list holds through sixteen, also at the old .61 threshold. In particular some new strong $K = 15$ or sixteen exists. The full conditional early-reference Lemma K.8 therefore gives a paid vector supported through K, of sum f_K, remaining individual P_K coefficient $> 1/9$, and effective energy

$$L_{15} > .94006, \qquad L_{16} > .95618, \quad f_{15} < 3.463,\ f_{16} < 3.590. \tag{B19early}$$

Effective energy means $2D_W - \|W\|^2$, where D_W is its actual pressure demand; it need not equal $\|W\|^2$.

At seventeen the right block is B_2, with simple-three at either end. The long-three row is strong or affine. For short three, own zero/four/six have actual affine principals, parallel has its positive future null, and finite own at least seven has directly strong ratio $\sqrt{2}g+1/2 > 83/50$. A full bridge has residual $1/34$ and numerator $\geq g + 1/34 + 1/(2\sqrt{2})$, strong even at own zero. Only own three and five remain.

For own five, let $g = \varphi/2$, $\alpha = \sqrt{2}g+1/2$. The LEFT actual H_4, using seventeen/sixteen/fifteen, has

$$\delta = (5-3\varphi)/8, \quad A^2 = 17\alpha^2 - 1/2, \quad V_1 = \cdots = V_{14} = 1, \quad V^2 < 17, \quad x \cdot V > G > 4.$$

The point is the positive finite projection, not its spacelike correction; every omitted-wall product is nonpositive. Use $\alpha > 1.643$, $\sqrt{\delta}/\alpha < .083$ for the last bound. An OLD strong eleven or twelve exists by the original bulk. Their positive references, in both old initial-five branches, have

$$q_{11} > .8472225, \; q_{12} > .8689525, \quad f_{11}^{\text{old}} < 2.886, \; f_{12}^{\text{old}} < 3.034, \quad \text{individual endpoint price} > 1/7.$$

These are the actual prefix construction with its holes retained, not an assertion that old five is strong. Their support lies in the flat first fourteen. One paid square gives

$$.8472225 + 2(4-2.886)/7 - 6/49 > 1.043, \qquad .8689525 + 2(4-3.034)/7 - 5/49 > 1.042.$$

For own three, the SAME cap has a finite D_{19} replacement after releasing the short old nineteen. Positive inverse projection gives an actual reciprocal short-edge endpoint, with

$$\cosh L_{19} = 17/2 + 19\sqrt{2}/4 > 15, \qquad S_{19}^2 > 133/8 > (4.07)^2.$$

Use (B19early), and the pressure $Z = P_{19}$. The legal $1/9$ endpoint payment gives

$$.94006 + 2(4.07-3.463)/9 - 4/81 > 1.025, \qquad .95618 + 2(4.07-3.590)/9 - 3/81 > 1.025.$$

Thus coverage reaches seventeen.

Nonadjacency is also actual at this boundary: replacing two adjacent helper caps at sixteen/seventeen retains the long A_{15} and the independent old B_2. Adding their actual far-one leaves gives the positive $D_{17} + B_2$. The later seventeen/eighteen pair instead gives $D_{18} + A_1$. The positive two-cap inverse and all-wall projection therefore apply. No long nineteenth old normal is invented. Every weak site five through seventeen is now (B19packet), except isolated dotted five.

M.2 An isolated long-end foot is a local geometric kernel

We record the point formulas once. They apply to any actual isolated long-end cap with initial long roots one/two/three as above, in B or D, whenever the listed own occurs. Write

$$c_L = Ap - \gamma\omega_1, \qquad A^2 = 2\gamma^2 - 1.$$

For each remaining own type the actual normalized pressure is $x \cdot Z \geq h$, $\sum Z_j = 1$:

own	Z	h	$R = Z^2$
5	$P_4/4$	$\sqrt{(\varphi-1)/4}$	$1/4$
finite ≥ 7	$(1-u/2)(e_1+e_2)+(u-1)e_3$	$\{\sqrt{2-u} - \sqrt{7u-4-3u^2}\}/\sqrt{2}$	$3-4u+3u^2/2$
dotted	$(1-u/2)e_1 + (u/2)e_2$	$\{\sqrt{2-u} - \sqrt{u(1-u)}\}/\sqrt{2}$	$1-u+u^2/2.$

(B19headfeet)

Here $u = \gamma^{-2}$, in $(1, \cos(\pi/7)^{-2}]$ for finite own, and $(0, 1)$ for dotted. All coordinates of Z are nonnegative.

At native five, the actual H_4 correction is $\gamma(3n_1/2+n_2+n_3/2)$, with residual $\delta = 1-3\gamma^2/2 > 0$. The future finite projected point $p + (A/\delta)(c_L + k)$ has square $-1 - A^2/\delta$ and every original wall product nonpositive. Its displacement is $(\gamma/4)\omega_4$, and its pressure is $H = AB/(B+\sqrt{\delta})$, $B = \gamma/\sqrt{2}$. The real native-five identity $4\gamma^2 = 2\gamma + 1$ gives $\sqrt{\delta} = (1-\gamma)/\sqrt{2}$, hence $H = \gamma A$ and the first row exactly. In particular $h > .393075$, by positive rational squaring.

For finite own at least seven, the actual negative core is

$$c_L + \gamma^{-1} n_1 + 2(\gamma^{-1} - \gamma) n_2.$$

Its cap/own products vanish, the next product is $(3 - 4\gamma^2)/(2\gamma) < 0$, and its square is $-(7 - 4\gamma^2 - 3/\gamma^2) < 0$. For dotted own use $c_L + \gamma^{-1} n_1$, square $-(1-u)$, zero cap product and negative own product. Both have positive future time and positive coefficients on actual normals. Every omitted original-wall product is nonpositive termwise. Their spatial displacements, divided by $\sqrt{2}\gamma$, give the other two rows on the SAME original x. This proves the kernel without an I spectator or an arithmetic hypothesis.

For every finite own at least seven and every dotted own,

$$.8h - .16R > .2893. \tag{longF}$$

At native seven, $.9009 < \gamma < .901$ gives $A > .7894$, $\sqrt{7 - 4\gamma^2 - 3/\gamma^2} < .2395$, $\sqrt{2}\gamma < 1.2743$; thus $h > .4315$. Also $1.2318 < u < 1.233$ and $R < .349$, giving $> .28936$.

For finite labels at least eight, $1 < u < 1.172$. Put $a = .8/\sqrt{2}$, $D = (u-1)(4-3u)$. The derivative of the left side is

$$a\left\{-\frac{1}{2\sqrt{2-u}} - \frac{7-6u}{2\sqrt{D}}\right\} + .64 - .48u.$$

On $(1, 7/6]$, the first term is $< -.28$, the second is nonpositive, and the last is at most .16. On $[7/6, 1.172)$, $\sqrt{D} > .284$, the positive second term is $< .032$, and the other terms are $< -.28 + .08$. Thus the scalar decreases with u; its minimum is at native eight. There the radical bounds are $> .910, < .289$, and $R < .373$. Hence $h > .4388$ and the scalar is $> .29136$.

For dotted own, $.5656 < a < .5657$. On $u \le .5$, use $\sqrt{2-u} \ge 1.4142 - .379u$, $R \le 1 - .75u$, giving

$$.8h - .16R > .63987152 - .0944u - .5657\sqrt{u(1-u)} > .30587152.$$

The circle bound uses $\sqrt{.0944^2 + .5657^2} < .5736$. On $u \ge .5$, use $\sqrt{2-u} \ge 1.4494 - .4494u$, $R \le .75 - .25u$, giving

$$.8h - .16R > .69978064 - .2142u - .5657\sqrt{u(1-u)} > .29023064,$$

with $\sqrt{.2142^2 + .5657^2} < .6049$. The chords lie below the concave root by their endpoint square checks. This proves the entire continuous dotted range.

M.3 The B_{19} long head and its paid gains

The actual long head is isolated. Indeed deleting one retains B_{18}, whose complete finite one-node list is isolation or simple-three at its long leaf two. A second contact makes a cycle; another four or

an additional fork is not finite; exceptional finite ranks are too small. The nonisolated cap would have

$$v = \gamma\omega_1 + \omega_2/2, \quad A^2 = 2\gamma^2 + 2\gamma, \quad \cosh L_1 = 2\gamma + 1.$$

Own zero has zero time and own three is the actual affine triangle, so $\gamma \geq 1/\sqrt{2}$ and $x_1 > \sqrt{\sqrt{2} - 1} > .643$.

The OLD actual bulk through fifteen has some strong $K = 14$ or fifteen. If old five is strong, its paid reference has energy $> .90715$, with every actual weak hole omitted. Graft .643 into its first-five block of mean $< .361$. The other four coordinates remain $> .2904$, above the outgoing slope $< .214$, and the gain is $(5/4)(.643 - .361)^2 = .099405$. If old five is weak, six is strong. Begin only with the FORMAL strong-five curve, whose scalar energy is E_K^{old}. Here $f_i^{\mathrm{old}} = \sqrt{i} - .61/\sqrt{2}$ and $E_K^{\mathrm{old}} = (f_5^{\mathrm{old}})^2/5 + \sum_{i=5}^{K-1} d_i^2$. Before assigning any pressure to it, replace its ENTIRE first-six block by .643 and five equal slopes of sum $f_6^{\mathrm{old}} - .643$. Their energy exceeds

$$.643^2 + (2.01815 - .643)^2/5 = .7916565045,$$

whereas the formal first-six energy is $E_6^{\mathrm{old}} < .6971$. The new slopes are ordered: $.643 > (f_6^{\mathrm{old}} - .643)/5 > .275$, above every outgoing slope $\leq d_6 < .197$. There is NO coefficient at five. All later actual weak knots lie in $7, \ldots, K-1$; at most four are nonadjacent, each costing $< 1/3200$. Their affected open intervals are disjoint from the first-six replacement, even for a hole at seven. Only now combine the actual pressures at one, six and the remaining strong knots. One square gives

$$\begin{aligned} \|x\|^2 &> E_K^{\mathrm{old}} - E_6^{\mathrm{old}} + .7916565045 - 4/3200 \\ &> .9084 - .6971 + .7916565045 - .00125 \\ &= 1.0017065045 > 1. \end{aligned}$$

Both references are ordered and paid by the original x. Thus the nonisolated head is impossible.

For the isolated cap, $A^2 = 2\gamma^2 - 1 > 0$ excludes own zero/three/four. Own six has the affine $6, 3$ principal on the actual first two roots; parallel has its feasible positive future null. The remaining owns are exactly native five, finite at least seven, or dotted. Their actual feet are (B19headfeet).

A native-five head forces true new strong five/six/seven. For a helper at $i \leq 7$,

$$c_L \cdot c_i = \gamma_0(2g + 1) - A_0 A_i, \qquad \gamma_0 = \varphi/2, \quad A_0^2 = (\varphi - 1)/2.$$

Since $\gamma_0^2/A_0^2 = \varphi^3/2 > 21/10$,

$$A_i^2 - \frac{\gamma_0^2}{A_0^2}(2g + 1)^2 < (28g^2 - 32g - 13)/5 < 0 \qquad (.8 < g < 1.2).$$

The convex quadratic is negative at both endpoints. The actual mutual would be positive. For the remaining weak isolated dotted five,

$$c_L \cdot c_5 = \varphi g - A_0\sqrt{10g^2 - 1}.$$

Its sign is strictly negative, since the squared difference in the opposite order is $(4\varphi - 6)g^2 - (\varphi - 1)/2 \geq (7\varphi - 11)/2 > 0$. Weakness gives $g < 1/.58 < 2$; $\varphi - \sqrt{5(\varphi - 1)} > -3/20$ then bounds the mutual in $(-3/10, 0)$, a forbidden native interval. This proves the stated three strong sites without a later terminal floor.

For any strong-five .58 reference with first-five mean $b = f_5/5 < .3652$ and individual P_5 coefficient $> .151$, a head correction $\mu(Z - P_5/5)$ costs $\mu/5$ before its floor is used. It is orthogonal to that constant block and has square $R - 1/5$. Its effective gain is

$$G_L = 2\mu(h - b) - \mu^2(R - 1/5).$$

The complete paid gains are

head	μ	$G_L >$
5	.56	.015456
finite ≥ 7	.4	.02914
dotted	$\sqrt{2}/5$	.03384.

(B19headgains)

The first uses $h > .393$; the second is (longF) minus $.8b$ plus .032. For the dotted row, expand its gain as

$$.4\{\sqrt{2-u} - \sqrt{u(1-u)} - \sqrt{2}b\} - .04(1.6 - 2u + u^2),$$

where $\sqrt{2}b < .5166$. For $u \leq .5$, its lower bound is $.29504 - .0916u - .4\sqrt{u(1-u)} > .04404$, using $\sqrt{.0916^2 + .4^2} < .4104$. For $u \geq .5$, it is $.32912 - .15976u - .4\sqrt{u(1-u)} > .03384$, using $\sqrt{.15976^2 + .4^2} < .4308$. The same two concave root chords prove these inequalities. All three individual prices fit.

M.4 Curved early references with the whole initial-five debt

Some subsequent pressures need a stronger early reference than (B19early). We give it by the actual initial five alternatives, retaining all holes and full cross terms. Each resulting vector is supported through $K = 15$ or sixteen, has sum f_K, and keeps endpoint coefficient $> 1/9$.

First record a useful exact scalar:

$$E_{17} > .9726645. \tag{B19E17}$$

Its first block is $f_5^2/5 = 1.03364 - .116\sqrt{10} > .6668157$, using $\sqrt{10} < 3.162278$. For $a = 4i + 2$, $u = d_i^2$ satisfies $u(a - u) = 1$. Three positive reciprocal substitutions give $u > (a^2 - 1)/(a(a^2 - 2))$. At denominator 10^7, the full list of lower numerators and cleared positive margins is

i	L_i	margin	i	L_i	margin
5	455488	5248	11	217494	13464
6	385186	536	12	200080	8000
7	333704	14240	13	185248	115712
8	294372	20208	14	172465	14860
9	263340	21360	15	161332	72272
10	238230	27080	16	151549	273164.

The margin is $10^7(a^2-1) - L_i a(a^2-2)$. The numerators sum to 3058488, proving (B19E17). This is a finite rational hand certificate, not a script-output premise. Also $d_{17}^2 > 1/70$, giving $E_{18} > .98659$.

If five is strong, its curved reference loses $< 1/1024$ over all nonadjacent internal holes. Adding the actual head pressure inside that reference gives

$$L_{15} > .955516, \qquad L_{16} > .971636.$$

These conservative values use the smallest gain in (B19headgains); the appropriate finite/dotted head is stronger.

If five is a weak helper, the actual packet gives $x_1 > .5455$ and six is strong. Use the first-six reference through these actual heights, followed by the ENTIRE curved retained window. Its first-six energy is $> .74385$, rather than the ordinary $< .7126$. Since $d_{15}^2 < .016134$, $d_{16}^2 < .015155$, (B19E17) gives $E_{15} > .9413755, E_{16} > .9575095$. After all holes,

$$L_{15} > .9716, \qquad L_{16} > .9877.$$

No long-head energy was added to this initial-coordinate gain.

If five is isolated dotted, the native-five head has already been excluded by its compatibility. For the other head types keep its actual foot $x \cdot Z \geq h$ and the initial-five foot

$$Z_5 = P_5 + (v/2)(e_6 - e_5), \quad x \cdot Z_5 > T_5 > 1.8185, \quad .4 < v < 1.$$

This is the actual dotted-five negative core; the weak condition forces $v > .4$ by the full continuous loss bound in the shared early-reference lemma. Choose a genuinely strong $J = 6$ or seven. The reference is constant $m = f_J/J$ on the first J coordinates and follows every retained curved knot through K. The complete conservative data are

J	$m <$	$w_{15}^2 >$	$w_{16}^2 >$	individual P_J coefficient $>$	$Q_5 <$
6	.3399	.9217	.9378	.1435	77/150
7	.31938	.9041	.9202	.1366	194/175.

For the energies, $f_6^2/6 > .69317$, $f_7^2/7 > .71400$. The bound $d_i^2 > 1/(4i+2)$ gives at denominator 10^6 the lower numerators

$$38461, 33333, 29411, 26315, 23809, 21739, 20000, 18518, 17241, 16129$$

for $i = 6, \ldots, 15$. Those from six through fourteen sum to 228827, and from seven to fourteen to 190366; $d_{15}^2 > .016129$. The maximal omission losses are $< .000267$ after six and $< .000181$ after seven, by the decreasing sums $1/[32(j-1)^3]$ over their earliest permitted nonadjacent sets. These prove every energy entry. The P_J price is $m - d_J$; the end-K price stays $> 1/9$. Finally

$$Q_5 = \|Z_5 - (5/J)P_J\|^2 = 5 - 25/J - v + v^2/2 < 4.68 - 25/J.$$

Let $Y = Z - P_J/J$, $Y_5 = Z_5 - (5/J)P_J$. Their first-J sums vanish, so they are orthogonal to w. Because Z is supported on the first three coordinates and $Z_5 = 1$ there,

$$Y^2 = R - 1/J, \qquad Y \cdot Y_5 = 1 - 5/J.$$

The one vector $w + .4Y + .09Y_5$ spends the COMBINED P_J price $.85/J$, which fits both rows. Its gain is

$$.8h - .16R - .8m + .16/J + .18(T_5 - 5m) - .0081Q_5 - .072(1 - 5/J).$$

The nonzero last cross is retained. Using (longF), the gain is $> .04930$ at six and $> .06699$ at seven. Thus both initial branches give

$$L_{15} > .9710, \qquad L_{16} > .9871.$$

These are effective energies of the same paid vector, not new independent norm summands.

The identical curved construction at a genuinely strong seventeen has initial reference squares $> .9530, .9354$ in the last case, so its gains give totals $> 1.00230, 1.00239$. For helper five it gives $> .97231 + .03125 - 1/1024 > 1.0025$. At a later actual strong eighteen, the extra endpoint increment dominates the possible additional hole, so the conclusions remain valid. These are conditional uses of an ACTUAL strong endpoint; none is used to assume that endpoint beforehand.

M.5 The zero-own short-only terminal walls are impossible

Suppose an actual last-long cap is

$$c_T = Bp - \delta P_{19}, \quad B^2 = 19\delta^2 - 1, \qquad \delta = \cos(\pi/m), \quad m = 5, 6, 7.$$

We do not yet assume that this exhausts the terminal possibilities. If seventeen were weak, its actual helper would have $A_H^2 = 34g^2 + 2g - 1/2$. The two walls are distinct by their old-one contacts. Put

$$M = A_H B - \delta(34g + 1)/\sqrt{2} = -c_H \cdot c_T.$$

For $b = \delta^2 > .65, g > .8$, the difference of the two positive squared terms is

$$(2b - 1)(34g^2 + 2g) + 1/2 - 10b.$$

It increases in both variables and at $(.65, .8)$ equals 1.008. Thus $M > 0$. Positive squaring of its derivatives also proves monotonicity in both: the respective positive differences are

$$(68b - 34)A_H^2 + 18(19b - 1), \qquad (38b + 17)A_H^2 - 171b + 9.$$

For the latter use $A_H^2 > 22.86$, $.65 < b < .812$. At $g = 6/5, \delta = .901$, rational square bounds give $A_H < 7.132, B < 3.798$, and $\delta(34g + 1)/\sqrt{2} > 26.6$. Consequently

$$0 < M < 7.132(3.798) - 26.6 = .487336 < 1/2,$$

a forbidden actual-wall product. Seventeen is therefore genuinely strong before any head split.

If the isolated long head were finite at least seven or dotted, the three initial-five constructions of Section M.4 at that actual strong seventeen would contradict the norm. For strong five, the reference square is $> .97133$ and its respective head gain is $> .02914$ or $> .03384$. The helper-five and dotted-five branches already exceed one by their full curved/joint references. Thus only a native-five long head remains.

Write it as

$$c_L = Ap - qe_1, \quad A^2 = (\varphi - 1)/2, \quad q = \varphi/\sqrt{2}, \quad x \cdot (P_4/4) \geq h = \sqrt{(\varphi - 1)/4} > .393075.$$

The true strong five/six/seven compatibility was proved before this terminal assumption. Use the ordinary reference through strong seventeen, omitting all actual internal weak helpers. The largest possible curvature loss is bounded by holes $8, 10, 12, 14, 16$:

$$\begin{aligned}\sum_{j=8,10,12,14,16} \frac{1}{32(j-1)^3} &< .000091108 + .000042867 + .000023479 \\ &\quad + .000014224 + .000009260 \\ &< .000181.\end{aligned}$$

The decreasing weights and nonadjacency make this earliest permitted independent set maximal. Hence the actual reference has

$$w^2 > .9724835, \quad b = f_5/5 < .365190, \quad f_{17} < 3.712985, \qquad \text{prices } P_5 > .151,\ P_{17} > .12. \tag{B19tightref}$$

No unproved strong eighteen was used.

At terminal short five, the actual mutual magnitude is exactly

$$-c_L \cdot c_T = \frac{\sqrt{4+15\varphi}-(\varphi+1)}{2\sqrt{2}} \in (.953, .956).$$

Use $1.618 < \varphi < 1.619$, $1.414 < \sqrt{2} < 1.415$ and positive squaring for this interval. But $\cos(\pi/10) < .952$, while $\cos(\pi/11) > 47/49 > .956$: the latter follows from $\pi < 22/7$ and $\cos t \geq 1 - t^2/2$. There is no finite native magnitude in this gap, and the magnitude is neither zero nor at least one. The two actual walls cannot have this product.

At short six put $\delta = \sqrt{3}/2$, $B = \sqrt{53}/2$ and $k = AB - q\delta$. The complete rational seals

$$.5558 < A < .556, \quad 1.1440 < q < 1.1442,$$
$$.8660 < \delta < .8661, \quad 3.6400 < B < 3.6401,$$
$$1.032 < k < 1.034$$

follow from positive squaring. The ACTUAL positive combination $\xi = c_L + c_T$ is future timelike, square $-2(k-1)$, with both selected cap products $1 - k < 0$. Every old and unknown wall product is nonpositive termwise. It gives the original pressure

$$U = qe_1 + \delta P_{19}, \qquad x \cdot U \geq H = A + B - \sqrt{2(k-1)} > 3.9348.$$

The reference pairing is $w \cdot U = qb + \delta f_{17} < 3.634$; its deficit exceeds .3. After paying BOTH anchors, the actual residual is

$$V = U - \delta P_{17} - (q/5)P_5 = q(e_1 - P_5/5) + \delta(e_{18} + e_{19}), \qquad w \cdot V = 0, \quad V^2 = (4/5)q^2 + 2\delta^2 < 2.548.$$

With $\nu = .1$, the individual prices are $< .023$ at five and $< .087$ at seventeen. A single square gives

$$\|x\|^2 > .9724835 + .06 - .02548 > 1.007.$$

No separate head-energy estimate was added here.

At short seven we need the full head/core cross. Let $t = 27/25$, $k = AB - q\delta$. The exact polynomial $8\delta^3 - 4\delta^2 - 4\delta + 1 = 0$ gives the rational seals

$$.9009688 < \delta < .9009690, \quad .5558929 < A < .5558931, \quad 1.1441227 < q < 1.1441231,$$
$$3.79778 < B < 3.79779, \qquad 1.08033 < k < 1.08035.$$

For the first bracket the cubic increases on $(.9, .901)$; at $.901 - e$ it equals $.000257608 - 8.275224e + 17.624e^2 - 8e^3$, whose signs at $e = .0000312, .0000310$ are negative/positive. The other seals are positive square/product comparisons.

Now $\xi = c_L + tc_T$ is future, with selected cap products $1 - tk < 0, t - k < 0$, and square $-s$, $s = 2tk - 1 - t^2 > 0$. All other original-wall products are nonpositive. The same original vector satisfies

$$U = qe_1 + t\delta P_{19}, \qquad x \cdot U \geq H = A + tB - \sqrt{s} > 4.2486453,$$

because $\sqrt{s} < .40885$. Also $w \cdot U = qb + t\delta f_{17} < 4.03074$, so the actual deficit is $> .2179$.

Use the independent actual native-five H_4 pressure through $Y = P_4/4 - P_5/5$. Its square is $1/20$, its cross with w is zero, and its deficit is $h - b > .027885$. For the two-cap point pay the residual

$$V = U - t\delta P_{17} - (q/5)P_5 = q(e_1 - P_5/5) + t\delta(e_{18} + e_{19}).$$

Its full products are

$$w \cdot V = 0, \quad V^2 = (4/5)q^2 + 2t^2\delta^2 < 2.941, \qquad Y \cdot V = q/20 < .057207.$$

Choose $\mu = .48, \nu = .065$ and use ONE vector $w + \mu Y + \nu V$. Its complete individual debits are

$$P_5 : \ (\mu + \nu q)/5 < .111 < .151, \qquad P_{17} : \ \nu t\delta < .064 < .12.$$

The actual primitive pressures, with these coefficients removed before their floors are used, give the single completed-square gain

$$.96(.027885) - .2304(.05) + .13(.2179) - .004225(2.941) - .0624(.057207) \\ = .0275811582.$$

The last term is the NONZERO head/core cross. Together with (B19tightref), the total is $1.0000646582 > 1$. Thus all three zero-own short-only native terminal walls are impossible.

M.6 The complete actual B_{19} last-long classification

Release eighteen. The retained old diagram is A_{17} on one through seventeen and the short A_1 at nineteen. A finite completion has at most one contact to A_{17}: simple-three at $1, 2, 16, 17$, or endpoint four at one/seventeen. Exceptional finite diagrams cannot contain A_{17}. A short-only contact can have ANY finite native label. If both retained components are joined, full rank-nineteen classification permits only a simple-three at left endpoint one/seventeen and short-three/four. An internal left attachment introduces a second inadmissible fork; an internal four is not a large finite B diagram. This exhausts the actual finite white list before any own cutoff.

Write the own product magnitude as $g \geq 0$. With no short contact, a white three at j has

$$\Delta = 1 - j(18 - j)/36, \quad \sigma = g + j/36;$$

a white endpoint four has $\Delta = 1/18$, $\sigma = g + j/(18\sqrt{2})$. The strong ratio target at $q = \sqrt{36} = 6$ is $< 33/20$.

For completeness keep the entire small-own list. Own-zero far three has negative time; far next-to-end three and far four have zero time. Near-three/next-to-end zero-own has the actual affine two-leaf fork ending in four, and near four has the $4, 3, 4$ principal. Own three with a white three has its actual affine cycle; far-four/own-three has the affine $3, 3, 4, 3$ path, while near four/own-three is directly strong. Own four without a short contact has the $4, 4$ principal through old eighteen/nineteen. Own six has the left $6, 3$ principal if its selected neighbour is uncontacted; the other near roles already have a directly strong ratio. Parallel has the positive future null $c + n_{18}$. Every affine null has a positive actual-cap coefficient, future time and nonpositive products with all omitted walls.

For every remaining $g \geq \varphi/2$, near three, next-to-end three and endpoint four are directly strong by the displayed residuals. Only the pure far-three helper (B19packet) can remain weak on the left. Its changed-loss cutoff is $g < 6/5$, reproved by the actual edge ratio, not inferred from an old weak set.

An isolated cap at this final index is also strong or impossible. Use the LEFT old roots eighteen/seventeen/sixteen. Native five has divided finite H_4 loss

$$6 - \sqrt{36 - u} + \sqrt{u - 3/2} < .13 + .168 < .58.$$

Native seven has the negative left-A_2 loss $6-\sqrt{36-u}+\sqrt{4/3-u}<.105+.319$. Finite labels at least eight have singleton loss $<.102+.415$. For dotted own,

$$6-\sqrt{36-u}+\sqrt{u(1-u)}<.084u+\sqrt{u(1-u)}<(.084+1.004)/2<.58.$$

The positive finite points or negative cores are the same actual constructions proved above, now at dual square thirty-six. Their products with short nineteen are nonpositive through the old eighteen term. The spatial $\sqrt{2}P_{19}$ used in a displacement is not an old long normal or its short dual.

If both a left endpoint $j=1,17$ and the short component are contacted, with short magnitude $\delta=1/2$ or $1/\sqrt{2}$, the exact data are

$$\Delta=19/36-\delta^2,\qquad \sigma=g+j/36+\delta/\sqrt{2}.$$

Short four is strong even at own zero. For short three and every positive own,

$$\sigma/\sqrt{\Delta}\geq\frac{1/2+1/36+1/(2\sqrt{2})}{\sqrt{5/18}}>33/20.$$

At own zero with $j=1$, the actual selected sixteen/seventeen/ eighteen/nineteen/cap path is affine F_4. At $j=17$, the cap is exactly $(\omega_{17}+\omega_{19})/2$, already excluded by its ACTUAL short-edge mate in the boundary proof. No mixed-left/short weak cap remains.

With no left contact write $v=g\omega_{18}+\delta\omega_{19}$. Its outgoing ratio is

$$\frac{g+\delta/\sqrt{2}}{\sqrt{1-\delta^2}}.$$

For own zero this is exactly $\cot(\pi/m)/\sqrt{2}$. Short three has the affine F_4 path and short four the double-four principal. At label eight the ratio is $1+1/\sqrt{2}>33/20$, increasing thereafter. The remaining short-five/six/seven rows were proved impossible in Section M.5.

For positive own and short at least five, the ratio is $>53/30>33/20$. For short four and own at least four it is $\geq 1+1/\sqrt{2}$; only own-three/short-four remains. For short three, own-three remains. Own four has a finite B_{19} replacement after releasing short nineteen, with actual reciprocal edge cosine $55/2>27$. Positive finite projection makes that endpoint actual, giving $S_{19}^2>247/14>(4.2)^2$. The early references (B19early) with the $1/9$ price exclude this pressure in both K rows. Parallel again has its actual positive feasible null. We next exclude every finite own at least five and dotted own in the remaining short-three row.

Put $\alpha=\sqrt{2}g+1/2$; its actual time is $A^2=18\alpha^2-3/4$. For finite $\varphi/2\leq g<1$, put $d=1-g^2$, $D=g/\sqrt{2}+1/2$, $\eta=d/D$, $\lambda=g+\eta/\sqrt{2}$. The actual positive combination

$$\xi=c+\lambda n_{18}+\eta n_{19}$$

has products zero with the cap and eighteen, and $-D+\eta/2<0$ with nineteen. Its square is $-k$, $k=d-\eta^2/2>0$, because $2D^2-d=\alpha^2-3/4>0$. It is future and satisfies every omitted wall termwise. Its normalized spatial displacement Z is one on the first seventeen coordinates, has square <18, and gives

$$x\cdot Z\geq G=(A-\sqrt{k})/\alpha>3.87.$$

Here is the full interval seal. For $\alpha>1.643$,

$$\frac{k}{\alpha^2}=\frac{d(2-1.5/\alpha^2)}{(\alpha+.5)^2}<.11,\qquad d=.875+\alpha/2-\alpha^2/2.$$

The numerator derivative is at most $-1.143(1.44)+.347(.677)<0$ and its denominator increases. At 1.643 the value is $<.347(1.445)/4.59<.11$. Also $A/\alpha>4.209$, while $\sqrt{.11}<.332$. Thus the stated bound is strict on every finite own in the range.

For ALL dotted $g>1$, use instead $\xi=c+g^{-1}n_{18}$. Its cap product is zero, both old eighteen/nineteen products are negative, its square is $-(1-g^{-2})$, and its time is positive. The normalized displacement again has first seventeen coordinates one and square <18. Since $\alpha>1.9$, $A/\alpha>4.21$. For $t=1/g\in(0,1)$,

$$\frac{\sqrt{1-g^{-2}}}{\alpha}=\frac{t\sqrt{1-t^2}}{\sqrt{2}+t/2}<1/3.$$

For $t\le 1/2$, its square is $\le 3/32$; for $t\ge 1/2$ it is $<100/1089$, both $<1/9$. Hence $G>3.876>3.87$ throughout the unbounded dotted interval.

Use the refined early references of Section M.4. Every earlier residual is supported through K, so $Z-P_K$ vanishes on its ENTIRE support. It is exactly orthogonal to all early head/dotted-five corrections, including their mutual nonzero cross already paid there. The remaining $1/9$ endpoint price is available. The new pure tail gain exceeds

$$\begin{cases}2(3.87-3.463)/9-3/81>.0534, & K=15,\\ 2(3.87-3.590)/9-2/81>.0375, & K=16.\end{cases}$$

Each exceeds one when added to EVERY strong/helper/isolated-dotted five effective-energy row above. This closes all short-three finite-own-at-least-five and dotted-own cases.

Thus the full classification leaves either strong eighteen, a pure far helper there, or precisely the two weak walls

$$(b)\quad v_T=(\omega_{18}+\omega_{19})/2,\qquad (c)\quad v_T=\omega_{18}/2+\omega_{19}/\sqrt{2}. \tag{B19survivors}$$

M.7 Strong eighteen and a weak far helper are also excluded

If eighteen is strong, the actual window through seventeen justifies the full curved references there. A helper at five gives $E_{18}+.03125-1/1024>1$. An isolated dotted five gives the two-early-foot $J=6,7$ contradictions at that ACTUAL strong endpoint. For strong five,

$$w^2>E_{18}-1/1024>.985613.$$

The smallest paid head gain $>.015456$ already contradicts the norm.

If eighteen is a weak far helper, seventeen is strong. At this boundary, nonadjacency uses the positive finite D_{18} replacement from the retained A_{16} and two simple leaves, together with the independent old short A_1. Both helper caps have zero contact to the short root. This checks the actual mixed two-cap hypotheses; no virtual long nineteenth root is used.

The full curved/early-foot constructions at strong seventeen exclude helper-five and isolated dotted-five. Hence five is strong. Its reference through seventeen has $w^2>.97133$, sum f_{17}, first-five mean $b<.3652$, P_5 price $>.151$ and P_{17} price $>1/9$.

For the actual weak eighteen helper $A^2=36g^2+2g-1/2$. Let $\omega_{19}^*=\sqrt{2}P_{19}$ denote a SPATIAL vector only. With $\alpha=\sqrt{2}g$, the complete actual pressure feet are

own	U	$x\cdot U\ge T$	$V^2,\ V=U-\alpha P_{17}$
5	$g(\omega_{17}+\omega_{19}^*)/2+\omega_3/6$	$H(A,2/3-g^2)$	$g^2+1/6$
7, 8	$g(\omega_{17}+\omega_{19}^*)/2+\omega_2/4$	$A-\sqrt{g^2-3/4}$	$g^2+1/4$
finite ≥ 9	$g(\omega_{17}+\omega_{19}^*)/2+\omega_1/2$	$H(A,1-g^2)$	$g^2+1/2$
dotted	$(g-g^{-1})\omega_{18}+(\omega_{17}+\omega_{19}^*)/(2g)+\omega_1/2$	$A-\sqrt{1-g^{-2}}$	$2g^2-3/2+g^{-2}$.

Here $H(A,\delta)$ is the actual positive finite-projection pressure. The normal corrections are respectively

$$gn_{18}+\tfrac{2}{3}n_1+\tfrac{1}{3}n_2,\quad gn_{18}+n_1/2,\quad gn_{18},\quad g^{-1}n_{18}.$$

They use ONLY actual long roots one/two/eighteen. At native five their finite H_4 residual is $2/3-g^2$; at seven/eight the core has negative square $-(g^2-3/4)$; the next finite pair has residual $1-g^2>0$, and the dotted core has square $-(1-g^{-2})$. Every selected point is future, and every omitted-wall product, including that with actual SHORT nineteen, is nonpositive. The finite pressure comes from a point, never from a spacelike correction by itself.

All four rows satisfy

$$T-w\cdot U>.21,\qquad V^2<2.1,\qquad w\cdot U=\alpha f_{17}+b/\sqrt{2}. \tag{B19helper18}$$

Use $\sqrt{2}f_{17}<5.251$, $b/\sqrt{2}<.259$. Native five has $A>4.9678,\sqrt{\delta}<.111,g<.81$, giving deficit $>.3444$. Native seven has $A>5.5245$, radical $<.249$, $g<.901$, giving $>.2853$. Native eight has $A>5.663$, radical $<.322$, $g<.9239$, giving $>.2306$. For finite labels at least nine, at $g=.939$, $A>5.75,\sqrt{A^2+\delta}>5.765$, $\sqrt{\delta}<.344$, so $H>5.42$ and deficit $>.2303$. The difference $H-5.251g$ increases thereafter: $A'>6,H_A>47/50,H_\delta<0$, and $\delta=1-g^2$ decreases. For dotted $1<g<6/5$, the difference decreases since $A'<6.1$ and the radical derivative >1. At $6/5$, $A^2=2687/50,A>7.3307$, radical $<.5528$, giving deficit $>.2177$. The finite V^2 bounds are immediate; the dotted square increases to $1867/900<2.1$. These positive-square/derivative seals cover every own interval.

Insert BOTH primitive head and helper pressures in one vector

$$W=w+\mu(Z-P_5/5)+\nu V,\qquad \nu=1/16,$$

with the head-specific μ in (B19headgains). The individual prices are $\mu/5$ at five and $\nu\alpha<.107<1/9$ at seventeen. The residual V has an initial piece $P_s/(s\sqrt{2})$, $s=3,2,1$. If ℓ_s is the mean of the first s coordinates of Z, then EXACTLY

$$(Z-P_5/5)\cdot V=(\ell_s-1/5)/\sqrt{2}.$$

The total subtracted cross is $<.002476$ for native-five head, $<.010608$ for other finite head, and $\le .02$ for dotted head. The helper gain from (B19helper18) is $>.018046875$. Thus the single squared-reference totals exceed respectively

$$\begin{aligned}.97133+.015456+.018046875-.002476&>1.0023,\\ .97133+.02914+.018046875-.010608&>1.0079,\\ .97133+.03384+.018046875-.020000&>1.0032.\end{aligned}$$

No independent head optimum was added to an already optimized helper energy. Strong eighteen and its pure-helper weak case are now both impossible.

M.8 Each terminal survivor supplies its own genuine strong seventeen

Only the two actual walls in (B19survivors) remain. Their first eighteen coordinates are a, last coordinate δ, with

$$(b):\ a=(\sqrt{2}+1)/2,\ \delta=1/2;\qquad (c):\ a=\sqrt{2},\ \delta=1/\sqrt{2}.$$

Their time square is $A_T^2=18a^2+\delta^2-1$.

If seventeen were weak, let its actual helper have parameter g, $A_H^2 = 34g^2 + 2g - 1/2$. The mutual magnitude is

$$M = A_H A_T - a(34g + 1)/\sqrt{2}.$$

The actual walls are distinct by their old-one products. With $\kappa = a^2 + \delta^2 - 1$, the positive squared difference is $A_H^2\kappa - 9a^2$. For $g > .8$, it is positive in both rows: in (b), $\kappa = 1/\sqrt{2}$ and it exceeds $22.86(.707) - 9(1.458) > 3$; in (c), it exceeds $22.86(1.5) - 18 > 16$. So $M > 0$. The squared derivative comparison $34\kappa A_H^2 + 18A_T^2 > 0$ shows that M increases with g.

For (b), $g < 1.2$ gives

$$M < 7.132(5.048) - .8533(41.8) = .334396 < 1/2,$$

using $A_T < 5.048, a/\sqrt{2} > .8533$. This violates the native gap. For (c), $A_T = \sqrt{71/2} < 5.959$ and $a/\sqrt{2} = 1$, so

$$0 < M < 7.132(5.959) - 41.8 = .699588 < .7.$$

The only possible native magnitude is $1/2$. But $M(1) = 1/2$ exactly, so strict monotonicity forces $g = 1$. The actual helper then has the nonzero positive future null $c_H + n_{17}$, impossible in a compact polytope. Thus seventeen is strong in BOTH survivors.

With a finite head at least seven or a dotted head, the actual strong-seventeen references now close every initial-five branch: helper five and isolated dotted five by their complete constructions, strong five by $.97133 + .02914 > 1$ or $.97133 + .03384 > 1$. Therefore both survivors force the isolated native-five long head, its H_4 pressure $x \cdot P_4/4 > .393075$, and the genuine strong five/six/seven/seventeen sites. These inputs precede the fresh changed-loss arguments below.

M.9 A common complete white list at the newly tested index seventeen

Deleting seventeen retains A_{16} on one through sixteen and B_2 on eighteen/nineteen. The full finite-white list is isolation; left simple-three at $1, 2, 15, 16$; left endpoint-four at one/sixteen; right simple-three at eighteen or nineteen; or a left endpoint-three joined to either right simple-three. Two contacts to one old tree make a cycle. A right-four or larger extension of B_2 is nonpositive. Full bridges have rank nineteen, so exceptional finite types, an internal four and a second fork supply no further alternatives. There are no spectator components.

For a unit actual cap $c = Ap - v$ write $g = -c \cdot n_{17} \geq 0$. The left-only exact data are

white	Δ	σ
$3@j,\ j = 1, 16$	$9/17$	$g + j/34$
$3@j,\ j = 2, 15$	$2/17$	$g + j/34$
$4@j,\ j = 1, 16$	$1/17$	$g + j/(17\sqrt{2})$.

A full bridge has residual $1/34$; even at own zero its numerator is $.5 + 1/34$ for right-long three, or $1/(2\sqrt{2}) + 1/34$ for right-short three. Both resulting ratios exceed two. This complete list will be tested AFRESH under each terminal wall; its old .58-strong labels do not imply the new floors.

The common actual small-own reductions, when not replaced by a direct mutual calculation below, are as follows. Zero-own far-three has negative time, far next-to-end three and far four have zero time. The near next-to-end zero-own has the affine two-leaf fork ending in four; near-four zero-own has the affine chain ending in four at both ends. Own-three with a white-three has the actual all-three cycle. When both right contacts vanish, own-four has the selected $4, 3, 4$ path through $c, 17, 18, 19$. When the contact with n_{18} vanishes, own-six has the $6, 3$ path through $c, 17, 18$. For right-short-three with own-four and zero contact at sixteen, the actual affine star instead has centre

n_{17}, orthogonal leaves c, n_{16}, n_{18} and weights $4, 3, 3$; its positive null is future feasible. Right-long-three with own-four or own-six is tested by its direct ratio below, without using a chorded affine path. Parallel has the positive feasible future null $c + n_{17}$. Each null has positive time through the actual cap and every omitted-wall product nonpositive. The endpoint-four positive-own rows are directly tested below, including own three.

M.10 Short-four terminal: a new .56 floor at the one actual index

Assume survivor (c), $v_T = \sqrt{2}P_{18} + e_{19}/\sqrt{2}$, $A_T^2 = 71/2$. We prove

$$S_{17} > F = \sqrt{17} - .56/\sqrt{2}. \tag{B19F56}$$

At $q = \sqrt{34} < 6$, its sufficient ratio target satisfies

$$T_{.56}(q) = 25/14 - \frac{q - .56}{q(2q - .56)} < 25/14 - 34/429 < 1.707.$$

The first function increases on the relevant range.

For isolated caps, the complete small-own/time/affine/null reductions leave five, seven, finite at least eight, or dotted. Use the actual LEFT corrections at seventeen/sixteen/fifteen. For native five, the divided finite-H_4 loss is

$$q - \sqrt{34 - u} + \sqrt{u - 3/2} < .133 + .168 < .56, \quad u = 8 - 4\varphi < 1.528.$$

For native seven the left-A_2 negative core gives loss

$$q - \sqrt{34 - u} + \sqrt{4/3 - u} < .107 + .319 < .56, \quad 1.2318 < u < 1.233.$$

For finite at least eight the actual pair projection gives $< .102 + .415 < .56$. For dotted own the actual core $c + g^{-1}n_{17}$, together with concavity, gives loss

$$q - \sqrt{34 - u} + \sqrt{u(1 - u)} < .087u + \sqrt{u(1 - u)} < (.087 + 1.004)/2 = .5455 < .56.$$

The core/finite-point signs are exactly those proved earlier; all corrections use actual long roots and have nonpositive product with short nineteen. Thus every isolated cap satisfies the new floor, including native eight.

For left-only rows, the small owns in the common list are removed first. For all remaining $g \geq \varphi/2 > .809$, near endpoint-three has ratio

$$(g + 8/17)/\sqrt{9/17} > 1.2795/.728 > 1.757.$$

The next-to-end rows exceed 2.5 and endpoint-four exceeds 3.5. Only far-three at one needs its actual terminal interaction.

For that actual cap, $v_H = g\omega_{17} + \omega_1/2$, $A_H^2 = 34g^2 + 2g - 1/2$, and

$$M(g) = A_H\sqrt{71/2} - (34g + 1) = -c_H \cdot c_T.$$

The two walls are distinct by their old-one products. For $g > .8$, its squared-product difference is $3A_H^2/2 - 18 > 0$, so $M > 0$. Its squared derivative comparison $51A_H^2 + 18(71/2) > 0$ proves strict increase. We have $M(1) = 1/2$. At $g = 243/200 = 1.215$,

$$A_H^2 = 52.12165 < (7.21954)^2, \quad \sqrt{71/2} < 5.9582,$$
$$M < 7.21954(5.9582) - 42.31$$
$$= .705463228 < .707 < \cos(\pi/4).$$

Thus for $g \le 1.215$ the only possible native magnitude is $1/2$; strict increase forces the forbidden parallel helper $g = 1$. Every remaining actual far helper has $g > 1.215$, hence

$$\frac{34g+1}{\sqrt{612}} > 42.31/24.739 > 1.71 > T_{.56}(q).$$

This tests the entire new cap domain; the earlier .58 cutoff $g < 1.2$ has not been imposed.

For right-long three, own zero has its affine star and own three its affine triangle. Every remaining own has $g \ge 1/\sqrt{2}$, ratio $\ge 1 + 1/\sqrt{2} > 1.707$. For right-short three, own zero/four/six/parallel has the actual affine/null obstruction. Own-three and own-five are impossible by the previous ACTUAL short-edge mate and left H_4 boundary arguments, which did not require the new seventeen floor. Every other finite own at least seven has ratio $\sqrt{2}g + 1/2 > 1.77$; dotted owns are stronger. Finally all full bridges have ratio > 2, including own zero. The complete new finite-white/ own domain is exhausted, proving (B19F56).

M.11 Short-four terminal: pool the tail before raising its endpoint

Use the old .58 reference through seventeen, but deliberately omit fifteen/sixteen as well as every actual weak knot. Because helpers are nonadjacent, the last retained K before seventeen is fourteen or thirteen. If fourteen is weak, it merges with the forced terminal cluster; all earlier omissions stay isolated.

All earlier holes cost $< .000181$. The terminal cluster averages at most four slopes $d_{13}, \dots, d_{16}$, whose successive gaps are

$$< 1/(4 \cdot 13\sqrt{13}) < .0054.$$

For four numbers with successive gaps at most κ, their variance about the mean is $\le 5\kappa^2$, by $(1/4)\sum_{i<j}(a_i - a_j)^2$ and distances one/two/three. The shorter three-slope cluster costs at most $2\kappa^2$. Thus the terminal loss is $< .000146$, and FULL loss $< .00035$. The affected open intervals are disjoint even when they share a retained endpoint.

Now raise only seventeen by $\Delta = .02/\sqrt{2}$, $.01414 < \Delta < .01415$. Both new terminal slopes are $< .133$:

$$\frac{\sqrt{17} - \sqrt{14} + \Delta}{3} < .133, \qquad \frac{\sqrt{17} - \sqrt{13} + \Delta}{4} < .133.$$

Their incoming slopes are at least $d_{13} > .136$ and $d_{12} > .14$, including any earlier isolated omission. The reference remains concave, with every coefficient nonnegative. Its endpoint coefficient is $> d_{16} > .123$; its first-five block and P_5 coefficient $> .151$ are unchanged.

The old terminal chord has slope $\ge d_{16} > .123$. Lifting its endpoint adds

$$2(\text{old slope})\Delta + \Delta^2/(17 - K) > .246(.01414) = .00347844.$$

Together with the scalar (B19E17),

$$w^2 > .9726645 - .00035 + .00347844 = .97579294, \qquad \sum_j w_j = F.$$

Only seventeen uses the new floor; all other active floors are original strict .58 floors.

The SAME actual terminal cap is also the short-nineteen mate. Deleting old nineteen and adjoining it to the retained A_{18} by its actual simple-three at eighteen gives positive finite A_{19}. Its positive inverse projection is future, strict against every unknown original wall. The omitted

old nineteen starts at zero and becomes strict through the positive cap coefficient and cap-short product $-1/\sqrt{2}$. Thus this is an actual endpoint of that original edge, with

$$\cosh L_{19} = 37/\sqrt{20}, \qquad S_{19} > 3.860.$$

For the last bound, using $\sqrt{20} < 4.47214$, the exact cleared comparison is

$$(19 - 14.8996)37 = 151.7148 > (19 + 14.8996)4.47214 = 151.603757144.$$

We do not assume this neighbouring A_{19} vertex is farthest.

Insert the native-five head pressure by $w_{\mathrm{head}} = w + .56(P_4/4 - P_5/5)$. The individual P_5 price is $.112 < .151$. Its residual has square $1/20$, zero sum on the first-five block and zero cross with w. Its gain is

$$1.12(.027885) - .56^2/20 = .0155512.$$

The modified reference still has support through seventeen, sum F, and P_{17} coefficient $> .123$.

Extend it by the last-two slope $r = (3.860 - F)/2$. The bounds $3.720 < F < 3.72713$ give $0 < r < .07$, so its P_{17} debit fits. The head residual and new tail residual have disjoint support and cross zero. One final square yields gain

$$.0155512 + (3.860 - F)^2/2 > .0155512 + (.13287)^2/2.$$

Consequently

$$\|x\|^2 > .97579294 + .0155512 + .00882721845 = 1.00017135845 > 1.$$

This excludes the entire short-four terminal survivor.

M.12 Short-three terminal: retain the orthogonal dotted exception

Assume survivor (b). Put $s = \sqrt{2}$, $a = (1 + s)/2$, $k = as = 1 + s/2$. Then

$$v_T = aP_{18} + \tfrac{1}{2}e_{19}, \qquad A_T^2 = 51/4 + 9s, \qquad 34A_T^2 - 289k^2 = 17s.$$

The actual index-seventeen cap is distinct from c_T: the latter contacts both retained B_2 roots with inverse cost $1 + s/2 > 1$, impossible for that cap's positive finite white block. Write

$$L = v \cdot v_T > 0, \quad M = AA_T - L = -c \cdot c_T, \qquad D = A^2A_T^2 - L^2.$$

Actual acuteness/native products give $M = 0$ or $M \geq 1/2$. Since $A, A_T, L > 0$, D has the sign of M. In particular $D < 0$ means a POSITIVE actual-wall product.

We prove the fresh floor

$$S_{17} > F = \sqrt{17} - .50/\sqrt{2}. \tag{B19F50}$$

Its sufficient ratio target is $T_{.50}(q) = 2 - (q - .50)/(q(2q - .50)) < 2$, $q = \sqrt{34}$.

For isolated caps,

$$v = g\omega_{17}, \quad A^2 = 34g^2 - 1, \quad L = 17kg, \qquad D = 17sg^2 - A_T^2.$$

Own zero has negative time. Every positive finite own and parallel has $g \leq 1$, giving $D \leq 8s - 51/4 < 0$; all violate acuteness. Thus an isolated cap is dotted.

The function $M(g) = A_T\sqrt{34g^2-1} - 17kg$ strictly increases: its squared derivative comparison is $578sg^2 + 289k^2 > 0$. If $M = 0$, the exact parameter is

$$u = g^{-2} = 17s/A_T^2, \qquad .94 < u < .95.$$

Orthogonality is RETAINED. The actual positive core $c + g^{-1}n_{17}$ has future time, square $-(1-u)$, zero cap product, negative own product and every omitted-wall product nonpositive. Its spatial displacement is

$$Y = g\{(1-u)\omega_{17} + (u/2)(\omega_{16} + \omega_{18})\}.$$

Original concavity gives $x \cdot Y \le gh_{17}$. Farthest comparison consequently gives

$$h_{17} \ge \sqrt{34-u} - \sqrt{u(1-u)}.$$

The loss below $\sqrt{34}$ is $< .087u + \sqrt{u(1-u)} < .083 + .238 = .321 < .50$, so the orthogonal case satisfies the new floor.

If $M > 0$, then $M \ge 1/2$. At $g = 7/4$,

$$(119k/4 + 1/2)^2 - A_T^2(825/8) = (684 - 451s)/16 > 0,$$

so $M(7/4) < 1/2$. Strict increase forces $g > 7/4$ and $u < 16/49$. The same actual foot has loss

$$< .087(16/49) + 4\sqrt{33}/49 < .029 + .469 = .498 < .50.$$

This covers every isolated dotted parameter and all mutual labels, including zero.

For left-only roles use the complete residual table of Section M.9. Keep its actual zero-own and affine/null reductions, except near endpoint-three at sixteen, which will be tested directly. Every positive-own endpoint-four has ratio $\ge \sqrt{17}/2 + 1/\sqrt{34} > 2$. Every remaining next-to-end three has $g \ge \varphi/2 > .809$ and ratio > 2.5.

For near endpoint-three at sixteen,

$$v = g\omega_{17} + \omega_{16}/2, \quad A^2 = 34g^2 + 32g + 7, \quad L = k(17g+8),$$
$$D = 17sg^2 + 16sg - 27/4 - s.$$

At $g = 0$, $D < 0$. For finite $g \ge 1/2$, $D > 0$, starting with $(45s - 27)/4 > 0$. The identity $(34g+16)^2 = 34A^2 + 18$ gives the positive derivative comparison $17sA^2 + 18A_T^2 > 0$ for M. At $g = 1$,

$$(25k + 1/2)^2 - 73A_T^2 = 32 - 39s/2 > 0.$$

Thus $0 < M < 1/2$ for all finite owns and parallel, impossible. Every remaining dotted $g > 1$ has ratio $> 25/\sqrt{153} > 2$, since $625 > 612$.

For far endpoint-three at one, the common small-own/null reductions leave $g \ge \varphi/2 > .8$. The exact data are

$$A^2 = 34g^2 + 2g - 1/2, \quad L = k(17g + 1/2),$$
$$D = 17sg^2 + sg - 27/4 - 19s/4.$$

This polynomial is positive at .8 and increases thereafter, so $M > 0$. The identity $(34g+1)^2 = 34A^2 + 18$ again gives the positive derivative comparison. At $g = 3/2$,

$$(26k + 1/2)^2 - 79A_T^2 = 33 - 22s > 0.$$

Thus every admissible remaining helper must have $g > 3/2$, whose ratio exceeds $52/\sqrt{612} > 2$. This tests the full role, not only its former .58-weak parameters.

For right-long three at eighteen, the exact data give

$$A^2 = 34g^2 + 34g + 8, \quad L = k(17g + 9),$$
$$D = 17sg^2 - (51/2)g - 39/2 - 9s.$$

The convex quadratic is negative at both zero and one; at one it is $8s - 45 < 0$. It is therefore negative on the WHOLE interval $[0, 1]$. Own zero, all finite owns and parallel violate acuteness. A remaining dotted $g > 1$ has ratio $\sqrt{2}g + 1/\sqrt{2} > 2$.

For right-short three at nineteen,

$$A^2 = 34g^2 + 17sg + 15/4, \quad L = 17kg + (19 + 18s)/4,$$
$$D = 17sg^2 - (17/2 + 17s)g - 61/4 - 9s.$$

This convex quadratic is negative at zero and $5/4$, the latter value being $-207/8 - 59s/16$. Thus only $g > 5/4$ can survive actual acuteness, with ratio $\sqrt{2}g + 1/2 > 2$. Finally all full bridges are directly strong by their ratio > 2, including own zero. This exhausts every actual white and own parameter, and proves (B19F50). The .56 floor from the other terminal case was not used.

M.13 Short-three terminal: pool every possible final weak cluster

Keep the old .58 floors, omit every actual weak knot, and deliberately omit thirteen through sixteen as well. Since at least one of eleven/ twelve is strong, the last retained K before seventeen is twelve if twelve is strong, otherwise eleven. This includes every old weak set.

Earlier isolated omissions cost $< .000181$. The terminal cluster averages at most six slopes $d_{11}, \dots, d_{16}$. Their consecutive gaps are

$$< \kappa = 1/(4 \cdot 11\sqrt{11}) < .00686.$$

For n such numbers the exact variance identity gives

$$\sum_i (a_i - \overline{a})^2 = \frac{1}{n} \sum_{i<j} (a_i - a_j)^2 \le n(n^2 - 1)\kappa^2/12.$$

At $n = 6$, this is $(35/2)\kappa^2 < .000824$; the five-slope case is smaller. The clusters have disjoint open edge intervals, so TOTAL loss is $< .00101$.

Lift seventeen by $\Delta = .08/\sqrt{2}$, $.05656 < \Delta < .05657$. Both outgoing terminal slopes are $< .144$:

$$\frac{\sqrt{17} - \sqrt{11} + \Delta}{6} < .144, \qquad \frac{\sqrt{17} - \sqrt{12} + \Delta}{5} < .144.$$

Their incoming slopes are $\ge d_{10} > .154$, respectively $\ge d_{11} > .147$, including any earlier isolated omission. Thus the final interpolation is concave, paid and supported on old strong floors, with only seventeen using the new .50 floor. No assertion is made about the unknown thirteen–sixteen floors.

The old last chord has slope $\ge d_{16} > .123$, so its endpoint lift adds $> .246(.05656) = .01391376$. The new reference satisfies

$$w^2 > .9726645 - .00101 + .01391376 = .98556826.$$

Its first-five mean is still $b < .365190$, with individual P_5 coefficient $> .151$. Insert the actual native-five H_4 pressure in the SAME vector

$$W = w + .56(P_4/4 - P_5/5).$$

Its price .112 fits. Its residual has square $1/20$, zero block sum and zero cross with w. The one original-vector square gives

$$\|x\|^2 > .98556826 + 1.12(.027885) - .56^2/20 = 1.00111946 > 1.$$

No short-edge energy or second farthest vector is added. This excludes survivor (b).

Proposition M.1. *A whole B_{19} forest cannot occur at the original farthest vertex of a compact Coxeter nineteen-polytope.*

Proof. The actual window and full last-long finite-white classification give strong eighteen, a weak pure far helper there, or the two walls (B19survivors). Every zero-own short-only exception was excluded using its own actual head/terminal points. The strong and helper end states are impossible. Each survivor supplies its genuine strong seventeen and isolated native-five head before the fresh single-index arguments use them. The separate short-three and short-four proofs then close both remaining walls. Every own, white, finite, parallel and dotted role has been kept, and every point is feasible against the entire original polytope. □

N The whole D_{19} case

N.1 The signed frame and the head reduction

Use unit actual roots and duals

$$n_i = (e_i - e_{i+1})/\sqrt{2}\ (i \le 18), \quad n_{19} = (e_{18} + e_{19})/\sqrt{2}, \quad \omega_i = \sqrt{2}P_i\ (i \le 17),$$
$$\omega_{18} = (P_{18} - e_{19})/\sqrt{2}, \qquad \omega_{19} = (P_{18} + e_{19})/\sqrt{2}.$$

The original vector satisfies $x_1 > \cdots > x_{18} > |x_{19}|$, $\|x\| < 1$. The last coordinate remains signed. The centre is seventeen and the two actual forks are eighteen/ nineteen; no artificial long fork normal is introduced.

At loss .58, the complete local whole-diagram lemma K.7 applies literally at five through fourteen: the retained right D_{19-i} has rank at least five. It retains every finite fork-to- E_6, E_7, E_8 extension; a surviving positive-own fork ratio is

$$> (.8 + .25)/\sqrt{3/8} > 83/50.$$

Thus every weak cap there is the pure far-three helper

$$v = g\omega_i + \omega_1/2, \quad A^2 = 2ig^2 + 2g - 1/2, \quad \varphi/2 \le g < 6/5,$$

with own five, finite at least seven or dotted, except for isolated dotted five. Actual helper nonadjacency uses the retained positive right block including BOTH fork normals. Hence some $K = 13$ or fourteen and some $J = 6$ or seven is strong, before any head or late-layer conclusion. No fifteen/sixteen/centre floor has yet been used.

Let $f_i = \sqrt{i} - .58/\sqrt{2}$, $E_J = f_5^2/5 + \sum_{i=5}^{J-1} d_i^2$. The old $E_{13} > .88989$, and changing .61 to .58 adds $-.00357 + .006\sqrt{10} > .0154$. Thus $E_{13} > .90529$. The elementary bounds needed below are

$$E_6 < .7126, \quad E_7 < .7510, \quad 2.0393 < f_6 < 2.0394, \quad 2.2356 < f_7 < 2.2357,$$
$$f_5/5 < .3652, \qquad d_6 < .197, \quad d_7 < .183.$$

They follow, for example, from $f_5^2/5 = 1.03364 - .116\sqrt{10} < .6668248$, $d_5^2 < .0456, d_6^2 < .03852$. All remaining internal holes cost

$$\sum \frac{1}{32(j-1)^3} < 1/4000 + 1/10976 + 1/23328 + 1/42592 < .000408 < 1/1024.$$

Only actual weak knots are omitted.

For a true first-coordinate floor $x_1 > X$, use the reference through $(1, X), (J, f_J)$, then every retained strong knot through the actual $K = 13, 14$. Its first slopes are X, then $m = (f_J - X)/(J-1)$. For $X = .643, .668$, they are positive decreasing and $m > d_J$. The single-reference energy is

$$> E_{13} - 1/1024 + C_J(X) - E_J, \qquad C_J(X) = X^2 + (f_J - X)^2/(J-1).$$

At $X = .643$ this gives

J	$C_J(X) >$	$q_J = w^2 >$	individual (P_1, P_J) prices $>$
6	.803379738	.99509	(.3637, .08226)
7	.8361781266	.98949	(.3775, .08243).

(D19headrefs)

Indeed the prices are $X - m$ and at least $m - d_J$; omitting the next knot increases the latter. At $X = .668$, the corresponding $C_6 > .822316738, C_7 > .8557856266$ already give reference energies $> 1.01403, 1.00909$. These are actual original-vector references, not a second energy budget.

Deleting old one retains D_{18}. Its complete finite one-node list is isolation or simple-three at its long leaf two. Two contacts make a cycle; an additional fork/internal double is not a large finite D extension; a short-fork attachment would require an exceptional finite rank nineteen. For a nonisolated head,

$$v_L = g_L\omega_1 + \omega_2/2, \quad A_L^2 = 2g_L^2 + 2g_L, \quad \cosh L_1 = 2g_L + 1.$$

Own zero has zero time, three has the actual affine triangle, and parallel has its feasible positive null. Every other own has $g_L \geq 1/\sqrt{2}$, hence $x_1 > \sqrt{\sqrt{2} - 1} > .643$. For finite own at least five and dotted own,

$$x_1^2 \geq g_L/(g_L + 1) > .668^2.$$

If new five is weak, choose the independently available strong $J = 6, 7$ and use the preceding $X = .668$ reference, contradiction. If new five is strong, even $X = .643$ suffices: graft it into the first-five block of mean $< .3652$, preserving its sum. The remaining coordinates stay $> .2957 > d_5$, and the gain is

$$(5/4)(.643 - .3652)^2 = .09646605.$$

The full reference then gives $.90529 - 1/1024 + .09646605 > 1.00077$. Thus only the nonisolated native-four head could remain:

$$g_L = 1/\sqrt{2}, \quad A_L^2 = 1 + \sqrt{2}, \quad x_1 > .643.$$

If five is a pure weak helper, let its own be r, with $A_5^2 = 10r^2+2r-1/2$. The actual distinct-wall magnitude is

$$M(r) = A_L A_5 - (2+\sqrt{2})r - (1+\sqrt{2})/2.$$

Its squared-product difference is

$$(4+6\sqrt{2})r^2 - (2+\sqrt{2})r - 5/4 - \sqrt{2},$$

positive already at .8 and increasing. Thus $M > 0$. The identity $(10r+1)^2 = 10A_5^2 + 6$ gives a positive squared derivative comparison $(4+6\sqrt{2})A_5^2 + 6(1+\sqrt{2}) > 0$. So M strictly increases.

At native seven,

$$M(r_7) < 1.5538(3.0693) - (3.4142(.9009) + 1.2071) = .48612556 < 1/2.$$

At native eight, positive squaring shows $M(r_8) > 1/2$: the required difference is equivalent to $3 + (5/2)\sqrt{2} > 2(2+\sqrt{2})r_8$, whose square difference is $3/2 + \sqrt{2} > 0$. Finally $M(1) < \sqrt{2}/2$, since the positive square difference is $11/4 - 3\sqrt{2}/2 > 0$. There is no own label between seven and eight. Every finite helper own therefore puts the mutual strictly in $(0, 1/2)$ or $(1/2, \cos(\pi/4))$, impossible. This is a discrete native-angle argument over the real numbers.

For dotted helper own $r > 1$, its exact weak ratio at five is $(10r+1)/\sqrt{60}$. Since

$$T(\sqrt{10}) < T(16/5) = 50/29 - 655/4656 < 317/200$$

and the ratio at $r = 1.13$ has square $15129/6000 > (317/200)^2$, weakness forces $r < 1.13$. Put $t = 1/r$, so $t > .8849$.

The ACTUAL positive future core $c_5 + r^{-1}n_5$ has square $-(1-t^2)$, zero cap product, negative own product and every other actual-wall product nonpositive, including that with the head. Its normalized pressure is

$$Z = P_5 + (t^2/2)(e_6 - e_5) + (t/2)P_1,$$
$$x \cdot Z \geq T(t) = \{\sqrt{10 + 2t - t^2/2} - t\sqrt{1-t^2}\}/\sqrt{2}.$$

Use the $J = 6$ reference of (D19headrefs); six is strong by helper nonadjacency. Projection to its first coordinate and constant next-five block gives

$$\Pi Z = (1/5 + t/2)P_1 + (4/5)P_6, \qquad Q = \|Z - \Pi Z\|^2 = 4/5 - t^2 + t^4/2 < .35.$$

Its true reference pairing is $w \cdot Z = .8f_6 + (.2 + t/2)(.643) < 1.76012 + .3215t$. The complete continuous deficit satisfies $T(t) - w \cdot Z > .0482$. Indeed $T(t) - .3215t$ increases for $t \geq 7/8$: the derivative contains $(2t^2-1)/(\sqrt{2}\sqrt{1-t^2}) > 1/\sqrt{2}$ and a positive further root derivative. At $t_0 = .8849$, the two radicals give $T(t_0) > 2.0929$. Thus the deficit is $> 2.0929 - .3215(.8849) - 1.76012 = .04828465$. The square checks use $.783 < t_0^2 < .784$, $10 + 2t_0 - t_0^2/2 > 3.373^2$ and $.784(.217) < .413^2$.

Insert $w + .1(Z - \Pi Z)$. It spends at most .07 at P_1 and exactly .08 at P_6, both below the separate available prices. The residual is orthogonal to this NONCONSTANT first reference. One square gives

$$\|x\|^2 > .99509 + .2(.0482) - .01(.35) = 1.00123 > 1.$$

If five is instead isolated dotted, its actual first-six core supplies

$$Z_0 = P_5 + (u/2)(e_6 - e_5), \qquad x \cdot Z_0 > T_0 > 1.8185, \quad 2/5 < u < 1.$$

These are the original local roots; no B_{20} conclusion has been transferred. At a strong $J = 6$,

$$\Pi_6 Z_0 = P_1/5 + 4P_6/5, \quad Q_6 < .48, \quad w \cdot Z_0 < 1.76012, \quad T_0 - w \cdot Z_0 > .05838.$$

The coefficient .1 costs .02 at one and .08 at six, giving

$$\|x\|^2 > .99509 + .2(.05838) - .01(.48) > 1.001966.$$

At strong $J = 7$,

$$\Pi_7 Z_0 = P_1/3 + 2P_7/3, \quad Q_7 < 76/75, \quad w \cdot Z_0 < 1.7048, \quad T_0 - w \cdot Z_0 > .1137.$$

Coefficient .12 costs .04 at one and .08 at seven, giving

$$\|x\|^2 > .98949 + .24(.1137) - .0144(76/75) > 1.002186.$$

The P_1 projections and debits are essential; a constant-block projection $Z_0 - (5/J)P_J$ would be wrong for these references.

The nonisolated head is now completely excluded. The head is isolated with time $A_L^2 = 2g_L^2 - 1 > 0$. Own zero/three/four has nonpositive time; six has the actual affine $6, 3$ path; parallel the nonzero positive future null. Its complete remaining own list is native five, finite at least seven, or dotted. No centre, tail or outgoing-fork floor was assumed in this proof.

N.2 An independent old strong-five floor in whole D_{19}

We next prove

$$S_5 > \sqrt{5} - .61/\sqrt{2} \tag{D19old5}$$

without any head hypothesis. It is the OLD .61 floor, not the new .58 floor.

Assume contrariwise old five is weak. The rank-valid generic inventory at $M = 11$, $19 \geq M + 6$, gives a pure helper

$$c_5 = Ap - g\omega_5 - \omega_1/2, \quad A^2 = 10g^2 + 2g - 1/2, \quad \varphi/2 \leq g < 6/5,$$

own five, finite at least seven, or dotted. Its actual first-six points give

$$x_1 > a_0 = .5965, \qquad S_6 > B_0 = 2.01815.$$

There is an independent old strong $K = 11$ or twelve. Let $f_i = \sqrt{i} - .61/\sqrt{2}$, $F_K = B_0 + \sqrt{K} - \sqrt{6} < f_K$. The interpolation through $(1, a_0), (6, B_0), (K, F_K)$ has positive decreasing slopes $a_0, .28433, \ldots, .28433, 1/(\sqrt{K} + \sqrt{6})$. Its precise paid data are

K	$q_K >$	$F_K <$	individual endpoint price $>$
11	.9104	2.886	1/7
12	.9300	3.034	1/7.

The first-six energy is .7600299945. The eleven tail exceeds $.8671^2/5$; the twelve tail exceeds .17. A pressure $V = 1$ through K, $x \cdot V \geq G, V^2 < R$, therefore adds legally

$$2(G - F_K)/7 - (R - K)/49. \tag{D19weak5ref}$$

This is payment at the actual anchor, not subtraction of two independent lower bounds.

Under this assumed weak five we can classify the two late layers afresh. Releasing fifteen retains $A_{14}+D_4$; releasing sixteen retains $A_{15}+D_3$. The right D_4 permits simple three at any leaf. The right $D_3=A_3$ permits middle-three, end-three/four/five. Two contacts to one right piece make a cycle. The long left path admits only the usual endpoint/next-to-end extensions, and the full rank-nineteen bridges are D_{19} or the end-three A_{19} bridge. No exceptional full diagram, internal four or second fork is missing. There is no foreign component.

Here are explicit one-sided seals before the new right rows. For $i=15,16$, an isolated own has $q^2=2i\ge 30$. The actual left H_4, left A_2, singleton and dotted cores give losses respectively

$$<.15+.168,\qquad <.12+.319,\qquad <.11+.415,\qquad <.093u+\sqrt{u(1-u)}<(.093+1.005)/2.$$

Each is below .61. The finite ranges are $u_5<1.528$, $1.2318<u_7<1.233$, and $1<u_{\ge 8}<1.172$; positive squaring at $q^2=30$ proves the first loss terms. The native-five concavity correction uses $i+1\le 17$, an ACTUAL long dual here.

For left-only contact at j, the exact data are $\sigma=g+j/(2i)$ and $\Delta=1-j(i-j)/(2i)$ for simple-three, or $\sigma=g+j/(\sqrt{2}i)$, $\Delta=1/i$ for endpoint-four. After the actual own-zero, three/four/six affine and parallel-null reductions, near endpoint-three has $\sigma>1.266,\sqrt{\Delta}<.731$; next-to-end three has $\sigma>.8,\sqrt{\Delta}<.366$; endpoint-four has a larger ratio. The positive-own-three near-four row is directly strong, while far-four/own-three retains its affine principal. These ratios exceed $25/16$. Only the pure far LEFT helper is retained. Own parallel has its feasible positive future null.

At fifteen the right LONG-leaf three has numerator $g+1/2$, residual $1/2$. Own zero has the actual affine double fork on $c,15,16,17,18,19$, kernel $(1,1,2,2,1,1)$. Own three is an actual triangle; every own at least four has ratio $\ge 1+1/\sqrt{2}$. At sixteen the right middle-three has its analogous own-zero four-leaf affine star, own-three triangle and same strong ratio for larger owns. Right end-four at own zero has the weighted affine principal on $c,f,17,\overline{f},16$, kernel $(\sqrt{2},2,2,1,1)$; positive own has ratio $\ge 1+1/\sqrt{2}$. End-five is the sealed finite H_4, directly strong EVEN at own zero. Full bridges have their own-zero cycles or middle-three star. For positive own, the worst end-three bridge has residual $5/32$, numerator $>3/4$; the other bridge residuals are smaller. The old target is $<25/16$ through these indices, because $T(\sqrt{32})<T(6)<25/16$. All these rows are strong.

The remaining one-sided right fork-three at fifteen or end-three at sixteen has own-zero affine E_8 with arms five/two/one. Own three gives the actual FIVE- cycle at fifteen, FOUR-cycle at sixteen. Own four has the affine star centred at the released root, with squared leaf weights $1/2,1/4,1/4$. Own six has its actual $6,3$ principal, and parallel its feasible positive null.

For native five use the LEFT actual H_4 projection on c,n_i,n_{i-1},n_{i-2}. Its positive correction gives

$$\alpha^2=\varphi+5/8>56/25,\quad \delta_5=(5-3\varphi)/8,\quad V_1=\cdots=V_{i-3}=1,\quad V^2<i,$$

$$x\cdot V>G>\sqrt{i-\kappa/\alpha^2}-.091,\qquad \kappa=1/2\ (i=15),\quad 5/8\ (i=16).$$

The actual finite point is future and all omitted-wall products are nonpositive. The displacement is not presumed to be a point by itself. The demands are $>3.753,3.873$, from $3.844^2<15-25/112$, $3.964^2<16-125/448$. They are flat on BOTH early reference supports. Even the conservative

$$.91+2(3.75-3.034)/7-5/49>1.0125$$

contradicts the norm.

At the D_4 row every finite own at least seven or dotted own is directly strong: $\sqrt{2}(g+1/4) > 1.626 > 25/16$. At the D_3 row its ratio is $(g+1/4)/\sqrt{5/8}$. Weakness forces $g<1$, so dotted own is directly strong. For the remaining $.9<g<1$, use the actual core

$$\xi = c + g^{-1}n_{16} + (2g)^{-1}n_{15}.$$

Its cap and fifteen products vanish, its sixteen product is $-g+3/(4g)<0$, and its square is $-k$, $k=1-3/(4g^2)>0$. It is future and satisfies BOTH original forks and every other wall termwise. At this actual rank,

$$\alpha=\sqrt{2}(g+1/4), \quad A^2=16\alpha^2-5/8, \quad V_1=\cdots=V_{14}=1, \quad V^2<16.$$

Since $\alpha^2 > 529/200$, $\sqrt{16-125/529} > 3.97$; also $\sqrt{k}/\alpha < .308$. Thus $G > 3.662$. The two early-reference bounds are

$$.9104+2(3.662-2.886)/7-5/49>1.0300, \qquad .9300+2(3.662-3.034)/7-4/49>1.0277.$$

Every non-left-helper possibility at fifteen/sixteen is excluded.

Those two remaining left helpers cannot both be weak. Their actual mutual satisfies

$$c_{15}\cdot c_{16} \geq 1/2-ab+(a/b+b/a)/4 \geq 1-ab > -11/25 > -1/2,$$

so distinct actual walls are orthogonal by the native gap. After retaining the left A_{14} and ENTIRE right D_3, their common helper cost is $14/30$, and their two-cap Schur matrix $I-(14/30)\mathbf{1}\mathbf{1}^T$ is positive with positive inverse. Its acute positive projection is an actual feasible point, with the original two signed forks in a positive retained block. The already proved mixed-packet formula, with $C_{ij}=2\min(i,j)$, times >2 and own parameters $>.8$, gives weighted loss $< 7/16+1/(4\sqrt{10}) < .61$. This contradicts two weak sites. Thus fifteen or sixteen is genuinely OLD strong, under the assumed weak five.

If sixteen is strong, the reference through one/six/sixteen already has

$$\|x\|^2 > .7600299945 + (4-\sqrt{6})^2/10 > 1.0004379945,$$

using $\sqrt{6}<2.44949$. Only strong fifteen remains.

Every FINITE weak-five own gives $x_1>.63$. Here $\overline{h}=\sqrt{10}-.61<2.553$. Native five uses the actual finite H_4, time $A>2.768$ and radical $<.111$, yielding

$$x_1 > 1.414(2.768-.111-.81(2.553)) > .63.$$

Native seven uses the actual negative core $c_5+gn_5+n_1/2$, with $A>3.0688$, radical $<.2486, g<.901$, giving $x_1>1.414(3.0688-.2486-.901(2.553))>.63$. Native eight is RETAINED and uses that same actual core: $A>3.1435$, radical $<.322$, $g<.9239$, giving $x_1>1.414(3.1435-.322-.9239(2.553))>.6543$. For every finite label at least nine, the actual pair projection gives

$$x_1 > \sqrt{2}\{A-\sqrt{1-g^2}-g\overline{h}\}.$$

The brace increases with g: $A'>\sqrt{10}>\overline{h}$, and the negative radical has positive derivative. At $g=.939$, $A>3.192$, $\sqrt{1-g^2}<.344$, giving $x_1>1.414(3.192-.344-.939(2.553))>.6373$. The elementary native-nine lower bracket follows by signs of $8g^3-6g-1$. All these feet are future/all-wall points constructed with original roots one through six.

The positive reference through $(1,.63),(6,B_0),(15,B_0+\sqrt{15}-\sqrt{6})$ has energy

$$.63^2+(B_0-.63)^2/5+(\sqrt{15}-\sqrt{6})^2/9>.7822920845+1.4234^2/9>1.0074.$$

Its heights lie below actual strong fifteen, its slopes $.63,.27763,1/(\sqrt{15}+\sqrt{6})$ decrease. Thus every finite weak-five own is impossible.

For dotted $1<g<6/5$, retain the actual foot BEFORE dropping the chamber difference $r=e_5-e_6$. The future core $c_5+g^{-1}n_5$ gives

$$T=P_1+2gP_5-g^{-1}r,\qquad x\cdot T\ge K=\sqrt{2}\{A-\sqrt{1-g^{-2}}\}.$$

The function $A-\sqrt{1-g^{-2}}-g\overline{h}$ decreases: $A'<13/4$, the radical derivative exceeds one and $\overline{h}>5/2$. At $6/5$ it gives the strict bound

$$K-2gf_5>\sqrt{2}\{\sqrt{163/10}-\sqrt{11}/6-(6/5)\overline{h}\}>.5965.$$

A rational certificate uses $\sqrt{163/10}>4.03732$, $\sqrt{11}/6<.552771$, $\sqrt{10}<3.162278$, $\sqrt{2}>1.41421$, whose lower product is $.596535556834$.

Set

$$F=1.80474,\quad B=3.44164,\quad X=.59647.$$

Positive square bounds give $1.80473<f_5<F$, $f_{15}>B$. Since $F-f_5<.00001$, the actual primitive pressure obeys $x\cdot T\ge K>X+2gF$. Define exact positive rationals

$$a=(7B-8F+X)/66=10.25003/66,\qquad t=(2F-3X+B)/22=5.26171/22,$$
$$y=2t-a=21.32023/66,\qquad \mu=X-y=18.04679/66.$$

Let

$$W=(X,y,y,y,t,t,\underbrace{a,\dots,a}_{9},0,0,0,0).$$

Then $W\cdot P_5=F$, $W\cdot P_{15}=B$, $W\cdot r=0$. For the ACTUAL unknown g, put

$$\lambda=y-a-2g\mu<0,\qquad \rho=a+\mu/g-t>0.$$

Indeed $y-a<.168$, $2g\mu>2(.2734)$, and $a>.1553,\mu>.2734,t<.2392,1/g>5/6$ give $\rho>.1439$. There is the exact original- coordinate identity

$$W=\mu T+\lambda P_5+aP_{15}+\rho r.$$

Its NEGATIVE coefficient is intentional and legal: the assumed old weakness is the upper bound $S_5\le f_5<F$. The other signs use the actual strong fifteen lower bound and the original strict chamber inequality $x\cdot r>0$. Consequently

$$x\cdot W>\mu(X+2gF)+\lambda F+aB=\mu X+(y-a)F+aB=W^2.$$

The last identity follows also by pairing the displayed coordinate decomposition with W, using $W\cdot T=X+2gF$. No pressure on a residual was invented and no independently optimized energy was added.

Finally $X>.5964,y>.3230,t>.2391,a>.1553$, so

$$W^2>.5964^2+3(.3230)^2+2(.2391)^2+9(.1553)^2=1.00008039>1.$$

The one original-vector square $\|x\|^2\ge 2x\cdot W-W^2>W^2$ contradicts $\|x\|<1$. This proves the head-independent OLD floor (D19old5). It asserts neither a .58 five floor nor centre/fork strength.

N.3 Native-five D_{19}: the fork bounds actually needed by the centre

Assume the isolated head has native five, and put

$$a = \varphi, \quad g = a/2, \quad c_H = A_H p - g\omega_1, \quad A_H^2 = (a-1)/2.$$

Its actual finite H_4 point, proved in Section M.2, gives $S_4 \geq 2\sqrt{2}A_H > 1.572$. We need two fork consequences, rather than the full outgoing- fork taxonomy.

First BOTH actual fork edge cosines are at most ten. Use the independent old strong- five floor (D19old5) and an actual old strong $K = 11$ or twelve. Its curved reference w_0 has first-five mean $b < .361$, individual P_5 price $> .146$, endpoint price $> 1/7$, and $q_{11} > .8472225, q_{12} > .8689525$. Insert the head ONCE through $w_H = w_0 + .64(P_4/4 - P_5/5)$. The price is .128 at five. The orthogonal residual square is $1/20$; its effective gain is $> 1.28(.393 - .361) - .4096/20 = .02048$.

If an actual fork had $C > 10$, its original-edge comparison would give the signed pressure

$$x \cdot V > \sqrt{171/11} > 3.942, \qquad V = P_{18} \pm e_{19}, \quad V^2 = 19.$$

The residual $V - P_K$ vanishes on the entire head-paid support. A legal $1/7$ endpoint payment gives the one-vector totals

$$.8472225 + .02048 + 2(3.942 - 2.886)/7 - 8/49 > 1.00615,$$
$$.8689525 + .02048 + 2(3.942 - 3.034)/7 - 7/49 > 1.00600.$$

Thus both actual fork cosines are at most ten, without a centre floor.

Second no ACTUAL wall whose spatial part is $d\omega_f$, for either fork f, can have finite own $d = \cos(\pi/m)$. This statement does not require the wall to be first on a fork edge. It also applies to a centre wall of own zero with one finite fork contact.

For such a wall, $A^2 = (19/2)d^2 - 1$, and

$$M = A_H\sqrt{(19/2)d^2 - 1} - (a/2)d.$$

The head and wall are distinct by their different old supports. For $1/2 \leq d \leq 1$, M is positive and increasing: at $1/2$ its squared-product difference is $(5a - 6)/8 > 0$; its derivative is positive since $A_H\sqrt{19/2} > a/2$. Furthermore

$$M < M(1) < .556(2.916) - .809 = .812296 < .82 < \cos(\pi/6).$$

The mutual is therefore FINITE, with exactly one of $M = 1/2, \sqrt{2}/2, a/2$. Its strict scalar range excluded orthogonality, parallel and dotted mutuals before algebra.

Only the elementary finite-angle fact is used next: $t = 2d = \zeta_{2m} + \zeta_{2m}^{-1}$ is an algebraic integer for EVERY Coxeter polytope. The fields below are fields of known real angle constants. They are NOT identified with a defining field of the reflection group, and no dotted product or invariant cycle is declared integral.

Put $L = \mathbb{Q}(a)$, $\mathcal{O}_L = \mathbb{Z}[a]$, $B = 19 - 10a$, $\mathrm{N}_L(B) = 71$. Squaring the positive mutual equation gives exactly

M	quadratic for integral t	discriminant
$1/2$	$Bt^2 - 2t + 2a - 6$	$12(45 - 26a)$
$\sqrt{2}/2$	$Bt^2 - 2\sqrt{2}t - 4$	$8(39 - 20a)$
$a/2$	$Bt^2 - 2at + 2a - 8$	$4(173 - 97a)$.

The first and last discriminant norms are $144(179)$, $16(3739)$. They are nonsquares, since $13^2 < 179 < 14^2$, $61^2 < 3739 < 62^2$. Thus the quadratics are irreducible over L. The relative norm of

integral t must be integral in L, yet its rational absolute norm would be $20/71$, respectively $44/71$, impossible.

For the middle row work in the genuine quadratic extension $L_2 = L(\sqrt{2})$. An element $D \in L$ can be square there only if D or $D/2$ is square in L: expand $(r + s\sqrt{2})^2$, whose mixed coefficient forces $rs = 0$. For $D = 8(39 - 20a)$, the two norms are $64(341), 16(341)$, both nonsquare as $18^2 < 341 < 19^2$. This quadratic is irreducible over L_2. Its integral root would have relative norm $-4/B$, rational absolute norm $256/71^2$, again impossible. Every finite own is excluded.

Own zero has negative time; parallel own has the nonzero feasible future null $c + n_f$. Consequently an isolated fork wall, when encountered, has dotted own. No integrality of that dotted own is used. These are the only outgoing- fork facts needed below.

N.4 Native-five D_{19}: one head-paid centre reference

The complete new bulk through fourteen is already available. The actual native-five mutual calculation uses only the first seven long roots, so the proof in Section M.3 applies here. Explicitly, for a helper with $r < 1.2$ and $i \le 7$,

$$A_H^2 A_i^2 - [a(r + 1/2)]^2 = [i(a - 1) - a^2]r^2 - 2r - a/2 < .0568r - .809 < 0.$$

It would have positive actual mutual with the head. The remaining isolated dotted five is excluded by the same actual native-gap calculation: weakness forces $1 < r < \sqrt{5/2}$, and its positive increasing head magnitude is $< .556(4.899) - 1.618(1.581) = .165786 < 1/2$. Thus five/six/seven are truly .58-strong.

Choose an actual strong $K = 13$ or fourteen. All internal weak helper holes cost $< .000158$, using the earliest permitted set $8, 10, 12$. The rational certificate (B19E17), restricted to its initial increments, gives

$$E_{13} > .9056051, \quad E_{14} > .9241299, \quad f_{13} < 3.1955, \quad f_{14} < 3.3316, \quad b = f_5/5 < .3652.$$

The reference w_0 has P_5 price $> .151$ and endpoint price $> 1/8$. Insert the head once in $w_H = w_0 + .14(P_4 - .8P_5)$. Its debit is .112 at five, orthogonal residual square .8, and effective gain

$$.28(1.572 - .8(1.826)) - .0196(.8) = .015456.$$

Hence the actual paid references have

$$L_{13} > .9209031, \qquad L_{14} > .9394279.$$

Their first four coordinates are $< .3932$, their sums remain f_K, and the individual end price stays $> 1/8$.

For an actual pressure $x \cdot V \ge G$, $V = 1$ through the chosen K, $V^2 \le R$, the residual $V - P_K$ vanishes on the ENTIRE head-paid support. One paid square gives

$$\|x\|^2 > L_K + \frac{16G - R - 16f_K + K}{64}. \tag{D19fivepay}$$

Therefore the uniform sufficient contradiction is

$$16G - R > 43.19021. \tag{D19fivebudget}$$

The exact thresholds for the two K's are $43.1902016, 43.1822144$. An ACTUAL .50-strong centre would have $S_{17} > 3.7695$; $V = P_{17}, R = 17$ already violates this budget. The following centre classification may thus assume .50 weakness as a CONTRARY hypothesis. No centre floor is being supplied in advance.

N.5 Native-five D_{19}: every isolated centre fails

Own zero has negative time. Own three has the actual four- leaf affine star centred at seventeen; own four the weighted star with both forks; six the $6,3$ path through sixteen; parallel the feasible positive null $c+n_{17}$. Each null is future through its positive cap coefficient and nonpositive against all omitted original walls.

At native five use the LEFT finite H_4 on the actual cap/seventeen/sixteen/fifteen. Its positive correction has residual $\delta=(5-3a)/8$; the actual projected point, normalized by $\alpha=\sqrt{2}g$, gives $V=1$ through fourteen, $R<17$, and

$$G>\sqrt{17-1/(2g^2)}-\sqrt{\delta}/\alpha>4.029-.119>3.91.$$

This contradicts (D19fivebudget).

At native seven/eight use $u=r^{-2}$ and the actual core $c+r^{-1}n_{17}+(2r)^{-1}n_{16}$. Its cap/sixteen products vanish, seventeen product is negative, and its square is $-(1-3/(4r^2))$. It is future and every other original-wall product is nonpositive. Its normalized displacement is

$$V=P_{15}+(1-u/4)(e_{16}+e_{17})+(u/2)e_{18},\quad R=17-u+3u^2/8<17,$$
$$G=\sqrt{17-u/2}-\sqrt{u/2-3u^2/8}>4.047-.27=3.777.$$

Indeed $1.168<u<100/81$; the second squared expression decreases and at 1.168 is $.072416<.27^2$. Thus $16G-R>43.432$.

For EVERY finite own at least nine, $r>.939$ and $r^{-2}<1.1352$. The actual finite pair with seventeen has normalized $V=P_{16}+(e_{17}+e_{18})/2$, $R=16.5$. Its TRUE projected-point demand is larger than

$$\sqrt{17-1/(2r^2)}-\sqrt{1/(2r^2)-1/2}>4.053-.26=3.793.$$

The native brackets used here are only ordinary real-angle square/cubic bounds. Every finite case violates the common budget.

For dotted $r>1$, $u=r^{-2}\in(0,1)$, the actual core $c+r^{-1}n_{17}$ gives

$$V=P_{16}+(1-u/2)e_{17}+(u/2)e_{18},\quad R=17-u+u^2/2,$$
$$G=\sqrt{17-u/2}-\sqrt{u(1-u)/2}.$$

The complete continuous estimate is $16G-R>43.19278$. To prove it, use the concave root chord, $\sqrt{17}>4.123105$ and $8/(\sqrt{17}+\sqrt{33/2})<.9775$, obtaining

$$16G-R>48.96968+.0225u-u^2/2-8\sqrt{2}\sqrt{u(1-u)}.$$

For $u\le.45$, $u^2\le.45u$ and monotonicity give the lower bound $48.96968-.091125-\sqrt{31.68}>43.249555$. For $.45\le u\le.55$, use $u^2\le u-.2475$ and the circle maximum

$$Au+B\sqrt{u(1-u)}\le(A+\sqrt{A^2+B^2})/2.$$

With $A=.4775, B=8\sqrt{2}$ and $\sqrt{128.22800625}<11.3238$, the lower bound is $48.96968+.12375-5.90065=43.19278$. For $u\ge.55$, use $u^2\le1.55u-.55$, $A=.7525$ and $\sqrt{128.56625625}<11.33872$, giving 43.19907. All radicals have positive rational square checks. Thus the ENTIRE dotted isolated centre case is excluded.

N.6 Native-five D_{19}: the complete finite-white centre list

Deleting seventeen leaves the old A_{16} and both isolated fork A_1's. The complete finite positive white list is:

isolation;
left-only $3@1,2,15,16$ or $4@1,16$;
one fork with any finite native contact;
two forks $(3,3),(3,4),(3,5)$, either orientation;
left endpoint-three $j=1,16$ and one fork-three/four, or both fork-threes.

A second left contact makes a cycle; a next-to-end attachment plus another fork has excessive exceptional arms; no exceptional finite diagram has the required full rank. The whole-forest hypothesis leaves no foreign component.

For direct .50 tests use $T_{.50}(\sqrt{34}) < 265/138 < 1.921$. For left-only three,

$$\rho = (\gamma + j/34)/\sqrt{1 - j(17-j)/34}.$$

At $j=2$, own zero has zero time, three an affine cycle; every own at least four is directly strong. At $j=15$, own zero has the actual positive affine kernel $(1,1,2,2,2,1,1)$ on $c,14,15,16,17,18,19$; every positive own is strong. At $j=16$, zero has kernel $(1,1,2,2,1,1)$ on $c,15,16,17,18,19$, three its triangle, four the weighted star using both forks, six its $6,3$ path. Finite owns at least nine and dotted owns are strong; five/seven/eight need the actual point below. At $j=1$, zero has negative time, three the actual long cycle, four the weighted star using sixteen and one fork, six its $6,3$ path. Its remaining far-helper owns are retained for a separate calculation.

For left-only four,

$$\rho = (\gamma + j/(17\sqrt{2}))/\sqrt{1 - j + j^2/17}.$$

Far own zero has zero time; all other far cases and even near own zero are directly strong.

For one fork β,

$$\rho = (\gamma + \beta/2)/\sqrt{1-\beta^2}.$$

Own zero is the actual finite- fork wall already excluded by Section N.3, whose proof did not require it to be first on a fork edge. At own three, deleting the contacted fork gives a positive finite D_{19} replacement. Actual positive inverse projection makes it the reciprocal fork endpoint, of cosine

$$C_f = (17 + 19\beta)/2 \geq 53/4 > 10.$$

This contradicts the actual fork bound. The centre and fork opposites are allowed to coincide; no false distinctness was assumed. Own four has the affine weighted star using sixteen and the OTHER fork; six its $6,3$ path; parallel its positive future null. For $\gamma \geq \varphi/2$ and $\beta \geq \cos(\pi/5)$, the ratio is > 2.06. Thus only $\beta = 1/2, 1/\sqrt{2}$ needs further point estimates.

For two forks β, η,

$$\rho = \frac{\gamma + (\beta+\eta)/2}{\sqrt{1-\beta^2-\eta^2}}.$$

Own zero of $(3,3)$ is the actual all-three square, three an affine triangle, six/parallel their actual null obstructions. Own four/five will be treated below; finite at least seven and dotted own are strong. Own zero of $(3,4)$ has an actual finite B_{19} reciprocal endpoint at the three-fork, with $C_f = 17/2 + (19/4)\sqrt{2} > 10$. Positive owns there, and even own zero in $(3,5)$, are directly strong.

For left endpoint-three at $j = 1, 16$ with fork magnitudes β, η, the exact data are

$$\delta = 9/17 - \beta^2 - \eta^2, \qquad \rho = \frac{\gamma + j/34 + (\beta + \eta)/2}{\sqrt{\delta}}.$$

One fork-three at own zero has the actual long/four-cycle. Near positive own is strong. Far own three has its fork triangle, four the weighted star using sixteen and the OTHER fork; larger owns are strong. One fork-four or two fork-threes has residual $1/34$, strong even at own zero. Both signed fork orientations and every mixed own-zero role are now included. Every positive affine/null combination is future through the cap and nonpositive against ALL omitted original walls.

N.7 Native-five D_{19}: every remaining centre point exception

For near-left three and own five/seven/eight, $v = \gamma\omega_{17} + \omega_{16}/2$, $A^2 = 34\gamma^2 + 32\gamma + 7$. Use the ACTUAL finite pair c, n_{17}, with residual $\delta = 1 - \gamma^2 > 0$ and projected-point pressure $H = AB/(B + \sqrt{\delta})$, $B^2 = A^2 + \delta$. Its normalized displacement is

$$V = P_{16} + \frac{\gamma}{2\gamma + 1}(e_{17} + e_{18}), \quad \alpha = \sqrt{2}(\gamma + 1/2), \quad R < 16.23.$$

At native five, $A^2 = (49a + 31)/2$, $B^2 = (97a + 65)/4$. Positive-square bounds $A > 7.425, B > 7.448, \sqrt{\delta} < .588, \alpha < 1.852$ give

$$G > 7.425(7.448)/(8.036(1.852)) > 3.715.$$

The sharper $R = 17 - a/2 < 16.2$ gives $16G - R > 43.24$. The weaker $A - \sqrt{\delta}$ pressure would be insufficient. At native seven/eight, $A/\alpha > 4.019$, $\sqrt{\delta}/\alpha < .221$; hence $G > 3.798$, which closes these rows as well.

For one/two fork helpers write

$$v = \gamma\omega_{17} + \beta\omega_f + \eta\omega_h, \quad \alpha = \sqrt{2}(\gamma + (\beta + \eta)/2), \quad A^2 = 17\alpha^2 - (1 - \beta^2 - \eta^2).$$

At own five, the LEFT actual finite H_4 has correction $g(3n_{17}/2 + n_{16} + n_{15}/2)$, residual $(5 - 3a)/8$. For a single helper $\beta = 1/2, 1/\sqrt{2}$, or two helpers $\beta = \eta = 1/2$, its actual projected point has $V = 1$ through fourteen, $R < 17$, and

$$G > 4.082 - .091 > 3.99.$$

Use $\alpha > 1.497$, $A/\alpha > 4.082$, $\sqrt{(5 - 3a)/8}/\alpha < .091$. The contacted fork walls remain outside the selected finite piece, with their actual products nonpositive. Every own-five helper exception is excluded.

For one fork-four and own finite at least seven or dotted, contrary .50 weakness forces $.9 < \gamma < 1.01$. The actual positive core

$$c + \gamma^{-1}n_{17} + (2\gamma)^{-1}n_{16}$$

has zero cap/sixteen products, negative seventeen product, square $-k$, $k = 1 - 3/(4\gamma^2) > 0$. It is future and every omitted wall product is nonpositive. With $\alpha = \sqrt{2}(\gamma + \sqrt{2}/4)$, V is flat through fifteen, $R < 17$, and

$$G = \sqrt{17 - (1/2)/\alpha^2} - \sqrt{k}/\alpha > 4.103 - .291 = 3.812.$$

The complete seals are $\alpha > 1.772, \alpha^2 > 3.141$, $k < .265, \sqrt{k} < .515$.

For one fork-three the entire contrary range is $.9 < \gamma < 1.42$. Use the SAME actual core, with $\alpha = \sqrt{2}(\gamma + 1/4)$, $A^2 = 17\alpha^2 - 3/4$. Its displacement is flat through fifteen and

$$G = \sqrt{17 - (3/4)/\alpha^2} - \sqrt{k}/\alpha > 3.756, \qquad R = 17 - (3/4 + k)/\alpha^2 < 16.8.$$

The continuous proof is explicit. For $\gamma \le 1.1$, the first root is > 4.088, the second $< .324$. The latter squared ratio increases up to 1.1, its derivative having sign $-2\gamma^3 + 3\gamma + 3/8 > 0$. For $\gamma \ge 1.1$, the first root is > 4.098. Uniformly $\sqrt{k}/\alpha < .342$: the inequality $k/\alpha^2 < 7/60$ is equivalent to

$$112\gamma^4 + 56\gamma^3 - 473\gamma^2 + 360 > 0.$$

At $t = \gamma - 5/4$ this becomes $15/4 - 45t + 787t^2 + 616t^3 + 112t^4$. Because $t \ge -.35$, it is at least $15/4 - 45t + (2857/5)t^2 > 0$, with negative discriminant. For the norm, $1.75 - .75/\gamma^2 - .4(\gamma + .25)^2$ is concave and positive at both $.9, 1.42$. Thus $16G - R > 43.296$ throughout, including all dotted parameters.

The two-fork-three/own-four row is a separate actual point. Let $s = \sqrt{2}, \gamma = s/2$ and

$$\xi = c + (4 - 2s)n_{17} + (2 - s)n_{16} + (3 - 2s)(n_{18} + n_{19}).$$

All coefficients are positive. Its cap/seventeen/sixteen products vanish, each fork product is $1/2 - s < 0$, and $\xi^2 = 11 - 8s < 0$. It is future and feasible against every original wall. For $\alpha = 1 + s/2$,

$$V = P_{15} + (5 - 3s)(e_{16} + e_{17} + e_{18}), \quad R = 144 - 90s < 16.721,$$

$$G = \sqrt{14 + 2s} - \sqrt{92s - 130} > 4.102 - .329 = 3.773.$$

Hence $16G - R > 43.647$. All non-far-helper centre cases have now been excluded at their actual changed threshold.

N.8 Native-five D_{19}: the only centre left is a dotted far helper

For the far-left-three cap, $v = \gamma\omega_{17} + \omega_1/2$, $A^2 = 34\gamma^2 + 2\gamma - 1/2$. Own zero/three/four/six/parallel already has its actual time, affine or positive-null obstruction. The remaining finite-own pressure table, normalized by $\sqrt{2}\gamma$, is

own	$V = P_{16} + Y$	actual point/core	$G >$
5	$a_3P_3 + (e_{17} + e_{18})/2$	finite H_4, $\delta = 2/3 - \gamma^2$	4.1158
7, 8	$a_2P_2 + (e_{17} + e_{18})/2$	$c + \gamma n_{17} + n_1/2$	3.963
finite ≥ 9	$a_1P_1 + (e_{17} + e_{18})/2$	finite pair c, n_{17}	3.965.

Here $a_3 = 1/(6\gamma) < 5/24$, $a_2 = 1/(4\gamma) < 5/18$, $a_1 = 1/(2\gamma) < .533$. All selected roots are actual. Their positive corrections give future points retaining every unknown wall and both original forks; the finite rows use projected POINTS.

All finite rows satisfy $A/(\sqrt{2}\gamma) \ge \sqrt{17.75} > 4.213$. The first two normalized radical subtractions are $< .097125, .25$. For finite at least nine, $\gamma \ge .95$ gives the crude pair lower bound $> 4.213 - .233 = 3.98$. For $.939 < \gamma \le .95$, the crude lower bound is $> 4.213 - .26$, and the TRUE finite pressure improves $A - \sqrt{\delta}$ by $\delta/(B + \sqrt{\delta})$. Since $\delta \ge .0975$, $\sqrt{2}\gamma < 1.344$, $B < 5.674, \sqrt{\delta} < .344$, its normalized improvement is $> .012$. Thus $G > 3.965$ in the whole finite-nine-and-higher range. Native eight was retained geometrically, not removed by a field condition.

The head overlap is NONZERO. For $Y = a_l P_l + (e_{17} + e_{18})/2$,

$$V - P_K = P_{16} - P_K + Y, \qquad w_H \cdot (V - P_K) = a_l(w_H \cdot P_l),$$
$$\|V - P_K\|^2 = 16 - K + la_l^2 + 1/2.$$

With the same legal 1/8 endpoint price, the sufficient barrier is therefore

$$16\{G - a_l(w_H \cdot P_l)\} - (16 + la_l^2 + 1/2) > 43.19021.$$

The true head-paid first coordinates give $w_H \cdot P_l < .3932l$. The three rows have respective lower bounds $45.29059, 43.25856, 43.30270$. All violate the norm budget. No aggregate lower bound was subtracted to manufacture a residual pressure.

For dotted $1 < \gamma \le 10/9$, let $a_0 = 1/(2\gamma) \in [.45, .5)$. The actual core $c + \gamma^{-1}n_{17}$ gives

$$V = P_{16} + a_0 P_1 + (1 - 2a_0^2)e_{17} + 2a_0^2 e_{18},$$
$$G = \sqrt{17 + 2a_0 - a_0^2} - a_0\sqrt{2 - 8a_0^2}.$$

The residual squared norm is $16 - K + 1 - 3a_0^2 + 8a_0^4 \le 16 - K + .75$. The function $G - .3932a_0$ strictly increases, since its derivative is

$$\frac{1 - a_0}{\sqrt{17 + 2a_0 - a_0^2}} + \frac{16a_0^2 - 2}{\sqrt{2 - 8a_0^2}} - .3932 > 0.$$

At $a_0 = .45$, $\sqrt{17.6975} > 4.2068$, $\sqrt{.38} < .6165$, giving $G - .3932a_0 > 3.752435$. The paid barrier is $> 16(3.752435) - 16.75 = 43.28896$, a contradiction on the full small-dotted interval.

For $\gamma \ge 11/8$, its exact edge ratio $(\gamma + 1/34)/\sqrt{9/17} > 1.928$ already gives the contradictory .50-strong centre floor. The ENTIRE remaining actual centre has consequently the profile

$$c = Ap - \gamma\omega_{17} - \omega_1/2, \qquad A^2 = 34\gamma^2 + 2\gamma - 1/2, \qquad 10/9 < \gamma < 11/8, \qquad \text{(D19fivecentre)}$$

with dotted own. No outgoing fork equality, isolation or symmetry was assumed.

N.9 Native-five D_{19}: classify short fork edges at the actual adjacent vertex

Replace old seventeen by its ACTUAL opposite (D19fivecentre). The other endpoint q of that original edge has finite forest

$$c \ - -3 - - n_1 \ - -3 - - \cdots - -3 - - n_{16}, \qquad n_f,\ n_h,$$

an A_{17} and two orthogonal A_1's. This is the actual neighbouring vertex, not a formal completion or a vertex assumed farthest. Set

$$B = 17\gamma + 1/2, \quad C = (34\gamma + 1)/(3\sqrt{2}), \quad s = A\sqrt{17}/3, \quad q = Cp - sP_{17}/\sqrt{17}, \quad C^2 - s^2 = 1.$$

Both forks are unchanged unit normals tangent at q. Put $\sigma = s/\sqrt{34} = A/(3\sqrt{2})$. The centre interval gives

$$A^2 < 2129/32, \qquad \sigma < 77/40 = 1.925, \qquad B > 349/18.$$

We need the products of the new path endpoint duals with the OLD actual wall n_{17}. Let

$$w = (e_1 - P_{17}/17)/\sqrt{2}, \quad \eta = c + w, \quad \eta^2 = 9/17, \quad d_{16} = \sqrt{2}(P_{16} - 16P_{17}/17).$$

These are tangent to q. The new path duals are $\Omega_c = (17/9)\eta$, $\Omega_{16} = d_{16} + \Omega_c/17$: η is orthogonal to old one through sixteen, and $d_{16} \cdot c = -1/17$. Directly,

$$\Omega_c \cdot n_{17} = -B/9, \qquad \Omega_{16} \cdot n_{17} = -(144 + B)/153. \qquad \text{(D19qduals)}$$

These are actual Lorentz tangent identities; no projected mirror has been presumed Coxeter.

Call a q-fork edge short if its actual length satisfies $\cosh L < \sqrt{3/2}$. Let r be its actual opposite, with positive own magnitude β, full finite-white cost $Q < 1$, and time $a_r > 0$. The unit A_1 inverse gives

$$a_r^2 = \beta^2 + Q - 1, \qquad 1 - Q < \beta^2 < (3/2)(1 - Q). \tag{D19qshort}$$

Own zero cannot satisfy this. Parallel own has the actual positive future null $r + n_f$ and is impossible. All finite and dotted owns remain in the displayed strip.

After releasing that A_1, the endpoint retains the ENTIRE new A_{17} and the other A_1. Its complete positive finite white list and costs are

contact outside the released A_1	Q
none	0
only other A_1, finite label $m \geq 3$	$\cos^2(\pi/m)$
A_{17} end-three	17/36
A_{17} next-to-end-three	8/9
A_{17} end-four	17/18
A_{17} end-three plus other-three	13/18
A_{17} end-three plus other-four	35/36.

A second path contact is a cycle. Its rank-eighteen one-node extensions are A, B, D, with no exceptional position. Joining both retained pieces allows only end-three followed by other-three/four, giving A_{19}, B_{19}. A next-to-end attachment plus another edge has excessive exceptional arms, and an end-four followed by another node has a nonterminal double. Thus every finite white type is included.

Applying the exact short strip leaves: isolated dotted own $1 < \beta^2 < 3/2$; path end-three with own five/ six; or other-A_1-only own-five/white-four, own-four/white-five, own-three/white-seven, or white-three with finite own at least seven or dotted, $3/4 < \beta^2 < 9/8$. Here are the complete label checks for the other-A_1 case: white three/four/five/ six/seven/at-least-eight gives respectively the own-square strips

$$(.75, 1.125), \quad (.5, .75), \quad ((3-a)/4, 3(3-a)/8), \quad (.25, .375),$$

a subinterval containing only native own-three at white seven, and upper bound $< .25$ at white at least eight. The other cost rows are empty: $Q \geq 8/9$ has own-square upper bound $< 1/6$, and $Q = 13/18$ has only possible native three below its positive- time lower bound. The mixed 35/36 row is smaller still. This is local finite classification at the ACTUAL q, with no farthest hypothesis there.

Every short opposite r is distinct from OLD n_{17}: that wall has dotted contact $-\gamma$ with new c, which cannot occur in its retained positive finite white endpoint. Actual acuteness/native products therefore apply to r, n_{17}.

For path end-three, $j = c$ or sixteen, write $r = a_r q - \beta n_f - \Omega_j/2$. Own is five/six, so $\beta > .809, a_r < .472$. Using (D19qduals),

$$r \cdot n_{17} = -a_r\sigma + \beta/2 - (\Omega_j \cdot n_{17})/2 > -.472(1.925) + .4045 + .533 = .0289 > 0.$$

For endpoint c the last contribution exceeds one; for endpoint sixteen it exceeds .533. Both are impossible.

For a cap supported on the two q-A_1's,

$$r = a_r q - \beta n_f - \eta n_h, \qquad r \cdot n_{17} = -a_r\sigma + (\beta + \eta)/2.$$

Own-three/white-seven has $a_r < .249$, half spatial sum $> .7$, so the product is $> .7 - .249(1.925) > .22$. Own-five/white-four and its reverse have $a_r^2 = (a-1)/4 < .3931^2$, $(\beta+\eta)/2 > .75805$. The product is $> .75805 - .3931(1.925) = .0013325 > 0$. These are strict actual-wall acuteness contradictions.

Only two short types remain:

I	$r = a_r q - \beta n_f, \quad 1 < \beta^2 < 3/2$
J	$r = a_r q - \beta n_f - n_h/2, \quad$ finite own ≥ 7 or dotted, $.9 < \beta < 1.061, \quad a_r^2 = \beta^2 - 3/4 < 3/8.$

For type J, $\rho = -r \cdot n_{17} = a_r\sigma - (\beta + 1/2)/2 \geq 0$. But $\rho < .613(1.925) - .7 = .480025 < 1/2$. The ACTUAL native gap forces $\rho = 0$. Hence this wall has zero products with ALL old one through seventeen; its exact OLD expansion is

$$r = H_\beta p - \beta\omega_f - \omega_h/2, \qquad H_\beta^2 = (19/2)\beta^2 + (17/2)\beta + 11/8. \tag{D19qJ}$$

It need not be first on an old fork edge; its old dual expansion follows just from the actual products and unit norm.

N.10 Native-five D_{19}: one q-fork edge is nonshort

Both q-fork edges cannot be short. The surviving types cover both same-wall and distinct-wall possibilities. An I wall has zero other contact and cannot close the other edge. A J wall has own $> .9$ and other contact $1/2$; in the reverse direction it does not have either surviving short type. Thus two short opposites in the remaining list are distinct.

Two type-I walls have product $-a_r a_t \in (-1/2, 0)$, because both time squares are $< 1/2$. This violates the native gap. For I at one fork and J at the other, the q-product is

$$-a_I a_J + \beta_I/2 > 1/2 - \sqrt{3}/4 > 0,$$

using $a_I^2 < 1/2, a_J^2 < 3/8$ and $\beta_I > 1$.

For two type-J walls, use their OLD expansions (D19qJ). Their negative mutual is

$$M(b,d) = H_b H_d - (17/2)bd - (19/4)(b+d) - 17/8.$$

This function increases in both independent parameters $b, d > .9$. Indeed $H_b' > \sqrt{19/2}$, and

$$(19/2)H_d^2 - [(17/2)d + 19/4]^2 = 18d^2 - 19/2 > 0.$$

Since BOTH actual parameters are < 1.061, monotonicity gives

$$M(b,d) < M(1.061, 1.061) = 1.061^2 - 1.061 - 3/4 = -.685279 < 0.$$

Their actual product would be positive. Equality of the two actual parameters was not assumed; the equal argument is only an upper bound after monotonicity.

At least one actual q-fork edge therefore has

$$D = \cosh L \geq \sqrt{3/2}. \tag{D19qnonshort}$$

No ordinary A_1 energy theorem or farthest-forest classification was applied at q.

N.11 Native-five D_{19}: compare that endpoint with the ORIGINAL farthest vertex

For the nonshort q-fork edge, let its ACTUAL other endpoint be $r' = Dq - \sqrt{D^2-1}\, n_f$, in whichever signed orientation has (D19qnonshort). Put

$$\tau = (-z \cdot q)/E = C - (s/\sqrt{17})S_{17}.$$

The ORIGINAL farthest property gives $0 < \tau \le 1$ and

$$\tau D - \sqrt{D^2-1}\, x \cdot n_f \le 1.$$

We retain τ; it is not replaced by one. For fixed $0 < \tau \le 1$, $(D\tau - 1)/\sqrt{D^2-1}$ increases with D: its derivative is $(D-\tau)/(D^2-1)^{3/2} > 0$. The lower endpoint $D_0 = \sqrt{3/2}$ is therefore legal, giving

$$x_{18} \pm x_{19} \ge \sqrt{6}\,\tau - 2.$$

Substitution yields the actual SAME-vector pressure

$$x \cdot V \ge G, \qquad V = P_{17} + t(e_{18} \pm e_{19}), \quad t = \sqrt{17}/(\sqrt{6}\, s),$$
$$G = \sqrt{17 + 6t^2} - 2t, \qquad R = 17 + 2t^2.$$

Every endpoint used is actual and hence satisfies every original and unknown wall. There is no formal replacement or independent energy at q.

Since $s^2 = 17A^2/9$ and $\gamma > 10/9$,

$$t^2 = 3/(2A^2) < 243/7079 < 9/256, \qquad 0 < t < 3/16.$$

The cleared comparison is $243(256) = 62208 < 63711 = 9(7079)$. The function $\sqrt{17+6t^2} - 2t$ decreases on this interval. At its upper endpoint,

$$G > \sqrt{17 + 27/128} - 3/8 > 4.1486 - .375 = 3.7736, \qquad R < 17 + 9/128 = 17.0703125.$$

The positive square check is $4.1486^2 = 17.21088196 < 17 + 27/128$. Thus

$$16G - R > 43.3072875 > 43.19021.$$

The displacement is one through the actual chosen $K = 13, 14$. Its residual is orthogonal to the head-paid reference, and its 1/8 individual endpoint price is still available. Equation (D19fivepay) gives

$$\|x\|^2 > 1 + (43.3072875 - 43.19021)/64 > 1.0018,$$

contradiction.

Proposition N.1. *A whole D_{19} forest at the original farthest vertex cannot have an isolated native-five long head.*

Proof. The complete centre reduction leaves exactly (D19fivecentre). At its actual adjacent vertex the full local short-edge classification and old-wall native gap force one nonshort fork edge. Its actual endpoint violates the single ORIGINAL-vector head-paid norm budget, with the exact distance slack retained. No equality or classification of the outgoing fork opposites was required. □

N.12 A reusable centre test for the remaining finite heads

The centre geometry in the preceding argument is useful beyond the native-five head. We separate it from the early payment that changes with the head. Throughout this section the forest is the SAME whole D_{19}, with its signed last coordinate and actual centre seventeen. No field restriction is used.

Suppose an actual strong $K \in \{13, 14\}$ has supplied a legally funded vector w, using only initial coordinates one through six to modify a curved reference. Write L for its effective energy: if its legally available pressure is $x \cdot w \geq d$, then $L = 2d - \|w\|^2$. In particular L need not equal $\|w\|^2$. Assume its INDIVIDUAL P_K coefficient is $> 1/8$, its demand there is $f_K = \sqrt{K} - .58/\sqrt{2}$, and all initial changes have zero coordinate sum. Consequently $w \cdot P_K = f_K$.

For any actual pressure $x \cdot V \geq G$, put

$$B = w \cdot (V - P_K), \qquad R_* = K + \|V - P_K\|^2.$$

Consume $1/8$ of the original P_K coefficient and use the ONE vector $w + (V - P_K)/8$. Its pressure follows from the remaining positive anchor/foot decomposition, rather than subtraction of unrelated lower bounds. Expanding its complete square gives

$$\|x\|^2 \geq L + \frac{16(G - B) - R_* - 16f_K + K}{64}. \tag{D19generalpay}$$

If V is one through K, its residual vanishes there and has no overlap with the early zero-sum changes. The curved reference vanishes after K, so $B = 0$ and $R_* = V^2$. For a nonflat displacement we retain BOTH displayed terms.

Lemma N.2. *Suppose the reference above makes* $16(G - B) - R_* > H$ *sufficient to contradict* $\|x\|^2 < 1$. *Either of the following sets of bounds suffices:*

H *at most*	w_1 *at most*	$w \cdot P_2$ *at most*	$w \cdot P_3$ *at most*
42.278	.478	.956	1.291
41.539	.54	.956	1.291.

Then the actual centre opposite is impossible.

Proof. Both ACTUAL fork cosines are less than ten. A fork of cosine at least ten gives $V = P_{18} \pm e_{19}$, $G \geq \sqrt{171/11} > 3.94$, $R = 19$, hence $16G - R > 44.04$. A .50-strong centre gives $V = P_{17}$, $G > 3.7695$, $R = 17$, hence $16G - R > 43.312$. Both clear either threshold. We may classify the centre under its actual .50-weak hypothesis.

The complete positive finite-white list and all direct, affine and reciprocal calculations are explicitly proved in Section N.6: isolation; left-only three at one, two, fifteen, sixteen or four at one/sixteen; one arbitrary finite fork; fork pairs $(3,3), (3,4), (3,5)$; and left endpoint-three joined to one fork-three/four or both fork-threes. Each formula depends on its ACTUAL old contacts, not the head. Own-three at one fork has reciprocal cosine $(17 + 19\beta)/2 \geq 53/4$; own-zero at a $(3,4)$ pair has reciprocal cosine $17/2 + (19/4)\sqrt{2} > 10$. The positive inverse proves these are actual endpoints even when a centre and fork opposite coincide. The affine kernels there preserve every original wall term by term. Thus all own-zero/three/four/six/parallel, mixed-left and direct-ratio decisions remain valid EXCEPT own-zero with one finite fork, which was separately removed for native five.

We prove that case without any head-specific obstruction. Let its fork be f, the other fork h, and $c = Ap - \beta\omega_f$, $A^2 = (19/2)\beta^2 - 1$. Fork labels three, four and six have the positive actual nulls

$$2c + 4n_f + 6n_{17} + 5n_{16} + 4n_{15} + 3n_{14} + 2n_{13} + n_{12} + 3n_h,$$
$$\sqrt{2}c + 2n_f + 2n_{17} + n_{16} + n_h,$$
$$\sqrt{3}c + 2n_f + n_{17}.$$

They are future through c, nonpositive against every omitted wall, so compactness excludes them. At label five the ACTUAL H_4 on c, n_f, n_{17}, n_{16} has positive correction $\beta(3n_f/2 + n_{17} + n_{16}/2)$, residual $(5 - 3\varphi)/8$. The finite projected POINT gives

$$V = P_{15} + (e_{16} + e_{17} + e_{18} \pm e_{19})/2, \quad R = 16, \quad G > \sqrt{3 + 8\varphi} - \varphi^{-3} > 3.75.$$

Indeed $1.618 < \varphi < 1.619$ gives first root > 3.992 and $\varphi^{-3} < .238$. Thus $16G - R > 44$. Both signed orientations are included.

For every finite fork label at least seven put $v = \beta^{-2}$. The positive actual core

$$c + \beta^{-1}n_f + 2(\beta^{-1} - \beta)n_{17}$$

has zero products with c, n_f, negative products with n_{17}, n_{16}, n_h, and square $-(1-\beta^2)(4\beta^2-3)/\beta^2 < 0$. All unknown-wall products are nonpositive and its time is positive. Its future normalization gives

$$V = P_{16} + (3 - 2v)e_{17} + (v - 1)e_{18} \pm (v - 1)e_{19},$$
$$G = \sqrt{19 - 2v} - \sqrt{-8 + 14v - 6v^2}, \quad R = 27 - 16v + 6v^2, \quad 1 < v < 100/81.$$

The line $4.364 - .242v$ lies below the first root: its endpoint square checks are $4.122^2 < 17$ and $329.284^2 < 1339(81)$, and concavity puts the chord below the root. With $z = v - 7/6$, the other root is $\sqrt{1/6 - 6z^2} \le 1/\sqrt{6} - 3\sqrt{6}z^2$, and $R = 16.5 - 2z + 6z^2$. Hence

$$16G - R > 42.2746 - 1.872z + 111.57z^2 > 42.2646. \qquad \text{(D19forkcontinuous)}$$

The completion uses $1.872^2 < .04(111.57)$. This bound alone is too weak for the first threshold; we do not round it up. Native seven has $.064 < z < .067$, giving > 42.60; every native label at least nine has $z < -.033$, giving > 42.45. For native eight, $v = 4 - 2\sqrt{2}$ gives

$$G = \sqrt{11 + 4\sqrt{2}} - \sqrt{68\sqrt{2} - 96} > 3.6732094,$$
$$R = 107 - 64\sqrt{2} < 16.490333, \qquad 16G - R > 42.2810174 > 42.278.$$

The positive square seals are $4.0812809^2 < 11 + 4(1.414213562)$, $.4080715^2 > 68(1.414213563) - 96$. These cases exhaust all finite labels without arithmeticity or the native-five finite-fork obstruction.

The remaining point exceptions are already constructed, with all original-wall checks, in Sections N.5 and N.7. We use their point identities and scalar inequalities, not their head-paid con-

clusions. Their complete weaker bounds are

actual centre profile	$16G - R$ greater than
isolated five	45.56
isolated seven/eight	43.432
isolated finite at least nine	44.188
isolated dotted	42.328
near-left-three, own five	43.24
near-left-three, own seven/eight	44.538
one/two fork-three/four, own five	46.84
one fork-four, larger finite/dotted own	43.992
one fork-three, larger finite/dotted own	43.296
two fork-threes, own four	43.647.

These vectors are flat through fourteen, hence have no early head cross. For isolated dotted, for example, the actual core gives

$$G = \sqrt{17 - u/2} - \sqrt{u(1-u)/2} > 4.062 - .354 = 3.708, \quad R = 17 - u + u^2/2 \le 17, \quad 0 < u < 1.$$

Near-left own five uses the finite pair's projected POINT $AB/(B + \sqrt{1-\gamma^2})$, not the smaller unprojected $A - \sqrt{1-\gamma^2}$. The one-fork continuous intervals $.9 < \gamma < 1.42$ and $.9 < \gamma < 1.01$ retain the polynomial/core proofs of Section N.7, including every dotted parameter left by weakness. All corrections are positive actual-root combinations or positive finite inverse projections, so neither signed fork nor any unknown wall is discarded.

Only far-left-three remains. The finite point identities of Section N.8 are local to $v = \gamma\omega_{17} + \omega_1/2$, not to the head:

own	$V = P_{16} + aP_l + (e_{17} + e_{18})/2$	G greater than
5	$l = 3,\ a < 5/24$	4.1158
7, 8	$l = 2,\ a < 5/18$	3.963
finite ≥ 9	$l = 1,\ a < .533$	3.953.

They are the actual H_4 point, the negative future core $c + \gamma n_{17} + n_1/2$, and the positive pair projection. Their normalized losses are $< .097125, < .25, < .26$ from a first term $> \sqrt{17.75} > 4.213$. For these NONFLAT displacements retain

$$B = a\, w \cdot P_l, \qquad R_* = 16 + la^2 + 1/2.$$

Under the first coordinate row the three tests are greater than $44.91, 42.50, 42.387527$; the last is exactly the safe expression $16(3.953 - .533(.478)) - (16.5 + .533^2)$. Under the second coordinate row the first two bounds still apply, and the last becomes

$$16(3.953 - .533(.54)) - (16.5 + .533^2) = 41.858791 > 41.539.$$

Every finite far own is removed with its actual head cross.

All dotted $\gamma \ge 11/8$ are directly .50-strong because $(\gamma + 1/34)/\sqrt{9/17} > 1.928$. For $1 < \gamma \le 10/9$, put $a = (2\gamma)^{-1} \in [.45, .5)$. The actual core $c + \gamma^{-1} n_{17}$ gives

$$V = P_{16} + aP_1 + (1 - 2a^2)e_{17} + 2a^2 e_{18},$$
$$G = \sqrt{17 + 2a - a^2} - a\sqrt{2 - 8a^2}, \qquad B = aw_1, \quad R_* \le 16.75.$$

The function $G - aw_1$ increases even for $w_1 = .54$: its derivative is

$$\frac{1-a}{\sqrt{17+2a-a^2}} + \frac{16a^2-2}{\sqrt{2-8a^2}} - w_1 > 0.$$

At $a = .45$, the positive square checks $\sqrt{17.6975} > 4.2068, \sqrt{.38} < .6165$ give $G > 3.929375$. The two coordinate rows yield left sides > 42.6784 and > 42.232, respectively. Both suffice.

The ONLY centre left is the literal far-three helper

$$c = Ap - \gamma\omega_{17} - \omega_1/2, \qquad A^2 = 34\gamma^2 + 2\gamma - 1/2, \qquad 10/9 < \gamma < 11/8.$$

Sections N.9–N.11 prove a local geometric statement from precisely this interval. Their derivation uses neither a native-five head nor head energy: the actual adjacent vertex has forest $A_{17}+A_1+A_1$; its two short fork edges are classified with their complete finite-white costs, old n_{17} acuteness and the actual native gap. At least one is nonshort. Comparing its ACTUAL endpoint with the ORIGINAL farthest vertex retains $0 < \tau \le 1$ and gives a displacement flat through seventeen, with

$$G > 3.7736, \quad R < 17.0703125, \quad 16G - R > 43.3072875.$$

This clears both thresholds in (D19generalpay). The adjacent vertex was never declared farthest. One original x pays every pressure, including all early and nonflat crosses, and the full centre list is exhausted. □

N.13 Changing the first knot without losing a weak-six omission

Put $f_i = \sqrt{i} - .58/\sqrt{2}$, $\epsilon = .03/\sqrt{2}$, and

$$F = f_5 - \epsilon = \sqrt{5} - .61/\sqrt{2}, \qquad b = F/5.$$

The independent OLD theorem supplies the INDIVIDUAL floor $S_5 > F$. Useful exact scalar bounds are

$$1.80473 < F < 1.804733, \quad .360946 < b < .3609466, \quad f_5/5 < .36519,$$
$$E_{13} > .9056051, \quad E_{14} > .9241299, \quad f_6 > 2.0393, \quad f_{13} < 3.1955, \quad f_{14} < 3.3316.$$

The E_K are the already proved curved-reference scalar values in (B19E17), not numerical enumeration outputs.

Start with the full .58 reference and lower its knot at five by ϵ, leaving the actual strong later knots. The energy change is $-2\epsilon\beta_5 + (6/5)\epsilon^2$, where

$$\beta_5 = (6/5)\sqrt{5} - \sqrt{6} - .116/\sqrt{2} < .15177.$$

The bounds $\sqrt{5} < 2.236068, \sqrt{6} > 2.449489, 1/\sqrt{2} > .7071$ give this seal, and $\epsilon < .021214$ gives total loss

$$2\epsilon\beta_5 - (6/5)\epsilon^2 < .042428(.15177) - .00054 = .00589929756 < .0059.$$

The new first-five mean is b; the first subsequent slope is at most $\sqrt{6} - \sqrt{5} + \epsilon < .234637$. Thus its original P_5 coefficient is $> .126$. If six is genuinely strong it is $> .126309$, while the INDIVIDUAL P_6 coefficient is $> .0375$, as used below.

Actual .58-weak knots at six through fourteen are pure far-three helpers and nonadjacent. Their ordinary disjoint averaging losses sum to $< .000408$. If six is weak, the interval five–seven changes its loss to $(2\sqrt{6} - \sqrt{5} - \sqrt{7} + \epsilon)^2/2$. Its EXTRA over the ordinary omission is

$$\epsilon(2\sqrt{6} - \sqrt{5} - \sqrt{7}) + \epsilon^2/2 < .021214(.0172) + .000225 = .0005898808.$$

Only this interval touches the changed knot. Hence the COMPLETE omission loss, including weak six, is $< .0010$. The mixed reference therefore has

$$q_{13} > .8987051, \qquad q_{14} > .9172299, \qquad \text{terminal price} > 1/8. \tag{D19mixedenergy}$$

If six is strong all omissions begin at seven; the sharper complete bound is

$$1/6912 + 1/16384 + 1/32000 + 1/55296 < .000256.$$

Omitting later knots only increases the individual capacities just identified. This calculation makes no assertion that five or six is new-strong unless its separate geometric argument proves that fact.

N.14 Native seven: exclude a weak six before making the reference

Let $g_0 = \cos(\pi/7)$, $S = g_0^2$, and

$$c_H = A_H p - g_0\omega_1, \qquad A_H^2 = 2S - 1, \qquad u = g_0^{-2}.$$

The polynomial $8g_0^3 - 4g_0^2 - 4g_0 + 1 = 0$, increasing above .9, gives

$$.90096886 < g_0 < .90096888, \quad 1.2319140 < u < 1.2319143, \quad .8116 < S < .8119, \quad A_H < .7897.$$

Expand at .901: value, derivative and quadratic coefficient are $.000257608, 8.275224, 17.624$. Increments $-.00003114, -.00003112$ give opposite signs, and reciprocal-square multiplications give the u interval. These are native-angle scalars, not a field census.

The positive core of Section M.2, $c_H + g_0^{-1}n_1 + 2(g_0^{-1} - g_0)n_2$, gives

$$\begin{aligned} Z &= (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad \sum Z_i = 1, \\ x \cdot Z \geq H &= \sqrt{1 - u/2} - \sqrt{(7u - 3u^2 - 4)/2} > .4318779, \\ R &= Z^2 = 3 - 4u + 3u^2/2 < .348763. \end{aligned}$$

Its selected products are zero or negative, its square negative, and positive actual-root coefficients check EVERY other wall. The seals use first root $> .6197119$, second $< .187834$, and $R(1.2319140) = .348762155094$.

If six were .58-weak, its actual bulk profile would be

$$c_6 = Ap - \gamma\omega_6 - \omega_1/2, \quad A^2 = 12\gamma^2 + 2\gamma - 1/2, \quad \varphi/2 \leq \gamma < 6/5.$$

It is distinct from c_H by its contact at six. Its mutual magnitude is $M_6 = A_H A - g_0(2\gamma + 1)$. The squared positive-term difference $(20S - 12)\gamma^2 - 2\gamma + 1/2 - 2S$ is positive and increasing: at $\gamma = .809$ it is greater than

$$4.232(.809)^2 - 1.618 - 1.1238 = .027963592.$$

Also $(12\gamma + 1)^2 = 12A^2 + 7$ proves M_6 increases, and

$$M_6 < .7897(4.38) - .9009(3.4) = .395826 < 1/2.$$

The actual native alphabet has no magnitude strictly between zero and one half. This excludes ALL finite and dotted weak-six parameters. Thus six is genuinely strong before any early reference is built; the complete omission loss now starts at seven and is $< .000256$.

A weak-five far helper has $A^2 = 10\gamma^2 + 2\gamma - 1/2$ and $M_5 = A_H A - g_0(2\gamma + 1)$. The derivative is positive, by the squared comparison $(16S - 10)A^2 + 6A_H^2 > 0$; and

$$M_5 < .7897(4.038) - .9009(3.4) = .1257486 < 1/2.$$

Acuteness and the native gap force $M_5 = 0$, hence $(16S-10)\gamma^2-2\gamma+1/2-2S = 0$. This polynomial increases above .809, is negative at one, and positive at 1.04, so $1 < \gamma < 1.04$. The actual core and weak UPPER bound at five, together with $x_5 > x_6$, give

$$x_1 > \sqrt{2}\{A - \sqrt{1 - \gamma^{-2}} - \gamma(\sqrt{10} - .58)\}.$$

The braces decrease: $A' < 13/4$, the radical derivative is greater than one, and $\sqrt{10} - .58 > 2.5$. At 1.04, $A > 3.52$, the radical is $< .275$, and $\sqrt{10} - .58 < 2.58228$, yielding $x_1 > .79114420896 > .79$. The legal reference through $(1, .79), (6, f_6), (K, f_K)$ has energy

$$.79^2 + (2.0393 - .79)^2/5 + 1.156^2/7 > 1.127.$$

Its slopes decrease, and $K = 14$ only raises the final chord energy. This contradicts the SAME norm. Thus weak five can ONLY be the actual isolated dotted cap; it has not been silently discarded.

For that cap put $v = \gamma_5^{-2} \in (0, 1)$. Its actual local first-six-root pressure, proved in the head reduction, is

$$Z_5 = P_5 + (v/2)(e_6 - e_5), \qquad x \cdot Z_5 \geq T_5 > 1.8185.$$

Weakness forces $v > .499$. For $v \leq .499$ the same core gives loss at most

$$\frac{v}{\sqrt{10} + \sqrt{10 - v}} + \sqrt{v(1 - v)} < .16015(.499) + .4999991 = .57991395 < .58.$$

Both upper terms increase there; the denominator seal uses $\sqrt{10} > 3.1622, \sqrt{9.5} > 3.0822$, and $.4999991^2 > .499(.501)$. Such a cap would be strong.

N.15 Native seven: pay the head and dotted-five foot jointly

If five is .58-strong, take the ordinary curved reference, with first-five mean $b = f_5/5 < .36519$, omission loss $< .000256$, P_5 price $> .15$ and P_K price $> 1/8$. Use $w = w_0 + .45(Z - P_5/5)$, paying .09 at five. The residual is orthogonal to w_0, and its full effective gain is

$$.9(H - b) - .2025(R - .2) > .0298946025.$$

Thus $L_{13} > .9352437025, L_{14} > .9537685025$.

If five is the retained weak isolated dotted cap, use the OLD floor F, the ACTUAL strong f_6, and every later actual strong knot. Section N.13 gives loss $< .0059 + .000256$, individual P_5 price $> .126309$, and P_6 price $> .0375$. Pay .094 at five and set $w_H = w_0 + .47(Z - P_5/5)$. The effective head gain $.94(H - F/5) - .2209(R - .2) > .0338136753$ gives intermediate energies $.9332627753, .9517875753$. The REMAINING P_5 price is $> .0323$, and exactly

$$w_H \cdot P_5 = F, \quad w_H \cdot P_6 = f_6, \quad T_5 - w_H \cdot Z_5 > .013767 + .01615v.$$

Project the second foot onto ONLY the legitimate P_5, P_6 space:

$$\Pi Z_5 = (1 - 3v/5)P_5 + (v/2)P_6, \qquad Q = \|Z_5 - \Pi Z_5\|^2 = v^2/5.$$

With $\nu = .046$, its INDIVIDUAL costs are $\nu(1-3v/5) < .0322276 < .0323$ and $\nu v/2 < .023 < .0375$. No head-induced P_3 coefficient is spent as an old anchor. The residual is $v/10$ on coordinates one through four, $-2v/5$ at five, zero at six. Its head cross is $w_H \cdot (Z_5 - \Pi Z_5) = .047v$, not zero. The ONE completed square includes it, giving extra effective gain

$$\begin{aligned} 2\nu(T_5 - w_H \cdot Z_5) - \nu^2 Q &> .001266564 + .0014858v - .0004232v^2 \\ &> .0019026009768 > .0019026. \end{aligned}$$

The polynomial increases on $[.499, 1]$. Final energies are therefore $L_{13} > .9351653753, L_{14} > .9536901753$. The foot is used once, not as an independent norm.

Both final references satisfy

$$w_1 < .453, \quad w \cdot P_2 < .906, \quad w \cdot P_3 < 1.291, \qquad \text{remaining } P_K > 1/8.$$

In the dotted branch the last residual adds at most .0046 to each of the first four coordinates; the preceding bounds were $.4475, .895, 1.2709$. The strong-five branch has $.4481, .8962, 1.2756$. All changes have zero sum within the first six. The exact sufficient thresholds in (D19generalpay), for $K = 13, 14$, are

strong-five	42.27240304	42.26441584
dotted-five	42.2774159808	42.2694287808.

The common 42.278 and first coordinate row of Lemma N.2 apply. This excludes the ENTIRE native-seven head branch in both actual states at five.

N.16 Native eight: the strong and weak six cases give different prices

Let $s = \sqrt{2}$, $g_0 = \cos(\pi/8)$, and

$$c_H = A_H p - g_0 \omega_1, \quad A_H^2 = s/2, \quad u = g_0^{-2} = 4 - 2s.$$

The same positive actual head core gives

$$\begin{aligned} Z &= (s-1)(e_1 + e_2) + (3 - 2s)e_3, \quad \sum Z_i = 1, \\ x \cdot Z &\geq H = \sqrt{s-1} - \sqrt{17s - 24} > .43955, \\ R &= 23 - 16s < .372584. \end{aligned}$$

All-wall feasibility and the future negative square follow from the local core proof in Section M.2. The numerical seals use $1.414213562 < s < 1.414213563$, with the first root $> .64359$ and second $< .20404$. Use ONLY the independent OLD five floor F, and split six by its ACTUAL .58 status.

If six is strong, the mixed curved reference has omissions beginning at seven, total $< .000256$, and P_5 price $> .126$. Its energies are $> .8994491, .9179739$ for $K = 13, 14$. Pay .09 at five and insert $.45(Z - P_5/5)$. The orthogonal residual gain is

$$.9(.43955 - .3609466) - .2025(.372584 - .2) = .0357948.$$

The final reference therefore satisfies

$$\begin{aligned} L_{13} > .9352439, \quad L_{14} > .9537687, \\ w_1 < .45735, \quad w \cdot P_2 < .9147, \quad w \cdot P_3 < 1.2629. \end{aligned} \tag{D19eightstrong}$$

Its remaining P_K price is $> 1/8$.

If six is weak, its ACTUAL wall is

$$c_6 = Ap - \gamma\omega_6 - \omega_1/2, \quad A^2 = 12\gamma^2 + 2\gamma - 1/2, \quad \varphi/2 \le \gamma < 6/5.$$

Its distinctness from the head is explicit in its six contact. The mutual magnitude $M = A_H A - g_0(2\gamma{+}1)$ is positive and increasing: the squared positive-term difference is $(5s{-}2)\gamma^2{-}2\gamma{-}(1{+}s)/2$, positive already at .8; the derivative comparison is $(5s-2)A^2 + 7s/2 > 0$. At 6/5,

$$M < .8409(4.38) - .9238(3.4) = .542222 < .55.$$

The actual native gap therefore forces $M = 1/2$. Since $M(1) < 1/2$ and $M(1.165) > .502559707$, we have $1 < \gamma < 1.165$.

The actual core and the weak UPPER bound $S_6 \le f_6$, with $x_6 > x_7$, now give

$$x_1 > \sqrt{2}\{A - \sqrt{1-\gamma^{-2}} - \gamma(\sqrt{12} - .58)\}.$$

The braces decrease: $A' < 15/4$, the radical derivative is > 1, and $\sqrt{12} - .58 > 2.8$. At 1.165, use $A > 4.2563$, the radical $< .5131$, and $\sqrt{12} - .58 < 2.88411$, obtaining $x_1 > .5419382 > .54$. The actual helper nonadjacency makes seven strong. No strong-six floor was inferred.

Take the legal reference through $(1, .54), (5, F), (7, f_7)$ and the later actual strong knots. Its initial slopes satisfy

$$m = (F - .54)/4 > .3161825, \quad n = (f_7 - F)/2 < .215485, \quad d_7 < .183.$$

Thus its INDIVIDUAL prices at P_1, P_5 are $> .2238, > .1006$. Relative to the mixed reference with six omitted, the first-coordinate gain is $(5/4)(.54 - b)^2 > .04005125$. Keeping the enlarged total omission budget $< .0010$, its energies are

$$q_{13} > .93875635, \quad q_{14} > .95728115.$$

This accounts for weak six before any head pressure is inserted.

The correct projection is onto BOTH available anchors P_1, P_5:

$$\Pi Z = (Z_1 - t)P_1 + tP_5, \quad t = (2-s)/4, \quad Q = \|Z - \Pi Z\|^2 = 37/2 - 13s < .115224.$$

Use $\mu = .3$. Its separate costs are $\mu(Z_1 - t) < .080331 < .2238$ and $\mu t < .043935 < .1006$. The old reference is constant only on coordinates two through five; we do not project onto a fictitious first-five constant block. Its actual pairing is

$$w_0 \cdot Z = .54(s-1) + m(2-s) < .409,$$

so the full head gain is $> .6(.03055) - .09(.115224) = .00795984$. The final weak-six reference satisfies

$$\begin{aligned} L_{13} > .94671619, \quad L_{14} > .96524099, \\ w_1 = .54, \quad w \cdot P_2 < .9366, \quad w \cdot P_3 < 1.2603. \end{aligned} \tag{D19eightweak}$$

It retains the P_K coefficient $> 1/8$.

In the strong-six case the exact sufficient thresholds in (D19generalpay) are 42.2723904, 42.2644032, so $H = 42.273$ suffices. In the weak-six case they are $41.53816384, 41.53017664$, so $H = 41.539$ suffices. The first and second coordinate rows of Lemma N.2, respectively, apply. In particular the weak-six case keeps its larger first coordinate in every nonflat centre pressure; it never borrows the strong-six bound. This excludes the ENTIRE native-eight head branch without a new-strong-five assumption or an independent head energy.

N.17 Native nine: one mixed reference covers every early omission

Put $g = \cos(\pi/9)$, $c_H = A_H p - g\omega_1$, $A_H^2 = 2g^2 - 1$, and $u = g^{-2}$. The polynomial $8g^3 - 6g - 1 = 0$, increasing above .9, gives

$$.9396 < g < .9398, \qquad 1.1322 < u < 1.1327.$$

The two exact cubic evaluations have the following numerators and denominators:

argument	numerator	denominator
2349/2500	−21983608	15625000000
4699/5000	203952792	125000000000.

The intended root is above .9 by the elementary cosine estimate. The reciprocal checks

$$.9398^2(1.1322) = .999986258088 < 1,$$
$$.9396^2(1.1327) = 1.000002110832 > 1$$

give the second interval.

The positive head core again supplies

$$Z = (1 - u/2)(e_1 + e_2) + (u - 1)e_3, \quad H = \sqrt{1 - u/2} - \sqrt{(7u - 3u^2 - 4)/2} > .45865,$$
$$R = 3 - 4u + 3u^2/2 < .3941.$$

Here the first root is $> .6585$, the second $< .19985$; the second radicand is increasing and R decreasing on the interval. Their endpoint values $.039936065, .39401526$ give the stated seals. The normalized core is future and satisfies EVERY original wall by the same actual positive-combination proof as before.

Use the mixed reference of Section N.13, including the enlarged weak-six omission. It uses only the OLD five floor, has $q_{13} > .8987051, q_{14} > .9172299$, individual P_5 price $> .126$, and terminal price $> 1/8$. Pay .1 at five and insert $.5(Z - P_5/5)$. The full effective gain is

$$(H - b) - .25(R - .2) > .049125.$$

Hence

$$L_{13} > .9478301, \quad L_{14} > .9663549,$$
$$w_1 = w_2 < .478, \quad w \cdot P_2 < .956, \quad w \cdot P_3 < 1.284.$$

The coordinate formulas are $w_1 = b + .4 - u/4$ and $w \cdot P_3 = 3b + .2$; all other omissions retain the same constant first-five block before this head change. The terminal coefficient is unchanged. The exact sufficient thresholds are $41.4668736, 41.4588864$; the common $H = 41.467$ satisfies the first row of Lemma N.2. Lemma N.2 therefore applies in its first row and excludes the actual centre opposite. Its proof includes the nonflat far-helper cases and the original-p adjacent-vertex conclusion, using the unchanged terminal coefficient $> 1/8$. This excludes every native-nine head.

N.18 Every finite long-head label at least ten

We use OLD loss .61 throughout this branch. Put $f_i = \sqrt{i} - .61/\sqrt{2}$, $b = f_5/5 < .361$. The independent OLD theorem gives $S_5 > f_5$. For an isolated head with $g = \cos(\pi/m)$, $m \geq 10$, put

$$c_H = A_H p - g\omega_1, \quad A_H^2 = 2g^2 - 1, \quad u = g^{-2}, \quad 1 < u < 1.106.$$

The positive core $c_H + g^{-1}n_1 + 2(g^{-1} - g)n_2$ has zero products with the cap and one, negative products with two/three, positive time, and negative square $-[7-4g^2-3g^{-2}]$. Every other original-wall product is nonpositive term by term. Its normalized pressure is

$$x \cdot Z \geq h(u) = \{\sqrt{2-u} - \sqrt{P(u)}\}/\sqrt{2}, \quad P(u) = (u-1)(4-3u),$$
$$Z = (1-u/2)(e_1+e_2) + (u-1)e_3, \quad \sum Z_i = 1, \quad R = 3 - 4u + 3u^2/2.$$

For any actual OLD reference constant b through five insert $.6Y_H$, where $Y_H = Z - P_5/5$. Its INDIVIDUAL five price is .12, smaller than the available $b - d_5 > .3609 - .214 > .146$. Orthogonality gives effective gain greater than

$$F(u) = 1.2(h(u) - .361) - .36(R - .2). \tag{D19largegain}$$

The function F decreases on $1 < u \leq 1.106$. Here $P' \geq .364, P < .073$, so the two negative terms in

$$h' = -\frac{1}{2\sqrt{2}\sqrt{2-u}} - \frac{P'}{2\sqrt{2}\sqrt{P}}$$

have magnitudes $> .35, > .45$. Since $-R' < 1$, $F' < -.96 + .36 < 0$. At $u = 1.106$, positive squares give $\sqrt{2-u} > .9455, \sqrt{P} < .26888$: $P = .072292 < .26888^2$, and $.4784(1.41422) = .676562848 < .9455 - .26888$. Thus $h > .4784$, $R - .2 = .210854$, and

$$F(u) > .06497256. \tag{D19largecommongain}$$

For every $m \geq 11$ we have $u < 1.087$. Since $\pi/11 < 2/7$, the alternating cosine lower bound gives

$$\cos(\pi/11) > 1 - 2/49 + 2/7203 - 4/5294205 > .9593.$$

The polynomial decreases on $[0, 2/7]$; $.9593^2(1.087) = 1.00031880463 > 1$, and cosine monotonicity covers all larger labels. At 1.087, $\sqrt{2-u} > .9555, \sqrt{P} < .25357$, because $P = .064293 < .25357^2$ and $.496(1.415) = .70184 < .9555 - .25357$. Hence $h > .496$, $R - .2 = .2243535$, and

$$F(u) > .08123274 \quad (m \geq 11). \tag{D19largerfinitegain}$$

Late coverage under this actual head. The OLD rank-valid bulk supplies an independent strong $K = 11$ or twelve, with ordinary curved energies $q_{11} > .8472225, q_{12} > .8689525$, endpoint price $> 1/7$. The ONE head payment gives

$$L_{11} > .91219506, \quad L_{12} > .93392506.$$

We establish late coverage with these references, not with a late conclusion from an earlier assumed-weak-five branch.

The complete rank19 finite-white classification and all-wall projection identities at fifteen/sixteen are printed in Section N.2. Before its numerical comparisons, those local proofs use only the retained $A_{14} + D_4$, respectively $A_{15} + D_3$, and actual released root. Its weak-five assumption supplied the earlier reference, not these identities. We now apply them with the new head-paid reference. For clarity the full coverage is as follows.

Isolated and left-only rows are .61-strong except pure far-left three helpers. The four isolated loss bounds at $q^2 \geq 30$, the near/next-to-end ratios and actual own-zero/three/four/six/parallel principals in that section are unchanged; native-five uses its ACTUAL long $i + 1$ dual, at most

seventeen. At right D_4, long-leaf-three has residual $1/2$; own-zero is the affine double fork, own-three the triangle, larger owns have ratio at least $1+1/\sqrt{2}$. The D_3 middle-three has the same decisions. End-four/own-zero has the printed weighted affine kernel, every positive own is strong, and end-five is finite H_4, directly strong even at own-zero. Both-sided bridges are endpoint-three; own-zero has the long cycle or middle star, while positive own has numerator $> 3/4$, residual at most $5/32$. Two right contacts make a cycle; there is no foreign component or exceptional full-rank bridge.

For right fork-three at fifteen or end-three at sixteen, own-zero has actual affine E_8 arms five/two/one. Own-three gives the FIVE-cycle at fifteen and FOUR-cycle at sixteen; four gives the affine star with squared weights $1/2, 1/4, 1/4$; six/parallel have their actual nulls. At D_4, all remaining finite owns at least seven and dotted owns are directly strong, with ratio $> 1.626 > 25/16$. At D_3, weakness forces a remaining finite $.9 < \gamma < 1$; dotted own is directly strong. There are just two point exceptions:

row	actual construction	$G >$	$V^2 <$
own five/right three at fifteen or sixteen	positive LEFT H_4 point	3.75	16
larger finite own/right end-three at sixteen	$c+\gamma^{-1}n_{16}+(2\gamma)^{-1}n_{15}$	3.662	16.

Section N.2 prints their full point/core formulas, finite inverse positivity and EVERY-wall check. They are flat through at least twelve, so have zero head-residual cross. With the ACTUAL separate endpoint data $f_{11} < 2.886$, $f_{12} < 3.034$, the legal $1/7$ payment gives

$$.91219506 + 2(3.662 - 2.886)/7 - 5/49 > 1.0318,$$
$$.93392506 + 2(3.662 - 3.034)/7 - 4/49 > 1.0317.$$

The H_4 demand is larger. Every old-weak late survivor is therefore a literal pure far-left-three helper, with own five, finite at least seven or dotted, parameter $< 6/5$.

Two adjacent such helpers have mutual $> -.44 > -1/2$, hence are orthogonal by the actual native gap. For fifteen/sixteen the retained A_{14} and ENTIRE right D_3 give $I - (14/30)\mathbf{1}\mathbf{1}^T$, positive with positive inverse; fourteen/fifteen retain $A_{13}+D_4$, cost $13/28$. The actual mixed-point calculation printed in Section N.2 has prefix inverse $2\min(i,j)$, weighted loss $< 7/16 + 1/(4\sqrt{10}) < .61$, and preserves BOTH forks and all unknown walls. Weak helpers are thus nonadjacent across the bulk/late join; an ACTUAL $J \in \{15, 16\}$ is OLD strong under the finite-head hypothesis.

Labels at least eleven. Use the ordinary OLD reference ending at that actual J. At most five nonadjacent omissions cost $< 5/3200$, giving energy

$$> E_{15} - 5/3200 > .92564 - .0015625 = .9240775.$$

Its five price is still $> .146$. Insert the head foot ONCE. Equation (D19largerfinitegain) yields

$$\|x\|^2 > .9240775 + .08123274 = 1.00531024 > 1.$$

The earlier references established coverage; their energies are not added to this final one.

Native ten. Here $u = 2 - (2/5)\sqrt{5} \in (1.1, 1.106)$, since $447/200 < \sqrt{5} < 9/4$. If sixteen is OLD strong, its ordinary reference has $q_{16} > E_{16} - 5/3200 > .9401975$; the common head gain gives $\|x\|^2 > 1.00517006$. Otherwise sixteen is the actual pure weak helper and fifteen strong. Take the

reference ending fifteen and insert the single head correction. It has effective energy $> .98905006$, remaining P_{15} coefficient $> d_{14} > 8/61 > .13$, and ACTUAL coordinate bounds

$$w_1 < .511, \quad w \cdot P_2 < 1.022, \quad w \cdot P_3 < 1.323.$$

They follow from $b + .6(.8 - u/2), 2b + .6(1.6 - u), 3b + .24$.

Write the actual weak helper

$$c = Ap - \gamma\omega_{16} - \omega_1/2, \qquad A^2 = 32\gamma^2 + 2\gamma - 1/2.$$

Its finite feet and negative cores give $x \cdot V \geq G$, $V = P_{15} + Y$, with COMPLETE data

own	Y	$G >$	$Y^2 <$	$w \cdot Y <$
5	$aP_3 + (e_{16} + e_{17})/2,\ a < 5/24$	3.9954	.631	.275625
$7, 8$	$aP_2 + (e_{16} + e_{17})/2,\ a < 5/18$	3.8426	.655	$5.11/18$
finite ≥ 9	$aP_1 + (e_{16} + e_{17})/2,\ a < .533$	3.8326	.785	.272363.

These are respectively the actual positive H_4 point, future core $c + \gamma n_{16} + n_1/2$, and finite pair point. Their first normalized term is at least $\sqrt{16.75} > 4.0926$; the losses are respectively $< .097125, .25, .26$. For the last, $\gamma > .939$ and $.939^2(1.1352) = 1.0009296792 > 1$. The displayed norms and overlaps follow by squaring Y and using the ACTUAL coordinate bounds, retaining every head/helper cross. All finite projections are points with positive coefficients in actual normals, so EVERY original wall is preserved.

For the entire dotted helper range, $a = (2\gamma)^{-1} \in (5/12, 1/2)$, and

$$Y = aP_1 + (1 - 2a^2)e_{16} + 2a^2 e_{17}, \quad Y^2 \leq .75,$$
$$G = \sqrt{16 + 2a - a^2} - a\sqrt{2 - 8a^2}, \quad G - w \cdot Y > 3.5429.$$

Indeed $.75 - Y^2 = (.25 - a^2)(8a^2 - 1) \geq 0$. The function $G - .511a$ increases, since the second term in G' is $(16a^2 - 2)/\sqrt{2 - 8a^2} > 7/9 > .511$, while the first is positive. At $a = 5/12$,

$$G - .511a = \sqrt{2399}/12 - 5\sqrt{22}/72 - .511(5/12) > 4.0816 - .325724 - .212917 = 3.542959.$$

Here $4.0816^2 < 2399/144$ and $\sqrt{22} < 4.69042$. This bounds the ACTUAL overlap, without a separately invented pressure on Y.

Pay $\nu = .13$ from the INDIVIDUAL P_{15} coefficient and use the ONE vector $w + \nu Y$. Its full square, including the already paid head, gives

$$\|x\|^2 > .98905006 + .26(G - f_{15} - w \cdot Y) - .0169Y^2.$$

Since $f_{15} < 3.442$, the finite/dotted deficits are respectively greater than $.2777, .1167, .1182, .1009$; their extra gains exceed $.0615381, .0192725, .0174655, .013559$. Every case satisfies $\|x\|^2 > .98905006 + .013559 = 1.00260906 > 1$. Own-three/four/six and parallel were already excluded by actual affine/null principals, so this helper list is complete. Native ten and every finite $m \geq 10$ are excluded, without borrowing late floors from a different hypothesis.

N.19 A dotted head: one exact parabola on its whole parameter interval

Let the isolated actual head be

$$c_H = A_H p - g\omega_1, \quad g > 1, \quad A_H^2 = 2g^2 - 1, \quad u = g^{-2} \in (0, 1).$$

The ACTUAL positive core $c_H + g^{-1}n_1$ has zero cap product, negative products with one/two, negative square $u-1$, positive time. Positivity of its coefficients checks EVERY other original wall. Its future normalization gives

$$x \cdot Z \geq H = \sqrt{1-u/2} - \sqrt{u(1-u)/2},$$
$$Z = (1-u/2)e_1 + (u/2)e_2, \quad \sum Z_i = 1, \qquad R = 1 - u + u^2/2.$$

There is no second-neighbour correction of either sign. We treat the WHOLE dotted interval, not a selected compact subinterval.

Put $s = \sqrt{6}$, $z = u - 2/3$, and $q = s/3 - sz/8 - 9z^2/100$. Here $-2/3 < z < 1/3$, and $q \geq s/4 - .04 > 0$. The exact identity

$$1 - u/2 - q^2 = z^2\{3s/50 - 3/32 - (9s/400)z - 81z^2/10000\}$$

is nonnegative: its bracket is at least $.045s - .09735 > 0$, using $|z| \leq 2/3$, $s > 2.4494$. Thus the first root is at least q. For $r = \sqrt{u(1-u)/2}$, the elementary square $(r - 1/3)^2 \geq 0$ gives $r \leq 1/3 - z/4 - 3z^2/4$. Consequently

$$H \geq (s-1)/3 + (2-s)z/8 + .66z^2, \qquad R = 5/9 - z/3 + z^2/2. \tag{D19headparabola}$$

This is a global square comparison, not a sampled minimum.

Return to mixed floors $f_i = \sqrt{i} - .58/\sqrt{2}$, $F = \sqrt{5} - .61/\sqrt{2}$, $b = F/5$. Section N.13 includes ALL actual omissions, including enlarged weak six, and supplies

$$q_{13} > .8987051, \quad q_{14} > .9172299, \quad P_5 \text{ price} > .126, \quad P_K \text{ price} > 1/8.$$

Weak isolated dotted five is retained through its OLD individual floor; neither five nor six has been upgraded to .58-strong. Use ONE vector $w = w_0 + .34(Z - P_5/5)$, consuming only .068 of the actual five coefficient. The effective head gain is

$$J = .68(H - b) - .1156(R - .2). \tag{D19dottedgain}$$

Using (D19headparabola), $b < .361$, and $2.4494897 < s < 2.4494898$, the constant/linear/quadratic terms give

$$J > .0419687 + A_J z + .391z^2, \quad 0 < A_J < .0004.$$

The exact linear coefficient is $A_J = .208533333\ldots - .085s$. Completing the square gives

$$J > .04196, \tag{D19dotteduniform}$$

since $.0004^2 < .00000011(4(.391))$. The ACTUAL coordinates retain their correlation with u:

$$w_1 = b + .34(.8 - u/2), \quad .462946 < w_1 < .633,$$
$$w \cdot P_2 = 2b + .204 < .926, \qquad w \cdot P_3 = 3b + .136 < 1.219. \tag{D19dottedcoordinates}$$

Every change has zero sum in the first five and preserves the INDIVIDUAL P_K coefficient. No artificial old P_2/P_3 pressure is paid.

N.20 The dotted-head centre: retain correlation in the nonflat row

The general square (D19generalpay) applies with the actual $L_K > q_K + J$. The two threshold upper bounds are

$$44.6108736 - 64J, \qquad 44.6028864 - 64J.$$

By (D19dotteduniform), the common sufficient test is

$$16(G - B) - R_* > 41.926. \tag{D19dottedtest}$$

An actual fork of cosine at least ten has test > 44.04; a .50-strong centre has test > 43.312. Thus both fork cosines are less than ten and the centre is .50-weak.

The complete finite-white classification, actual affine/nulls, reciprocal endpoint computations and flat point proofs of Lemma N.2 apply before its nonflat coordinate tests. They use the head only through the paid threshold. Every flat point bound is at least 42.328. Own-zero/single-fork with finite label at least seven has the COMPLETE continuous estimate (D19forkcontinuous), > 42.2646, sufficient now without discrete refinements; three/four/six have the actual nulls and five the positive H_4 point. Thus every isolated, near, fork-only, mixed bridge and reciprocal exception is excluded. The actual signed forks and unknown-wall checks are those printed in that shared proof; no native-five-only fork theorem is used.

The remaining far-left-three wall has

$$c = Ap - \gamma\omega_{17} - \omega_1/2, \quad A^2 = 34\gamma^2 + 2\gamma - 1/2.$$

Its complete finite feet are those constructed in the shared proof:

$$V = P_{16} + aP_l + (e_{17} + e_{18})/2, \quad B = a\,w \cdot P_l, \quad R_* = 16 + la^2 + 1/2.$$

Own-five and seven/eight have respectively $(l, a, G) = (3, < 5/24, > 4.1158)$ and $(2, < 5/18, > 3.963)$. The ACTUAL coordinate bounds (D19dottedcoordinates) give tests $> 45.15, 42.638$, sufficient in (D19dottedtest).

For finite own at least nine, $l = 1, a < .533, G > 3.953$. A separately maximized head coordinate would be unnecessarily weak. Retain the ACTUAL correlation: its generalized test is greater than

$$S_9 = 16(3.953 - .533w_1) - (16.5 + .533^2) = 46.463911 - 8.528w_1.$$

Subtract the LARGER unrounded threshold above and substitute the actual J, w_1, using only $b < .361$:

$$\begin{aligned} S_9 - (44.6108736 - 64J) &> D(u), \\ D(u) &= 43.52H - 7.3984R + 1.44976u - 17.7762266. \end{aligned} \tag{D19correlated}$$

This is the complete nonzero head/centre cross calculation. The global parabola (D19headparabola) gives

$$D(u) > .1073 + A_D z + 25.024z^2, \quad A_D = 14.795893333\ldots - 5.44s, \quad 0 < A_D < 1.471.$$

The same two positive square bounds for $\sqrt{6}$ give the constant and linear seals; the quadratic coefficient is exactly 25.024. Therefore

$$D(u) > .1073 - \frac{A_D^2}{4(25.024)} > .1073 - .022 = .0853.$$

The exact last check is $1.471^2 = 2.163841 < .022(100.096) = 2.202112$. This ENTIRE finite-at-least-nine row violates the actual completed square; it need not clear the coarser standalone test41.926.

For a dotted far helper with $1 < \gamma \le 10/9$, put $a = (2\gamma)^{-1} \in [.45, .5)$. Its same actual core gives

$$V = P_{16} + aP_1 + (1 - 2a^2)e_{17} + 2a^2 e_{18},$$
$$G = \sqrt{17 + 2a - a^2} - a\sqrt{2 - 8a^2}, \quad B = aw_1, \quad R_* \le 16.75.$$

Since $w_1 < .633$, $G - aw_1$ increases on the whole interval: the first derivative term is positive, and $(16a^2 - 2)/\sqrt{2 - 8a^2} - w_1 \ge 1.24/.617 - .633 > 0$. At .45, $G > 3.929375$, as proved in the shared centre lemma. The generalized test is thus greater than

$$S_{\text{dot}} = 46.12 - 7.2w_1.$$

Pointwise it dominates S_9, since

$$S_{\text{dot}} - S_9 = -.343911 + 1.328w_1 > -.343911 + 1.328(.462946) > 0.$$

The correlated contradiction therefore covers every small dotted far helper without substituting unrelated maxima.

Every $\gamma \ge 11/8$ is directly .50-strong. The ONLY centre left is the literal interval $10/9 < \gamma < 11/8$. The head-independent actual-q proof in Sections N.9–N.11 gives a flat-through-seventeen pressure with $16G - R > 43.3072875 > 41.926$. Its 1/8 terminal payment is still available and its residual has zero cross with the head-paid support. The original slack $0 < \tau \le 1$, both actual fork orientations and actual other endpoint are retained; the adjacent q is never called farthest. Equation (D19generalpay) contradicts the ONE original norm. Every $g > 1$ dotted-head parameter is excluded.

N.21 The whole classical alternatives at rank nineteen

Proposition N.3. *At the original farthest vertex of a compact hyperbolic Coxeter* 19*-polytope the finite forest cannot be a whole* D_{19}.

Proof. Section N.1 first excludes every nonisolated head, then isolated own-zero/three/four/six and parallel. Its remaining head list is disjoint and exhaustive: native five, seven, eight, nine, finite label at least ten, or dotted. The independent OLD five floor in Section N.2 relies on none of these later exclusions.

Proposition N.1 excludes native five. Sections N.14–N.15 exclude native seven in both actual states at five. Section N.16 excludes native eight in both actual states at six. Section N.17 excludes native nine while retaining every weak-six omission. Section N.18 proves its own fresh late coverage and excludes the entire finite-label tail. Sections N.19–N.20 exclude the entire continuous dotted interval with correlated payments. Each branch uses actual walls at this SAME rank, one ORIGINAL farthest vector, every original-wall check and legal individual prices. No face is presumed Coxeter, no adjacent vertex presumed farthest, and no arithmeticity or stronger-bound census enters. The complete head partition is exhausted. □

Together with Propositions L.1 and M.1, this excludes all three whole classical alternatives at nineteen. Proper forests with spectators and the necessary forest cover have separate proofs; this appendix does not replace those obligations. Its conclusions enter that SAME-rank composition, not a rank-decrement principle.

N.22 The complete higher-dimensional conjunction

Proof of Theorem 5.1. Suppose a compact Coxeter polytope has dimension at least nineteen. The published compact ceiling puts its dimension at most twenty-nine [1, 2]. Proposition K.5 excludes all dimensions from twenty-four through twenty-nine. Proposition K.12 excludes twenty-three and twenty-two. Propositions K.20 and K.28 exclude twenty-one and twenty. At dimension nineteen, Proposition K.35 leaves only the whole forests A_{19}, B_{19}, D_{19} at the same original farthest vertex. Propositions L.1, M.1 and N.3 exclude them all. Every dimension in the required interval is therefore impossible. Each cover is proved at its own actual rank, and each whole-component consumer is used only after that cover has supplied its hypothesis. No dimension is inferred by passing to a face, and no arithmetic result enters the conjunction. □

O Proof dependencies and reproducibility

The proof has one common geometric foundation and three genuine branch points. Ordinary components and actual defects are treated in Appendix A. The chain and prefix arguments use that foundation but have different coordinate models and terminal classifications. Their component bounds are combined only after an actual-rank forest cover has fixed the allowed spectators.

At exact eighteen the complete cover in Section 5 uses the whole-A_{18} and whole-B_{18} proofs in Appendices G and H. Appendix I supplies the remaining head and fork cases; the main text closes the final dotted-head near/far case. The independent higher tail is proved in Appendices K–N, then joined in Section N.22. There is no dependency on the arithmetic companion's dimension bound.

The accompanying proof map records the source version of each local argument, its exact hypotheses and its verification records. It distinguishes written proof, finite arithmetic replay, conditional Lean deductions and PDF production. Historical intermediate bounds and unused numerical refinements are preserved as provenance rather than additional cases of this proof.

P External mathematical inputs

The external inputs are spherical and affine Coxeter classification, acute-angled compact-polytope simplicity and wall acuteness, the Lannér order bound of five, and the published compact dimension bound 29 for the tail beyond 29 [1, 2]. The actual-wall sign and positive projection conventions are those of [2, p. 42, §2.1]; all project-specific component and terminal statements are proved internally. The elementary real-cyclotomic degree formula $[\mathbb{Q}(\cos(\pi/m)) : \mathbb{Q}] = \phi(2m)/2$ for $m \geq 3$ is a standard algebraic input in the literal native-cosine analysis; it imposes no bound on a polytope ground field. Arithmeticity, field restrictions, integral dotted products, arbitrary Coxeter-face inheritance, and abstract Gram completions being realizable are not premises.

References

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[2] E. B. Vinberg, *Hyperbolic reflection groups*, Russian Math. Surveys **40**:1 (1985), 31–75, doi:10.1070/RM1985v040n01ABEH003527.